

Product Technical Appendix

Enterprise GenAI Full-Stack Architecture, October 2026

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This appendix holds the full record for every product assessed: current facts with their claim labels and source IDs (resolve in 06_References/bibliography.xlsx), the assessment, the classification and the scorecard (generic and regulated-FS weights). Fact cells marked *Not publicly verified* could not be confirmed from a public source and were never guessed. Disclosure: researched and drafted by an Anthropic model; Anthropic-related items were scored on the same rubric, and tiers set by the reader at checkpoints are noted in the relevant chapter.

L9: Evaluation and observability

MLflow (GenAI capabilities: tracing, evaluation, prompt registry, AI Gateway) (L9-mlflow-genai)

Tier: Strategic · **Flags:** none · **Original graphic label:** not in graphic

Rationale: Apache-2.0, Linux Foundation-hosted and OTel-native, with managed options on the major clouds; the most portable platform of record where an ML platform already exists ^{A1}.

Criterion	Score	Rationale
Technical	4	Tracing, evaluation, monitoring, prompt registry and gateway; red-teaming not native.
Enterprise readiness	4	Scored on the managed route the decision tree recommends. On Amazon SageMaker and Azure ML (A1-S103), platform IAM, SSO and audit logging are presumed under rule 6 (CP2 Q1), so 4 (within the rule 2 cap of 4); platform controls presumed (CP2 Q1); confirm per service. The Databricks, Nebius and OpenShift AI routes are not covered by the presumption, and the open-source tracking server's own controls remain NPV.
Security and compliance	3	Inherits host controls; clear Apache-2.0 licence and foundation hosting; managed-service certifications must be checked per provider.
Deployment flexibility	5	Five managed services, self-host and on-premises clusters.
Ecosystem	5	OTel- and MCP-native; managed by multiple clouds; integrated by other L9 tools (Langfuse guide).
Reliability and maturity	5	Since 2018, foundation-hosted.
Cost / TCO	4	Free; managed pricing per provider; tracking-server operations if self-hosted.

Criterion	Score	Rationale
Lock-in / portability	5	Apache-2.0, Linux Foundation governance, OTel.
Total (generic / FS)	4.25 / 4.25	

Evidence rules applied: Rule 2 (self-hosted software) applied: NPV cap not applied; enterprise_readiness and security_compliance scored on what the software enables and on project hygiene, capped at 4; inherits host controls; Rule 6 (CP2 Q1, hyperscaler presumption) applied at CP2 rework (8 October 2026) to the SageMaker and Azure ML managed routes: enterprise_readiness 3 -> 4; platform controls presumed (CP2 Q1); confirm per service

Capabilities. Apache-2.0 AI engineering platform: OTel-based tracing of LLM applications and agents, evaluation and quality monitoring, prompt registry and optimisation, and an AI Gateway, alongside the classic ML lifecycle ^{vi1}.

Strengths

- Managed offerings on Databricks, Amazon SageMaker, Azure ML, Nebius and Red Hat OpenShift AI, plus self-hosting and on-premises clusters ^{vi2}
- Tracing built on OpenTelemetry; native MCP integration ^{vi3}
- Hosted by the Linux Foundation since 2020 under a vendor-neutral open governance model ^{vi4}
- Years of production use; first release June 2018 ^{vi5}

Limitations and risks

- Strong Databricks association; code copyright Databricks ^{vi6}
- Security and access-control features of the open-source tracking server not verified in the fact base ^{NPV}
- GenAI evaluation depth relative to specialist tools not assessed in primary sources ^{NPV}
- Spans gateway (C1) and prompt management (C5), which can blur ownership boundaries ^{vi7}

Choose when

- MLflow already holds the firm's model inventory and lifecycle on Databricks, SageMaker or Azure ML, so GenAI evidence lands beside classic model evidence ^{AI}
- You want the most neutral governance and licence in the layer ^{AI}

Avoid when

¹ Verified fact — A1-S103: MLflow README (MLflow project (Databricks-originated) (GitHub)) <https://raw.githubusercontent.com/mlflow/mlflow/master/README.md> accessed 2026-10-07

² Verified fact — A1-S103: MLflow README (MLflow project (Databricks-originated) (GitHub)) <https://raw.githubusercontent.com/mlflow/mlflow/master/README.md> accessed 2026-10-07

³ Verified fact — A1-S103: MLflow README (MLflow project (Databricks-originated) (GitHub)) <https://raw.githubusercontent.com/mlflow/mlflow/master/README.md> accessed 2026-10-07

⁴ Verified fact — B-L9-S004: The MLflow Project Joins Linux Foundation (The Linux Foundation) <https://www.linuxfoundation.org/press/press-release/the-mlflow-project-joins-linux-foundation> accessed 2026-10-08 [search-tool extract]

⁵ Verified fact — A1-S019: PyPI JSON metadata: mlflow (Python Package Index) <https://pypi.org/pypi/mlflow/json> accessed 2026-10-07

⁶ Verified fact — A1-S019: PyPI JSON metadata: mlflow (Python Package Index) <https://pypi.org/pypi/mlflow/json> accessed 2026-10-07

⁷ Verified fact — A1-S103: MLflow README (MLflow project (Databricks-originated) (GitHub)) <https://raw.githubusercontent.com/mlflow/mlflow/master/README.md> accessed 2026-10-07

- You would self-host the open-source server without putting it behind the firm's identity and access controls ^{AJ}
- Teams need richer agent-debugging UX than the platform offers and will not run a second tool ^{AJ}

Nearest competitors: L9-langfuse, L9-wandb-weave, L9-opik

Regulated-FS note. For firms whose model risk inventory already lives in MLflow, this is the shortest path to SS1/23-style monitoring evidence for LLM components; on SageMaker or Azure ML the platform's IAM, SSO and audit logging are presumed under CP2 Q1 but must be confirmed per service, and certifications and residency must be verified per provider ^{AJ}.

Facts (as of 2026-10-07)

Field	Value	Label	Sources
Company	MLflow open-source project; code copyright Databricks, Inc.	Verified fact	A1-S019, A1-S103
Category	Open-source AI engineering platform for agents, LLMs and ML models	Verified fact	A1-S103
Version / lineup	mlflow 3.17.0, released 7 October 2026	Verified fact	A1-S019
Licence	Apache-2.0	Verified fact	A1-S019
Strategic direction	Repositioned as 'The Open Source AI Engineering Platform for Agents, LLMs & Models' with OTel-based tracing, evaluation and monitoring, prompt optimisation and an AI Gateway	Verified fact	A1-S103
What it does	Captures traces of LLM applications and agents (built on OpenTelemetry), evaluates and monitors quality, manages and optimises prompts, and offers an AI Gateway for cost and model access control, alongside classic ML lifecycle features.	Verified fact	A1-S103
Stack position	Cross-cutting L9 plus gateway (C1) and prompt management (C5) functions	Verified fact	A1-S103
Integration	Python, TypeScript/JavaScript, Java SDKs; natively integrates with OpenTelemetry and MCP	Verified fact	A1-S103
Dependencies	Tracking server (local or managed); deployable on Docker, Kubernetes, Azure ML, SageMaker	Verified fact	A1-S103
Certifications	Not publicly verified	Not public	

Field	Value	Label	Sources
		y verifie d	
GDPR / residency	Not publicly verified	Not publicl y verifie d	
Security features	Not publicly verified	Not publicl y verifie d	
Access controls	Not publicly verified	Not publicl y verifie d	
Enterprise support	Not publicly verified	Not publicl y verifie d	
Pricing	Open source free; managed pricing set by each cloud provider	Verifie d fact	A1-S019, A1-S103
Infrastructure cost	Not publicly verified	Not publicl y verifie d	
Ecosystem	Managed offerings on Databricks, SageMaker, Azure ML, Nebius, Red Hat OpenShift AI; tracing integrations incl. Databricks AI Gateway	Verifie d fact	A1-S103
Adoption signals	Not publicly verified	Not publicl y verifie d	
Maturity	First PyPI release 4 June 2018; 190 releases	Verifie d fact	A1-S019
Deployment	saas: Yes (managed by Databricks, Amazon SageMaker, Azure ML, Nebius, Red Hat OpenShift AI); managed_cloud: Yes; vpc_byoc: Not publicly verified; private_cloud: Yes; self_hosted: Yes; on_prem: Yes (on-premises clusters)		

Langfuse (L9-langfuse)

Tier: Strategic · **Flags:** Acquired · **Original graphic label:** Langfuse – open source

Rationale: MIT core, OTel ingestion, self-hosting and certified EU Cloud make it a credible platform of record; ClickHouse ownership and the enterprise-gated audit/RBAC modules are the caveats ^{A1}.

Criterion	Score	Rationale
Technical	4	Covers tracing, prompts, LLM-as-a-judge, datasets, experiments and annotation; no native red-teaming or documented trajectory metrics.
Enterprise readiness	4	SSO, project-level RBAC, audit logs and SCIM are documented, with SCIM, audit logs and project RBAC Enterprise-licensed when self-hosted (A1-S033): all three controls plus SCIM reach 4 under rule 7 (CP2 Q2). Not 5: SLA terms are not published and pages disagree on which tier carries SSO.
Security and compliance	4	SOC 2 Type II, ISO 27001, HIPAA BAA and isolated EU region on Cloud; no CMK/BYOK documented, so not 5 under the layer calibration.
Deployment flexibility	4	SaaS, private cloud, self-host and on-prem; BYOC not verified; air-gap not documented.
Ecosystem	4	OTel-native with broad integration guides; HTTP-only OTLP.
Reliability and maturity	4	GA, 555 PyPI releases since July 2023; ownership changed to ClickHouse in January 2026 with a stated unchanged roadmap.
Cost / TCO	4	Free MIT core and transparent Cloud pricing; ClickHouse operations and additive Enterprise pricing when self-hosted.
Lock-in / portability	3	Base 4 (MIT core, OTel ingest, own data model) reduced by 1 for the 2026 ownership change; governance is company-controlled.
Total (generic / FS)	3.95 / 3.85	

Evidence rules applied: Rule 7 (CP2 Q2) applied at CP2 rework (8 October 2026): SSO, RBAC and audit logs plus SCIM verified (A1-S033); enterprise_readiness 3 -> 4

Capabilities. Open-core LLM engineering platform: OTel-based tracing of LLM calls, retrieval and agent actions; prompt versioning; LLM-as-a-judge and code evaluators over ingested traces; user feedback and manual labelling; datasets, experiments and a playground; public API ^{VF8}.

Strengths

- MIT core covers tracing, evals, prompt management, experiments and annotation; self-hosting on Docker, Kubernetes (Helm) or Terraform templates for AWS, Azure and GCP ^{VF9}
- Native OTLP/HTTP endpoint and OTel-based SDKs, so instrumentation is not proprietary ^{VF10}
- Langfuse Cloud has SOC 2 Type II and ISO 27001, isolated EU, US, HIPAA and Japan regions, and a DPA ^{VF11}

⁸ Verified fact — A1-S073: Langfuse README (Langfuse / ClickHouse (GitHub)) <https://raw.githubusercontent.com/langfuse/langfuse/main/README.md> accessed 2026-10-07; A1-S032: OpenTelemetry (OTEL) for LLM Observability - Langfuse (Langfuse (ClickHouse)) <https://langfuse.com/integrations/native/opentelemetry> accessed 2026-10-07 [search-tool extract]

⁹ Verified fact — A1-S023: Langfuse repository LICENSE (Langfuse / ClickHouse, Inc.) <https://raw.githubusercontent.com/langfuse/langfuse/main/LICENSE> accessed 2026-10-07; A1-S033: Self-Hosted Pricing - Langfuse (Langfuse (ClickHouse)) <https://langfuse.com/pricing-self-host> accessed 2026-10-07 [search-tool extract]; A1-S073: Langfuse README (Langfuse / ClickHouse (GitHub)) <https://raw.githubusercontent.com/langfuse/langfuse/main/README.md> accessed 2026-10-07

¹⁰ Verified fact — A1-S032: OpenTelemetry (OTEL) for LLM Observability - Langfuse (Langfuse (ClickHouse)) <https://langfuse.com/integrations/native/opentelemetry> accessed 2026-10-07 [search-tool extract]

¹¹ Verified fact — A1-S029: Security & Compliance - Langfuse (Langfuse (ClickHouse)) <https://langfuse.com/security> accessed 2026-10-07 [search-tool extract]; A1-S030: Data Regions & Availability - Langfuse (Langfuse (ClickHouse)) <https://langfuse.com/security/data-regions> accessed 2026-10-07 [search-tool extract]

- Published, unit-based Cloud pricing ^{VF12}

Limitations and risks

- Owned by ClickHouse, Inc. (announced 16 January 2026); self-hosted Enterprise tier is bundled with ClickHouse commercial offerings ^{VF13}
- SCIM, audit logging, retention policies, project-level RBAC and server-side ingestion masking need an Enterprise licence key when self-hosted ^{VF14}
- OTLP over HTTP only (no gRPC); Langfuse-specific attributes take precedence over gen_ai.* ^{VF15}
- Self-hosted telemetry is on by default (opt-out) ^{VF16}
- SLA terms not published in sources reviewed ^{VF17}

Choose when

- You want a self-hosted platform of record for traces, evals and prompts inside your own estate, with an MIT core and an enterprise licence for the audit and RBAC modules ^{AJ}
- Framework-neutral instrumentation through OTEL matters more than deep integration with one agent framework ^{AJ}

Avoid when

- You need SCIM, audit logs and project RBAC but will not buy the Enterprise licence ^{AJ}
- You cannot operate a ClickHouse-backed service and do not want Langfuse Cloud ^{AJ}

Nearest competitors: L9-langsmith, L9-opik, L9-arize-phoenix, L9-mlflow-genai

Regulated-FS note. Best fit for a UK/EU manager that must keep traces (which contain client data) in-region or in-estate; budget for the Enterprise licence because audit logs and project RBAC are evidence controls, not extras; disable default telemetry ^{AJ}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Langfuse, acquired by ClickHouse, Inc. (announced January 2026); repository copyright now 'ClickHouse, Inc.'	Verified fact	A1-S021, A1-S022, A1-S023, A1-S073, V1-S007

¹² Verified fact — A1-S031: Pricing - Langfuse (Langfuse (ClickHouse)) <https://langfuse.com/pricing> accessed 2026-10-07 [search-tool extract]

¹³ Verified fact — A1-S021: ClickHouse raises \$400M Series D, acquires Langfuse, launches Postgres (ClickHouse, Inc.) <https://clickhouse.com/blog/clickhouse-raises-400-million-series-d-acquires-langfuse-launches-postgres> accessed 2026-10-07 [search-tool extract]; A1-S033: Self-Hosted Pricing - Langfuse (Langfuse (ClickHouse)) <https://langfuse.com/pricing-self-host> accessed 2026-10-07 [search-tool extract]; V2-S041: ClickHouse welcomes Langfuse (with langfuse.com 'Langfuse joins ClickHouse', Orrick) <https://clickhouse.com/blog/clickhouse-acquires-langfuse-open-source-llm-observability> accessed 2026-10-08 [search-tool extract]

¹⁴ Verified fact — A1-S033: Self-Hosted Pricing - Langfuse (Langfuse (ClickHouse)) <https://langfuse.com/pricing-self-host> accessed 2026-10-07 [search-tool extract]

¹⁵ Verified fact — A1-S032: OpenTelemetry (OTEL) for LLM Observability - Langfuse (Langfuse (ClickHouse)) <https://langfuse.com/integrations/native/opentelemetry> accessed 2026-10-07 [search-tool extract]

¹⁶ Verified fact — A1-S033: Self-Hosted Pricing - Langfuse (Langfuse (ClickHouse)) <https://langfuse.com/pricing-self-host> accessed 2026-10-07 [search-tool extract]

¹⁷ Verified fact — A1-S031: Pricing - Langfuse (Langfuse (ClickHouse)) <https://langfuse.com/pricing> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
Category	Open-core LLM engineering platform: tracing/observability, prompt management, evaluations, datasets and playground	Verified fact	A1-S073
Version / lineup	Python SDK langfuse 4.17.0, released 5 October 2026 (server release number not verified)	Verified fact	A1-S001
Licence	Open core: MIT for all product capabilities (tracing, evals, prompt management, experiments, annotation, playground); code under ee/directories needs a commercial licence key when self-hosted (SCIM, audit logging, data-retention policies, project-level RBAC, server-side ingestion masking)	Verified fact	A1-S023, A1-S033
Status events	Acquired by ClickHouse, Inc.; announced with ClickHouse's US\$400M Series D (announced 16 January 2026 per the ClickHouse blog); terms undisclosed	Verified fact	A1-S021, A1-S022, A1-S073, V1-S007
Strategic direction	Joined ClickHouse in January 2026; founders state the roadmap is unchanged and the commitment to open source and self-hosting remains; self-hosted Enterprise tier is now bundled with ClickHouse Cloud, BYOC or Private and priced additively	Verified fact	A1-S021, A1-S022, A1-S033
What it does	Instruments LLM applications and ingests traces of LLM calls, retrieval, embedding and agent actions; provides prompt versioning, LLM-as-a-judge and code evaluators, user feedback and manual labelling, datasets and experiments, a playground and a public API.	Verified fact	A1-S073
Stack position	Cross-cutting observability and evaluation platform spanning development and production; graphic places it in L9 as 'open source', which is accurate for the core but omits the open-core split	Verified fact	A1-S073, A1-S033
Integration	Native OTLP/HTTP endpoint (/api/public/otel, JSON and protobuf; no gRPC); Python SDK v3+ and JS SDK v4+ built on OpenTelemetry; maps gen_ai.* attributes to generations but Langfuse-specific attributes take precedence; OpenAPI spec and typed SDKs	Verified fact	A1-S032, A1-S073
Dependencies	Built on ClickHouse (analytics store); self-hosted via Docker Compose, Kubernetes (Helm, preferred for production) or Terraform templates for AWS, Azure and GCP	Verified fact	A1-S073
Certifications	Langfuse Cloud: SOC 2 Type II and ISO 27001 (reports on request, Pro/Team/Enterprise); HIPAA-ready region with BAA on eligible plans; annual third-party penetration tests	Verified fact	A1-S029
GDPR / residency	Isolated regions: EU (AWS eu-west-1, Ireland), US (us-west-2), HIPAA (us-west-2) and Japan (ap-northeast-1); DPA available; Postgres backups copied within the same jurisdiction; self-hosting for full isolation	Verified fact	A1-S029, A1-S030
Security features	Annual third-party penetration tests (Cloud); server-side ingestion masking (Enterprise module); self-hosted telemetry on by default (opt-out, but always on with an EE licence key)	Verified fact	A1-S029, A1-S033
Access controls	SCIM, audit logs, project-level RBAC and retention policies are Enterprise modules when self-hosted; pages are inconsistent on whether SSO/RBAC sit in the open-source tier	Verified fact	A1-S033
Enterprise support	Enterprise plan (US\$2,499/month on Cloud); SLA terms not published in sources reviewed	Verified fact	A1-S031
Pricing	Cloud: Hobby free (50k units/month, 30 days, 2 users); Core US\$29/month; Pro US\$199/month; Enterprise US\$2,499/month; 100k	Verified fact	A1-S031

Field	Value	Label	Sources
	units included on paid plans, then US\$8 per 100k units, falling to US\$6 per 100k above 50M; unit = trace + observation + score		
Infrastructure cost	Self-hosted Enterprise pricing is additive to a ClickHouse commercial plan; infrastructure sizing not verified	Verified fact	A1-S033
Ecosystem	OpenTelemetry-based; integration guides for OpenLLMetry, MLflow, vLLM, Spring AI, GitHub Copilot and Go/Java/C#/Ruby via OTel	Verified fact	A1-S032
Adoption signals	Over 20K GitHub stars and more than 26M SDK installs per month at end of 2025 (ClickHouse figures)	Verified fact	A1-S021
Maturity	First PyPI release 12 July 2023; 555 PyPI releases to 5 October 2026 (frequent cadence)	Verified fact	A1-S001
Deployment	saas: Yes; managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Yes; self_hosted: Yes; on_prem: Yes		

LangSmith (L9-langsmith)

Tier: Strategic · **Flags:** none · **Original graphic label:** LangSmith – trace & eval

Rationale: Strongest commercial option for LangGraph estates with the certifications and deployment options FS needs. Strategic is conditional: only inside a LangGraph estate, and only with OTel dual-instrumentation as the exit route, because lock-in scores 2 ^{AJ}.

Criterion	Score	Rationale
Technical	4	Offline and online evals, pytest plugin, prompt management and failure clustering; Engine still beta; red-teaming not native.
Enterprise readiness	4	SSO, RBAC, ABAC, gateway audit logging and support SLA on Enterprise; SCIM not documented in the fact base.
Security and compliance	4	SOC 2 Type II, ISO 27001:2022, HIPAA BAA, EU region; no CMK/BYOK documented, so not 5 under the layer calibration.
Deployment flexibility	5	SaaS (US/EU), BYOC (AWS), self-hosted and air-gapped self-host on Enterprise.
Ecosystem	4	Large LangChain ecosystem plus OTLP ingest; native format preferred.
Reliability and maturity	4	GA since 2023, 536 SDK releases, independent and funded (US\$125M Series B, October 2025).

Criterion	Score	Rationale
Cost / TCO	3	Transparent but per-seat plus per-trace; heavy self-host footprint.
Lock-in / portability	2	Proprietary platform, native format recommended, growing bundling with the agent runtime.
Total (generic / FS)	3.95 / 3.80	

Capabilities. Proprietary observability, evaluation and deployment platform: native and OTEL trace ingest, offline and online evaluation on datasets, prompt management, pytest plugin, and hosted agent deployment; now also Engine (beta), Fleet, Sandboxes and an LLM Gateway ^{vr18}.

Strengths

- Deepest coupling to LangGraph/LangChain, the orchestration framework many teams already use ^{vr19}
- SOC 2 Type II, ISO 27001:2022, HIPAA (BAA on Enterprise) and GDPR; US or EU managed regions on all tiers ^{vr20}
- SaaS, BYOC on AWS, self-hosted and air-gapped self-host on Enterprise ^{vr21}
- Enterprise SSO, RBAC and ABAC; support SLA ^{vr22}

Limitations and risks

- Native tracing format is recommended over OTEL for performance, which pulls instrumentation towards a proprietary format ^{vr23}
- Platform increasingly bundles the LangChain agent runtime (Fleet, Deployment), widening the blast radius of a vendor change ^{vr24}
- Self-hosting needs at least 16 vCPU / 64 GB plus PostgreSQL, Redis and ClickHouse ^{vr25}
- Per-seat Plus pricing plus per-trace coverage ^{vr26}

¹⁸ Verified fact — A1-S034: Set up LangSmith (deployment options) - Docs by LangChain (LangChain, Inc.) <https://docs.langchain.com/langsmith/platform-setup> accessed 2026-10-07 [search-tool extract]; A1-S037: Introducing End-to-End OpenTelemetry Support in LangSmith (LangChain, Inc.) <https://www.langchain.com/blog/end-to-end-opentelemetry-langsmith> accessed 2026-10-07 [search-tool extract]; A1-S039: Everything we shipped at Interrupt (LangChain, Inc.) <https://www.langchain.com/blog/interrupt-2026-overview> accessed 2026-10-07 [search-tool extract]; A1-S108: LangSmith Python SDK README (LangChain, Inc. (GitHub)) <https://raw.githubusercontent.com/langchain-ai/langsmith-sdk/main/python/README.md> accessed 2026-10-07

¹⁹ Verified fact — A1-S037: Introducing End-to-End OpenTelemetry Support in LangSmith (LangChain, Inc.) <https://www.langchain.com/blog/end-to-end-opentelemetry-langsmith> accessed 2026-10-07 [search-tool extract]; A1-S039: Everything we shipped at Interrupt (LangChain, Inc.) <https://www.langchain.com/blog/interrupt-2026-overview> accessed 2026-10-07 [search-tool extract]

²⁰ Verified fact — A1-S036: LangSmith Regions FAQ - Docs by LangChain (LangChain, Inc.) <https://docs.langchain.com/langsmith/regions-faq> accessed 2026-10-07 [search-tool extract]; A1-S126: LangSmith Engine security - Docs by LangChain (LangChain, Inc.) <https://docs.langchain.com/langsmith/engine-security> accessed 2026-10-07 [search-tool extract]

²¹ Verified fact — A1-S034: Set up LangSmith (deployment options) - Docs by LangChain (LangChain, Inc.) <https://docs.langchain.com/langsmith/platform-setup> accessed 2026-10-07 [search-tool extract]

²² Verified fact — A1-S035: LangSmith Plans and Pricing (LangChain, Inc.) <https://www.langchain.com/pricing> accessed 2026-10-07 [search-tool extract]

²³ Verified fact — A1-S037: Introducing End-to-End OpenTelemetry Support in LangSmith (LangChain, Inc.) <https://www.langchain.com/blog/end-to-end-opentelemetry-langsmith> accessed 2026-10-07 [search-tool extract]

²⁴ Verified fact — A1-S039: Everything we shipped at Interrupt (LangChain, Inc.) <https://www.langchain.com/blog/interrupt-2026-overview> accessed 2026-10-07 [search-tool extract]

²⁵ Verified fact — A1-S034: Set up LangSmith (deployment options) - Docs by LangChain (LangChain, Inc.) <https://docs.langchain.com/langsmith/platform-setup> accessed 2026-10-07 [search-tool extract]

²⁶ Verified fact — A1-S035: LangSmith Plans and Pricing (LangChain, Inc.) <https://www.langchain.com/pricing> accessed 2026-10-07 [search-tool extract]

- LangSmith Engine is public beta ^{vr27}

Choose when

- LangGraph is your orchestration standard and you want one supported platform for traces, evals and deployment ^{AJ}
- You need a commercial vendor with SOC 2, ISO 27001 and an EU region, or BYOC/self-host for residency ^{AJ}

Avoid when

- Your agents are framework-heterogeneous and you want a neutral trace store ^{AJ}
- You do not want the observability vendor to also be your agent runtime vendor ^{AJ}

Nearest competitors: L9-langfuse, L9-braintrust, L9-arize-ax

Regulated-FS note. Use the EU region or self-host for client data; instrument with OTel GenAI conventions even if native SDKs are also used, so the evidence trail survives a platform exit; keep Engine-proposed fixes behind change control ^{AJ}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	LangChain, Inc. (independent; US\$125M Series B at US\$1.25B valuation, 20 October 2025)	Verified fact	A1-S038
Category	Proprietary agent and LLM observability, evaluation and deployment platform (now also hosting LangSmith Fleet, Engine, Sandboxes and an LLM Gateway)	Verified fact	A1-S034, A1-S039
Version / lineup	SaaS (continuously released); Python SDK langsmith 0.14.4, released 2 October 2026	Verified fact	A1-S002
Licence	Proprietary platform; SDK MIT-licensed	Verified fact	A1-S002, A1-S108
Status events	Agent Builder renamed LangSmith Fleet (2026); LangSmith Deployment billing moved to usage-based model; existing customers migrated 1 October 2026	Verified fact	A1-S035, A1-S039, V1-S016
Strategic direction	Interrupt 2026 (13-14 May): LangSmith Engine (public beta; clusters production failures and proposes fixes, can open PRs), SmithDB (claimed up to 15x faster; simpler self-hosting), Sandboxes GA, LLM Gateway with audit logging; Agent Builder renamed LangSmith Fleet; new Fleet pricing from 15 July 2026	Verified fact	A1-S039

²⁷ Verified fact — A1-S039: Everything we shipped at Interrupt (LangChain, Inc.) <https://www.langchain.com/blog/interrupt-2026-overview> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
What it does	Captures traces of LLM and agent runs, runs offline and online evaluations on datasets, manages prompts and hosts agent deployments; ingests native and OpenTelemetry traces.	Verified fact	A1-S034, A1-S037, A1-S108
Stack position	Cross-cutting observability/evaluation plus agent deployment runtime; broader than the graphic's 'trace & eval' label	Verified fact	A1-S034, A1-S039
Integration	Native SDKs (Python/JS); OTLP endpoint for any OTel exporter (x-api-key header); original OTel launch required OpenLLMetry conventions, newer guides use OTel GenAI conventions; native format recommended when LangSmith is sole backend; pytest plugin in SDK	Verified fact	A1-S037, A1-S108
Dependencies	Self-hosted: Kubernetes (min. 16 vCPU, 64 GB) or Docker, PostgreSQL, Redis, ClickHouse, optional blob storage	Verified fact	A1-S034
Certifications	SOC 2 Type II; ISO 27001:2022 (stated in LangChain resource page and LangSmith Engine security docs); HIPAA (BAA on Enterprise plan only); GDPR; reports via trust.langchain.com	Verified fact	A1-S036, A1-S126, V1-S068
GDPR / residency	Managed cloud in US or EU (EU data stored in the Netherlands per academy page) on all tiers at no extra cost; DPA on request; trust centre trust.langchain.com	Verified fact	A1-S036
Security features	Not publicly verified	Not publicly verified	
Access controls	Enterprise: custom SSO, ABAC and RBAC; audit logging for LLM Gateway administrative actions	Verified fact	A1-S035, A1-S039
Enterprise support	Enterprise plan with support SLA, custom seats and workspaces	Verified fact	A1-S035
Pricing	Developer US\$0 (1 seat, 5k base traces/month); Plus US\$39 per seat/month with 10k traces included, overage per trace; Enterprise custom, invoiced annually	Verified fact	A1-S035
Infrastructure cost	Self-hosting requires a Kubernetes cluster of at least 16 vCPU / 64 GB plus PostgreSQL, Redis and ClickHouse	Verified fact	A1-S034
Ecosystem	Works with LangChain/LangGraph and any framework via SDK or OTel; guides for Temporal, VS Code Copilot Chat and Microsoft Agent Framework	Verified fact	A1-S037
Adoption signals	LangChain Series B investors include IVP (lead), Sequoia, Benchmark, CapitalG, Sapphire, ServiceNow, Workday, Cisco, Datadog and Databricks	Verified fact	A1-S038
Maturity	First langsmith SDK release on PyPI 26 June 2023; 536 releases to 2 October 2026	Verified fact	A1-S002
Deployment	saas: Yes; managed_cloud: Not publicly verified; vpc_byoc: Yes (BYOC on AWS, Enterprise only); private_cloud: Yes (self-hosted in own cloud, Enterprise); self_hosted: Yes (Enterprise); on_prem: Yes (air-gapped self-host positioned)		

Opik (L9-opik)

Tier: Tactical • **Flags:** none • **Original graphic label:** Opik – comet

Rationale: Most permissive full-platform licence in the layer, but missing user management in the open edition and a young codebase keep it below the foundational tier ^{AJ}.

Criterion	Score	Rationale
Technical	4	Tracing, offline and online evals, guardrails and optimiser; red-teaming not native.
Enterprise readiness	3	Enterprise SAML/OIDC SSO, organisation and workspace roles with Enterprise custom roles (B-REV-S022); audit logs only on marketing pages; open edition has no user management.
Security and compliance	4	SOC 2 Type 2 and ISO 27001:2022 at Comet level; residency NPV; no CMK documented.
Deployment flexibility	4	Comet SaaS, self-host on Kubernetes, on-prem as part of Comet; BYOC not verified.
Ecosystem	3	OTel HTTP, PyTest, MCP and a moderate integration list.
Reliability and maturity	3	635 releases since September 2024; young and fast-moving.
Cost / TCO	5	Free full-platform self-hosting and low cloud entry price.
Lock-in / portability	4	Apache-2.0 platform with OTel ingest; company-governed, so not 5.
Total (generic / FS)	3.75 / 3.75	

Evidence rules applied: Rule 8 (CP2 Q4) checked at CP2 rework (8 October 2026): SOC 2 Type 2 and ISO 27001 are listed against the Opik Enterprise plan (A1-S122), so product scope is stated; security_compliance 4 kept

Capabilities. Apache-2.0 platform from Comet for tracing LLM calls and agents, datasets and experiments with LLM-as-a-judge metrics (hallucination, moderation, RAG), online evaluation rules on production traces, dashboards, guardrails and an agent/prompt optimiser; PyTest integration, OTel over HTTP, MCP server ^{vi28}.

Strengths

- The whole platform, including online evaluation and the optimiser, is Apache-2.0 ^{vi29}
- Online evaluation rules apply LLM-as-a-judge to production traces ^{vi30}
- Comet Trust Center lists SOC 2 Type 2 and ISO/IEC 27001:2022 ^{vi31}
- Low published cloud pricing and free self-hosting ^{vi32}

Limitations and risks

²⁸ Verified fact — A1-S067: Opik README (Comet ML (GitHub)) <https://raw.githubusercontent.com/comet-ml/opik/main/README.md> accessed 2026-10-07; A1-S070: Opik OpenTelemetry integration docs (Comet ML (Opik docs source)) <https://raw.githubusercontent.com/comet-ml/opik/main/apps/opik-documentation/documentation/fern/docs-v2/integrations/opentelemetry.mdx> accessed 2026-10-07; A1-S072: Opik Online Evaluation rules (Comet ML (Opik docs source)) <https://raw.githubusercontent.com/comet-ml/opik/main/apps/opik-documentation/documentation/fern/docs-v2/production/online-evaluation/rules.mdx> accessed 2026-10-07

²⁹ Verified fact — A1-S050: Opik LICENSE (Apache-2.0, Comet ML) (Comet ML (GitHub)) <https://raw.githubusercontent.com/comet-ml/opik/main/LICENSE> accessed 2026-10-07; A1-S067: Opik README (Comet ML (GitHub)) <https://raw.githubusercontent.com/comet-ml/opik/main/README.md> accessed 2026-10-07

³⁰ Verified fact — A1-S072: Opik Online Evaluation rules (Comet ML (Opik docs source)) <https://raw.githubusercontent.com/comet-ml/opik/main/apps/opik-documentation/documentation/fern/docs-v2/production/online-evaluation/rules.mdx> accessed 2026-10-07

³¹ Verified fact — A1-S122: Comet Trust Center (Comet ML, Inc.) <https://trust.comet.com/> accessed 2026-10-07 [search-tool extract]

³² Verified fact — A1-S123: Comet pricing (Opik plans) (Comet ML, Inc.) <https://www.comet.com/site/pricing/> accessed 2026-10-07 [search-tool extract]

- Self-hosted open source has no user management; SSO, custom roles and service accounts come with Enterprise ^{VF33}
- Local Docker install is not production-ready; Kubernetes/Helm required ^{VF34}
- OTel over HTTP only ^{VF35}
- EU region and data residency not publicly verified ^{NPV}
- About two years old with very frequent releases ^{VF36}

Choose when

- You want a fully Apache-2.0 platform with online evaluation and are willing to buy Comet Enterprise for user management ^{AJ}
- You already run Comet for ML experiment tracking ^{AJ}

Avoid when

- You plan to self-host the open-source edition for multiple teams without an identity layer ^{AJ}
- You need a verified EU SaaS region ^{AJ}

Nearest competitors: L9-langfuse, L9-arize-phoenix, L9-mlflow-genai

Regulated-FS note. Self-host for residency, but only with Enterprise user management or a fronting identity proxy; without it the trace store has no access control or attribution ^{AJ}.

Facts (as of 2026-10-07)

Field	Value	Label	Sources
Company	Comet ML, Inc. (Comet)	Verified fact	A1-S050, A1-S071, A1-S006
Category	Open-source LLM observability, evaluation, prompt/agent optimisation and guardrails platform	Verified fact	A1-S067
Version / lineup	opik 2.2.94 (Python SDK), released 7 October 2026	Verified fact	A1-S006
Licence	Apache-2.0 for the full platform (backend, UI, evaluation, online evaluation, optimiser)	Verified fact	A1-S050, A1-S067
Strategic	Agent Optimizer SDK, Opik Guardrails, online evaluation rules, MCP	Verified	A1-S067,

³³ Verified fact — A1-S069: Self-Hosting Opik: Overview (Comet ML (Opik docs source)) <https://raw.githubusercontent.com/comet-ml/opik/main/apps/opik-documentation/documentation/fern/docs-v2/self-host/overview.mdx> accessed 2026-10-07; A1-S122: Comet Trust Center (Comet ML, Inc.) <https://trust.comet.com/> accessed 2026-10-07 [search-tool extract]; A1-S123: Comet pricing (Opik plans) (Comet ML, Inc.) <https://www.comet.com/site/pricing/> accessed 2026-10-07 [search-tool extract]; B-REV-S022: Opik: Roles and Permissions (and SSO pages) (Comet ML, Inc.) https://www.comet.com/docs/opik/administration/roles_and_permissions accessed 2026-10-08 [search-tool extract]

³⁴ Verified fact — A1-S069: Self-Hosting Opik: Overview (Comet ML (Opik docs source)) <https://raw.githubusercontent.com/comet-ml/opik/main/apps/opik-documentation/documentation/fern/docs-v2/self-host/overview.mdx> accessed 2026-10-07

³⁵ Verified fact — A1-S070: Opik OpenTelemetry integration docs (Comet ML (Opik docs source)) <https://raw.githubusercontent.com/comet-ml/opik/main/apps/opik-documentation/documentation/fern/docs-v2/integrations/opentelemetry.mdx> accessed 2026-10-07

³⁶ Verified fact — A1-S006: PyPI JSON metadata: opik (Python Package Index) <https://pypi.org/pypi/opik/json> accessed 2026-10-07

Field	Value	Label	Sources
direction	server; positioned as free-to-self-host alternative to platforms gating self-hosting behind Enterprise plans	d fact	A1-S072
What it does	Traces LLM calls and agent activity, runs datasets/experiments with LLM-as-a-judge metrics (hallucination, moderation, RAG), scores production traces with online evaluation rules, and provides dashboards, guardrails and prompt optimisation.	Verified fact	A1-S067, A1-S072
Stack position	Cross-cutting observability and evaluation; can run standalone or inside the Comet MLOps platform	Verified fact	A1-S071
Integration	Python and TypeScript SDKs, REST API, OpenTelemetry (HTTP transport only), PyTest integration for CI, MCP server	Verified fact	A1-S067, A1-S070
Dependencies	Self-host via local Docker install (not production-ready) or Kubernetes/Helm; ClickHouse cold-tier and backup guides exist	Verified fact	A1-S069, A1-S067
Certifications	Comet Trust Center: SOC 2 Type 2, ISO/IEC 27001:2022, ISO 9001:2015; HIPAA and GDPR listed as Enterprise-plan features (not certifications)	Verified fact	A1-S122
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Self-hosted open source has no user management; Enterprise adds SSO and service accounts	Verified fact	A1-S069, A1-S122, A1-S123
Enterprise support	Enterprise plan: dedicated support and SLAs, flexible deployments	Verified fact	A1-S122, A1-S123
Pricing	Self-hosted free; Free cloud (25k spans/month, 10 members, 60-day retention); Pro Cloud US\$19/month (100k spans, 50 members); Enterprise custom	Verified fact	A1-S123, A1-S067
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Integrations incl. Google ADK, Autogen, Flowise AI, Cloudflare Workers AI and OpenTelemetry	Verified fact	A1-S067
Adoption signals	20,000+ GitHub stars (vendor README); claims scale of 40M+ traces/day	Verified fact	A1-S067
Maturity	First PyPI release 2 September 2024; 635 releases to 7 October 2026	Verified fact	A1-S006
Deployment	saas: Yes (Comet-hosted); managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Yes; self_hosted: Yes; on_prem: Yes (as part of self-hosted Comet platform)		

Arize AX (L9-arize-ax)

Tier: Tactical · **Flags:** Acquired, Duplicated · **Original graphic label:** Arize – RAG metrics

Rationale: Technically strong and FS-deployable, but acquired seven days ago with a convergence roadmap; re-assess for Strategic once Dynatrace publishes product plans ^{vi}.

Criterion	Score	Rationale
Technical	4	Online and offline evals, monitoring, alerting and failure-pattern grouping at volume; red-teaming not native.
Enterprise readiness	4	SAML SSO, RBAC, audit trails, multi-tenancy, SLAs; SCIM not documented in the fact base.
Security and compliance	4	SOC 2 Type II, PCI DSS 4.0, HIPAA, EU region and self-host; ISO 27001 unconfirmed on the official compliance page, so 4.
Deployment flexibility	5	SaaS, VPC, customer-managed cloud, self-host and air-gap.
Ecosystem	4	OpenInference catalogue shared with Phoenix; increasingly tied to Dynatrace.
Reliability and maturity	3	Commercial since 2020, but ownership changed on 1 October 2026 with platform convergence planned.
Cost / TCO	3	Transparent Free and Pro tiers, no per-seat charges; Enterprise custom.
Lock-in / portability	2	Base 3 (proprietary backend on portable instrumentation) reduced by 1 for the 2026 ownership change.
Total (generic / FS)	3.85 / 3.70	

Capabilities. Commercial agent observability and evaluation: managed tracing, online and offline evaluation, monitoring and alerting at high volume, managed evaluation compute, multi-tenancy via organisations and spaces, and Signal, a scheduled worker that groups recurring failure patterns into issues ^{vi37}.

Strengths

- Same OTel/OpenInference instrumentation as Phoenix, with a documented Phoenix-to-AX migration by dual-writing ^{vi38}
- SaaS, VPC, customer-managed cloud, self-hosted and air-gapped deployment ^{vi39}
- SAML SSO with RBAC mapping, audit trails, dedicated support and SLAs on Enterprise ^{vi40}
- SOC 2 Type II, PCI DSS 4.0 Level 1, HIPAA (Enterprise) and an EU data region in Belgium ^{vi41}

³⁷ Verified fact — A1-S046: Arize Phoenix or Arize AX - Phoenix docs (Arize AI) <https://arize.com/docs/phoenix/resources/frequently-asked-questions/what-is-the-difference-between-phoenix-and-arize> accessed 2026-10-07 [search-tool extract]; A1-S047: Arize AX Pricing: AI agent observability and evaluation (Arize AI) <https://arize.com/pricing/> accessed 2026-10-07 [search-tool extract]

³⁸ Verified fact — A1-S046: Arize Phoenix or Arize AX - Phoenix docs (Arize AI) <https://arize.com/docs/phoenix/resources/frequently-asked-questions/what-is-the-difference-between-phoenix-and-arize> accessed 2026-10-07 [search-tool extract]; A1-S049: OpenInference LICENSE (Apache-2.0) (Arize AI (GitHub)) <https://raw.githubusercontent.com/Arize-ai/openinference/main/LICENSE> accessed 2026-10-07

³⁹ Verified fact — A1-S046: Arize Phoenix or Arize AX - Phoenix docs (Arize AI) <https://arize.com/docs/phoenix/resources/frequently-asked-questions/what-is-the-difference-between-phoenix-and-arize> accessed 2026-10-07 [search-tool extract]; A1-S047: Arize AX Pricing: AI agent observability and evaluation (Arize AI) <https://arize.com/pricing/> accessed 2026-10-07 [search-tool extract]

⁴⁰ Verified fact — A1-S046: Arize Phoenix or Arize AX - Phoenix docs (Arize AI) <https://arize.com/docs/phoenix/resources/frequently-asked-questions/what-is-the-difference-between-phoenix-and-arize> accessed 2026-10-07 [search-tool extract]; A1-S124: Compliance - Arize AX Docs (Arize AI) <https://arize.com/docs/ax/security-and-settings/compliance> accessed 2026-10-07 [search-tool extract]

Limitations and risks

- Dynatrace completed the acquisition on 1 October 2026 and plans to converge the product into its platform ^{VF42}
- Runs on the proprietary adb OLAP database ^{VF43}
- ISO 27001 appears on marketing pages but not on the official AX compliance page; request the certificate ^{VF44}
- Enterprise pricing is custom ^{VF45}

Choose when

- You run Dynatrace already, or you need managed high-volume online evaluation with air-gapped or self-hosted options ^{AJ}
- Phoenix is in use for development and you want a supported production tier on the same instrumentation ^{AJ}

Avoid when

- You need roadmap certainty in the next two to three quarters while Dynatrace integration plans settle ^{AJ}
- ISO 27001 is a hard procurement gate and the certificate cannot be produced ^{AJ}

Nearest competitors: L9-langsmith, L9-datadog-agent-observability, L9-braintrust

Regulated-FS note. Self-hosted AX (Arize stores nothing) answers trace residency; re-run third-party due diligence and the DORA register entry after the ownership change, and obtain the ISO certificate before relying on it ^{AJ}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Arize AI, a Dynatrace company (acquisition completed 1 October 2026)	Verified fact	A1-S044, A1-S045, V1-S005
Category	Commercial AI agent observability and evaluation platform (SaaS or licensed private deployment)	Verified fact	A1-S046, A1-S047
Version / lineup	SaaS; Python SDK arize 8.58.0, released 7 October 2026; tiers AX Free, AX Pro, AX Enterprise	Verified fact	A1-S007, A1-S047
Licence	Proprietary; self-hosting requires a commercial licence; Python SDK	Verified fact	A1-S046,

⁴¹ Verified fact — A1-S124: Compliance - Arize AX Docs (Arize AI) <https://arize.com/docs/ax/security-and-settings/compliance> accessed 2026-10-07 [search-tool extract]; A1-S047: Arize AX Pricing: AI agent observability and evaluation (Arize AI) <https://arize.com/pricing/> accessed 2026-10-07 [search-tool extract]

⁴² Verified fact — A1-S045: Dynatrace Completes Acquisition of Arize, Extending AI Observability Across the Full Development Lifecycle (Dynatrace, Inc.) <https://ir.dynatrace.com/news-events/press-releases/detail/439/dynatrace-completes-acquisition-of-arize-extending-ai-observability-across-the-full-development-lifecycle> accessed 2026-10-07 [search-tool extract]; V1-S005: Dynatrace Completes Acquisition of Arize (IR press release) (Dynatrace, Inc.) <https://ir.dynatrace.com/news-events/press-releases/detail/439/dynatrace-completes-acquisition-of-arize-extending-ai-observability-across-the-full-development-lifecycle> accessed 2026-10-08 [search-tool extract]

⁴³ Verified fact — A1-S046: Arize Phoenix or Arize AX - Phoenix docs (Arize AI) <https://arize.com/docs/phoenix/resources/frequently-asked-questions/what-is-the-difference-between-phoenix-and-arize> accessed 2026-10-07 [search-tool extract]

⁴⁴ Verified fact — A1-S124: Compliance - Arize AX Docs (Arize AI) <https://arize.com/docs/ax/security-and-settings/compliance> accessed 2026-10-07 [search-tool extract]; V1-S084: Compliance - Arize AX Docs; Arize AX pricing; self-hosted page (Arize AI) <https://arize.com/docs/ax/security-and-settings/compliance> accessed 2026-10-08 [search-tool extract]

⁴⁵ Verified fact — A1-S047: Arize AX Pricing: AI agent observability and evaluation (Arize AI) <https://arize.com/pricing/> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
	Apache-2.0	d fact	A1-S007
Status events	Dynatrace definitive agreement to acquire Arize, 13 August 2026 (US\$915M, approx. US\$815M cash); Acquisition completed 1 October 2026	Verified fact	A1-S044, A1-S045, V1-S005
Strategic direction	Signal (scheduled worker grouping recurring failure patterns into issues), Alyx assistant, ADB data fabric; post-acquisition integration into Dynatrace planned	Verified fact	A1-S046, A1-S047, A1-S045
What it does	Managed tracing, online and offline evaluation, monitoring and alerting for LLM and agent applications at high volume, with managed evaluation compute and multi-tenancy via organisations and spaces.	Verified fact	A1-S046
Stack position	Cross-cutting production observability and evaluation; graphic's 'RAG metrics' descriptor understates scope	Verified fact	A1-S046, A1-S047
Integration	OpenTelemetry and OpenInference instrumentation shared with Phoenix; documented Phoenix-to-AX migration via dual-writing	Verified fact	A1-S046, A1-S049
Dependencies	Runs on Arize's proprietary adb OLAP database	Verified fact	A1-S046
Certifications	SOC 2 Type II; PCI DSS 4.0 Level 1 Service Provider; CSA STAR Level 1 self-assessment; HIPAA (Enterprise tier only per pricing table; docs say compliant, self-hosted page says HIPAA-ready); ISO/IEC 27001 (A-LIGN, announced c. August 2026) is stated on marketing pages but is NOT listed on the official AX compliance docs page - confirm the certificate before relying on it	Verified fact	A1-S124, A1-S047, V1-S084
GDPR / residency	EU data region in Belgium; self-hosted (customer Kubernetes and object storage, Arize stores nothing) and air-gapped options; GDPR compliance claimed	Verified fact	A1-S124
Security features	Not publicly verified	Not publicly verified	
Access controls	SSO, RBAC and audit trails; self-hosted AX supports any SAML 2.0 IdP (Okta, Entra, OneLogin, Ping) with RBAC role mapping; 'Enterprise SSO' on Enterprise tier; multi-tenancy via organisations and spaces	Verified fact	A1-S046, A1-S124
Enterprise support	Dedicated support and SLAs (Enterprise)	Verified fact	A1-S046, A1-S047
Pricing	AX Free US\$0 (25k spans, 1 GB, 15-day retention); AX Pro US\$50/month (50k spans, 10 GB, 30 days, overages); AX Enterprise custom; no per-seat charges, unlimited users and evaluations	Verified fact	A1-S047
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Same integration catalogue as Phoenix via OpenInference	Verified fact	A1-S046, A1-S066
Adoption signals	Acquisition value US\$915M (Dynatrace)	Verified fact	A1-S044

Field	Value	Label	Sources
Maturity	arize SDK first published on PyPI 25 March 2020; 369 releases	Verified fact	A1-S007
Deployment	saas: Yes; managed_cloud: Not publicly verified; vpc_byoc: Yes (VPC deployment, Enterprise); private_cloud: Yes (customer-managed cloud, commercial licence); self_hosted: Yes (commercial licence); on_prem: Yes (on-prem / air-gapped, Enterprise)		

DeepEval (L9-deepeval)

Tier: Tactical · **Flags:** none · **Original graphic label:** DeepEval – LLM unit tests

Rationale: The most complete open metric library for the layer’s questions, but it is a substitutable component of an in-house eval harness, not a platform ^{A1}.

Criterion	Score	Rationale
Technical	4	Broadest metric coverage (RAG, agentic, trajectory, G-Eval, DAG); no tracing store of its own.
Enterprise readiness	3	Self-hosted library (rule 2): enables CI evaluation in-estate; commercial platform exists but its access controls are not verified.
Security and compliance	3	Inherits host controls; clear Apache-2.0 licence; Confident AI SOC 2/HIPAA are vendor statements with gating conflicts.
Deployment flexibility	5	Library runs anywhere, including air-gapped with local judges; Confident AI SaaS, VPC and self-host options.
Ecosystem	4	Many framework integrations; OTel export mainly to Confident AI.
Reliability and maturity	3	527 releases since August 2023; fast-changing metric set; funding NPV.
Cost / TCO	4	Free; judge-token spend dominates.
Lock-in / portability	4	Apache-2.0 library; company-governed, with commercial steering.
Total (generic / FS)	3.75 / 3.70	

Evidence rules applied: Rule 2 (self-hosted software) applied: NPV cap not applied; enterprise_readiness and security_compliance scored on what the software enables and on project hygiene, capped at 4; inherits host controls

Capabilities. Apache-2.0 Python evaluation framework, ‘similar to Pytest but specialised for unit testing LLM apps’: LLM-as-a-judge, statistical and NLP metrics, including G-Eval, DAG, RAG faithfulness and contextual recall/precision, and agentic metrics (task completion, tool correctness, plan adherence) with trajectory evaluation; runs in any CI/CD ^{vr46}.

Strengths

- Covers the layer’s research questions directly: faithfulness, contextual recall/precision, tool correctness and trajectory ^{vr47}

⁴⁶ Verified fact — A1-S068: DeepEval README (Confident AI (GitHub)) <https://raw.githubusercontent.com/confident-ai/deepeval/main/README.md> accessed 2026-10-07

- Runs locally with any LLM as judge or with local NLP models, so evaluation can stay inside the estate ^{vf-48}
- Broad framework integrations (LangGraph, Pydantic AI, CrewAI, OpenAI Agents, Google ADK, LlamaIndex and others) ^{vf-49}
- Apache-2.0 ^{vf-50}

Limitations and risks

- Commercial features steer to the Confident AI platform ^{vf-51}
- OTel export is primarily to Confident AI's OTLP/HTTP endpoint; export to arbitrary collectors not confirmed ^{vf-52}
- Confident AI certifications are vendor statements with inconsistent plan gating ^{vf-53}
- LLM-judge token spend is the main cost driver ^{AI}
- Maintainer funding not publicly verified ^{NPV}

Choose when

- You want a code-first metric library inside pytest and CI, owned and versioned by the engineering team ^{AI}
- You need agent trajectory and tool-correctness metrics without adopting a platform ^{AI}

Avoid when

- You expect a production observability system; DeepEval is a test framework ^{AI}
- You would rely on its LLM-judge metrics as the sole evidence for validation without calibrating them against human labels ^{AI}

Nearest competitors: L9-promptfoo, L9-mlflow-genai, L9-opik

Regulated-FS note. Run it inside the firm's CI with an approved judge model behind the gateway; pin the library and judge versions in the evidence pack so results are reproducible ^{AI}.

Facts (as of 2026-10-07)

Field	Value	Label	Sources
Company	Confident AI (maintainer; also sells the Confident AI evals and observability platform)	Verified fact	A1-S068, A1-S051
Category	Open-source LLM evaluation framework ('similar to Pytest but specialised for unit testing LLM apps')	Verified fact	A1-S068

⁴⁷ Verified fact — A1-S068: DeepEval README (Confident AI (GitHub)) <https://raw.githubusercontent.com/confident-ai/deepeval/main/README.md> accessed 2026-10-07

⁴⁸ Verified fact — A1-S068: DeepEval README (Confident AI (GitHub)) <https://raw.githubusercontent.com/confident-ai/deepeval/main/README.md> accessed 2026-10-07

⁴⁹ Verified fact — A1-S068: DeepEval README (Confident AI (GitHub)) <https://raw.githubusercontent.com/confident-ai/deepeval/main/README.md> accessed 2026-10-07

⁵⁰ Verified fact — A1-S051: DeepEval LICENSE (Apache-2.0) (Confident AI (GitHub)) <https://raw.githubusercontent.com/confident-ai/deepeval/main/LICENSE.md> accessed 2026-10-07

⁵¹ Verified fact — A1-S068: DeepEval README (Confident AI (GitHub)) <https://raw.githubusercontent.com/confident-ai/deepeval/main/README.md> accessed 2026-10-07

⁵² Verified fact — A1-S121: Confident AI: OpenTelemetry integration <https://www.confident-ai.com/docs/integrations/opentelemetry> accessed 2026-10-07 [search-tool extract]

⁵³ Verified fact — A1-S119: Confident AI: Data Residency <https://www.confident-ai.com/docs/settings/data-residency> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
Version / lineup	deepeval 4.2.8, released 2 October 2026	Verified fact	A1-S005
Licence	Apache-2.0 (framework); Confident AI platform is commercial	Verified fact	A1-S051, A1-S005, A1-S068
Strategic direction	Agentic metrics (task completion, tool correctness, plan adherence), trajectory evaluation, JevEval metric; Confident AI positioned for organisation-wide quality standards, native red teaming and an MCP server	Verified fact	A1-S068
What it does	Provides ready-made LLM-as-a-judge, statistical and NLP metrics (G-Eval, DAG, RAG metrics such as faithfulness and contextual recall/precision, agentic metrics) that run as test cases, including in CI/CD.	Verified fact	A1-S068
Stack position	Developer-side evaluation library feeding CI; production observability comes from Confident AI or another platform. Graphic label 'LLM unit tests' is accurate.	Verified fact	A1-S068
Integration	Python and TypeScript integrations (LangChain/LangGraph callbacks, Pydantic AI, CrewAI, OpenAI Agents, AWS AgentCore, Google ADK, AI SDK, Mastra, LlamaIndex, Strands); runs in any CI/CD; OpenTelemetry: ConfidentSpanExporter and framework OTel span processors export to Confident AI's OTLP/HTTP endpoint (no gRPC), regional endpoint via CONFIDENT_OTEL_URL, run_otel exports evaluation-run spans to an OTLP endpoint	Verified fact	A1-S068, A1-S121
Dependencies	Python library; metrics use any LLM as judge or local NLP models	Verified fact	A1-S068
Certifications	Confident AI states SOC 2 and HIPAA compliance (HIPAA BAA); plan gating conflicts between pages (Team-and-above vs Enterprise/Premium add-on); self-hosted compliance depends on customer environment	Verified fact	A1-S119
GDPR / residency	Confident AI regions: US (default, North Carolina), EU (Frankfurt) and AU; GDPR handling applies to EU-region data; EU hosting may need a higher plan or add-on (pages conflict)	Verified fact	A1-S119
Security features	Encryption at rest and TLS in transit (Confident AI); self-hosted keeps keys and data in customer cloud account	Verified fact	A1-S119
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	DeepEval framework free (Apache-2.0); Confident AI: Free (2 seats, 1 project, 1 GB-month spans), Starter US\$200/month (unlimited seats, 5 projects, 5 GB-months, then US\$1 per GB-month), Team	Verified fact	A1-S051, A1-S120

Field	Value	Label	Sources
	US\$2,000/month (75 GB-months), Enterprise custom		
Infrastructure cost	Main cost driver is LLM-judge token spend per evaluation run	Architectural judgment	
Ecosystem	Broad framework integrations listed above	Verified fact	A1-S068
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	First PyPI release 15 August 2023; 527 releases to 2 October 2026	Verified fact	A1-S005
Deployment	saas: Yes (Confident AI platform); managed_cloud: Not publicly verified; vpc_byoc: Yes (Enterprise: deploy in own VPC via Terraform module); private_cloud: Yes (AWS, Azure, GCP); self_hosted: Yes (library runs locally; platform self-host on Enterprise); on_prem: Yes (on-prem on Enterprise, per vendor homepage)		

Promptfoo (L9-promptfoo)

Tier: Tactical · **Flags:** Acquired · **Original graphic label:** Promptfoo – red-teaming

Rationale: Valuable eval and red-team harness under MIT, but now owned by a model vendor with an unpublished closing date; use as one of two red-team tools, not the sole independent check ⁴¹.

Criterion	Score	Rationale
Technical	4	Regression evals, red-teaming, model comparison and PR code scanning; not a production trace store.
Enterprise readiness	3	Enterprise RBAC, team management, SLA and professional services; SSO and audit logs not documented.
Security and compliance	3	Scored as a locally run MIT CLI (rule 2): inherits host controls, local-only execution. Enterprise SaaS certifications are NPV and would cap that edition at 2.
Deployment flexibility	5	Local CLI, Enterprise SaaS, and On-Prem on AWS, Azure or GCP with a dedicated runner.
Ecosystem	4	CI integrations, OTLP receiver, many providers, OWASP/NIST reports.
Reliability and maturity	3	Frequent releases since 2023 but pre-1.0; acquisition announced 9 March 2026, closing not published.
Cost / TCO	4	Free MIT CLI; attack-generation and grader model spend.

Criterion	Score	Rationale
Lock-in / portability	3	Base 4 (MIT, portable configs) reduced by 1 for the announced acquisition; governance is vendor-controlled.
Total (generic / FS)	3.70 / 3.55	

Evidence rules applied: Rule 2 (self-hosted software) applied: NPV cap not applied; enterprise_readiness and security_compliance scored on what the software enables and on project hygiene, capped at 4; inherits host controls

Capabilities. MIT CLI and library for automated evaluation of prompts, models and agents, red-teaming and vulnerability scanning, side-by-side model comparison, CI/CD checks and pull-request code scanning for LLM security issues; Promptfoo Enterprise as SaaS or On-Prem ^{vi54}.

Strengths

- Combines regression evals and red-teaming in one declarative config that runs in GitHub Actions, GitLab CI and Jenkins ^{vi55}
- Evals run locally ('your prompts never leave your machine'); On-Prem edition with a dedicated runner and network isolation ^{vi56}
- Built-in OTLP receiver using OTEL GenAI conventions ^{vi57}
- OWASP and NIST compliance reports; providers include OpenAI, Anthropic, Azure, Bedrock and Ollama ^{vi58}

Limitations and risks

- OpenAI announced the acquisition on 9 March 2026; the closing date has not been published, and the README says Promptfoo is part of OpenAI ^{vi59}
- OpenAI plans to integrate it into OpenAI Frontier, so the roadmap follows a model vendor's priorities ^{vi60}

⁵⁴ Verified fact — A1-S062: Promptfoo README (Promptfoo (OpenAI) (GitHub)) <https://raw.githubusercontent.com/promptfoo/promptfoo/main/README.md> accessed 2026-10-07; A1-S063: Promptfoo Enterprise docs (SaaS and On-Prem) (Promptfoo (OpenAI) (GitHub docs source)) <https://raw.githubusercontent.com/promptfoo/promptfoo/main/site/docs/enterprise/index.md> accessed 2026-10-07

⁵⁵ Verified fact — A1-S062: Promptfoo README (Promptfoo (OpenAI) (GitHub)) <https://raw.githubusercontent.com/promptfoo/promptfoo/main/README.md> accessed 2026-10-07; A1-S065: Promptfoo CI/CD integration docs (Promptfoo (OpenAI) (GitHub docs source)) <https://raw.githubusercontent.com/promptfoo/promptfoo/main/site/docs/integrations/ci-cd.md> accessed 2026-10-07

⁵⁶ Verified fact — A1-S062: Promptfoo README (Promptfoo (OpenAI) (GitHub)) <https://raw.githubusercontent.com/promptfoo/promptfoo/main/README.md> accessed 2026-10-07; A1-S063: Promptfoo Enterprise docs (SaaS and On-Prem) (Promptfoo (OpenAI) (GitHub docs source)) <https://raw.githubusercontent.com/promptfoo/promptfoo/main/site/docs/enterprise/index.md> accessed 2026-10-07

⁵⁷ Verified fact — A1-S064: Promptfoo tracing (OpenTelemetry/OTLP) docs (Promptfoo (OpenAI) (GitHub docs source)) <https://raw.githubusercontent.com/promptfoo/promptfoo/main/site/docs/tracing.md> accessed 2026-10-07

⁵⁸ Verified fact — A1-S062: Promptfoo README (Promptfoo (OpenAI) (GitHub)) <https://raw.githubusercontent.com/promptfoo/promptfoo/main/README.md> accessed 2026-10-07; A1-S063: Promptfoo Enterprise docs (SaaS and On-Prem) (Promptfoo (OpenAI) (GitHub docs source)) <https://raw.githubusercontent.com/promptfoo/promptfoo/main/site/docs/enterprise/index.md> accessed 2026-10-07; A1-S065: Promptfoo CI/CD integration docs (Promptfoo (OpenAI) (GitHub docs source)) <https://raw.githubusercontent.com/promptfoo/promptfoo/main/site/docs/integrations/ci-cd.md> accessed 2026-10-07

⁵⁹ Verified fact — A1-S024: OpenAI to acquire Promptfoo <https://openai.com/index/openai-to-acquire-promptfoo/> accessed 2026-10-07 [search-tool extract]; A1-S062: Promptfoo README (Promptfoo (OpenAI) (GitHub)) <https://raw.githubusercontent.com/promptfoo/promptfoo/main/README.md> accessed 2026-10-07; V1-S006: OpenAI to acquire Promptfoo <https://openai.com/blog/openai-to-acquire-promptfoo> accessed 2026-10-08 [search-tool extract]; V2-S042: OpenAI to acquire Promptfoo (with TechCrunch via Yahoo Finance, SDxCentral, Security Boulevard) <https://openai.com/blog/openai-to-acquire-promptfoo> accessed 2026-10-08 [search-tool extract]

- Pre-1.0 version numbering (0.124.0) ⁶¹
- Enterprise edition certifications not publicly verified ^{NPV}

Choose when

- You need red-team and regression suites as code in CI, run locally against any provider ^{AJ}
- You want OWASP-mapped test evidence for C7 security testing ^{AJ}

Avoid when

- A model vendor owning your independent red-team tool is unacceptable to your model-risk function without a second, independent tool ^{AJ}
- You would buy Enterprise SaaS before certifications are evidenced ^{AJ}

Nearest competitors: L9-deepeval, L9-braintrust, L9-langsmith

Regulated-FS note. Keep using the MIT CLI locally, pin the version, and keep test configs in the firm's repo so they are portable; pair it with an independent tool when the system under test uses OpenAI models, to preserve effective challenge ^{AJ}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Promptfoo, now part of OpenAI (acquisition agreement announced 9 March 2026; README states 'Promptfoo is now part of OpenAI')	Verified fact	A1-S024, A1-S025, A1-S062
Category	Open-source CLI and library for LLM evals and red teaming, plus Promptfoo Enterprise (SaaS and On-Prem)	Verified fact	A1-S062, A1-S063
Version / lineup	promptfoo 0.124.0 (npm), published 6 October 2026	Verified fact	A1-S020
Licence	MIT (open source); Enterprise editions commercial	Verified fact	A1-S052, A1-S020, A1-S062, A1-S063
Status events	Acquisition by OpenAI announced 9 March 2026 (terms undisclosed; subject to closing conditions); the README now says "part of OpenAI", but no dated closing notice was found as of 8 October 2026	Verified fact	A1-S024, A1-S025, A1-S062, V1-S006
Strategic direction	OpenAI plans to integrate Promptfoo into OpenAI Frontier for automated red-teaming, security review of agentic workflows and risk/compliance monitoring, while continuing the open-source project	Verified fact	A1-S024, A1-S025

⁶⁰ Verified fact — A1-S024: OpenAI to acquire Promptfoo <https://openai.com/index/openai-to-acquire-promptfoo/> accessed 2026-10-07 [search-tool extract]

⁶¹ Verified fact — A1-S020: npm registry metadata: promptfoo <https://registry.npmjs.org/promptfoo> accessed 2026-10-07

Field	Value	Label	Sources
What it does	Runs automated evaluations of prompts, models and agents, red-teaming and vulnerability scanning, side-by-side model comparison, CI/CD checks and code scanning of pull requests for LLM security issues.	Verified fact	A1-S062
Stack position	Spans pre-production evaluation and AI security testing (C7); the graphic's 'red-teaming' descriptor omits its general eval role	Verified fact	A1-S062, A1-S063
Integration	CLI/library; CI/CD (GitHub Actions, GitLab CI, Jenkins); built-in OTLP receiver ingesting spans with OTel GenAI semantic conventions; API (Enterprise)	Verified fact	A1-S064, A1-S065, A1-S063
Dependencies	Node.js CLI (also brew and pip); evals run locally ('your prompts never leave your machine'); Enterprise On-Prem includes a dedicated runner	Verified fact	A1-S062, A1-S063
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	Community edition runs locally; Enterprise On-Prem keeps scans inside the network perimeter	Verified fact	A1-S062, A1-S063
Security features	Network isolation and dedicated runner in On-Prem edition	Verified fact	A1-S063
Access controls	RBAC and team management in Enterprise editions only	Verified fact	A1-S063
Enterprise support	Enterprise: SLA, professional services, dedicated Slack	Verified fact	A1-S063
Pricing	Community free (MIT); Enterprise contact sales	Verified fact	A1-S063
Infrastructure cost	Cost driver is model/API spend for generated attacks and graders	Architectural judgment	
Ecosystem	Providers incl. OpenAI, Anthropic, Azure, Bedrock, Ollama; SIEM and issue-tracker integrations in Enterprise; OWASP/NIST compliance reports	Verified fact	A1-S062, A1-S063, A1-S065
Adoption signals	Over 25% of Fortune 500 use Promptfoo (company claim); ~US\$23M raised, US\$86M valuation after July 2025 round (PitchBook)	Reported	A1-S026
Maturity	npm package created 3 May 2023; frequent releases	Verified fact	A1-S020
Deployment	saas: Yes (Promptfoo Enterprise SaaS); managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Yes (On-Prem on AWS, Azure, GCP); self_hosted: Yes; on_prem: Yes (Enterprise On-Prem)		

Arize Phoenix (L9-arize-phoenix)

Tier: Tactical • **Flags:** Acquired • **Original graphic label:** Phoenix – Atrace

Rationale: Useful, portable workbench with strong instrumentation; ELv2, no audit trail, community-only support and new Dynatrace ownership make it unsuitable as the production record ^{A1}.

Criterion	Score	Rationale
Technical	4	Tracing, retrieval and response evals, datasets, experiments, prompt management and pytest; online production monitoring is the AX proposition.
Enterprise readiness	3	Scored as self-hosted software (rule 2). With authentication enabled, Phoenix provides admin, member and viewer roles and login through OAuth2 identity providers (A1-S105): two of SSO, RBAC and audit logs verified, so 3 under rule 7 (CP2 Q2). No audit trail, SCIM or commercial support for Phoenix itself, so not above 3.
Security and compliance	3	Inherits host controls; licence is clear, telemetry can be disabled; auth off by default is a hygiene weakness. Certifications NPV for Phoenix Cloud, not used.
Deployment flexibility	4	Self-host, private-cloud templates and air-gap; Phoenix Cloud low confidence; no BYOC.
Ecosystem	4	OpenInference is accepted by other backends; broad framework and provider auto-instrumentation.
Reliability and maturity	4	700 releases since January 2023; Dynatrace completed acquisition 1 October 2026 with a support commitment.
Cost / TCO	4	Free with no caps; one instance per team and PostgreSQL to operate.
Lock-in / portability	3	Base 4 (portable OTel/OpenInference, source-available) reduced by 1 for the 2026 ownership change; ELv2 is not permissive.
Total (generic / FS)	3.65 / 3.50	

Evidence rules applied: Rule 2 (self-hosted software) applied: NPV cap not applied; enterprise_readiness and security_compliance scored on what the software enables and on project hygiene, capped at 4; inherits host controls; Rule 7 (CP2 Q2) applied at CP2 rework (8 October 2026) for consistency with Opik and Unstructured: roles and OAuth2 IdP login verified (A1-S105); enterprise_readiness 2 -> 3; no audit trail

Capabilities. Self-hosted tracing and evaluation built on OpenTelemetry and OpenInference: LLM-based response and retrieval evaluations, versioned datasets, experiments, prompt playground and management, REST API, MCP endpoint and a pytest plugin for eval-in-CI ^{vi62}.

Strengths

- OpenInference instrumentation (Apache-2.0) routes the same spans to Phoenix, Arize AX or other backends, including Datadog ^{vi63}
- No feature gates and no usage caps when self-hosted ^{vi64}

⁶² Verified fact — A1-S066: Arize Phoenix README (Arize AI (GitHub)) <https://raw.githubusercontent.com/Arize-ai/phoenix/main/README.md> accessed 2026-10-07; A1-S049: OpenInference LICENSE (Apache-2.0) (Arize AI (GitHub)) <https://raw.githubusercontent.com/Arize-ai/openinference/main/LICENSE> accessed 2026-10-07; A1-S106: Phoenix: Eval CI with pytest (docs source) (Arize AI (GitHub docs source)) <https://raw.githubusercontent.com/Arize-ai/phoenix/main/docs/phoenix/datasets-and-experiments/how-to-experiments/eval-ci-with-pytest.mdx> accessed 2026-10-07

⁶³ Verified fact — A1-S049: OpenInference LICENSE (Apache-2.0) (Arize AI (GitHub)) <https://raw.githubusercontent.com/Arize-ai/openinference/main/LICENSE> accessed 2026-10-07; A1-S066: Arize Phoenix README (Arize AI (GitHub)) <https://raw.githubusercontent.com/Arize-ai/phoenix/main/README.md> accessed 2026-10-07; A1-S098: Datadog LLM Observability: OpenTelemetry instrumentation (Datadog, Inc.) https://docs.datadoghq.com/llm_observability/instrumentation/otel_instrumentation/ accessed 2026-10-07

⁶⁴ Verified fact — A1-S046: Arize Phoenix or Arize AX - Phoenix docs (Arize AI) <https://arize.com/docs/phoenix/resources/frequently-asked-questions/what-is-the-difference-between-phoenix-and-arize> accessed 2026-10-07 [search-tool extract]; A1-S047: Arize AX Pricing: AI agent observability and evaluation (Arize AI) <https://arize.com/pricing/> accessed 2026-10-07 [search-tool extract]

- Data stays in the customer environment; default web-analytics telemetry carries no trace data and can be disabled ^{vf65}
- Air-gapped deployment supported; cloud templates for AWS, Azure and Google Cloud Run ^{vf66}

Limitations and risks

- Elastic License 2.0: source-available, not OSI open source; cannot be offered to third parties as a managed service ^{vf67}
- Authentication is optional and off by default ^{vf68}
- Single-tenant per instance; full SSO, RBAC and audit trails are provided through AX, not Phoenix ^{vf69}
- Community support only ^{vf70}
- Owned by Dynatrace since 1 October 2026; capabilities to be integrated into Dynatrace over time ^{vf71}

Choose when

- You want a free, self-hosted tracing and eval workbench for development and validation teams, with OpenInference as the portable instrumentation ^{ai}
- Validators need an isolated, air-gapped instance per model review ^{ai}

Avoid when

- You need a multi-team production system of record with SSO, audit trails and support ^{ai}
- Your open-source policy admits only OSI-approved licences ^{ai}

Nearest competitors: L9-langfuse, L9-opik, L9-mlflow-genai

Regulated-FS note. Good for independent validation sandboxes; switch authentication on, disable telemetry, and do not treat it as the audit store, because it has no audit trail ^{ai}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Arize AI, a Dynatrace company (Dynatrace completed the acquisition on 1 October 2026)	Verified fact	A1-S044, A1-S045,

⁶⁵ Verified fact — A1-S066: Arize Phoenix README (Arize AI (GitHub)) <https://raw.githubusercontent.com/Arize-ai/phoenix/main/README.md> accessed 2026-10-07

⁶⁶ Verified fact — A1-S046: Arize Phoenix or Arize AX - Phoenix docs (Arize AI) <https://arize.com/docs/phoenix/resources/frequently-asked-questions/what-is-the-difference-between-phoenix-and-arize> accessed 2026-10-07 [search-tool extract]; A1-S066: Arize Phoenix README (Arize AI (GitHub)) <https://raw.githubusercontent.com/Arize-ai/phoenix/main/README.md> accessed 2026-10-07

⁶⁷ Verified fact — A1-S048: Arize Phoenix LICENSE (Elastic License 2.0) (Arize AI (GitHub)) <https://raw.githubusercontent.com/Arize-ai/phoenix/main/LICENSE> accessed 2026-10-07; A1-S107: Phoenix self-hosting: License (docs source) (Arize AI (GitHub docs source)) <https://raw.githubusercontent.com/Arize-ai/phoenix/main/docs/phoenix/self-hosting/license.mdx> accessed 2026-10-07

⁶⁸ Verified fact — A1-S105: Phoenix self-hosting: Authentication (docs source) (Arize AI (GitHub docs source)) <https://raw.githubusercontent.com/Arize-ai/phoenix/main/docs/phoenix/self-hosting/features/authentication.mdx> accessed 2026-10-07

⁶⁹ Verified fact — A1-S046: Arize Phoenix or Arize AX - Phoenix docs (Arize AI) <https://arize.com/docs/phoenix/resources/frequently-asked-questions/what-is-the-difference-between-phoenix-and-arize> accessed 2026-10-07 [search-tool extract]

⁷⁰ Verified fact — A1-S046: Arize Phoenix or Arize AX - Phoenix docs (Arize AI) <https://arize.com/docs/phoenix/resources/frequently-asked-questions/what-is-the-difference-between-phoenix-and-arize> accessed 2026-10-07 [search-tool extract]

⁷¹ Verified fact — A1-S045: Dynatrace Completes Acquisition of Arize, Extending AI Observability Across the Full Development Lifecycle (Dynatrace, Inc.) <https://ir.dynatrace.com/news-events/press-releases/detail/439/dynatrace-completes-acquisition-of-arize-extending-ai-observability-across-the-full-development-lifecycle> accessed 2026-10-07 [search-tool extract]; V1-S005: Dynatrace Completes Acquisition of Arize (IR press release) (Dynatrace, Inc.) <https://ir.dynatrace.com/news-events/press-releases/detail/439/dynatrace-completes-acquisition-of-arize-extending-ai-observability-across-the-full-development-lifecycle> accessed 2026-10-08 [search-tool extract]

Field	Value	Label	Sources
			V1-S005
Category	Source-available (ELv2), self-hosted AI observability and evaluation platform	Verified fact	A1-S048, A1-S066, A1-S107
Version / lineup	arize-phoenix 20.19.0, released 1 October 2026	Verified fact	A1-S004
Licence	Elastic License 2.0: free self-hosting, modification and redistribution, but may not be provided to third parties as a hosted or managed service; not OSI-approved although marketed as 'open source'; no feature gates	Verified fact	A1-S048, A1-S046, A1-S107, V1-S013
Status events	Dynatrace signed a definitive agreement to acquire Arize, 13 August 2026 (US\$915M cash and stock); Acquisition completed 1 October 2026 (Dynatrace investor-relations release, 4:05 p.m. EDT)	Verified fact	A1-S044, A1-S045, V1-S005
Strategic direction	Arize will continue to support Phoenix after the Dynatrace acquisition and integrate capabilities into Dynatrace over time; recent additions include PXI (built-in AI engineering agent) and a remote MCP server	Verified fact	A1-S045, A1-S066
What it does	Traces LLM applications with OpenTelemetry-based instrumentation and runs LLM-based response and retrieval evaluations, versioned datasets, experiments, a prompt playground and prompt management.	Verified fact	A1-S066
Stack position	Cross-cutting tracing and evaluation for development and self-hosted production; it is the open product, while Arize AX is the commercial platform (graphic shows both as separate tiles)	Verified fact	A1-S046, A1-S066
Integration	OpenTelemetry with OpenInference instrumentation (Apache-2.0); same instrumentation routes to Phoenix or AX; REST API; built-in MCP endpoint; pytest plugin for eval-in-CI	Verified fact	A1-S049, A1-S066, A1-S106
Dependencies	SQLite by default; PostgreSQL 14+ recommended for production; single-tenant per instance	Verified fact	A1-S046
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	Self-hosted: data remains in the customer environment; default web-analytics telemetry collects no trace or evaluation data and can be disabled (PHOENIX_TELEMETRY_ENABLED=false)	Verified fact	A1-S066
Security features	Authentication optional and off by default; when enabled supports secure cookies and Phoenix acts as an OAuth2 authorisation server for MCP and other clients	Verified fact	A1-S046, A1-S105
Access controls	Roles admin, member and viewer when auth enabled; login via OAuth2 identity providers; Arize states full SSO/RBAC/audit trails are provided through AX	Verified fact	A1-S105, A1-S046
Enterprise support	Community support only; dedicated support and SLAs via Arize AX	Verified fact	A1-S046
Pricing	Free to self-host, no usage caps	Verified fact	A1-S047, A1-S107
Infrastructure	Cost is the customer's compute and PostgreSQL; multi-team use	Architecture	

Field	Value	Label	Sources
cost	requires multiple instances	ctural judge ment	
Ecosystem	Out-of-the-box tracing for OpenAI Agents SDK, Claude Agent SDK, LangGraph, Vercel AI SDK, Mastra, CrewAI, LlamaIndex, DSPy and providers incl. OpenAI, Anthropic, Google GenAI, Bedrock, OpenRouter, LiteLLM	Verifie d fact	A1-S066
Adoption signals	Not publicly verified	Not publicl y verifie d	
Maturity	First PyPI release 18 January 2023; 700 releases to 1 October 2026	Verifie d fact	A1-S004
Deployment	saas: Yes (Phoenix Cloud referenced); managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Yes (AWS CloudFormation, Azure, Google Cloud Run templates); self_hosted: Yes; on_prem: Yes (air-gapped supported)		

W&B Weave (Weights & Biases) (L9-wandb-weave)

Tier: Tactical · **Flags:** Acquired · **Original graphic label:** not in graphic

Rationale: Good security posture and deployment options, but evaluation depth is less evidenced and the product follows a compute vendor's platform strategy ^{AJ}.

Criterion	Score	Rationale
Technical	3	Agent tracing and evaluation with SDK autopatching; RAG metrics, online rules and red-teaming not evidenced.
Enterprise readiness	4	SSO, automated provisioning, custom roles, audit logs and an enterprise support package.
Security and compliance	4	The Weave documentation states SOC 2 Type II for both managed platforms, HIPAA on Dedicated Cloud and customer-managed keys on Enterprise (A1-S137, A1-S138); ISO/IEC 27001/27017/27018 are listed at W&B site level with Weave scope not stated, so 4 under rule 8 (CP2 Q4): 5 needs SOC 2 Type II and ISO 27001 within the product's scope.
Deployment flexibility	4	Multi-tenant SaaS, Dedicated Cloud, Self-Managed and on-prem; air-gap not documented.

Criterion	Score	Rationale
Ecosystem	3	OTLP ingest and agent SDK autopatching; strongest inside the W&B estate.
Reliability and maturity	3	Three years of releases, pre-1.0 SDK; owned by CoreWeave since May 2025.
Cost / TCO	3	Pro from US\$60/month plus usage-based ingestion; Enterprise by quote.
Lock-in / portability	2	Base 3 (proprietary platform, OTel ingest, Apache-2.0 SDK) reduced by 1 for the 2025 ownership change.
Total (generic / FS)	3.40	
	/	
	3.35	

Evidence rules applied: Rule 8 (CP2 Q4) applied at CP2 rework (8 October 2026): ISO certifications are W&B site-level with Weave scope not stated; security_compliance 5 -> 4

Capabilities. Tracing and evaluation toolkit for agents and LLM applications from Weights & Biases: traces conversations, turns, LLM and tool calls; evaluates agents; autopatches agent SDKs including OpenAI Agents SDK, Claude Agent SDK and Google ADK; OTLP/HTTP endpoint plus a dedicated agents endpoint ^{vr72}.

Strengths

- SOC 2 Type II, ISO/IEC 27001, 27017 and 27018; HIPAA on Dedicated Cloud with BAA ^{vr73}
- Dedicated Cloud in a customer-chosen cloud and region, with isolated resources, per-instance keys and private connectivity; customer-managed keys on Enterprise ^{vr74}
- Enterprise SSO, automated user provisioning, custom roles and audit logs ^{vr75}
- Self-managed and on-premises options with ingest sampling for cost control ^{vr76}

Limitations and risks

- Part of CoreWeave since 5 May 2025 and now inside CoreWeave Forge ^{vr77}
- Multi-tenant cloud is North America only ^{vr78}
- Pre-1.0 SDK versioning ^{vr79}

⁷² Verified fact — A1-S104: Weave by Weights & Biases README (Weights & Biases (GitHub)) <https://raw.githubusercontent.com/wandb/weave/master/README.md> accessed 2026-10-07; A1-S139: Send OpenTelemetry Traces to Weave (Weights & Biases (CoreWeave)) <https://docs.wandb.ai/weave/guides/tracking/otel> accessed 2026-10-07 [search-tool extract]

⁷³ Verified fact — A1-S137: Weave: Deployment options and security features - Weights & Biases Documentation (Weights & Biases (CoreWeave)) <https://docs.wandb.ai/weave/guides/platform> accessed 2026-10-07 [search-tool extract]

⁷⁴ Verified fact — A1-S137: Weave: Deployment options and security features - Weights & Biases Documentation (Weights & Biases (CoreWeave)) <https://docs.wandb.ai/weave/guides/platform> accessed 2026-10-07 [search-tool extract]; A1-S138: CoreWeave Forge Pricing (Weights & Biases (CoreWeave)) <https://wandb.ai/site/pricing> accessed 2026-10-07 [search-tool extract]

⁷⁵ Verified fact — A1-S138: CoreWeave Forge Pricing (Weights & Biases (CoreWeave)) <https://wandb.ai/site/pricing> accessed 2026-10-07 [search-tool extract]

⁷⁶ Verified fact — A1-S137: Weave: Deployment options and security features - Weights & Biases Documentation (Weights & Biases (CoreWeave)) <https://docs.wandb.ai/weave/guides/platform> accessed 2026-10-07 [search-tool extract]

⁷⁷ Verified fact — A1-S131: CoreWeave Completes Acquisition of Weights & Biases (CoreWeave, Inc.) <https://investors.coreweave.com/news/news-details/2025/CoreWeave-Completes-Acquisition-of-Weights-Biases/default.aspx> accessed 2026-10-07 [search-tool extract]; A1-S138: CoreWeave Forge Pricing (Weights & Biases (CoreWeave)) <https://wandb.ai/site/pricing> accessed 2026-10-07 [search-tool extract]

⁷⁸ Verified fact — A1-S137: Weave: Deployment options and security features - Weights & Biases Documentation (Weights & Biases (CoreWeave)) <https://docs.wandb.ai/weave/guides/platform> accessed 2026-10-07 [search-tool extract]

⁷⁹ Verified fact — A1-S018: PyPI JSON metadata: weave (Python Package Index) <https://pypi.org/pypi/weave/json> accessed 2026-10-07

- Hosted platform is proprietary ^{vi,80}
- Evaluation depth for RAG and red-teaming not evidenced in the fact base ^{NPV}

Choose when

- W&B is already the firm's ML experiment platform and you want agent traces and evals in the same governed estate ^{AJ}
- You need a dedicated single-tenant cloud in a chosen region with customer-managed keys ^{AJ}

Avoid when

- You would use the multi-tenant cloud for EU or UK client data ^{AJ}
- You do not otherwise use W&B; the platform cost is hard to justify for L9 alone ^{AJ}

Nearest competitors: L9-mlflow-genai, L9-langsmith, L9-arize-ax

Regulated-FS note. Use Dedicated Cloud in an EU/UK region or Self-Managed; include CoreWeave in the concentration analysis if it also supplies GPU capacity ^{AJ}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Weights & Biases, part of CoreWeave, Inc. (acquisition completed 5 May 2025); now a named product within CoreWeave Forge	Verified fact	A1-S131, A1-S138
Category	Tracing and evaluation toolkit for AI agents and LLM applications	Verified fact	A1-S104
Version / lineup	weave 0.53.11 (Python), released 25 September 2026	Verified fact	A1-S018
Licence	SDK Apache-2.0; hosted W&B platform proprietary (Pro/Enterprise)	Verified fact	A1-S018, A1-S138
Status events	Weights & Biases acquired by CoreWeave (announced 4 March 2025, completed 5 May 2025)	Verified fact	A1-S131, V1-S011
Strategic direction	Agent-focused tracing (Agents view, dedicated OTLP endpoint for agent spans) inside CoreWeave Forge	Verified fact	A1-S104, A1-S139, A1-S138
What it does	Traces agent conversations, turns, LLM and tool calls, and evaluates agents and LLM applications; autopatches agent SDKs such as OpenAI Agents SDK, Claude Agent SDK and Google ADK.	Verified fact	A1-S104
Stack position	Cross-cutting L9 tracing/evaluation within the broader W&B ML	Verified	A1-S104,

⁸⁰ Verified fact — A1-S138: CoreWeave Forge Pricing (Weights & Biases) (Weights & Biases (CoreWeave)) <https://wandb.ai/site/pricing> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
Integration	platform	d fact	A1-S137
	Python/TypeScript SDK; OTLP/HTTP protobuf endpoint (trace.wandb.ai/otel/v1/traces) and agents endpoint; OTel Collector forwarding	Verified fact	A1-S104, A1-S139
Dependencies	Self-managed W&B Server: Kubernetes, MySQL 8.4.x, S3-compatible object storage, Redis; Dedicated Cloud stores Weave data in a dedicated ClickHouse Cloud cluster	Verified fact	A1-S137
Certifications	SOC 2 Type II (multi-tenant and Dedicated Cloud); ISO/IEC 27001:2022, 27017:2015, 27018:2019; HIPAA on Dedicated Cloud with BAA (Weave HIPAA status on other deployments conflicting)	Verified fact	A1-S137
GDPR / residency	Multi-tenant cloud in North America; Dedicated Cloud in customer-chosen cloud and region	Verified fact	A1-S137
Security features	Dedicated Cloud: isolated network/compute/storage, per-instance encryption key, IP allowlisting, private connectivity; Enterprise: customer-managed encryption keys	Verified fact	A1-S137, A1-S138
Access controls	Enterprise: SSO, automated user provisioning, custom roles, audit logs	Verified fact	A1-S138
Enterprise support	Enterprise support package; Enterprise invoiced annually upfront	Verified fact	A1-S138
Pricing	Pro from US\$60/month; Enterprise by quote; Weave data ingestion billed monthly in arrears by usage; extra storage US\$0.03/GB	Verified fact	A1-S138
Infrastructure cost	Self-managed instances can use ingest sampling to control cost at high trace volume	Verified fact	A1-S137
Ecosystem	Integrations incl. OpenAI Agents SDK, Claude Agent SDK, Google ADK, Agno, Vercel AI SDK	Verified fact	A1-S104, A1-S139
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	weave first on PyPI 7 June 2023; 147 releases	Verified fact	A1-S018
Deployment	saas: Yes (multi-tenant on Google Cloud, North America); managed_cloud: Yes (Dedicated Cloud on AWS, Google Cloud, Azure); vpc_byoc: Not publicly verified; private_cloud: Yes (Self-Managed); self_hosted: Yes; on_prem: Yes (on-premises Self-Managed)		

Braintrust (L9-braintrust)

Tier: Tactical · **Flags:** none · **Original graphic label:** Braintrust – evals platform

Rationale: Strong evaluation workflow and a sensible hybrid model, but proprietary eval assets and no ISO 27001 keep it out of the foundational tier for FS ^{AJ}.

Criterion	Score	Rationale
Technical	4	Leading offline-eval ergonomics plus production logging, clustering and gateway; red-teaming not native.

Criterion	Score	Rationale
Enterprise readiness	3	SSO/SAML (Okta, Entra ID, Google Workspace) and OIDC; built-in groups plus Enterprise custom, project and object-level permissions; Enterprise audit logs queryable by SQL (B-REV-S020, B-REV-S021); SCIM and SLA not documented.
Security and compliance	3	SOC 2 Type II, encryption, CMK in hybrid and EU data plane; no ISO 27001 found and HIPAA wording inconsistent.
Deployment flexibility	4	SaaS, hybrid data plane in customer cloud, self-host on Enterprise; air-gap low confidence.
Ecosystem	4	OTLP ingest, MCP endpoint, GitHub Action, several agent framework integrations.
Reliability and maturity	3	GA since 2023 with frequent releases; venture-funded and repositioning towards observability.
Cost / TCO	3	Published Pro price (US\$249/month plus usage); Enterprise custom.
Lock-in / portability	2	Proprietary platform holding eval logic and datasets; OTel ingest only partially mitigates.
Total (generic / FS)	3.40 / 3.20	

Capabilities. Proprietary evaluation and observability platform: experiments, datasets and scorers; production trace logging; Topics trace clustering; Loop AI-assisted prompt/model optimisation; a gateway that routes and traces model calls; GitHub Action posting eval results on pull requests ^{vr81}.

Strengths

- Strong offline evaluation workflow with PR-level CI feedback ^{vr82}
- Hybrid deployment: Braintrust-hosted control plane with the data plane in the customer's AWS, GCP or Azure; customer-managed KMS keys and private networking ^{vr83}
- US or EU data plane chosen at organisation creation; EU processing in the DPA ^{vr84}
- OTLP trace endpoints in US and EU ^{vr85}

Limitations and risks

- No ISO 27001 certification found; SOC 2 Type II reports under mutual NDA ^{vr86}

⁸¹ Verified fact — A1-S043: Trace keynote recap: See it, improve it, optimize it (Braintrust Data, Inc.) <https://www.braintrust.dev/blog/trace-keynote> accessed 2026-10-07 [search-tool extract]; A1-S040: Braintrust pricing (Braintrust Data, Inc.) <https://www.braintrust.dev/pricing> accessed 2026-10-07 [search-tool extract]; A1-S109: Braintrust Eval Action README (Braintrust Data, Inc. (GitHub)) <https://raw.githubusercontent.com/braintrustdata/eval-action/main/README.md> accessed 2026-10-07

⁸² Verified fact — A1-S109: Braintrust Eval Action README (Braintrust Data, Inc. (GitHub)) <https://raw.githubusercontent.com/braintrustdata/eval-action/main/README.md> accessed 2026-10-07

⁸³ Verified fact — A1-S041: Security is a choice: how Braintrust lets you decide where your AI data lives (Braintrust Data, Inc.) <https://www.braintrust.dev/blog/security-data-control> accessed 2026-10-07 [search-tool extract]

⁸⁴ Verified fact — A1-S042: OpenTelemetry (OTel) - Braintrust docs (Braintrust Data, Inc.) <https://www.braintrust.dev/docs/integrations/opentelemetry> accessed 2026-10-07 [search-tool extract]

⁸⁵ Verified fact — A1-S042: OpenTelemetry (OTel) - Braintrust docs (Braintrust Data, Inc.) <https://www.braintrust.dev/docs/integrations/opentelemetry> accessed 2026-10-07 [search-tool extract]

⁸⁶ Verified fact — A1-S041: Security is a choice: how Braintrust lets you decide where your AI data lives (Braintrust Data, Inc.) <https://www.braintrust.dev/blog/security-data-control> accessed 2026-10-07 [search-tool extract]; A1-S125: Security - Braintrust docs <https://www.braintrust.dev/docs/security> accessed 2026-10-07 [search-tool extract]

- Evaluation logic and datasets live in the proprietary platform ^{vf⁸⁷}
- Region cannot be changed after organisation creation ^{vf⁸⁸}
- Hybrid control plane still holds metadata and hashed keys, and uses Clerk for auth ^{vf⁸⁹}
- Valuation figures are press-reported only and are not a decision input ^{r⁹⁰}

Choose when

- Evaluation-driven development is the main need and teams want rich experiment comparison in pull requests ^{AJ}
- You want trace data in your own cloud account but are content with a vendor-run control plane ^{AJ}

Avoid when

- Your procurement baseline requires ISO 27001 ^{AJ}
- You need a fully self-contained, air-gapped deployment with no external control plane (air-gap is vendor-mentioned only, low confidence) ^{AJ}

Nearest competitors: L9-langsmith, L9-langfuse, L9-arize-ax

Regulated-FS note. The hybrid data plane answers the residency question for traces, but the control plane's metadata must still be assessed under SYSC 8/DORA; choose the EU data plane at creation because it cannot be changed ^{AJ}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Braintrust (independent, venture-backed; no acquisition found)	Verified fact	A1-S028
Category	Proprietary AI evaluation and observability platform, marketed as 'the active observability platform for agents'	Verified fact	A1-S043
Version / lineup	SaaS; Python SDK braintrust 0.45.0, released 7 October 2026	Verified fact	A1-S003
Licence	Proprietary platform; Python SDK MIT	Verified fact	A1-S003
Strategic direction	US\$80M Series B (17 February 2026) to become the observability layer for production AI; Trace conference (February 2026) introduced Topics (trace clustering), Loop (AI-assisted prompt/model optimisation) and later Patterns; Braintrust Gateway routes and traces model calls	Verified fact	A1-S028, A1-S043
What it does	Runs offline evaluations (experiments, datasets, scorers) and logs production traces; clusters traces into topics and uses an AI assistant (Loop) to propose prompt or model changes; a gateway routes LLM calls while recording traces.	Verified fact	A1-S043, A1-S040

⁸⁷ Verified fact — A1-S042: OpenTelemetry (OTel) - Braintrust docs (Braintrust Data, Inc.) <https://www.braintrust.dev/docs/integrations/opentelemetry> accessed 2026-10-07 [search-tool extract]

⁸⁸ Verified fact — A1-S042: OpenTelemetry (OTel) - Braintrust docs (Braintrust Data, Inc.) <https://www.braintrust.dev/docs/integrations/opentelemetry> accessed 2026-10-07 [search-tool extract]

⁸⁹ Verified fact — A1-S041: Security is a choice: how Braintrust lets you decide where your AI data lives (Braintrust Data, Inc.) <https://www.braintrust.dev/blog/security-data-control> accessed 2026-10-07 [search-tool extract]

⁹⁰ Reported — A1-S027: AI observability Braintrust raises \$80 million at \$800 million (Axios Pro) <https://www.axios.com/pro/enterprise-software-deals/2026/02/17/ai-observability-braintrust-80-million-800-million> accessed 2026-10-07

Field	Value	Label	Sources
Stack position	Cross-cutting evaluation and observability; graphic's 'evals platform' understates the production tracing and gateway scope	Verified fact	A1-S043
Integration	SDKs; OTLP trace endpoint (api.braintrust.dev/otel, EU api-eu.braintrust.dev/otel) with Bearer auth; MCP endpoint; GitHub Action for evals on pull requests	Verified fact	A1-S042, A1-S109
Dependencies	Hybrid: Braintrust-hosted control plane (UI, metadata, auth via Clerk) with data plane in customer AWS, GCP or Azure deployed by Terraform	Verified fact	A1-S041
Certifications	SOC 2 Type II (reports via Trust Center under mutual NDA); HIPAA support with BAA on Enterprise plans; no ISO 27001 certification found	Verified fact	A1-S041, A1-S125
GDPR / residency	US or EU data plane chosen at organisation creation and not changeable; DPA states EU storage and processing of data plane for Pro customers electing EU; control plane holds hashed keys and metadata	Verified fact	A1-S042
Security features	AES-256 at rest, TLS 1.2 in transit, API keys stored as one-way hashes; customer-managed KMS keys and private networking in hybrid	Verified fact	A1-S041
Access controls	Pro: more flexible permission roles; Enterprise: RBAC, deployment approvals, retention controls	Verified fact	A1-S040
Enterprise support	Enterprise: custom pricing, dedicated support	Verified fact	A1-S040
Pricing	Starter free (usage-based, no platform fee); Pro US\$249/month incl. 5 GB processed data (+US\$3/GB) and 50k scores (+US\$1.50 per 1k), 30-day retention; Enterprise custom	Verified fact	A1-S040
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	OTel ingest from any framework; integrations listed for Cloudflare Agents, LiveKit Agents, Claude Code (OTel)	Verified fact	A1-S042
Adoption signals	Seed US\$5.1M (Greylock), Series A US\$36M (Oct 2024), Series B US\$80M led by ICONIQ (Feb 2026)	Verified fact	A1-S028, V1-S017
Maturity	First braintrust SDK on PyPI 24 May 2023; 290 releases to 7 October 2026	Verified fact	A1-S003
Deployment	saas: Yes; managed_cloud: Not publicly verified; vpc_byoc: Yes (hybrid data plane in customer cloud); private_cloud: Yes; self_hosted: Yes (Enterprise); on_prem: Yes (air-gapping mentioned by vendor)		

Datadog Agent Observability (documented under the LLM Observability URL path) (L9-datadog-agent-observability)

Tier: Tactical • **Flags:** Renamed • **Original graphic label:** not in graphic

Rationale: Right answer for production monitoring in Datadog estates, with strong company-level certifications; SaaS-only and per-span metering make it a complement to, not the home of, evaluation evidence ^{AJ}.

Criterion	Score	Rationale
Technical	3	Strong production tracing, cost, redaction and online evals; offline experiments and CI evaluation not evidenced.
Enterprise readiness	4	SAML SSO, RBAC with default and custom roles, a Roles API for managing roles and permissions, and Audit Trail (B-L9-S003, B-L9-S005): all three controls plus an admin API, so 4 under rule 7 (CP2 Q2); SLA and support terms not captured.
Security and compliance	4	SOC 2 Type 2, ISO 27001, ISO 42001, HIPAA and FedRAMP High at company level (B-L9-S001); module-level scope not stated, so one point below the anchor under the cross-layer scope rule applied to Gemini, Cohere and Mistral.
Deployment flexibility	2	SaaS only, though across isolated regional sites including EU1 (Germany).
Ecosystem	4	OTel GenAI 1.37+ and OpenInference ingest; native APM integration.
Reliability and maturity	3	Established platform, but the module's maturity and rename date are not verified.
Cost / TCO	2	Per-span metering on top of the Datadog platform; rates not captured in the fact base.
Lock-in / portability	3	Proprietary SaaS on standard ingest interfaces; export not verified.
Total (generic / FS)	3.15 / 3.20	

Evidence rules applied: Rule 7 (CP2 Q2) checked at CP2 rework (8 October 2026): enterprise_readiness 4 kept, re-evidenced with the Roles API (B-L9-S003); rule 8 (CP2 Q4) scope rule confirmed for security_compliance 4

Capabilities. Datadog module (documented as Agent Observability under the LLM Observability path) that traces each agent step and LLM call with latency, token usage, cost and errors; scans and redacts sensitive data; identifies prompt injection; runs quality, privacy and safety evaluations; and offers automated Insights⁹¹.

Strengths

- Correlates LLM traces with APM and infrastructure telemetry already in Datadog⁹²
- Accepts OTel traces using OTel 1.37+ GenAI conventions or OpenInference⁹³
- Built-in sensitive-data scanning and redaction in traces⁹⁴
- Datadog Trust Center lists SOC 2 Type 2, ISO/IEC 27001, 27017, 27018, 27701 and 42001, HIPAA and FedRAMP High; EU1 site in Germany; SAML SSO, RBAC with custom roles and an Audit Trail⁹⁵

Limitations and risks

⁹¹ Verified fact — A1-S097: Datadog LLM Observability docs (Datadog, Inc.) https://docs.datadoghq.com/llm_observability/ accessed 2026-10-07; A1-S099: Datadog LLM Observability: Evaluations (Datadog, Inc.) https://docs.datadoghq.com/llm_observability/evaluations/ accessed 2026-10-07

⁹² Verified fact — A1-S097: Datadog LLM Observability docs (Datadog, Inc.) https://docs.datadoghq.com/llm_observability/ accessed 2026-10-07

⁹³ Verified fact — A1-S098: Datadog LLM Observability: OpenTelemetry instrumentation (Datadog, Inc.) https://docs.datadoghq.com/llm_observability/instrumentation/otel_instrumentation/ accessed 2026-10-07

⁹⁴ Verified fact — A1-S097: Datadog LLM Observability docs (Datadog, Inc.) https://docs.datadoghq.com/llm_observability/ accessed 2026-10-07

⁹⁵ Verified fact — B-L9-S001: Datadog Trust Center (Datadog, Inc.) <https://trust.datadoghq.com/> accessed 2026-10-08; B-L9-S002: Getting Started with Datadog Sites (Datadog, Inc.) https://docs.datadoghq.com/getting_started/site/ accessed 2026-10-08; B-L9-S003: Access Control (Datadog RBAC) (Datadog, Inc.) https://docs.datadoghq.com/account_management/rbac/ accessed 2026-10-08; B-L9-S005: Configuring Single Sign-On With SAML (Datadog, Inc.) https://docs.datadoghq.com/account_management/saml/configuration/ accessed 2026-10-08

- SaaS only; self-hosted and VPC deployment not publicly verified ^{NPV}
- Requires the Datadog platform ^{VF⁹⁶}
- Metered per LLM span; rates are on the Datadog pricing page and were not captured ^{VF⁹⁷}
- Whether the Trust Center listings cover this module specifically is not stated ^{NPV}
- Offline/CI evaluation features not documented in the fact base ^{NPV}

Choose when

- Datadog is the firm's APM standard and production LLM monitoring should sit with on-call operations ^{AJ}
- You want sensitive-data redaction applied at the observability tier ^{AJ}

Avoid when

- You need self-hosting or air-gap for trace data ^{AJ}
- It would become the only evaluation tool; pair it with a CI eval harness ^{AJ}

Nearest competitors: L9-arize-ax, L9-langsmith, L9-langfuse

Regulated-FS note. Pin the EU1 site at organisation creation (sites are isolated), turn on sensitive-data scanning before any client data flows, and confirm in the contract that LLM Observability is within the certified scope ^{AJ}.

Facts (as of 2026-10-07)

Field	Value	Label	Sources
Company	Datadog, Inc.	Verified fact	A1-S097
Category	APM-vendor LLM/agent observability and evaluation module	Verified fact	A1-S097
Version / lineup	SaaS module of the Datadog platform (continuous release)	Verified fact	A1-S097
Licence	Proprietary SaaS	Verified fact	A1-S097
Status events	Product documented as 'Agent Observability' at the llm_observability docs path (rename date not verified)	Verified fact	A1-S097
Strategic direction	Agent-centric monitoring with automated Insights (root cause, impact, recommended fix), evaluations of quality, privacy and safety, sensitive-data scanning and prompt-injection detection	Verified fact	A1-S097, A1-S099
What it does	Monitors, troubleshoots and evaluates LLM-powered applications: traces each agent step and LLM call with latency, token usage, cost and errors, scans and redacts sensitive data and runs evaluations.	Verified fact	A1-S097, A1-S099
Stack position	Cross-cutting production observability integrated with Datadog APM and infrastructure monitoring	Verified fact	A1-S097
Integration	Accepts OpenTelemetry traces using OTel 1.37+ GenAI semantic	Verified fact	A1-S098,

⁹⁶ Verified fact — A1-S097: Datadog LLM Observability docs (Datadog, Inc.) https://docs.datadoghq.com/llm_observability/ accessed 2026-10-07

⁹⁷ Verified fact — A1-S097: Datadog LLM Observability docs (Datadog, Inc.) https://docs.datadoghq.com/llm_observability/ accessed 2026-10-07

Field	Value	Label	Sources
	conventions or OpenInference conventions; auto-instrumentation for OpenAI, LangChain, Bedrock, Anthropic	d fact	A1-S097
Dependencies	Datadog platform; Datadog SDK auto-instrumentation or OTel	Verified fact	A1-S097, A1-S098
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Automatic sensitive-data scanning and redaction; prompt-injection identification	Verified fact	A1-S097
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Metered on number of LLM spans ingested (rates on Datadog pricing page)	Verified fact	A1-S097
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Integrates with OpenAI, LangChain, AWS Bedrock, Anthropic; OTel and OpenInference ingestion	Verified fact	A1-S097, A1-S098
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Not publicly verified	Not publicly verified	
Deployment	saas: Yes; managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Not publicly		

Field	Value	Label	Sources
	verified; on_prem: Not publicly verified		

L8: Data extraction, ingestion and web

Docling (L8-docling)

Tier: Strategic · **Flags:** none · **Original graphic label:** Docling – doc parser

Rationale: MIT, foundation-governed and self-hostable, so it can anchor the firm’s canonical document model; enterprise controls are built around it.

Criterion	Score	Rationale
Technical	4	Broad formats, OCR, VLM and chunking; no ACL, PII or lineage, and table accuracy not independently verified.
Enterprise readiness	3	Library rule: inherits host controls; docling-serve gives an API boundary; commercial managed option via IBM watsonx.
Security and compliance	4	Library rule: MIT; SECURITY.md with private vulnerability reporting, OpenSSF Best Practices badge participation and cryptographically signed releases on PyPI, Quay.io and GHCR ^{vf98} ; IBM offers a managed service.
Deployment flexibility	4	Self-hosted, private cloud, on-prem and an IBM-managed service; air-gap not explicitly documented.
Ecosystem	4	MCP server, REST server and agent-framework integrations; LF AI & Data project.
Reliability and maturity	4	Graduate-tier foundation project with steady releases; rapid minor-version churn.
Cost / TCO	5	Free; CPU sufficient for standard pipelines, GPU only for VLM/ASR paths ^{vi} .
Lock-in / portability	5	MIT, open format, neutral governance.
Total (generic / FS)	4.00 / 4.05	

Evidence rules applied: Library rule applied (enterprise_readiness and security_compliance scored on what the library enables and project hygiene; inherits host controls)

Capabilities. Open-source conversion of PDF, Office, legacy Office, HTML, images, audio and video into a unified DoclingDocument, with OCR, VLM pipelines, ASR and hybrid chunking; runs as a library, CLI, docling-serve REST API or MCP server ^{vf99}.

Strengths

- MIT licence and LF AI & Data governance (donated March 2025; Graduate tier August 2026) ^{vf100}
- Local or self-hosted processing keeps documents in the firm’s estate ^{vi}

⁹⁸ Verified fact — B-REV-S001: Security and Disclosure Information Policy for the Docling Project (Docling Project (LF AI & Data)) <https://raw.githubusercontent.com/docling-project/docling/main/.github/SECURITY.md> accessed 2026-10-08

⁹⁹ Verified fact — A1-S057: Docling README (Docling Project (GitHub)) <https://raw.githubusercontent.com/docling-project/docling/main/README.md> accessed 2026-10-07; A1-S111: Docling supported formats (docs source) (Docling Project (GitHub docs source)) https://raw.githubusercontent.com/docling-project/docling/main/docs/usage/supported_formats.md accessed 2026-10-07; A1-S082: Docling Serve README (Docling Project (GitHub)) <https://raw.githubusercontent.com/docling-project/docling-serve/main/README.md> accessed 2026-10-07

- Broad format coverage including PPTX, XLSX and legacy Office ^{VF-101}
- A documented canonical representation (DoclingDocument) that can serve as the firm’s internal document model ^{AJ}

Limitations and risks

- No ACL, PII or lineage features in the README or docs navigation ^{VF-102}
- Very fast release cadence (2.135.0 on 7 October 2026; 223 releases in about two years) makes version pinning mandatory ^{VF-103}
- Individual models carry their own licences, which need separate review ^{VF-104}
- A security policy with private vulnerability reporting, OpenSSF Best Practices badge participation and signed releases is published ^{VF-105}

Choose when

- Documents must not leave the firm’s estate
- You want an open canonical document model that survives a change of parser or OCR engine

Avoid when

- You need managed, permission-aware connectors out of the box
- You have no platform team to run, pin and patch a Python service

Nearest competitors: L8-unstructured, L8-mineru, L8-llamaparse

Regulated-FS note. Default self-hosted parser for client and confidential documents; pin the docling version and each model version in a parse manifest so that a parse can be reproduced for audit ^{REC}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Docling Project, hosted by the LF AI & Data Foundation; started by IBM Research Zurich	Verified fact	A1-S057, V1-S091
Category	Open-source document conversion and parsing toolkit (PDF, Office, HTML, images, audio) with VLM pipelines and chunking	Verified fact	A1-S057, A1-S111
Version / lineup	docling 2.135.0, released 7 October 2026; docling-serve has a stable v1 API	Verified fact	A1-S008, A1-S082
Licence	MIT (codebase); individual models carry their own licences	Verified fact	A1-S057, A1-S008

¹⁰⁰ Verified fact — A1-S057: Docling README (Docling Project (GitHub)) <https://raw.githubusercontent.com/docling-project/docling/main/README.md> accessed 2026-10-07; V1-S091: Docling - LF AI & Data project page; LF press release 29 April 2025 (IBM contributes Docling, Data Prep Kit, BeeAI) (LF AI & Data Foundation) <https://lfaidata.foundation/projects/docling/> accessed 2026-10-08 [search-tool extract]

¹⁰¹ Verified fact — A1-S111: Docling supported formats (docs source) (Docling Project (GitHub docs source)) https://raw.githubusercontent.com/docling-project/docling/main/docs/usage/supported_formats.md accessed 2026-10-07

¹⁰² Verified fact — A1-S057: Docling README (Docling Project (GitHub)) <https://raw.githubusercontent.com/docling-project/docling/main/README.md> accessed 2026-10-07

¹⁰³ Verified fact — A1-S008: PyPI JSON metadata: docling (Python Package Index) <https://pypi.org/pypi/docling/json> accessed 2026-10-07

¹⁰⁴ Verified fact — A1-S057: Docling README (Docling Project (GitHub)) <https://raw.githubusercontent.com/docling-project/docling/main/README.md> accessed 2026-10-07

¹⁰⁵ Verified fact — B-REV-S001: Security and Disclosure Information Policy for the Docling Project (Docling Project (LF AI & Data)) <https://raw.githubusercontent.com/docling-project/docling/main/.github/SECURITY.md> accessed 2026-10-08

Field	Value	Label	Sources
Status events	March 2025: IBM donated Docling to the LF AI & Data Foundation as an Incubation-stage project (formal induction announced 29 April 2025); August 2026: graduated to Graduate-tier LF AI & Data project; Repository now under the docling-project organisation	Verified fact	A1-S057, V1-S091
Strategic direction	VLM support including IBM GraniteDocling, MCP server for agents, docling-serve API server, managed 'Docling for IBM watsonx'	Verified fact	A1-S057, A1-S082, A1-S110
What it does	Parses many document formats into a unified DoclingDocument representation and exports Markdown, JSON and other formats; supports OCR, VLM pipelines, ASR for audio/video and hybrid chunking.	Verified fact	A1-S057, A1-S111
Stack position	L8 document parsing; runs locally or as an API service	Verified fact	A1-S057, A1-S082
Integration	Python library and CLI, REST API (docling-serve), MCP server; integrations with agent frameworks	Verified fact	A1-S057, A1-S082
Dependencies	Python; LibreOffice for legacy Office/RTF; optional ASR extra and ffmpeg for media; optional VLMs	Verified fact	A1-S111
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	Local/self-hosted processing keeps documents in the customer environment	Architectural judgment	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Commercial managed option via IBM watsonx	Verified fact	A1-S110
Pricing	Free (MIT); managed IBM watsonx pricing not captured	Verified fact	A1-S057, A1-S110
Infrastructure cost	CPU is sufficient for standard pipelines; VLM and ASR pipelines benefit from GPU	Architectural judgment	
Ecosystem	Format coverage: PDF, DOCX/XLSX/PPTX, legacy Office, ODF, EPUB, Apple iWork, Markdown, AsciiDoc, LaTeX, HTML, MHTML, CSV, images, audio, video	Verified fact	A1-S111
Adoption	Not publicly verified	Not	

Field	Value	Label	Sources
signals		publicly verified	
Maturity	First PyPI release 15 July 2024; 223 releases to 7 October 2026	Verified fact	A1-S008
Deployment	saas: Not publicly verified; managed_cloud: Yes (Docling for IBM watsonx managed service); vpc_byoc: Not publicly verified; private_cloud: Yes; self_hosted: Yes; on_prem: Yes		

Unstructured: open-source ‘unstructured’ library and ‘unstructured-ingest’ connectors; commercial Transform v2 API / platform (L8-unstructured)

Tier: Strategic · **Flags:** Renamed · **Original graphic label:** Unstructured – ETL for docs

Rationale: The open-source ingest connectors are the only verified source of ACL metadata and incremental reprocessing in L8, with an in-VPC commercial path.

Criterion	Score	Rationale
Technical	4	Partitioning, chunking and the only verified ACL-aware incremental connectors; best table quality is API-only.
Enterprise readiness	3	SAML 2.0/OIDC SSO and account- and workspace-level roles mapped from IdP groups on Business deployments ^{vr106} ; no workspace audit log documented (stdout logging for the customer’s own collection), so 3.
Security and compliance	4	Vendor-listed SOC 2 Type 2, ISO 27001, HIPAA and CMMC 2.0 Level 2 (medium confidence); FedRAMP wording is ‘alignment’ only and is not relied on; data region not verified.
Deployment flexibility	5	Shared SaaS, dedicated, in-VPC, self-hosted open source and vendor-marketed on-prem/air-gapped.
Ecosystem	4	Large connector catalogue into vector stores and warehouses.
Reliability and maturity	3	Four years of releases, but pre-1.0 library and a PyPI/changelog version mismatch.
Cost / TCO	3	Open source free; API pricing published but conflicting.
Lock-in / portability	4	Apache-2.0 library and connectors; proprietary API for best models.
Total (generic / FS)	3.80 / 3.85	

Evidence rules applied: NPV cap lifted at CP2 review (8 October 2026): SSO and RBAC verified (B-REV-S016); audit-log feature not documented; Rule 7 (CP2 Q2) checked at CP2 rework: SSO and RBAC verified, audit logs not documented, so 3 kept (maximum without all three controls)

¹⁰⁶ Verified fact — B-REV-S016: Unstructured: IdP overview, Configure an identity provider, Security and compliance overview (Unstructured Technologies) <https://docs.unstructured.io/business/idp/overview> accessed 2026-10-08 [search-tool extract]

Capabilities. Apache-2.0 ‘unstructured’ partitioning library and ‘unstructured-ingest’ connectors (SharePoint including Teams channel files, OneDrive, Confluence, Salesforce, S3 and others) that move data to vector stores and warehouses; commercial Transform v2 API with shared, dedicated and in-VPC tiers ^{vf-107}.

Strengths

- Only L8 product with verified permission-aware ingestion: a permissions_version SHA-256 ACL digest that triggers reprocessing on ACL-only change ^{vf-108}
- Distinguishes an unavailable permission fetch from an empty permission set ^{vf-109}
- Vendor lists SOC 2 Type 2, ISO 27001, HIPAA and CMMC 2.0 Level 2 ^{vf-110}
- Open-source library and connectors stay free, so the pipeline can run in the firm’s estate ^{vf-111}

Limitations and risks

- No workspace audit-log feature is documented; logs go to stdout for the customer’s own collection ^{vf-112}
- Highest-quality models only via the proprietary API ^{vf-113}
- Library still pre-1.0 after four years (0.27.16) ^{vf-114}
- Per-page price conflict (US\$0.015 vs US\$0.03) ^{vf-115}
- Library sends usage analytics by default ^{vf-116}

Choose when

- You need connector-level ACL capture and incremental reprocessing from SharePoint, OneDrive or Confluence
- You want one pipeline that can run as open source in-estate or as an in-VPC commercial tier

Avoid when

¹⁰⁷ Verified fact — A1-S075: Unstructured open-source library README (Unstructured Technologies (GitHub)) <https://raw.githubusercontent.com/Unstructured-IO/unstructured/main/README.md> accessed 2026-10-07; A1-S094: unstructured-ingest CHANGELOG (Unstructured Technologies (GitHub)) <https://raw.githubusercontent.com/Unstructured-IO/unstructured-ingest/main/CHANGELOG.md> accessed 2026-10-07; A1-S117: Unstructured Pricing Plans for Data Processing <https://unstructured.io/pricing> accessed 2026-10-07 [search-tool extract]; V1-S019: Unstructured Transform: Parse your first document (API v2) <https://docs.unstructured.io/transform/first-request> accessed 2026-10-08 [search-tool extract]

¹⁰⁸ Verified fact — A1-S094: unstructured-ingest CHANGELOG (Unstructured Technologies (GitHub)) <https://raw.githubusercontent.com/Unstructured-IO/unstructured-ingest/main/CHANGELOG.md> accessed 2026-10-07

¹⁰⁹ Verified fact — A1-S094: unstructured-ingest CHANGELOG (Unstructured Technologies (GitHub)) <https://raw.githubusercontent.com/Unstructured-IO/unstructured-ingest/main/CHANGELOG.md> accessed 2026-10-07

¹¹⁰ Verified fact — A1-S117: Unstructured Pricing Plans for Data Processing <https://unstructured.io/pricing> accessed 2026-10-07 [search-tool extract]; V1-S090: Unstructured API overview (compliance list); pricing; CMMC 2.0 Level 2 blog <https://docs.unstructured.io/api-reference/overview> accessed 2026-10-08 [search-tool extract]

¹¹¹ Verified fact — A1-S075: Unstructured open-source library README (Unstructured Technologies (GitHub)) <https://raw.githubusercontent.com/Unstructured-IO/unstructured/main/README.md> accessed 2026-10-07

¹¹² Verified fact — B-REV-S016: Unstructured: IdP overview, Configure an identity provider, Security and compliance overview (Unstructured Technologies) <https://docs.unstructured.io/business/idp/overview> accessed 2026-10-08 [search-tool extract]

¹¹³ Verified fact — A1-S075: Unstructured open-source library README (Unstructured Technologies (GitHub)) <https://raw.githubusercontent.com/Unstructured-IO/unstructured/main/README.md> accessed 2026-10-07

¹¹⁴ Verified fact — A1-S011: PyPI JSON metadata: unstructured (Python Package Index) <https://pypi.org/pypi/unstructured/json> accessed 2026-10-07

¹¹⁵ Verified fact — A1-S117: Unstructured Pricing Plans for Data Processing <https://unstructured.io/pricing> accessed 2026-10-07 [search-tool extract]; A1-S118: Unstructured Transform billing <https://docs.unstructured.io/transform/billing> accessed 2026-10-07 [search-tool extract]

¹¹⁶ Verified fact — A1-S075: Unstructured open-source library README (Unstructured Technologies (GitHub)) <https://raw.githubusercontent.com/Unstructured-IO/unstructured/main/README.md> accessed 2026-10-07

- You need a documented, exportable audit log on the hosted service today
- Your corpus is mostly scanned or table-heavy and you will not pay for the API models ^{AJ}

Nearest competitors: L8-docling, L8-llamaparse, L8-reducto

Regulated-FS note. The connector ACL digest is the nearest thing in this layer to entitlement-preserving ingestion; disable default analytics and confirm the hosted data region before any client data is sent ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Unstructured (Unstructured-IO)	Verified fact	A1-S075, A1-S012
Category	Document partitioning and ETL for LLMs: open-source library plus commercial hosted API	Verified fact	A1-S075
Version / lineup	unstructured 0.27.16 (5 October 2026); unstructured-ingest 1.11.19 on PyPI (23 September 2026; changelog lists 1.11.21); unstructured-client SDK 0.46.2	Verified fact	A1-S011, A1-S095, A1-S094, A1-S012
Licence	Open source Apache-2.0 (library and ingest); commercial API proprietary; client SDK MIT	Verified fact	A1-S011, A1-S095, A1-S012, A1-S075
Status events	Commercial API now branded 'Transform v2' (date not verified)	Verified fact	A1-S075
Strategic direction	Open-source library 'is, and will stay, completely free'; higher-quality models offered via the Transform v2 API, which the vendor claims gives 2x table content accuracy and 58% less invented content than the library	Verified fact	A1-S075
What it does	Partitions PDFs, HTML, Word and many other formats into structured elements, with chunking, and moves data from source connectors to destinations (vector DBs, warehouses).	Verified fact	A1-S075, A1-S094
Stack position	L8 ingestion/ETL, including connector-level permissions metadata relevant to H7	Verified fact	A1-S094
Integration	Python library, ingest connectors (SharePoint incl. Teams channel files, OneDrive, Confluence, Salesforce, Databricks, S3, Teradata, Milvus and others), hosted API with client SDK	Verified fact	A1-S094, A1-S012, A1-S075
Dependencies	Python; tesseract and poppler for PDFs/images; Docker images available	Verified fact	A1-S075

Field	Value	Label	Sources
Certifications	SOC 2 Type 2, HIPAA, ISO 27001 and GDPR listed on pricing page and API docs; CMMC 2.0 Level 2 certification (110/110, announced c. November 2025); API docs also claim “adherence” to FedRAMP, but no FedRAMP authorisation was found - do not state FedRAMP; certificates via trust.unstructured.io	Verified fact	A1-S117, V1-S090
GDPR / residency	DPA available; GDPR claim scoped to SaaS hosted product; data region not verified	Verified fact	A1-S117
Security features	Ingest connectors emit a permissions_version ACL digest (SHA-256) so ACL-only changes trigger reprocessing; credential redaction in connector errors; library sends lightweight usage analytics by default	Verified fact	A1-S094, A1-S075
Access controls	Connector-level permission metadata captured from SharePoint, OneDrive and Confluence (ACL digest); platform RBAC/SSO not verified	Verified fact	A1-S094
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Let's Go free (10,000 pages once per account); Pay-As-You-Go US\$0.015 per page; Business (dedicated or in-VPC) custom. Conflict: Terms of Service and older pages state US\$0.03 per page after 15,000 free pages/month	Verified fact	A1-S117, A1-S118
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Large connector catalogue in unstructured-ingest	Verified fact	A1-S094
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	unstructured first on PyPI 6 September 2022; 237 releases	Verified fact	A1-S011
Deployment	saas: Yes (hosted API; Let's Go / Pay-As-You-Go shared); managed_cloud: Yes (dedicated instance in Unstructured's cloud, Business plan); vpc_byoc: Yes (in-VPC in customer cloud, Business plan); private_cloud: Yes (open source); self_hosted: Yes (open source); on_prem: Yes (open source; vendor markets on-prem/air-gapped)		

Google Cloud Document AI (L8-google-document-ai)

Tier: Strategic • **Flags:** none • **Original graphic label:** not in graphic

Rationale: Strategic, conditional: where Google Cloud is your primary cloud (CP3 Q2, rubric rule 10). Google's lead document-processing service; confirm the processor region and keep outputs in the firm's canonical document model to limit processor lock-in ^{A1}.

Criterion	Score	Rationale
Technical	4	OCR, form parsing, layout parsing with chunking, custom extraction and classification.
Enterprise readiness	4	Consumed through Google Cloud: IAM including deny policies, VPC Service Controls, IAM-governed Cloud Audit Logs and SAML/OIDC federation at platform level ^{vf117} ; rule 6 (CP2 Q1) presumes platform controls, and no product-specific gap is evidenced, so 4; platform controls presumed (CP2 Q1); confirm per service. Not 5: Document AI's predefined roles and SLA terms were not read.
Security and compliance	4	In scope for Google Cloud ISO 27001/27017/27018, SOC 1/2/3 and PCI DSS (A1-S102), plus CMEK and US/EU data residency (B-REV-S010, B-REV-S011); CMEK needs allowlisting in some regions. Held at 4, not 5, under rule 8: the scope page names a 'SOC 2 Report' without stating Type II, as for AgentCore Memory (L5); the same Google scope evidence gives 4 for Sensitive Data Protection, Model Armor and Apigee.
Deployment flexibility	2	Managed SaaS only; us and eu multi-regions and regional processors (B-REV-S010, B-REV-S011); no VPC or self-hosted option.
Ecosystem	3	Google Cloud ecosystem; integrations not captured.
Reliability and maturity	3	Hyperscaler managed service; maturity facts not captured ^{AI} .
Cost / TCO	4	Transparent per-page pricing with volume tiers.
Lock-in / portability	2	Google-specific processors and outputs; adds to hyperscaler concentration.
Total (generic / FS)	3.40	
	/	
	3.25	

Evidence rules applied: NPV cap lifted at CP2 review (8 October 2026): platform IAM, Cloud Audit Logs and federated SSO verified (B-REV-S010, S012); Rule 6 (CP2 Q1, hyperscaler presumption) applied at CP2 rework: enterprise_readiness 3 -> 4; platform controls presumed (CP2 Q1); confirm per service; Rule 8 (CP2 Q4) re-checked at CP3 rework (8 October 2026): certifications are product-scoped (A1-S102) with CMEK, but the SOC 2 report type is not stated, so security_compliance 5 -> 4 for consistency with other hyperscaler services

Capabilities. Google Cloud managed OCR, parsing, extraction and classification through processors: Enterprise Document OCR, Form Parser, Layout Parser (with initial chunking), Custom Extractor, classifier/splitter and Summarizer ^{vf118}.

Strengths

- In scope for Google Cloud ISO 27001/27017/27018, SOC 1/2/3 and PCI DSS ^{vf119}
- Published per-page pricing; Enterprise Document OCR at US\$1.50 per 1,000 pages ^{vf120}
- Runs on a provider designated under DORA and the UK CTP regime (Google Cloud EMEA Limited) ^{vf121}

Limitations and risks

¹¹⁷ Verified fact — B-REV-S010: Document AI security and compliance (Google Cloud) <https://cloud.google.com/document-ai/docs/security-cmek-logging> accessed 2026-10-08 [search-tool extract]; B-REV-S012: Set up user access to the console (federated) (Google Cloud) <https://docs.cloud.google.com/iam/docs/workforce-console-ssso> accessed 2026-10-08 [search-tool extract]

¹¹⁸ Verified fact — A1-S101: Document AI pricing (Google Cloud) <https://cloud.google.com/document-ai/pricing> accessed 2026-10-07

¹¹⁹ Verified fact — A1-S102: Google Cloud services in scope for compliance <https://cloud.google.com/security/compliance/services-in-scope> accessed 2026-10-07

¹²⁰ Verified fact — A1-S101: Document AI pricing (Google Cloud) <https://cloud.google.com/document-ai/pricing> accessed 2026-10-07

- Processor outputs are Google-specific and require Google Cloud ^{VF}¹²²
- IAM, VPC Service Controls, us/eu multi-regions and CMEK were evidenced at CP2 review ^{VF}¹²³; support and SLA terms not captured ^{NPV}; platform controls presumed (CP2 Q1); confirm per service ^{Rec}
- SaaS only; no self-hosting ^{AJ}
- Form Parser and Custom Extractor at US\$30 per 1,000 pages ^{VF}¹²⁴

Choose when

- Google Cloud is already the firm’s approved cloud and processing region is confirmed
- You want hyperscaler certifications and consolidated third-party oversight

Avoid when

- Documents must stay on-prem or in a non-GCP estate
- You are trying to reduce hyperscaler concentration

Nearest competitors: L8-mistral-ocr, L8-reducto, Azure AI Document Intelligence (not in fact base), AWS Textract (not in fact base)

Regulated-FS note. Natural choice for a GCP-centred firm; confirm the processor region before use and keep outputs in the firm’s canonical document model to avoid processor lock-in ^{Rec}.

Facts (as of 2026-10-07)

Field	Value	Label	Sources
Company	Google Cloud (Alphabet)	Verified fact	A1-S101
Category	Managed document OCR, parsing, extraction and classification service	Verified fact	A1-S101
Version / lineup	Processors incl. Enterprise Document OCR, Form Parser, Layout Parser (with initial chunking), Custom Extractor, classifier/splitter, Summarizer	Verified fact	A1-S101
Licence	Proprietary cloud service	Verified fact	A1-S101
Strategic direction	Not publicly verified	Not publicly verified	
What it does	Extracts text, structure and entities from documents via pre-trained and custom processors, including layout parsing with chunking for RAG.	Verified fact	A1-S101
Stack position	L8 OCR/parsing as a hyperscaler managed service	Verified	A1-S101

¹²¹ Verified fact — A8-S020: The ESAs designate critical ICT third-party providers under DORA (with List_of_designated_CTPPs.pdf, 18 Nov 2025) (EBA / ESMA / EIOPA (Joint ESAs)) <https://www.eba.europa.eu/publications-and-media/press-releases/european-supervisory-authorities-designate-critical-ict-third-party-providers-under-digital> accessed 2026-10-07 [search-tool extract]; A8-S023: The Critical Third Parties (Designation) Regulations 2026 (SI 2026/777); BoE/FCA/HMT announcements (HM Treasury / legislation.gov.uk / Bank of England) <https://www.legislation.gov.uk/uksi/2026/777/made> accessed 2026-10-07 [search-tool extract]

¹²² Verified fact — A1-S101: Document AI pricing (Google Cloud) <https://cloud.google.com/document-ai/pricing> accessed 2026-10-07

¹²³ Verified fact — B-REV-S010: Document AI security and compliance (Google Cloud) <https://cloud.google.com/document-ai/docs/security-cmek-logging> accessed 2026-10-08 [search-tool extract]; B-REV-S011: Document AI customer-managed encryption keys (CMEK) (Google Cloud) <https://docs.cloud.google.com/document-ai/docs/cmek> accessed 2026-10-08 [search-tool extract]

¹²⁴ Verified fact — A1-S101: Document AI pricing (Google Cloud) <https://cloud.google.com/document-ai/pricing> accessed 2026-10-07

Field	Value	Label	Sources
Integration	Not publicly verified	Not publicly verified	
Dependencies	Google Cloud project	Verified fact	A1-S101
Certifications	In scope for Google Cloud ISO 27001/27017/27018, SOC 1/2/3, PCI DSS and penetration testing	Verified fact	A1-S102
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Enterprise Document OCR US\$1.50 per 1,000 pages (1k-5M pages/month; US\$0.60 above 5M; first 1,000 free); Form Parser and Custom Extractor US\$30 per 1,000 pages (first 1M, US\$20 above); Layout Parser US\$10 per 1,000 pages; discounts for consumption models	Verified fact	A1-S101
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
Maturity	Not publicly verified	verified Not publicly verified	
Deployment	saas: Yes; managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Not publicly verified; on_prem: Not publicly verified		

Mistral OCR 4.1 (alias mistral-ocr-latest), part of Mistral Document AI (L8-mistral-ocr)

Tier: Tactical · **Flags:** none · **Original graphic label:** Mistral OCR – OCR

Rationale: A good EU or self-managed OCR engine to sit behind an interface, but short model-retirement windows make it a component, not a foundation.

Criterion	Score	Rationale
Technical	3	Strong OCR feature set with confidence scores; covers only the OCR step of the pipeline.
Enterprise readiness	4	Enterprise plan: SAML SSO, Admin/Billing/Member roles with isolated Workspaces and SCIM provisioning, audit logs on by default ^{v125} ; all three controls plus SCIM, so 4 under rule 7 (CP2 Q2). Not 5: audit-log export is not supported, OIDC is not supported and SLA terms are NPV.
Security and compliance	3	Company-level SOC 2 Type II, ISO 27001 and ISO 27701; OCR scope unconfirmed.
Deployment flexibility	4	EU SaaS, private cloud, self-managed container and on-prem under enterprise agreement.
Ecosystem	3	Mistral SDKs with opt-in OTEL spans; usable inside other parsers.
Reliability and maturity	2	OCR 4.0 retired about five weeks after 4.1 reached GA; frequent version replacement.

¹²⁵ Verified fact — B-REV-S014: Mistral docs: SAML SSO, SSO and Admin overview (Mistral AI) <https://docs.mistral.ai/admin/set-up-organization/sign-in-method/saml-ssso> accessed 2026-10-08 [search-tool extract]; B-REV-S015: Mistral docs: Audit logs (Mistral AI) <https://docs.mistral.ai/admin/security-access/audit-logs> accessed 2026-10-08 [search-tool extract]

Criterion	Score	Rationale
Cost / TCO	4	US\$4 per 1,000 pages, US\$2 batch; self-host pricing unpublished.
Lock-in / portability	3	Proprietary model, but text/JSON output is easy to swap behind an interface.
Total (generic / FS)	3.30 / 3.25	

Evidence rules applied: NPV cap lifted at CP2 review (8 October 2026): SSO, RBAC and audit logs verified (B-REV-S014, S015); Rule 7 (CP2 Q2) applied at CP2 rework: all three controls plus SCIM; enterprise_readiness 3 -> 4; audit-log export not supported (stated as a due-diligence condition); Rule 8 (CP2 Q4) confirmed: company-level certifications with OCR scope not confirmed score one below the anchor (3)

Capabilities. Mistral OCR 4.1 (alias mistral-ocr-latest), part of Mistral Document AI: OCR endpoint returning structured output with bounding boxes, block labels, table formatting, header/footer extraction and block-level confidence scores ^{vf126}.

Strengths

- EU-hosted managed API ^{vf127}
- Self-managed single-container, private-cloud and on-prem deployment for enterprise customers ^{vf128}
- Low published price: US\$4 per 1,000 pages, US\$2 in batch ^{vf129}
- Block-level confidence scores support quarantine thresholds ^{vf130}

Limitations and risks

- Model-version churn: OCR 4.1 GA 26 August 2026 and OCR 4.0 retired 30 September 2026; the original model is no longer maintained ^{vf131}
- Certifications are company-level; OCR scope not confirmed ^{vf132}
- Self-hosting licence terms and hardware sizing not published ^{NPV}
- OCR only: no connectors, ACL, PII or lineage ^{AJ}
- Audit logs cannot be exported ^{vf133}, so the firm cannot hold them in its own archive without a vendor export route ^{AJ}

¹²⁶ Verified fact — A1-S083: Mistral Python SDK: OCR operation docs (Mistral AI (GitHub)) <https://raw.githubusercontent.com/mistralai/client-python/main/docs/sdks/ocr/README.md> accessed 2026-10-07; A1-S130: OCR 4.1 - Mistral AI docs <https://docs.mistral.ai/models/ocr-4-1> accessed 2026-10-07 [search-tool extract]

¹²⁷ Verified fact — A1-S135: Mistral OCR 4 : SOTA OCR for Document Intelligence (Mistral AI) <https://mistral.ai/news/ocr-4/> accessed 2026-10-07 [search-tool extract]

¹²⁸ Verified fact — A1-S135: Mistral OCR 4 : SOTA OCR for Document Intelligence (Mistral AI) <https://mistral.ai/news/ocr-4/> accessed 2026-10-07 [search-tool extract]

¹²⁹ Verified fact — A1-S130: OCR 4.1 - Mistral AI docs <https://docs.mistral.ai/models/ocr-4-1> accessed 2026-10-07 [search-tool extract]; V1-S010: OCR 4.1 model card / changelog / pricing (Mistral AI) <https://docs.mistral.ai/models/ocr-4-1> accessed 2026-10-08 [search-tool extract]

¹³⁰ Verified fact — A1-S130: OCR 4.1 - Mistral AI docs <https://docs.mistral.ai/models/ocr-4-1> accessed 2026-10-07 [search-tool extract]

¹³¹ Verified fact — A1-S130: OCR 4.1 - Mistral AI docs <https://docs.mistral.ai/models/ocr-4-1> accessed 2026-10-07 [search-tool extract]; V1-S014: Mistral docs Changelog; OCR 4.0 model page; OCR 4 governance page (Mistral AI) <https://docs.mistral.ai/resources/changelogs> accessed 2026-10-08 [search-tool extract]

¹³² Verified fact — A1-S136: Do you have SOC 2 or ISO 27001 certification? - Mistral AI Help Center <https://help.mistral.ai/en/articles/347638-do-you-have-soc-2-or-iso-27001-certification> accessed 2026-10-07 [search-tool extract]

¹³³ Verified fact — B-REV-S015: Mistral docs: Audit logs (Mistral AI) <https://docs.mistral.ai/admin/security-access/audit-logs> accessed 2026-10-08 [search-tool extract]

Choose when

- You need an EU-resident or self-managed OCR engine behind your own parsing pipeline
- Scanned documents dominate and cost per page matters

Avoid when

- You cannot re-validate within a deprecation window of weeks
- You call the 'latest' alias in production

Nearest competitors: L8-google-document-ai, L8-docling, L8-reducto

Regulated-FS note. Pin the dated model ID (not mistral-ocr-latest), store OCR outputs with the model ID, and treat each model retirement as a change-control event with re-validation ^{Rec.}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Mistral AI	Verified fact	A1-S083, A1-S084
Category	Hosted OCR / document understanding model API	Verified fact	A1-S083
Version / lineup	OCR 4.1 (mistral-ocr-4-1; aliases mistral-ocr-latest and mistral-ocr-4): released July 2026 (model page 16 July; changelog 26 July) and marked Generally Available in the changelog on 26 August 2026 (governance page gives a 13 August 2026 release date); OCR 4.0 deprecated late September and retired 30 September 2026; OCR 3 (25.12) still available for existing integrations; original Mistral OCR no longer maintained	Verified fact	A1-S130, V1-S014, V1-S010
Licence	Proprietary model; hosted API, with self-managed deployment for enterprise customers under agreement	Verified fact	A1-S135
Status events	OCR 4.0 retired 30 September 2026 (replaced by OCR 4.1 at same price); Original Mistral OCR no longer maintained	Verified fact	A1-S130, V1-S014
Strategic direction	OCR 4 positioned for document intelligence: paragraph-level bounding boxes, structural block labels, block-level confidence scores; compact single-container self-hosting for residency and sovereignty	Verified fact	A1-S130, A1-S135
What it does	OCR endpoint that processes a document (URL or file) and returns structured output, with optional bounding-box and document annotations in a requested format, table formatting, header/footer extraction, layout blocks and confidence scores.	Verified fact	A1-S083
Stack position	L8 OCR; also usable as a model inside other parsers	Verified fact	A1-S083

Field	Value	Label	Sources
Integration	REST /v1/ocr via Mistral SDKs; SDK can emit opt-in OpenTelemetry traces following GenAI semantic conventions	Verified fact	A1-S083, A1-S084
Dependencies	Mistral API (La Plateforme)	Verified fact	A1-S083
Certifications	Mistral AI company-level SOC 2 Type II, ISO/IEC 27001:2022 and ISO/IEC 27701:2019 (Trust Center); scope for OCR not confirmed	Verified fact	A1-S136
GDPR / residency	Managed API runs on Mistral infrastructure hosted in the EU; self-hosting keeps documents in customer environment	Verified fact	A1-S135
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	US\$4 per 1,000 pages (cached input US\$0.40); annotated pages US\$5 per 1,000; 50% batch discount (US\$2 per 1,000); no-code Document AI US\$5 per 1,000 pages	Verified fact	A1-S130, A1-S135, V1-S010
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Python SDK mistralai 3.1.0 (6 October 2026)	Verified fact	A1-S017
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Not publicly verified	Not publicly verified	
Deployment	saas: Yes (EU-hosted serverless API); managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Yes (private cloud, enterprise); self_hosted: Yes (single-container self-managed, enterprise); on_prem: Yes (on-premises, enterprise)		

LlamaParse (now the name of LlamaIndex’s whole document platform: Parse, Extract, Index, Split, Agents; previously marketed as LlamaCloud) (L8-11lamaparse)

Tier: Tactical · **Flags:** Renamed · **Original graphic label:** LlamaParse – PDF / documents

Rationale: Capable managed parser with EU and BYOC options, but proprietary credits, SDK churn and a platform that reaches into L6/L3 argue against a foundational dependency.

Criterion	Score	Rationale
Technical	4	Parse, extract, split, classify and index across 130+ formats; no ACL, PII or lineage.
Enterprise readiness	3	SSO through SAML and OIDC on the managed service and the Enterprise plan is verified (A1-S116, B-REV-S023): one verified control lifts the NPV cap to 3 under rule 7 (CP2 Q2). Hosted organisations are flat (every member has the same access) ^{vf134} , named roles exist only when self-hosted, and audit logs are not documented, so not above 3.
Security and compliance	3	SOC 2 Type II and HIPAA BAA (medium confidence); no ISO 27001 found.
Deployment flexibility	4	SaaS NA/EU, BYOC, self-hosted and on-prem; air-gap not stated.
Ecosystem	4	LlamaIndex framework, MCP server, Python/TypeScript SDKs.
Reliability and maturity	3	Rename and SDK migration in 2026; company focus has shifted onto this product.
Cost / TCO	3	Published credits, but agentic tiers cost up to 45 credits per page.
Lock-in / portability	2	Proprietary credit model and API; managed index and agents deepen dependency.
Total (generic / FS)	3.40 / 3.20	

Evidence rules applied: NPV cap lifted at CP2 rework (8 October 2026) under rule 7 (CP2 Q2): SSO verified (A1-S116, B-REV-S023); enterprise_readiness 2 -> 3, held at 3 because hosted roles are flat and audit logs are not documented

Capabilities. LlamaParse is now LlamaIndex’s whole document platform (formerly LlamaCloud): agentic OCR and parsing of 130+ formats in priced tiers, structured extraction, splitting, classification, managed ingest/index/RAG pipelines and document agents ^{vf135}.

Strengths

- EU region with in-region storage and processing, EU DPA and SCCs ^{vf136}

¹³⁴ Verified fact — B-REV-S023: LlamaParse: Access & Identity (and Organizations, User Roles) (LlamaIndex, Inc.) <https://developers.llamaindex.ai/llamaparse/general/enterprise-readiness/access-identity/> accessed 2026-10-08 [search-tool extract]

¹³⁵ Verified fact — A1-S080: LlamaIndex README (company focus shift to LlamaParse) (LlamaIndex, Inc. (GitHub)) https://raw.githubusercontent.com/run-llama/llama_index/main/README.md accessed 2026-10-07; A1-S081: Llama Cloud Python SDK README (LlamaIndex, Inc. (GitHub)) <https://raw.githubusercontent.com/run-llama/llama-cloud-py/main/README.md> accessed 2026-10-07; V1-S008: LlamaParse (formerly LlamaCloud) (LlamaIndex) https://developers.llamaindex.ai/python/framework/llama_cloud/ accessed 2026-10-08 [search-tool extract]

¹³⁶ Verified fact — A1-S115: Compliance - LlamaParse documentation (Enterprise Readiness) (LlamaIndex, Inc.) <https://developers.llamaindex.ai/llamaparse/general/enterprise-readiness/compliance/> accessed 2026-10-07 [search-tool extract]

- BYOC on Azure, AWS and GCP, self-hosting via Helm, and on-premises ^{VF137}
- SOC 2 Type II and a HIPAA pipeline with BAA on request ^{VF138}
- Transparent credit pricing by parse tier ^{VF139}

Limitations and risks

- Permission sync, PII detection and lineage not found ^{VF140}
- SDK churn: legacy llama-cloud-services and llama-parse packages deprecated in 2026 ^{VF141}
- Platform scope spans L8, L6 and L3, which invites a single-vendor knowledge stack ^{AJ}
- Whether SOC 2 covers BYOC or self-hosted installs is not stated ^{NPV}
- On the hosted service every organisation member has the same access to projects and resources; named roles exist only in self-hosted installs ^{VF142}

Choose when

- You want a managed parser with an EU region or BYOC and per-page tier choice
- Complex layouts justify a higher-cost agentic tier on a subset of documents ^{AJ}

Avoid when

- You would let the vendor’s Index become the system of record for entitled content
- You need ACL-preserving connectors

Nearest competitors: L8-reducto, L8-unstructured, L8-docling

Regulated-FS note. Use Parse and Extract behind the firm’s own document model and keep indexing in the firm’s L6 store; record the parse tier and SDK version with every output ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	LlamaIndex, Inc.	Verified fact	A1-S080
Category	Proprietary ‘enterprise platform for agentic OCR, parsing, extraction, indexing’	Verified fact	A1-S080, A1-S081
Version / lineup	SaaS; current Python SDK llama-cloud 2.17.0 (7 October 2026); legacy llama-cloud-services / llama-parse packages deprecated, maintained until 1 May 2026	Verified fact	A1-S014, A1-S013, A1-S081

¹³⁷ Verified fact — A1-S115: Compliance - LlamaParse documentation (Enterprise Readiness) (LlamaIndex, Inc.) <https://developers.llamaindex.ai/llamaparse/general/enterprise-readiness/compliance/> accessed 2026-10-07 [search-tool extract]

¹³⁸ Verified fact — A1-S115: Compliance - LlamaParse documentation (Enterprise Readiness) (LlamaIndex, Inc.) <https://developers.llamaindex.ai/llamaparse/general/enterprise-readiness/compliance/> accessed 2026-10-07 [search-tool extract]

¹³⁹ Verified fact — A1-S116: Pricing - LlamaParse documentation (LlamaIndex, Inc.) <https://developers.llamaindex.ai/llamaparse/general/pricing/> accessed 2026-10-07 [search-tool extract]

¹⁴⁰ Verified fact — A1-S080: LlamaIndex README (company focus shift to LlamaParse) (LlamaIndex, Inc. (GitHub)) https://raw.githubusercontent.com/run-llama/llama_index/main/README.md accessed 2026-10-07

¹⁴¹ Verified fact — A1-S013: PyPI JSON metadata: llama-cloud-services (Python Package Index) <https://pypi.org/pypi/llama-cloud-services/json> accessed 2026-10-07; A1-S014: PyPI JSON metadata: llama-cloud (Python Package Index) <https://pypi.org/pypi/llama-cloud/json> accessed 2026-10-07

¹⁴² Verified fact — B-REV-S023: LlamaParse: Access & Identity (and Organizations, User Roles) (LlamaIndex, Inc.) <https://developers.llamaindex.ai/llamaparse/general/enterprise-readiness/access-identity/> accessed 2026-10-08 [search-tool extract]

Field	Value	Label	Sources
Licence	Proprietary SaaS; SDKs MIT; companion open-source LiteParse is Apache-2.0	Verified fact	A1-S014, A1-S080
Status events	Platform branding consolidated under LlamaParse (README: 'LlamaParse is its own platform'); llama-cloud-services and llama-parse SDKs deprecated (maintenance to 1 May 2026)	Verified fact	A1-S080, A1-S013, V1-S008
Strategic direction	LlamaIndex states its primary focus has shifted from the OSS framework to LlamaParse, LiteParse and parsing benchmarks (ParseBench, ExtractBench)	Verified fact	A1-S080
What it does	Agentic OCR and parsing of 130+ formats with parse tiers, structured extraction, document splitting, managed ingest/index/RAG pipelines and deployable document agents.	Verified fact	A1-S080, A1-S081
Stack position	L8 parsing plus managed indexing (L6) and document agents (L3); broader than the graphic's 'PDF / documents' tile	Verified fact	A1-S080
Integration	REST API, Python and TypeScript SDKs, MCP server; usable with or without the LlamaIndex framework	Verified fact	A1-S081, A1-S080
Dependencies	LlamaParse cloud (NA or EU) and API key; self-hosted/BYOC: Helm chart on Kubernetes (EKS, AKS, GKE or conformant) plus customer-selected LLM providers	Verified fact	A1-S081, A1-S115
Certifications	SOC 2 Type II (report via Trust Center); HIPAA-compliant pipeline for Enterprise with BAA on request	Verified fact	A1-S115
GDPR / residency	EU region (api.cloud.eu.llamaindex.ai): data stored and processed in EU; EU DPA, SCCs and Article 27 representative; region-specific API keys; self-hosted LLM provider choice affects residency	Verified fact	A1-S115
Security features	Not publicly verified	Not publicly verified	
Access controls	Self-hosted: OIDC with Microsoft Entra ID, Okta; Enterprise plan includes SSO	Verified fact	A1-S115, A1-S116
Enterprise support	Enterprise: negotiated credits and rate limits (5x), dedicated support, flexible deployment; Pro priority support	Verified fact	A1-S116
Pricing	Free (10k credits/month), Starter US\$50/month (40k credits), Pro US\$500/month (400k credits), Enterprise custom; overage US\$1.25 per 1,000 credits; parse 1-45 credits/page by tier (Fast 1, Cost-effective 3, Agentic 10, Agentic Plus 45)	Verified fact	A1-S116
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	LlamaIndex framework, MCP clients, LiteParse (open-source local parser)	Verified fact	A1-S080, A1-S081
Adoption signals	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
		d	
Maturity	llama-parse first on PyPI 2 February 2024; new llama-cloud SDK line >=1.0	Verified fact	A1-S013, A1-S014
Deployment	saas: Yes (NA and EU regions); managed_cloud: Not publicly verified; vpc_byoc: Yes (BYOC on Azure, AWS, GCP); private_cloud: Yes; self_hosted: Yes; on_prem: Yes (on-premises)		

Firecrawl (L8-firecrawl)

Tier: Tactical · **Flags:** none · **Original graphic label:** Firecrawl – web to LLM-ready

Rationale: The most enterprise-ready web acquisition option, but AGPL open core, US-only data location and unverified security evidence keep it tactical.

Criterion	Score	Rationale
Technical	4	Full web acquisition surface plus parsing, PII redaction and prompt-injection check.
Enterprise readiness	4	SAML/OIDC SSO and SCIM on Enterprise, Admin and Member roles, SIEM audit logging of scrape events to Sentinel ¹⁴³ ; all three controls plus SCIM, so 4 under rule 7 (CP2 Q2). Not 5: built-in roles only (no custom roles), team-management events not stated as logged, SLA NPV.
Security and compliance	2	Capped: SOC 2 Type II and HIPAA wording are vendor-page claims flagged by the V1 verifier.
Deployment flexibility	3	Cloud plus a reduced-feature self-hosted server.
Ecosystem	4	SDKs, CLI, webhooks; Alexandria plugins for major assistants.
Reliability and maturity	3	Two and a half years of SDK releases and recent Series B; API moved to v2.
Cost / TCO	3	Published credits; self-hosting means owning auth, TLS, persistence and anti-bot

¹⁴³ Verified fact — B-REV-S024: Firecrawl: Enterprise (and SIEM Audit Logging) <https://docs.firecrawl.dev/enterprise> accessed 2026-10-08 [search-tool extract]

Criterion	Score	Rationale
		services.
Lock-in / portability	2	AGPL server and Cloud-only features create SaaS dependency.
Total (generic / FS)	3.25 / 3.00	

Evidence rules applied: security_compliance capped at 2: Firecrawl security claims are vendor-only and flagged in V1 verification_log section 4; Rule 7 (CP2 Q2) applied at CP2 rework (8 October 2026): all three controls plus SCIM (B-REV-S024); enterprise_readiness 3 -> 4

Capabilities. Web data API (v2) for scrape, crawl, map, search, extract and parse into Markdown or JSON, plus Cloud-only Agent, Browser and Interact; respects robots.txt by default; Alexandria agent data library launched September 2026 ^{vf-144}.

Strengths

- Per-request PII redaction, Zero Data Retention and an opt-in prompt-injection check for JSON extraction ^{vf-145}
- Enterprise SSO (SAML, OIDC), SCIM, IP and key restrictions and a DPA ^{vf-146}
- Self-hostable server for public-web crawling inside the firm's estate ^{vf-147}
- Transparent credit pricing ^{vf-148}

Limitations and risks

- Server is AGPL-3.0; enterprise controls, Agent and Browser are Cloud-only ^{vf-149}
- Privacy policy places servers and stored data in the US ^{vf-150}
- Security claims are vendor pages only; verifier says do not rely on them ^{vf-151}
- No document-level ACL or lineage ^{AI}

¹⁴⁴ Verified fact — A1-S053: Firecrawl README (Firecrawl (GitHub)) <https://raw.githubusercontent.com/firecrawl/firecrawl/main/README.md> accessed 2026-10-07; A1-S077: Firecrawl Billing (docs source) (Firecrawl (GitHub docs source)) <https://raw.githubusercontent.com/firecrawl/firecrawl-docs/main/billing.mdx> accessed 2026-10-07; A1-S079: Open source or Firecrawl Cloud (docs source) (Firecrawl (GitHub docs source)) <https://raw.githubusercontent.com/firecrawl/firecrawl-docs/main/contributing/open-source-or-cloud.mdx> accessed 2026-10-07; A1-S129: Introducing Alexandria and our \$75M Series B (Firecrawl) <https://www.firecrawl.dev/blog/introducing-alexandria-series-b> accessed 2026-10-07 [search-tool extract]

¹⁴⁵ Verified fact — A1-S076: Firecrawl Enterprise (docs source) (Firecrawl (GitHub docs source)) <https://raw.githubusercontent.com/firecrawl/firecrawl-docs/main/enterprise.mdx> accessed 2026-10-07; A1-S077: Firecrawl Billing (docs source) (Firecrawl (GitHub docs source)) <https://raw.githubusercontent.com/firecrawl/firecrawl-docs/main/billing.mdx> accessed 2026-10-07

¹⁴⁶ Verified fact — A1-S076: Firecrawl Enterprise (docs source) (Firecrawl (GitHub docs source)) <https://raw.githubusercontent.com/firecrawl/firecrawl-docs/main/enterprise.mdx> accessed 2026-10-07

¹⁴⁷ Verified fact — A1-S078: Firecrawl self-hosting guide (docs source) (Firecrawl (GitHub docs source)) <https://raw.githubusercontent.com/firecrawl/firecrawl-docs/main/contributing/self-host.mdx> accessed 2026-10-07

¹⁴⁸ Verified fact — A1-S127: Firecrawl Pricing <https://www.firecrawl.dev/pricing.md> accessed 2026-10-07 [search-tool extract]

¹⁴⁹ Verified fact — A1-S053: Firecrawl README (Firecrawl (GitHub)) <https://raw.githubusercontent.com/firecrawl/firecrawl/main/README.md> accessed 2026-10-07; A1-S079: Open source or Firecrawl Cloud (docs source) (Firecrawl (GitHub docs source)) <https://raw.githubusercontent.com/firecrawl/firecrawl-docs/main/contributing/open-source-or-cloud.mdx> accessed 2026-10-07; V1-S002: Firecrawl LICENSE (AGPL-3.0) (Firecrawl (GitHub)) <https://raw.githubusercontent.com/firecrawl/firecrawl/main/LICENSE> accessed 2026-10-08

¹⁵⁰ Verified fact — A1-S134: Firecrawl Privacy Policy <https://www.firecrawl.dev/privacy-policy> accessed 2026-10-07 [search-tool extract]

¹⁵¹ Verified fact — A1-S076: Firecrawl Enterprise (docs source) (Firecrawl (GitHub docs source)) <https://raw.githubusercontent.com/firecrawl/firecrawl-docs/main/enterprise.mdx> accessed 2026-10-07

Choose when

- Batch acquisition of approved public web sources, where US processing is acceptable or the self-hosted server is used
- You want PII redaction and ZDR as per-request switches on web content

Avoid when

- You plan to modify and expose the server as a network service without AGPL review
- Data must be processed in the EU or UK

Nearest competitors: L8-crawl4ai, L8-apify

Regulated-FS note. Use only for public, approved sources; route client or confidential data elsewhere, and obtain the SOC 2 report before relying on the Cloud for anything beyond public content ^{Rec.}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Firecrawl (PyPI author listed as Mendable.ai); venture-backed (Series B US\$75M, September 2026)	Verified fact	A1-S015, A1-S053, A1-S129
Category	Web data API for AI: scrape, crawl, map, search, extract, parse, browser interaction and an autonomous research agent	Verified fact	A1-S077, A1-S079
Version / lineup	Self-host guide pins server release v2.11.162; Python SDK firecrawl-py 4.49.3 released 7 October 2026; API v2	Verified fact	A1-S078, A1-S015
Licence	Open core: server AGPL-3.0; SDKs and some UI components MIT; Agent, Browser, Interact, dashboards and enterprise controls are Cloud-only	Verified fact	A1-S053, A1-S079, V1-S002
Strategic direction	Launched Alexandria (September 2026): one interface over official data providers (e.g. Wikimedia Enterprise), custom connectors, Firecrawl indexes and the live web for agents; available as plugin in ChatGPT, Codex, Claude and Claude Code (October 2026); plus Agent, Interact, PII redaction and enterprise controls	Verified fact	A1-S129, A1-S076, A1-S079
What it does	Turns websites and documents into LLM-ready markdown or structured JSON via scrape/crawl/map/search endpoints, with LLM extraction, PDF parsing and an agent for web research; respects robots.txt by default.	Verified fact	A1-S053, A1-S077
Stack position	L8 web acquisition and parsing; overlaps L4 (search/browser tools for agents)	Verified fact	A1-S077, A1-S079
Integration	REST API v2, Python/JS SDKs, CLI; webhooks	Verified fact	A1-S053, A1-S077
Dependencies	Self-host: Docker Compose stack with PostgreSQL (pg_cron) and	Verified fact	A1-S078,

Field	Value	Label	Sources
	supporting services; LLM features need an OpenAI-compatible provider or Ollama; no verified minimum host size	d fact	A1-S079
Certifications	SOC 2 Type II	Verified fact	A1-S076, V1-S080
GDPR / residency	Privacy policy states servers and stored data are in the United States; vendor pages conflict (one older blog claims EU residency; 2026 page says US data residency); DPA and Zero Data Retention available	Verified fact	A1-S134, A1-S076
Security features	Zero Data Retention (+1 credit/page), PII redaction (+4 credits/page), opt-in prompt-injection check for JSON extraction, static IP allowlisting, IP and API-key restrictions	Verified fact	A1-S076, A1-S077
Access controls	SSO (SAML, OIDC), SCIM directory sync, per-key restrictions, spend limits per key/team (Enterprise)	Verified fact	A1-S076
Enterprise support	Enterprise plans with dedicated support, custom credits, reserved concurrency	Verified fact	A1-S076
Pricing	Free 1,000 credits/month; Hobby US\$19/month (5k credits); Standard US\$99/month (100k); Growth US\$399/month (500k); Scale and Enterprise above; annual discounts; credits: scrape 1/page, JSON extraction +4, ZDR +1, PII redaction +4	Verified fact	A1-S127, A1-S077
Infrastructure cost	Self-hosting means owning authentication, TLS, persistence, monitoring, capacity and upgrades; advanced anti-bot services must be run separately	Verified fact	A1-S079
Ecosystem	SDKs, CLI, X/LinkedIn routed via third-party providers with separate pricing	Verified fact	A1-S077, A1-S053
Adoption signals	Series A US\$14.5M (Nexus, 19 August 2025; >350,000 developers, 48k GitHub stars then); Series B US\$75M led by Smash Capital (announced 22 September 2026 on the Firecrawl blog; changelog entry dated 20 August 2026; Form D shows US\$82.06M preferred sold, first sale 31 August 2026)	Verified fact	A1-S128, A1-S129, V1-S015, V1-S009
Maturity	firecrawl-py first published 12 April 2024; 209 releases	Verified fact	A1-S015
Deployment	saas: Yes; managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Yes (self-host); self_hosted: Yes (reduced feature set); on_prem: Not publicly verified		

Crawl4AI (L8-crawl4ai)

Tier: Experimental · **Flags:** none · **Original graphic label:** Crawl4AI – open crawler

Rationale: Useful and permissive in practice, but pre-1.0, single-maintainer and carrying a non-standard licence term.

Criterion	Score	Rationale
Technical	3	Covers crawling, Markdown and extraction; no PII, ACL or provenance features.
Enterprise readiness	2	Library rule: token auth by default; no commercial enterprise support verified.
Security and compliance	2	Library rule: SECURITY.md and GitHub advisories exist (B-REV-S003), but eight advisories, five rated HIGH (SSRF, arbitrary file write, secret leakage, XSS), were

Criterion	Score	Rationale
		fixed in 0.9.3-0.9.4 (August-September 2026) and only 0.9.x is supported; attribution term on the licence.
Deployment flexibility	4	Library, Docker server, private cloud, on-prem and hosted cloud.
Ecosystem	3	MCP for common coding assistants; smaller ecosystem than commercial peers.
Reliability and maturity	2	Pre-1.0 versioning and single-maintainer origin.
Cost / TCO	4	Free; operations cost is browsers and proxies.
Lock-in / portability	4	Open source, but the attribution clause is not plain Apache-2.0.
Total (generic / FS)	2.90	
	/	
	2.90	

Evidence rules applied: Library rule applied (enterprise_readiness and security_compliance capped at 4; scored on project hygiene; inherits host controls); Rule 7 (CP2 Q2) checked at CP2 rework: no SSO, RBAC or audit control verified (token auth only, A1-S074); enterprise_readiness 2 kept under the library rule

Capabilities. Open-source crawler and scraper producing LLM-ready Markdown or structured data with browser automation, batch crawling and LLM extraction; Python library, CLI, self-hosted REST server, MCP and a hosted Crawl4AI Cloud ^{vr152}.

Strengths

- Self-hosted Docker server, with an API token required on every endpoint by default ^{vr153}
- Free library; cost is browser compute and proxies ^{vr154}
- Runs entirely inside the firm's estate ^{vr155}

Limitations and risks

- Apache-2.0 plus an appended attribution requirement covering distributions and public uses; legal review advised ^{vr156}
- Pre-1.0 (0.9.4) with a single-maintainer origin ^{vr157}
- No disclosed venture funding (aggregator only) ^{vr158}

¹⁵² Verified fact — A1-S074: Crawl4AI README (UncleCode / Crawl4AI (GitHub)) <https://raw.githubusercontent.com/unclecode/crawl4ai/main/README.md> accessed 2026-10-07

¹⁵³ Verified fact — A1-S074: Crawl4AI README (UncleCode / Crawl4AI (GitHub)) <https://raw.githubusercontent.com/unclecode/crawl4ai/main/README.md> accessed 2026-10-07

¹⁵⁴ Verified fact — A1-S074: Crawl4AI README (UncleCode / Crawl4AI (GitHub)) <https://raw.githubusercontent.com/unclecode/crawl4ai/main/README.md> accessed 2026-10-07

¹⁵⁵ Verified fact — A1-S074: Crawl4AI README (UncleCode / Crawl4AI (GitHub)) <https://raw.githubusercontent.com/unclecode/crawl4ai/main/README.md> accessed 2026-10-07

¹⁵⁶ Verified fact — A1-S056: Crawl4AI LICENSE (Apache-2.0 with attribution requirement) (UncleCode / Crawl4AI (GitHub)) <https://raw.githubusercontent.com/unclecode/crawl4ai/main/LICENSE> accessed 2026-10-07; V1-S001: Crawl4AI LICENSE (Apache-2.0 plus Attribution Requirement) (UncleCode / Crawl4AI (GitHub)) <https://raw.githubusercontent.com/unclecode/crawl4ai/main/LICENSE> accessed 2026-10-08

¹⁵⁷ Verified fact — A1-S009: PyPI JSON metadata: crawl4ai (Python Package Index) <https://pypi.org/pypi/crawl4ai/json> accessed 2026-10-07; A1-S074: Crawl4AI README (UncleCode / Crawl4AI (GitHub)) <https://raw.githubusercontent.com/unclecode/crawl4ai/main/README.md> accessed 2026-10-07

¹⁵⁸ Reported — A1-S132: Crawl4AI - Company Profile (Tracxn) https://tracxn.com/d/companies/crawl4ai/_EgBR0RlX6L34MHENV_ItNiaxrP5zsR1oR97-bT7LWmg accessed 2026-10-07 [search-tool extract]

- No certifications or enterprise support found ^{NPV}

Choose when

- Self-hosted crawling of a small set of approved public sources by a team that can own it
- Prototyping web acquisition before committing to a commercial service

Avoid when

- It would be a critical production dependency without a support route
- Legal cannot accept the attribution clause for your use

Nearest competitors: L8-firecrawl, L8-apify

Regulated-FS note. Acceptable for public data in a controlled estate; record the attribution clause in the open-source register and do not use the hosted cloud for anything confidential ^{REC}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Open-source project by UncleCode, now also offering hosted Crawl4AI Cloud	Verified fact	A1-S009, A1-S056, A1-S074
Category	Open-source web crawler/scrapper producing LLM-ready Markdown, plus a hosted API	Verified fact	A1-S074
Version / lineup	v0.9.4, 23 September 2026	Verified fact	A1-S009, A1-S074
Licence	Apache-2.0 with an additional attribution requirement for distributions and public uses	Verified fact	A1-S056, V1-S001
Status events	Crawl4AI Cloud hosted service launched (date not verified)	Verified fact	A1-S074
Strategic direction	Launched Crawl4AI Cloud (search, answer, extraction without own LLM key, bot-wall handling, MCP) alongside the free library and Docker server	Verified fact	A1-S074
What it does	Crawls and scrapes websites into clean Markdown or structured data for RAG and agents, with browser automation, batch crawling and LLM-based extraction.	Verified fact	A1-S074
Stack position	L8 web acquisition; overlaps L4 search tools via the cloud API	Verified fact	A1-S074
Integration	Python library, CLI, self-hosted REST API (/md, /crawl, /screenshot, /pdf, /execute_js), MCP, hosted API	Verified fact	A1-S074
Dependencies	Python with headless browsers; Docker server image (AMD64/ARM64)	Verified fact	A1-S074
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	Not publicly verified	Not publicly	

Field	Value	Label	Sources
		y verifie d	
Security features	Self-hosted server requires CRAWL4AI_API_TOKEN on every endpoint by default	Verifie d fact	A1-S074
Access controls	Not publicly verified	Not publicl y verifie d	
Enterprise support	Not publicly verified	Not publicl y verifie d	
Pricing	Library free; Crawl4AI Cloud pay-as-you-go with free starting credit; 1 credit = US\$0.001	Report ed	A1-S132, A1-S074
Infrastructure cost	Self-hosting cost is browser compute and proxies	Archite ctural judge ment	
Ecosystem	MCP for Claude Code, Codex, Cursor, OpenCode	Verifie d fact	A1-S074
Adoption signals	No venture funding found; aggregator lists Crawl4AI as unfunded (Singapore, founded 2024); project funded via sponsorships and Crawl4AI Cloud	Report ed	A1-S132
Maturity	First PyPI release 25 September 2024; pre-1.0 (0.9.x)	Verifie d fact	A1-S009
Deployment	saas: Yes (Crawl4AI Cloud); managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Yes; self_hosted: Yes; on_prem: Yes		

Reducto (L8-reducto)

Tier: Tactical · Flags: none · Original graphic label: Reducto – enterprise docs

Rationale: Technically strong managed extraction with flexible deployment on paper, but security evidence is vendor-only and the API is proprietary.

Criterion	Score	Rationale
Technical	4	Parse, extract, split, classify, edit and pipelines with webhooks; no governance metadata.
Enterprise readiness	3	SSO/SAML and RBAC listed as Enterprise features in the product docs ^{vr159} ; audit logging appears only on a marketing page; support terms NPV.
Security and	2	Capped: SOC 2/HIPAA/ZDR claims are vendor-only and flagged by the V1 verifier.

¹⁵⁹ Verified fact — B-REV-S017: Reducto: Enterprise Readiness <https://docs.reducto.ai/enterprise/enterprise-readiness> accessed 2026-10-08 [search-tool extract]

Criterion	Score	Rationale
compliance		
Deployment flexibility	4	SaaS with EU/AU endpoints, VPC, on-prem and air-gap, all vendor-stated at medium confidence.
Ecosystem	2	REST API and Python SDK; wider integrations not verified.
Reliability and maturity	3	Well funded and in production use per vendor (about 1 billion pages, October 2025); SDK still 0.x.
Cost / TCO	3	Transparent per-page rates; Parse at US\$10 per 1,000 pages is above OCR-only engines.
Lock-in / portability	2	Proprietary API and schema; no open format.
Total (generic / FS)	3.05 / 2.90	

Evidence rules applied: NPV cap on enterprise_readiness lifted at CP2 review: SSO/SAML and RBAC on Enterprise per docs.reducto.ai (B-REV-S017); security_compliance capped at 2 (kept): certifications are vendor-only claims flagged in V1 verification_log section 4; Rule 7 (CP2 Q2) checked at CP2 rework: SSO and RBAC verified, audit logging only on a marketing page, so 3 kept

Capabilities. Proprietary document ingestion API with Parse, Extract, Split, Edit, Classify and Pipeline resources, asynchronous jobs and webhooks; EU and AU endpoints ^{vr160}.

Strengths

- Vendor-stated VPC, on-prem and air-gapped deployment with offline model updates ^{vr161}
- Retention controls: auto-deletion within 24 hours on Growth/Enterprise and retention=0 on Enterprise ^{vr162}
- Published per-page pricing by operation ^{vr163}
- Funded (US\$108M total, Series B led by a16z, October 2025) ^{vr164}

Limitations and risks

- Security and deployment claims come from vendor-authored pages with internal inconsistencies; verifier says do not rely on them without the SOC 2 report ^{vr165}
- SSO/SAML and RBAC are Enterprise-only ^{vr166}; audit logging and enterprise support terms not verified ^{NPV}

¹⁶⁰ Verified fact — A1-S093: Reducto Python SDK API reference (api.md) (Reducto (GitHub)) <https://raw.githubusercontent.com/reductoai/reducto-python-sdk/main/api.md> accessed 2026-10-07; A1-S092: Reducto Python SDK client (environment base URLs) (Reducto (GitHub)) https://raw.githubusercontent.com/reductoai/reducto-python-sdk/main/src/reducto/_client.py accessed 2026-10-07

¹⁶¹ Verified fact — A1-S112: Reducto Trust Center: Zero Data Retention, on-prem, air-gapped <https://llms.reducto.ai/trust-center-on-prem-zero-retention-hipaa-soc2-baa> accessed 2026-10-07 [search-tool extract]

¹⁶² Verified fact — A1-S112: Reducto Trust Center: Zero Data Retention, on-prem, air-gapped <https://llms.reducto.ai/trust-center-on-prem-zero-retention-hipaa-soc2-baa> accessed 2026-10-07 [search-tool extract]

¹⁶³ Verified fact — A1-S113: Reducto Pricing <https://reducto.ai/pricing> accessed 2026-10-07 [search-tool extract]

¹⁶⁴ Verified fact — A1-S114: Reducto Announces \$108M in Funding to Define the Future of AI Document Intelligence <https://reducto.ai/blog/reducto-series-b-funding> accessed 2026-10-07 [search-tool extract]; V1-S018: Reducto Secures \$75 Million Series B (Cooley LLP) [https://www.cooley.com/news/coverage/2025/2025-10-14-reducto-secures-\\$75-million-series-b](https://www.cooley.com/news/coverage/2025/2025-10-14-reducto-secures-$75-million-series-b) accessed 2026-10-08 [search-tool extract]

¹⁶⁵ Verified fact — A1-S112: Reducto Trust Center: Zero Data Retention, on-prem, air-gapped <https://llms.reducto.ai/trust-center-on-prem-zero-retention-hipaa-soc2-baa> accessed 2026-10-07 [search-tool extract]

- No PII, ACL or lineage features found ^{VF-167}
- Proprietary API and output schema ^{VF-168}

Choose when

- Complex financial or legal layouts where a managed extraction API justifies its price, and you can obtain the SOC 2 report and contract terms
- You need an air-gapped commercial parser and will validate the claim in due diligence

Avoid when

- Due diligence cannot obtain the SOC 2 report and BAA/DPA
- Extraction output would be treated as authoritative numbers

Nearest competitors: L8-llamaparse, L8-google-document-ai, L8-unstructured

Regulated-FS note. Promising for complex documents, but treat every security claim as unverified until the SOC 2 report and contract are in hand; pin to the EU endpoint or on-prem install for EU/UK client data ^{Rec.}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Reducto (private; US\$108M total funding after US\$75M Series B led by a16z, October 2025)	Verified fact	A1-S114, V1-S018
Category	Proprietary document ingestion API: parse, extract, split, edit, classify and pipelines	Verified fact	A1-S093
Version / lineup	SaaS API; Python SDK reductoai 0.24.0, released 8 September 2026	Verified fact	A1-S090
Licence	Proprietary API; Python SDK Apache-2.0	Verified fact	A1-S090
Strategic direction	Series B used for model research, product and enterprise adoption; launched pay-as-you-go tier with 15k free credits (October 2025)	Verified fact	A1-S114
What it does	API that parses documents into structured output and extracts, splits, edits and classifies documents, with asynchronous jobs, pipelines and webhooks.	Verified fact	A1-S093, A1-S091
Stack position	L8 parsing and structured extraction	Verified fact	A1-S093
Integration	REST API; Python SDK (sync/async); webhooks	Verified fact	A1-S091, A1-S093
Dependencies	Reducto hosted API	Verified fact	A1-S092
Certifications	SOC 2 Type I and Type II completed (report on request); HIPAA BAAs offered	Verified fact	A1-S112

¹⁶⁶ Verified fact — B-REV-S017: Reducto: Enterprise Readiness <https://docs.reducto.ai/enterprise/enterprise-readiness> accessed 2026-10-08 [search-tool extract]

¹⁶⁷ Verified fact — A1-S092: Reducto Python SDK client (environment base URLs) (Reducto (GitHub)) https://raw.githubusercontent.com/reductoai/reducto-python-sdk/main/src/reducto/_client.py accessed 2026-10-07; A1-S093: Reducto Python SDK API reference (api.md) (Reducto (GitHub)) <https://raw.githubusercontent.com/reductoai/reducto-python-sdk/main/api.md> accessed 2026-10-07

¹⁶⁸ Verified fact — A1-S093: Reducto Python SDK API reference (api.md) (Reducto (GitHub)) <https://raw.githubusercontent.com/reductoai/reducto-python-sdk/main/api.md> accessed 2026-10-07

Field	Value	Label	Sources
GDPR / residency	EU and AU regional endpoints (Enterprise option); API data auto-deleted within 24 hours on Growth/Enterprise, retention=0 (immediate deletion) for Enterprise; customer data not used for training on Growth/Enterprise	Verified fact	A1-S092, A1-S112
Security features	Zero/short data retention controls; VPC deployment with no external storage; air-gapped install with offline model updates	Verified fact	A1-S112
Access controls	SSO/SAML supported (VPC deployment)	Verified fact	A1-S112
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Per 1,000 pages: Parse US\$10, Extract US\$20, Deep Extract US\$40, Split US\$20, Classify US\$7.50, Edit US\$60; Standard pay-as-you-go with US\$150 free usage (page also says 15,000 free credits); Enterprise custom	Verified fact	A1-S113
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	Close to 1 billion pages processed; named customers Harvey, Rogo, Scale AI (October 2025)	Verified fact	A1-S114
Maturity	SDK first published on PyPI 12 October 2024	Verified fact	A1-S090
Deployment	saas: Yes; managed_cloud: Not publicly verified; vpc_byoc: Yes (customer VPC); private_cloud: Yes (customer VPC); self_hosted: Yes; on_prem: Yes (on-prem and air-gapped)		

MinerU (L8-mineru)

Tier: Experimental · **Flags:** none · **Original graphic label:** MinerU – PDF parser

Rationale: Technically capable and local-first, but a custom licence with revenue thresholds, an unpublished commercial price and fresh major-version change.

Criterion	Score	Rationale
Technical	4	Four parsing tiers, broad formats and citation locators; no ACL, PII or lineage.
Enterprise readiness	2	Library rule: no enterprise support offer or access controls found.
Security and	2	Library rule: local-first and telemetry opt-out, but custom licence with

Criterion	Score	Rationale
compliance		termination clause and no verified CVE process.
Deployment flexibility	4	Local, self-hosted, on-prem and an optional remote service.
Ecosystem	2	Integrations not verified.
Reliability and maturity	3	GA 4.0 line with frequent releases, but young (first PyPI June 2025) and a 2026 major migration.
Cost / TCO	2	Free below thresholds; commercial price above them is unpublished, plus GPU cost.
Lock-in / portability	2	Custom licence with revenue/MAU thresholds and automatic termination.
Total (generic / FS)	2.80	
	/	
	2.70	

Evidence rules applied: Library rule applied (enterprise_readiness and security_compliance capped at 4; scored on project hygiene; inherits host controls); Rule 7 (CP2 Q2) checked at CP2 rework: no SSO, RBAC or audit control verified; enterprise_readiness 2 kept under the library rule

Capabilities. MinerU 4.0: local-first parsing of PDF, images, Office, OpenDocument, EPUB, OFD, HTML and CSV into Markdown/JSON with four parsing tiers, a document library with citation locators for agents, and a multi-service router ^{vf169}.

Strengths

- Local by default; documents are not uploaded unless remote parsing is set explicitly ^{vf170}
- Citation locators give stable references for agent reading ^{vf171}
- Telemetry is anonymous, excludes document content and file names, and can be disabled ^{vf172}

Limitations and risks

- Custom ‘MinerU Open Source License’: separate commercial licence above 100M MAU or US\$20M monthly revenue (group-consolidated); rights terminate on breach ^{vf173}
- Commercial licence price not published ^{vf174}
- Standard/Advanced tiers need a GPU with 8 GB+ VRAM ^{vf175}
- Major-version migration (3.x to 4.0) in 2026 ^{vf176}

¹⁶⁹ Verified fact — A1-S055: MinerU README (MinerU 4.0) (OpenDataLab MinerU Team (GitHub)) <https://raw.githubusercontent.com/opendatalab/MinerU/master/README.md> accessed 2026-10-07; A1-S010: PyPI JSON metadata: mineru (Python Package Index) <https://pypi.org/pypi/mineru/json> accessed 2026-10-07

¹⁷⁰ Verified fact — A1-S055: MinerU README (MinerU 4.0) (OpenDataLab MinerU Team (GitHub)) <https://raw.githubusercontent.com/opendatalab/MinerU/master/README.md> accessed 2026-10-07

¹⁷¹ Verified fact — A1-S055: MinerU README (MinerU 4.0) (OpenDataLab MinerU Team (GitHub)) <https://raw.githubusercontent.com/opendatalab/MinerU/master/README.md> accessed 2026-10-07

¹⁷² Verified fact — A1-S055: MinerU README (MinerU 4.0) (OpenDataLab MinerU Team (GitHub)) <https://raw.githubusercontent.com/opendatalab/MinerU/master/README.md> accessed 2026-10-07

¹⁷³ Verified fact — A1-S054: MinerU Open Source License (OpenDataLab MinerU Team (GitHub)) <https://raw.githubusercontent.com/opendatalab/MinerU/master/LICENSE.md> accessed 2026-10-07; V1-S003: MinerU Open Source License (OpenDataLab (GitHub)) <https://raw.githubusercontent.com/opendatalab/MinerU/master/LICENSE.md> accessed 2026-10-08

¹⁷⁴ Verified fact — A1-S054: MinerU Open Source License (OpenDataLab MinerU Team (GitHub)) <https://raw.githubusercontent.com/opendatalab/MinerU/master/LICENSE.md> accessed 2026-10-07

Choose when

- Below the licence thresholds, or with a commercial licence in place, and local GPU parsing is acceptable
- As a second parser for reconciliation of hard tables ^{AJ}

Avoid when

- A large firm whose group revenue exceeds the threshold has no commercial licence
- You need vendor support or certifications

Nearest competitors: L8-docling, L8-unstructured, L8-llamaparse

Regulated-FS note. Large asset managers are likely to exceed the group-consolidated revenue threshold, so legal must clear the licence before any production use ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	OpenDataLab MinerU Team	Verified fact	A1-S054, A1-S055
Category	Open-source document parsing tool and local document library with VLM-based parsing tiers	Verified fact	A1-S055, A1-S010
Version / lineup	MinerU 4.0 line; mineru 4.0.10 released 29 September 2026	Verified fact	A1-S010, A1-S055, V1-S004
Licence	'MinerU Open Source License': Apache-2.0 plus additional terms - separate commercial licence required above 100M MAU or US\$20M monthly revenue (group-consolidated); attribution required for online services; rights terminate automatically on breach	Verified fact	A1-S054, A1-S010, V1-S003
Status events	MinerU 4.0 released (3.x to 4.0 migration guide)	Verified fact	A1-S055
Strategic direction	MinerU 4.0 adds four parsing tiers (Flash/Basic/Standard/Advanced), a document library with citation locators for agent reading, multi-format input and a router for multiple services	Verified fact	A1-S055
What it does	Converts PDF, images, Office, OpenDocument, EPUB, OFD, HTML and CSV into Markdown/JSON and other formats, caches and searches results, and serves them to agents with stable citations.	Verified fact	A1-S055
Stack position	L8 parsing/OCR; local-first with optional remote service (mineru.net)	Verified fact	A1-S055
Integration	Python SDK, V1 API, CLI, Gradio WebUI, multi-service Router; remote parsing only with explicit -remote	Verified fact	A1-S055
Dependencies	ONNX/PyTorch small models; llama.cpp, vLLM or LMDeploy for the VLM; Standard/Advanced tiers require GPU/MPS with 8 GB+ VRAM	Verified fact	A1-S055
Certifications	Not publicly verified	Not publicly	

¹⁷⁵ Verified fact — A1-S055: MinerU README (MinerU 4.0) (OpenDataLab MinerU Team (GitHub)) <https://raw.githubusercontent.com/opendatalab/MinerU/master/README.md> accessed 2026-10-07

¹⁷⁶ Verified fact — A1-S055: MinerU README (MinerU 4.0) (OpenDataLab MinerU Team (GitHub)) <https://raw.githubusercontent.com/opendatalab/MinerU/master/README.md> accessed 2026-10-07

Field	Value	Label	Sources
		verified	
GDPR / residency	Parses locally by default; documents are not uploaded to the official service unless remote parsing is explicitly configured	Verified fact	A1-S055
Security features	Anonymous aggregated telemetry (no document content or file names), can be disabled	Verified fact	A1-S055
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Free under MinerU licence below thresholds; commercial licence above thresholds (price not published)	Verified fact	A1-S054
Infrastructure cost	GPU with 8 GB+ VRAM needed for higher-quality tiers	Verified fact	A1-S055
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	mineru package first on PyPI 13 June 2025; 96 releases	Verified fact	A1-S010
Deployment	saas: Yes (mineru.net web app / remote API); managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Yes; self_hosted: Yes; on_prem: Yes		

Apify (L8-apify)

Tier: Tactical • **Flags:** none • **Original graphic label:** Apify – scrapers

Rationale: Convenient for public long-tail scraping, but single US region, marketplace code risk and platform-specific Actors.

Criterion	Score	Rationale
Technical	3	Scraping, automation, proxy and storage; no parsing depth or governance metadata.
Enterprise	3	Organisation accounts with customisable roles and per-resource permissions,

Criterion	Score	Rationale
readiness		enforced 2FA and session limits; SSO stated in Apify's shared-responsibility documentation ^{vf177} ; no audit log found, so 3.
Security and compliance	3	SOC 2 Type II, encryption in transit and at rest, tenant isolation.
Deployment flexibility	1	One SaaS region only (AWS us-east-1).
Ecosystem	3	Store Actors, MCP server and connectors.
Reliability and maturity	3	Client library on PyPI since 2021 with frequent releases.
Cost / TCO	3	Published plans and compute-unit rates.
Lock-in / portability	2	Platform-specific Actors and storage; proprietary API.
Total (generic / FS)	2.65	
	/	
	2.55	

Evidence rules applied: NPV cap lifted at CP2 review (8 October 2026): roles and SSO documented (B-REV-S025); no audit log found; Rule 7 (CP2 Q2) checked at CP2 rework: roles and SSO verified, no audit log found, so 3 kept

Capabilities. Serverless 'Actor' platform for scrapers and automations with Apify Proxy, datasets, key-value stores and request queues, the Apify Store marketplace and an MCP server ^{vf178}.

Strengths

- Large marketplace of ready-made Actors ^{vf179}
- SOC 2 Type II with report via trust.apify.com ^{vf180}
- Transparent compute-unit pricing ^{vf181}

Limitations and risks

- Hosted only in AWS us-east-1; no EU region documented ^{vf182}

¹⁷⁷ Verified fact — B-REV-S025: Apify: Organization account (and permissions, shared responsibility) (Apify Technologies s.r.o.) <https://docs.apify.com/platform/collaboration/organization-account> accessed 2026-10-08 [search-tool extract]

¹⁷⁸ Verified fact — A1-S087: Apify platform architecture (docs source) (Apify Technologies s.r.o. (GitHub docs source)) <https://raw.githubusercontent.com/apify/apify-docs/master/sources/platform/security/architecture.md> accessed 2026-10-07; A1-S089: Apify MCP server (docs source) (Apify Technologies s.r.o. (GitHub docs source)) <https://raw.githubusercontent.com/apify/apify-docs/master/sources/platform/integrations/ai/mcp.md> accessed 2026-10-07

¹⁷⁹ Verified fact — A1-S089: Apify MCP server (docs source) (Apify Technologies s.r.o. (GitHub docs source)) <https://raw.githubusercontent.com/apify/apify-docs/master/sources/platform/integrations/ai/mcp.md> accessed 2026-10-07

¹⁸⁰ Verified fact — A1-S085: Apify Security (docs source) (Apify Technologies s.r.o. (GitHub docs source)) <https://raw.githubusercontent.com/apify/apify-docs/master/sources/platform/security/index.md> accessed 2026-10-07; V1-S089: Apify Security; Shared responsibility model; 'Yes, Apify is SOC 2 Type II compliant!' blog <https://docs.apify.com/security> accessed 2026-10-08 [search-tool extract]

¹⁸¹ Verified fact — A1-S133: Apify pricing - flexible plan + pay as you go (Apify Technologies s.r.o.) <https://apify.com/pricing> accessed 2026-10-07 [search-tool extract]

¹⁸² Verified fact — A1-S086: Apify shared responsibility model (docs source) (Apify Technologies s.r.o. (GitHub docs source)) <https://raw.githubusercontent.com/apify/apify-docs/master/sources/platform/security/shared-responsibility.md> accessed 2026-10-07; V1-S089: Apify Security; Shared responsibility model; 'Yes, Apify is SOC 2 Type II compliant!' blog <https://docs.apify.com/security> accessed 2026-10-08 [search-tool extract]

- Third-party Store Actors are outside Apify's control under its shared-responsibility model ¹⁸³
- Actors and storage are platform-specific ¹⁸⁴
- Access controls and enterprise support not verified ^{NPV}

Choose when

- Long-tail public-web acquisition where a maintained Actor already exists and the data is public
- Short-lived research or market-data collection that never touches client data

Avoid when

- Any personal, client or confidential data is involved
- Data must be processed in the EU or UK

Nearest competitors: L8-firecrawl, L8-crawl4ai

Regulated-FS note. US-only processing and third-party Actors make it unsuitable for anything beyond public data; review each Actor as third-party code ^{Rec.}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Apify Technologies s.r.o.	Verified fact	A1-S016
Category	Web scraping and automation platform with an Actor marketplace (Apify Store), managed proxy and storage	Verified fact	A1-S087, A1-S089
Version / lineup	SaaS platform; apify-client (Python) 3.2.1, released 25 September 2026	Verified fact	A1-S016
Licence	Proprietary platform; Python API client Apache-2.0	Verified fact	A1-S016
Strategic direction	Exposes Store Actors to AI agents through the Apify MCP server (mcp.apify.com)	Verified fact	A1-S089
What it does	Runs serverless 'Actors' (scrapers and automations) in isolated containers, routes traffic through Apify Proxy, and stores results in datasets, key-value stores and request queues reachable via API.	Verified fact	A1-S087
Stack position	L8 web acquisition; overlaps L4 tools via MCP	Verified fact	A1-S087, A1-S089
Integration	REST API v2, JavaScript and Python clients, webhooks, MCP server, third-party connectors	Verified fact	A1-S087, A1-S089
Dependencies	Hosted on AWS in a single region (us-east-1) across multiple Availability Zones; MongoDB Atlas, S3 and DynamoDB	Verified fact	A1-S086, A1-S087
Certifications	SOC 2 Type II (report via trust.apify.com)	Verified fact	A1-S085, V1-S089
GDPR / residency	Platform data hosted in AWS us-east-1 (no EU region documented)	Verified fact	A1-S086, V1-S089
Security	Encryption in transit and at rest; tenant isolation; shared-responsibility	Verified	A1-S086,

¹⁸³ Verified fact — A1-S086: Apify shared responsibility model (docs source) (Apify Technologies s.r.o. (GitHub docs source)) <https://raw.githubusercontent.com/apify/apify-docs/master/sources/platform/security/shared-responsibility.md> accessed 2026-10-07

¹⁸⁴ Verified fact — A1-S087: Apify platform architecture (docs source) (Apify Technologies s.r.o. (GitHub docs source)) <https://raw.githubusercontent.com/apify/apify-docs/master/sources/platform/security/architecture.md> accessed 2026-10-07

Field	Value	Label	Sources
features	model documented	d fact	A1-S085
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Free US\$0 (US\$5 prepaid usage, CU US\$0.20); Starter US\$19/month (CU US\$0.20); Scale US\$199/month (CU US\$0.16); Business US\$999/month (CU US\$0.13); 1 CU = 1 GB RAM for 1 hour	Verified fact	A1-S133, A1-S088
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Apify Store Actors; MCP clients	Verified fact	A1-S089
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	apify-client first on PyPI 17 May 2021; 237 releases	Verified fact	A1-S016
Deployment	saas: Yes; managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Not publicly verified; on_prem: Not publicly verified		

L7: Embeddings and reranking

Cohere Embed 5 (embed-v5.0-pro, embed-v5.0-fast) and Cohere Rerank 4 (rerank-v4.0-pro, rerank-v4.0-fast) (L7-cohere)

Tier: Tactical · **Flags:** none · **Original graphic label:** Cohere – Embed v3 + Rerank

Rationale: The widest deployment range among the hosted vendors, with FS-grade certifications. FS 3.70 on the hyperscaler route meets the numeric guide for Strategic, but it stays Tactical until its stated condition is met: per-service due diligence confirming the Foundry or SageMaker controls (the CP2 Q1 presumption is not that evidence), with the pending Aleph Alpha combination recorded as an ownership event. Upgrade candidate ^{AJ}.

Criterion	Score	Rationale
Technical	4	Current embed and rerank, multimodal, 128K context, Matryoshka and binary; no domain-specific models evidenced.
Enterprise readiness	4	Scored on the Microsoft Foundry and Amazon SageMaker routes (A2-S012), which the FS recommendation uses for VPC and in-region deployment: platform IAM, SSO and audit logging presumed under rule 6 (CP2 Q1), so 4; platform controls presumed (CP2 Q1); confirm per service. The direct hosted API documents only Owner and User team roles (B-REV-S013), one verified control, and would score 3 under rule 7 (CP2 Q2); SSO and audit logs for it remain NPV.
Security and compliance	4	SOC 2 Type II, ISO 27001, ISO 42001, ZDR on request and private deployment; CMK and EU hosted region unverified.
Deployment flexibility	4	SaaS, partner clouds, VPC, single-tenant and on-prem; air-gap not verified.
Ecosystem	4	Microsoft Foundry, SageMaker, LangChain, and Cohere Rerank hosted inside Pinecone.
Reliability and maturity	3	Mature, fast-moving line; pending Aleph Alpha combination adds governance uncertainty.
Cost / TCO	3	Embed prices published (Pro US\$0.12, Fast US\$0.08 per 1M text tokens); rerank price not found.
Lock-in / portability	3	Proprietary, but deployable in the firm's estate; rule 3 not applied because Cohere is the combining party, revisit on close.
Total (generic / FS)	3.75	

Criterion	Score	Rationale
	/	
	3.70	

Evidence rules applied: NPV cap lifted at CP2 rework (8 October 2026): rule 6 (CP2 Q1) on the Foundry and SageMaker routes gives 4 (platform controls presumed (CP2 Q1); confirm per service); the direct hosted API, with only Owner and User roles verified (B-REV-S013), would be 3 under rule 7 (CP2 Q2); Rule 8 (CP2 Q4) confirmed: company-level SOC 2 Type II, ISO 27001 and ISO 42001 with product scope not stated, security_compliance 4

Capabilities. Embed 5 (embed-v5.0-pro, embed-v5.0-fast; 30 September 2026): text, images and mixed text-image inputs into one vector, 100+ languages, 128K-token context, Matryoshka 256-2,048 dims, float/int8/binary outputs, Pro and Fast in one shared space. Rerank 4 (Pro and Fast; 11 December 2025): 32K-token context, multilingual, JSON documents ^{vf185}.

Strengths

- Embed and rerank from one vendor, both current generations ^{vf186}
- Widest deployment range among hosted vendors: SaaS, Microsoft Foundry, Amazon SageMaker, VPC, Model Vault single-tenant, on-premises ^{vf187}
- SOC 2 Type II, ISO 27001 and ISO 42001 ^{vf188}; 30-day default log deletion and ZDR on request for enterprise customers ^{vf189}
- Storage arithmetic is published: Cohere's example shrinks 100M chunks from about 819 GB to 3.2 GB with reduced dimensions and binary output ^{vf190}

Limitations and risks

- The hosted platform documents only Owner and User team roles ^{vf191}; SSO/SAML and audit logs for the direct hosted API were not found ^{NPV}; on Microsoft Foundry and SageMaker the platform's controls are presumed under CP2 Q1 and must be confirmed per service ^{AJ}; enterprise support terms NPV
- Rerank per-search price not found ^{vf192}
- Aleph Alpha business combination signed 16 September 2026, pending regulatory approval ^{vf193}; valuation figures conflict and are not used ^{AJ}

¹⁸⁵ Verified fact — A2-S012: Announcing Cohere's Embed 5 Models <https://docs.cohere.com/changelog/embed-v5> accessed 2026-10-07 [search-tool extract]; A2-S009: Rerank 4: Cohere's Most Powerful Reranker Yet <https://cohere.com/blog/rerank-4> accessed 2026-10-07 [search-tool extract]; A2-S010: Cohere's Rerank v4.0 Model is Here! <https://docs.cohere.com/changelog/rerank-v4.0> accessed 2026-10-07 [search-tool extract]

¹⁸⁶ Verified fact — A2-S012: Announcing Cohere's Embed 5 Models <https://docs.cohere.com/changelog/embed-v5> accessed 2026-10-07 [search-tool extract]; A2-S010: Cohere's Rerank v4.0 Model is Here! <https://docs.cohere.com/changelog/rerank-v4.0> accessed 2026-10-07 [search-tool extract]

¹⁸⁷ Verified fact — A2-S012: Announcing Cohere's Embed 5 Models <https://docs.cohere.com/changelog/embed-v5> accessed 2026-10-07 [search-tool extract]; A2-S015: Deployment Options - SaaS, Cloud API, Virtual Private Cloud, On-prem (Cohere) <https://cohere.com/deployment-options> accessed 2026-10-07 [search-tool extract]

¹⁸⁸ Verified fact — A2-S014: Cohere Trust Center <https://trustcenter.cohere.com/> accessed 2026-10-07 [search-tool extract]

¹⁸⁹ Verified fact — A2-S145: Enterprise Data Commitments (Cohere) <https://cohere.com/enterprise-data-commitments> accessed 2026-10-07 [search-tool extract]

¹⁹⁰ Verified fact — A2-S013: Embed 5: Frontier Embedding Models for Enterprise (Cohere) <https://cohere.com/blog/embed-5> accessed 2026-10-07 [search-tool extract]

¹⁹¹ Verified fact — B-REV-S013: Teams and Roles on the Cohere Platform <https://docs.cohere.com/reference/teams-and-roles> accessed 2026-10-08 [search-tool extract]

¹⁹² Verified fact — A2-S013: Embed 5: Frontier Embedding Models for Enterprise (Cohere) <https://cohere.com/blog/embed-5> accessed 2026-10-07 [search-tool extract]

- EU hosted-region details not verified ^{NPV}; CMK for the hosted API not found ^{NPV}
- Vendor claims of 'highest average' scores are not decision inputs ^{R194}

Choose when

- Embedding and reranking must run inside the firm's VPC or data centre with a commercial vendor behind it
- Long documents (128K context) or mixed text-image pages need one vector

Avoid when

- Public evidence of SSO/RBAC/audit for the direct hosted API is a hard procurement gate, the Foundry or SageMaker route is not available, and the evidence cannot be obtained in due diligence
- Ownership stability through the Aleph Alpha approval period is a hard requirement

Nearest competitors: L7-voyage, L7-jina, L7-nvidia-nemo-retriever, L7-gemini-embedding

Regulated-FS note. The private and Model Vault routes keep client-data embeddings off shared infrastructure, and Cohere states it has no access to prompts in private or partner deployments ^{VF195}. Track the Aleph Alpha approval as an ownership event in the third-party register ^{REC}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Cohere Inc. (pending business combination with Aleph Alpha, signed 16 September 2026)	Verified fact	A2-S018
Category	Embedding and reranking models (hosted API, cloud marketplaces, private deployment)	Verified fact	A2-S011, A2-S015
Version / lineup	Embed 5 (30 September 2026): Pro and Fast tiers in a shared embedding space; embed-v4.0 still listed. Rerank 4 (11 December 2025): Pro and Fast; Rerank 3.5 still available on Azure.	Verified fact	A2-S012, A2-S010, A2-S011, V1-S027, V1-S033
Licence	Proprietary models (hosted API, marketplace and private deployments)	Verified fact	A2-S015
Status events	2025: Embed v4 (multimodal, 128K context) released; August 2025: US\$500M raise at US\$6.8B valuation; 11 December 2025: Rerank v4.0 released; 24 April 2026: Cohere and Aleph Alpha announce plan to combine; 16 September 2026: definitive business combination agreement signed, subject to regulatory approval; 30 September 2026: Embed 5 released	Verified fact	A2-S011, A2-S017, A2-S010, A2-S018, A2-S012, V1-S028
Strategic direction	Enterprise/sovereign AI positioning: private on-prem/VPC deployment, Model Vault single-tenant hosting, and a transatlantic combination with Aleph Alpha (Berlin/Toronto HQs)	Verified fact	A2-S015, A2-S012, A2-S018

¹⁹³ Verified fact — A2-S018: Cohere & Aleph Alpha: Transatlantic Sovereign AI <https://cohere.com/blog/cohere-and-aleph-alpha-sign-agreement> accessed 2026-10-07 [search-tool extract]; V1-S028: Cohere & Aleph Alpha: Transatlantic Sovereign AI (definitive agreement) <https://cohere.com/blog/cohere-and-aleph-alpha-sign-agreement> accessed 2026-10-08 [search-tool extract]

¹⁹⁴ Reported — A2-S013: Embed 5: Frontier Embedding Models for Enterprise (Cohere) <https://cohere.com/blog/embed-5> accessed 2026-10-07 [search-tool extract]

¹⁹⁵ Verified fact — A2-S015: Deployment Options - SaaS, Cloud API, Virtual Private Cloud, On-prem (Cohere) <https://cohere.com/deployment-options> accessed 2026-10-07 [search-tool extract]; A2-S145: Enterprise Data Commitments (Cohere) <https://cohere.com/enterprise-data-commitments> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
What it does	Embed 5 embeds text, images and mixed text-image inputs (e.g. PDF pages) into one vector; 100+ languages; 128K-token context; Matryoshka 256-2,048 dims; float, int8 and binary outputs. Rerank 4 reorders candidates by relevance with 32K-token context and handles multilingual text and JSON documents.	Verified fact	A2-S012, A2-S009
Stack position	L7, covering embedding and reranking from one vendor; correctly placed.	Architectural judgment	
Integration	Cohere Embed and Rerank APIs (v2), SDKs, Azure AI Foundry and SageMaker endpoints; LangChain integration	Verified fact	A2-S011, A2-S012
Dependencies	Cohere API, or partner clouds (Microsoft Foundry, Amazon SageMaker), or customer infrastructure for private deployments	Verified fact	A2-S012, A2-S015
Certifications	SOC 2 Type II (annual; report under mNDA), ISO 27001, ISO 42001, UK Cyber Essentials; HIPAA listed as an FAQ entry (BAA not confirmed)	Verified fact	A2-S014
GDPR / residency	Enterprise default: logged prompts/generations deleted after 30 days; Zero Data Retention available to enterprise customers on request (usage data still received); no Cohere access to prompts in private/partner deployments; EU hosted-region details not verified	Verified fact	A2-S145, A2-S014
Security features	Private on-prem/VPC deployments with no Cohere access to customer data; Encrypted Vault product for regulated workloads	Verified fact	A2-S015, A2-S014
Access controls	SSO/SAML, RBAC and audit-log export for the hosted API not found in public sources (7 October 2026); Model Vault attestation produces a signed record usable for audit	Verified fact	A2-S145
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Embed 5 Pro US\$0.12 and Fast US\$0.08 per 1M text tokens; images US\$0.40 per 1M tokens (both). Rerank billed per search (rate not found). Model Vault: e.g. Embed 5 Pro Small US\$3.00/hour or US\$2,000/month; Rerank 4 Pro Large US\$10.00/hour or US\$6,500/month per instance.	Verified fact	A2-S013, A2-S016
Infrastructure cost	Dedicated (Vault) instances billed per instance-hour or month	Verified fact	A2-S016
Ecosystem	Available on Microsoft Foundry/Azure, Amazon SageMaker; LangChain integration	Verified fact	A2-S011, A2-S012
Adoption signals	Funding: US\$500M at US\$6.8B (August 2025); press reports (September 2026) of talks at a US\$20B valuation (unconfirmed)	Verified fact	A2-S017, A2-S019
Maturity	Mature product line; frequent generations (Embed v4 2025, Rerank 4 Dec 2025, Embed 5 Sep 2026)	Verified fact	A2-S011, A2-S010, A2-S012
Deployment	saas: Yes; managed_cloud: Yes (Microsoft Foundry, Amazon SageMaker; Model Vault single-tenant); vpc_byoc: Yes (VPC deployment); private_cloud: Yes (Model Vault single-tenant); self_hosted: Yes (private deployment); on_prem: Yes		

Sentence Transformers (sentence-transformers library; SBERT.net)

(L7-sentence-transformers)

Tier: Strategic · **Flags:** none · **Original graphic label:** SBERT – Sentence transformers

Rationale: The portable, self-hosted toolkit for embeddings, reranking, sparse and late interaction; the firm's exit route and fine-tuning base.

Criterion	Score	Rationale
Technical	4	Covers dense, cross-encoder, sparse and multi-vector with fine-tuning; quality depends on the chosen model.
Enterprise readiness	3	Rule 2: enables in-estate serving and fine-tuning; no commercial support verified (capped at 4).
Security and compliance	3	Rule 2: Apache-2.0; SECURITY.md with private vulnerability reporting and CVE issuance through GitHub advisories (B-REV-S002); release signing not verified; inherits host controls.
Deployment flexibility	4	Self-host, on-prem and air-gap; no managed option of its own.
Ecosystem	5	Over 15,000 models on Hugging Face; many MTEB models load through it.
Reliability and maturity	4	Since 2019, regular 2026 releases, Hugging Face stewardship.
Cost / TCO	4	Free; compute and operations are the cost.
Lock-in / portability	4	Apache-2.0 and model-agnostic; stewarded by one company rather than a neutral foundation.
Total (generic / FS)	3.80 / 3.70	

Evidence rules applied: Rule 2 (self-hosted library): enterprise_readiness and security_compliance capped at 4; NPV cap not applied

Capabilities. Apache-2.0 Python library (6.1.0, 18 September 2026) for dense embeddings, Cross-Encoder rerankers, Sparse Encoders and Multi-Vector Encoders (ColBERT-style late interaction), with training and fine-tuning; over 15,000 pre-trained models on Hugging Face ^{VF196}. Maintained by Hugging Face ^{VF197}.

Strengths

- One toolkit for the whole retrieval-optimisation layer: bi-encoder, cross-encoder, sparse and late interaction ^{VF198}
- Fine-tuning on in-domain pairs is the main route to domain adaptation without a vendor ^{VF199 AJ}
- Model-agnostic and permissively licensed, so it is the natural exit route from any hosted embedding API ^{AJ}

¹⁹⁶ Verified fact — A2-S029: sentence-transformers - PyPI (PyPI / Hugging Face (maintainer)) <https://pypi.org/project/sentence-transformers/> accessed 2026-10-07; A2-S032: sentence-transformers PyPI JSON API <https://pypi.org/pypi/sentence-transformers/json> accessed 2026-10-07; A2-S028: Sentence Transformers is joining Hugging Face! <https://huggingface.co/blog/sentence-transformers-joins-hf> accessed 2026-10-07 [search-tool extract]

¹⁹⁷ Verified fact — A2-S028: Sentence Transformers is joining Hugging Face! <https://huggingface.co/blog/sentence-transformers-joins-hf> accessed 2026-10-07 [search-tool extract]; V1-S092: Sentence Transformers is joining Hugging Face! <https://huggingface.co/blog/sentence-transformers-joins-hf> accessed 2026-10-08 [search-tool extract]

¹⁹⁸ Verified fact — A2-S029: sentence-transformers - PyPI (PyPI / Hugging Face (maintainer)) <https://pypi.org/project/sentence-transformers/> accessed 2026-10-07

Limitations and risks

- A library, not a model: quality depends on the model chosen and on the firm’s evaluation ^{AJ}
- No commercial support offer verified ^{NPV}; a security policy with private reporting and CVE issuance exists ^{VF200}, but release signing is not verified ^{NPV}
- Depends on Hugging Face stewardship and a small maintainer team ^{AJ}
- Model licences on the Hub vary; each model needs its own licence check (for example, Jina weights are CC-BY-NC-4.0) ^{VF201 AJ}

Choose when

- Embeddings or reranking must run inside the firm’s estate, including air-gapped
- Domain fine-tuning or a reproducible, pinned retrieval stack is required

Avoid when

- There is no team to operate model serving, patching and evaluation
- A vendor SLA is mandatory for the retrieval path

Nearest competitors: L7-nvidia-nemo-retriever, L7-qwen3-embedding, L7-cohere

Regulated-FS note. Inherits host controls: residency, access and logging are whatever the firm’s platform provides ^{AJ}. Pin library and model versions and record both in the evidence pack ^{REC}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Hugging Face (maintainer Tom Aarsen); originally developed by UKP Lab, TU Darmstadt (author Nils Reimers)	Verified fact	A2-S029, A2-S028, V1-S092
Category	Open-source Python library for embedding, reranker (cross-encoder), sparse-encoder and multi-vector models	Verified fact	A2-S029
Version / lineup	6.1.0, released 18 September 2026 (6.0.0 on 18 August 2026)	Verified fact	A2-S029, A2-S032
Licence	Open source, Apache-2.0	Verified fact	A2-S029, A2-S028
Status events	Late 2023: Tom Aarsen took over maintainership; c. late 2025: project formally transitioned from UKP Lab to Hugging Face (date not confirmed); 18 August 2026: v6.0.0; 18 September 2026: v6.1.0	Verified fact	A2-S028, A2-S032, V1-S092
Strategic direction	Broadening from dense embeddings to Cross-Encoder rerankers (v4), Sparse Encoders (v5) and Multi-Vector Encoders for ColBERT-style late interaction (v6)	Verified fact	A2-S028, A2-S029
What it does	Computes embeddings, cross-encoder similarity scores, sparse embeddings and token-level multi-vector embeddings, and trains/fine-tunes such models; over 15,000 pre-trained models on Hugging Face.	Verified fact	A2-S029

¹⁹⁹ Verified fact — A2-S029: sentence-transformers - PyPI (PyPI / Hugging Face (maintainer)) <https://pypi.org/project/sentence-transformers/> accessed 2026-10-07

²⁰⁰ Verified fact — B-REV-S002: sentence-transformers Security Policy (Hugging Face) <https://raw.githubusercontent.com/huggingface/sentence-transformers/main/SECURITY.md> accessed 2026-10-08

²⁰¹ Verified fact — A2-S024: Legal Information - Jina AI (Jina AI (Elastic)) <https://jina.ai/legal/> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
Stack position	L7 tooling (library/runtime for self-hosted models), not a hosted model; complements rather than competes with API vendors.	Architectural judgment	
Integration	Python library; models from Hugging Face Hub	Verified fact	A2-S029
Dependencies	Python >=3.10; PyTorch/transformers ecosystem; optional image and audio extras	Verified fact	A2-S029
Certifications	Not applicable (open-source library)	Architectural judgment	
GDPR / residency	Not applicable (runs in customer environment)	Architectural judgment	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Free, open source (Apache-2.0)	Verified fact	A2-S029
Infrastructure cost	CPU/GPU compute for inference and fine-tuning; cost scales with model size and corpus volume	Architectural judgment	
Ecosystem	Hugging Face Hub; many MTEB leaderboard models load via the library	Verified fact	A2-S029
Adoption signals	Over 15,000 pre-trained Sentence Transformers models on Hugging Face	Verified fact	A2-S029
Maturity	Introduced 2019; eight releases between May and September 2026	Verified fact	A2-S028, A2-S032
Deployment	saas: No (library); managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes; on_prem: Yes		

Gemini Embedding 2 (model id gemini-embedding-2) (L7-gemini-embedding)

Tier: Strategic · **Flags:** none · **Original graphic label:** Gemini – Embedding 2

Rationale: Strategic, conditional: where Google Cloud is your primary cloud (CP3 Q2, rubric rule 10), consumed through Vertex AI. Google’s lead embedding model and the leading multimodal option; not where UK-only processing is mandatory, because the eu multi-region excludes the UK ^{AJ}.

Criterion	Score	Rationale
Technical	4	Multimodal, multilingual, Matryoshka 128-3,072; task types unsupported; Google’s reranker is the separate Vertex ranking API (B-REV-S026).
Enterprise readiness	4	Consumed through Google Cloud Vertex AI: IAM predefined, custom and endpoint-level roles, Cloud Audit Logs (Data Access logs for predict calls must be enabled) and SAML/OIDC federation verified at platform level ²⁰² ; rule 6 (CP2 Q1) presumes platform controls, and no product-specific gap is evidenced, so 4; platform controls presumed (CP2 Q1); confirm per service. Not 5: SCIM and support terms are not evidenced.
Security and compliance	4	Strong platform certifications including ISO 42001 and FedRAMP High; generative AI on the platform supports CMEK, VPC-SC and data residency (B-REV-S009), but per-model scope for embedding calls is not confirmed, so 4.
Deployment flexibility	2	Gemini API and Vertex AI only (global, us, eu); no private or self-hosted route.
Ecosystem	3	Google platform integration and one named adopter (Box); wider ecosystem NPV.
Reliability and	3	GA since April 2026 (about 5.5 months); pricing pages lagged the GA status.

²⁰² Verified fact — B-REV-S007: Agent Platform access control with IAM (Google Cloud) <https://docs.cloud.google.com/vertex-ai/docs/general/access-control> accessed 2026-10-08 [search-tool extract]; B-REV-S008: Agent Platform audit logging information (Google Cloud) <https://docs.cloud.google.com/vertex-ai/docs/general/audit-logging> accessed 2026-10-08 [search-tool extract]; B-REV-S012: Set up user access to the console (federated) (Google Cloud) <https://docs.cloud.google.com/iam/docs/workforce-console-ss> accessed 2026-10-08 [search-tool extract]

Criterion	Score	Rationale
maturity		
Cost / TCO	3	Transparent, but text pricing above peers; multimodal priced per modality.
Lock-in / portability	2	Proprietary hosted API plus Google platform coupling; re-embedding to switch.
Total (generic / FS)	3.30	
	/	
	3.20	

Evidence rules applied: NPV cap lifted at CP2 review (8 October 2026): platform IAM, Cloud Audit Logs and federated SSO verified (B-REV-S007, S008, S012); Rule 6 (CP2 Q1, hyperscaler presumption) applied at CP2 rework: enterprise_readiness 3 -> 4; platform controls presumed (CP2 Q1); confirm per service; Rule 8 (CP2 Q4) confirmed: platform-level certifications with per-model scope not confirmed, security_compliance 4

Capabilities. gemini-embedding-2 (GA 22 April 2026): one embedding space for text, images, video, audio and PDFs across 100+ languages; 8,192 text tokens; Matryoshka output 128-3,072 dims ^{vr203}. Hosted on the Gemini API and Vertex AI (being renamed Gemini Enterprise Agent Platform) ^{vr204}.

Strengths

- Natively multimodal with Matryoshka dimensions, useful for scanned documents and charts ^{vr205}
- Platform certifications for Generative AI on Vertex AI: SOC 2, ISO/IEC 27001, ISO/IEC 42001, HIPAA, FedRAMP High ^{vr206}
- 'eu' multi-region endpoint keeps data in EU member states ^{vr207}; Batch at 50% of standard price ^{vr208}

Limitations and risks

- EU multi-region excludes the UK and Switzerland ^{vr209}
- Per-model certification coverage and CMEK applicability for embedding calls not confirmed ^{vr210}
- Hosted only, no weights; coupling to Google platform services (File Search, Memory Bank) ^{AI}

²⁰³ Verified fact — A2-S004: Gemini Embedding 2 is now generally available (Google) <https://blog.google/innovation-and-ai/models-and-research/gemini-models/gemini-embedding-2-generally-available/> accessed 2026-10-07 [search-tool extract]; A2-S005: Embeddings (Gemini API docs) / Gemini Embedding 2 preview model (Google) <https://ai.google.dev/gemini-api/docs/embeddings> accessed 2026-10-07 [search-tool extract]

²⁰⁴ Verified fact — A2-S039: Data residency (Vertex AI generative AI) (Google Cloud) <https://docs.cloud.google.com/vertex-ai/generative-ai/docs/learn/data-residency> accessed 2026-10-07 [search-tool extract]

²⁰⁵ Verified fact — A2-S004: Gemini Embedding 2 is now generally available (Google) <https://blog.google/innovation-and-ai/models-and-research/gemini-models/gemini-embedding-2-generally-available/> accessed 2026-10-07 [search-tool extract]; A2-S005: Embeddings (Gemini API docs) / Gemini Embedding 2 preview model (Google) <https://ai.google.dev/gemini-api/docs/embeddings> accessed 2026-10-07 [search-tool extract]

²⁰⁶ Verified fact — A2-S043: Cloud compliance and regulations resources (Vertex AI) (Google Cloud) <https://cloud.google.com/vertex-ai/docs/general/vertexai-compliance> accessed 2026-10-07 [search-tool extract]

²⁰⁷ Verified fact — A2-S039: Data residency (Vertex AI generative AI) (Google Cloud) <https://docs.cloud.google.com/vertex-ai/generative-ai/docs/learn/data-residency> accessed 2026-10-07 [search-tool extract]

²⁰⁸ Verified fact — A2-S005: Embeddings (Gemini API docs) / Gemini Embedding 2 preview model (Google) <https://ai.google.dev/gemini-api/docs/embeddings> accessed 2026-10-07 [search-tool extract]; A2-S038: Gemini Developer API pricing (Google) <https://ai.google.dev/gemini-api/docs/pricing> accessed 2026-10-07 [search-tool extract]

²⁰⁹ Verified fact — A2-S039: Data residency (Vertex AI generative AI) (Google Cloud) <https://docs.cloud.google.com/vertex-ai/generative-ai/docs/learn/data-residency> accessed 2026-10-07 [search-tool extract]

²¹⁰ Verified fact — A2-S043: Cloud compliance and regulations resources (Vertex AI) (Google Cloud) <https://cloud.google.com/vertex-ai/docs/general/vertexai-compliance> accessed 2026-10-07 [search-tool extract]; A2-S039: Data residency (Vertex AI generative AI) (Google Cloud) <https://docs.cloud.google.com/vertex-ai/generative-ai/docs/learn/data-residency> accessed 2026-10-07 [search-tool extract]

- task_type parameter not supported ^{vr211}; Google's reranker is a separate service, the Vertex ranking API (semantic-ranker models; version 005 in preview from 1 September 2026) ^{vr212}
- Text price US\$0.20 per 1M tokens is higher than most hosted peers (as of 7 October 2026) ^{vr213}

Choose when

- The estate is Google Cloud-centred and EU processing is sufficient
- Retrieval must span images, audio, video or PDF pages in one vector space

Avoid when

- UK processing is mandatory
- Self-hosting or an exit route that keeps the same vectors is required

Nearest competitors: L7-cohere, L7-voyage, L7-jina, L7-openai

Regulated-FS note. Google Cloud EMEA Limited is a designated DORA CTPP and UK CTP ^{vr214}; consuming the model through Vertex AI places it with a designated provider, which does not remove the firm's own outsourcing duties ^{vi}. The 'eu' endpoint does not cover the UK ^{vr215}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Google (Google DeepMind / Google Cloud)	Verified fact	A2-S004
Category	Hosted multimodal embedding model API	Verified fact	A2-S004, A2-S005
Version / lineup	gemini-embedding-2: preview 10 March 2026 (gemini-embedding-2-preview), GA 22 April 2026; text-only gemini-embedding-001 remains available	Verified fact	A2-S004, A2-S005, A2-S039, V1-S073
Licence	Proprietary; hosted API only (Gemini API and Vertex AI / Gemini Enterprise Agent Platform)	Verified fact	A2-S004
Status events	10 March 2026: gemini-embedding-2-preview released; 22 April 2026: gemini-embedding-2 GA on Gemini API and Vertex AI; 2026: Google Cloud documentation renames Vertex AI generative AI docs to 'Gemini Enterprise Agent Platform'	Verified fact	A2-S004, A2-S005, A2-S039
Strategic	Google's first natively multimodal embedding model: text, images,	Verified	A2-S004,

²¹¹ Verified fact — A2-S005: Embeddings (Gemini API docs) / Gemini Embedding 2 preview model (Google) <https://ai.google.dev/gemini-api/docs/embeddings> accessed 2026-10-07 [search-tool extract]

²¹² Verified fact — B-REV-S026: Improve search and RAG quality with ranking API (Google Cloud) <https://docs.cloud.google.com/generative-ai-app-builder/docs/ranking> accessed 2026-10-08 [search-tool extract]

²¹³ Verified fact — A2-S038: Gemini Developer API pricing (Google) <https://ai.google.dev/gemini-api/docs/pricing> accessed 2026-10-07 [search-tool extract]

²¹⁴ Verified fact — A8-S020: The ESAs designate critical ICT third-party providers under DORA (with List_of_designated_CTPPs.pdf, 18 Nov 2025) (EBA / ESMA / EIOPA (Joint ESAs)) <https://www.eba.europa.eu/publications-and-media/press-releases/european-supervisory-authorities-designate-critical-ict-third-party-providers-under-digital> accessed 2026-10-07 [search-tool extract]; A8-S023: The Critical Third Parties (Designation) Regulations 2026 (SI 2026/777); BoE/FCA/HMT announcements (HM Treasury / legislation.gov.uk / Bank of England) <https://www.legislation.gov.uk/uksi/2026/777/made> accessed 2026-10-07 [search-tool extract]

²¹⁵ Verified fact — A2-S039: Data residency (Vertex AI generative AI) (Google Cloud) <https://docs.cloud.google.com/vertex-ai/generative-ai/docs/learn/data-residency> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
direction	video, audio and documents in one embedding space; positioned for agentic multimodal RAG; Gemini File Search API gained multimodal support	d fact	A2-S005
What it does	Maps text, images (up to 6 per request), video, audio (no transcription step) and PDFs (up to 6 pages) into a shared vector space across 100+ languages; 8,192 text-token input; Matryoshka output 128-3,072 dims (3,072/1,536/768 recommended) with automatic normalisation; multiple inputs in one request produce one aggregated embedding; task_type parameter not supported.	Verified fact	A2-S004, A2-S005
Stack position	L7 embedding model; placement correct. Also reached through Google platform services (File Search, Memory Bank), so it can be consumed implicitly inside higher layers.	Architectural judgement	
Integration	Gemini API and Vertex AI SDK/REST; Batch API at 50% of standard price	Verified fact	A2-S005, A2-S038
Dependencies	Google Gemini API or Google Cloud Vertex AI (Gemini Enterprise Agent Platform)	Verified fact	A2-S004, A2-S039
Certifications	Generative AI on Vertex AI: SOC 2, ISO/IEC 27001, ISO/IEC 42001, HIPAA, FedRAMP High (scope per product/feature; per-model coverage for gemini-embedding-2 not confirmed)	Verified fact	A2-S043
GDPR / residency	Available on the 'eu' multi-region endpoint, which keeps data within EU member states (UK and Switzerland excluded); batch supported in global, us and eu	Verified fact	A2-S039
Security features	Vertex AI supports customer-managed encryption keys (CMEK) with location rules; applicability to gemini-embedding-2 calls not confirmed	Verified fact	A2-S039
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Gemini API paid tier per 1M tokens: text US\$0.20, image US\$0.45, audio US\$6.50, video US\$12.00; Batch: text US\$0.10, image US\$0.225, audio US\$3.25, video US\$6.00; free tier available. Vertex AI: text US\$0.20 online / US\$0.10 batch, image US\$0.45 online.	Verified fact	A2-S038
Infrastructure cost	Not applicable: hosted only	Architectural judgement	
Ecosystem	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
		d	
Adoption signals	Box is integrating Gemini Embedding 2 multimodal capabilities into the Box Agentic Platform (Google Cloud blog)	Verified fact	A2-S047
Maturity	GA since 22 April 2026 (about 5.5 months)	Verified fact	A2-S004
Deployment	saas: Yes (Gemini Developer API); managed_cloud: Yes (Google Cloud Vertex AI / Gemini Enterprise Agent Platform; global, us and eu endpoints); vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: No (no downloadable weights); on_prem: Not publicly verified		

OpenAI embeddings: text-embedding-3-large and text-embedding-3-small (L7-openai)

Tier: Tactical · **Flags:** none · **Original graphic label:** OpenAI – Embeddings 3

Rationale: Competent, well-controlled text embeddings, but hosted-only, text-only and without a reranker; a baseline, not a foundation.

Criterion	Score	Rationale
Technical	3	Core text embedding with shortening via 'dimensions'; no reranker, no multimodal, 2024 generation.
Enterprise readiness	4	SAML/OIDC SSO, tenant SCIM for the API Platform (from 17 September 2026), organisation and project roles with custom roles, Admin API and audit logs of administrative events ²¹⁶ ; support/SLA terms not verified and audit logs exclude request content with best-effort retention, so not 5.
Security and compliance	4	SOC 2 Type 2, ISO 27001/27701, HIPAA BAA, ZDR and EU residency for the endpoint; CMK not verified.
Deployment flexibility	2	Hosted SaaS only, with US and EU processing regions; no VPC or self-host verified.
Ecosystem	3	Standard REST endpoint, SDKs and Batch API; broader ecosystem evidence NPV in the dataset.
Reliability and maturity	4	GA since January 2024 and unchanged for about 2.7 years.

²¹⁶ Verified fact — B-REV-S004: Single sign-on (SSO) setup for OpenAI products <https://help.openai.com/en/articles/9534785-configuring-sso> accessed 2026-10-08 [search-tool extract]; B-REV-S005: Manage permissions in the OpenAI platform <https://platform.openai.com/docs/guides/rbac> accessed 2026-10-08 [search-tool extract]; B-REV-S006: Admin and Audit Logs API for the API Platform (OpenAI) <https://help.openai.com/en/articles/9687866-admin-and-audit-logs-api-for-the-api-platform> accessed 2026-10-08 [search-tool extract]

Criterion	Score	Rationale
Cost / TCO	4	Transparent per-token pricing at the low end of the hosted market; Batch halves 3-large.
Lock-in / portability	2	Proprietary hosted API, no weights; switching forces full re-embedding.
Total (generic / FS)	3.30 / 3.20	

Evidence rules applied: NPV cap lifted at CP2 review (8 October 2026): SSO, RBAC and audit logs verified (B-REV-S004 to S006); Rules 7 and 8 checked at CP2 rework (8 October 2026): enterprise_readiness 4 and security_compliance 4 kept

Capabilities. Hosted text embeddings: text-embedding-3-large (3,072 dims by default, shortened with the 'dimensions' parameter) and text-embedding-3-small, 8,192-token input, released 25 January 2024^{vi217}. No first-party reranker was found^{vi218}.

Strengths

- Stable, unchanged generation since January 2024, so a pinned model has not been forced to move^{vi219} ^{AJ}
- Strong documented controls for the endpoint: SOC 2 Type 2, ISO/IEC 27001 and 27701, ZDR-eligible /v1/embeddings, EU storage and processing residency with Modified Abuse Monitoring or ZDR, HIPAA BAA^{vi220}
- Low, transparent price: 3-small US\$0.02 and 3-large US\$0.13 per 1M tokens, Batch at half price for 3-large (as of 7 October 2026)^{vi221}
- API Platform identity and access: SAML/OIDC SSO, SCIM, organisation and project roles with custom roles, Admin API and audit logs^{vi222}

Limitations and risks

- Text only; no reranker, so a second vendor or library is needed for two-stage retrieval^{vi223} ^{AJ}
- Hosted only, no downloadable weights; vectors are model-specific, so exit means re-embedding the corpus^{AJ}

²¹⁷ Verified fact — A2-S001: text-embedding-3-large Model (OpenAI) <https://developers.openai.com/api/docs/models/text-embedding-3-large> accessed 2026-10-07 [search-tool extract]; A2-S002: New embedding models and API updates (OpenAI) <https://openai.com/index/new-embedding-models-and-api-updates/> accessed 2026-10-07 [search-tool extract]

²¹⁸ Verified fact — A2-S001: text-embedding-3-large Model (OpenAI) <https://developers.openai.com/api/docs/models/text-embedding-3-large> accessed 2026-10-07 [search-tool extract]

²¹⁹ Verified fact — A2-S002: New embedding models and API updates (OpenAI) <https://openai.com/index/new-embedding-models-and-api-updates/> accessed 2026-10-07 [search-tool extract]

²²⁰ Verified fact — A2-S036: Security and privacy at OpenAI <https://openai.com/security-and-privacy/> accessed 2026-10-07 [search-tool extract]; A2-S144: Data controls in the OpenAI platform <https://developers.openai.com/api/docs/guides/your-data> accessed 2026-10-07 [search-tool extract]

²²¹ Verified fact — A2-S001: text-embedding-3-large Model (OpenAI) <https://developers.openai.com/api/docs/models/text-embedding-3-large> accessed 2026-10-07 [search-tool extract]; A2-S035: text-embedding-3-small Model (OpenAI) <https://developers.openai.com/api/docs/models/text-embedding-3-small> accessed 2026-10-07 [search-tool extract]

²²² Verified fact — B-REV-S004: Single sign-on (SSO) setup for OpenAI products <https://help.openai.com/en/articles/9534785-configuring-ssso> accessed 2026-10-08 [search-tool extract]; B-REV-S005: Manage permissions in the OpenAI platform <https://platform.openai.com/docs/guides/rbac> accessed 2026-10-08 [search-tool extract]; B-REV-S006: Admin and Audit Logs API for the API Platform (OpenAI) <https://help.openai.com/en/articles/9687866-admin-and-audit-logs-api-for-the-api-platform> accessed 2026-10-08 [search-tool extract]

²²³ Verified fact — A2-S001: text-embedding-3-large Model (OpenAI) <https://developers.openai.com/api/docs/models/text-embedding-3-large> accessed 2026-10-07 [search-tool extract]

- UK region is storage-only for embeddings; processing in Europe needs Modified Abuse Monitoring or ZDR ^{VF-224}
- Audit logs record administrative events only, not request content, and are kept on a best-effort basis, so they must be exported to the firm's archive ^{VF-225}; CMK/BYOK not verified ^{NPV}
- Ageing generation with no announced successor; successor timing unknown ^{AJ}

Choose when

- OpenAI is already the contracted L1 provider with ZDR and EU residency approved, and the corpus is text-only
- A low-cost English or multilingual text baseline is needed for an in-domain bake-off

Avoid when

- Data must be processed in the UK or inside the firm's own estate
- Multimodal retrieval (scanned factsheets, charts) or a single-vendor embed-plus-rerank pair is required

Nearest competitors: L7-gemini-embedding, L7-cohere, L7-voyage

Regulated-FS note. Usable for EU-processed client data only under ZDR or Modified Abuse Monitoring on an EU project ^{VF-226}; UK processing is not offered for this endpoint ^{VF-227}. A direct OpenAI contract is an ICT third-party arrangement outside the DORA CTPP perimeter, so oversight sits with the firm ^{VF-228} ^{AJ}.

Facts (as of 2026-10-07)

Field	Value	Label	Sources
Company	OpenAI	Verified fact	A2-S002
Category	Hosted text-embedding model API	Verified fact	A2-S003
Version / lineup	text-embedding-3-large (most capable; 3,072 dims default) and text-embedding-3-small, both released 25 January 2024; text-embedding-ada-002 still listed. No newer OpenAI embedding model found as of 7 October 2026.	Verified fact	A2-S001, A2-S002
Licence	Proprietary; available only as a hosted API	Verified fact	A2-S001, A2-S003
Status events	25 January 2024: text-embedding-3-small/-large launched; ada-002 not deprecated at launch	Verified fact	A2-S002
Strategic	Not publicly verified	Not	

²²⁴ Verified fact — A2-S144: Data controls in the OpenAI platform <https://developers.openai.com/api/docs/guides/your-data> accessed 2026-10-07 [search-tool extract]

²²⁵ Verified fact — B-REV-S006: Admin and Audit Logs API for the API Platform (OpenAI) <https://help.openai.com/en/articles/9687866-admin-and-audit-logs-api-for-the-api-platform> accessed 2026-10-08 [search-tool extract]

²²⁶ Verified fact — A2-S144: Data controls in the OpenAI platform <https://developers.openai.com/api/docs/guides/your-data> accessed 2026-10-07 [search-tool extract]

²²⁷ Verified fact — A2-S144: Data controls in the OpenAI platform <https://developers.openai.com/api/docs/guides/your-data> accessed 2026-10-07 [search-tool extract]

²²⁸ Verified fact — A8-S021: Digital Operational Resilience Act (DORA) pages and RoI reporting FAQ (ESMA / EBA / EIOPA) <https://www.esma.europa.eu/esmas-activities/digital-finance-and-innovation/digital-operational-resilience-act-dora> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
direction		publicly verified	
What it does	Converts text into dense vectors for search, clustering and recommendation. Output defaults to 3,072 dimensions for 3-large and can be shortened via the 'dimensions' parameter; maximum input 8,192 tokens.	Verified fact	A2-S001, A2-S003
Stack position	L7 embedding model; placement in the graphic is correct. Embedding only (no first-party reranker found), so a separate reranker is needed for two-stage retrieval.	Architectural judgment	
Integration	REST /v1/embeddings endpoint and OpenAI SDKs; Batch API at half the standard per-token price for 3-large	Verified fact	A2-S003, A2-S035
Dependencies	OpenAI API platform (cloud service)	Verified fact	A2-S003
Certifications	SOC 2 Type 2 (API and business products); ISO/IEC 27001:2022 and 27701:2019; CSA STAR Level 1; /v1/embeddings on HIPAA-eligible list with BAA and Modified Retention	Verified fact	A2-S036, A2-S144
GDPR / residency	/v1/embeddings: Zero Data Retention eligible (prior approval); data residency per project - US and Europe support storage and processing (Europe requires Modified Abuse Monitoring or ZDR); UK, Australia, Canada, Japan, India, Singapore, South Korea storage only; Europe projects handle requests in-region with ZDR; DPA available	Verified fact	A2-S144, A2-S037
Security features	AES-256 at rest, TLS 1.2+ in transit; customer API data not used for training by default. CMK/BYOK for the API not verified.	Verified fact	A2-S037
Access controls	SSO/MFA advertised for the API platform; SCIM, RBAC granularity and audit-log scope not verified in this run	Verified fact	A2-S036
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	text-embedding-3-large US\$0.13 per 1M tokens (Batch US\$0.065); text-embedding-3-small US\$0.02 per 1M tokens	Verified fact	A2-S001, A2-S035
Infrastructure cost	Not applicable: hosted only; cost is per token	Architectural judgment	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
		verified	
Maturity	GA since January 2024; no successor model announced in the ~2.7 years since	Verified fact	A2-S002, A2-S001
Deployment	saas: Yes; managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: No (no downloadable weights published); on_prem: No		

Jina AI search foundation models (jina-embeddings-v5-text, jina-embeddings-v5-omni, jina-reranker-v3.5), part of Elastic (L7-jina)

Tier: Tactical · **Flags:** Acquired · **Original graphic label:** Jina AI – Embeddings v3

Rationale: Natural choice inside Elastic estates; non-commercial weights and an opaque price list make it a poor neutral default.

Criterion	Score	Rationale
Technical	4	Multilingual, multimodal, long-context embeddings and listwise and multimodal rerankers.
Enterprise readiness	3	Scored on the recommended Elastic Inference Service route: Elastic Cloud SAML SSO and RBAC at platform level ^{v1.229} ; audit logging of EIS calls not evidenced. The standalone Jina API (API keys only) would stay capped at 2.
Security and compliance	3	Elastic Cloud SOC 2 Type II and ISO 27001 cover the EIS route; hosted Jina API scope unconfirmed.
Deployment flexibility	4	SaaS API, Elastic Inference Service, on-prem Docker with air-gap; VPC NPV.
Ecosystem	4	Default in Elasticsearch semantic_text; Hugging Face and vLLM recipes.
Reliability and maturity	3	Three releases in February-July 2026; ownership changed October 2025.
Cost / TCO	2	API rates not retrieved, on-prem by quote, CC-BY-NC weights: opaque for

²²⁹ Verified fact — B-REV-S019: Elastic Cloud Feature Matrix (and EIS, Jina models docs) (Elastic N.V.) <https://www.elastic.co/subscriptions/cloud> accessed 2026-10-08 [search-tool extract]

Criterion	Score	Rationale
		planning.
Lock-in / portability	2	CC-BY-NC weights route commercial use through Elastic (would be 3); reduced by 1 for the 2025 acquisition.
Total (generic / FS)	3.30	
	/	
	3.15	

Evidence rules applied: NPV cap lifted at CP2 review for the Elastic Inference Service route (B-REV-S019); the standalone Jina API remains capped at 2 (API keys only); Rule 7 (CP2 Q2) checked at CP2 rework: SSO and RBAC verified on Elastic Cloud, audit logging of EIS calls not evidenced, so 3 kept; rule 8 (CP2 Q4) confirmed for security 3 (Elastic Cloud certifications, Jina API scope unconfirmed)

Capabilities. Jina AI, part of Elastic since 9 October 2025^{vi-230}; jina-embeddings-v5-text (February 2026; 32,768-token context, 1,024 dims), v5-omni (May 2026; text, image, audio, video, PDF, with text vectors identical to v5-text), jina-reranker-v3.5 (July 2026, listwise) and jina-reranker-m0 (multimodal)^{vi-231}.

Strengths

- Default embedding model behind Elasticsearch semantic_text, so it is the low-effort path for Elastic estates^{vi-232}
- Commercial on-premises licence with Docker containers and air-gapped deployment^{vi-233}
- v5-omni text vectors match v5-text, so adding modalities need not force text re-embedding^{vi-234}

Limitations and risks

- Weights are CC-BY-NC-4.0: commercial self-hosting needs the Jina API, a marketplace, Elastic Inference Service or an on-prem licence^{vi-235}
- Acquired by Elastic (October 2025); Jina's own policies predate the acquisition^{vi-236}

²³⁰ Verified fact — A2-S023: Elastic Completes Acquisition of Jina AI, a Leader in Frontier Models for Multimodal and Multilingual Search <https://ir.elastic.co/News--Events/news/news-details/2025/Elastic-Completes-Acquisition-of-Jina-AI-a-Leader-in-Frontier-Models-for-Multimodal-and-Multilingual-Search/default.aspx> accessed 2026-10-07 [search-tool extract]; V1-S025: Elastic Completes Acquisition of Jina AI (Elastic N.V.) <https://ir.elastic.co/News--Events/news/news-details/2025/Elastic-Completes-Acquisition-of-Jina-AI-a-Leader-in-Frontier-Models-for-Multimodal-and-Multilingual-Search/default.aspx> accessed 2026-10-08 [search-tool extract]

²³¹ Verified fact — A2-S025: jina-embeddings-v5-omni: Embeddings for Text, Image, Audio and Video (Jina AI (Elastic)) <https://jina.ai/news/jina-embeddings-v5-omni-multimodal-embeddings-for-text-image-audio-and-video/> accessed 2026-10-07 [search-tool extract]; A2-S026: jina-reranker-v3.5: Faster Listwise Reranking with Hybrid Attention and Self-Distillation (Jina AI (Elastic)) <https://jina.ai/en-US/news/jina-reranker-v3-5-faster-listwise-reranking-hybrid-attention-self-distillation/> accessed 2026-10-07 [search-tool extract]; V1-S026: jina-embeddings-v5 model pages (text-small, text-nano, omni-small, omni-nano); Jina llms.txt (Jina AI (Elastic)) <https://jina.ai/models/jina-embeddings-v5-text-small/> accessed 2026-10-08 [search-tool extract]

²³² Verified fact — A2-S133: Elastic 9.5: Columnar, VectorDB index mode & auto-calibration, and AI-driven alert triage <https://www.elastic.co/blog/whats-new-elastic-9-5-0> accessed 2026-10-07 [search-tool extract]

²³³ Verified fact — A2-S042: Jina models (Elastic docs) <https://www.elastic.co/docs/explore-analyze/machine-learning/nlp/ml-nlp-jina> accessed 2026-10-07 [search-tool extract]; A2-S045: Elastic Trust Center <https://www.elastic.co/trust> accessed 2026-10-07 [search-tool extract]

²³⁴ Verified fact — A2-S025: jina-embeddings-v5-omni: Embeddings for Text, Image, Audio and Video (Jina AI (Elastic)) <https://jina.ai/news/jina-embeddings-v5-omni-multimodal-embeddings-for-text-image-audio-and-video/> accessed 2026-10-07 [search-tool extract]

²³⁵ Verified fact — A2-S024: Legal Information - Jina AI (Jina AI (Elastic)) <https://jina.ai/legal/> accessed 2026-10-07 [search-tool extract]; A2-S042: Jina models (Elastic docs) <https://www.elastic.co/docs/explore-analyze/machine-learning/nlp/ml-nlp-jina> accessed 2026-10-07 [search-tool extract]

²³⁶ Verified fact — A2-S023: Elastic Completes Acquisition of Jina AI, a Leader in Frontier Models for Multimodal and Multilingual Search <https://ir.elastic.co/News--Events/news/news-details/2025/Elastic-Completes-Acquisition-of-Jina-AI-a-Leader-in-Frontier-Models-for-Multimodal-and-Multilingual-Search/default.aspx> accessed 2026-10-07 [search-tool extract]; A2-S024: Legal Information - Jina AI (Jina AI (Elastic)) <https://jina.ai/legal/> accessed 2026-10-07 [search-tool extract]

- Hosted Jina API SOC 2 scope not confirmed; Elastic Cloud holds ISO 27001/27017/27018 and SOC 2 Type II ^{ver²³⁷}
- API per-token rates not retrieved; on-prem licence priced via sales ^{ver²³⁸}
- Vendor-reported 63.20 nDCG@10 on BEIR is not a decision input ^{rep²³⁹}

Choose when

- Elasticsearch is the L6 store and the firm wants embedding and reranking inside the same platform
- Air-gapped multimodal retrieval is needed under a commercial licence

Avoid when

- The plan assumes free commercial self-hosting of the weights
- Store and embedding vendor must be independent for concentration reasons

Nearest competitors: L7-voyage, L7-cohere, L7-qwen3-embedding, L7-sentence-transformers

Regulated-FS note. Data processing is now under Elastic's terms; region details for the hosted API not verified ^{ver²⁴⁰}. Confirm the licence route in writing before any self-hosted production use ^{rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Jina AI GmbH, acquired by Elastic N.V. (completed October 2025)	Verified fact	A2-S023, A2-S024
Category	Embedding and reranking models (hosted API, Elastic Inference Service, on-prem licence, open weights for non-commercial use)	Verified fact	A2-S042, A2-S024
Version / lineup	jina-embeddings-v5-text small (677M params, 32,768-token context, 1,024 dims) and nano (239M) - February 2026; jina-embeddings-v5-omni small and nano - May 2026 (default API embedding model); jina-reranker-v3.5 (0.6B listwise) - July 2026; jina-reranker-m0 (multimodal) also listed	Verified fact	A2-S025, A2-S026, V1-S026
Licence	Model weights CC-BY-NC-4.0 (commercial use via Jina API, cloud marketplaces, Elastic Inference Service or on-prem licence); jina-embeddings-v4 under Qwen Research Licence (no commercial use)	Verified fact	A2-S024, A2-S042, V1-S026
Status events	9 October 2025: Elastic completes acquisition of Jina AI (IR release); February 2026: jina-embeddings-v5-text; May 2026: jina-embeddings-v5-omni; June 2026: v5-omni small/nano available on Elastic Inference Service; July 2026: jina-reranker-v3.5	Verified fact	A2-S023, A2-S025, A2-S027, A2-S026, V1-S025
Strategic direction	Being integrated into Elastic: Jina models served on Elastic Inference Service alongside ELSER; rerankers for visual and long-context multilingual documents	Verified fact	A2-S023, A2-S027, A2-S042

²³⁷ Verified fact — A2-S045: Elastic Trust Center <https://www.elastic.co/trust> accessed 2026-10-07 [search-tool extract]

²³⁸ Verified fact — A2-S042: Jina models (Elastic docs) <https://www.elastic.co/docs/explore-analyze/machine-learning/nlp/ml-nlp-jina> accessed 2026-10-07 [search-tool extract]

²³⁹ Reported — A2-S026: jina-reranker-v3.5: Faster Listwise Reranking with Hybrid Attention and Self-Distillation (Jina AI (Elastic)) <https://jina.ai/en-US/news/jina-reranker-v3-5-faster-listwise-reranking-hybrid-attention-self-distillation/> accessed 2026-10-07 [search-tool extract]

²⁴⁰ Verified fact — A2-S024: Legal Information - Jina AI (Jina AI (Elastic)) <https://jina.ai/legal/> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
What it does	Multilingual and multimodal embedding models (text, image, audio, video, PDF in one space) and listwise rerankers; v3.5 reranker reports 63.20 nDCG@10 on BEIR and up to 1.56x faster than v3 on long documents (vendor-reported).	Verified fact	A2-S025, A2-S026
Stack position	L7 embed + rerank; now also a native component of Elasticsearch (L6) via Elastic Inference Service.	Architectural judgment	
Integration	Jina Embedding and Reranker APIs (token-billed); Elasticsearch open inference API / EIS; Hugging Face weights; vLLM recipes	Verified fact	A2-S042, A2-S025
Dependencies	Jina hosted API; or Elastic Cloud (EIS); or Docker containers on customer infrastructure (on-prem licence)	Verified fact	A2-S042, A2-S045
Certifications	Elastic Cloud: ISO 27001, ISO 27017, ISO 27018, SOC 2 Type II; coverage of the hosted Jina API not confirmed	Verified fact	A2-S045
GDPR / residency	Jina data processing now governed by Elastic's data processing terms; region details not verified	Verified fact	A2-S024
Security features	Customer data encrypted in transit and at rest with per-customer keys; request data not used for training (Jina legal page)	Verified fact	A2-S045
Access controls	API-key access managed by customer; other controls not verified	Verified fact	A2-S045
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Jina API billed per token processed (rates not retrieved); EIS billed per million input tokens on Elastic Cloud; on-prem annual licence via Elastic Sales. Rate limits: Free 100 RPM/100K TPM, Paid 500 RPM/2M TPM, Premium 5,000 RPM/50M TPM.	Verified fact	A2-S042
Infrastructure cost	On-prem licence priced by models deployed and inference hardware, not per token	Verified fact	A2-S042
Ecosystem	Elasticsearch integration via open inference API and EIS; Hugging Face; vLLM	Verified fact	A2-S042, A2-S025
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Rapid cadence: three model releases in Feb-Jul 2026 after acquisition	Verified fact	A2-S025, A2-S026
Deployment	saas: Yes (Jina API); managed_cloud: Yes (Elastic Inference Service on Elastic Cloud; cloud marketplaces); vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes (commercial on-prem licence; Docker containers, air-gapped supported); on_prem: Yes		

Qwen3-Embedding and Qwen3-Reranker (open weights); Qwen3-VL-Embedding / Qwen3-VL-Reranker (multimodal); hosted as text-embedding-v4 and qwen3-rerank on Alibaba Cloud Model Studio (L7-qwen3-embedding)

Tier: Tactical · Flags: none · Original graphic label: Qwen3 – Embeddings

Rationale: Capable Apache 2.0 embed-plus-rerank pair for self-hosting; usable only after provenance review and with in-house operations.

Criterion	Score	Rationale
Technical	4	Embed and rerank in three sizes, 32K context, 119 languages, multimodal VL variants.
Enterprise readiness	2	Rule 2 (self-hosted open weights; hosted Model Studio route not recommended): enables in-estate serving only; no commercial support for self-hosting verified, so 2.
Security and compliance	2	Rule 2: Apache-2.0 licence is clear, but no security policy or signed-release evidence was verified and a model-provenance review is outstanding, so 2; inherits host controls.
Deployment flexibility	4	Self-host, on-prem and disconnected use via open weights, plus SaaS outside the EU.
Ecosystem	3	Hugging Face and ModelScope weights; OpenAI-compatible batch API on Model Studio; wider ecosystem NPV.
Reliability and maturity	3	Released June 2025 with a January 2026 multimodal successor; long-term support NPV.
Cost / TCO	4	Free weights; cost is GPU and operations, light for 0.6B, heavier for 8B.
Lock-in / portability	4	Apache 2.0 and self-hostable; single-vendor governance and model-specific vectors keep it below 5.
Total (generic / FS)	3.20 / 3.15	

Evidence rules applied: Rule 2 (self-hosted open weights) applied instead of the NPV cap, because the recommendation is self-host only; scores unchanged at 2 on project hygiene and support, not on missing hosted certifications; Rule 7 (CP2 Q2) checked at CP2 rework: no SSO, RBAC or audit control verified; rule 2 scores kept

Capabilities. Open-weight Qwen3-Embedding and Qwen3-Reranker in 0.6B, 4B and 8B (June 2025; 32K context; 119 languages; Apache 2.0) and Qwen3-VL-Embedding/-Reranker (January 2026)²⁴¹. Hosted as text-embedding-v4 (64-2,048 dims) and qwen3-rerank on Alibaba Cloud Model Studio²⁴².

Strengths

²⁴¹ Verified fact — A2-S020: Qwen3 Embedding: Advancing Text Embedding and Reranking Through Foundation Models (Alibaba Qwen) <https://qwen.ai/blog?id=qwen3-embedding> accessed 2026-10-07 [search-tool extract]; A2-S021: Qwen3-VL-Embedding and Qwen3-VL-Reranker: A Unified Framework for State-of-the-Art Multimodal Retrieval and Ranking (Alibaba Qwen (arXiv)) <https://arxiv.org/abs/2601.04720> accessed 2026-10-07 [search-tool extract]; V1-S093: Qwen3-VL-Embedding and Qwen3-VL-Reranker (paper page, arXiv 2601.04720); Qwen blog (Qwen team (Alibaba) / Hugging Face) <https://huggingface.co/papers/2601.04720> accessed 2026-10-08 [search-tool extract]

²⁴² Verified fact — A2-S022: Embedding - Alibaba Cloud Model Studio <https://www.alibabacloud.com/help/en/model-studio/embedding> accessed 2026-10-07 [search-tool extract]

- Apache 2.0 embed and rerank pair that can run fully inside the firm’s estate, including disconnected environments ^{vr²⁴³ AJ}
- Three sizes allow a cost/quality trade-off; 0.6B runs on modest hardware ^{AJ}
- Multimodal successors (VL) in the same family ^{vr²⁴⁴}

Limitations and risks

- Chinese-origin model; FS firms will require provenance and supply-chain review before use ^{AJ}
- Hosted Model Studio regions are Singapore, Hong Kong and Beijing; no EU region verified ^{vr²⁴⁵}
- No commercial support or certifications verified ^{NPV}; Qwen3-VL embedding licence not verified ^{NPV}
- MTEB No.1 claim dates from 5 June 2025 and is vendor-reported; current rank not verified ^{r²⁴⁶}
- Self-hosting the 8B models needs datacentre GPUs and an in-house serving team ^{AJ}

Choose when

- A self-hosted, permissively licensed embed-plus-rerank pair is required and provenance review approves it
- Multilingual corpora need in-estate processing at low marginal cost

Avoid when

- Policy excludes Chinese-origin model weights, or only the hosted API would be used
- There is no capacity to operate GPU inference

Nearest competitors: L7-sentence-transformers, L7-nvidia-nemo-retriever, L7-jina, L7-voyage

Regulated-FS note. Self-hosted weights keep client data in the firm’s estate; the hosted API is not a fit for EU/UK client data on verified regions ^{vr²⁴⁷}. Hosted prices come from Alibaba’s own pricing page ^{vr²⁴⁸}, but the hosted route is not the recommended one for client data, so they are not a decision input ^{AJ}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Alibaba Group (Qwen team; Alibaba Cloud)	Verifie d fact	A2-S020, A2-S022
Category	Open-weight embedding and reranking models, also offered as hosted API	Verifie d fact	A2-S020, A2-S022
Version /	Qwen3-Embedding and Qwen3-Reranker in 0.6B, 4B and 8B (June 2025;	Verifie	A2-S020,

²⁴³ Verified fact — A2-S020: Qwen3 Embedding: Advancing Text Embedding and Reranking Through Foundation Models (Alibaba Qwen) <https://qwen.ai/blog?id=qwen3-embedding> accessed 2026-10-07 [search-tool extract]

²⁴⁴ Verified fact — A2-S021: Qwen3-VL-Embedding and Qwen3-VL-Reranker: A Unified Framework for State-of-the-Art Multimodal Retrieval and Ranking (Alibaba Qwen (arXiv)) <https://arxiv.org/abs/2601.04720> accessed 2026-10-07 [search-tool extract]

²⁴⁵ Verified fact — A2-S022: Embedding - Alibaba Cloud Model Studio <https://www.alibabacloud.com/help/en/model-studio/embedding> accessed 2026-10-07 [search-tool extract]

²⁴⁶ Reported — A2-S020: Qwen3 Embedding: Advancing Text Embedding and Reranking Through Foundation Models (Alibaba Qwen) <https://qwen.ai/blog?id=qwen3-embedding> accessed 2026-10-07 [search-tool extract]

²⁴⁷ Verified fact — A2-S022: Embedding - Alibaba Cloud Model Studio <https://www.alibabacloud.com/help/en/model-studio/embedding> accessed 2026-10-07 [search-tool extract]

²⁴⁸ Verified fact — A2-S022: Embedding - Alibaba Cloud Model Studio <https://www.alibabacloud.com/help/en/model-studio/embedding> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
lineup	32K context; 119 languages); Qwen3-VL-Embedding and Qwen3-VL-Reranker (January 2026); hosted text-embedding-v4, qwen3.7-text-embedding, qwen3-rerank, qwen3-vl-rerank	d fact	A2-S021, A2-S022
Licence	Open weights, Apache 2.0 (Qwen3-Embedding/Reranker); hosted API proprietary service	Verified fact	A2-S020
Status events	June 2025: Qwen3-Embedding and Qwen3-Reranker released (Apache 2.0); January 2026 (paper 8 January 2026): Qwen3-VL-Embedding and Qwen3-VL-Reranker (2B and 8B) released	Verified fact	A2-S020, A2-S021, V1-S093
Strategic direction	Extending to multimodal retrieval (Qwen3-VL-Embedding/Reranker) and newer hosted embedding versions (qwen3.7-text-embedding, 201 languages)	Verified fact	A2-S021, A2-S022
What it does	Text embedding and reranking models built on Qwen3 foundation models; hosted text-embedding-v4 supports 64-2,048 dims (1,024 default).	Verified fact	A2-S020, A2-S022
Stack position	L7; family covers both embed and rerank, so the graphic understates it.	Architectural judgement	
Integration	Hugging Face / ModelScope weights; Model Studio API including OpenAI-compatible batch API	Verified fact	A2-S020, A2-S022
Dependencies	Self-host on GPU/CPU from Hugging Face/ModelScope weights, or Alibaba Cloud Model Studio	Verified fact	A2-S020, A2-S022
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	Model Studio pricing lists Singapore, Hong Kong and Beijing regions; no EU region verified	Verified fact	A2-S022
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Model Studio: text-embedding-v4 US\$0.07 per 1M tokens (Singapore; US\$0.072 Beijing); qwen3.7-text-embedding US\$0.07 (Singapore); qwen3-rerank US\$0.10 per 1M input tokens (Singapore), output free;	Verified fact	A2-S022

Field	Value	Label	Sources
	1M free tokens for 90 days. Open weights free.		
Infrastructure cost	Self-hosting cost driven by GPU size: 0.6B runs on modest hardware; 8B needs a datacentre GPU for production throughput	Architectural judgment	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	8B embedding ranked No.1 on MTEB multilingual (score 70.58) as of 5 June 2025 (vendor-reported); Jina reports its v3.5 reranker beating Qwen3-Reranker-4B on BEIR	Verified fact	A2-S020, A2-S026
Maturity	Open-weight release June 2025; multimodal successor January 2026	Verified fact	A2-S020, A2-S021
Deployment	saas: Yes (Alibaba Cloud Model Studio); managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes (open weights); on_prem: Yes (open weights)		

Voyage AI by MongoDB (Voyage 4 embedding family; Rerank 3 rerankers) (L7-voyage)

Tier: Tactical · **Flags:** Acquired · **Original graphic label:** Voyage AI – Voyage-3

Rationale: Technically the deepest lineup, but certifications are unverified, key APIs are in preview and ownership now couples it to MongoDB.

Criterion	Score	Rationale
Technical	5	Leading breadth in primary docs: domain, contextual, multimodal, shared space, quantisation and rerankers.
Enterprise readiness	3	Scored on the Atlas route recommended for MongoDB estates: Voyage model API keys are governed by Atlas organisation and project roles (Organization Owner, Project Owner, Project Model Owner and read-only roles), with an Admin API endpoint for model keys ^{vr249} . One verified control lifts the NPV cap to 3 under rule 7 (CP2 Q2); SSO and audit logging of model-key use are not stated. The AWS Marketplace in-VPC route (A2-S044) would score 4 under rule 6 (CP2 Q1).
Security and compliance	2	SOC 2/HIPAA appear only as a reported homepage listing; trust-portal scope unverified.
Deployment	3	SaaS, Atlas, AWS/Azure marketplaces and in-VPC SageMaker; only nano is self-

²⁴⁹ Verified fact — B-REV-S028: Manage Voyage AI Model API Keys (and Organization and Project Access) (MongoDB, Inc.) <https://www.mongodb.com/docs/voyageai/management/api-keys/> accessed 2026-10-08 [search-tool extract]

Criterion	Score	Rationale
flexibility		hostable.
Ecosystem	4	Native in MongoDB, on AWS and Azure marketplaces; listed in Anthropic's embeddings docs (author conflict noted).
Reliability and maturity	3	Frequent generations; MongoDB-integrated APIs and rerank-3 in preview; ownership changed in February 2025.
Cost / TCO	4	Transparent, low per-token pricing with a large free allowance.
Lock-in / portability	2	Proprietary API (except nano) on standard interfaces would be 3; reduced by 1 for the 2025 acquisition.
Total (generic / FS)	3.40 / 3.05	

Evidence rules applied: NPV cap on enterprise_readiness lifted at CP2 rework (8 October 2026) under rule 7 (CP2 Q2): Atlas organisation and project roles govern model API keys (B-REV-S028); SSO and audit logging for key use not stated, so 3; security_compliance capped at 2 (kept): certifications only Reported from a homepage listing; MongoDB's SOC 2 scope page does not name Voyage and excludes preview features (B-REV-S027)

Capabilities. Voyage AI by MongoDB: Voyage 4 family (voyage-4-large, voyage-4, voyage-4-lite, open-weight voyage-4-nano) in one shared embedding space; voyage-code-4, voyage-context-4 (contextualised chunks), voyage-multimodal-3.5, domain models voyage-finance-2 and voyage-law-2; rerank-3 and rerank-3-lite (30 September 2026) with 32K-token context ^{vi250}. rerank-3 is listed as Preview on MongoDB's model lifecycle page ^{vi251}.

Strengths

- Broadest lineup against this layer's research questions: domain (finance), contextual, code, multimodal, quantisation including binary, and instruction-following rerankers ^{vi252}
- Shared Voyage 4 space lets a team index with a large model and query with a small one without re-embedding ^{vi253}
- Low published prices: voyage-4 US\$0.06 per 1M tokens with 200M free tokens per model; rerank-3 US\$0.05, rerank-3-lite US\$0.02 per 1M tokens (as of 7 October 2026) ^{vi254}

²⁵⁰ Verified fact — A2-S006: The Voyage 4 model family: shared embedding space with MoE architecture (Voyage AI (MongoDB)) <https://blog.voyageai.com/2026/01/15/voyage-4/> accessed 2026-10-07 [search-tool extract]; A2-S007: Voyage AI Embedding and Reranking Models Overview - Voyage AI by MongoDB <https://www.mongodb.com/docs/voyageai/models/> accessed 2026-10-07 [search-tool extract]; A2-S034: rerank-3 and rerank-3-lite: the next generation of Voyage rerankers (Voyage AI (MongoDB)) <https://blog.voyageai.com/2026/09/30/rerank-3/> accessed 2026-10-07 [search-tool extract]

²⁵¹ Verified fact — V1-S024: rerank-3 and rerank-3-lite: the next generation of Voyage rerankers; MongoDB Voyage model lifecycle page (Voyage AI by MongoDB) <https://blog.voyageai.com/2026/09/30/rerank-3/> accessed 2026-10-08 [search-tool extract]

²⁵² Verified fact — A2-S006: The Voyage 4 model family: shared embedding space with MoE architecture (Voyage AI (MongoDB)) <https://blog.voyageai.com/2026/01/15/voyage-4/> accessed 2026-10-07 [search-tool extract]; A2-S007: Voyage AI Embedding and Reranking Models Overview - Voyage AI by MongoDB <https://www.mongodb.com/docs/voyageai/models/> accessed 2026-10-07 [search-tool extract]; A2-S034: rerank-3 and rerank-3-lite: the next generation of Voyage rerankers (Voyage AI (MongoDB)) <https://blog.voyageai.com/2026/09/30/rerank-3/> accessed 2026-10-07 [search-tool extract]

²⁵³ Verified fact — A2-S006: The Voyage 4 model family: shared embedding space with MoE architecture (Voyage AI (MongoDB)) <https://blog.voyageai.com/2026/01/15/voyage-4/> accessed 2026-10-07 [search-tool extract]

²⁵⁴ Verified fact — A2-S007: Voyage AI Embedding and Reranking Models Overview - Voyage AI by MongoDB <https://www.mongodb.com/docs/voyageai/models/> accessed 2026-10-07 [search-tool extract]; A2-S034: rerank-3 and rerank-3-lite: the next generation of Voyage rerankers (Voyage AI (MongoDB)) <https://blog.voyageai.com/2026/09/30/rerank-3/> accessed 2026-10-07 [search-tool extract]

- In-VPC route via the AWS Marketplace model package in the customer's account ²⁵⁵

Limitations and risks

- Acquired by MongoDB (closed 17 February 2025) ²⁵⁶; roadmap now tied to MongoDB ^{AI}
- Certifications rest on a homepage listing; trust-portal report scope not verified ²⁵⁷; MongoDB points standalone Voyage to a separate trust portal ²⁵⁸
- rerank-3 is Preview on MongoDB's lifecycle page; Atlas Embedding and Reranking API and Automated Embedding are public preview ²⁵⁹
- Only voyage-4-nano can be self-hosted ²⁶⁰
- Vendor claims that rerank-3 beats competitors are not decision inputs ²⁶¹
- Atlas roles govern who can create, edit or delete model API keys ²⁶², but SSO and audit logging of key use are not stated ^{NPV}

Choose when

- MongoDB is the L6 store, or the firm accepts MongoDB as a strategic supplier
- Finance-domain vocabulary and contextualised chunks matter, subject to in-domain evaluation
- EU processing via the Atlas Europe Geography is acceptable

Avoid when

- UK processing or on-premises hosting of the main models is mandatory
- Independence between the retrieval-store vendor and the embedding vendor is a concentration requirement

Nearest competitors: L7-cohere, L7-jina, L7-qwen3-embedding, L7-openai

²⁵⁵ Verified fact — A2-S044: MongoDB Voyage AI Models in AWS (Marketplace model package) (Voyage AI (MongoDB)) <https://docs.voyageai.com/docs/aws-marketplace-model-package> accessed 2026-10-07 [search-tool extract]

²⁵⁶ Verified fact — A2-S033: MongoDB Announces Acquisition of Voyage AI to Enable Organizations to Build Trustworthy AI Applications <https://investors.mongodb.com/news-releases/news-release-details/mongodb-announces-acquisition-voyage-ai-enable-organizations> accessed 2026-10-07 [search-tool extract]; V1-S023: MongoDB, Inc. Form 10-K FY2026 (Voyage AI acquisition note) and IR release 24 Feb 2025 (MongoDB, Inc. (SEC EDGAR)) <https://www.sec.gov/Archives/edgar/data/1441816/000162828026016799/mdb-20260131.htm> accessed 2026-10-08 [search-tool extract]

²⁵⁷ Reported — A2-S044: MongoDB Voyage AI Models in AWS (Marketplace model package) (Voyage AI (MongoDB)) <https://docs.voyageai.com/docs/aws-marketplace-model-package> accessed 2026-10-07 [search-tool extract]

²⁵⁸ Verified fact — A2-S138: MongoDB Atlas Trust Center <https://www.mongodb.com/products/platform/trust> accessed 2026-10-07 [search-tool extract]

²⁵⁹ Verified fact — V1-S024: rerank-3 and rerank-3-lite: the next generation of Voyage rerankers; MongoDB Voyage model lifecycle page (Voyage AI by MongoDB) <https://blog.voyageai.com/2026/09/30/rerank-3/> accessed 2026-10-08 [search-tool extract]; A2-S077: Introducing the Embedding and Reranking API on MongoDB Atlas <https://www.mongodb.com/company/blog/product-release-announcements/introducing-the-embedding-and-reranking-api-on-mongodb-atlas> accessed 2026-10-07 [search-tool extract]; A2-S078: Unlocking AI Search: Introducing Automated Embedding in MongoDB Vector Search <https://www.mongodb.com/company/blog/product-release-announcements/unlocking-ai-search-introducing-automated-embedding-in-mongodb-vector-search> accessed 2026-10-07 [search-tool extract]

²⁶⁰ Verified fact — A2-S071: Embeddings - Claude Platform Docs (Anthropic) <https://docs.claude.com/en/docs/build-with-claude/embeddings> accessed 2026-10-07

²⁶¹ Reported — A2-S141: Context Engineering Has a Retrieval Issue: Introducing rerank-3 and Native Reranking (MongoDB) <https://www.mongodb.com/company/blog/product-release-announcements/introducing-rerank-3-and-native-rerank> accessed 2026-10-07 [search-tool extract]

²⁶² Verified fact — B-REV-S028: Manage Voyage AI Model API Keys (and Organization and Project Access) (MongoDB, Inc.) <https://www.mongodb.com/docs/voyageai/management/api-keys/> accessed 2026-10-08 [search-tool extract]

Regulated-FS note. The Atlas Embedding and Reranking API offers an EEA Geography with no fallback outside it, at a 10% premium ^{vf263}; Automated Embedding and Native Reranking do not yet support Geography targeting ^{vf264}. Request the Voyage SOC 2 report and its scope before use with client data ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Voyage AI, acquired by MongoDB, Inc. (closed 17 February 2025)	Verified fact	A2-S033
Category	Embedding and reranking models (hosted API, marketplace model packages, one open-weight model)	Verified fact	A2-S007, A2-S044
Version / lineup	Voyage 4 (launched 15 January 2026): voyage-4-large, voyage-4, voyage-4-lite, voyage-4-nano (open weights); plus voyage-code-4, voyage-context-4 (contextualised chunk embeddings), voyage-multimodal-3.5 (text, image, video); domain models voyage-finance-2 and voyage-law-2. Rerankers: rerank-3 and rerank-3-lite (announced 30 September 2026; MongoDB model-lifecycle page lists them as Preview), rerank-2.5 series retained for existing users.	Verified fact	A2-S006, A2-S007, A2-S034, A2-S071, V1-S024
Licence	Proprietary hosted models; voyage-4-nano is open-weight under Apache 2.0	Verified fact	A2-S006, A2-S071
Status events	17 February 2025: acquired by MongoDB (announced 24 February 2025); August 2025: rerank-2.5 / rerank-2.5-lite with instruction following; 15 January 2026: Voyage 4 family launched; Voyage 3.x becomes previous generation; 30 September 2026: rerank-3 / rerank-3-lite launched; rerank-2.5 becomes previous generation	Verified fact	A2-S033, A2-S006, A2-S034, V1-S023, V1-S024
Strategic direction	Being folded into MongoDB: Embedding and Reranking API on Atlas (public preview) and Automated Embedding in MongoDB Vector Search (public preview on Community Edition; Atlas 'coming soon' as of January 2026). Voyage 4 models share one embedding space so documents and queries can use different model sizes.	Verified fact	A2-S008, A2-S077, A2-S078, A2-S006
What it does	General-purpose, code, domain-specific, contextualised and multimodal embedding models plus instruction-following rerankers (32K-token context). voyage-4: 32,000-token context, 1,024 default dims (256/512/2,048 options), multiple quantisation options including binary.	Verified fact	A2-S007, A2-S006, A2-S034

²⁶³ Verified fact — A2-S142: Europe Data Residency for Atlas Embedding and Reranking (MongoDB) <https://www.mongodb.com/company/blog/product-release-announcements/europe-data-residency-for-atlas-embedding-reranking> accessed 2026-10-07 [search-tool extract]

²⁶⁴ Verified fact — A2-S142: Europe Data Residency for Atlas Embedding and Reranking (MongoDB) <https://www.mongodb.com/company/blog/product-release-announcements/europe-data-residency-for-atlas-embedding-reranking> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
Stack position	L7, covering both embedding and reranking; increasingly also embedded inside L6 (MongoDB Vector Search automated embedding).	Architectural judgment	
Integration	REST API and SDKs; Atlas Embedding and Reranking API; AWS and Azure Marketplace; automated embedding inside MongoDB Vector Search	Verified fact	A2-S033, A2-S077, A2-S078
Dependencies	Voyage/MongoDB hosted API, or AWS SageMaker (GPU instances such as ml.g5.xlarge) for marketplace model packages; voyage-4-nano runs anywhere (Hugging Face weights)	Verified fact	A2-S044, A2-S071
Certifications	SOC 2 and HIPAA listed in Voyage AI homepage/terms; MongoDB Trust Portal refers standalone Voyage to a separate VoyageAI Trust Portal (report scope not verified)	Reported	A2-S044, A2-S143, A2-S138
GDPR / residency	Atlas Embedding and Reranking API: Europe Geography (EEA regions) and US Geography processing boundaries, no fallback outside the Geography, 10% price premium (announced 1 September 2026, public preview at launch); Voyage-hosted API: opt-out of training gives zero-day retention	Verified fact	A2-S142, A2-S143
Security features	In-VPC marketplace deployment keeps data flow and API access in the customer's account (AWS acts as sub-processor)	Verified fact	A2-S044
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	voyage-4 US\$0.06 per 1M tokens, first 200M tokens free per model; rerank-3 US\$0.05 and rerank-3-lite US\$0.02 per 1M tokens	Verified fact	A2-S007, A2-S034
Infrastructure cost	Marketplace self-deployment requires SageMaker GPU instances (e.g. ml.g5.xlarge); default quotas are often zero	Verified fact	A2-S044
Ecosystem	Recommended in Anthropic's Claude embeddings documentation; available via AWS and Azure marketplaces; native in MongoDB	Verified fact	A2-S071, A2-S033, A2-S078
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Frequent model generations (rerank-2.5 Aug 2025, Voyage 4 Jan 2026, rerank-3 Sep 2026); MongoDB-integrated APIs still in public preview	Verified fact	A2-S034, A2-S006, A2-S077
Deployment	saas: Yes; managed_cloud: Yes (AWS Marketplace, Azure Marketplace, MongoDB Atlas); vpc_byoc: Yes (model package deployed in the		

Field	Value	Label	Sources
	customer's AWS account and VPC via SageMaker); private_cloud: Not publicly verified; self_hosted: Partial (voyage-4-nano open weights only); on_prem: Not publicly verified		

NVIDIA NeMo Retriever embedding and reranking NIM microservices (Nemotron embedding/reranking models) (L7-nvidia-nemo-retriever)

Tier: Tactical · **Flags:** Renamed · **Original graphic label:** NVIDIA – Embed

Rationale: Sound self-hosted runtime for NVIDIA-standardised estates; licence cost and GPU coupling make it a poor neutral default.

Criterion	Score	Rationale
Technical	3	Embed and rerank including multimodal; shorter context and fixed dimensions; Matryoshka not evidenced.
Enterprise readiness	3	Commercial subscription with Helm/Kubernetes deployment; inherits host IAM; support terms NPV.
Security and compliance	3	Self-hosted container (certifications not applicable); licence clear; container hardening evidence NPV.
Deployment flexibility	4	Any cloud or data centre with supported GPUs, NVIDIA API Catalog for development; air-gap NPV.
Ecosystem	3	NeMo Retriever pipeline and Helm chart; wider ecosystem NPV.
Reliability and maturity	3	Frequent NIM releases; inconsistent version dating across pages.
Cost / TCO	2	Per-GPU licence on top of GPU infrastructure.
Lock-in / portability	2	Tied to NVIDIA GPUs and the AI Enterprise licence; per-model licences vary.
Total (generic / FS)	3.00	
	/	
	2.95	

Evidence rules applied: NPV cap not applied to security_compliance: self-hosted container, vendor certifications not applicable; scored on in-estate deployment and licence clarity; Rule 6 (CP2 Q1) not applied at CP2 rework: no hyperscaler-consumed route is in the fact base; scores kept

Capabilities. NVIDIA NeMo Retriever embedding and reranking NIM microservices: Embedding NIM 2.3 (nemotron-3-embed-1b added in 2.2; llama-nemotron-embed-1b-v2; llama-nemotron-embed-vl-1b-v2) and Reranking NIM 2.0.0 (llama-nemotron-rerank-vl-1b-v2, rerank-1b-v2, rerank-500m-v2), deployed by Helm on Kubernetes or Docker ²⁶⁵.

Strengths

- Embed and rerank, text and multimodal, as GPU-optimised containers that run in any cloud or data centre with supported GPUs ²⁶⁶
- Client data stays in the firm's environment when self-hosted ^{AJ}

²⁶⁵ Verified fact — A2-S030: Release Notes for NVIDIA NeMo Retriever Reranking NIM <https://docs.nvidia.com/nim/nemo-retriever/text-reranking/latest/release-notes.html> accessed 2026-10-07 [search-tool extract]; A2-S031: Get Started With NVIDIA NeMo Retriever Embedding NIM (2.3) <https://docs.nvidia.com/nim/nemo-retriever/embedding/2.3/getting-started.html> accessed 2026-10-07 [search-tool extract]; V1-S094: NeMo Retriever Embedding NIM release notes / overview / support matrix (NVIDIA) <https://docs.nvidia.com/nim/nemo-retriever/embedding/2.2/release-notes.html> accessed 2026-10-08 [search-tool extract]

- Fits estates that already standardise on NVIDIA AI Enterprise for L2 serving ^{AJ}

Limitations and risks

- Production use requires NVIDIA AI Enterprise, from US\$4,500 per GPU per year or about US\$1 per GPU per hour in cloud (as of 7 October 2026) ^{VF267}
- Two-layer licensing (container agreement plus per-model licences that vary) ^{VF268}
- Shorter limits than API peers: nemotron-3-embed-1b is text-only, fixed 2,048 dims, 4,096-token maximum ^{VF269}
- Helm chart warns that a text-only reranker silently degrades multimodal reranking ^{VF270}
- Release dates inconsistent across NGC and Hugging Face pages; container security features NPV

Choose when

- The firm already licenses NVIDIA AI Enterprise and runs GPU Kubernetes
- A vendor-supported, self-hosted embed-plus-rerank runtime is required

Avoid when

- Retrieval volumes do not justify per-GPU licences
- The firm wants to avoid deepening GPU-vendor concentration

Nearest competitors: L7-sentence-transformers, L7-qwen3-embedding, L7-cohere

Regulated-FS note. Self-hosting removes cross-border transfer questions for the retrieval path ^{AJ}; the AI Enterprise subscription becomes the third-party arrangement to register ^{AJ}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	NVIDIA	Verified fact	A2-S030
Category	Containerised embedding and reranking inference microservices (NIM) with NVIDIA models	Verified fact	A2-S031, A2-S040
Version / lineup	Embedding NIM 2.3: nemotron-3-embed-1b, llama-nemotron-embed-1b-v2, llama-nemotron-embed-vl-1b-v2 (multimodal). Reranking NIM 2.0.0: llama-nemotron-rerank-vl-1b-v2 (multimodal, NGC update 3 August 2026), llama-nemotron-rerank-1b-v2, llama-nemotron-rerank-	Verified fact	A2-S031, A2-S030

²⁶⁶ Verified fact — A2-S031: Get Started With NVIDIA NeMo Retriever Embedding NIM (2.3) <https://docs.nvidia.com/nim/nemo-retriever/embedding/2.3/getting-started.html> accessed 2026-10-07 [search-tool extract]; A2-S040: General NIM FAQ (NVIDIA) <https://docs.api.nvidia.com/nim/docs/product> accessed 2026-10-07 [search-tool extract]

²⁶⁷ Verified fact — A2-S040: General NIM FAQ (NVIDIA) <https://docs.api.nvidia.com/nim/docs/product> accessed 2026-10-07 [search-tool extract]

²⁶⁸ Verified fact — A2-S041: Governing Terms for NVIDIA NeMo Retriever Embedding NIM <https://docs.nvidia.com/nim/nemo-retriever/embedding/2.3/eula.html> accessed 2026-10-07 [search-tool extract]

²⁶⁹ Verified fact — V1-S094: NeMo Retriever Embedding NIM release notes / overview / support matrix (NVIDIA) <https://docs.nvidia.com/nim/nemo-retriever/embedding/2.2/release-notes.html> accessed 2026-10-08 [search-tool extract]; A2-S031: Get Started With NVIDIA NeMo Retriever Embedding NIM (2.3) <https://docs.nvidia.com/nim/nemo-retriever/embedding/2.3/getting-started.html> accessed 2026-10-07 [search-tool extract]

²⁷⁰ Verified fact — A2-S031: Get Started With NVIDIA NeMo Retriever Embedding NIM (2.3) <https://docs.nvidia.com/nim/nemo-retriever/embedding/2.3/getting-started.html> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
	500m-v2 (8,192-token max).		
Licence	Proprietary container (NVIDIA Software License Agreement + AI product terms); models under NVIDIA Open Model License (varies by model); production use requires NVIDIA AI Enterprise	Verified fact	A2-S041, A2-S040
Status events	Reranking NIM 1.11.0: multimodal reranker llama-nemotron-rerank-vl-1b-v2 introduced; Reranking NIM 2.0.0: major runtime upgrade; Embedding NIM 2.2: nemotron-3-embed-1b added (text-only, fixed 2,048 dims, 4,096-token max); NIM 2.3 current	Verified fact	A2-S030, A2-S031, V1-S094
Strategic direction	Moving to multimodal (text + image) embedding and reranking as defaults in the NeMo Retriever Helm chart	Verified fact	A2-S031, A2-S030
What it does	GPU-optimised microservices that serve NVIDIA embedding and reranking models behind an API, deployable via Helm on Kubernetes or Docker, as part of the NeMo Retriever RAG pipeline.	Verified fact	A2-S031, A2-S030
Stack position	L7 runtime + models (embed and rerank) - straddles L7 and L2 (serving). Graphic label 'Embed' understates the reranking component.	Architectural judgement	
Integration	NIM REST APIs; NVIDIA API catalog Retrieval APIs for hosted trial; Helm chart	Verified fact	A2-S031
Dependencies	NVIDIA GPUs; NGC container registry; Kubernetes/Helm for the NeMo Retriever chart	Verified fact	A2-S031
Certifications	Not applicable to self-hosted containers; NVIDIA-hosted service certifications not researched	Architectural judgement	
GDPR / residency	Runs in customer environment when self-hosted	Architectural judgement	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Free for research/development/test via NVIDIA Developer Program (up to 16 GPUs); production requires NVIDIA AI Enterprise from US\$4,500 per GPU per year or ~US\$1 per GPU per hour in cloud; 90-day trial	Verified fact	A2-S040

Field	Value	Label	Sources
Infrastructure cost	GPU-bound: licence is per GPU plus GPU infrastructure	Verified fact	A2-S040
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Frequent releases (NIM 2.x; Helm 26.08.1 recommended)	Verified fact	A2-S031
Deployment	saas: Yes for development (NVIDIA API catalog); partner-hosted serverless NIM for production; managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes (any cloud or data centre with supported GPUs); on_prem: Yes		

Not publicly verified (L7-ethicalagents)

Tier: Not scored · **Flags:** Not publicly verified · **Original graphic label:** EthicalAgents – embeddings

Rationale: Could not be verified; removed per CP1 Q4(a).

Capabilities. Not publicly verified: no product of this name found after six searches by Stage A and a fresh search by the verifier ^{vf271}.

Limitations and risks

- No evidence the product exists ^{NPV}

Avoid when

- Do not select: removed per CP1 decision Q4(a)

Regulated-FS note. Removed from the analysis per CP1 decision Q4(a).

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Not publicly verified	Not publicly verified	
Category	Not publicly verified	Not	

²⁷¹ Verified fact — A2-S079: Privacy Policy - Ethical Agentic AI (Ethical Agentic AI Inc.) <https://www.ethicalagentic.ai/privacy-policy> accessed 2026-10-07 [search-tool extract]; V1-S036: Null-result searches for 'EthicalAgents' embeddings and 'Ragoos' reranker (V1 re-check) (Web search tool (V1)) <search-tool-null-result (no URL)> accessed 2026-10-08 [search-tool extract]

Field	Value	Label	Sources
		publicly verified	
Version / lineup	Not publicly verified	Not publicly verified	
Licence	Not publicly verified	Not publicly verified	
Strategic direction	Not publicly verified	Not publicly verified	
What it does	Not publicly verified	Not publicly verified	
Stack position	Not publicly verified	Not publicly verified	
Integration	Not publicly verified	Not publicly verified	
Dependencies	Not publicly verified	Not publicly verified	
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Not publicly verified	Not publicly verified	
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Not publicly verified	Not publicly verified	
Deployment	saas: Not publicly verified; managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Not publicly verified; on_prem: Not publicly verified		

Not publicly verified (L7 - ragoos)

Tier: Not scored · **Flags:** Not publicly verified · **Original graphic label:** Ragoos – RAG re-rankers

Rationale: Could not be verified; removed per CP1 Q4(a).

Capabilities. Not publicly verified: no product of this name found after six searches by Stage A and a fresh search by the verifier ²⁷².

Limitations and risks

- No evidence the product exists ^{NPV}

Avoid when

- Do not select: removed per CP1 decision Q4(a)

Regulated-FS note. Removed from the analysis per CP1 decision Q4(a).

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Not publicly verified	Not publicly verified	
Category	Not publicly verified	Not publicly verified	
Version / lineup	Not publicly verified	Not publicly verified	
Licence	Not publicly verified	Not publicly verified	

²⁷² Verified fact — A2-S080: Ragie <https://www.ragie.ai/> accessed 2026-10-07 [search-tool extract]; V1-S036: Null-result searches for 'EthicalAgents' embeddings and 'Ragoos' reranker (V1 re-check) (Web search tool (V1)) <search-tool-null-result (no URL)> accessed 2026-10-08 [search-tool extract]

Field	Value	Label	Sources
Strategic direction	Not publicly verified	Not publicly verified	
What it does	Not publicly verified	Not publicly verified	
Stack position	Not publicly verified	Not publicly verified	
Integration	Not publicly verified	Not publicly verified	
Dependencies	Not publicly verified	Not publicly verified	
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
Pricing	Not publicly verified	Not publicly verified	
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Not publicly verified	Not publicly verified	
Deployment	saas: Not publicly verified; managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Not publicly verified; on_prem: Not publicly verified		

L6: Retrieval and knowledge stores

Elasticsearch (Elastic Search AI Platform) (L6-elasticsearch)

Tier: Strategic · **Flags:** none · **Original graphic label:** Elasticsearch – hybrid search

Rationale: Strategic, conditional: where Elastic is already operated, because cost scores 2 ^{AI}.

Criterion	Score	Rationale
Technical	5	BM25, quantised vectors, RRF/linear fusion, restrictive-filter optimisation and document-level security cover the layer's research questions
Enterprise readiness	4	SAML/OIDC SSO, RBAC, DLS/FLS and audit logging (A2-S134; Elastic Cloud SSO/RBAC B-REV-S019), plus the Elastic Cloud API with role-bearing organisation

Criterion	Score	Rationale
		API keys (B-REVA-S003): rule 7 allows 4; SCIM not found, Cloud audit tier and SLA not confirmed
Security and compliance	5	Elastic Cloud ISO 27001 and SOC 2 Type II within the product, plus CMK
Deployment flexibility	4	Hosted and Serverless on three clouds, self-managed and on-prem; no BYOC evidenced
Ecosystem	5	De-facto search standard; REST; OpenSearch fork
Reliability and maturity	5	Years of production use; frequent minors
Cost / TCO	2	Paid tiers for key security features; cluster pricing not verified; operations heavy
Lock-in / portability	3	AGPL option but paid features proprietary; SSPL/ELv2 alternatives (rule 4)
Total (generic / FS)	4.30	
	/	
	4.25	

Capabilities. Distributed search engine combining BM25, dense/sparse vectors (BBQ/DiskBBQ default since 9.1), RRF and linear hybrid retrievers, semantic_text auto-embedding (Jina v5 default) and document-level security; Elastic 9.5 GA 4 August 2026 ^{vi273}.

Strengths

- Mature lexical plus vector and hybrid retrieval in one engine ^{vi274 AJ}
- Field- and document-level security maps to entitlement-aware retrieval ^{vi275}
- Elastic Cloud ISO 27001/27017/27018, SOC 2 Type II; CMK on Hosted Enterprise; Private Link incl. London ^{vi276}
- AGPLv3 option and OpenSearch fork as exit routes ^{vi277}

Limitations and risks

- Security features in paid tiers (Platinum/Enterprise) ^{vi278}
- Cluster pricing not verified; heavy operations when self-managed ^{NPV AJ}

²⁷³ Verified fact — A2-S133: Elastic 9.5: Columnar, VectorDB index mode & auto-calibration, and AI-driven alert triage <https://www.elastic.co/blog/whats-new-elastic-9-5-0> accessed 2026-10-07 [search-tool extract]; A2-S134: Use a customer-managed encryption key - Elastic Cloud <https://www.elastic.co/docs/deploy-manage/security/encrypt-deployment-with-customer-managed-encryption-key> accessed 2026-10-07 [search-tool extract]; V1-S085: Elastic 9.5: Columnar, VectorDB index mode & auto-calibration; Elastic community newsletter August 2026 <https://www.elastic.co/blog/whats-new-elastic-9-5-0> accessed 2026-10-08 [search-tool extract]

²⁷⁴ Verified fact — A2-S133: Elastic 9.5: Columnar, VectorDB index mode & auto-calibration, and AI-driven alert triage <https://www.elastic.co/blog/whats-new-elastic-9-5-0> accessed 2026-10-07 [search-tool extract]

²⁷⁵ Verified fact — A2-S134: Use a customer-managed encryption key - Elastic Cloud <https://www.elastic.co/docs/deploy-manage/security/encrypt-deployment-with-customer-managed-encryption-key> accessed 2026-10-07 [search-tool extract]

²⁷⁶ Verified fact — A2-S045: Elastic Trust Center <https://www.elastic.co/trust> accessed 2026-10-07 [search-tool extract]; A2-S134: Use a customer-managed encryption key - Elastic Cloud <https://www.elastic.co/docs/deploy-manage/security/encrypt-deployment-with-customer-managed-encryption-key> accessed 2026-10-07 [search-tool extract]

²⁷⁷ Verified fact — A2-S132: Elasticsearch Is Open Source. Again! <https://www.elastic.co/blog/elasticsearch-is-open-source-again> accessed 2026-10-07 [search-tool extract]; A2-S136: Release Schedule and Maintenance Policy - OpenSearch (OpenSearch project (Linux Foundation)) <https://opensearch.org/releases/> accessed 2026-10-07 [search-tool extract]

²⁷⁸ Verified fact — A2-S134: Use a customer-managed encryption key - Elastic Cloud <https://www.elastic.co/docs/deploy-manage/security/encrypt-deployment-with-customer-managed-encryption-key> accessed 2026-10-07 [search-tool extract]

- Bundled Jina embedding couples L6 and L7 ^{VF279}
- VectorDB index mode only in technical preview ^{VF280}

Choose when

- Elastic or OpenSearch is already an operated platform, or the corpus is lexical-heavy ^{AJ}

Avoid when

- It would be introduced only for vectors with no search team ^{AJ}

Nearest competitors: L6-mongodb-atlas-vector-search, L6-weaviate, OpenSearch (no record)

Regulated-FS note. Use document-level security for entitlements; pin the semantic_text model; choose AGPL distribution only after legal review ^{REC}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Elastic N.V. (NYSE: ESTC)	Verified fact	A2-S023
Category	Search engine with full-text (BM25), vector and hybrid retrieval; general-purpose search platform	Verified fact	A2-S023, A2-S042
Version / lineup	Elastic Stack 9.5 (GA 4 August 2026; patch 9.5.4); Python client 9.5.1 (9 September 2026)	Verified fact	A2-S133, A2-S057, V1-S085
Licence	Source: triple licence - AGPLv3 (OSI open source, added 2024; core under AGPL since 8.16), SSPL 1.0 or Elastic License 2.0; paid features proprietary; clients Apache-2.0	Verified fact	A2-S132
Status events	9 October 2025: Elastic completes acquisition of Jina AI; June 2026: jina-embeddings-v5-omni on Elastic Inference Service	Verified fact	A2-S023, A2-S027, V1-S025
Strategic direction	'Search AI Company': acquired Jina AI for multimodal/multilingual embeddings and rerankers complementing ELSER; Elastic Inference Service hosts Jina v5-omni models	Verified fact	A2-S023, A2-S027, A2-S042
What it does	Distributed search engine combining BM25, dense/sparse vectors (BBQ/DiskBBQ quantisation, default since 9.1) and hybrid retrievers (RRF, linear with l2_norm), semantic_text fields with automatic chunking and embedding (default Jina v5), MMR diversification; 9.5 adds VectorDB index mode (preview) and multimodal semantic field (preview).	Verified fact	A2-S133, A2-S042, A2-S023
Stack position	L6 retrieval store with built-in L7 inference; a 'retrieval/knowledge store' rather than a pure vector DB.	Architectural judgement	
Integration	REST API, official language clients (e.g. Python 9.5.1), open inference	Verified fact	A2-S057,

²⁷⁹ Verified fact — A2-S023: Elastic Completes Acquisition of Jina AI, a Leader in Frontier Models for Multimodal and Multilingual Search <https://ir.elastic.co/News--Events/news/news-details/2025/Elastic-Completes-Acquisition-of-Jina-AI-a-Leader-in-Frontier-Models-for-Multimodal-and-Multilingual-Search/default.aspx> accessed 2026-10-07 [search-tool extract]

²⁸⁰ Verified fact — V1-S085: Elastic 9.5: Columnar, VectorDB index mode & auto-calibration; Elastic community newsletter August 2026 <https://www.elastic.co/blog/whats-new-elastic-9-5-0> accessed 2026-10-08 [search-tool extract]

Field	Value	Label	Sources
Dependencies	API, Elastic Inference Service	d fact	A2-S042
	Not publicly verified	Not publicly verified	
Certifications	Elastic Cloud: ISO 27001, ISO 27017, ISO 27018, SOC 2 Type II (report via Trust Center)	Verified fact	A2-S045
GDPR / residency	EU/UK hosted regions: AWS Frankfurt, Zurich, Stockholm, Milan, Ireland, London, Paris; GCP Finland, Belgium, London, Frankfurt, Netherlands, Paris; Azure Ireland, UK South, Netherlands; Serverless in AWS Frankfurt, Ireland, London and others	Verified fact	A2-S135
Security features	Encryption at rest by default; customer-managed keys (AWS KMS, Azure Key Vault, Google Cloud KMS) on Elastic Cloud Hosted with Enterprise subscription (Serverless BYOK on roadmap); AWS PrivateLink, Azure Private Link, GCP Private Service Connect incl. London	Verified fact	A2-S134
Access controls	SAML/OIDC SSO, RBAC, field/document-level security and audit logging (self-managed: Platinum tier); Elastic Cloud tier mapping not confirmed	Verified fact	A2-S134
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Elastic Inference Service billed per million tokens on Elastic Cloud (consumption-based); cluster pricing not verified	Verified fact	A2-S042
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Jina models and ELSER via Elastic Inference Service; OpenSearch fork as alternative (OpenSearch 3.9.0, 29 September 2026; opensearch-py 3.2.0, Apache-2.0)	Verified fact	A2-S023, A2-S136, A2-S058
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Mature; frequent minors (9.0-9.5 in ~16 months)	Verified fact	A2-S133
Deployment	saas: Yes (Elastic Cloud Hosted and Serverless); managed_cloud: Yes (AWS, GCP, Azure); vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes (self-managed; AGPL/SSPL/ELv2 source); on_prem: Yes		

Milvus (open source) and Zilliz Cloud (managed Milvus) (L6-milvus-zilliz)

Tier: Strategic · **Flags:** none · **Original graphic label:** Milvus – vector search

Rationale: Foundation-governed, permissively licensed engine with a certified managed and BYOC route; scale option behind Qdrant ^{AJ}.

Criterion	Score	Rationale
Technical	4	Dense/sparse, BM25, JSON indexing, lake-native storage; multi-tenancy and filtered performance not evidenced
Enterprise readiness	4	SSO, audit logs and 99.95% SLA on Enterprise Dedicated (A2-S120), plus organisation, project and cluster RBAC with custom roles and SCIM (B-REVA-S001): all three controls plus SLA and SCIM (rule 7)
Security and compliance	4	Combined record: Zilliz Cloud has product-scoped SOC 2 Type II and ISO 27001 plus CMEK (would reach 5 under rule 8), but the self-hosted Milvus route is scored on rule 2 hygiene, which is not evidenced; 4 reflects the recommended mix of self-host and BYOC
Deployment flexibility	5	SaaS on three clouds, BYOC, self-host, on-prem
Ecosystem	4	pymilvus; mainstream integrations
Reliability and maturity	4	LF AI & Data graduated; 2.6 maintained; 3.0 GA only ten weeks
Cost / TCO	3	OSS free but distributed operations heavy; Zilliz BYOC pricing unpublished
Lock-in / portability	5	Apache-2.0 under neutral foundation governance
Total (generic / FS)	4.10 / 4.25	

Capabilities. Apache-2.0 distributed vector database (LF AI & Data graduated) with dense/sparse vectors, BM25, JSON path indexing and online schema changes; Milvus 3.0 ‘lake-native’ GA 29 July 2026; Zilliz Cloud as managed and BYOC ‘Vector Lakebase’ ^{VF281}.

Strengths

- Foundation governance and Apache-2.0 ^{VF282}
- Zilliz Cloud SOC 2 Type II (incl. BYOC) and ISO 27001; CMEK on Business Critical ^{VF283}
- Enterprise Dedicated with SSO, audit logs and 99.95% SLA ^{VF284}

²⁸¹ Verified fact — A2-S118: Zilliz Announces Milvus 3.0, Making the World’s Most Adopted Open-Source Vector Database Lake-Native <https://zilliz.com/news/milvus-3-0-lake-native-vector-database> accessed 2026-10-07 [search-tool extract]; A2-S117: Release Notes - Milvus (Milvus project) https://milvus.io/docs/release_notes.md accessed 2026-10-07 [search-tool extract]; A2-S054: milvus-lite - PyPI (PyPI / Milvus contributors) <https://pypi.org/project/milvus-lite/> accessed 2026-10-07; A2-S119: Zilliz Trust Center <https://zilliz.com/trust-center> accessed 2026-10-07 [search-tool extract]

²⁸² Verified fact — A2-S118: Zilliz Announces Milvus 3.0, Making the World’s Most Adopted Open-Source Vector Database Lake-Native <https://zilliz.com/news/milvus-3-0-lake-native-vector-database> accessed 2026-10-07 [search-tool extract]

²⁸³ Verified fact — A2-S119: Zilliz Trust Center <https://zilliz.com/trust-center> accessed 2026-10-07 [search-tool extract]; A2-S120: Zilliz Cloud Pricing <https://zilliz.com/pricing> accessed 2026-10-07 [search-tool extract]

²⁸⁴ Verified fact — A2-S120: Zilliz Cloud Pricing <https://zilliz.com/pricing> accessed 2026-10-07 [search-tool extract]

- Same API from Lite to Distributed to Zilliz Cloud ^{VF285}
- Organisation, project and cluster RBAC with custom roles, IdP group mapping and SCIM on Zilliz Cloud ^{VF286}

Limitations and risks

- 3.0 not guaranteed compatible with 2.6 servers ^{VF287}
- Milvus Lite has no authentication, roles or TLS ^{VF288}
- Distributed Milvus is a heavy operational commitment ^{AJ}
- Zilliz last disclosed raise 2022 ^{VF289}

Choose when

- Very large corpora, open-source portability with managed or BYOC options, or vectors alongside lakehouse data ^{AJ}

Avoid when

- A small team must self-operate Distributed Milvus ^{AJ}
- 3.0 features are needed before the 3.x line matures ^{AJ}

Nearest competitors: L6-qdrant, L6-pinecone, L6-weaviate

Regulated-FS note. Stay on 2.6 or a validated 3.0.x until regression tests pass; Zilliz BYOC for residency; never expose Lite beyond a laptop ^{REC}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Milvus: open-source project, graduated LF AI & Data project; Zilliz (creator and main maintainer) offers Zilliz Cloud	Verified fact	A2-S118, A2-S053
Category	Distributed open-source vector database ('lake-native' from 3.0) plus managed 'Vector Lakebase' (Zilliz Cloud)	Verified fact	A2-S118
Version / lineup	Milvus 3.0.2 (20 September 2026); 3.0.0 GA 29 July 2026; 2.6.24 (16 September 2026) still maintained; PyMilvus 3.0.2; Milvus Lite 3.2.1. Zilliz Cloud: Milvus 3.0.x in Private Review for on-demand compute	Verified fact	A2-S117, A2-S118, A2-S053, A2-S054, V1-S031
Licence	Open source, Apache-2.0 (Milvus 3.0); Zilliz Cloud commercial	Verified fact	A2-S118
Status events	June 2025: Milvus 2.6 (BM25 full-text speedups, JSON path indexing); 16	Verified fact	A2-S117,

²⁸⁵ Verified fact — A2-S054: milvus-lite - PyPI (PyPI / Milvus contributors) <https://pypi.org/project/milvus-lite/> accessed 2026-10-07

²⁸⁶ Verified fact — B-REVA-S001: Access Control Explained (and Manage Platform Users) (Zilliz) <https://docs.zilliz.com/docs/access-control> accessed 2026-10-08 [search-tool extract]

²⁸⁷ Verified fact — A2-S117: Release Notes - Milvus (Milvus project) https://milvus.io/docs/release_notes.md accessed 2026-10-07 [search-tool extract]

²⁸⁸ Verified fact — A2-S054: milvus-lite - PyPI (PyPI / Milvus contributors) <https://pypi.org/project/milvus-lite/> accessed 2026-10-07

²⁸⁹ Verified fact — A2-S122: Vector Database Company Zilliz Raises \$60 Million Series B Extension <https://zilliz.com/news/vector-database-company-zilliz-series-b-extension> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
	July 2026: Milvus 3.0 announced; 29 July 2026: Milvus 3.0.0 GA (not guaranteed compatible with 2.6 servers); 20 September 2026: 3.0.2	d fact	A2-S118
Strategic direction	Repositioning as 'lake-native': indexes over vectors kept in object storage and open formats (Loon engine, Vortex columnar format), External Collections for lakehouse workflows; Zilliz Cloud as 'Vector Lakebase'	Verified fact	A2-S118, A2-S117
What it does	Vector database with dense/sparse vectors (SINDI sparse index), BM25 full-text, JSON path indexing, faceted search, online schema changes; tiers from Milvus Lite to Distributed and Zilliz Cloud.	Verified fact	A2-S117, A2-S054
Stack position	L6 dedicated vector database; placement correct.	Architectural judgement	
Integration	pymilvus SDK (gRPC); same API across Lite, Standalone, Distributed and Zilliz Cloud	Verified fact	A2-S054
Dependencies	Not publicly verified	Not publicly verified	
Certifications	Zilliz Cloud: SOC 2 Type II (scope incl. Free, Serverless, Dedicated, BYOC; report under NDA) and ISO/IEC 27001; GDPR-ready and HIPAA-ready (HIPAA-eligible on Business Critical)	Verified fact	A2-S119, A2-S120
GDPR / residency	EU regions: AWS Frankfurt and Ireland, GCP Frankfurt, Azure Germany West Central and North Europe (Ireland)	Verified fact	A2-S121
Security features	CMEK and HIPAA eligibility on Business Critical; private endpoints on Enterprise; BYOC shared-responsibility model (customer manages VPC and storage encryption)	Verified fact	A2-S120, A2-S119
Access controls	SSO and audit logs on Enterprise Dedicated	Verified fact	A2-S120
Enterprise support	Enterprise Dedicated with 99.95% uptime SLA	Verified fact	A2-S120
Pricing	Zilliz Cloud: Serverless from US\$0/month; Enterprise Dedicated from US\$197/month; storage standardised at US\$0.04/GB-month from 1 January 2026 (announced late 2025); BYOC not published. Milvus open source free.	Verified fact	A2-S120
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption	Zilliz total funding US\$113M (Series B extension US\$60M, August 2022);	Verified fact	A2-S122

Field	Value	Label	Sources
signals	no 2025-2026 round found	d fact	
Maturity	Mature 2.x line still patched; 3.0 GA July 2026 with breaking compatibility	Verified fact	A2-S117
Deployment	saas: Yes (Zilliz Cloud Serverless/Dedicated); managed_cloud: Yes (AWS, GCP, Azure); vpc_byoc: Yes (Zilliz BYOC: data plane in customer cloud account); private_cloud: Not publicly verified; self_hosted: Yes (Standalone, Distributed); on_prem: Yes		

PostgreSQL with the pgvector extension (L6-pgvector)

Tier: Strategic · **Flags:** none · **Original graphic label:** Postgres – + pgvector

Rationale: Lowest-increment, permissively licensed, transactionally integrated option on every hyperscaler; the default answer to ‘do we need a dedicated vector DB?’ ^{AI}.

Criterion	Score	Rationale
Technical	4	Dense+sparse, HNSW/IVFFlat, filter-aware scans and SQL joins; no native BM25 and scale limits keep it below 5
Enterprise readiness	4	Rule 2: inherits host controls; consumed via RDS/Aurora, Azure or Cloud SQL/AlloyDB so platform controls presumed (CP2 Q1); confirm per service
Security and compliance	3	Rule 2 hygiene: permissive licence and prompt CVE fixes, but two 2026 memory-safety CVEs (one CVSS 8.8) and host patch lag; signed releases not verified
Deployment flexibility	5	Self-host, on-prem and air-gap, plus managed services on AWS, Azure and Google Cloud
Ecosystem	5	SQL; supported by all three hyperscalers’ managed PostgreSQL and Bedrock Knowledge Bases via Aurora
Reliability and maturity	4	Active 0.8.x maintenance (five 2026 releases) on a long-established database; still pre-1.0 versioning
Cost / TCO	5	Free; cost is PostgreSQL compute and RAM already operated
Lock-in / portability	5	PostgreSQL License, SQL interface, multiple managed hosts

Criterion	Score	Rationale
Total (generic / FS)	4.25 / 4.20	

Evidence rules applied: Rule 2 (self-hosted software): security scored on project hygiene; enterprise readiness on host controls with CP2 Q1 presumption for managed hyperscaler PostgreSQL

Capabilities. PostgreSQL extension adding vector types (incl. sparsevec) and HNSW/IVFFlat ANN indexes, so vectors sit beside relational data and SQL filters; iterative index scans and filter-aware index selection since 0.8.0; 0.8.7 released 1 October 2026 ^{VF290}.

Strengths

- Transactional integration: vectors, metadata and entitlement tables in one ACID database ^{AJ}
- PostgreSQL License (permissive, OSI) ^{VF291}
- Managed on Amazon RDS/Aurora, Azure Database for PostgreSQL, Cloud SQL and AlloyDB ^{VF292}
- Security fixes shipped promptly in 2026 (0.8.2, 0.8.7) ^{VF293}

Limitations and risks

- CVE-2026-103484 (CVSS 8.8, IVFFlat build, possible code execution) fixed in 0.8.7; CVE-2026-3172 fixed in 0.8.2 ^{VF294}
- Managed hosts lag upstream (latest RDS release seen carries 0.8.2) and Azure manages the version ^{VF295}
- No BM25 ranking in the extension; hybrid needs sparsevec or app-side fusion ^{VF296 AJ}
- HNSW indexes are memory-hungry; large-scale filtered behaviour not evidenced ^{AJ NPV}

Choose when

²⁹⁰ Verified fact — A2-S063: pgvector CHANGELOG.md (pgvector project) <https://github.com/pgvector/pgvector/blob/master/CHANGELOG.md> accessed 2026-10-07 [search-tool extract]; A2-S061: pgvector 0.8.7 Released (PostgreSQL Global Development Group (news post by pgvector p) <https://www.postgresql.org/about/news/pgvector-087-released-3392/> accessed 2026-10-07 [search-tool extract]; V1-S020: pgvector CHANGELOG.md (pgvector (GitHub)) <https://raw.githubusercontent.com/pgvector/pgvector/master/CHANGELOG.md> accessed 2026-10-08; V1-S022: pgvector 0.8.7 Released (PostgreSQL Global Development Group) <https://www.postgresql.org/about/news/pgvector-087-released-3392/> accessed 2026-10-08 [search-tool extract]

²⁹¹ Verified fact — V1-S021: pgvector LICENSE (pgvector (GitHub)) <https://raw.githubusercontent.com/pgvector/pgvector/master/LICENSE> accessed 2026-10-08

²⁹² Verified fact — B-L6-S004: Extension versions for Amazon RDS for PostgreSQL; RDS PostgreSQL release notes; Aurora pgvector announcements (Amazon Web Services) <https://docs.aws.amazon.com/AmazonRDS/latest/PostgreSQLReleaseNotes/postgresql-extensions.html> accessed 2026-10-08 [search-tool extract]; B-L6-S005: Vector search in Azure Database for PostgreSQL flexible server; Cloud SQL vector embeddings; AlloyDB vector search overview (Microsoft; Google Cloud) <https://learn.microsoft.com/en-us/azure/postgresql/extensions/how-to-use-pgvector> accessed 2026-10-08 [search-tool extract]

²⁹³ Verified fact — V1-S020: pgvector CHANGELOG.md (pgvector (GitHub)) <https://raw.githubusercontent.com/pgvector/pgvector/master/CHANGELOG.md> accessed 2026-10-08; V1-S022: pgvector 0.8.7 Released (PostgreSQL Global Development Group) <https://www.postgresql.org/about/news/pgvector-087-released-3392/> accessed 2026-10-08 [search-tool extract]

²⁹⁴ Verified fact — V1-S022: pgvector 0.8.7 Released (PostgreSQL Global Development Group) <https://www.postgresql.org/about/news/pgvector-087-released-3392/> accessed 2026-10-08 [search-tool extract]; A2-S062: pgvector 0.8.2 Released (PostgreSQL Global Development Group (news post by pgvector p) <https://www.postgresql.org/about/news/pgvector-082-released-3245/> accessed 2026-10-07 [search-tool extract]

²⁹⁵ Verified fact — B-L6-S004: Extension versions for Amazon RDS for PostgreSQL; RDS PostgreSQL release notes; Aurora pgvector announcements (Amazon Web Services) <https://docs.aws.amazon.com/AmazonRDS/latest/PostgreSQLReleaseNotes/postgresql-extensions.html> accessed 2026-10-08 [search-tool extract]; B-L6-S005: Vector search in Azure Database for PostgreSQL flexible server; Cloud SQL vector embeddings; AlloyDB vector search overview (Microsoft; Google Cloud) <https://learn.microsoft.com/en-us/azure/postgresql/extensions/how-to-use-pgvector> accessed 2026-10-08 [search-tool extract]

²⁹⁶ Verified fact — A2-S063: pgvector CHANGELOG.md (pgvector project) <https://github.com/pgvector/pgvector/blob/master/CHANGELOG.md> accessed 2026-10-07 [search-tool extract]

- PostgreSQL is a firm standard and the corpus is millions, not billions, of chunks ^{AJ}
- Entitlements already live in relational tables ^{AJ}

Avoid when

- Thousands of isolated tenants with independent scaling are needed ^{AJ}
- Low-latency search over a very large corpus under selective filters ^{AJ}

Nearest competitors: L6-mongodb-atlas-vector-search, L6-elasticsearch, L6-qdrant

Regulated-FS note. Default for UK/EU managed PostgreSQL; track host pgvector version against CVEs; restrict index-creation rights; one schema per segregated mandate ^{Reg.}

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	pgvector open-source project (lead author Andrew Kane per Python client) on PostgreSQL	Verified fact	A2-S050, A2-S061
Category	Vector search extension for a general-purpose relational database	Verified fact	A2-S061
Version / lineup	pgvector 0.8.7 (released 1 October 2026 per CHANGELOG; announced on postgresql.org 5 October 2026); earlier 2026 releases 0.8.2 (Feb), 0.8.3 (17 Jun), 0.8.4 (30 Jun), 0.8.5 (8 Jul), 0.8.6 (29 Jul). Python client pgvector 0.5.0 (6 July 2026).	Verified fact	A2-S061, A2-S062, A2-S063, A2-S050, V1-S020, V1-S022
Licence	Open source: PostgreSQL License (permissive, OSI-approved) for the extension; pgvector-python is MIT	Verified fact	A2-S050, V1-S021
Status events	February 2026: 0.8.2 fixes CVE-2026-3172 (parallel HNSW build buffer overflow); 1 October 2026 (announced 5 October): 0.8.7 fixes an IVFFlat index-build buffer overflow (CVE-2026-103484, CVSS 8.8) that can allow arbitrary code execution by a user able to create IVFFlat indexes; Adjacent: pgvecto.rs deprecated in favour of VectorChord (early 2025); Timescale renamed Tiger Data (17 June 2025), pgvectorscale continues	Verified fact	A2-S062, A2-S061, A2-S066, A2-S067, A2-S064, A2-S065, V1-S020, V1-S022
Strategic	Maintenance and security fixes on the 0.8.x line in 2026 (HNSW	Verified	A2-S063,

Field	Value	Label	Sources
direction	vacuuming, IVFFlat memory); supports dense and sparse (sparsevec) vectors	d fact	A2-S061
What it does	Adds vector data types (including sparsevec) and HNSW and IVFFlat approximate-nearest-neighbour indexes to PostgreSQL, so vectors live alongside relational data and SQL filters.	Verified fact	A2-S063, A2-S061
Stack position	L6 retrieval store inside the operational/analytical database tier; strong fit where transactional integration and SQL filtering matter.	Architectural judgement	
Integration	SQL; client libraries such as pgvector-python	Verified fact	A2-S050
Dependencies	PostgreSQL server (Postgres 18 referenced in 2026 fixes)	Verified fact	A2-S063
Certifications	Depends on the hosting Postgres service; not applicable to the extension	Architectural judgement	
GDPR / residency	Depends on hosting choice	Architectural judgement	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Free, open source	Verified fact	A2-S050, A2-S061
Infrastructure cost	Cost is Postgres compute/RAM (HNSW indexes are memory-hungry) and storage; no separate vector service	Architectural judgement	
Ecosystem	Extensions: pgvector (Tiger Data; up to 16,000 dims) and VectorChord (successor to pgvector.rs)	Verified fact	A2-S064, A2-S066
Adoption signals	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
		verified	
Maturity	Active 0.8.x maintenance with five 2026 releases	Verified fact	A2-S061, A2-S062, A2-S063
Deployment	saas: Not publicly verified; managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes; on_prem: Yes		

Qdrant (open-source vector search engine; Qdrant Cloud, Hybrid Cloud, Private Cloud) (L6-qdrant)

Tier: Strategic · **Flags:** none · **Original graphic label:** Qdrant – open source

Rationale: Permissive licence, filter-aware multi-tenant engine and air-gap-capable deployment make it the default dedicated engine when a trigger is met ⁴.

Criterion	Score	Rationale
Technical	4	Dense/sparse, ACORN filtering, RRF, multitenancy, quantisation; BM25 not evidenced
Enterprise readiness	4	Cloud RBAC with custom roles, SSO (Premium add-on) and audit access logging (B-L6-S007), plus the Cloud Management API and a 99.9% (99.95% multi-AZ) SLA on Premium (B-REVA-S002): rule 7 allows 4; SCIM not verified
Security and compliance	3	SOC 2 Type II and HIPAA but no ISO 27001: maximum 3 in this layer
Deployment flexibility	5	SaaS, hybrid, private incl. air-gap, self-host
Ecosystem	4	REST/gRPC with OpenAPI, FastEmbed, three clouds
Reliability and maturity	3	Frequent releases but breaking storage migration at 1.16; vendor-governed

Criterion	Score	Rationale
Cost / TCO	4	Free OSS; managed from US\$25 (as captured)
Lock-in / portability	4	Apache-2.0 and self-hostable; single-vendor governance
Total (generic / FS)	3.90 / 3.85	

Capabilities. Apache-2.0 Rust vector search engine with dense and sparse vectors, payload filtering (ACORN), weighted RRF fusion, tiered multitenancy, quantisation and audit access logging; server 1.19.2 (5 October 2026)^{VF-297}.

Strengths

- Filter-aware search and tiered multitenancy suit entitlement-heavy, many-tenant retrieval^{VF-298 AJ}
- Managed, Hybrid (customer network) and Private Cloud incl. air-gapped, plus self-host^{VF-299}
- Cloud RBAC with custom roles, SSO add-on, collection-scoped JWT keys^{VF-300}
- Apache-2.0; US\$50M Series B March 2026^{VF-301}
- Cloud Management API with management keys; 99.9% uptime SLA on Premium (99.95% multi-AZ)^{VF-302}

Limitations and risks

- No ISO 27001 found; HIPAA BAA documented for Managed Cloud only^{VF-303}
- Upgrade cannot skip 1.16 (storage migration)^{VF-304}
- SSO is a Premium add-on^{VF-305}
- Managed pricing evidence thin and possibly stale^{VF-306}
- Management and database keys are not revoked when the creating user leaves; offboarding must revoke them^{VF-307}

²⁹⁷ Verified fact — A2-S108: qdrant/LICENSE (Qdrant) <https://github.com/qdrant/qdrant/blob/master/LICENSE> accessed 2026-10-07 [search-tool extract]; A2-S103: Qdrant 1.17 - Relevance Feedback & Search Latency Improvements <https://qdrant.tech/blog/qdrant-1.17.x/> accessed 2026-10-07 [search-tool extract]; A2-S102: Releases - qdrant/qdrant (Qdrant) <https://github.com/qdrant/qdrant/releases> accessed 2026-10-07 [search-tool extract]

²⁹⁸ Verified fact — A2-S103: Qdrant 1.17 - Relevance Feedback & Search Latency Improvements <https://qdrant.tech/blog/qdrant-1.17.x/> accessed 2026-10-07 [search-tool extract]

²⁹⁹ Verified fact — A2-S106: Qdrant Hybrid Cloud: Flexible Deployment, Data Privacy, and Cost Efficiency <https://qdrant.tech/hybrid-cloud/> accessed 2026-10-07 [search-tool extract]

³⁰⁰ Verified fact — B-L6-S007: Qdrant Cloud RBAC role management; enterprise SSO announcement; database security (JWT RBAC) <https://qdrant.tech/documentation/cloud-rbac/role-management/> accessed 2026-10-08 [search-tool extract]

³⁰¹ Verified fact — A2-S108: qdrant/LICENSE (Qdrant) <https://github.com/qdrant/qdrant/blob/master/LICENSE> accessed 2026-10-07 [search-tool extract]; A2-S107: We Raised \$50M to Build Composable Vector Search as Core Infrastructure (Qdrant) <https://qdrant.tech/blog/series-b-announcement/> accessed 2026-10-07 [search-tool extract]

³⁰² Verified fact — B-REVA-S002: Qdrant Cloud API (and Qdrant pricing page) <https://qdrant.tech/documentation/cloud-api/> accessed 2026-10-08 [search-tool extract]

³⁰³ Verified fact — V1-S069: Qdrant Achieves SOC 2 Type II and HIPAA Certifications; Qdrant Security; Cloud Security docs <https://qdrant.tech/blog/soc-2-type-ii-hipaa/> accessed 2026-10-08 [search-tool extract]

³⁰⁴ Verified fact — A2-S102: Releases - qdrant/qdrant (Qdrant) <https://github.com/qdrant/qdrant/releases> accessed 2026-10-07 [search-tool extract]

³⁰⁵ Verified fact — B-L6-S007: Qdrant Cloud RBAC role management; enterprise SSO announcement; database security (JWT RBAC) <https://qdrant.tech/documentation/cloud-rbac/role-management/> accessed 2026-10-08 [search-tool extract]

³⁰⁶ Verified fact — A2-S106: Qdrant Hybrid Cloud: Flexible Deployment, Data Privacy, and Cost Efficiency <https://qdrant.tech/hybrid-cloud/> accessed 2026-10-07 [search-tool extract]

Choose when

- A dedicated engine is needed in your own estate or cloud account, with strong filtering and tenant isolation ^{AJ}

Avoid when

- ISO 27001 is a hard gate for a vendor-operated service ^{AJ}

Nearest competitors: L6-milvus-zilliz, L6-weaviate, L6-pinecone

Regulated-FS note. Self-host or Hybrid Cloud for client data; collection-scoped keys per service; enable audit access logging ^{REC}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Qdrant (Qdrant Solutions GmbH, Berlin-headquartered)	Verified fact	A2-S105, A2-S107
Category	Open-source vector search engine with managed, hybrid and private cloud offerings	Verified fact	A2-S107, A2-S106
Version / lineup	Server v1.19.2 (5 October 2026; 1.19.0 on 5 August 2026); Private Cloud validated 1.19.0 (4 September 2026); Python client 1.19.1 (16 September 2026)	Verified fact	A2-S102, A2-S104, A2-S052
Licence	Open source, Apache-2.0 (server and client); managed/hybrid/private cloud commercial	Verified fact	A2-S108, A2-S052
Status events	1.16: RocksDB to Gridstore storage migration (cannot be skipped on upgrade); 12 March 2026: US\$50M Series B led by AVP; 27 March 2026: 1.17.1; 17 July 2026: 1.18.3; 5 August 2026: 1.19.0; 5 October 2026: 1.19.2	Verified fact	A2-S102, A2-S103, A2-S107
Strategic direction	'Composable vector search as core infrastructure' from agentic workflows to edge devices (Series B, March 2026); recent releases add relevance-feedback queries, weighted RRF, audit access logging, tiered multitenancy, TurboQuant quantisation and low-memory mode	Verified fact	A2-S107, A2-S103, A2-S102
What it does	Rust vector search engine with dense and sparse vectors, payload	Verified fact	A2-S103,

³⁰⁷ Verified fact — B-REVA-S002: Qdrant Cloud API (and Qdrant pricing page) <https://qdrant.tech/documentation/cloud-api/> accessed 2026-10-08 [search-tool extract]

Field	Value	Label	Sources
	filtering (ACORN filtered search), hybrid fusion (weighted RRF), multitenancy, quantisation; client local mode and FastEmbed; Qdrant Cloud server-side inference on paid plans.	d fact	A2-S102, A2-S052
Stack position	L6 dedicated vector search engine; placement correct.	Architectural judgement	
Integration	REST/gRPC API (OpenAPI), Python client with sync/async, FastEmbed (ONNX) embeddings, Qdrant Cloud inference	Verified fact	A2-S052
Dependencies	Self-hosted Docker/Kubernetes, or Qdrant Cloud on AWS, GCP, Azure	Verified fact	A2-S052, A2-S105
Certifications	SOC 2 Type II (renewed over 12-month period); HIPAA compliance with BAA (Managed Cloud / on request); ISO 27001 not found	Verified fact	A2-S105, V1-S069
GDPR / residency	DPA for GDPR; EU customers can keep data in EU regions exclusively; Cloud Inference runs in the EU for EU-region clusters	Verified fact	A2-S105
Security features	Paid managed clusters on dedicated resources; Hybrid/Private Cloud keep data in customer environment; v1.19.2 warns at startup when no API keys configured	Verified fact	A2-S106, A2-S102
Access controls	Audit access logging added in 1.17; SSO/RBAC for Qdrant Cloud not verified	Verified fact	A2-S103
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Managed Cloud priced on vCPU, RAM, disk and inference tokens, 'from US\$25'; Hybrid Cloud figure US\$0.014 per hour (unit unclear); Private Cloud and Enterprise on request	Verified fact	A2-S106
Infrastructure cost	Self-hosting cost driven by RAM (or disk with on-disk/inline storage) and CPU; quantisation reduces memory	Architectural judgement	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	US\$50M Series B (12 March 2026; AVP, Bosch Ventures, Unusual Ventures, Spark Capital, 42CAP); total funding US\$87.8M (TechTarget); earlier US\$28M Series A and US\$7.5M seed; Canva quoted as customer	Verified fact	A2-S107, V1-S032
Maturity	Monthly-ish minor releases (1.16-1.19 in about a year); breaking storage migration at 1.16	Verified fact	A2-S102
Deployment	saas: Yes (Qdrant Managed Cloud); managed_cloud: Yes (AWS, GCP, Azure); vpc_byoc: Yes (Hybrid Cloud: clusters in customer network, managed via Qdrant Cloud UI; Enterprise plan); private_cloud: Yes (Private Cloud, incl. air-gapped); self_hosted: Yes; on_prem: Yes		

Pinecone (vector database; Pinecone Nexus knowledge engine) (L6-pinecone)

Tier: Tactical · **Flags:** none · **Original graphic label:** Pinecone – managed

Rationale: Scores 3.80 FS but lock-in is 2 and it would be a net-new proprietary dependency whose vendor is widening into L5/L8 (Nexus); a criterion at 2 is accepted as a Strategic condition only where it rides on an existing platform commitment (Elastic, MongoDB), so Pinecone is held at Tactical until an exit route is decided ^{AI}.

Criterion	Score	Rationale
Technical	4	Dense+sparse cascading retrieval, filters, namespaces, DRN scale; BM25 not evidenced
Enterprise readiness	4	SSO, SCIM, RBAC and audit logs verified on Enterprise; SLA not verified
Security and compliance	5	SOC 2 Type II and ISO 27001 from the product's own trust centre plus CMEK
Deployment flexibility	4	SaaS on three clouds plus BYOC; no self-host or air-gap
Ecosystem	3	Proprietary REST/SDK; OneLake and Marketplace integrations
Reliability and maturity	4	Core service mature; SDK v10 breaking; Nexus GA only since August 2026
Cost / TCO	3	Transparent unit pricing but Standard/Enterprise minimums and idle BYOC nodes
Lock-in / portability	2	Proprietary API and format; Nexus/KnowQL coupling
Total (generic / FS)	3.85 /	

Criterion	Score	Rationale
	3.80	

Capabilities. Fully managed vector/document index with metadata filtering, namespaces, server-side embedding, hosted rerankers and a sparse model for cascading retrieval; Dedicated Read Nodes GA April 2026; Nexus knowledge engine GA 6 August 2026 ^{vf308}.

Strengths

- SOC 2 Type II, ISO 27001:2022, HIPAA BAA, project-level CMEK, PrivateLink ^{vf309}
- SAML SSO, SCIM, RBAC for users, service accounts and API keys, audit logs on Enterprise ^{vf310}
- BYOC GA on AWS, GCP and Azure with no vendor inbound access ^{vf311}
- EU serverless regions in Ireland, Frankfurt and the Netherlands ^{vf312}

Limitations and risks

- Proprietary API and service; Nexus/KnowQL deepen coupling ^{vf313 AJ}
- BYOC needs Enterprise and supports Dedicated Read Nodes only; some features absent in BYOC as of late August 2026 ^{vf314}
- SDK v10 breaking changes (September 2026) ^{vf315}
- Last disclosed funding 2023; CEO change September 2025 ^{vf316}
- BYOC nodes billed whether idle or busy ^{vf317}

³⁰⁸ Verified fact — A2-S051: pinecone (Python SDK) - PyPI (PyPI / Pinecone Systems, Inc.) <https://pypi.org/project/pinecone/> accessed 2026-10-07; A2-S101: Introducing cascading retrieval: Unifying dense and sparse with reranking (Pinecone) <https://www.pinecone.io/blog/cascading-retrieval/> accessed 2026-10-07 [search-tool extract]; A2-S072: ZoomInfo and Pinecone Bring Real-Time, AI-Powered Contact Recommendations to Go-to-Market Teams <https://www.pinecone.io/newsroom/zoominfo-pinecone-drm/> accessed 2026-10-07 [search-tool extract]; A2-S073: General Availability of Pinecone Nexus Proves Knowledge Drives Real Outcomes for Agentic AI <https://www.pinecone.io/newsroom/general-availability-of-pinecone-nexus-proves-knowledge-drives-real-outcomes-for-agentic-ai/> accessed 2026-10-07 [search-tool extract]

³⁰⁹ Verified fact — A2-S095: Pinecone Trust and Security Center <https://security.pinecone.io/> accessed 2026-10-07 [search-tool extract]; A2-S096: Trust and Security (Pinecone) <https://www.pinecone.io/security/> accessed 2026-10-07 [search-tool extract]; V1-S078: Pinecone Trust and Security Center; Trust and Security page <https://security.pinecone.io/> accessed 2026-10-08 [search-tool extract]

³¹⁰ Verified fact — A2-S097: Security overview - Pinecone Docs <https://docs.pinecone.io/guides/production/security-overview> accessed 2026-10-07 [search-tool extract]; A2-S098: Pinecone BYOC: Trusted AI Knowledge in the Customer Cloud <https://www.pinecone.io/blog/byoc-generally-available/> accessed 2026-10-07 [search-tool extract]

³¹¹ Verified fact — A2-S069: Bring Your Own Cloud (BYOC) - Pinecone Docs <https://docs.pinecone.io/guides/production/bring-your-own-cloud> accessed 2026-10-07 [search-tool extract]; V1-S034: Pinecone BYOC: Trusted AI Knowledge in the Customer Cloud (GA); BYOC docs; 2026 release notes <https://www.pinecone.io/blog/byoc-generally-available/> accessed 2026-10-08 [search-tool extract]

³¹² Verified fact — B-L6-S003: Pinecone 2026 release notes; available cloud regions; Singapore region announcement <https://docs.pinecone.io/release-notes/2026> accessed 2026-10-08 [search-tool extract]

³¹³ Verified fact — A2-S051: pinecone (Python SDK) - PyPI (PyPI / Pinecone Systems, Inc.) <https://pypi.org/project/pinecone/> accessed 2026-10-07; A2-S073: General Availability of Pinecone Nexus Proves Knowledge Drives Real Outcomes for Agentic AI <https://www.pinecone.io/newsroom/general-availability-of-pinecone-nexus-proves-knowledge-drives-real-outcomes-for-agentic-ai/> accessed 2026-10-07 [search-tool extract]

³¹⁴ Verified fact — V1-S034: Pinecone BYOC: Trusted AI Knowledge in the Customer Cloud (GA); BYOC docs; 2026 release notes <https://www.pinecone.io/blog/byoc-generally-available/> accessed 2026-10-08 [search-tool extract]

³¹⁵ Verified fact — A2-S051: pinecone (Python SDK) - PyPI (PyPI / Pinecone Systems, Inc.) <https://pypi.org/project/pinecone/> accessed 2026-10-07

³¹⁶ Verified fact — A2-S075: Pinecone Raises \$100M in Series B Funding to Provide Long-Term Memory for AI <https://www.pinecone.io/newsroom/pinecone-raises-usd100m-in-series-b-funding-to-provide-long-term-memory-for-ai/> accessed 2026-10-07 [search-tool extract]; A2-S068: Pinecone Founder Edo Liberty to Spearhead Pinecone's Growing AI Ambitions; Appoints Ash Ashutosh as CEO <https://www.pinecone.io/newsroom/next-chapter/> accessed 2026-10-07 [search-tool extract]

³¹⁷ Verified fact — A2-S069: Bring Your Own Cloud (BYOC) - Pinecone Docs <https://docs.pinecone.io/guides/production/bring-your-own-cloud> accessed 2026-10-07 [search-tool extract]

Choose when

- A managed dedicated store at scale with enterprise identity, keys and BYOC is required ^{AJ}

Avoid when

- Self-hosting or air-gap is required ^{AJ}
- Nexus would become the only home of curated knowledge ^{AJ}

Nearest competitors: L6-milvus-zilliz, L6-qdrant, L6-turbopuffer

Regulated-FS note. Use Enterprise BYOC or an EU region; keep source of truth and raw text outside; treat Nexus as a separate decision with its own exit plan ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Pinecone Systems, Inc. (New York); CEO Ash Ashutosh since September 2025, founder Edo Liberty Chief Scientist	Verified fact	A2-S051, A2-S072, A2-S068
Category	Managed vector database / knowledge infrastructure for AI	Verified fact	A2-S051, A2-S073
Version / lineup	Managed service (serverless on-demand and Dedicated Read Nodes; BYOC); Python SDK 10.0.0 (3 September 2026) introduces schema-based document indexes and a deployment parameter; Nexus GA 6 August 2026	Verified fact	A2-S051, A2-S072, A2-S069, A2-S073
Licence	Proprietary managed service; Python SDK Apache-2.0	Verified fact	A2-S051
Status events	September 2025: Ash Ashutosh appointed CEO; April 2026: Dedicated Read Nodes GA; May 2026: Nexus early access, Marketplace launch, US\$20/month Builder tier; 3 June 2026: Nexus + Microsoft OneLake integration; July 2026: Nexus public preview; 6 August 2026: Nexus GA; 3 September 2026: Python SDK v10 (breaking API changes); Late September 2026: BYOC announced generally available on AWS, GCP and Azure (Enterprise plan; some features such as Assistant, Inference and on-demand indexes were not yet available in BYOC as of late August)	Verified fact	A2-S068, A2-S072, A2-S074, A2-S073, A2-S051, V1-S029, V1-S034

Field	Value	Label	Sources
Strategic direction	Repositioning from vector database to knowledge infrastructure for agents: Nexus context compiler + composable retriever with KnowQL query language, field-level access control, citations, PII-aware ingestion and lineage; Microsoft OneLake integration (June 2026); Pinecone Marketplace (May 2026)	Verified fact	A2-S073, A2-S074
What it does	Fully managed vector/document index with metadata filtering, namespaces and server-side embedding (create_for_model), plus Nexus for compiling enterprise data into task-specific knowledge artifacts for agents.	Verified fact	A2-S051, A2-S073
Stack position	L6; with Nexus it also reaches into L5 memory/knowledge and L8 ingestion territory.	Architectural judgment	
Integration	REST API and SDKs (Python v10); integrated inference: server-side embedding, hosted rerankers (pinecone-rerank-v0, cohere-rerank 3.5, bge-reranker-v2-m3) and sparse model pinecone-sparse-english-v0 for hybrid/cascading retrieval; Marketplace; OneLake connector for Nexus	Verified fact	A2-S051, A2-S101, A2-S073, A2-S074
Dependencies	Pinecone cloud on AWS, GCP or Azure; BYOC runs in the customer's Kubernetes cluster with FoundationDB for metadata	Verified fact	A2-S069
Certifications	SOC 2 Type II (2025 report, zero deviations); ISO/IEC 27001:2022 (annual surveillance audit); HIPAA with BAA on request; GDPR-ready; CCPA	Verified fact	A2-S095, A2-S096, V1-S078
GDPR / residency	Frankfurt cloud region announced (newsroom headline; date not retrieved); Nexus BYOC data-residency reference exists	Reported	A2-S076
Security features	AES-256 at rest, TLS in transit; CMEK at project level (irreversible toggle); AWS PrivateLink GA for serverless indexes; BYOC data plane in customer VPC with no vendor SSH/VPN/inbound access	Verified fact	A2-S096, A2-S097, A2-S069
Access controls	SSO (SAML; Okta, Google Workforce) for Enterprise; SCIM + SAML; RBAC for users, service accounts and API keys (control/data plane roles); audit logs on Enterprise or HIPAA add-on (30-minute JSON batches); Nexus field-level access control	Verified fact	A2-S097, A2-S098, A2-S073
Enterprise support	Standard and Enterprise plans; Enterprise unlocks SSO, audit logs, BYOC	Verified fact	A2-S072, A2-S097, A2-S099
Pricing	Builder tier US\$20/month (May 2026); Standard minimum US\$50/month, Enterprise minimum US\$500/month; committed-use rates (vary by cloud/region): Standard storage US\$0.33/GB-month, writes US\$4-4.50 per million WU, reads US\$16-18 per million RU; Enterprise writes US\$6-6.75, reads US\$24-27 per million; query = 1 RU per GB of namespace (min 0.25); BYOC: platform fee + per-DRN rate + own cloud costs	Verified fact	A2-S074, A2-S099, A2-S100, A2-S069
Infrastructure cost	BYOC nodes billed whether idle or busy	Verified fact	A2-S069
Ecosystem	Microsoft OneLake (Nexus); Pinecone Marketplace	Verified fact	A2-S073, A2-S074
Adoption signals	US\$138M total funding (Andreessen Horowitz, ICONIQ, Menlo, Wing); Series B US\$100M at US\$750M (2023); ZoomInfo DRN customer story	Verified fact	A2-S072, A2-S075

Field	Value	Label	Sources
	(April 2026)		
Maturity	Core service mature; Nexus GA since August 2026; SDK v10 is a breaking change	Verified fact	A2-S073, A2-S051
Deployment	saas: Yes; managed_cloud: Yes (AWS, GCP, Azure regions); vpc_byoc: Yes (Enterprise; AWS, GCP, Azure; Dedicated Read Node indexes only; zero vendor access); private_cloud: Not publicly verified; self_hosted: No; on_prem: No		

Weaviate (database; Weaviate Cloud: Shared, Dedicated, BYOC) (L6-weaviate)

Tier: Tactical · **Flags:** none · **Original graphic label:** Weaviate – open source

Rationale: Capable and ISO-certified, but the open-core licence transition is incomplete and unannounced ^{AJ}.

Criterion	Score	Rationale
Technical	4	Hybrid search, filtering, MMR, Boost, quantisation; multi-tenancy not evidenced in the fact base
Enterprise readiness	3	RBAC, OIDC and audit logs verified for the database; SCIM, SLA and Cloud console RBAC not
Security and compliance	4	SOC 2 Type II and ISO 27001; CMEK not verified; BYOC scope not stated
Deployment flexibility	5	SaaS, dedicated, BYOC, self-host
Ecosystem	4	Mainstream clients and integrations
Reliability and maturity	3	Frequent minors, three-minor support window, licence change in flight
Cost / TCO	3	Open core; enterprise features behind licence keys; pricing naming conflicts

Criterion	Score	Rationale
Lock-in / portability	3	BSD core but commercial enterprise features (rule 4)
Total (generic / FS)	3.75 / 3.70	

Capabilities. Vector database combining vector search with structured filtering and hybrid search; Boost API and MMR GA in 1.39; v1.39.7 (25 September 2026); open core in transition ^{vi318}.

Strengths

- SOC 2 Type II and ISO 27001:2022; HIPAA on Premium Dedicated (AWS) ^{vi319}
- RBAC with custom roles, OIDC users and groups, automatic RBAC audit logging ^{vi320}
- SaaS, Dedicated (~40 regions), BYOC and self-host ^{vi321}

Limitations and risks

- Enterprise features moving to commercial 'wl/' licence (1.40 RC; unannounced) ^{vi322}
- Only latest three minors supported ^{vi323}
- Weaviate Cloud console RBAC not yet available; BYOC certification scope not stated ^{vi324}
- Pricing tier names conflict; last confirmed funding 2023 ^{vi325}, ^{vi326}

Choose when

- Weaviate is already in use, or hybrid search with an ISO-certified dedicated service is needed ^{ai}

Avoid when

- Licence policy needs certainty about which features remain BSD ^{ai}

³¹⁸ Verified fact — A2-S110: Weaviate 1.39 Release <https://weaviate.io/blog/weaviate-1-39-release> accessed 2026-10-07 [search-tool extract]; A2-S109: Releases - weaviate/weaviate (Weaviate) <https://github.com/weaviate/weaviate/releases> accessed 2026-10-07 [search-tool extract]; A2-S115: weaviate/weaviate (README and licence files; PR #13360) (Weaviate) <https://github.com/weaviate/weaviate> accessed 2026-10-07 [search-tool extract]

³¹⁹ Verified fact — A2-S111: Vector Database Pricing - Weaviate <https://weaviate.io/pricing> accessed 2026-10-07 [search-tool extract]; A2-S112: Weaviate is now ISO 27001 compliant <https://weaviate.io/blog/weaviate-iso-compliant> accessed 2026-10-07 [search-tool extract]; A2-S113: Secure AI for Healthcare: HIPAA-compliant vector search with Weaviate <https://weaviate.io/blog/weaviate-hipaa-compliant> accessed 2026-10-07 [search-tool extract]

³²⁰ Verified fact — B-L6-S009: Weaviate RBAC configuration; Authentication (OIDC); Logging (audit) <https://docs.weaviate.io/deploy/configuration/configuring-rbac> accessed 2026-10-08 [search-tool extract]

³²¹ Verified fact — A2-S111: Vector Database Pricing - Weaviate <https://weaviate.io/pricing> accessed 2026-10-07 [search-tool extract]; A2-S114: Weaviate BYOC <https://weaviate.io/products/byoc> accessed 2026-10-07 [search-tool extract]; A2-S115: weaviate/weaviate (README and licence files; PR #13360) (Weaviate) <https://github.com/weaviate/weaviate> accessed 2026-10-07 [search-tool extract]

³²² Verified fact — A2-S115: weaviate/weaviate (README and licence files; PR #13360) (Weaviate) <https://github.com/weaviate/weaviate> accessed 2026-10-07 [search-tool extract]; V1-S070: weaviate/weaviate README (Community vs Enterprise Edition), releases, PR #13360, PR #12942 (Weaviate (GitHub)) <https://github.com/weaviate/weaviate> accessed 2026-10-08 [search-tool extract]

³²³ Verified fact — A2-S109: Releases - weaviate/weaviate (Weaviate) <https://github.com/weaviate/weaviate/releases> accessed 2026-10-07 [search-tool extract]

³²⁴ Verified fact — B-L6-S009: Weaviate RBAC configuration; Authentication (OIDC); Logging (audit) <https://docs.weaviate.io/deploy/configuration/configuring-rbac> accessed 2026-10-08 [search-tool extract]; A2-S114: Weaviate BYOC <https://weaviate.io/products/byoc> accessed 2026-10-07 [search-tool extract]

³²⁵ Verified fact — A2-S111: Vector Database Pricing - Weaviate <https://weaviate.io/pricing> accessed 2026-10-07 [search-tool extract]

³²⁶ Reported — A2-S116: Weaviate reels in \$50M for its AI-optimized vector database (SiliconANGLE) <https://siliconangle.com/2023/04/21/weaviate-reels-50m-ai-optimized-vector-database/> accessed 2026-10-07 [search-tool extract]

Nearest competitors: L6-qdrant, L6-milvus-zilliz, L6-elasticsearch

Regulated-FS note. Obtain a written statement of which security features fall under the commercial licence before standardising ^{Rec.}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Weaviate B.V.	Verified fact	A2-S115, A2-S116
Category	Vector database with hybrid search, open core	Verified fact	A2-S110, A2-S115
Version / lineup	v1.39.7 (25 September 2026); v1.40.0-rc.1 (19 September 2026); supported minors 1.37-1.39; Python client 4.23.1	Verified fact	A2-S109, A2-S055
Licence	Open core: most code BSD-3-Clause; 'wl/' enterprise features under a separate commercial Weaviate licence with licence key (2026 change, in 1.40 RC; merge/stable status not confirmed)	Verified fact	A2-S115, A2-S109, V1-S070
Status events	April 2023: US\$50M Series B (secondary source); 2026: enterprise 'wl/' licence and licence-key switch added (1.40.0-rc.1, 19 September 2026); 25 September 2026: v1.39.7	Verified fact	A2-S116, A2-S109, A2-S115
Strategic direction	Introducing an Enterprise Edition behind licence keys (2026); 1.39 GA of Boost API and MMR diversity, 4-bit Rotational Quantization preview, experimental Search REST API	Verified fact	A2-S115, A2-S110
What it does	Stores objects and vectors and combines vector search with structured filtering; features include Boost API, MMR diversity, rotational quantisation.	Verified fact	A2-S110, A2-S115
Stack position	L6 dedicated vector database; placement correct.	Architectural judgement	
Integration	Python client (v4)	Verified fact	A2-S055

Field	Value	Label	Sources
Dependencies	Not publicly verified	Not publicly verified	
Certifications	SOC 2 Type II (via Drata); ISO 27001:2022; HIPAA on Premium Dedicated (AWS) with signed BAA; BYOC certification scope not stated	Verified fact	A2-S111, A2-S112, A2-S113, A2-S114
GDPR / residency	Shared Cloud on AWS US East and Europe (Frankfurt); Premium Dedicated in ~40 regions	Verified fact	A2-S111
Security features	Encryption in transit and at rest, immutable backups (HIPAA offering)	Verified fact	A2-S113
Access controls	Role-based access controls	Verified fact	A2-S113
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Free; Flex pay-as-you-go from US\$45/month; Premium (Dedicated) priced by region; BYOC metered CPU/memory rates; September 2026 backup-rate update. A blog describes a 'Plus' tier from US\$280/month (naming conflict)	Verified fact	A2-S111
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	US\$50M Series B (April 2023, led by Index Ventures), total US\$67.7M; no 2025-2026 round confirmed	Reported	A2-S116
Maturity	Frequent minor releases; only latest three minors supported	Verified fact	A2-S109
Deployment	saas: Yes (Shared and Dedicated cloud); managed_cloud: Yes (~40 regions across GCP, AWS, Azure for Premium Dedicated); vpc_byoc: Yes (Weaviate manages data plane in customer VPC); private_cloud: Not publicly verified; self_hosted: Yes; on_prem: Yes		

MongoDB Vector Search (Atlas; self-managed Community Edition and Enterprise Advanced) (L6-mongodb-atlas-vector-search)

Tier: Strategic · **Flags:** none · **Original graphic label:** MongoDB – Atlas vector

Rationale: Strategic, conditional: only where MongoDB is already the operational store, because lock-in scores 2; the condition is an existing platform commitment, which is why it differs from Pinecone ^{AI}.

Criterion	Score	Rationale
Technical	4	Vector, full-text and hybrid GA; rerank and auto-embedding preview
Enterprise readiness	4	SSO, roles and auditing verified (B-L6-S001) plus the Atlas Admin API (B-REV-S028); rule 7 allows 4; SCIM and SLA not verified
Security and compliance	4	Strong platform certifications but scope rule (CP2 Q4): platform-level, preview features excluded
Deployment flexibility	4	Atlas on three clouds plus self-managed; no BYOC
Ecosystem	4	Official drivers and aggregation API; Voyage integration
Reliability and maturity	4	Atlas mature; self-managed search GA only since c. July 2026
Cost / TCO	3	Hourly Search Nodes; Community free; price tables conflict
Lock-in / portability	2	SSPL and proprietary Atlas; Voyage coupling
Total (generic / FS)	3.80	
	/	

Criterion	Score	Rationale
	3.65	

Capabilities. Vector and full-text search inside MongoDB via \$vectorSearch/\$search with hybrid fusion GA; \$rerank (Voyage, preview) and Automated Embedding (preview); GA on Atlas and self-managed Community/Enterprise Advanced 8.2+ ^{vf327}.

Strengths

- Transactional integration where documents already live in MongoDB ^{aj}
- Atlas ISO 27001:2022, SOC 2 Type II, PCI DSS, HITRUST, CSA STAR; FedRAMP Moderate for Gov ^{vf328}
- SAML/OIDC SSO, org/project and custom database roles, database auditing on M10+ ^{vf329}
- Self-managed option on 8.2+ ^{vf330}

Limitations and risks

- SSPL for self-managed; Atlas proprietary; Voyage-native features couple L6 and L7 ^{vf331}
- CMK covers search indexes only on dedicated Search Nodes ^{vf332}
- Native rerank and Automated Embedding are preview and lack Geography targeting ^{vf333}
- SOC 2 scope excludes preview features; Atlas log retention 30 days ^{vf334}

Choose when

- MongoDB is the system of record for the documents being retrieved ^{aj}

Avoid when

³²⁷ Verified fact — A2-S137: MongoDB Search and Vector Search Now Run Anywhere <https://www.mongodb.com/company/blog/product-release-announcements/mongodb-search-vector-search-now-run-anywhere> accessed 2026-10-07 [search-tool extract]; A2-S141: Context Engineering Has a Retrieval Issue: Introducing rerank-3 and Native Reranking (MongoDB) <https://www.mongodb.com/company/blog/product-release-announcements/introducing-rerank-3-and-native-rerank> accessed 2026-10-07 [search-tool extract]; A2-S078: Unlocking AI Search: Introducing Automated Embedding in MongoDB Vector Search <https://www.mongodb.com/company/blog/product-release-announcements/unlocking-ai-search-introducing-automated-embedding-in-mongodb-vector-search> accessed 2026-10-07 [search-tool extract]; V1-S035: MongoDB Delivers Accurate AI Retrieval Wherever Enterprise Data Lives; Community Edition product page (MongoDB, Inc.) <https://investors.mongodb.com/news-releases/news-release-details/mongodb-delivers-accurate-ai-retrieval-wherever-enterprise-data> accessed 2026-10-08 [search-tool extract]

³²⁸ Verified fact — A2-S138: MongoDB Atlas Trust Center <https://www.mongodb.com/products/platform/trust> accessed 2026-10-07 [search-tool extract]

³²⁹ Verified fact — B-L6-S001: Manage Identity Providers (Atlas federated authentication); Guidance for Atlas Authorization; Set Up Database Auditing (MongoDB) <https://www.mongodb.com/docs/atlas/security/manage-federated-auth/> accessed 2026-10-08 [search-tool extract]

³³⁰ Verified fact — A2-S137: MongoDB Search and Vector Search Now Run Anywhere <https://www.mongodb.com/company/blog/product-release-announcements/mongodb-search-vector-search-now-run-anywhere> accessed 2026-10-07 [search-tool extract]

³³¹ Verified fact — A2-S137: MongoDB Search and Vector Search Now Run Anywhere <https://www.mongodb.com/company/blog/product-release-announcements/mongodb-search-vector-search-now-run-anywhere> accessed 2026-10-07 [search-tool extract]; A2-S141: Context Engineering Has a Retrieval Issue: Introducing rerank-3 and Native Reranking (MongoDB) <https://www.mongodb.com/company/blog/product-release-announcements/introducing-rerank-3-and-native-rerank> accessed 2026-10-07 [search-tool extract]

³³² Verified fact — A2-S139: Encryption at Rest using Customer Key Management - Atlas (MongoDB) <https://www.mongodb.com/docs/atlas/security-kms-encryption/> accessed 2026-10-07 [search-tool extract]

³³³ Verified fact — A2-S141: Context Engineering Has a Retrieval Issue: Introducing rerank-3 and Native Reranking (MongoDB) <https://www.mongodb.com/company/blog/product-release-announcements/introducing-rerank-3-and-native-rerank> accessed 2026-10-07 [search-tool extract]; A2-S142: Europe Data Residency for Atlas Embedding and Reranking (MongoDB) <https://www.mongodb.com/company/blog/product-release-announcements/europe-data-residency-for-atlas-embedding-reranking> accessed 2026-10-07 [search-tool extract]

³³⁴ Verified fact — B-REV-S027: SOC Compliance - MongoDB Atlas (MongoDB, Inc.) <https://www.mongodb.com/products/platform/trust/soc> accessed 2026-10-08 [search-tool extract]; B-L6-S001: Manage Identity Providers (Atlas federated authentication); Guidance for Atlas Authorization; Set Up Database Auditing (MongoDB) <https://www.mongodb.com/docs/atlas/security/manage-federated-auth/> accessed 2026-10-08 [search-tool extract]

- MongoDB would be adopted only for retrieval ^{AI}
- Preview reranking or embedding would sit in regulated flows ^{AI}

Nearest competitors: L6-pgvector, L6-elasticsearch, L6-pinecone

Regulated-FS note. Dedicated Search Nodes so CMK covers indexes; export audit logs to the firm’s archive; Voyage features off until GA and Geography-scoped ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	MongoDB, Inc. (owner of Voyage AI since February 2025)	Verified fact	A2-S033
Category	Vector search built into a general-purpose document database	Verified fact	A2-S078
Version / lineup	Atlas Vector Search; self-managed Search/Vector Search GA on MongoDB 8.2+ (mongot); hybrid search GA; \$rerank native reranking (public preview, 8.3+); Automated Embedding (public preview); PyMongo 4.18.2 (24 September 2026)	Verified fact	A2-S137, A2-S141, A2-S078, A2-S059, V1-S035
Licence	Atlas: proprietary managed service; Community Edition and mongot under SSPL (source-available); Enterprise Advanced commercial	Verified fact	A2-S137
Status events	17 February 2025: acquires Voyage AI; January 2026: Automated Embedding (public preview, Community); c. July 2026: Search and Vector Search GA for Community and Enterprise Advanced; c. July 2026: hybrid search GA; \$rerank public preview; 1 September 2026: Europe Geography for Atlas Embedding and Reranking API; c. 30 September 2026: rerank-3 announced for native reranking	Verified fact	A2-S033, A2-S008, A2-S137, A2-S141, A2-S142, V1-S023, V1-S035
Strategic	Vertical integration of retrieval: Voyage AI embeddings/rerankers native	Verified	A2-S008,

Field	Value	Label	Sources
direction	in the database (automated embedding, Atlas Embedding and Reranking API); announcement on native reranking and hybrid search for agent retrieval	d fact	A2-S077, A2-S078
What it does	Vector and full-text search inside the document database via \$vectorSearch/\$search, hybrid fusion (\$rankFusion RRF, \$scoreFusion), index-less \$rerank with Voyage cross-encoders (max 1,000 docs), and automated embedding with Voyage models.	Verified fact	A2-S137, A2-S141, A2-S078
Stack position	L6 inside the operational database; with Voyage it spans L6 and L7.	Architectural judgement	
Integration	MongoDB query API/aggregation via official drivers (e.g. PyMongo); Atlas Embedding and Reranking API	Verified fact	A2-S059, A2-S077
Dependencies	MongoDB Atlas or MongoDB Community Edition; Voyage AI models for automated embedding	Verified fact	A2-S078
Certifications	Atlas: ISO/IEC 27001:2022, SOC 2 Type II, PCI DSS, HITRUST, CSA STAR, HIPAA examination by third-party assessor; FedRAMP Moderate for Atlas for Government	Verified fact	A2-S138
GDPR / residency	Atlas cluster region chosen by customer; Atlas Embedding and Reranking API offers Europe (EEA) Geography processing boundary; Automated Embedding and Native Reranking do not yet support Geography targeting	Verified fact	A2-S142, A2-S139
Security features	AES-256 at rest by default; customer key management with AWS KMS, Azure Key Vault, Google Cloud KMS (not Free/Flex); search indexes covered by CMK only with dedicated Search Nodes	Verified fact	A2-S139
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Vector Search runs on cluster or dedicated Search Nodes (2-32 nodes, hourly, M10+); indicative Search Node tiers S20 US\$0.12/hour to S80 US\$3.26/hour; Flex US\$8-30/month; Community self-managed free (SSPL); Geography-scoped Embedding API +10%	Verified fact	A2-S140, A2-S137, A2-S142
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Not publicly verified	Not publicly verified	
Deployment	saas: Yes (MongoDB Atlas); managed_cloud: Yes (Atlas on AWS, Azure, GCP); vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes (Community Edition and Enterprise Advanced, MongoDB 8.2+, GA); on_prem: Yes		

Amazon S3 Vectors (L6-s3-vectors)

Tier: Tactical · **Flags:** none · **Original graphic label:** S3 Vectors – AWS

Rationale: Tactical, default in AWS estates as the low-cost vector tier. Not AWS's lead vector-search service: AWS positions it as a durable tier behind OpenSearch ³³⁵, so rubric rule 10 (CP3 Q2) does not make it Strategic ³⁴.

Criterion	Score	Rationale
Technical	3	Vector similarity with metadata filters at large scale; no lexical or hybrid
Enterprise readiness	4	Hyperscaler presumption: platform controls presumed (CP2 Q1); confirm per service; IAM and CloudTrail data events verified
Security and compliance	4	CMK per index and PrivateLink; platform certifications without stated S3 Vectors scope: one below anchor (CP2 Q4)
Deployment flexibility	2	AWS regional service only
Ecosystem	3	Bedrock Knowledge Bases and OpenSearch; proprietary API
Reliability and maturity	3	GA about ten months; evolving quotas

³³⁵ Verified fact — A2-S092: Amazon S3 Vectors (feature page) (Amazon Web Services) <https://aws.amazon.com/s3/features/vectors/> accessed 2026-10-07 [search-tool extract]

Criterion	Score	Rationale
Cost / TCO	5	Pay-per-use storage and query, clearly efficient
Lock-in / portability	2	AWS-only API; designated CTPP concentration
Total (generic / FS)	3.30 / 3.15	

Evidence rules applied: security_compliance at 4 (not 5) under CP2 Q4: S3 Vectors compliance scope not stated

Capabilities. Object storage with native vector indexes in vector buckets: up to 2 billion vectors per index, 50 metadata keys, 10,000 top-K, metadata filters; no BM25; Bedrock Knowledge Bases and OpenSearch integration; GA December 2025 ^{vr336}.

Strengths

- Very low storage cost with no provisioned compute ^{vr337}
- SSE-KMS CMK per bucket or index, PrivateLink, IAM ^{vr338}
- CloudTrail data events for vector operations ^{vr339}
- About 34 Regions incl. European Sovereign Cloud and GovCloud ^{vr340}

Limitations and risks

- AWS-only proprietary API ^{vr341}
- No BM25/hybrid; pairs with OpenSearch ^{vr342}
- Compliance-programme scope and HIPAA eligibility not confirmed ^{vr343}

³³⁶ Verified fact — A2-S083: Limitations and restrictions - Amazon S3 Vectors (Amazon Web Services) <https://docs.aws.amazon.com/AmazonS3/latest/userguide/s3-vectors-limitations.html> accessed 2026-10-07 [search-tool extract]; A2-S086: Amazon S3 Vectors now supports up to 10,000 similarity search results per query (Amazon Web Services) <https://aws.amazon.com/about-aws/whats-new/2026/06/s3-vectors-supports-10000-search-results-per-query/> accessed 2026-10-07 [search-tool extract]; A2-S089: Prerequisites for using a vector store you created for a knowledge base - Amazon Bedrock (Amazon Web Services) <https://docs.aws.amazon.com/bedrock/latest/userguide/knowledge-base-setup.html> accessed 2026-10-07 [search-tool extract]; A2-S093: Selecting a vector store for Amazon Bedrock Knowledge Bases (Amazon Web Services (AWS Machine Learning Blog)) <https://aws.amazon.com/blogs/machine-learning/selecting-a-vector-store-for-amazon-bedrock-knowledge-bases/> accessed 2026-10-07 [search-tool extract]; V1-S030: Amazon S3 Vectors is now generally available with 40 times the scale of preview; S3 Vectors limitations; March 2026 region expansion (Amazon Web Services) <https://aws.amazon.com/about-aws/whats-new/2025/12/amazon-s3-vectors-generally-available/> accessed 2026-10-08 [search-tool extract]

³³⁷ Verified fact — A2-S087: Amazon S3 Pricing (S3 Vectors section) (Amazon Web Services) <https://aws.amazon.com/s3/pricing/> accessed 2026-10-07 [search-tool extract]

³³⁸ Verified fact — A2-S090: Data protection and encryption in S3 Vectors (Amazon Web Services) <https://docs.aws.amazon.com/AmazonS3/latest/userguide/s3-vectors-data-encryption.html> accessed 2026-10-07 [search-tool extract]; A2-S091: VPC endpoints for S3 Vectors (Amazon Web Services) <https://docs.aws.amazon.com/AmazonS3/latest/userguide/s3-vectors-privatelink.html> accessed 2026-10-07 [search-tool extract]

³³⁹ Verified fact — B-L6-S006: Logging with AWS CloudTrail for S3 Vectors; Compliance validation for Amazon S3 (Amazon Web Services) <https://docs.aws.amazon.com/AmazonS3/latest/userguide/s3-vectors-logging.html> accessed 2026-10-08 [search-tool extract]

³⁴⁰ Verified fact — A2-S084: AWS Regions, endpoints, and quotas for S3 Vectors (Amazon Web Services) <https://docs.aws.amazon.com/AmazonS3/latest/userguide/s3-vectors-regions-quotas.html> accessed 2026-10-07 [search-tool extract]

³⁴¹ Verified fact — A2-S092: Amazon S3 Vectors (feature page) (Amazon Web Services) <https://aws.amazon.com/s3/features/vectors/> accessed 2026-10-07 [search-tool extract]

³⁴² Verified fact — A2-S083: Limitations and restrictions - Amazon S3 Vectors (Amazon Web Services) <https://docs.aws.amazon.com/AmazonS3/latest/userguide/s3-vectors-limitations.html> accessed 2026-10-07 [search-tool extract]; A2-S093: Selecting a vector store for Amazon Bedrock Knowledge Bases (Amazon Web Services (AWS Machine Learning Blog)) <https://aws.amazon.com/blogs/machine-learning/selecting-a-vector-store-for-amazon-bedrock-knowledge-bases/> accessed 2026-10-07 [search-tool extract]

- Limits and prices changed several times in 2026 ^{VR344}

Choose when

- AWS-centred firms needing a cheap durable tier or Bedrock Knowledge Base ^{AI}

Avoid when

- Lexical search, low tail latency or a multi-cloud exit is required ^{AI}

Nearest competitors: L6-turbopuffer, L6-milvus-zilliz, L6-pgvector

Regulated-FS note. Enable CloudTrail data events on regulated indexes; set KMS key at index creation; record under the AWS CTPP concentration entry ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Amazon Web Services	Verified fact	A2-S081
Category	Object storage with native vector storage and similarity query (vector buckets and indexes)	Verified fact	A2-S094, A2-S092
Version / lineup	Generally available since December 2025 (preview July 2025); up to 2 billion vectors per index, 10,000 indexes per bucket; 10,000 top-K per query since June 2026	Verified fact	A2-S081, A2-S094, A2-S086, V1-S030
Licence	Proprietary AWS managed service	Verified fact	A2-S092
Status events	July 2025: preview (5 Regions); December 2025: GA in 14 Regions; March 2026: +17 Regions (31 total); June 2026: 10,000 results per query; query data-processed charges cut up to 80% for >10M-vector indexes; July 2026: AWS GovCloud (US) Regions	Verified fact	A2-S094, A2-S081, A2-S085, A2-S086, A2-S088, A2-S084
Strategic direction	Positioned as low-cost, durable vector tier integrated with Bedrock Knowledge Bases and OpenSearch (tiering hot vectors to OpenSearch); rapid regional expansion and price cuts for large indexes in 2026	Verified fact	A2-S092, A2-S093, A2-S085, A2-S088
What it does	Stores vectors (1-4,096 dims) with up to 50 metadata keys in vector indexes inside vector buckets and serves similarity queries with metadata filters via S3 Vectors APIs (PutVectors, QueryVectors); up to 1,000 write requests/s per index.	Verified fact	A2-S083
Stack position	L6 cost-optimised vector tier on object storage; not a full retrieval engine (no native BM25/hybrid) - pairs with OpenSearch for hybrid/low-	Architectural	

³⁴³ Verified fact — A2-S090: Data protection and encryption in S3 Vectors (Amazon Web Services) <https://docs.aws.amazon.com/AmazonS3/latest/userguide/s3-vectors-data-encryption.html> accessed 2026-10-07 [search-tool extract]; B-L6-S006: Logging with AWS CloudTrail for S3 Vectors; Compliance validation for Amazon S3 (Amazon Web Services) <https://docs.aws.amazon.com/AmazonS3/latest/userguide/s3-vectors-logging.html> accessed 2026-10-08 [search-tool extract]

³⁴⁴ Verified fact — A2-S086: Amazon S3 Vectors now supports up to 10,000 similarity search results per query (Amazon Web Services) <https://aws.amazon.com/about-aws/whats-new/2026/06/s3-vectors-supports-10000-search-results-per-query/> accessed 2026-10-07 [search-tool extract]; A2-S088: Amazon S3 Vectors reduces query charges by up to 80% for large vector indexes (Amazon Web Services) <https://aws.amazon.com/about-aws/whats-new/2026/06/s3-vectors-reduces-query-charges-80-percent-large-indexes/> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
	latency; supports H5 'object-store vectors'.	judgement	
Integration	S3 Vectors API/SDKs; Bedrock Knowledge Bases vector store (35 metadata keys, 1 KB metadata limit); OpenSearch managed clusters can use S3 Vectors as storage engine, or export to OpenSearch Serverless; CloudFormation support	Verified fact	A2-S089, A2-S093, A2-S092
Dependencies	AWS account and Region; optionally Bedrock Knowledge Bases or Amazon OpenSearch Service	Verified fact	A2-S089, A2-S093
Certifications	HIPAA eligibility and other compliance scope for S3 Vectors specifically not confirmed (general S3 programmes do not establish S3 Vectors scope)	Verified fact	A2-S090
GDPR / residency	Regional service: about 34 Regions listed incl. AWS European Sovereign Cloud (Germany); data stays in chosen Region	Verified fact	A2-S084, A2-S085
Security features	SSE-S3 by default; SSE-KMS with customer-managed keys per bucket or index (immutable after index creation; covers metadata); AWS PrivateLink interface endpoints with endpoint policies; IAM-based access	Verified fact	A2-S090, A2-S091
Access controls	AWS IAM policies (S3 Vectors actions plus KMS permissions); CloudTrail monitoring of KMS key usage; S3 Vectors CloudTrail data-event coverage not confirmed	Verified fact	A2-S090
Enterprise support	Covered by AWS Support plans (not verified specifically)	Architectural judgement	
Pricing	Storage US\$0.06/GB-month; PUT US\$0.20 per logical GB (128 KB minimum per PUT); queries US\$2.50 per million requests plus data processed US\$0.004/TB (first 100K vectors), US\$0.002/TB (100K-10M), US\$0.0004/TB (10M+), data returned US\$0.01/GB (first 512 KB/query free); up to 80% lower data-processed charges for >10M-vector indexes from June 2026	Verified fact	A2-S087, A2-S088, V1-S030
Infrastructure cost	Pay-per-use storage and query; no provisioned compute; AWS worked example: 50M vectors (4.1 KB each), 1M queries x 10,000 results = US\$178.61/month	Verified fact	A2-S087
Ecosystem	Amazon Bedrock Knowledge Bases, Amazon OpenSearch Service, Aurora PostgreSQL (AWS blog on SQL over S3 Vectors)	Verified fact	A2-S089, A2-S093
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	GA ~10 months (December 2025); limits and pricing changed several times in 2026	Verified fact	A2-S081, A2-S086, A2-S088
Deployment	saas: Yes (AWS managed); managed_cloud: Yes (AWS Regions incl. GovCloud and European Sovereign Cloud); vpc_byoc: Not applicable (runs in customer's AWS account as a regional service); private_cloud: Not publicly verified; self_hosted: No; on_prem: No		

turbopuffer (L6-turbopuffer)

Tier: Tactical · **Flags:** none · **Original graphic label:** turbopuffer – cloud vector store

Rationale: Distinctive object-storage design and per-namespace keys, but coarse key scoping, no ISO 27001, an Enterprise entry price well above peers and a proprietary API; author conflict disclosed and borderline calls resolved against it ^{AJ}.

Criterion	Score	Rationale
Technical	4	Vector + BM25 + filters with namespace isolation at scale; cold latency trade-off
Enterprise readiness	3	Dashboard SSO verified (lifts cap, CP2 Q2); no documented roles, admin-only BYOC keys, audit logs in beta
Security and compliance	3	SOC 2 Type 2, HIPAA, per-namespace CMEK, but no ISO 27001: maximum 3
Deployment flexibility	4	SaaS on three clouds, dedicated clusters and BYOC; no self-host
Ecosystem	3	HTTP API and generated SDK; integrations not evidenced
Reliability and maturity	3	Young company; named customers and profitability only via press
Cost / TCO	3	Published minimums and object-storage pricing, but the Enterprise tier needed for CMEK and private networking starts at US\$4,096 per month with a 35% usage premium (Pinecone Enterprise from US\$500); 3, resolved against the Anthropic-related item (CP3 review A)
Lock-in / portability	2	Proprietary API and storage format
Total (generic / FS)	3.30 / 3.15	

Capabilities. Proprietary serverless search on object storage: namespaced ANN vector (SPFresh) and BM25 full-text with filters; stateless compute with SSD/memory caches; SaaS and BYOC on AWS, GCP and Azure ^{VP345}. Conflict of interest: Anthropic, the author’s developer, is reported as a customer ⁿ³⁴⁶.

Strengths

- Object-storage economics for very large, many-tenant corpora ^{VP347 AJ}
- Per-namespace CMEK and private networking on Enterprise; HIPAA BAA on Scale+ ^{VP348}
- BYOC with manual customer approval of vendor operations ^{VP349}
- Region-pinned data incl. Frankfurt ^{VP350}

³⁴⁵ Verified fact — A2-S056: turbopuffer (Python SDK) - PyPI (PyPI / turbopuffer) <https://pypi.org/project/turbopuffer/> accessed 2026-10-07; A2-S124: Architecture - turbopuffer <https://turbopuffer.com/docs/architecture> accessed 2026-10-07 [search-tool extract]; A2-S125: turbopuffer BYOC <https://turbopuffer.com/docs/byoc> accessed 2026-10-07 [search-tool extract]

³⁴⁶ Reported — A2-S126: Ex-Shopify engineers raise fresh financing to scale Turbopuffer’s AI search (BetaKit) <https://betakit.com/ex-shopify-engineers-raise-fresh-financing-to-scale-turbopuffers-ai-search/> accessed 2026-10-07 [search-tool extract]

³⁴⁷ Verified fact — A2-S124: Architecture - turbopuffer <https://turbopuffer.com/docs/architecture> accessed 2026-10-07 [search-tool extract]

³⁴⁸ Verified fact — A2-S123: turbopuffer pricing <https://turbopuffer.com/pricing> accessed 2026-10-07 [search-tool extract]; V1-S077: turbopuffer Security & Compliance; pricing; CMEK; BYOC; roadmap <https://turbopuffer.com/docs/security> accessed 2026-10-08 [search-tool extract]

³⁴⁹ Verified fact — A2-S125: turbopuffer BYOC <https://turbopuffer.com/docs/byoc> accessed 2026-10-07 [search-tool extract]

Limitations and risks

- No built-in document-level RBAC; BYOC keys are admin keys ^{VF}³⁵¹
- Audit logs with SIEM only an opt-in beta (March 2026) ^{VF}³⁵²
- No ISO 27001 found ^{NPV}
- Cold-query latency much higher than cached (vendor figures) ^R³⁵³
- Funding and revenue only from press/aggregator ^R³⁵⁴

Choose when

- Very many tenants or very large corpora with skewed access, BYOC and per-tenant keys ^{AJ}

Avoid when

- Fine-grained key scopes, ISO 27001 or self-hosting are required ^{AJ}

Nearest competitors: L6-pinecone, L6-milvus-zilliz, L6-s3-vectors

Regulated-FS note. BYOC only for client data; one namespace and key per client; warm-up policy for latency-sensitive tenants; independent alternative Zilliz BYOC ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	turbopuffer (Ottawa; CEO Simon Eskildsen per press)	Verified fact	A2-S056, A2-S126
Category	Serverless vector and full-text search on object storage	Verified fact	A2-S124, A2-S056
Version / lineup	Managed service; Python SDK 2.11.0 (7 October 2026)	Verified fact	A2-S056
Licence	Proprietary service; Python SDK MIT	Verified fact	A2-S056
Status events	19 December 2025: undisclosed raise from Lachy Groom and Thrive Capital (press)	Verified fact	A2-S126
Strategic direction	Object-storage-first search engine targeting very large multi-tenant workloads; BYOC on AWS, GCP and Azure	Verified fact	A2-S124, A2-S125
What it does	Namespaced search API with ANN vector (SPFresh centroid index) and BM25 full-text ranking plus filters; all durable state in object storage with SSD/memory caches (cold p50 874 ms vs cached p50 14 ms on 1M docs).	Verified fact	A2-S056, A2-S124

³⁵⁰ Verified fact — B-L6-S002: turbopuffer Regions; Security; Permissions; Roadmap & Changelog; Private Networking <https://turbopuffer.com/docs/regions> accessed 2026-10-08 [search-tool extract]

³⁵¹ Verified fact — B-L6-S002: turbopuffer Regions; Security; Permissions; Roadmap & Changelog; Private Networking <https://turbopuffer.com/docs/regions> accessed 2026-10-08 [search-tool extract]

³⁵² Verified fact — B-L6-S002: turbopuffer Regions; Security; Permissions; Roadmap & Changelog; Private Networking <https://turbopuffer.com/docs/regions> accessed 2026-10-08 [search-tool extract]

³⁵³ Reported — A2-S124: Architecture - turbopuffer <https://turbopuffer.com/docs/architecture> accessed 2026-10-07 [search-tool extract]

³⁵⁴ Reported — A2-S126: Ex-Shopify engineers raise fresh financing to scale Turbopuffer's AI search (BetaKit) <https://betakit.com/ex-shopify-engineers-raise-fresh-financing-to-scale-turbopuffers-ai-search/> accessed 2026-10-07 [search-tool extract]; A2-S127: Turbopuffer revenue, funding & growth rate (Sacra) <https://sacra.com/c/turbopuffer/> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
Stack position	L6 retrieval store (vector + full-text), placement correct; the BM25 support supports the H5 'retrieval store' framing.	Architectural judgement	
Integration	HTTP API; generated Python SDK (sync and async)	Verified fact	A2-S056
Dependencies	Object storage (S3, GCS, Azure Blob) as system of record; Kubernetes for BYOC	Verified fact	A2-S124, A2-S125
Certifications	SOC 2 Type 2 (report on request); HIPAA BAA on Scale and Enterprise tiers	Verified fact	A2-S125, A2-S123, V1-S077
GDPR / residency	DPA for GDPR/CCPA; region chosen per client (e.g. gcp-us-central1); full EU region list not verified	Verified fact	A2-S123, A2-S056
Security features	TLS 1.2+ in transit; AES-256 at rest including SSD cache; Enterprise: per-namespace customer-managed encryption keys and private networking	Verified fact	A2-S125, A2-S123
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Minimum monthly usage: Launch US\$16, Scale US\$256, Enterprise >=US\$4,096 (35% usage premium); storage on logical bytes; queries on data scanned (min 1.28 GB/namespace) and returned, with volume discounts	Verified fact	A2-S123, V1-S077
Infrastructure cost	Object-storage economics; cold-query latency trade-off	Verified fact	A2-S124
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	Named customers include Anthropic, Atlassian, Cursor, Notion, Legora; Sacra estimates ~US\$100M annualised revenue (March 2026); <US\$1M primary capital raised per CEO	Reported	A2-S126, A2-S127, A2-S125
Maturity	Very frequent SDK releases (2.11.0 on 7 October 2026)	Verified fact	A2-S056
Deployment	saas: Yes; managed_cloud: Not publicly verified; vpc_byoc: Yes (customer Kubernetes on AWS, GCP, Azure; manual customer approval for operations); private_cloud: Not publicly verified; self_hosted: No (BYOC only); on_prem: Not publicly verified		

Chroma (open source) and Chroma Cloud (L6-chroma)

Tier: Tactical · **Flags:** none · **Original graphic label:** Chroma – open source

Rationale: Useful for prototypes and harnesses; enterprise controls undocumented and release cadence uncertain ^{AJ}.

Criterion	Score	Rationale
Technical	3	Core vector store with filters; hybrid and full-text only on Cloud; scale not evidenced
Enterprise readiness	2	Customer-facing SSO, roles and audit logs NPV; only tenant-scoped access and DB-scoped keys found
Security and compliance	3	SOC 2 Type II and CMEK; no ISO 27001: maximum 3
Deployment flexibility	4	Embedded, self-host, Cloud, single-tenant and BYOC, but self-host lacks auth and only one EU cloud region
Ecosystem	3	Python/JS clients; ecosystem not evidenced
Reliability and maturity	2	Rust rewrite 2025; five months without a PyPI release; seed-funded
Cost / TCO	4	Free OSS; transparent cloud usage pricing
Lock-in / portability	4	Apache-2.0, vendor-governed
Total (generic / FS)	3.05 / 3.10	

Evidence rules applied: enterprise_readiness capped at 2: customer SSO, RBAC and audit logs NPV

Capabilities. Apache-2.0 embedded or client-server database for documents and embeddings with a small API; Chroma Cloud adds serverless vector, hybrid and full-text search, BYOC and single-tenant options ^{vr³⁵⁵}.

Strengths

- Fastest path from notebook to working retrieval ^{AJ}
- Apache-2.0 ^{vr³⁵⁶}
- Chroma Cloud SOC 2 Type II and CMEK GA ^{vr³⁵⁷}
- Transparent usage pricing ^{vr³⁵⁸}

Limitations and risks

- Open-source build has no built-in auth since 1.0 ^{vr³⁵⁹}
- Customer SSO, roles and audit logs not documented ^{NPV}

³⁵⁵ Verified fact — A2-S060: chromadb - PyPI (PyPI / Chroma) <https://pypi.org/project/chromadb/> accessed 2026-10-07; A2-S128: Pricing - Chroma <https://www.trychroma.com/pricing> accessed 2026-10-07 [search-tool extract]

³⁵⁶ Verified fact — A2-S060: chromadb - PyPI (PyPI / Chroma) <https://pypi.org/project/chromadb/> accessed 2026-10-07

³⁵⁷ Verified fact — A2-S130: Security - Chroma <https://www.trychroma.com/security> accessed 2026-10-07 [search-tool extract]; B-L6-S008: Chroma Security; architecture overview; CMEK changelog <https://www.trychroma.com/security> accessed 2026-10-08 [search-tool extract]

³⁵⁸ Verified fact — A2-S128: Pricing - Chroma <https://www.trychroma.com/pricing> accessed 2026-10-07 [search-tool extract]

³⁵⁹ Verified fact — A2-S131: Migration - Chroma Docs <https://docs.trychroma.com/docs/overview/migration> accessed 2026-10-07 [search-tool extract]

- No PyPI release since 5 May 2026 at time of research ^{VF-360}
- Multi-tenant cloud only in us-east-1 and GCP Belgium; no UK region ^{VF-361}

Choose when

- Prototypes, evaluation harnesses, embedded single-user tools ^{AJ}

Avoid when

- Client-confidential data or production-critical systems ^{AJ}

Nearest competitors: L6-pgvector, L6-qdrant, L6-turbopuffer

Regulated-FS note. Never self-host the open-source server without an authenticating proxy; no promotion of prototypes without a re-platforming decision ^{REC}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Chroma (Jeff Huber, Anton Troynikov listed as authors)	Verified fact	A2-S060
Category	Open-source embedding/vector database with serverless cloud service	Verified fact	A2-S060
Version / lineup	chromadb 1.5.9 (5 May 2026)	Verified fact	A2-S060
Licence	Open source, Apache-2.0	Verified fact	A2-S060
Status events	1 March 2025: Chroma 1.0 (Rust rewrite; built-in auth implementations removed); 5 May 2026: chromadb 1.5.9	Verified fact	A2-S131, A2-S060
Strategic direction	Positions itself as ‘the open-source data infrastructure for AI’; Chroma Cloud offers serverless vector, hybrid and full-text search	Verified fact	A2-S060
What it does	Embedded or client-server database for documents and embeddings with a small API (create collection, add, query); Chroma Cloud adds serverless hybrid and full-text search.	Verified fact	A2-S060
Stack position	L6; strong for prototyping/embedded use, Cloud targets production.	Architectural judgement	
Integration	Python and JavaScript clients; in-memory, persistent or client-server mode	Verified fact	A2-S060

³⁶⁰ Verified fact — A2-S060: chromadb - PyPI (PyPI / Chroma) <https://pypi.org/project/chromadb/> accessed 2026-10-07

³⁶¹ Verified fact — A2-S129: Chroma Cloud - Chroma Docs <https://docs.trychroma.com/cloud/getting-started> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
Dependencies	Not publicly verified	Not publicly verified	
Certifications	Chroma Cloud SOC 2 Type II	Verified fact	A2-S130, A2-S128, V1-S086
GDPR / residency	Multi-tenant regions: AWS us-east-1 and GCP europe-west1 (Belgium); data stays in chosen region	Verified fact	A2-S129
Security features	Customer-managed encryption keys changelog entry exists (content not retrieved); open-source build has no built-in auth since 1.0	Reported	A2-S130, A2-S131
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Starter US\$0 + usage (US\$5 credit): US\$2.50/GiB written, US\$0.33/GiB-month stored, US\$0.0075/TiB queried, US\$0.09/GiB returned; Team US\$250/month + usage; single-tenant/BYOC custom	Verified fact	A2-S128, V1-S086
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	US\$18M seed round	Verified fact	A2-S131
Maturity	Not publicly verified	Not publicly verified	
Deployment	saas: Yes (Chroma Cloud, multi-tenant); managed_cloud: Not publicly verified; vpc_byoc: Yes (BYOC clusters, fixed minimum cost); private_cloud: Yes (single-tenant clusters, custom pricing); self_hosted: Yes; on_prem: Yes		

L5: Memory

Zep (managed context-graph platform, 'Zep Cloud') and Graphiti (open-source temporal knowledge-graph framework) (L5-zep)

Tier: Tactical • **Flags:** Deprecated • **Original graphic label:** Zep – graph memory

Rationale: Technically the strongest memory model for audit and temporal reasoning; tier-gated compliance, a proprietary managed engine and pre-1.0 Graphiti keep it Tactical ⁴¹.

Criterion	Score	Rationale
Technical	5	Bi-temporal graph, episodes with provenance, hybrid retrieval, users and threads: leading depth on the layer's memory-type questions
Enterprise readiness	4	IdP sign-in, RBAC + ABAC and audit logs with one-year retention, plus a guaranteed SLA on Enterprise (A3-S107, A3-S089): rule 7 allows 4 (CP3 review A, consistent with Mistral OCR); condition: audit logs cover web-app actions only, not API or SDK calls
Security and compliance	3	SOC 2 Type II + HIPAA + BYOK on Enterprise; no ISO 27001 found (same calibration as Braintrust in L9)
Deployment flexibility	4	SaaS and BYOC (AWS/GCP/Azure); Graphiti self-host; no supported self-host of Zep itself
Ecosystem	4	Broad framework integrations; Graphiti 31,529 stars; MCP server
Reliability and maturity	3	Graphiti pre-1.0 with frequent releases; SDK 4.0 beta; Community Edition deprecated; no acquisition found
Cost / TCO	3	Published credit pricing (Flex US\$125, Flex Plus US\$375 per month); Graphiti cost is graph DB + LLM ingestion
Lock-in / portability	2	Zep Cloud proprietary engine and API; Graphiti (Apache-2.0) is a partial exit with no documented migration

Criterion	Score	Rationale
Total (generic / FS)	3.75 / 3.50	

Capabilities. Zep Cloud managed context-graph platform on a proprietary Context Graph Engine, with Graphiti (Apache-2.0) as the open-source temporal knowledge-graph framework: entities, facts with validity windows, episodes with provenance, communities; invalidation rather than deletion; hybrid semantic, keyword and graph retrieval ^{VF362}.

Strengths

- Bi-temporal facts and per-episode provenance suit audit and how-facts-changed questions ^{VF363}
- SOC 2 Type II, HIPAA BAA, BYOK and BYOC on Enterprise ^{VF364}
- Right-to-be-forgotten and time-based retention purge (Zep Archive); EU residency on request; DPAs ^{VF365}
- Integrations with ADK, Microsoft Agent Framework, LangGraph, Strands and others; Graphiti MCP server ^{VF366}

Limitations and risks

- Community Edition deprecated; Graphiti is the only OSS path ^{VF367}
- Compliance only on Enterprise, not Flex or Flex Plus ^{VF368}
- Audit logs cover web-app member actions, not API or SDK calls ^{VF369}
- Graphiti pre-1.0; zep-cloud SDK 4.0 in beta ^{VF370}
- Proprietary graph engine in Zep Cloud; migration path to Graphiti not documented ^{VF371} ^{NPV}

Choose when

³⁶² Verified fact — A3-S003: PyPI JSON metadata: graphiti-core (Python Package Index (PyPI)) <https://pypi.org/pypi/graphiti-core/json> accessed 2026-10-07; A3-S059: getzep/zep README: Zep Cloud Examples & Integrations (Zep AI (GitHub)) <https://raw.githubusercontent.com/getzep/zep/main/README.md> accessed 2026-10-07

³⁶³ Verified fact — A3-S003: PyPI JSON metadata: graphiti-core (Python Package Index (PyPI)) <https://pypi.org/pypi/graphiti-core/json> accessed 2026-10-07

³⁶⁴ Verified fact — A3-S089: Zep pricing <https://www.getzep.com/pricing/> accessed 2026-10-07 [search-tool extract]; A3-S090: Zep Enterprise (and Emerging Companies programme page) <https://www.getzep.com/enterprise/> accessed 2026-10-07 [search-tool extract]

³⁶⁵ Verified fact — A3-S107: Zep docs: Governance (and Audit logging page, Zep Archive blog) <https://help.getzep.com/governance> accessed 2026-10-07 [search-tool extract]

³⁶⁶ Verified fact — A3-S059: getzep/zep README: Zep Cloud Examples & Integrations (Zep AI (GitHub)) <https://raw.githubusercontent.com/getzep/zep/main/README.md> accessed 2026-10-07; A3-S003: PyPI JSON metadata: graphiti-core (Python Package Index (PyPI)) <https://pypi.org/pypi/graphiti-core/json> accessed 2026-10-07

³⁶⁷ Verified fact — A3-S059: getzep/zep README: Zep Cloud Examples & Integrations (Zep AI (GitHub)) <https://raw.githubusercontent.com/getzep/zep/main/README.md> accessed 2026-10-07; V1-S044: Announcing a New Direction for Zep's Open Source Strategy; Zep FAQ <https://blog.getzep.com/announcing-a-new-direction-for-zeps-open-source-strategy/> accessed 2026-10-08 [search-tool extract]

³⁶⁸ Verified fact — A3-S089: Zep pricing <https://www.getzep.com/pricing/> accessed 2026-10-07 [search-tool extract]

³⁶⁹ Verified fact — A3-S107: Zep docs: Governance (and Audit logging page, Zep Archive blog) <https://help.getzep.com/governance> accessed 2026-10-07 [search-tool extract]

³⁷⁰ Verified fact — A3-S003: PyPI JSON metadata: graphiti-core (Python Package Index (PyPI)) <https://pypi.org/pypi/graphiti-core/json> accessed 2026-10-07; A3-S002: PyPI JSON metadata: zep-cloud (Python Package Index (PyPI)) <https://pypi.org/pypi/zep-cloud/json> accessed 2026-10-07

³⁷¹ Verified fact — A3-S003: PyPI JSON metadata: graphiti-core (Python Package Index (PyPI)) <https://pypi.org/pypi/graphiti-core/json> accessed 2026-10-07

- Temporal validity and provenance of memory matter (mandate or terminology changes over time) ^{AJ}
- Graphiti self-hosted on Neo4j or Neptune for in-estate data ^{AJ}

Avoid when

- You need API-level audit logs from the vendor, ISO 27001, or a supported self-hosted full platform ^{AJ}

Nearest competitors: L5-mem0, L5-cognee, L5-gcp-vertex-memory-bank

Regulated-FS note. Prefer self-hosted Graphiti on an approved graph store and test that erasure removes invalidated facts and episodes; use Zep Cloud only on Enterprise with EU residency requested ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Zep (getzep)	Verified fact	A3-S059
Category	Managed agent memory / context-graph infrastructure; Graphiti is the open-source temporal knowledge-graph library underneath	Verified fact	A3-S003
Version / lineup	zep-cloud Python SDK 3.30.0 (24 September 2026), 4.0.0b1 pre-release (6 October 2026); graphiti-core 0.30.2 (8 September 2026)	Verified fact	A3-S002, A3-S003
Licence	Zep platform proprietary; Graphiti Apache-2.0; Zep Community Edition deprecated and unsupported	Verified fact	A3-S003, A3-S059
Status events	Zep Community Edition deprecated; code moved to legacy/ folder (status on 7 October 2026; date of deprecation not verified); Graphiti: Kuzu backend deprecated because upstream Kuzu project is unmaintained	Verified fact	A3-S059, A3-S003, V1-S044
Strategic direction	Positions Zep as 'context infrastructure' powered by a proprietary Context Graph Engine built for millions of per-user/entity context graphs; ships agent plugins for Claude Code, Codex, Cursor, Claude Desktop/Cowork and ChatGPT	Verified fact	A3-S003, A3-S059
What it does	Graphiti builds temporal context graphs: entities, facts with validity windows (bi-temporal), episodes with provenance and communities; old facts are invalidated rather than deleted; retrieval is hybrid semantic, keyword and graph search. Zep adds managed users, threads and message storage, dashboard and pre-configured retrieval (vendor claims sub-200ms)	Verified fact	A3-S003
Stack position	Spans L5 memory and L6 knowledge store: a temporal knowledge graph	Verified fact	A3-S003

Field	Value	Label	Sources
	that ingests chat history and business data; the graphic's 'graph memory' descriptor matches Graphiti's design	d fact	
Integration	Zep SDKs for Python, TypeScript and Go; framework integrations for Google ADK, Microsoft Agent Framework, AutoGen, AG2, CrewAI, LangGraph, LiveKit, Pydantic AI, Strands Agents, Mastra and Vercel AI SDK; Graphiti offers an MCP server and FastAPI REST service	Verified fact	A3-S059, A3-S003
Dependencies	Graphiti requires Neo4j 5.26, FalkorDB 1.1.2 or Amazon Neptune (with OpenSearch Serverless for full text); Kuzu deprecated; defaults to OpenAI for LLM and embeddings and needs an LLM with structured output. Zep Cloud uses its proprietary graph engine (no third-party graph DB)	Verified fact	A3-S003
Certifications	SOC 2 Type II and HIPAA BAA on Enterprise plan only (not on Flex/Flex Plus); reports via Zep Trust Center	Verified fact	A3-S089, A3-S090
GDPR / residency	Signs DPAs with EU customers; US hosting by default, EU residency available on request; right-to-be-forgotten and time-based retention/purge features (Zep Archive)	Verified fact	A3-S107, A3-S089
Security features	BYOK (cloud with your own keys) and BYOC (your own VPC on AWS, GCP or Azure) on Enterprise; BYOK via AWS KMS in emerging-companies programme	Verified fact	A3-S089, A3-S090
Access controls	IdP-based member sign-in (SAML not named); RBAC for dashboard (account/project roles) and ABAC policies for data reads/writes; audit logs cover web-app member actions only, not API/SDK calls; Enterprise retains audit and API logs 1 year	Verified fact	A3-S107
Enterprise support	Enterprise: custom pricing, negotiated credit rates, guaranteed SLA	Verified fact	A3-S089
Pricing	Flex US\$125/month (50,000 credits); Flex Plus US\$375/month (200,000 credits); Enterprise custom; emerging-companies programme US\$13,000 first year	Verified fact	A3-S089, A3-S090
Infrastructure cost	Graphiti self-hosting cost is driven by the graph database (Neo4j/FalkorDB/Neptune) and LLM calls during ingestion; README warns of LLM rate-limit (429) errors at default concurrency	Verified fact	A3-S003
Ecosystem	Integrations across major agent frameworks; Graphiti backends Neo4j, FalkorDB, Neptune	Verified fact	A3-S059, A3-S003
Adoption signals	getzep/graphiti 31,529 GitHub stars; getzep/zep 4,951 stars (7 October 2026). Backers named as Y Combinator and Engineering Capital; no 2025-2026 round found	Verified fact	A3-S054, A3-S090
Maturity	graphiti-core first released August 2024, 194 releases, still pre-1.0 (0.30.x); zep-cloud SDK on 3.x with a 4.0 beta in progress	Verified fact	A3-S003, A3-S002
Deployment	saas: Yes; managed_cloud: Not publicly verified; vpc_byoc: Yes (BYOC on Enterprise; AWS, GCP, Azure); private_cloud: Not publicly verified; self_hosted: Graphiti only (self-hosted framework); Zep Community Edition is deprecated; on_prem: Not publicly verified		

Mem0 (open-source library and self-hosted server; Mem0 Platform managed service) (L5-mem0)

Tier: Tactical · **Flags:** none · **Original graphic label:** Mem0 – memory layer

Rationale: Broad, store-agnostic memory API useful behind a firm-owned wrapper; certification evidence and OSS/Platform divergence keep it off the Strategic tier ^{A1}.

Criterion	Score	Rationale
Technical	4	Covers extraction, scoping, hybrid recall, deletion and history; graph and decay only on Platform
Enterprise readiness	4	Enterprise plan lists SSO, audit logs and SLA support (A3-S049); Platform org/project member roles and an event feed (A3-S080): all three controls plus SLA reach 4 under rule 7 (CP3 review A, consistent with Firecrawl and Mistral OCR); condition: evidence is a pricing page and OSS has no org concept
Security and compliance	2	SOC 2 Type I confirmed by vendor; Type II claims conflict; no report seen; HIPAA and BYOK vendor-stated
Deployment flexibility	4	SaaS, self-host and on-prem verified; VPC and air-gap are vendor claims from a marketing guide
Ecosystem	4	Wide vector-store support, LangGraph and CrewAI integrations; MCP repository archived
Reliability and maturity	3	Two years of releases but breaking changes in April 2026 (ADD-only, graph removed from OSS); no acquisition found
Cost / TCO	4	Apache-2.0 OSS; published tiers (Starter US\$19, Pro US\$249 per month); self-host cost is LLM + embedder + vector store
Lock-in / portability	3	Apache-2.0 OSS on standard stores, but key features (incl. export) Platform-only
Total (generic / FS)	3.55 / 3.35	

Evidence rules applied: security_compliance at 2: SOC 2 Type II status conflicting on vendor pages, only Type I confirmed (treated as below the 3 anchor)

Capabilities. Open-core agent memory layer: extracts facts from conversations and agent actions, scopes them by user_id, agent_id and run_id, retrieves with semantic + BM25 + entity signals; add/search/get/update/delete/delete_all and per-memory history in OSS and Platform; storage in SQL + third-party vector store + entity store ^{vi372}.

Strengths

- Supports 14+ vector stores incl. PGVector, Elasticsearch, OpenSearch, MongoDB, so the canonical record can sit in the firm's L6 store ^{vi373}

³⁷² Verified fact — A3-S001: PyPI JSON metadata: mem0ai (Python Package Index (PyPI)) <https://pypi.org/pypi/mem0ai/json> accessed 2026-10-07; A3-S080: Mem0 docs source: Platform vs Open Source (Mem0 (GitHub)) <https://raw.githubusercontent.com/mem0ai/mem0/main/docs/platform/platform-vs-oss.mdx> accessed 2026-10-07; A3-S053: Mem0 docs: Platform vs Open Source (and OSS v2-to-v3 migration notes) <https://docs.mem0.ai/platform/platform-vs-oss> accessed 2026-10-07 [search-tool extract]

³⁷³ Verified fact — A3-S080: Mem0 docs source: Platform vs Open Source (Mem0 (GitHub)) <https://raw.githubusercontent.com/mem0ai/mem0/main/docs/platform/platform-vs-oss.mdx> accessed 2026-10-07; A3-S053: Mem0 docs: Platform vs Open Source (and OSS v2-to-v3 migration notes) <https://docs.mem0.ai/platform/platform-vs-oss> accessed 2026-10-07 [search-tool extract]

- Apache-2.0 OSS server with authentication on by default ^{VF374}
- Platform has org/project roles and a project-wide event feed of add/search/delete events usable for audit ^{VF375}
- Largest community in the layer (66,777 GitHub stars, 7 October 2026) ^{VF376}

Limitations and risks

- SOC 2 status conflicts on vendor pages (Type I; Type II in progress; Type II for Enterprise); trust-centre report not seen ^{VF377}
- No managed EU region found ^{VF378}
- OSS v2.0.0 (14 April 2026) removed external graph stores; graph memory, decay, temporal reasoning, Dream, webhooks and export are Platform-only ^{VF379}
- ADD-only OSS extraction: memories accumulate unless pruned ^{VF380}
- Vendor benchmark figures (LoCoMo 92.5, LongMemEval 94.4) are vendor-reported only ^{VF381}

Choose when

- You want a framework-neutral memory API self-hosted on your own approved vector store, behind your own policy gate ^{AJ}

Avoid when

- You would put regulated data on the managed Platform before a SOC 2 Type II report is obtained ^{AJ}
- You need a queryable graph in open source ^{AJ}

Nearest competitors: L5-zep, L5-supermemory, L5-aws-agentcore-memory, L5-cognee

Regulated-FS note. Self-host the OSS server against the approved vector store, implement the subject index and write gate in a firm-owned wrapper, and obtain the SOC 2 report before any Platform use ^{Rec}.

Facts (as of 2026-10-08)

³⁷⁴ Verified fact — A3-S001: PyPI JSON metadata: mem0ai (Python Package Index (PyPI)) <https://pypi.org/pypi/mem0ai/json> accessed 2026-10-07

³⁷⁵ Verified fact — A3-S080: Mem0 docs source: Platform vs Open Source (Mem0 (GitHub)) <https://raw.githubusercontent.com/mem0ai/mem0/main/docs/platform/platform-vs-oss.mdx> accessed 2026-10-07

³⁷⁶ Verified fact — A3-S054: GitHub repository metadata (stars, forks, created, archived) retrieved via GitHub MCP search_repositories tool, 7 Oct 2026 <https://api.github.com/search/repositories> accessed 2026-10-07 [search-tool extract]

³⁷⁷ Verified fact — A3-S051: Enterprise AI Memory Options That Pass Security Review (Mem0) <https://mem0.ai/guide/enterprise-ai-memory-security-review-options> accessed 2026-10-07 [search-tool extract]; A3-S106: Mem0 guide: Solving SOC 2 Compliance for Enterprise AI Memory (and related mem0.ai pages) <https://mem0.ai/guide/solving-soc-2-compliance-enterprise-ai-memory> accessed 2026-10-07 [search-tool extract]

³⁷⁸ Verified fact — A3-S106: Mem0 guide: Solving SOC 2 Compliance for Enterprise AI Memory (and related mem0.ai pages) <https://mem0.ai/guide/solving-soc-2-compliance-enterprise-ai-memory> accessed 2026-10-07 [search-tool extract]

³⁷⁹ Verified fact — A3-S081: Mem0 docs source: Open Source migration to the new memory algorithm (Mem0 (GitHub)) <https://raw.githubusercontent.com/mem0ai/mem0/main/docs/migration/oss-v2-to-v3.mdx> accessed 2026-10-07; V1-S043: Open Source: Migrating to the New Memory Algorithm; SDK changelog; Highlights (Mem0) <https://docs.mem0.ai/migration/oss-v2-to-v3> accessed 2026-10-08 [search-tool extract]; A3-S080: Mem0 docs source: Platform vs Open Source (Mem0 (GitHub)) <https://raw.githubusercontent.com/mem0ai/mem0/main/docs/platform/platform-vs-oss.mdx> accessed 2026-10-07

³⁸⁰ Verified fact — A3-S081: Mem0 docs source: Open Source migration to the new memory algorithm (Mem0 (GitHub)) <https://raw.githubusercontent.com/mem0ai/mem0/main/docs/migration/oss-v2-to-v3.mdx> accessed 2026-10-07

³⁸¹ Reported — A3-S081: Mem0 docs source: Open Source migration to the new memory algorithm (Mem0 (GitHub)) <https://raw.githubusercontent.com/mem0ai/mem0/main/docs/migration/oss-v2-to-v3.mdx> accessed 2026-10-07

Field	Value	Label	Sources
Company	Mem0 (start-up; Y Combinator S24 per project README)	Verified fact	A3-S001
Category	Agent memory layer: open-source memory library/server plus managed memory platform	Verified fact	A3-S001, A3-S080
Version / lineup	mem0ai 2.2.1 on PyPI (25 September 2026). New memory algorithm announced April 2026; open-source v2.0.0 (14 April 2026) introduced breaking changes	Verified fact	A3-S001, A3-S053, A3-S081
Licence	Open core: open-source SDK/server under Apache-2.0; Mem0 Platform is proprietary and carries Platform-only features (graph memory, memory decay, temporal reasoning, Dream consolidation, webhooks, memory export)	Verified fact	A3-S001, A3-S080
Status events	14 April 2026: open-source v2.0.0 removed all external graph store backends (Neo4j, Memgraph, Kuzu, Apache AGE, Neptune) and the enableGraph option (SDK changelog files the deletions under v2.0.1); OSS now uses built-in entity linking (a parallel entity collection that boosts retrieval, not a queryable graph); graph memory is a Platform-only feature; October 2025: US\$24M combined seed and Series A led by Basis Set Ventures (Reported); mem0ai/mem0-mcp repository archived (status on 7 October 2026)	Verified fact	A3-S053, A3-S081, A3-S052, A3-S054, V1-S043, V1-S081
Strategic direction	April 2026 algorithm: single-pass ADD-only extraction, entity linking, multi-signal retrieval (semantic + BM25 + entity) and temporal reasoning; vendor-reported LoCoMo 92.5 and LongMemEval 94.4 on the managed platform. Graph memory moved to Platform-only. Skills for coding agents include an 'OSS-to-Platform' migration skill	Verified fact	A3-S001, A3-S081
What it does	Extracts facts from conversations and agent actions, stores them as memories scoped by user_id, agent_id and run_id, and retrieves them with entity-aware ranking. Core operations add, search, get, update, delete, delete_all and per-memory history exist in both the open-source Memory class and the hosted MemoryClient	Verified fact	A3-S080
Stack position	Memory service sitting on top of retrieval infrastructure: stores facts in a SQL database, embeddings in a third-party vector database and entities in an entity store; the open-source build defaults to a local Qdrant instance	Verified fact	A3-S053, A3-S080
Integration	Python and JavaScript SDKs plus REST API (self-hosted server or hosted); integrations documented for LangGraph and CrewAI; vector store support includes Qdrant, Chroma, PGVector, Pinecone, Oracle, Milvus, MongoDB, Redis, Valkey, Elasticsearch, OpenSearch, Supabase, Upstash and Azure (Python)	Verified fact	A3-S001, A3-S080, A3-S053
Dependencies	Requires an LLM and an embedder; Python package depends on openai, qdrant-client and sqlalchemy; you provision and pay for the vector store, LLM and embedder when self-hosting	Verified fact	A3-S001, A3-S080
Certifications	Vendor pages conflict: SOC 2 Type I (homepage/About), 'Type II audit in progress' (one article) and 'Type II for Enterprise' (one guide); HIPAA claimed; Trust Center report not seen	Verified fact	A3-S051, A3-S106
GDPR / residency	Vendor states GDPR compliant; no managed EU region found; residency addressed through self-hosted/private-cloud/air-gapped deployment	Verified fact	A3-S051, A3-S106

Field	Value	Label	Sources
Security features	Vendor states BYOK encryption across deployment models; self-hosted server has authentication on by default	Verified fact	A3-S051, A3-S001
Access controls	Enterprise plan lists SSO and audit logs; Platform has multi-organisation/multi-project structure with member roles and a project-wide event feed (add/search/delete events) usable for audit; open source has no org/project concept and only per-memory history	Verified fact	A3-S049, A3-S080
Enterprise support	Enterprise plan with SLA support; Pro plan includes private Slack support	Verified fact	A3-S049
Pricing	Hobby free (10,000 adds / 1,000 retrievals per month); Starter US\$19 per month; Pro US\$249 per month (graph memory, Dream); Enterprise custom (on-prem, SSO, audit logs, SLA)	Verified fact	A3-S049
Infrastructure cost	Self-hosting cost is driven by the vector store, LLM calls for extraction and the embedder, all provisioned and paid for by the customer	Verified fact	A3-S080
Ecosystem	Broad vector-store support (14+ in Python); TypeScript SDK adds Amazon S3 Vectors and Neptune Analytics; LangGraph and CrewAI integrations	Verified fact	A3-S053, A3-S001
Adoption signals	66,777 GitHub stars (7 October 2026). Reported at Series A (October 2025): 14M downloads, API calls grew from 35M (Q1) to 186M (Q3 2025)	Verified fact	A3-S054, A3-S052
Maturity	First PyPI release 18 May 2024; 192 releases; v2.x with frequent releases (eight between 4 August and 25 September 2026)	Verified fact	A3-S001
Deployment	saas: Yes; managed_cloud: Yes (vendor states private Kubernetes deployment inside customer VPC); vpc_byoc: Yes (vendor claim); private_cloud: Yes (vendor claim: private Kubernetes); self_hosted: Yes (docker compose server; library); on_prem: Yes (Enterprise on-prem; vendor also claims air-gapped)		

Cognee (L5-cognee)

Tier: Tactical · **Flags:** none · **Original graphic label:** Cognee – knowledge graphs

Rationale: Credible in-estate graph-plus-vector memory with strong deployment options; no certification and seed-stage funding limit it to scoped, self-hosted use ^{A1}.

Criterion	Score	Rationale
Technical	4	Graph + vector memory with remember/recall/improve/forget and local extraction
Enterprise readiness	3	Rule 2 basis: authenticated multi-tenant users, Enterprise SSO, SLAs and a dedicated support engineer (A3-S005, A3-S095); one verified control gives 3 under rule 7 (CP3 review A, consistent with LlamaParse and Browserbase); RBAC and audit logs NPV
Security and compliance	2	Scored as self-hosted software (vendor-recommended) under rule 2: encryption, private mode, air-gap verified; security policy and signed releases NPV. Cognee Cloud alone would score 1 (no certification by vendor statement)

Criterion	Score	Rationale
Deployment flexibility	5	Cloud, BYOC, self-host, on-prem and air-gap
Ecosystem	3	MCP server and community plugins; 31,561 stars; optional standard graph and vector stores
Reliability and maturity	3	Post-1.0 (1.6.x), 178 releases since March 2024; seed-stage company; no acquisition found
Cost / TCO	3	Free OSS locally; Cloud US\$1 per 1M tokens; production Postgres graph licensed
Lock-in / portability	3	Apache-2.0 core on standard stores, but production Postgres graph is licensed
Total (generic / FS)	3.35 / 3.25	

Evidence rules applied: Rule 2 applied (self-hosted software; inherits host controls): security_compliance and enterprise_readiness scored on self-hosted basis, capped at 4; enterprise_readiness: NPV cap lifted to 3 by verified SSO (rule 7)

Capabilities. Open-source memory platform building knowledge graphs plus vectors from documents, code and conversations; remember, recall, improve and forget; retrieval selects graph, vector or code context; local small-model extraction; MCP server ^{vr382}.

Strengths

- Runs fully in-estate or air-gapped with zero external API calls ^{vr383}
- Backs onto Postgres/pgvector, Neo4j or Neptune ^{vr384}
- EU company with GDPR process audit, DPA, erasure and portability support; BYOC in customer EU cloud account ^{vr385}
- Enterprise SSO, SLAs and dedicated support engineer ^{vr386}

Limitations and risks

- States it holds no SOC 2, ISO 27001 or equivalent certification ^{vr387}
- Production Postgres graph store is a licensed product ^{vr388}
- RBAC and audit logs not documented ^{npv}
- Seed-stage funding; aggregators disagree on amount ^{vr389}

³⁸² Verified fact — A3-S005: PyPI JSON metadata: cognee (Python Package Index (PyPI)) <https://pypi.org/pypi/cognee/json> accessed 2026-10-07

³⁸³ Verified fact — A3-S095: Cognee Cloud: Security & Data Protection (docs) and cognee.ai FAQ/GDPR pages <https://docs.cognee.ai/cognee-cloud/functionality/data-and-security> accessed 2026-10-07 [search-tool extract]

³⁸⁴ Verified fact — A3-S005: PyPI JSON metadata: cognee (Python Package Index (PyPI)) <https://pypi.org/pypi/cognee/json> accessed 2026-10-07

³⁸⁵ Verified fact — A3-S095: Cognee Cloud: Security & Data Protection (docs) and cognee.ai FAQ/GDPR pages <https://docs.cognee.ai/cognee-cloud/functionality/data-and-security> accessed 2026-10-07 [search-tool extract]

³⁸⁶ Verified fact — A3-S095: Cognee Cloud: Security & Data Protection (docs) and cognee.ai FAQ/GDPR pages <https://docs.cognee.ai/cognee-cloud/functionality/data-and-security> accessed 2026-10-07 [search-tool extract]

³⁸⁷ Verified fact — A3-S095: Cognee Cloud: Security & Data Protection (docs) and cognee.ai FAQ/GDPR pages <https://docs.cognee.ai/cognee-cloud/functionality/data-and-security> accessed 2026-10-07 [search-tool extract]

³⁸⁸ Verified fact — A3-S005: PyPI JSON metadata: cognee (Python Package Index (PyPI)) <https://pypi.org/pypi/cognee/json> accessed 2026-10-07

³⁸⁹ Reported — A3-S094: Cognee: \$7.5 Million Seed Funding Raised (Feb 2026) (Pulse 2.0) <https://pulse2.com/cognee-7-5-million-seed-funding-raised-for-building-enterprise-grade-memory-layer-for-ai-agents/> accessed 2026-10-07 [search-tool extract]

Choose when

- Memory or graph-backed knowledge must be fully in-estate or air-gapped in the EU and you can supply the controls ^{AI}

Avoid when

- You want a managed service to carry certification evidence ^{AI}

Nearest competitors: L5-zep, L5-mem0, L5-supermemory

Regulated-FS note. Self-host only, on an approved Postgres or Neo4j, behind the firm's identity proxy and audit logging; treat Cognee Cloud as out of scope for client data until certified ^{Reg.}

Facts (as of 2026-10-07)

Field	Value	Label	Sources
Company	Cognee (Berlin-based; GitHub org topoteretes)	Verified fact	A3-S005, A3-S095
Category	Open-source AI memory platform building knowledge graphs plus vectors from documents, code and conversations	Verified fact	A3-S005
Version / lineup	cognee 1.6.3 (7 October 2026)	Verified fact	A3-S005
Licence	Open core: Apache-2.0 library; production-ready Postgres-as-graph store is a licensed product; Cognee Cloud managed service	Verified fact	A3-S005
Strategic direction	Since 1.0, the whole memory layer can run on one Postgres instance (demo feature in OSS); small-model local extraction (GLiNER) without an LLM key; plugins and MCP for agents	Verified fact	A3-S005
What it does	Turns documents, code and conversations into a self-hosted knowledge graph agents can search; operations include remember, recall, improve (enrich/apply feedback) and forget (remove an item or dataset); retrieval selects graph, vector or code context	Verified fact	A3-S005
Stack position	Spans L5 memory and L6/L8 (ingestion into graph and vector stores); graphic's 'knowledge graphs' descriptor is accurate	Verified fact	A3-S005
Integration	Python library, CLI, REST API (port 8000), UI and MCP server (port 8001) via Docker	Verified fact	A3-S005

Field	Value	Label	Sources
Dependencies	Default embedded stores: SQLite (relational), LanceDB (vector) and the 'ladybug' graph package; optional Neo4j, Amazon Neptune and Postgres/PGVector; Redis-compatible cache	Verified fact	A3-S005
Certifications	Cognee states it does not currently hold SOC 2, ISO 27001 or equivalent third-party certification; recommends self-hosting in customer's certified environment	Verified fact	A3-S095
GDPR / residency	EU company; GDPR-aligned processes audited with heyData; DPA on request; access, erasure and portability rights supported; Enterprise BYOC data plane in customer's EU cloud account; Cloud tenants each get a dedicated managed Postgres project	Verified fact	A3-S095
Security features	Encryption at rest and in transit; private deployment mode with zero external API calls; air-gapped self-hosting	Verified fact	A3-S095
Access controls	Multi-tenant mode with authenticated users by default (API); Enterprise SSO	Verified fact	A3-S005, A3-S095
Enterprise support	Enterprise: SSO, SLAs, dedicated support engineer, BYOC (fixed-scope engagement, custom price)	Verified fact	A3-S095
Pricing	Cloud free tier (1M tokens, one workspace); Standard US\$1.00 per 1M tokens processed plus US\$5 per extra workspace; Enterprise custom; self-hosted OSS free	Verified fact	A3-S095
Infrastructure cost	Can run free locally with small models; production graph on Postgres requires a licence	Verified fact	A3-S005
Ecosystem	Community plugins repository (cognee-community); MCP server	Verified fact	A3-S005, A3-S054
Adoption signals	topoteretes/cognee 31,561 GitHub stars (7 October 2026); US\$7.5M seed led by Pebblebed (February 2026, Reported) after US\$1.5M (November 2024)	Verified fact	A3-S054, A3-S094
Maturity	First PyPI release 13 March 2024; 178 releases; post-1.0 (1.6.x)	Verified fact	A3-S005
Deployment	saas: Yes (Cognee Cloud); managed_cloud: Not publicly verified; vpc_byoc: (Enterprise BYOC); private_cloud: Not publicly verified; self_hosted: Yes; on_prem: Yes (on-prem and air-gapped self-hosting)		

Amazon Bedrock AgentCore Memory (L5-aws-agentcore-memory)

Tier: Strategic · **Flags:** none · **Original graphic label:** not in graphic

Rationale: Strategic, conditional: where AWS is your primary cloud and the agent runtime is AgentCore (CP3 Q2, rubric rule 10). AWS's lead managed agent memory service with good lifecycle documentation; a customer-managed key at creation and a pruner from day one are part of the condition, and deployment and lock-in at 2 are accepted under rule 11 ¹⁰.

Criterion	Score	Rationale
Technical	4	Short- and long-term memory with multiple extraction strategies, namespaces and self-managed pipelines
Enterprise readiness	4	Hyperscaler presumption (CP2 Q1): AWS IAM, SSO federation and CloudTrail; AWS guidance on restricting memory APIs. Platform controls presumed (CP2 Q1); confirm per service

Criterion	Score	Rationale
Security and compliance	4	Service-level SOC 1/2/3 and ISO 27001 scope plus CMK; SOC 2 report type not stated and Memory not named separately, so 4 not 5 (rule 8)
Deployment flexibility	2	AWS managed only, with VPC/PrivateLink connectivity; no self-host
Ecosystem	3	Strands SDK and AgentCore APIs; wider ecosystem NPV
Reliability and maturity	3	GA October 2025 with steady feature additions
Cost / TCO	3	Published unit pricing for short- and long-term memory (B-REVA-S005), changed on 6 October 2026; extraction LLM cost on top; transparent, so 3 (consistent with Memory Bank)
Lock-in / portability	2	Proprietary AWS API; records listable for export; hyperscaler lock-in
Total (generic / FS)	3.30 / 3.20	

Capabilities. Managed memory in Amazon Bedrock AgentCore: short-term raw session events; long-term records via semantic, summary, user-preference, episodic or custom strategies, grouped by namespaces and retrieved by semantic search; self-managed strategy; record streaming; JSON-payload extraction (August 2026) ^{vf390}.

Strengths

- Detailed lifecycle documentation incl. per-user namespace erasure, batch delete and AWS guidance on retention policies ^{vf391}
- KMS encryption with optional customer-managed key; AWS Config rule and Security Hub control flag memories without CMK ^{vf392}
- AgentCore in AWS scope for SOC 1/2/3 and ISO/IEC 27001; HIPAA eligible ^{vf393}
- VPC and PrivateLink ^{vf394}

Limitations and risks

- No built-in TTL for long-term records; firm must build a pruner ^{vf395}

³⁹⁰ Verified fact — A3-S111: Designing lifecycle policies for AgentCore memory (and AgentCore Memory docs: harness-memory, delete-memory, UpdateMemory, memory-record-streaming) (Amazon Web Services) <https://aws.amazon.com/blogs/machine-learning/designing-lifecycle-policies-for-agentcore-memory/> accessed 2026-10-07 [search-tool extract]; A3-S047: Amazon Bedrock AgentCore is now generally available (Oct 2025) (Amazon Web Services) <https://aws.amazon.com/about-aws/whats-new/2025/10/amazon-bedrock-agentcore-available/> accessed 2026-10-07 [search-tool extract]; V1-S087: Amazon Bedrock AgentCore is now generally available; AgentCore FAQs; release notes (Amazon Web Services) <https://aws.amazon.com/about-aws/whats-new/2025/10/amazon-bedrock-agentcore-available/> accessed 2026-10-08 [search-tool extract]

³⁹¹ Verified fact — A3-S111: Designing lifecycle policies for AgentCore memory (and AgentCore Memory docs: harness-memory, delete-memory, UpdateMemory, memory-record-streaming) (Amazon Web Services) <https://aws.amazon.com/blogs/machine-learning/designing-lifecycle-policies-for-agentcore-memory/> accessed 2026-10-07 [search-tool extract]

³⁹² Verified fact — B-L5-S002: Encrypt your Amazon Bedrock AgentCore Memory (and AWS Config / Security Hub / SRA pages) (Amazon Web Services) <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/storage-encryption.html> accessed 2026-10-08 [search-tool extract]

³⁹³ Verified fact — B-L5-S003: Compliance validation for Amazon Bedrock AgentCore (and AWS Services in Scope pages) (Amazon Web Services) <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/compliance-validation.html> accessed 2026-10-08 [search-tool extract]

³⁹⁴ Verified fact — A3-S047: Amazon Bedrock AgentCore is now generally available (Oct 2025) (Amazon Web Services) <https://aws.amazon.com/about-aws/whats-new/2025/10/amazon-bedrock-agentcore-available/> accessed 2026-10-07 [search-tool extract]

³⁹⁵ Verified fact — A3-S111: Designing lifecycle policies for AgentCore memory (and AgentCore Memory docs: harness-memory, delete-memory, UpdateMemory, memory-record-streaming) (Amazon Web Services) <https://aws.amazon.com/blogs/machine-learning/designing-lifecycle-policies-for-agentcore-memory/> accessed 2026-10-07 [search-tool extract]

- Harness-managed memory cannot be deleted via the Memory APIs ^{vi-396}
- Customer-managed key cannot be changed after creation for harness-managed memory ^{vi-397}
- AWS-only proprietary API ^{vi-398}; memory pricing changed on 6 October 2026 (short-term now per GB) ^{vi-399}; EU Region availability for Memory not individually verified ^{vi-400} NPV

Choose when

- The agent runs on AgentCore or Strands in AWS ^{AJ}

Avoid when

- You need memory portable across clouds, or the runtime is elsewhere ^{AJ}

Nearest competitors: L5-gcp-vertex-memory-bank, L5-mem0, L5-zep

Regulated-FS note. Set a customer-managed key at creation, deploy the pruner on day one, confirm the Region list for UK/EU data, and keep a periodic export to the firm’s store; platform controls presumed (CP2 Q1), confirm per service ^{Rec.}

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Amazon Web Services	Verified fact	A3-S047
Category	Cloud-platform managed agent memory service	Verified fact	A3-S047
Version / lineup	Generally available since October 2025 as part of Amazon Bedrock AgentCore; AgentCore harness GA auto-provisions managed memory	Verified fact	A3-S047, A3-S048
Licence	Proprietary managed service	Verified fact	A3-S047
Status events	13 October 2025: AgentCore GA (Memory included; nine Regions at launch, fifteen by October 2026); August 2026: extraction from non-conversational JSON payloads	Verified fact	A3-S047, A3-S111, V1-S087
Strategic direction	Self-managed memory strategy gives control of extraction and consolidation pipelines; bundled into AgentCore harness	Verified fact	A3-S047, A3-S048
What it does	Short-term memory stores raw session events (messages, tool calls);	Verified	A3-S111

³⁹⁶ Verified fact — A3-S111: Designing lifecycle policies for AgentCore memory (and AgentCore Memory docs: harness-memory, delete-memory, UpdateMemory, memory-record-streaming) (Amazon Web Services) <https://aws.amazon.com/blogs/machine-learning/designing-lifecycle-policies-for-agentcore-memory/> accessed 2026-10-07 [search-tool extract]

³⁹⁷ Verified fact — B-L5-S002: Encrypt your Amazon Bedrock AgentCore Memory (and AWS Config / Security Hub / SRA pages) (Amazon Web Services) <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/storage-encryption.html> accessed 2026-10-08 [search-tool extract]

³⁹⁸ Verified fact — A3-S047: Amazon Bedrock AgentCore is now generally available (Oct 2025) (Amazon Web Services) <https://aws.amazon.com/about-aws/whats-new/2025/10/amazon-bedrock-agentcore-available/> accessed 2026-10-07 [search-tool extract]

³⁹⁹ Verified fact — B-REVA-S005: Amazon Bedrock AgentCore Pricing (Memory) (Amazon Web Services) <https://aws.amazon.com/bedrock/agentcore/pricing/> accessed 2026-10-08 [search-tool extract]

⁴⁰⁰ Verified fact — A3-S048: Release notes for Amazon Bedrock AgentCore (Amazon Web Services) <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/release-notes.html> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
	long-term memory extracts records via strategies (semantic, summary, user preference, episodic, custom model/prompt) retrievable by semantic search, grouped by namespaces	d fact	
Stack position	Platform-native alternative to independent L5 memory products	Architectural judgment	
Integration	AgentCore APIs (CreateMemory, ListMemoryRecords, DeleteMemoryRecord, BatchDeleteMemoryRecords); record streaming events; Strands Agents SDK integration; JSON-payload extraction (August 2026)	Verified fact	A3-S111
Dependencies	AWS account; part of Amazon Bedrock AgentCore	Verified fact	A3-S047
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	AgentCore available in 15 AWS Regions (EU regions not individually verified)	Verified fact	A3-S048
Security features	VPC, PrivateLink, CloudFormation and tagging across AgentCore; event expiry for short-term events (7-365 days per API reference, minimum value conflict); no built-in TTL for long-term records (AWS recommends a pruner using timestamp filters); per-user erasure by listing and deleting namespace records; deleting a memory resource removes all events and records; harness-managed memory cannot be deleted via Memory APIs	Verified fact	A3-S047, A3-S111
Access controls	AWS IAM (assumed from platform; not separately verified for Memory)	Architectural judgment	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Not publicly verified	Not publicly verified	
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicl	

Field	Value	Label	Sources
		y verifie d	
Adoption signals	Not publicly verified	Not publicl y verifie d	
Maturity	GA October 2025	Verifie d fact	A3-S047, V1-S087
Deployment	saas: Yes (AWS managed); managed_cloud: Yes; vpc_byoc: VPC and AWS PrivateLink supported; private_cloud: Not publicly verified; self_hosted: No; on_prem: No		

Vertex AI Agent Engine Memory Bank (newer docs: ‘Agent Platform Memory Bank’ in Gemini Enterprise Agent Platform) (L5-gcp-vertex-memory-bank)

Tier: Strategic · Flags: Renamed · Original graphic label: not in graphic

Rationale: Strategic, conditional: where Google Cloud is your primary cloud and the agent runtime is Agent Engine (CP3 Q2, rubric rule 10). Google’s lead managed agent memory service with a strong scope model; written confirmation of data residency (the feature table and residency terms conflict) and a regional CMEK endpoint are part of the condition ¹⁰.

Criterion	Score	Rationale
Technical	4	Extraction, consolidation, scoping, TTL and revisions
Enterprise readiness	4	Hyperscaler presumption (CP2 Q1): Google Cloud IAM, Cloud Audit Logs and SSO federation. Platform controls presumed (CP2 Q1); confirm per service
Security and compliance	4	Platform-level SOC 2, ISO 27001, ISO 42001 with product scope not stated (rule 8: one below the anchor of 5), plus CMEK and VPC-SC; residency conflict noted
Deployment flexibility	2	Google Cloud managed only; no self-host
Ecosystem	3	ADK, LangGraph and CrewAI samples
Reliability and maturity	3	Preview July 2025, GA December 2025, billing from January 2026; rebranding under way
Cost / TCO	3	Published per-memory pricing; extraction LLM billed separately
Lock-in / portability	2	Proprietary API; memories tied to Agent Runtime instance; Gemini dependency
Total (generic / FS)	3.30 / 3.20	

Capabilities. Managed long-term memory in Google's agent platform: Gemini-based asynchronous extraction of facts and preferences, consolidation with contradiction resolution, exact-scope retrieval with optional similarity search; TTL optional; revisions; purge by filter ^{vr401}.

Strengths

- Immutable scope with exact-scope retrieval ^{vr402}
- Consolidation resolves contradictions; purge by filter with dry run ^{vr403}
- Transparent unit pricing: US\$0.25 per 1,000 stored, US\$0.50 per 1,000 retrieved (LLM billed separately) ^{vr404}
- VPC-SC, CMEK and data residency at rest listed for Memory Bank ^{vr405}

Limitations and risks

- CMEK not available on the global endpoint; data residency terms list Memory Bank as excluded, conflicting with the feature table ^{vr406}
- TTL default none; memory revisions kept 365 days by default ^{vr407}
- Memories deleted with the Agent Runtime instance; extraction depends on Gemini ^{vr408}
- Platform certifications (SOC 2, ISO 27001, ISO 42001) stated per product or feature; Memory Bank inclusion not stated ^{vr409} ^{NPV}

Choose when

- The agent runs on ADK or Agent Engine in Google Cloud ^{AI}

Avoid when

⁴⁰¹ Verified fact — A3-S108: Vertex AI Memory Bank in public preview (8 Jul 2025) (Google Cloud Blog) <https://cloud.google.com/blog/products/ai-machine-learning/vertex-ai-memory-bank-in-public-preview> accessed 2026-10-07; A3-S109: Vertex AI Agent Engine Memory Bank overview / generate memories / Memory Bank REST reference (docs.cloud.google.com, now 'Agent Platform Memory Bank') (Google Cloud) <https://docs.cloud.google.com/vertex-ai/generative-ai/docs/agent-engine/memory-bank/overview> accessed 2026-10-07 [search-tool extract]

⁴⁰² Verified fact — A3-S109: Vertex AI Agent Engine Memory Bank overview / generate memories / Memory Bank REST reference (docs.cloud.google.com, now 'Agent Platform Memory Bank') (Google Cloud) <https://docs.cloud.google.com/vertex-ai/generative-ai/docs/agent-engine/memory-bank/overview> accessed 2026-10-07 [search-tool extract]

⁴⁰³ Verified fact — A3-S109: Vertex AI Agent Engine Memory Bank overview / generate memories / Memory Bank REST reference (docs.cloud.google.com, now 'Agent Platform Memory Bank') (Google Cloud) <https://docs.cloud.google.com/vertex-ai/generative-ai/docs/agent-engine/memory-bank/overview> accessed 2026-10-07 [search-tool extract]

⁴⁰⁴ Verified fact — V1-S088: Vertex AI release notes (Agent Engine Sessions and Memory Bank GA; pricing) (Google Cloud) <https://docs.cloud.google.com/vertex-ai/docs/release-notes> accessed 2026-10-08 [search-tool extract]

⁴⁰⁵ Verified fact — B-L5-S004: Scale your agents / Set up Memory Bank / release notes / data residency terms (Google Cloud) <https://docs.cloud.google.com/gemini-enterprise-agent-platform/scale> accessed 2026-10-08 [search-tool extract]

⁴⁰⁶ Verified fact — B-L5-S004: Scale your agents / Set up Memory Bank / release notes / data residency terms (Google Cloud) <https://docs.cloud.google.com/gemini-enterprise-agent-platform/scale> accessed 2026-10-08 [search-tool extract]

⁴⁰⁷ Verified fact — A3-S109: Vertex AI Agent Engine Memory Bank overview / generate memories / Memory Bank REST reference (docs.cloud.google.com, now 'Agent Platform Memory Bank') (Google Cloud) <https://docs.cloud.google.com/vertex-ai/generative-ai/docs/agent-engine/memory-bank/overview> accessed 2026-10-07 [search-tool extract]

⁴⁰⁸ Verified fact — A3-S109: Vertex AI Agent Engine Memory Bank overview / generate memories / Memory Bank REST reference (docs.cloud.google.com, now 'Agent Platform Memory Bank') (Google Cloud) <https://docs.cloud.google.com/vertex-ai/generative-ai/docs/agent-engine/memory-bank/overview> accessed 2026-10-07 [search-tool extract]; A3-S108: Vertex AI Memory Bank in public preview (8 Jul 2025) (Google Cloud Blog) <https://cloud.google.com/blog/products/ai-machine-learning/vertex-ai-memory-bank-in-public-preview> accessed 2026-10-07

⁴⁰⁹ Verified fact — A2-S043: Cloud compliance and regulations resources (Vertex AI) (Google Cloud) <https://cloud.google.com/vertex-ai/docs/general/vertexai-compliance> accessed 2026-10-07 [search-tool extract]

- You need cross-cloud portability or confirmed UK/EU at-rest residency before Google confirms it in writing ^{AJ}

Nearest competitors: L5-aws-agentcore-memory, L5-mem0, L5-zep

Regulated-FS note. Use a regional endpoint with CMEK, set TTL and a shorter revision retention, and obtain written residency confirmation; platform controls presumed (CP2 Q1), confirm per service ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Google Cloud	Verified fact	A3-S108
Category	Cloud-platform managed agent long-term memory service	Verified fact	A3-S108
Version / lineup	Public preview 8 July 2025; release notes state Sessions and Memory Bank generally available (date not captured); usage charging from 28 January 2026	Verified fact	A3-S108, A3-S109
Licence	Proprietary managed service	Verified fact	A3-S108
Status events	8 July 2025: public preview; December 2025: Agent Engine Sessions and Memory Bank GA (Vertex AI release notes; exact day not captured); 28 January 2026: billing starts	Verified fact	A3-S108, A3-S109, V1-S088
Strategic direction	Integrated with Agent Development Kit and Agent Engine Sessions; samples for LangGraph and CrewAI; rebranding under Gemini Enterprise Agent Platform	Verified fact	A3-S108, A3-S109
What it does	Uses Gemini models to extract facts, preferences and context asynchronously from conversation history, consolidates them with existing memories (resolving contradictions), and retrieves them by scope (e.g. user ID) with optional embedding similarity search	Verified fact	A3-S108, A3-S109
Stack position	Platform-native alternative to independent L5 memory products	Architectural judgement	

Field	Value	Label	Sources
Integration	REST API and Vertex AI SDK; ADK memory service; LangGraph and CrewAI samples	Verified fact	A3-S108, A3-S109
Dependencies	Google Cloud project; Agent Engine Sessions as a source; Gemini models	Verified fact	A3-S108
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Scope is immutable and retrieval returns only exact-scope matches; optional TTL (default none; memory revisions default 365 days); delete by name and purge by filter with dry run; consolidation may delete contradicted memories or honour explicit 'forget' instructions; deleting the Agent Runtime instance deletes built-in memories	Verified fact	A3-S109
Access controls	Google Cloud IAM (assumed from platform; not separately verified)	Architectural judgement	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	US\$0.25 per 1,000 memories stored and US\$0.50 per 1,000 memories retrieved; LLM usage for memory generation billed separately; charging began 28 January 2026	Verified fact	A3-S109, V1-S088
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	ADK, LangGraph, CrewAI	Verified fact	A3-S108
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Preview 8 July 2025; GA December 2025; billing from 28 January 2026	Verified fact	A3-S108, A3-S109, V1-S088
Deployment	saas: Yes (Google Cloud managed); managed_cloud: Yes; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: No;		

Field	Value	Label	Sources
	on_prem: No		

Supermemory (L5-supermemory)

Tier: Experimental · Flags: none · Original graphic label: Supermemory – memory API

Rationale: Fast-moving combined memory-and-RAG API with a proprietary server, seed funding and an API that changed major version this week ^{AJ}.

Criterion	Score	Rationale
Technical	4	Memory, profiles, forgetting and hybrid RAG in one API
Enterprise readiness	3	SSO on Enterprise verified (protocols undocumented) (A3-S096): one verified control lifts the cap to 3 under rule 7 (CP3 review A, consistent with LlamaParse and Browserbase); RBAC and audit logs NPV
Security and compliance	2	SOC 2 claimed from Scale tier with report type unconfirmed; HIPAA Enterprise; DPA
Deployment flexibility	5	SaaS, dedicated, customer cloud, self-host (Scale+), air-gap (Enterprise)
Ecosystem	4	Framework wrappers, MCP, coding-agent plugins
Reliability and maturity	2	SDK first released April 2025; v5 breaking change 6 October 2026; seed only; no acquisition found
Cost / TCO	3	Published tiers (Pro US\$19 to Scale US\$399 per month); self-hosting only on higher tiers
Lock-in / portability	2	Proprietary server binary and API; namespace API just changed
Total (generic / FS)	3.30 / 3.05	

Evidence rules applied: security_compliance at 2: SOC 2 report type not confirmed (pricing page only); enterprise_readiness: NPV cap lifted to 3 by verified SSO (rule 7)

Capabilities. Memory and context engine API: fact extraction, user profiles, temporal and contradiction handling, automatic forgetting, hybrid search over documents and memories, connectors and file processing; v5 namespace-first API ^{vr410}.

Strengths

- Explicit forget operations: forget, forget_matching, namespace delete, automatic expiry ^{vr411}
- Wide deployment range incl. customer-cloud dedicated deployments and air-gapped Enterprise ^{vr412}
- Many framework wrappers and an MCP server ^{vr413}

⁴¹⁰ Verified fact — A3-S064: supermemoryai/supermemory README (Supermemory (GitHub)) <https://raw.githubusercontent.com/supermemoryai/supermemory/main/README.md> accessed 2026-10-07; A3-S012: PyPI JSON metadata: supermemory (Python Package Index (PyPI)) <https://pypi.org/pypi/supermemory/json> accessed 2026-10-07

⁴¹¹ Verified fact — A3-S012: PyPI JSON metadata: supermemory (Python Package Index (PyPI)) <https://pypi.org/pypi/supermemory/json> accessed 2026-10-07; A3-S064: supermemoryai/supermemory README (Supermemory (GitHub)) <https://raw.githubusercontent.com/supermemoryai/supermemory/main/README.md> accessed 2026-10-07

⁴¹² Verified fact — A3-S122: Supermemory docs: Security & compliance (and product page) <https://supermemory.ai/docs/overview/security> accessed 2026-10-07 [search-tool extract]; A3-S096: Supermemory pricing <https://supermemory.ai/pricing/> accessed 2026-10-07 [search-tool extract]

Limitations and risks

- Self-hosted server binary not open source; hosted API proprietary ^{VF414}
- SOC 2 report type not confirmed on pricing page; no managed EU region found ^{VF415}
- SSO protocols undocumented; RBAC and audit logs not documented ^{VF416 NPV}
- v5 breaking API change on 6 October 2026 ^{VF417}
- Bundles memory and knowledge behind one API, blurring owner-curated and agent-written content ^{AJ}
- Benchmark leadership is a vendor claim ^{R418}

Choose when

- A team prototype needs memory plus RAG quickly, outside regulated data ^{AJ}

Avoid when

- Memory and knowledge need different owners, approvals and retention, or you need a stable API ^{AJ}

Nearest competitors: L5-mem0, L5-zep, L5-cognee

Regulated-FS note. Not for client or personal data until SOC 2 report type, audit logging and EU residency are confirmed ^{REC}.

Facts (as of 2026-10-07)

Field	Value	Label	Sources
Company	Supermemory	Verified fact	A3-S064
Category	Memory and context engine API: memory extraction, user profiles, hybrid RAG search, connectors and file processing	Verified fact	A3-S064
Version / lineup	Python SDK 5.0.0 (6 October 2026) for the namespace-first v5 API (breaking change from 3.x)	Verified fact	A3-S012
Licence	Open core: repository MIT and SDK Apache-2.0, but the self-hosted server binary is not open source (free within a lite licence); hosted API proprietary	Verified fact	A3-S066, A3-S012, A3-S096
Status events	6 October 2026: v5 SDK moves to namespace-first API (MIGRATION from 3.x required)	Verified fact	A3-S012
Strategic direction	Bundles memory with RAG, connectors and multimodal extractors in one API; plugins for Claude Code, Cursor, Codex, OpenCode and others;	Verified fact	A3-S064

⁴¹³ Verified fact — A3-S064: supermemoryai/supermemory README (Supermemory (GitHub)) <https://raw.githubusercontent.com/supermemoryai/supermemory/main/README.md> accessed 2026-10-07

⁴¹⁴ Verified fact — A3-S066: Supermemory LICENSE (MIT) (Supermemory (GitHub)) <https://raw.githubusercontent.com/supermemoryai/supermemory/main/LICENSE> accessed 2026-10-07; A3-S096: Supermemory pricing <https://supermemory.ai/pricing/> accessed 2026-10-07 [search-tool extract]

⁴¹⁵ Verified fact — A3-S096: Supermemory pricing <https://supermemory.ai/pricing/> accessed 2026-10-07 [search-tool extract]; A3-S122: Supermemory docs: Security & compliance (and product page) <https://supermemory.ai/docs/overview/security> accessed 2026-10-07 [search-tool extract]

⁴¹⁶ Verified fact — A3-S096: Supermemory pricing <https://supermemory.ai/pricing/> accessed 2026-10-07 [search-tool extract]

⁴¹⁷ Verified fact — A3-S012: PyPI JSON metadata: supermemory (Python Package Index (PyPI)) <https://pypi.org/pypi/supermemory/json> accessed 2026-10-07

⁴¹⁸ Reported — A3-S064: supermemoryai/supermemory README (Supermemory (GitHub)) <https://raw.githubusercontent.com/supermemoryai/supermemory/main/README.md> accessed 2026-10-07

Field	Value	Label	Sources
	vendor claims #1 on LongMemEval, LoCoMo and ConvoMem		
What it does	Extracts facts from conversations, maintains user profiles, handles temporal changes and contradictions, automatically forgets expired information and returns context via hybrid search over documents and memories	Verified fact	A3-S064
Stack position	Combines L5 memory, L6 retrieval and L8 ingestion (connectors, OCR, transcription) behind one API	Verified fact	A3-S064
Integration	REST API, Python/TypeScript SDKs, hosted MCP server, framework wrappers (Vercel AI SDK, LangChain, LangGraph, OpenAI Agents SDK, Mastra, Agno, Claude Memory Tool, n8n); connectors for Google Drive, Gmail, Notion, OneDrive and GitHub	Verified fact	A3-S064, A3-S012
Dependencies	Self-hosted build embeds a graph engine and local embeddings (bge-base-en-v1.5 by default) and stores data in a local directory; any OpenAI-compatible LLM	Verified fact	A3-S064
Certifications	SOC 2 (from Scale tier) and HIPAA on Enterprise per pricing page; blog posts say SOC 2 Type 2 (report type not confirmed on pricing page)	Verified fact	A3-S096
GDPR / residency	Security page states GDPR compliant with access and erasure workflows; DPA signed on request; no managed EU region found; dedicated deployments in customer AWS/GCP/Azure for residency	Verified fact	A3-S122, A3-S096
Security features	Not publicly verified	Not publicly verified	
Access controls	SSO on Enterprise tier (protocols not documented); namespaces isolate content	Verified fact	A3-S096, A3-S012
Enterprise support	Enterprise: custom contracts, DPA, SSO, custom integrations	Verified fact	A3-S096
Pricing	Free (US\$5 monthly credits); Pro US\$19; Max US\$100; Scale US\$399 per month; Enterprise custom	Verified fact	A3-S096
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Plugins for coding agents and MCP clients; framework wrappers listed above	Verified fact	A3-S064
Adoption signals	supermemoryai/supermemory 31,146 GitHub stars (7 October 2026); seed US\$2.6M (October 2025, later reported as US\$3M) led by Susa Ventures, Browder Capital and SF1.vc (Reported)	Verified fact	A3-S054, A3-S097
Maturity	Python SDK first released 29 April 2025; 87 releases; v5 API just released	Verified fact	A3-S012
Deployment	saas: Yes; managed_cloud: Not publicly verified; vpc_byoc: Yes (dedicated deployment in customer AWS, GCP or Azure account); private_cloud: Yes (dedicated infrastructure run by Supermemory); self_hosted: Yes: Supermemory local free for individuals within a lite licence (server binary not open source); organisational self-hosting on		

Field	Value	Label	Sources
	Scale and Enterprise; on_prem: Yes (Enterprise air-gapped self-hosting)		

Letta (formerly MemGPT); current product is Letta Code, a stateful agent harness, with Letta Cloud (L5-letta)

Tier: Experimental · **Flags:** Superseded · **Original graphic label:** Letta – stateful agents

Rationale: Repositioned as an agent harness with changing architecture and no certification; belongs with L3 harness evaluation ^{AJ}.

Criterion	Score	Rationale
Technical	3	Memory blocks and MemFS within its own harness; not usable as a memory layer for other frameworks
Enterprise readiness	3	RBAC and SAML/OIDC SSO on Enterprise (two of three); audit logs NPV; dedicated support
Security and compliance	2	No certification found
Deployment flexibility	3	Letta Cloud SaaS and self-hosted server; on-prem and VPC NPV
Ecosystem	3	Model-agnostic; skills ecosystem; but a harness rather than an integration point
Reliability and maturity	2	Pre-1.0 Letta Code launched October 2025; architecture changed twice; no acquisition found
Cost / TCO	4	Apache-2.0; Pro US\$20 per month; free with own model keys
Lock-in / portability	3	Apache-2.0 and git-backed memory files are portable, but agent state defaults to Letta Cloud and AgentFile export was removed
Total (generic / FS)	2.85 / 2.75	

Evidence rules applied: security_compliance capped at 2: certifications NPV (none found, A3-S118)

Capabilities. Stateful agent harness (Letta Code) with memory blocks and all context tracked in a git-backed memory filesystem (MemFS) that can sync to GitHub; Letta Cloud default backend or self-hosted 'letta server'; not a standalone memory layer ^{vf419}.

Strengths

- Memory as version-controlled files: every change is a reviewable diff ^{vf420}
- Model-agnostic; works with OpenAI, Anthropic, Z.ai and local models ^{vf421}
- Enterprise RBAC and SAML/OIDC SSO ^{vf422}

⁴¹⁹ Verified fact — A3-S073: letta-ai/letta-code README (Letta (GitHub)) <https://raw.githubusercontent.com/letta-ai/letta-code/main/README.md> accessed 2026-10-07; A3-S060: letta-ai/letta README (Letta (GitHub)) <https://raw.githubusercontent.com/letta-ai/letta/main/README.md> accessed 2026-10-07; A3-S093: Letta's Next Phase (Mar 2026) <https://www.letta.com/blog/our-next-phase/> accessed 2026-10-07 [search-tool extract]

⁴²⁰ Verified fact — A3-S073: letta-ai/letta-code README (Letta (GitHub)) <https://raw.githubusercontent.com/letta-ai/letta-code/main/README.md> accessed 2026-10-07

⁴²¹ Verified fact — A3-S073: letta-ai/letta-code README (Letta (GitHub)) <https://raw.githubusercontent.com/letta-ai/letta-code/main/README.md> accessed 2026-10-07

- Apache-2.0 licence ^{VF423}

Limitations and risks

- Pivot in March 2026: server memory tools, templates, server-side sleep-time agents and tool rules deprecated; V1 server retired ^{VF424}
- AgentFile export/import removed ^{VF425}
- No SOC 2 or trust centre found; privacy policy states no specific retention periods ^{VF426}
- Pre-1.0 with very rapid releases (eight in six days) ^{VF427}
- Git history retains deleted content unless rewritten, complicating erasure ^{AJ}

Choose when

- Evaluating a stateful coding or operations agent outside regulated client data ^{AJ}

Avoid when

- You need a memory service for an existing LangGraph, ADK or Strands workflow ^{AJ}
- Certifications are a gate ^{AJ}

Nearest competitors: L5-langmem, L5-aws-agentcore-memory, L5-gcp-vertex-memory-bank

Regulated-FS note. Copy the git-backed memory idea for governed style memory in the firm’s own repository; do not adopt the product as a regulated-workflow dependency today ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Letta (letta-ai)	Verified fact	A3-S060
Category	Stateful agent harness/runtime with self-editing memory; not a standalone memory library	Verified fact	A3-S073
Version / lineup	Letta Code 0.34.4 (4 October 2026; PyPI ‘letta’ and npm ‘@letta-ai/letta-code’); letta-client Python SDK 1.12.1 (2 June 2026)	Verified fact	A3-S004, A3-S077, A3-S014
Licence	Open source Apache-2.0 (Letta Code); Letta Cloud hosted service	Verified fact	A3-S004, A3-S077, A3-S073

⁴²² Verified fact — A3-S091: Letta pricing (docs) <https://docs.letta.com/pricing> accessed 2026-10-07 [search-tool extract]

⁴²³ Verified fact — A3-S004: PyPI JSON metadata: letta (Python Package Index (PyPI)) <https://pypi.org/pypi/letta/json> accessed 2026-10-07; A3-S077: npm registry metadata: @letta-ai/letta-code <https://registry.npmjs.org/@letta-ai/letta-code> accessed 2026-10-07

⁴²⁴ Verified fact — A3-S093: Letta’s Next Phase (Mar 2026) <https://www.letta.com/blog/our-next-phase/> accessed 2026-10-07 [search-tool extract]; A3-S060: letta-ai/letta README (Letta (GitHub)) <https://raw.githubusercontent.com/letta-ai/letta/main/README.md> accessed 2026-10-07

⁴²⁵ Verified fact — A3-S073: letta-ai/letta-code README (Letta (GitHub)) <https://raw.githubusercontent.com/letta-ai/letta-code/main/README.md> accessed 2026-10-07

⁴²⁶ Verified fact — A3-S118: Letta Privacy Policy (and docs on agent state persistence) <https://www.letta.com/privacy-policy/> accessed 2026-10-07 [search-tool extract]

⁴²⁷ Verified fact — A3-S077: npm registry metadata: @letta-ai/letta-code <https://registry.npmjs.org/@letta-ai/letta-code> accessed 2026-10-07; A3-S004: PyPI JSON metadata: letta (Python Package Index (PyPI)) <https://pypi.org/pypi/letta/json> accessed 2026-10-07

Field	Value	Label	Sources
Status events	Letta V1 API server retired to an 'archive' branch; active source moved to letta-ai/letta-code (status on 7 October 2026); PyPI package 'letta' now installs the Letta Code CLI, 'not a Python SDK or API server'; AgentFile (.af) export/import deprecated and removed from Letta Code; lettabot repository archived, replaced by Letta Code channels	Verified fact	A3-S060, A3-S004, A3-S073, A3-S054, V1-S081
Strategic direction	'Letta's Next Phase' (March 2026): Letta Code as model-agnostic agent harness with git-backed memory (MemFS); legacy server memory tools (e.g. core_memory_replace), templates, server-side sleep-time agents and tool rules deprecated. Letta Code launched December 2025; desktop app April 2026	Verified fact	A3-S093, A3-S073, V1-S042
What it does	Runs long-lived agents with memory, identity and conversation history; memory blocks and all context are tracked in a git-backed memory filesystem (MemFS) that can sync to a GitHub repository; supports message search across agents, subagents, hooks, permissions and secrets	Verified fact	A3-S073
Stack position	Straddles L3 (agent harness) and L5 (memory); the graphic's 'stateful agents' descriptor is accurate, but Letta is not a drop-in memory layer for other frameworks	Verified fact	A3-S073
Integration	Letta Agent SDK (TypeScript), letta-client for the Letta API (Python), 'letta server' App Server, Agent Skills folders (.agents/skills), messaging channels	Verified fact	A3-S060, A3-S004, A3-S073
Dependencies	Bring-your-own LLM provider keys; Letta Cloud is the default backend storing agent state; local mode keeps agents on disk; Python wheel bundles a Node.js runtime	Verified fact	A3-S004, A3-S073
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	Privacy policy: model inputs/outputs retained only for hosted services; no specific retention periods; agent state (memories, messages, tool calls) persisted in a database and retrievable after compaction	Verified fact	A3-S118
Security features	Secrets feature exposes secrets as environment variables while obfuscating values from context (requires Letta sign-in)	Verified fact	A3-S073
Access controls	Enterprise tier: RBAC and SAML/OIDC SSO; Letta Code permission modes for agent actions	Verified fact	A3-S091, A3-S073
Enterprise support	Enterprise tier with dedicated support	Verified fact	A3-S091
Pricing	Pro US\$20/month with usage quota and pay-as-you-go overage (up to 20 stateful agents); LLM usage at underlying token cost; free with own model keys; Enterprise volume pricing	Verified fact	A3-S091
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Works with OpenAI, Anthropic, Z.ai and local models; installs skills from	Verified fact	A3-S073

Field	Value	Label	Sources
	GitHub, ClawHub and Hermes Skills Hub	d fact	
Adoption signals	letta-ai/letta 25,070 and letta-code 3,541 GitHub stars (7 October 2026). US\$10M seed led by Felicis (September 2024, Reported); no later round found	Verified fact	A3-S054, A3-S092
Maturity	Letta Code first published 24 October 2025; pre-1.0 with very rapid releases (eight releases 29 September to 4 October 2026)	Verified fact	A3-S077, A3-S004
Deployment	saas: Yes (Letta Cloud); managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes ('letta server' for local or self-hosted agents); on_prem: Not publicly verified		

LangMem (L5-langmem)

Tier: Experimental • **Flags:** Not publicly verified • **Original graphic label:** LangMem – long-term

Rationale: Pre-1.0 and apparently stalled; useful as patterns, not as a dependency ^{AJ}.

Criterion	Score	Rationale
Technical	3	Extraction, consolidation and procedural memory, but no lifecycle or erasure features documented
Enterprise readiness	2	Rule 2: enables memory inside LangGraph; no commercial support for LangMem itself verified
Security and compliance	2	Rule 2: security policy, CVE handling and signed releases NPV
Deployment flexibility	4	Library runs wherever LangGraph and its store run, incl. in-estate
Ecosystem	3	LangGraph and LangChain only
Reliability and maturity	1	Pre-1.0 (0.0.x) with no release for about 11 months
Cost / TCO	4	MIT and free; cost is store plus extraction LLM; a stalled library may need in-house maintenance
Lock-in / portability	3	MIT, but bound to LangGraph store abstractions
Total (generic / FS)	2.75 / 2.65	

Evidence rules applied: Rule 2 applied (self-hosted library; inherits host controls): security_compliance and enterprise_readiness capped at 4

Capabilities. MIT library of memory utilities for LangGraph: extraction, hot-path memory tools, background memory manager and prompt refinement (procedural memory); persists via LangGraph BaseStore ^{vr⁴²⁸}.

Strengths

- Clear example of procedural memory (prompt refinement) and background consolidation ^{vr⁴²⁹}

⁴²⁸ Verified fact — A3-S006: PyPI JSON metadata: langmem (Python Package Index (PyPI)) <https://pypi.org/pypi/langmem/json> accessed 2026-10-07

⁴²⁹ Verified fact — A3-S006: PyPI JSON metadata: langmem (Python Package Index (PyPI)) <https://pypi.org/pypi/langmem/json> accessed 2026-10-07

- No service to operate; MIT licence ^{VF}⁴³⁰

Limitations and risks

- 0.0.30, no PyPI release since 27 October 2025 ^{VF}⁴³¹
- Tied to LangGraph store abstractions; InMemoryStore loses data on restart ^{VF}⁴³²
- Whether LangChain has superseded it is not verified ^{NPV}
- Security and access control inherit entirely from the store and platform ^{AJ}

Choose when

- Already on LangGraph and want reference patterns, accepting you may maintain the code ^{AJ}

Avoid when

- You need a supported dependency ^{AJ}

Nearest competitors: L5-mem0, L5-zep, L5-letta

Regulated-FS note. Never let automatic prompt refinement change production prompts; route proposals through C5 change control ^{Rec}.

Facts (as of 2026-10-07)

Field	Value	Label	Sources
Company	LangChain	Verified fact	A3-S006
Category	Open-source memory utilities library for LangGraph agents	Verified fact	A3-S006
Version / lineup	langmem 0.0.30 (27 October 2025); no newer PyPI release as of 7 October 2026	Verified fact	A3-S006
Licence	Open source, MIT	Verified fact	A3-S006
Strategic direction	Not publicly verified	Not publicly verified	
What it does	Extracts information from conversations, provides hot-path memory management/search tools for agents, a background memory manager that consolidates knowledge, and prompt refinement to optimise agent behaviour	Verified fact	A3-S006
Stack position	Library inside L3 (LangGraph) using LangGraph’s long-term memory	Verified	A3-S006

⁴³⁰ Verified fact — A3-S006: PyPI JSON metadata: langmem (Python Package Index (PyPI)) <https://pypi.org/pypi/langmem/json> accessed 2026-10-07

⁴³¹ Verified fact — A3-S006: PyPI JSON metadata: langmem (Python Package Index (PyPI)) <https://pypi.org/pypi/langmem/json> accessed 2026-10-07

⁴³² Verified fact — A3-S006: PyPI JSON metadata: langmem (Python Package Index (PyPI)) <https://pypi.org/pypi/langmem/json> accessed 2026-10-07

Field	Value	Label	Sources
	store; not a separate memory service	d fact	
Integration	Python API; native integration with LangGraph long-term memory store, available in LangGraph Platform deployments	Verified fact	A3-S006
Dependencies	langgraph, langchain-core, langsmith and trustcall; persistence via LangGraph BaseStore (InMemoryStore loses data on restart)	Verified fact	A3-S006
Certifications	Not applicable (library); inherits controls of the hosting platform and store	Architectural judgment	
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Free (MIT library)	Verified fact	A3-S006
Infrastructure cost	Cost driven by the chosen LangGraph store backend and LLM calls for extraction	Verified fact	A3-S006
Ecosystem	LangGraph/LangChain ecosystem	Verified fact	A3-S006
Adoption signals	langchain-ai/langmem 1,696 GitHub stars (7 October 2026)	Verified fact	A3-S054
Maturity	Pre-1.0 (0.0.x); last PyPI release 27 October 2025, about 11 months before review date	Verified fact	A3-S006
Deployment	saas: No standalone SaaS (runs inside LangGraph Platform deployments); managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes (library); on_prem: Yes (library)		

L4: Tools, protocols and connectivity

Agent2Agent (A2A) Protocol (L4-a2a)

Tier: Strategic · Flags: none · Original graphic label: A2A – agent-to-agent

Rationale: Strategic, conditional: where cross-team or cross-vendor agent delegation is in scope (CP3 Q1); signed Agent Cards verified, and delegated authority scoped and revocable by the firm, because the protocol does not define it ^{AJ}.

Criterion	Score	Rationale
Technical	3	Covers agent-to-agent interop well; not the main tool-invocation path and delegation semantics incomplete ^{AJ}
Enterprise readiness	3	Open spec (rule 2): authenticated extended cards and declared security schemes; support via implementers ^{vr433}
Security and compliance	3	Rule 2 hygiene: modern OAuth flows, optional signing, undefined revocation semantics ^{vr434}
Deployment flexibility	5	Implementations self-hostable anywhere ^{AJ}
Ecosystem	4	150+ organisations and cloud availability (April 2026) ^{vr435}
Reliability and maturity	3	1.0 only since March 2026 with breaking changes; foundation-hosted ^{vr436}
Cost / TCO	4	Free specification; operations cost in implementations ^{AJ}
Lock-in / portability	5	Apache-2.0 with neutral multi-vendor governance ^{vr437}
Total (generic / FS)	3.60 / 3.70	

Capabilities. Open protocol (Apache-2.0) for communication between opaque agents: Agent Card discovery, messages, long-running tasks with streaming and push notifications; v1.0 adds multi-tenancy, multiple bindings, optional JWS-signed Agent Cards and modernised OAuth flows ^{vr438}.

⁴³³ Verified fact — A3-S078: A2A Protocol Specification (source, latest released version 1.0.0) (A2A Project (GitHub)) <https://raw.githubusercontent.com/a2aproject/A2A/main/docs/specification.md> accessed 2026-10-07

⁴³⁴ Verified fact — A3-S078: A2A Protocol Specification (source, latest released version 1.0.0) (A2A Project (GitHub)) <https://raw.githubusercontent.com/a2aproject/A2A/main/docs/specification.md> accessed 2026-10-07; A3-S079: A2A CHANGELOG (A2A Project (GitHub)) <https://raw.githubusercontent.com/a2aproject/A2A/main/CHANGELOG.md> accessed 2026-10-07

⁴³⁵ Verified fact — A3-S025: A2A Protocol Surpasses 150 Organizations, Lands in Major Cloud Platforms, and Sees Enterprise Production Use in First Year (Linux Foundation) <https://www.linuxfoundation.org/press/a2a-protocol-surpasses-150-organizations-lands-in-major-cloud-platforms-and-sees-enterprise-production-use-in-first-year> accessed 2026-10-07 [search-tool extract]

⁴³⁶ Verified fact — A3-S079: A2A CHANGELOG (A2A Project (GitHub)) <https://raw.githubusercontent.com/a2aproject/A2A/main/CHANGELOG.md> accessed 2026-10-07; A3-S116: A2A joins AaIF's open agentic stack (17 Aug 2026) (Agentic AI Foundation (AaIF)) <https://aaif.io/blog/a2a-joins-aaif> accessed 2026-10-07 [search-tool extract]

⁴³⁷ Verified fact — A3-S075: PyPI JSON metadata: a2a-sdk (Python Package Index (PyPI)) <https://pypi.org/pypi/a2a-sdk/json> accessed 2026-10-07; A3-S065: A2A GOVERNANCE.md (A2A Project (GitHub)) <https://raw.githubusercontent.com/a2aproject/A2A/main/GOVERNANCE.md> accessed 2026-10-07

⁴³⁸ Verified fact — A3-S078: A2A Protocol Specification (source, latest released version 1.0.0) (A2A Project (GitHub)) <https://raw.githubusercontent.com/a2aproject/A2A/main/docs/specification.md> accessed 2026-10-07; A3-S030: A2A Protocol Ships v1.0: Production-Ready Standard (A2A Project (Linux Foundation)) <https://a2a-protocol.org/latest/announcing-1.0/> accessed 2026-10-07 [search-tool extract]; A3-S079: A2A CHANGELOG (A2A Project (GitHub)) <https://raw.githubusercontent.com/a2aproject/A2A/main/CHANGELOG.md> accessed 2026-10-07

Strengths

- Multi-vendor TSC of eight companies and AAIF Growth Stage status ^{vf⁴³⁹}
- Designed for delegated long-running tasks between agents ^{vf⁴⁴⁰}
- Signed Agent Cards available ^{vf⁴⁴¹}

Limitations and risks

- Card signing optional; clients SHOULD verify only when present ^{vf⁴⁴²}
- No defined scope, validity or revocation semantics for authorisation obtained in AUTH_REQUIRED ^{vf⁴⁴³}
- 1.0 broke the interaction protocol; some SDKs lag ^{vf⁴⁴⁴ r⁴⁴⁵}

Choose when

- Agents owned by different teams or vendors must delegate work to each other ^{aj}

Avoid when

- A deterministic workflow calling tools would do ^{aj}

Nearest competitors: L4-mcp, C1-agentgateway, L4-aws-agentcore-gateway-identity

Regulated-FS note. Require signed Agent Cards and verify them; scope and revoke delegated authority in your own gateway and IdP ^{Rec.}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	A2A Project: contributed by Google to the Linux Foundation (2025); accepted as a Growth Stage project of the Agentic AI Foundation (AAIF announcement 17 August 2026; A2A blog 27 August 2026); TSC seats: Google, Microsoft, Cisco, AWS, Salesforce, ServiceNow, SAP, IBM	Verified fact	A3-S065, A3-S116, A3-S117, A3-S025, V1-S039
Category	Open protocol for communication and interoperability between opaque agent applications	Verified fact	A3-S078
Version /	Specification 1.0.0 (12 March 2026), patch 1.0.1 (26 May 2026); A2A	Verified fact	A3-S079,

⁴³⁹ Verified fact — A3-S065: A2A GOVERNANCE.md (A2A Project (GitHub)) <https://raw.githubusercontent.com/a2aproject/A2A/main/GOVERNANCE.md> accessed 2026-10-07; A3-S117: A New Chapter for A2A: Joining the Agentic AI Foundation (27 Aug 2026) (A2A Project) <https://a2a-protocol.org/latest/blog/2026/08/27/a-new-chapter-for-a2a-joining-the-agentic-ai-foundation/> accessed 2026-10-07 [search-tool extract]

⁴⁴⁰ Verified fact — A3-S078: A2A Protocol Specification (source, latest released version 1.0.0) (A2A Project (GitHub)) <https://raw.githubusercontent.com/a2aproject/A2A/main/docs/specification.md> accessed 2026-10-07

⁴⁴¹ Verified fact — A3-S078: A2A Protocol Specification (source, latest released version 1.0.0) (A2A Project (GitHub)) <https://raw.githubusercontent.com/a2aproject/A2A/main/docs/specification.md> accessed 2026-10-07

⁴⁴² Verified fact — A3-S078: A2A Protocol Specification (source, latest released version 1.0.0) (A2A Project (GitHub)) <https://raw.githubusercontent.com/a2aproject/A2A/main/docs/specification.md> accessed 2026-10-07

⁴⁴³ Verified fact — A3-S078: A2A Protocol Specification (source, latest released version 1.0.0) (A2A Project (GitHub)) <https://raw.githubusercontent.com/a2aproject/A2A/main/docs/specification.md> accessed 2026-10-07

⁴⁴⁴ Verified fact — A3-S079: A2A CHANGELOG (A2A Project (GitHub)) <https://raw.githubusercontent.com/a2aproject/A2A/main/CHANGELOG.md> accessed 2026-10-07

⁴⁴⁵ Reported — A3-S030: A2A Protocol Ships v1.0: Production-Ready Standard (A2A Project (Linux Foundation)) <https://a2a-protocol.org/latest/announcing-1.0/> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
lineup	Python SDK a2a-sdk 1.2.2 (5 October 2026)	d fact	A3-S078, A3-S075, V1-S039
Licence	Open specification and SDKs, Apache-2.0	Verified fact	A3-S075, A3-S025
Status events	August 2025: IBM's Agent Communication Protocol (ACP) merged into A2A; IBM joined the TSC; 12 March 2026: A2A 1.0.0, first stable release (changelog; AAIF and Google also say March 2026; the 9 April 2026 Linux Foundation release is the one-year anniversary announcement); 26 May 2026: A2A 1.0.1; 17 August 2026: AAIF announced A2A joining; A2A blog (27 August 2026) confirms acceptance as AAIF Growth Stage project	Verified fact	A3-S032, A3-S065, A3-S079, A3-S116, A3-S026, A3-S025, A3-S117, V1-S039, V1-S038
Strategic direction	v1.0 adds multi-tenancy, multi-protocol bindings, verb-style operations, signed Agent Cards and modernised OAuth flows; joined AAIF alongside MCP as separate project with own TSC (August 2026)	Verified fact	A3-S030, A3-S040
What it does	Lets a client agent discover a remote agent via its Agent Card (/well-known/agent-card.json), send messages and manage asynchronous, long-running tasks with streaming and push notifications; version negotiated via A2A-Version header	Verified fact	A3-S078, A3-S030
Stack position	L4 agent-to-agent interoperability, complementary to MCP (agent-to-tool)	Verified fact	A3-S040
Integration	Official SDKs (Python, JavaScript and others); Agent Cards advertise supportedInterfaces with protocol binding and version	Verified fact	A3-S075, A3-S031, A3-S054
Dependencies	HTTP-based bindings (JSON-RPC, gRPC, REST); security via declared securitySchemes incl. OAuth 2.0 and OpenID Connect	Verified fact	A3-S078
Certifications	Not applicable (open specification)	Architectural judgement	
GDPR / residency	Not applicable (open specification)	Architectural judgement	
Security features	Agent Cards MAY be signed with JWS (RFC 7515) over RFC 8785-canonicalised JSON; clients SHOULD verify when present. OAuth flows modernised (device code, PKCE; implicit/password removed). The protocol does not define scope, validity or revocation semantics for authorisation obtained in the AUTH_REQUIRED task state	Verified fact	A3-S078, A3-S079, A3-S031
Access controls	Authenticated extended Agent Card available only after client authenticates with a declared scheme	Verified fact	A3-S078
Enterprise support	Not applicable (community specification)	Architectural judgement	

Field	Value	Label	Sources
Pricing	Free open specification	Verified fact	A3-S075
Infrastructure cost	Not applicable	Architectural judgement	
Ecosystem	More than 150 supporting organisations and availability in major cloud platforms (April 2026)	Verified fact	A3-S025
Adoption signals	a2aproject/A2A 26,049 GitHub stars (7 October 2026); 150+ organisations (April 2026)	Verified fact	A3-S054, A3-S025
Maturity	First public spec 2025; 1.0 stable March 2026 with breaking changes to the interaction protocol (Agent Card backward-compatible)	Verified fact	A3-S079, A3-S030
Deployment	saas: No (open specification; deployment depends on each implementation); managed_cloud: No (open specification; deployment depends on each implementation); vpc_byoc: No (open specification; deployment depends on each implementation); private_cloud: No (open specification; deployment depends on each implementation); self_hosted: Yes (implementations can be self-hosted; the specification itself is not a hosted service); on_prem: Yes (implementations can be self-hosted; the specification itself is not a hosted service)		

Model Context Protocol (MCP) (L4-mcp)

Tier: Strategic · **Flags:** none · **Original graphic label:** MCP – tools standard

Rationale: Strategic, conditional: only behind a gateway with mandatory authorisation, an allow-list and pinned tool definitions. Tier set by the reader at Checkpoint 3 (CP3 Q1), treating MCP and its authorisation profile (C4-mcp-authorization) as one decision; FS 3.55 is below the 3.6 guide and security scores 2, which is why the condition is mandatory. Conflict of interest: MCP originated at Anthropic and the author is an Anthropic model ^{AJ}.

Criterion	Score	Rationale
Technical	4	Covers tool invocation, enterprise authorisation and long-running tasks; security enforcement (allow-listing, signing) left outside the spec ^{AJ}
Enterprise	3	Open spec (rule 2): enables central IdP policy via EMA; no commercial support

Criterion	Score	Rationale
readiness		from the project itself ^{AJ}
Security and compliance	2	Rule 2 hygiene: advisories are published with fixes (B-REVA-S006), but the specification leaves authorisation optional, has no tool-definition signing, carries a documented tool-poisoning attack class, and the TypeScript SDK had several High advisories in 2026, including one sending OAuth credentials to a server-chosen authorisation server ^{VF} ⁴⁴⁶ . Below A2A (optional card signing) on integrity; borderline 3/2 resolved against the Anthropic-originated item (CP3 review A)
Deployment flexibility	5	Implementations run anywhere, including stdio locally and air-gapped ^{AJ}
Ecosystem	5	De facto standard with broad client and launch-partner support ^{VF} ⁴⁴⁷
Reliability and maturity	3	Foundation-hosted with a 12-month deprecation policy, but breaking changes in July 2026 and Registry still preview ^{VF} ⁴⁴⁸
Cost / TCO	4	Free specification and SDKs; cost lies in operating servers, gateway and authorisation server ^{AJ}
Lock-in / portability	4	MIT under AAIF, but technical direction concentrated in Anthropic maintainers, so not fully neutral governance ^{VF} ⁴⁴⁹
Total (generic / FS)	3.70 / 3.55	

Capabilities. Open protocol (MIT) for connecting agent hosts to tools, resources and prompts over JSON-RPC; spec 2026-07-28 is stateless with Mcp-Method/Mcp-Name headers for gateway routing, Tasks for long-running work, an optional OAuth 2.1 authorisation profile (RFC 8707, RFC 9728, RFC 9207, PKCE, no token passthrough) and the stable Enterprise-Managed Authorization extension (ID-JAG) ^{VF}⁴⁵⁰. Conflict of interest: originated at Anthropic; author is an Anthropic model ^{AJ}.

Strengths

⁴⁴⁶ Verified fact — A3-S055: MCP Specification 2026-07-28: Authorization (Model Context Protocol project (GitHub)) <https://raw.githubusercontent.com/modelcontextprotocol/modelcontextprotocol/main/docs/specification/2026-07-28/basic/authorization/index.mdx> accessed 2026-10-07; A3-S023: MCPTox: A Benchmark for Tool Poisoning Attack on Real-World MCP Servers (arXiv (academic preprint; AAAI 2026)) <https://arxiv.org/pdf/2508.14925> accessed 2026-10-07 [search-tool extract]; B-REVA-S006: modelcontextprotocol/typescript-sdk security advisories (GitHub Advisory Database) (Model Context Protocol project (GitHub)) <https://github.com/modelcontextprotocol/typescript-sdk/security/advisories> accessed 2026-10-08 [search-tool extract]

⁴⁴⁷ Verified fact — A3-S015: The 2026-07-28 Specification (Model Context Protocol project (blog)) <https://blog.modelcontextprotocol.io/posts/2026-07-28/> accessed 2026-10-07; A3-S018: MCP joins the Agentic AI Foundation (9 Dec 2025) (Model Context Protocol project (blog)) <https://blog.modelcontextprotocol.io/posts/2025-12-09-mcp-joins-agentic-ai-foundation/> accessed 2026-10-07

⁴⁴⁸ Verified fact — A3-S015: The 2026-07-28 Specification (Model Context Protocol project (blog)) <https://blog.modelcontextprotocol.io/posts/2026-07-28/> accessed 2026-10-07; A3-S042: modelcontextprotocol/registry (GitHub README) (Model Context Protocol project) <https://github.com/modelcontextprotocol/registry> accessed 2026-10-07 [search-tool extract]

⁴⁴⁹ Verified fact — A3-S082: Expanding the MCP Maintainer Team (8 Apr 2026) (Model Context Protocol project (blog)) <https://blog.modelcontextprotocol.io/posts/2026-04-08-maintainer-update/> accessed 2026-10-07

⁴⁵⁰ Verified fact — A3-S015: The 2026-07-28 Specification (Model Context Protocol project (blog)) <https://blog.modelcontextprotocol.io/posts/2026-07-28/> accessed 2026-10-07; A3-S057: MCP Specification 2026-07-28: Key Changes (Model Context Protocol project (GitHub)) <https://raw.githubusercontent.com/modelcontextprotocol/modelcontextprotocol/main/docs/specification/2026-07-28/changelog.mdx> accessed 2026-10-07; A3-S055: MCP Specification 2026-07-28: Authorization (Model Context Protocol project (GitHub)) <https://raw.githubusercontent.com/modelcontextprotocol/modelcontextprotocol/main/docs/specification/2026-07-28/basic/authorization/index.mdx> accessed 2026-10-07; A3-S056: MCP Specification 2026-07-28: Authorization Security Considerations (Model Context Protocol project (GitHub)) <https://raw.githubusercontent.com/modelcontextprotocol/modelcontextprotocol/main/docs/specification/2026-07-28/basic/authorization/security-considerations.mdx> accessed 2026-10-07; A3-S017: Enterprise-Managed Authorization: Zero-touch OAuth for MCP (18 Jun 2026) (Model Context Protocol project (blog)) <https://blog.modelcontextprotocol.io/posts/enterprise-managed-auth/> accessed 2026-10-07

- De facto agent-to-tool contract with Tier 1 SDKs in TypeScript, Python, Go and C# and client support across major vendors ^{vf451}
- Header-based routing makes gateway enforcement practical ^{vf452}
- EMA gives central IdP policy and one audit trail ^{vf453}
- Hosted by AAIIF (Linux Foundation) since 9 December 2025 ^{vf454}

Limitations and risks

- Authorisation is OPTIONAL in the specification ^{vf455}
- Tool annotations are untrusted hints ^{vf456}
- Tool poisoning: 36.5% average attack success across 20 models in MCPTox ^{vf457}; OWASP MCP03 tool poisoning ^{vf458}
- IDE clients (including Claude Code) had high-severity auto-execution issues ^{vf459}
- Both Lead Maintainers are Anthropic staff; AAIIF does not set technical direction ^{vf460}
- Official Registry still preview, API v0.1, denylist moderation ^{vf461}
- 2026-07-28 contained breaking changes and deprecations (Roots, Sampling, Logging, DCR) ^{vf462}

⁴⁵¹ Verified fact — A3-S015: The 2026-07-28 Specification (Model Context Protocol project (blog)) <https://blog.modelcontextprotocol.io/posts/2026-07-28/> accessed 2026-10-07; A3-S018: MCP joins the Agentic AI Foundation (9 Dec 2025) (Model Context Protocol project (blog)) <https://blog.modelcontextprotocol.io/posts/2025-12-09-mcp-joins-agentic-ai-foundation/> accessed 2026-10-07

⁴⁵² Verified fact — A3-S015: The 2026-07-28 Specification (Model Context Protocol project (blog)) <https://blog.modelcontextprotocol.io/posts/2026-07-28/> accessed 2026-10-07

⁴⁵³ Verified fact — A3-S017: Enterprise-Managed Authorization: Zero-touch OAuth for MCP (18 Jun 2026) (Model Context Protocol project (blog)) <https://blog.modelcontextprotocol.io/posts/enterprise-managed-auth/> accessed 2026-10-07

⁴⁵⁴ Verified fact — A3-S018: MCP joins the Agentic AI Foundation (9 Dec 2025) (Model Context Protocol project (blog)) <https://blog.modelcontextprotocol.io/posts/2025-12-09-mcp-joins-agentic-ai-foundation/> accessed 2026-10-07; V1-S038: Linux Foundation Announces the Formation of the Agentic AI Foundation (AAIIF) (The Linux Foundation) <https://www.linuxfoundation.org/press/linux-foundation-announces-the-formation-of-the-agentic-ai-foundation> accessed 2026-10-08 [search-tool extract]

⁴⁵⁵ Verified fact — A3-S055: MCP Specification 2026-07-28: Authorization (Model Context Protocol project (GitHub)) <https://raw.githubusercontent.com/modelcontextprotocol/modelcontextprotocol/main/docs/specification/2026-07-28/basic/authorization/index.mdx> accessed 2026-10-07

⁴⁵⁶ Verified fact — A3-S020: Tool Annotations as Risk Vocabulary (16 Mar 2026) (Model Context Protocol project (blog)) <https://blog.modelcontextprotocol.io/posts/2026-03-16-tool-annotations/> accessed 2026-10-07

⁴⁵⁷ Verified fact — A3-S023: MCPTox: A Benchmark for Tool Poisoning Attack on Real-World MCP Servers (arXiv (academic preprint; AAAI 2026)) <https://arxiv.org/pdf/2508.14925> accessed 2026-10-07 [search-tool extract]

⁴⁵⁸ Verified fact — A3-S045: OWASP MCP Top 10: MCP03:2025 Tool Poisoning <https://owasp.org/www-project-mcp-top-10/2025/MCP03-2025%E2%80%9393Tool-Poisoning> accessed 2026-10-07 [search-tool extract]

⁴⁵⁹ Reported — A3-S022: MCP Attack Surface: Tool Poisoning and IDE Auto-Execution (CSA research note, Jul 2026) (Cloud Security Alliance Labs) <https://labs.cloudsecurityalliance.org/research/csa-research-note-mcp-tool-poisoning-auto-execution-20260701/> accessed 2026-10-07 [search-tool extract]

⁴⁶⁰ Verified fact — A3-S082: Expanding the MCP Maintainer Team (8 Apr 2026) (Model Context Protocol project (blog)) <https://blog.modelcontextprotocol.io/posts/2026-04-08-maintainer-update/> accessed 2026-10-07; A3-S018: MCP joins the Agentic AI Foundation (9 Dec 2025) (Model Context Protocol project (blog)) <https://blog.modelcontextprotocol.io/posts/2025-12-09-mcp-joins-agentic-ai-foundation/> accessed 2026-10-07

⁴⁶¹ Verified fact — A3-S019: Introducing the MCP Registry (8 Sep 2025) (Model Context Protocol project (blog)) <https://blog.modelcontextprotocol.io/posts/2025-09-08-mcp-registry-preview/> accessed 2026-10-07; A3-S042: modelcontextprotocol/registry (GitHub README) (Model Context Protocol project) <https://github.com/modelcontextprotocol/registry> accessed 2026-10-07 [search-tool extract]; A3-S036: Official MCP Registry API v0.1 sample response (v0.2 and v1 paths returned HTTP 404 on 7 Oct 2026) (Model Context Protocol project) <https://registry.modelcontextprotocol.io/v0.1/servers?limit=1> accessed 2026-10-07

⁴⁶² Verified fact — A3-S015: The 2026-07-28 Specification (Model Context Protocol project (blog)) <https://blog.modelcontextprotocol.io/posts/2026-07-28/> accessed 2026-10-07; V2-S031: MCP specification 2026-07-28 changelog (Model Context Protocol project) <https://raw.githubusercontent.com/modelcontextprotocol/modelcontextprotocol/main/docs/specification/2026-07-28/changelog.mdx> accessed 2026-

- 2026 TypeScript SDK advisories include GHSA-6qxp-vccf-f47h (High: OAuth credentials sent to a server-chosen authorisation server; the fix also needs expectedIssuer configured), GHSA-22jm-h49p-29qw (High) and CVE-2026-25536 (High)⁴⁶³

Choose when

- One tool contract is needed across several agent frameworks and model vendors^{AJ}

Avoid when

- Clients would connect directly to third-party servers without a gateway, private registry and pinned definitions^{AJ}

Nearest competitors: OpenAPI tool definitions via gateway, L4-a2a, L4-aws-agentcore-gateway-identity

Regulated-FS note. Adopt the protocol, not the public ecosystem: internal servers behind a firm-owned gateway with authorisation mandatory and definitions hash-pinned; independent alternative is OpenAPI-described tools exposed through a gateway^{Rec.}

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Open-source project hosted by the Agentic AI Foundation (AAIF), a directed fund under the Linux Foundation; originated at Anthropic (created by David Soria Parra and Justin Spahr-Summers) and donated on 9 December 2025	Verified fact	A3-S018, A3-S058, V1-S038
Category	Open protocol specification for connecting agents/LLM applications to tools, resources and prompts	Verified fact	A3-S058, A3-S015
Version / lineup	Specification 2026-07-28 (released 28 July 2026; previous 2025-11-25). Tier 1 SDKs: TypeScript, Python, Go, C#; Python 'mcp' 2.3.0 (2 October 2026); TypeScript '@modelcontextprotocol/sdk' 1.32.1 (5 October 2026)	Verified fact	A3-S015, A3-S057, A3-S011, A3-S074, V1-S037
Licence	Open specification; specification repository and SDKs MIT-licensed	Verified fact	A3-S058, A3-S011, A3-S074
Status events	9 December 2025: donated by Anthropic to the Agentic AI Foundation (Linux Foundation); maintainer structure unchanged; 8 September	Verified fact	A3-S015, A3-S017,

⁴⁶³ Verified fact — B-REVA-S006: modelcontextprotocol/typescript-sdk security advisories (GitHub Advisory Database) (Model Context Protocol project (GitHub)) <https://github.com/modelcontextprotocol/typescript-sdk/security/advisories> accessed 2026-10-08 [search-tool extract]

Field	Value	Label	Sources
	2025: official MCP Registry launched in preview; API frozen at v0.1 (24 October 2025); Registry working-group charter lists 'Registry API v1 GA' as Ideating with no target date; /v0.1 live and /v1 absent on 7 October 2026; 18 June 2026: Enterprise-Managed Authorization extension declared stable; 28 July 2026: spec 2026-07-28 removes sessions and the initialize handshake (stateless core), adds Multi Round-Trip Requests, header-based routing, cacheable lists, extensi		A3-S018, A3-S019, A3-S036, A3-S042, A3-S057, A3-S082, A3-S114, V1-S037, V1-S038, V1-S045
Strategic direction	Roadmap (22 August 2026): agentic messaging primitives (server-initiated events, Tasks), HTTP-native transport unification, agent identity and enterprise security (DPoP, Workload Identity Federation, ID-JAG, token exchange; engagement with IETF OAuth and WIMSE), progressive tool discovery for large catalogues, SDK conformance	Verified fact	A3-S016
What it does	Defines how clients (agent hosts) discover and call server-exposed tools, resources and prompts over JSON-RPC; as of 2026-07-28 each request is self-describing and stateless, with optional server/discover, and long-running work via the Tasks extension	Verified fact	A3-S015, A3-S057
Stack position	L4 connectivity standard; the 2026-07-28 release puts method and tool names in Mcp-Method/Mcp-Name HTTP headers so gateways, rate limiters and WAFs can route and authorise without parsing bodies	Verified fact	A3-S015
Integration	Open protocol with official SDKs (TypeScript, Python, Go, C# Tier 1; Rust beta); first-class client support reported in ChatGPT, Claude, Cursor, Gemini, Microsoft Copilot and VS Code (December 2025)	Verified fact	A3-S015, A3-S018
Dependencies	Transports stdio and Streamable HTTP; remote authorisation relies on an external OAuth 2.1 authorisation server	Verified fact	A3-S055, A3-S057
Certifications	Not applicable (open specification)	Architectural judgement	
GDPR / residency	Not applicable (open specification)	Architectural judgement	
Security features	Authorisation is OPTIONAL. When used: OAuth 2.1 (draft-13) with PKCE; servers MUST publish Protected Resource Metadata (RFC 9728); clients MUST send Resource Indicators (RFC 8707); servers MUST validate token audience; token passthrough forbidden; RFC 9207 issuer validation; DCR deprecated for Client ID Metadata Documents. Tool annotations are untrusted hints	Verified fact	A3-S055, A3-S056, A3-S057, A3-S020
Access controls	Enterprise-Managed Authorization extension (stable 18 June 2026): IdP issues an Identity Assertion JWT Authorization Grant (ID-JAG) exchanged for MCP access tokens, giving central policy and audit; Okta (Cross App Access) first IdP; adopted by Anthropic clients, VS Code and servers incl. Asana, Atlassian, Canva, Figma, Linear and Supabase	Verified fact	A3-S017
Enterprise support	Not applicable (community specification); support comes from implementers	Architectural	

Field	Value	Label	Sources
		judgement	
Pricing	Free open specification and SDKs	Verified fact	A3-S058, A3-S011
Infrastructure cost	Cost lies in hosting MCP servers, gateways and the authorisation server	Architectural judgement	
Ecosystem	Launch partners for 2026-07-28 include AWS (AgentCore; Tasks extension contributed by AWS), Cloudflare, Google Cloud, Microsoft Foundry, Figma, Netlify, Supabase, Honeycomb, FastMCP/Horizon and Runlayer; official Registry with public and private sub-registries	Verified fact	A3-S015, A3-S019
Adoption signals	97M monthly SDK downloads and 10,000 active servers (December 2025); close to 0.5bn monthly downloads across Tier 1 SDKs and >1bn total each for TypeScript and Python (July 2026, project claim); modelcontextprotocol/servers 91,068 GitHub stars	Verified fact	A3-S018, A3-S015, A3-S054
Maturity	Initial release November 2024 (first SDK on PyPI 20 November 2024); formal deprecation policy with 12-month minimum window; 2026-07-28 contained breaking changes	Verified fact	A3-S011, A3-S015
Deployment	saas: No (open specification; deployment depends on each implementation); managed_cloud: No (open specification; deployment depends on each implementation); vpc_byoc: No (open specification; deployment depends on each implementation); private_cloud: No (open specification; deployment depends on each implementation); self_hosted: Yes (implementations can be self-hosted; the specification itself is not a hosted service); on_prem: Yes (implementations can be self-hosted; the specification itself is not a hosted service)		

Amazon Bedrock AgentCore Gateway and AgentCore Identity (L4-aws-agentcore-gateway-identity)

Tier: Strategic · **Flags:** none · **Original graphic label:** not in graphic

Rationale: Strategic, conditional: where AWS is your primary cloud (CP3 Q2, rubric rule 10). AWS's lead managed tool gateway and agent identity service; deployment 2 (AWS-managed only) is accepted under rule 11 because the condition is an existing platform commitment. Same tier as C1-aws-agentcore-gateway ^{AJ}.

Criterion	Score	Rationale
Technical	4	Tool conversion, inbound identity, outbound vault and Cedar policy ^{vi-464}
Enterprise readiness	4	Hyperscaler presumption: platform controls presumed (CP2 Q1); confirm per service ^{AJ}
Security and compliance	4	SOC 1/2/3 scope listed; ISO listed with inconsistent wording; CMK for vault; one point below 5 under rule 8 ^{vi-465}

⁴⁶⁴ Verified fact — A6-S021: Amazon Bedrock AgentCore Pricing; AgentCore GA announcement (Oct 2025); AgentCore Gateway docs (Amazon Web Services) <https://aws.amazon.com/bedrock/agentcore/pricing/> accessed 2026-10-07 [search-tool extract]; A6-S026: Policy in Amazon Bedrock AgentCore is now generally available; AgentCore Policy docs; Why Policy in AgentCore chose Cedar (Amazon Web Services) <https://aws.amazon.com/about-aws/whats-new/2026/03/policy-amazon-bedrock-agentcore-generally-available/> accessed 2026-10-07 [search-tool extract]

Criterion	Score	Rationale
Deployment flexibility	2	Managed AWS service only, with VPC and PrivateLink connectivity; no self-host (A3-S047, A3-S048); 2, consistent with C1-aws-agentcore-gateway and every other AWS-only managed service (CP3 review A)
Ecosystem	4	MCP on both sides, OAuth and IAM integration ^{vf466}
Reliability and maturity	3	GA October 2025 with regular feature releases (A3-S047, A3-S048); about one year GA, so 3, consistent with AgentCore Memory (L5) (CP3 review A)
Cost / TCO	4	Low published per-call pricing ^{vf467}
Lock-in / portability	3	Proprietary AWS API but standard MCP/OAuth interfaces; Cedar policies portable ^{vf468}
Total (generic / FS)	3.55 / 3.45	

Capabilities. Managed MCP tool gateway (APIs, Lambda and MCP servers as tools; IAM or JWT inbound; 3LO outbound) plus agent identity and refresh-token vault; pairs with Cedar-based AgentCore Policy for pre-execution authorisation ^{vf469}. Same service also profiled as C1-aws-agentcore-gateway ^{AI}.

Strengths

- Most complete managed implementation of the tool-governance sub-layer found ^{AI}
- Supports MCP 2026-07-28 and earlier revisions ^{vf470}
- Customer-managed KMS key for the token vault ^{vf471}
- Published per-call pricing ^{vf472}

⁴⁶⁵ Verified fact — B-L4-S001: AWS Services in Scope by Compliance Program: SOC (and AgentCore release notes per search summary) (Amazon Web Services) <https://aws.amazon.com/compliance/services-in-scope/SOC/> accessed 2026-10-08 [search-tool extract]; B-L4-S002: Compliance validation for Amazon Bedrock AgentCore (Amazon Web Services) <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/compliance-validation.html> accessed 2026-10-08 [search-tool extract]; A6-S074: AWS API model: Amazon Bedrock AgentCore control plane (botocore service-2.json) (Amazon Web Services (via boto/botocore)) <https://raw.githubusercontent.com/boto/botocore/develop/botocore/data/bedrock-agentcore-control/2023-06-05/service-2.json> accessed 2026-10-07

⁴⁶⁶ Verified fact — A3-S047: Amazon Bedrock AgentCore is now generally available (Oct 2025) (Amazon Web Services) <https://aws.amazon.com/about-aws/whats-new/2025/10/amazon-bedrock-agentcore-available/> accessed 2026-10-07 [search-tool extract]

⁴⁶⁷ Verified fact — A6-S021: Amazon Bedrock AgentCore Pricing; AgentCore GA announcement (Oct 2025); AgentCore Gateway docs (Amazon Web Services) <https://aws.amazon.com/bedrock-agentcore/pricing/> accessed 2026-10-07 [search-tool extract]

⁴⁶⁸ Verified fact — A6-S045: Cedar README (GitHub) (Cedar project) <https://raw.githubusercontent.com/cedar-policy/cedar/main/README.md> accessed 2026-10-07

⁴⁶⁹ Verified fact — A3-S047: Amazon Bedrock AgentCore is now generally available (Oct 2025) (Amazon Web Services) <https://aws.amazon.com/about-aws/whats-new/2025/10/amazon-bedrock-agentcore-available/> accessed 2026-10-07 [search-tool extract]; A3-S048: Release notes for Amazon Bedrock AgentCore (Amazon Web Services) <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/release-notes.html> accessed 2026-10-07 [search-tool extract]; A6-S021: Amazon Bedrock AgentCore Pricing; AgentCore GA announcement (Oct 2025); AgentCore Gateway docs (Amazon Web Services) <https://aws.amazon.com/bedrock-agentcore/pricing/> accessed 2026-10-07 [search-tool extract]; A6-S074: AWS API model: Amazon Bedrock AgentCore control plane (botocore service-2.json) (Amazon Web Services (via boto/botocore)) <https://raw.githubusercontent.com/boto/botocore/develop/botocore/data/bedrock-agentcore-control/2023-06-05/service-2.json> accessed 2026-10-07; A6-S026: Policy in Amazon Bedrock AgentCore is now generally available; AgentCore Policy docs; Why Policy in AgentCore chose Cedar (Amazon Web Services) <https://aws.amazon.com/about-aws/whats-new/2026/03/policy-amazon-bedrock-agentcore-generally-available/> accessed 2026-10-07 [search-tool extract]

⁴⁷⁰ Verified fact — A6-S105: AgentCore Gateway: Inference targets; Inference connector/provider targets; Add rate limits to a gateway (Amazon Web Services) <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/gateway-targets-inference.html> accessed 2026-10-07 [search-tool extract]

⁴⁷¹ Verified fact — A6-S074: AWS API model: Amazon Bedrock AgentCore control plane (botocore service-2.json) (Amazon Web Services (via boto/botocore)) <https://raw.githubusercontent.com/boto/botocore/develop/botocore/data/bedrock-agentcore-control/2023-06-05/service-2.json> accessed 2026-10-07

Limitations and risks

- AWS only; no self-hosted option ^{vf473}
- ISO scope wording inconsistent across AWS pages ('aligns' pending third-party review) ^{vf474}
- Adds to AWS concentration ^{AJ}

Choose when

- AWS is the primary agent platform ^{AJ}

Avoid when

- Tools and agents span several clouds and one policy point is required ^{AJ}

Nearest competitors: C1-azure-apim-ai-gateway, C1-agentgateway, C1-kong-ai-gateway, L4-composio

Regulated-FS note. Sits within AWS, a designated CTP under DORA and the UK regime; include in AWS concentration analysis; platform controls presumed (CP2 Q1), confirm per service ^{Rec.}.

Facts (as of 2026-10-07)

Field	Value	Label	Sources
Company	Amazon Web Services	Verified fact	A3-S047
Category	Managed MCP tool gateway and agent identity/credential service	Verified fact	A3-S047
Version / lineup	GA October 2025; Gateway three-legged OAuth for MCP targets GA and VPC egress (April 2026); supports MCP 2026-07-28 stateless core	Verified fact	A3-S047, A3-S048, A3-S015
Licence	Proprietary managed service	Verified fact	A3-S047
Status events	October 2025: GA; April 2026: Gateway 3LO for MCP targets GA; VPC egress for Gateway and Identity	Verified fact	A3-S047, A3-S048
Strategic direction	AWS contributed the MCP Tasks extension and supports the stateless 2026-07-28 spec in AgentCore	Verified fact	A3-S015

⁴⁷² Verified fact — A6-S021: Amazon Bedrock AgentCore Pricing; AgentCore GA announcement (Oct 2025); AgentCore Gateway docs (Amazon Web Services) <https://aws.amazon.com/bedrock/agentcore/pricing/> accessed 2026-10-07 [search-tool extract]

⁴⁷³ Verified fact — A3-S047: Amazon Bedrock AgentCore is now generally available (Oct 2025) (Amazon Web Services) <https://aws.amazon.com/about-aws/whats-new/2025/10/amazon-bedrock-agentcore-available/> accessed 2026-10-07 [search-tool extract]

⁴⁷⁴ Verified fact — B-L4-S002: Compliance validation for Amazon Bedrock AgentCore (Amazon Web Services) <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/compliance-validation.html> accessed 2026-10-08 [search-tool extract]

Field	Value	Label	Sources
What it does	Gateway converts APIs and Lambda functions into agent tools and fronts existing MCP servers with IAM or OAuth authorisation; Identity provides identity-aware authorisation, a vault for refresh tokens and integrations with OAuth services	Verified fact	A3-S047
Stack position	L4 governance sub-layer (MCP gateway) plus C4 agent identity	Architectural judgement	
Integration	MCP (server-facing and client-facing), OAuth, IAM	Verified fact	A3-S047
Dependencies	AWS account and Amazon Bedrock AgentCore	Verified fact	A3-S047
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	15 AWS Regions (EU coverage not individually verified)	Verified fact	A3-S048
Security features	Refresh-token vault; IAM or OAuth for agent-to-tool calls; private networking	Verified fact	A3-S047
Access controls	Identity-aware authorisation; per-user tokens via 3LO for MCP targets	Verified fact	A3-S047, A3-S048
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Not publicly verified	Not publicly verified	
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
Maturity	GA October 2025	d Verified fact	A3-S047
Deployment	saas: Yes; managed_cloud: Yes; vpc_byoc: VPC, PrivateLink and VPC egress supported; private_cloud: Not publicly verified; self_hosted: No; on_prem: No		

Browserbase (cloud browser infrastructure) and Stagehand (open-source browser automation SDK) (L4-browserbase)

Tier: Tactical · **Flags:** none · **Original graphic label:** Browserbase – cloud browsers

Rationale: Well-controlled browser runtime for the narrow case where no API exists ^{A1}.

Criterion	Score	Rationale
Technical	4	Isolated browsers plus AI interaction layer ^{vf475}
Enterprise readiness	3	SAML SSO verified (CP2 rule 7: max 3) ^{vf476}
Security and compliance	3	SOC 2 Type II, pen test, DPA; no ISO 27001 ^{vf477}
Deployment flexibility	3	SaaS plus VPC/private-cloud options for large enterprise ^{vf478}
Ecosystem	4	Stagehand widely used; CDP/Playwright standard ^{vf479}
Reliability and	3	Since April 2024, 46 SDK releases; MCP repo archived ^{vf480}

⁴⁷⁵ Verified fact — A3-S013: PyPI JSON metadata: browserbase (Python Package Index (PyPI)) <https://pypi.org/pypi/browserbase/json> accessed 2026-10-07

⁴⁷⁶ Verified fact — B-L4-S004: Browserbase docs: SSO with SAML (and Enterprise security page) <https://docs.browserbase.com/account/team/sso> accessed 2026-10-08 [search-tool extract]

⁴⁷⁷ Verified fact — A3-S102: Browserbase Trust Center (and docs enterprise security / plans, pricing pages) <https://trust.browserbase.com/> accessed 2026-10-07 [search-tool extract]

⁴⁷⁸ Verified fact — A3-S102: Browserbase Trust Center (and docs enterprise security / plans, pricing pages) <https://trust.browserbase.com/> accessed 2026-10-07 [search-tool extract]

⁴⁷⁹ Verified fact — A3-S054: GitHub repository metadata (stars, forks, created, archived) retrieved via GitHub MCP search_repositories tool, 7 Oct 2026 <https://api.github.com/search/repositories> accessed 2026-10-07 [search-tool extract]

⁴⁸⁰ Verified fact — A3-S013: PyPI JSON metadata: browserbase (Python Package Index (PyPI)) <https://pypi.org/pypi/browserbase/json> accessed 2026-10-07; A3-S054: GitHub repository metadata (stars, forks, created, archived) retrieved via GitHub MCP search_repositories tool, 7 Oct 2026 <https://api.github.com/search/repositories> accessed 2026-10-07 [search-tool extract]

Criterion	Score	Rationale
maturity		
Cost / TCO	3	Transparent tiered pricing ^{VF481}
Lock-in / portability	4	Client-side tooling open and portable; service proprietary ^{AJ}
Total (generic / FS)	3.35	
	/	
	3.35	

Capabilities. Managed headless browsers driven over CDP by Playwright/Puppeteer, with Stagehand (MIT) for AI-driven interaction and extraction ^{VF482}.

Strengths

- SOC 2 Type II (2025, 2026), pen test, HIPAA BAA, DPA, US/EU/Asia residency for large enterprise ^{VF483}
- SAML SSO on Enterprise; recordings to customer S3 ^{VF484}
- Portable client side (Playwright, CDP, MIT Stagehand) ^{VF485}

Limitations and risks

- Standalone MCP server repository archived ^{VF486}
- No admin audit log found; plan naming inconsistent ^{VF487 NPV}
- Browser automation is fragile and hard to audit compared with APIs ^{AJ}

Choose when

- An agent must operate a website that has no API ^{AJ}

Avoid when

- An API or read-only data tool exists ^{AJ}

Nearest competitors: L8-firecrawl, L8-apify, L4-e2b

Regulated-FS note. Allow-list target domains and never give browser sessions credentials to client-facing systems ^{Rec}.

⁴⁸¹ Verified fact — A3-S102: Browserbase Trust Center (and docs enterprise security / plans, pricing pages) <https://trust.browserbase.com/> accessed 2026-10-07 [search-tool extract]

⁴⁸² Verified fact — A3-S013: PyPI JSON metadata: browserbase (Python Package Index (PyPI)) <https://pypi.org/pypi/browserbase/json> accessed 2026-10-07; A3-S068: Stagehand LICENSE (MIT, Browserbase Inc.) (Browserbase (GitHub)) <https://raw.githubusercontent.com/browserbase/stagehand/main/LICENSE> accessed 2026-10-07

⁴⁸³ Verified fact — A3-S102: Browserbase Trust Center (and docs enterprise security / plans, pricing pages) <https://trust.browserbase.com/> accessed 2026-10-07 [search-tool extract]

⁴⁸⁴ Verified fact — B-L4-S004: Browserbase docs: SSO with SAML (and Enterprise security page) <https://docs.browserbase.com/account/team/sso> accessed 2026-10-08 [search-tool extract]

⁴⁸⁵ Verified fact — A3-S013: PyPI JSON metadata: browserbase (Python Package Index (PyPI)) <https://pypi.org/pypi/browserbase/json> accessed 2026-10-07; A3-S068: Stagehand LICENSE (MIT, Browserbase Inc.) (Browserbase (GitHub)) <https://raw.githubusercontent.com/browserbase/stagehand/main/LICENSE> accessed 2026-10-07

⁴⁸⁶ Verified fact — A3-S054: GitHub repository metadata (stars, forks, created, archived) retrieved via GitHub MCP search_repositories tool, 7 Oct 2026 <https://api.github.com/search/repositories> accessed 2026-10-07 [search-tool extract]; V1-S081: GitHub repository metadata (archive flags) for 8 repos (GitHub (via GitHub MCP search_repositories)) <https://api.github.com/search/repositories> accessed 2026-10-08 [search-tool extract]

⁴⁸⁷ Verified fact — B-L4-S004: Browserbase docs: SSO with SAML (and Enterprise security page) <https://docs.browserbase.com/account/team/sso> accessed 2026-10-08 [search-tool extract]

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Browserbase Inc.	Verified fact	A3-S068
Category	Managed headless-browser infrastructure for agents	Verified fact	A3-S013
Version / lineup	browserbase Python SDK 1.20.0 (24 September 2026); Stagehand 4.1.0 (9 September 2026)	Verified fact	A3-S013, A3-S076
Licence	SDK Apache-2.0; Stagehand MIT; browser service proprietary	Verified fact	A3-S013, A3-S068
Status events	browserbase/mcp-server-browserbase repository archived (status on 7 October 2026)	Verified fact	A3-S054, V1-S081
Strategic direction	Agent skills collection and Stagehand integrations with coding agents; standalone MCP server repository archived	Verified fact	A3-S054
What it does	Creates remote browser sessions that Playwright/Puppeteer connect to over CDP; Stagehand adds AI-driven interaction and data extraction	Verified fact	A3-S013, A3-S054
Stack position	L4 tool runtime (browser execution); overlaps L8 web extraction	Verified fact	A3-S013
Integration	REST API, Python/Node SDKs, CDP connect URL, Stagehand SDK, agent skills	Verified fact	A3-S013, A3-S054
Dependencies	Browserbase hosted service; Playwright/CDP client	Verified fact	A3-S013
Certifications	SOC 2 Type II reports 2025 and 2026; 2026 penetration test; HIPAA with BAA on Scale/Enterprise	Verified fact	A3-S102
GDPR / residency	Data residency controls US/EU/Asia for large enterprise customers; DPA available	Verified fact	A3-S102
Security features	Isolated browsers; VPC connectivity and private cloud options for large enterprises	Verified fact	A3-S102
Access	SSO on enterprise tier	Verified	A3-S102

Field	Value	Label	Sources
controls		d fact	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Free; Developer US\$20/month; Startup US\$99/month; Scale/Enterprise custom	Verified fact	A3-S102
Infrastructure cost	Not applicable (SaaS)	Architectural judgment	
Ecosystem	Stagehand works with Claude Code, Codex, Mastra and others (repo description)	Verified fact	A3-S054
Adoption signals	browserbase/stagehand 25,562 GitHub stars (7 October 2026); US\$40M Series B led by Notable Capital (June 2025, reported US\$300M valuation)	Verified fact	A3-S054, A3-S103
Maturity	Python SDK first released April 2024; 46 releases	Verified fact	A3-S013
Deployment	saas: Yes; managed_cloud: Not publicly verified; vpc_byoc: VPC connectivity for large enterprise customers; private_cloud: Yes (private cloud options for large enterprise customers); self_hosted: Not publicly verified; on_prem: Not publicly verified		

E2B (L4-e2b)

Tier: Tactical · **Flags:** none · **Original graphic label:** E2B – code sandboxes

Rationale: Reference sandbox pattern; missing enterprise identity controls keep it short of Strategic ^{AJ}.

Criterion	Score	Rationale
Technical	4	Leading isolation, egress and secrets design for code execution ^{vf488}
Enterprise readiness	2	Capped at 2: SSO, SCIM and RBAC planned, no verified customer audit log ^{vf489}
Security and compliance	3	SOC 2 Type II (control plane), HIPAA BAA, pen test; no ISO 27001 ^{vf490}
Deployment flexibility	4	Cloud with EU region, BYOC AWS/GCP, dedicated; no production self-host or air-gap ^{vf491}

⁴⁸⁸ Verified fact — A3-S062: E2B Runtime README (e2b-dev/infra) (E2B (GitHub)) <https://raw.githubusercontent.com/e2b-dev/infra/main/README.md> accessed 2026-10-07

⁴⁸⁹ Verified fact — B-L4-S003: E2B docs: lifecycle events API (and Enterprise page and changelog per search summary) <https://docs.e2b.dev/sandbox/lifecycle-events-api> accessed 2026-10-08 [search-tool extract]

⁴⁹⁰ Verified fact — A3-S104: E2B Security (and e2b.dev/enterprise, pricing, docs BYOC) <https://e2b.dev/security> accessed 2026-10-07 [search-tool extract]; A3-S120: E2B Enterprise (and docs: regions, BYOC; trust.e2b.dev controls) <https://e2b.dev/enterprise> accessed 2026-10-07 [search-tool extract]

⁴⁹¹ Verified fact — A3-S120: E2B Enterprise (and docs: regions, BYOC; trust.e2b.dev controls) <https://e2b.dev/enterprise> accessed 2026-10-07 [search-tool extract]; A3-S062: E2B Runtime README (e2b-dev/infra) (E2B (GitHub)) <https://raw.githubusercontent.com/e2b-dev/infra/main/README.md> accessed 2026-10-07

Criterion	Score	Rationale
Ecosystem	4	SDKs, OTel export, broad agent use ^{vf-492}
Reliability and maturity	3	Since August 2023, 365 SDK releases; Series A stage ^{vf-493}
Cost / TCO	3	Published pricing but high Enterprise minimum and heavy self-host stack ^{vf-494}
Lock-in / portability	4	Apache-2.0 runtime and MIT SDK; proprietary cloud ^{vf-495}
Total (generic / FS)	3.35	
	/	
	3.35	

Evidence rules applied: enterprise_readiness capped at 2: SSO/SCIM/RBAC listed as planned; no verified SSO, RBAC or customer audit log (B-L4-S003)

Capabilities. Firecracker microVM sandboxes for AI-generated code with per-sandbox egress firewall, workload identity tokens and secrets kept out of API/logs/spans; Apache-2.0 runtime; Cloud, BYOC (AWS, GCP) and dedicated deployments ^{vf-496}.

Strengths

- Strong isolation and egress model ^{vf-497}
- Open-source runtime as an exit route ^{vf-498}
- EU region and BYOC keep data in customer VPC ^{vf-499}

Limitations and risks

- SSO, SCIM and RBAC still 'planned'; no customer audit-log export found ^{vf-500}
- Self-hosting is evaluation-only per vendor ^{vf-501}
- SOC 2 scope excludes customer BYOC accounts ^{vf-502}

⁴⁹² Verified fact — A3-S062: E2B Runtime README (e2b-dev/infra) (E2B (GitHub)) <https://raw.githubusercontent.com/e2b-dev/infra/main/README.md> accessed 2026-10-07; A3-S054: GitHub repository metadata (stars, forks, created, archived) retrieved via GitHub MCP search_repositories tool, 7 Oct 2026 <https://api.github.com/search/repositories> accessed 2026-10-07 [search-tool extract]

⁴⁹³ Verified fact — A3-S007: PyPI JSON metadata: e2b (Python Package Index (PyPI)) <https://pypi.org/pypi/e2b/json> accessed 2026-10-07; A3-S105: We Raised \$21M to Give Fortune 100 Cloud for AI Agents (Jul 2025) (E2B) <https://e2b.dev/blog/series-a> accessed 2026-10-07 [search-tool extract]

⁴⁹⁴ Verified fact — A3-S104: E2B Security (and e2b.dev/enterprise, pricing, docs BYOC) <https://e2b.dev/security> accessed 2026-10-07 [search-tool extract]; A3-S062: E2B Runtime README (e2b-dev/infra) (E2B (GitHub)) <https://raw.githubusercontent.com/e2b-dev/infra/main/README.md> accessed 2026-10-07

⁴⁹⁵ Verified fact — A3-S062: E2B Runtime README (e2b-dev/infra) (E2B (GitHub)) <https://raw.githubusercontent.com/e2b-dev/infra/main/README.md> accessed 2026-10-07

⁴⁹⁶ Verified fact — A3-S062: E2B Runtime README (e2b-dev/infra) (E2B (GitHub)) <https://raw.githubusercontent.com/e2b-dev/infra/main/README.md> accessed 2026-10-07; A3-S007: PyPI JSON metadata: e2b (Python Package Index (PyPI)) <https://pypi.org/pypi/e2b/json> accessed 2026-10-07; A3-S120: E2B Enterprise (and docs: regions, BYOC; trust.e2b.dev controls) <https://e2b.dev/enterprise> accessed 2026-10-07 [search-tool extract]

⁴⁹⁷ Verified fact — A3-S062: E2B Runtime README (e2b-dev/infra) (E2B (GitHub)) <https://raw.githubusercontent.com/e2b-dev/infra/main/README.md> accessed 2026-10-07

⁴⁹⁸ Verified fact — A3-S062: E2B Runtime README (e2b-dev/infra) (E2B (GitHub)) <https://raw.githubusercontent.com/e2b-dev/infra/main/README.md> accessed 2026-10-07

⁴⁹⁹ Verified fact — A3-S120: E2B Enterprise (and docs: regions, BYOC; trust.e2b.dev controls) <https://e2b.dev/enterprise> accessed 2026-10-07 [search-tool extract]; A3-S104: E2B Security (and e2b.dev/enterprise, pricing, docs BYOC) <https://e2b.dev/security> accessed 2026-10-07 [search-tool extract]

⁵⁰⁰ Verified fact — B-L4-S003: E2B docs: lifecycle events API (and Enterprise page and changelog per search summary) <https://docs.e2b.dev/sandbox/lifecycle-events-api> accessed 2026-10-08 [search-tool extract]

- Enterprise minimum US\$3,000 per month ^{W503}

Choose when

- Any generated code must run, including derived calculations ^{AI}

Avoid when

- You need console SSO and RBAC today and cannot compensate with BYOC and your cloud IAM ^{AI}

Nearest competitors: L4-aws-agentcore-gateway-identity, LangSmith Sandboxes, Daytona / Modal (not profiled)

Regulated-FS note. Use BYOC or the EU region with default-deny egress; reconcile sandbox outputs to authoritative engines before use ^{Rec.}

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	E2B	Verified fact	A3-S062
Category	Secure cloud sandboxes (Firecracker microVMs) for running AI-generated code	Verified fact	A3-S062
Version / lineup	e2b SDK 2.53.1 and e2b-code-interpreter 2.10.3 (6 October 2026)	Verified fact	A3-S007
Licence	Open source: SDK MIT; E2B Runtime (control plane, orchestrator, in-VM agent) Apache-2.0; E2B Cloud hosted service	Verified fact	A3-S007, A3-S062
Strategic direction	Positions itself as ‘the AI agent cloud’: snapshot-resume microVMs, pause/resume, forking running sandboxes, persistent volumes, secrets and workload identity	Verified fact	A3-S062
What it does	Starts isolated sandboxes from pre-booted templates, runs commands and code, exposes ports via sandbox URLs and pauses idle sandboxes	Verified fact	A3-S062, A3-S007
Stack position	L4 tool runtime (code execution); also a C7 sandboxing control	Verified fact	A3-S062
Integration	Python/JavaScript SDKs and CLI; REST API; envd in-VM API over Connect RPC/REST	Verified fact	A3-S062
Dependencies	Self-hosting needs Linux with KVM; runtime uses PostgreSQL, Redis, ClickHouse and object storage	Verified fact	A3-S062
Certifications	SOC 2 Type II (covers E2B software and control plane; BYOC cloud account is customer’s responsibility); HIPAA BAA on Enterprise; pen test and DPA in trust centre	Verified fact	A3-S104, A3-S120, V1-S096
GDPR / residency	Managed regions US (default), EU and APAC (EU/APAC on Pro and above, enabled by support); regions do not share state; BYOC on AWS and GCP keeps traffic and logs in customer VPC	Verified fact	A3-S120, A3-S104
Security	One Firecracker microVM per sandbox with own cgroup and network	Verified fact	A3-S062

⁵⁰¹ Verified fact — A3-S062: E2B Runtime README (e2b-dev/infra) (E2B (GitHub)) <https://raw.githubusercontent.com/e2b-dev/infra/main/README.md> accessed 2026-10-07

⁵⁰² Verified fact — A3-S104: E2B Security (and e2b.dev/enterprise, pricing, docs BYOC) <https://e2b.dev/security> accessed 2026-10-07 [search-tool extract]; V1-S096: E2B Security; Enterprise; BYOC docs <https://e2b.dev/security> accessed 2026-10-08 [search-tool extract]

⁵⁰³ Verified fact — A3-S104: E2B Security (and e2b.dev/enterprise, pricing, docs BYOC) <https://e2b.dev/security> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
features	namespace; per-sandbox nftables egress firewall with domain allow/deny lists; secrets never cross API/logs/spans; short-lived workload identity tokens; per-sandbox access tokens for URLs	d fact	
Access controls	SSO, SCIM and RBAC listed as planned on Enterprise page (page about 436 days old); Enterprise telemetry to customer OTLP endpoint and signed lifecycle webhooks	Verified fact	A3-S120
Enterprise support	Enterprise dedicated deployments offered	Verified fact	A3-S062
Pricing	Hobby free (one-time US\$100 credit); Pro US\$150/month plus usage; Enterprise custom with US\$3,000/month minimum; per-second billing, paused sandboxes not billed	Verified fact	A3-S104
Infrastructure cost	Self-host cost drivers: KVM hosts, object storage for templates/snapshots, PostgreSQL, Redis and ClickHouse	Verified fact	A3-S062
Ecosystem	OpenTelemetry export built in	Verified fact	A3-S062
Adoption signals	e2b-dev/E2B 14,219 GitHub stars (7 October 2026); US\$21M Series A led by Insight Partners (July 2025); vendor claims 88% of Fortune 100 signed up	Verified fact	A3-S054, A3-S105
Maturity	SDK first released August 2023; 365 releases	Verified fact	A3-S007
Deployment	saas: Yes (E2B Cloud); managed_cloud: Yes (dedicated deployment run by E2B inside customer account); vpc_byoc: Yes (BYOC on AWS and GCP; Azure not yet per docs, listed on Enterprise page - conflict); private_cloud: Not publicly verified; self_hosted: Yes for evaluation (E2B Embed single host; GCP Terraform and Kubernetes manifests) - vendor says not a production pattern; on_prem: Not publicly verified		

Agent Skills (SKILL.md open format) (L4-agent-skills)

Tier: Tactical · Flags: none · Original graphic label: Agent Skills – reusable skills

Rationale: Useful format, but vendor-led governance, no versioning and no provenance keep it out of foundational use ^{AJ}.

Criterion	Score	Rationale
Technical	3	Packages procedural knowledge competently; no versioning, dependency or signing model ^{AJ}
Enterprise readiness	2	Open format (rule 2): enterprise provisioning is a vendor feature, not part of the standard ^{vf-504}
Security and	2	Rule 2 hygiene: executable content without signing or provenance ^{vf-505}

⁵⁰⁴ Verified fact — A3-S029: Skills for organizations, partners, the ecosystem (18 Dec 2025) (Anthropic) <https://claude.com/blog/organization-skills-and-directory> accessed 2026-10-07

⁵⁰⁵ Verified fact — A3-S037: Agent Skills Specification (Agent Skills project (agentskills.io)) <https://agentskills.io/specification> accessed 2026-10-07 [search-tool extract]; A3-S072: Equipping agents for the real world with Agent Skills (Anthropic) <https://www.anthropic.com/engineering/equipping-agents-for-the-real-world-with-agent-skills> accessed 2026-10-07

Criterion	Score	Rationale
compliance		
Deployment flexibility	5	Plain files usable anywhere ^{vr506}
Ecosystem	4	46 clients including major vendors, but young ^{vr507}
Reliability and maturity	2	Under one year as an open standard; no tagged releases ^{vr508}
Cost / TCO	5	Free, no operations ^{vr509}
Lock-in / portability	3	Portable Markdown, but no neutral governance ^{vr510}
Total (generic / FS)	3.20	
	/	
	3.00	

Capabilities. Open packaging format (SKILL.md plus optional scripts, references, assets) with progressive disclosure; adopted by 46 listed clients including OpenAI Codex and Gemini CLI ^{vr511}. Conflict of interest: originated at Anthropic and maintained by Anthropic staff; author is an Anthropic model ^{AI}.

Strengths

- Plain, reviewable files portable across clients ^{vr512}
- Independent adoption confirmed by OpenAI and Google documentation ^{vr513}

Limitations and risks

- No neutral governance body; maintainers are Anthropic employees; AAIF proposal not accepted on available evidence ^{vr514}

⁵⁰⁶ Verified fact — A3-S061: agentskills/agentskills README (Agent Skills project (GitHub)) <https://raw.githubusercontent.com/agentskills/agentskills/main/README.md> accessed 2026-10-07

⁵⁰⁷ Verified fact — A3-S113: Agent Skills client showcase list (docs/snippets/clients.jsx) (Agent Skills project (GitHub)) <https://raw.githubusercontent.com/agentskills/agentskills/main/docs/snippets/clients.jsx> accessed 2026-10-07

⁵⁰⁸ Verified fact — A3-S029: Skills for organizations, partners, the ecosystem (18 Dec 2025) (Anthropic) <https://claude.com/blog/organization-skills-and-directory> accessed 2026-10-07; A3-S115: (Project Proposal) Agent Skills (aaif/project-proposals issue #47) (Agentic AI Foundation (GitHub)) <https://github.com/aaif/project-proposals/issues/47> accessed 2026-10-07 [search-tool extract]

⁵⁰⁹ Verified fact — A3-S061: agentskills/agentskills README (Agent Skills project (GitHub)) <https://raw.githubusercontent.com/agentskills/agentskills/main/README.md> accessed 2026-10-07

⁵¹⁰ Verified fact — V1-S046: Skills Over MCP (AAIF blog); Agent Plugins 1.0 post; agentskills.io specification (Agentic AI Foundation / agentskills.io) <https://aaif.io/blog/skills-over-mcp> accessed 2026-10-08 [search-tool extract]

⁵¹¹ Verified fact — A3-S061: agentskills/agentskills README (Agent Skills project (GitHub)) <https://raw.githubusercontent.com/agentskills/agentskills/main/README.md> accessed 2026-10-07; A3-S113: Agent Skills client showcase list (docs/snippets/clients.jsx) (Agent Skills project (GitHub)) <https://raw.githubusercontent.com/agentskills/agentskills/main/docs/snippets/clients.jsx> accessed 2026-10-07; A3-S033: Codex: Build skills (OpenAI) <https://developers.openai.com/codex/skills> accessed 2026-10-07 [search-tool extract]; A3-S034: Gemini CLI: Agent Skills (Google (Gemini CLI docs)) <https://gemini-cli.com/docs/cli/skills/> accessed 2026-10-07 [search-tool extract]

⁵¹² Verified fact — A3-S061: agentskills/agentskills README (Agent Skills project (GitHub)) <https://raw.githubusercontent.com/agentskills/agentskills/main/README.md> accessed 2026-10-07

⁵¹³ Verified fact — A3-S033: Codex: Build skills (OpenAI) <https://developers.openai.com/codex/skills> accessed 2026-10-07 [search-tool extract]; A3-S034: Gemini CLI: Agent Skills (Google (Gemini CLI docs)) <https://gemini-cli.com/docs/cli/skills/> accessed 2026-10-07 [search-tool extract]; V1-S047: Codex: Build skills (OpenAI) <https://developers.openai.com/codex/skills> accessed 2026-10-08 [search-tool extract]

⁵¹⁴ Verified fact — V1-S046: Skills Over MCP (AAIF blog); Agent Plugins 1.0 post; agentskills.io specification (Agentic AI Foundation / agentskills.io) <https://aaif.io/blog/skills-over-mcp> accessed 2026-10-08 [search-tool extract]; A3-S115: (Project Proposal) Agent Skills (aaif/project-proposals issue #47) (Agentic AI Foundation (GitHub)) <https://github.com/aaif/project-proposals/issues/47> accessed 2026-10-07 [search-tool extract]; A3-S112: Agent Skills

- Living document with no tagged releases ^{vi}⁵¹⁵
- No signing or provenance; skills can carry executable scripts; vendor warns of exfiltration ^{vi}⁵¹⁶

Choose when

- Packaging internally authored house procedures for several agent clients ^{ai}

Avoid when

- Skills would come from outside the firm or carry scripts that run with agent credentials ^{ai}

Nearest competitors: AGENTS.md, C5 prompt packages (Git-based), L4-mcp

Regulated-FS note. Internally authored, Git-reviewed, script-free skills only until signing, versioning and neutral governance exist; independent alternatives are C5 prompt packages and AGENTS.md ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Originated at Anthropic (launched 16 October 2025; published as an open standard 18 December 2025); specification maintained in the agentskills/agentskills repository and agentskills.io; no published governance body or charter, and no evidence that it is an AAIF-hosted project (AAIF material treats the format as owned by the Agent Skills specification; secondary claims that Anthropic stewards it through AAIF are unsupported)	Verified fact	A3-S028, A3-S029, A3-S061, V1-S046
Category	Open packaging format for procedural knowledge: folders with SKILL.md (name and description required) plus optional scripts, references and assets	Verified fact	A3-S061, A3-S037
Version / lineup	No versioned releases: specification is a 'living document'; tagged releases with changelog are planned (AAIF proposal); no CHANGELOG.md in repository root on 7 October 2026	Verified fact	A3-S115, A3-S112
Licence	Open specification; repository code Apache-2.0, documentation CC-BY-4.0	Verified fact	A3-S061
Status events	18 December 2025: published as an open standard	Verified fact	A3-S029, V1-S040
Strategic direction	Portable across agent products; AAIF 'Skills Over MCP' working group exploring delivery of skills via MCP; Claude adds org-wide admin provisioning and a partner skills directory	Verified fact	A3-S038, A3-S029
What it does	Agents load only skill names/descriptions at start-up, read the full SKILL.md when a task matches (progressive disclosure) and may execute bundled scripts	Verified fact	A3-S061
Stack position	Sits between L3 agent harness and L4 tools: packaged instructions plus code, often wrapping MCP tools	Architectural	

CONTRIBUTING.md (Agent Skills project (GitHub)) <https://raw.githubusercontent.com/agentskills/agentskills/main/CONTRIBUTING.md> accessed 2026-10-07

⁵¹⁵ Verified fact — A3-S115: (Project Proposal) Agent Skills (aaif/project-proposals issue #47) (Agentic AI Foundation (GitHub)) <https://github.com/aaif/project-proposals/issues/47> accessed 2026-10-07 [search-tool extract]

⁵¹⁶ Verified fact — A3-S037: Agent Skills Specification (Agent Skills project (agentskills.io)) <https://agentskills.io/specification> accessed 2026-10-07 [search-tool extract]; A3-S061: agentskills/agentskills README (Agent Skills project (GitHub)) <https://raw.githubusercontent.com/agentskills/agentskills/main/README.md> accessed 2026-10-07; A3-S072: Equipping agents for the real world with Agent Skills (Anthropic) <https://www.anthropic.com/engineering/equipping-agents-for-the-real-world-with-agent-skills> accessed 2026-10-07

Field	Value	Label	Sources
		judgement	
Integration	Filesystem convention (.agents/skills and vendor paths); Claude API /v1/skills endpoint; adopted by OpenAI Codex and Google Gemini CLI	Verified fact	A3-S029, A3-S033, A3-S034
Dependencies	A skills-compatible agent with file access; scripts require a code-execution environment (Claude requires Code Execution and File Creation)	Verified fact	A3-S029, A3-S061
Certifications	Not applicable (open format)	Architectural judgement	
GDPR / residency	Not applicable (open format)	Architectural judgement	
Security features	No signing or provenance in the core format found; Anthropic warns malicious skills may exfiltrate data and advises installing only from trusted sources; Gemini CLI docs advise inspecting third-party skills	Verified fact	A3-S072, A3-S034, A3-S037
Access controls	Claude Team/Enterprise admins can provision skills centrally (vendor feature, not part of the standard)	Verified fact	A3-S029
Enterprise support	Not applicable (open format)	Architectural judgement	
Pricing	Free open format	Verified fact	A3-S061
Infrastructure cost	Not applicable	Architectural judgement	
Ecosystem	46 client products listed on agentskills.io showcase (e.g. ChatGPT & Codex, Gemini CLI, GitHub Copilot, VS Code, Cursor, Goose, Kiro, Junie, Databricks Genie Code, Snowflake Cortex Code, Mistral AI Vibe, Spring AI, Letta); OpenAI and Google primary docs confirm support	Verified fact	A3-S027, A3-S033, A3-S034, A3-S113, V1-S047
Adoption signals	agentskills/agentskills 25,954 GitHub stars (7 October 2026)	Verified fact	A3-S054
Maturity	Specification repository created 16 December 2025; no versioned releases found	Verified fact	A3-S054
Deployment	saas: No (open specification; deployment depends on each implementation); managed_cloud: No (open specification; deployment depends on each implementation); vpc_byoc: No (open specification; deployment depends on each implementation); private_cloud: No (open specification; deployment depends on each implementation); self_hosted: Yes (implementations can be self-hosted; the specification itself is not a hosted service); on_prem: Yes (implementations can be		

Field	Value	Label	Sources
	self-hosted; the specification itself is not a hosted service)		

Composio (L4-composio)

Tier: Experimental · **Flags:** none · **Original graphic label:** Composio – integrations

Rationale: Credential-broker model, vendor-only security evidence and the May 2026 token-exposure incident make it unsuitable as a critical dependency ^{AJ}.

Criterion	Score	Rationale
Technical	4	Broad connectors, per-user auth, gateway and policy features ^{VF517}
Enterprise readiness	3	SSO, SCIM, RBAC, audit logs and SLA are vendor-stated; one-or-more controls lift the cap to 3 under CP2 rule 7 ^{VF518}
Security and compliance	2	Capped at 2: certifications vendor-marketing only (V1 section 4); May 2026 incident ^{VF519}
Deployment flexibility	4	SaaS plus VPC, self-host and on-prem on Enterprise; no air-gap evidence ^{VF520}
Ecosystem	4	Adapters for major frameworks and remote MCP server ^{VF521}
Reliability and maturity	2	Pre-1.0 SDK, 2.0 beta, recent security incident ^{VF522}
Cost / TCO	3	Metered per-call pricing published (low confidence) ^{VF523}
Lock-in / portability	2	Proprietary API holding user tokens; hard exit ^{AJ}
Total (generic / FS)	3.15 / 2.90	

Evidence rules applied: security_compliance capped at 2: Composio security claims are vendor marketing only (V1 verification log section 4)

⁵¹⁷ Verified fact — A3-S063: ComposioHQ/composio README (Composio (GitHub)) <https://raw.githubusercontent.com/ComposioHQ/composio/next/README.md> accessed 2026-10-07; A3-S119: Composio: Two questions every security review asks us (and MCP gateway/governance pages) <https://composio.dev/blog/two-questions-every-security-review-asks-us> accessed 2026-10-07 [search-tool extract]

⁵¹⁸ Verified fact — A3-S119: Composio: Two questions every security review asks us (and MCP gateway/governance pages) <https://composio.dev/blog/two-questions-every-security-review-asks-us> accessed 2026-10-07 [search-tool extract]; A3-S098: Composio Enterprise (and related composio.dev content pages) <https://composio.dev/enterprise> accessed 2026-10-07 [search-tool extract]

⁵¹⁹ Verified fact — B-L4-S007: Composio May 2026 security incident update <https://composio.dev/blog/composio-may-2026-security-incident> accessed 2026-10-08 [search-tool extract]

⁵²⁰ Verified fact — A3-S098: Composio Enterprise (and related composio.dev content pages) <https://composio.dev/enterprise> accessed 2026-10-07 [search-tool extract]

⁵²¹ Verified fact — A3-S063: ComposioHQ/composio README (Composio (GitHub)) <https://raw.githubusercontent.com/ComposioHQ/composio/next/README.md> accessed 2026-10-07

⁵²² Verified fact — A3-S008: PyPI JSON metadata: composio (Python Package Index (PyPI)) <https://pypi.org/pypi/composio/json> accessed 2026-10-07; B-L4-S007: Composio May 2026 security incident update <https://composio.dev/blog/composio-may-2026-security-incident> accessed 2026-10-08 [search-tool extract]

⁵²³ Verified fact — A3-S098: Composio Enterprise (and related composio.dev content pages) <https://composio.dev/enterprise> accessed 2026-10-07 [search-tool extract]

Capabilities. Managed tool-integration platform: 1000+ toolkits, per-user sessions, OAuth/connected-account brokering, per-session tool restriction, and an MCP Gateway with vendor-stated SSO, SCIM, policy-as-code and per-call audit logs ^{VF-524}.

Strengths

- Broadest SaaS connector catalogue with per-user authentication ^{VF-525}
- Self-hosted, VPC and on-premises options on Enterprise ^{VF-526}

Limitations and risks

- May 2026 incident: unauthorised access to internal systems; about 0.3% of connections and 5,241 API keys possibly exposed; token revocations and customer rotation required ^{VF-527}
- Security claims are vendor marketing only (V1 log section 4) ^{NIPV}
- Holds end users' OAuth tokens: high switching cost ^{AJ}
- Managed cloud US-hosted; EU residency needs self-hosting ^{VF-528}
- SDK pre-1.0 with 2.0 in beta ^{VF-529}

Choose when

- A low-risk internal productivity agent needs many SaaS connectors quickly, self-hosted ^{AJ}

Avoid when

- The agent touches client data or you cannot self-host in-region ^{AJ}

Nearest competitors: L4-aws-agentcore-gateway-identity, C4-okta-auth0-ai-agents, C1-kong-ai-gateway

Regulated-FS note. Do not let a third party hold staff OAuth tokens to firm systems in its multi-tenant cloud; if used, self-host and keep the token vault in your estate ^{ROC}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Composio (SDK copyright Sampark Inc.; San Francisco)	Verified fact	A3-S063, A3-S067,

⁵²⁴ Verified fact — A3-S063: ComposioHQ/composio README (Composio (GitHub)) <https://raw.githubusercontent.com/ComposioHQ/composio/next/README.md> accessed 2026-10-07; A3-S008: PyPI JSON metadata: composio (Python Package Index (PyPI)) <https://pypi.org/pypi/composio/json> accessed 2026-10-07; A3-S119: Composio: Two questions every security review asks us (and MCP gateway/governance pages) <https://composio.dev/blog/two-questions-every-security-review-asks-us> accessed 2026-10-07 [search-tool extract]; A3-S098: Composio Enterprise (and related composio.dev content pages) <https://composio.dev/enterprise> accessed 2026-10-07 [search-tool extract]

⁵²⁵ Verified fact — A3-S063: ComposioHQ/composio README (Composio (GitHub)) <https://raw.githubusercontent.com/ComposioHQ/composio/next/README.md> accessed 2026-10-07

⁵²⁶ Verified fact — A3-S098: Composio Enterprise (and related composio.dev content pages) <https://composio.dev/enterprise> accessed 2026-10-07 [search-tool extract]

⁵²⁷ Verified fact — B-L4-S007: Composio May 2026 security incident update <https://composio.dev/blog/composio-may-2026-security-incident> accessed 2026-10-08 [search-tool extract]

⁵²⁸ Verified fact — A3-S119: Composio: Two questions every security review asks us (and MCP gateway/governance pages) <https://composio.dev/blog/two-questions-every-security-review-asks-us> accessed 2026-10-07 [search-tool extract]

⁵²⁹ Verified fact — A3-S008: PyPI JSON metadata: composio (Python Package Index (PyPI)) <https://pypi.org/pypi/composio/json> accessed 2026-10-07; V1-S098: PyPI JSON metadata: composio (0.25.0; 2.0.0b0 pre-release 31 Aug 2026) (Python Package Index) <https://pypi.org/pypi/composio/json> accessed 2026-10-08

Field	Value	Label	Sources
			A3-S099
Category	Managed tool-integration platform for agents: 1000+ pre-authenticated toolkits, per-user sessions, authentication, triggers and a sandbox	Verified fact	A3-S063
Version / lineup	Python SDK composio 0.25.0 (29 September 2026); 2.0.0b0 pre-release (31 August 2026)	Verified fact	A3-S008, V1-S098
Licence	SDKs MIT; platform is a hosted proprietary service (API key from dashboard)	Verified fact	A3-S067, A3-S063
Strategic direction	Session-based meta tools; markets an MCP Gateway product with governance (policy, audit, SSO)	Verified fact	A3-S008, A3-S119
What it does	Creates per-user sessions whose tools an agent calls across 1000+ apps; handles OAuth/connected accounts and restricts toolkits, tools, auth configs and connected accounts per session	Verified fact	A3-S063, A3-S008
Stack position	L4 integration/tool-hub; also acts as a managed credential broker for third-party apps	Verified fact	A3-S063
Integration	Python/TypeScript SDKs, CLI, provider adapters for OpenAI Agents, Claude Agent SDK, Vercel AI SDK, LangChain; remote MCP server	Verified fact	A3-S063, A3-S054
Dependencies	Composio hosted API	Verified fact	A3-S063
Certifications	Vendor states SOC 2 Type II and ISO/IEC 27001:2022; evidence via trust centre (security.composio.dev)	Verified fact	A3-S098
GDPR / residency	Managed cloud hosted in the US; EU-only residency requires self-hosted Enterprise deployment	Verified fact	A3-S119
Security features	Credentials AES-256 encrypted and isolated from app code and LLM context; KMS key management on Enterprise; Zero Data Retention on Pro and above	Verified fact	A3-S098, A3-S119
Access controls	SAML 2.0/OIDC SSO (Okta, Entra ID, Google Workspace) and SCIM 2.0 on Enterprise; action-level policy-as-code RBAC; per-call audit logs incl. denied calls, 7-day to 1-year retention, SIEM export; per-session tool restrictions	Verified fact	A3-S119, A3-S098, A3-S008
Enterprise support	Enterprise: MSA/DPA/SLA, dedicated support, committed volume	Verified fact	A3-S098
Pricing	Metered by tool call: free 20,000 calls/month; US\$29/month for 200,000; US\$229/month for 2M; Enterprise custom (as reported on Composio comparison pages)	Verified fact	A3-S098
Infrastructure cost	Not applicable (SaaS)	Architectural judgment	
Ecosystem	Provider adapters for major agent frameworks; maintains awesome-claude-skills list (76,670 stars)	Verified fact	A3-S063, A3-S054
Adoption signals	ComposioHQ/composio 30,464 GitHub stars (7 October 2026); US\$25M Series A led by Lightspeed (July 2025; US\$29M total, Reported)	Verified fact	A3-S054, A3-S099
Maturity	Current Python SDK line first published December 2024; pre-1.0 with 2.0 in beta	Verified fact	A3-S008
Deployment	saas: Yes; managed_cloud: Yes (regional managed cloud handling); vpc_byoc: Yes (Enterprise VPC deployment); private_cloud: Not publicly		

Field	Value	Label	Sources
	verified; self_hosted: Yes (Enterprise fully self-hosted); on_prem: Yes (on-premise option on Enterprise)		

Exa (L4-exa)

Tier: Tactical · Flags: none · Original graphic label: Exa – search API

Rationale: Competent, certified search API; SaaS-only with no EU region ^{AJ}.

Criterion	Score	Rationale
Technical	3	Covers search and extraction competently ^{vf530}
Enterprise readiness	3	SSO and SCIM verified; RBAC and audit logs not found (CP2 rule 7: max 3) ^{vf531}
Security and compliance	3	SOC 2 Type II, DPA; no ISO 27001 found ^{vf532}
Deployment flexibility	1	One SaaS service, no region choice found ^{vf533}
Ecosystem	3	SDKs and MCP server; mainstream ^{vf534}
Reliability and maturity	3	Steady releases since January 2024; well funded ^{vf535} ^{re536}
Cost / TCO	3	Transparent per-request pricing with a page conflict ^{vf537}

⁵³⁰ Verified fact — A3-S010: PyPI JSON metadata: exa-py (Python Package Index (PyPI)) <https://pypi.org/pypi/exa-py/json> accessed 2026-10-07

⁵³¹ Verified fact — A3-S121: Exa Security Overview (and exa.ai/enterprise, exa.ai/dpa) <https://exa.ai/docs/reference/security> accessed 2026-10-07 [search-tool extract]

⁵³² Verified fact — A3-S100: Exa Trust Center (and exa.ai pricing/enterprise/FAQ pages) <https://trust.exa.ai/> accessed 2026-10-07 [search-tool extract]; A3-S121: Exa Security Overview (and exa.ai/enterprise, exa.ai/dpa) <https://exa.ai/docs/reference/security> accessed 2026-10-07 [search-tool extract]

⁵³³ Verified fact — A3-S121: Exa Security Overview (and exa.ai/enterprise, exa.ai/dpa) <https://exa.ai/docs/reference/security> accessed 2026-10-07 [search-tool extract]

⁵³⁴ Verified fact — A3-S054: GitHub repository metadata (stars, forks, created, archived) retrieved via GitHub MCP search_repositories tool, 7 Oct 2026 <https://api.github.com/search/repositories> accessed 2026-10-07 [search-tool extract]

⁵³⁵ Verified fact — A3-S010: PyPI JSON metadata: exa-py (Python Package Index (PyPI)) <https://pypi.org/pypi/exa-py/json> accessed 2026-10-07

⁵³⁶ Reported — A3-S101: Latham & Watkins advises Exa in Series C (May 2026) <https://www.lw.com/en/news/2026/05/latham-advises-exa-in-series-c> accessed 2026-10-07 [search-tool extract]

⁵³⁷ Verified fact — A3-S100: Exa Trust Center (and exa.ai pricing/enterprise/FAQ pages) <https://trust.exa.ai/> accessed 2026-10-07 [search-tool extract]

Criterion	Score	Rationale
Lock-in / portability	3	Proprietary index, but trivially swappable behind one interface ^{AJ}
Total (generic / FS)	2.70 / 2.70	

Capabilities. Web search API for AI with filters, page contents, answers and structured output; MCP server ^{VF-538}.

Strengths

- SOC 2 Type II with 2026 bridge letter; ZDR on Enterprise; DPA with SCCs and UK Addendum ^{VF-539}
- Dashboard SSO and SCIM on Enterprise ^{VF-540}

Limitations and risks

- SaaS only; no EU processing region found ^{VF-541}
- Pricing conflict US\$7 vs US\$4 per 1,000 searches ^{VF-542}
- Search queries can leak confidential intent ^{AJ}

Choose when

- Public-web research on non-confidential questions ^{AJ}

Avoid when

- Queries could contain client, holdings or deal information ^{AJ}

Nearest competitors: L4-tavily, L8-firecrawl, L8-apify

Regulated-FS note. Enable ZDR, route through the egress proxy with DLP, and register as an ICT third-party service ^{REC}.

Facts (as of 2026-10-07)

Field	Value	Label	Sources
Company	Exa (Exa Labs, Inc.)	Verified fact	A3-S010, A3-S121
Category	Web search API for AI (search, contents, answer, streaming)	Verified fact	A3-S010
Version / lineup	exa-py 2.25.0 (1 October 2026)	Verified fact	A3-S010

⁵³⁸ Verified fact — A3-S010: PyPI JSON metadata: exa-py (Python Package Index (PyPI)) <https://pypi.org/pypi/exa-py/json> accessed 2026-10-07; A3-S054: GitHub repository metadata (stars, forks, created, archived) retrieved via GitHub MCP search_repositories tool, 7 Oct 2026 <https://api.github.com/search/repositories> accessed 2026-10-07 [search-tool extract]

⁵³⁹ Verified fact — A3-S100: Exa Trust Center (and exa.ai pricing/enterprise/FAQ pages) <https://trust.exa.ai/> accessed 2026-10-07 [search-tool extract]; A3-S121: Exa Security Overview (and exa.ai/enterprise, exa.ai/dpa) <https://exa.ai/docs/reference/security> accessed 2026-10-07 [search-tool extract]

⁵⁴⁰ Verified fact — A3-S121: Exa Security Overview (and exa.ai/enterprise, exa.ai/dpa) <https://exa.ai/docs/reference/security> accessed 2026-10-07 [search-tool extract]

⁵⁴¹ Verified fact — A3-S121: Exa Security Overview (and exa.ai/enterprise, exa.ai/dpa) <https://exa.ai/docs/reference/security> accessed 2026-10-07 [search-tool extract]

⁵⁴² Verified fact — A3-S100: Exa Trust Center (and exa.ai pricing/enterprise/FAQ pages) <https://trust.exa.ai/> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
Licence	SDK MIT; search API proprietary	Verified fact	A3-S010
Strategic direction	Search with structured output schemas and streamed answers	Verified fact	A3-S010
What it does	Searches the web with date/domain filters, returns page contents/highlights, generates answers and supports structured output via output_schema	Verified fact	A3-S010
Stack position	L4 tool (web retrieval) feeding agents; overlaps L8 web extraction	Verified fact	A3-S010
Integration	Python and JavaScript SDKs; MCP server (exa-mcp-server)	Verified fact	A3-S010, A3-S054
Dependencies	Exa hosted API (API key)	Verified fact	A3-S010
Certifications	SOC 2 Type II (security, confidentiality, availability); 2026 bridge letter; HIPAA BAA for eligible Enterprise customers	Verified fact	A3-S100, A3-S121
GDPR / residency	DPA with EU SCCs (2021/914) and UK Addendum; GDPR/CCPA rights honoured; zero data retention on Enterprise; no EU processing region stated	Verified fact	A3-S121, A3-S100
Security features	Not publicly verified	Not publicly verified	
Access controls	SSO and SCIM for the Exa Dashboard on Enterprise	Verified fact	A3-S121
Enterprise support	Enterprise: custom MSA/DPA, volume discounts, custom datasets	Verified fact	A3-S100, A3-S121
Pricing	Search US\$7 per 1,000 requests; Monitors US\$15 per 1,000; one page says from US\$4 per 1,000 (conflict); Enterprise custom	Verified fact	A3-S100
Infrastructure cost	Not applicable (SaaS)	Architectural judgment	
Ecosystem	MCP server for agent clients	Verified fact	A3-S054
Adoption signals	exa-mcp-server 5,091 GitHub stars (7 October 2026); US\$85M Series B at US\$700M (September 2025, Benchmark) and US\$250M Series C at US\$2.2B (May 2026, a16z) (Reported)	Verified fact	A3-S054, A3-S101
Maturity	exa-py first released January 2024; 130 releases	Verified fact	A3-S010
Deployment	saas: Yes; managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Not publicly verified; on_prem: Not publicly verified		

Tavily (L4-tavily)

Tier: Tactical • **Flags:** Acquired • **Original graphic label:** Tavily – search API

Rationale: Useful search API with good proxy support; SaaS-only and newly acquired ^{AJ}.

Criterion	Score	Rationale
Technical	3	Search, extract, crawl and research endpoints ^{VF} ⁵⁴³
Enterprise readiness	3	RBAC roles verified (CP2 rule 7 lifts cap to 3); SSO and audit logs not found ^{VF} ⁵⁴⁴
Security and compliance	3	SOC 2 Type II per Trust Center; GDPR and ZDR stated; no ISO ^{VF} ⁵⁴⁵
Deployment flexibility	1	SaaS only, no region choice found ^{VF} ⁵⁴⁶
Ecosystem	3	SDKs and MCP server ^{VF} ⁵⁴⁷
Reliability and maturity	3	Releases since 2023; acquired by Nebius February 2026 ^{VF} ⁵⁴⁸
Cost / TCO	3	Transparent credit pricing ^{VF} ⁵⁴⁹
Lock-in / portability	2	Proprietary but swappable; reduced by 1 for ownership change (rule 3) ^{AJ}
Total (generic / FS)	2.65	/

⁵⁴³ Verified fact — A3-S009: PyPI JSON metadata: tavily-python (Python Package Index (PyPI)) <https://pypi.org/pypi/tavily-python/json> accessed 2026-10-07

⁵⁴⁴ Verified fact — B-L4-S006: Tavily help centre: Team Roles and Permissions (Owner/Admin/Member) <https://help.tavily.com/articles/7788853762-team-roles-and-permissions-owner-admin-member> accessed 2026-10-08 [search-tool extract]

⁵⁴⁵ Verified fact — B-L4-S005: Tavily Trust Center FAQ <https://trust.tavily.com/faq> accessed 2026-10-08 [search-tool extract]; A3-S088: Tavily pricing, FAQ and help-centre security pages <https://www.tavily.com/pricing> accessed 2026-10-07 [search-tool extract]

⁵⁴⁶ Verified fact — A3-S088: Tavily pricing, FAQ and help-centre security pages <https://www.tavily.com/pricing> accessed 2026-10-07 [search-tool extract]

⁵⁴⁷ Verified fact — A3-S054: GitHub repository metadata (stars, forks, created, archived) retrieved via GitHub MCP search_repositories tool, 7 Oct 2026 <https://api.github.com/search/repositories> accessed 2026-10-07 [search-tool extract]

⁵⁴⁸ Verified fact — A3-S009: PyPI JSON metadata: tavily-python (Python Package Index (PyPI)) <https://pypi.org/pypi/tavily-python/json> accessed 2026-10-07; V1-S041: Nebius Group N.V. Form 20-F FY2025 (subsequent event: Tavily acquisition) (Nebius Group N.V. (SEC EDGAR)) <https://www.sec.gov/Archives/edgar/data/0001513845/000110465926052948/nbis-20251231x20f.htm> accessed 2026-10-08 [search-tool extract]

⁵⁴⁹ Verified fact — A3-S088: Tavily pricing, FAQ and help-centre security pages <https://www.tavily.com/pricing> accessed 2026-10-07 [search-tool extract]

Criterion	Score	Rationale
	2.55	

Capabilities. Search, extract, crawl, map and research API for agents; custom HTTP session injection for proxying through an enterprise gateway; MCP server ^{VF-550}.

Strengths

- Gateway-proxy support built into the SDK ^{VF-551}
- SOC 2 Type II via Trust Center; Owner/Admin/Member roles; Enterprise SLAs ^{VF-552}

Limitations and risks

- Acquired by Nebius (closed 19 February 2026); roadmap tied to a neocloud ^{VF-553}
- No EU processing region found; SSO not found ^{VF-554} ^{NPV}

Choose when

- Search plus extraction in one API, proxied through your gateway ^{AJ}

Avoid when

- You need EU processing or roadmap independence from Nebius ^{AJ}

Nearest competitors: L4-exa, L8-firecrawl

Regulated-FS note. Refresh third-party due diligence after the change of control and plan contract changes around PS7/26 notification lead times ^{REC}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Tavily, a wholly owned subsidiary of Nebius Group N.V. (acquisition announced 10 February 2026; closed 19 February 2026)	Verified fact	A3-S084, A3-S085, A3-S086, V1-S041
Category	Search, extract, crawl, map and research API for agents	Verified fact	A3-S009
Version / lineup	tavily-python 0.8.5 (6 October 2026); @tavily/core 0.7.14 (6 October 2026)	Verified fact	A3-S009, A3-S083

⁵⁵⁰ Verified fact — A3-S009: PyPI JSON metadata: tavily-python (Python Package Index (PyPI)) <https://pypi.org/pypi/tavily-python/json> accessed 2026-10-07; A3-S054: GitHub repository metadata (stars, forks, created, archived) retrieved via GitHub MCP search_repositories tool, 7 Oct 2026 <https://api.github.com/search/repositories> accessed 2026-10-07 [search-tool extract]

⁵⁵¹ Verified fact — A3-S009: PyPI JSON metadata: tavily-python (Python Package Index (PyPI)) <https://pypi.org/pypi/tavily-python/json> accessed 2026-10-07

⁵⁵² Verified fact — B-L4-S005: Tavily Trust Center FAQ <https://trust.tavily.com/faq> accessed 2026-10-08 [search-tool extract]; B-L4-S006: Tavily help centre: Team Roles and Permissions (Owner/Admin/Member) <https://help.tavily.com/articles/7788853762-team-roles-and-permissions-owner-admin-member> accessed 2026-10-08 [search-tool extract]; A3-S088: Tavily pricing, FAQ and help-centre security pages <https://www.tavily.com/pricing> accessed 2026-10-07 [search-tool extract]

⁵⁵³ Verified fact — V1-S041: Nebius Group N.V. Form 20-F FY2025 (subsequent event: Tavily acquisition) (Nebius Group N.V. (SEC EDGAR)) <https://www.sec.gov/Archives/edgar/data/0001513845/000110465926052948/nbis-20251231x20f.htm> accessed 2026-10-08 [search-tool extract]; A3-S084: Tavily is Joining Nebius (10 Feb 2026) <https://www.tavily.com/blog/tavily-is-joining-nebius> accessed 2026-10-07 [search-tool extract]

⁵⁵⁴ Verified fact — A3-S088: Tavily pricing, FAQ and help-centre security pages <https://www.tavily.com/pricing> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
Licence	Proprietary API; JavaScript SDK MIT; Python SDK licence not stated in PyPI metadata	Verified fact	A3-S083, A3-S009
Status events	10 February 2026: Tavily announced it is joining Nebius; 19 February 2026: acquisition closed, Nebius acquiring 100% ownership (Nebius FY2025 Form 20-F); Nebius 6-K describes a merger making Tavily a wholly owned subsidiary, cash consideration; Nebius Q1 2026 filing accounts for the acquisition as completed: fair value of consideration US\$189.7M (US\$177.2M cash) plus an ARR-based earnout measured December 2026 and March 2027; Press reported a US\$275M deal value (not disclosed by Nebius)	Verified fact	A3-S084, A3-S085, A3-S086, A3-S087, V1-S041
Strategic direction	Integration into Nebius AI cloud (Token Factory reported); Tavily states API, data policies and zero data retention unchanged; adds asynchronous research tasks and keyless trial mode	Verified fact	A3-S084, A3-S086, A3-S009
What it does	Provides web search, content extraction, site crawl/map and multi-step research endpoints via API	Verified fact	A3-S009
Stack position	L4 tool (web retrieval) for agents; overlaps L8	Verified fact	A3-S009
Integration	Python/JS SDKs; supports custom HTTP session injection so enterprises can proxy traffic through an API gateway for central auth, logging and policy; MCP server	Verified fact	A3-S009, A3-S054
Dependencies	Tavily hosted API	Verified fact	A3-S009
Certifications	Vendor states SOC 2 certified; report type and period not confirmed (trust.tavily.com)	Verified fact	A3-S088
GDPR / residency	Vendor states GDPR and CCPA compliant; zero data retention described as core; EU processing region not found	Verified fact	A3-S088, A3-S084
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Enterprise plan (custom): custom rate limits, dedicated account manager, Slack channel, AI engineer support, uptime and support SLAs	Verified fact	A3-S088
Pricing	Free 1,000 credits per month; pay-as-you-go US\$0.008 per credit; monthly plans US\$0.0075-0.005 per credit; Enterprise custom	Verified fact	A3-S088, A3-S009
Infrastructure cost	Not applicable (SaaS)	Architectural judgment	
Ecosystem	MCP server (tavily-mcp)	Verified fact	A3-S054
Adoption	tavily-mcp 2,422 and tavily-python 1,420 GitHub stars (7 October 2026)	Verified fact	A3-S054

Field	Value	Label	Sources
signals		d fact	
Maturity	tavily-python first released September 2023; 69 releases	Verified fact	A3-S009
Deployment	saas: Yes; managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Not publicly verified; on_prem: Not publicly verified		

L3: Agent frameworks and orchestration

LangGraph (open-source framework); managed runtime now branded LangSmith Deployment (formerly LangGraph Platform) (L3-langgraph)

Tier: Strategic · **Flags:** Renamed · **Original graphic label:** LangGraph – workflow

Rationale: Deepest coverage of the L3 questions with a stable 1.x line; Renamed refers to LangGraph Platform -> LangSmith Deployment.

Criterion	Score	Rationale
Technical	5	Durable execution, interrupts, time travel, memory, deterministic plus LLM nodes, both Functional and Graph APIs ^{vr555}
Enterprise readiness	4	LangSmith Enterprise: SAML/OIDC SSO, SCIM, custom-role RBAC, audit logs on self-hosted ^{vr556}
Security and compliance	4	LangSmith SOC 2 Type II and HIPAA; ISO 27001 claimed, scope unconfirmed, so 4 not 5 (rule 8); OSS advisories published and patched ^{vr557}
Deployment flexibility	5	Library anywhere; LangSmith Deployment SaaS, BYOC/hybrid, self-hosted and on-prem Kubernetes ^{vr558}
Ecosystem	5	LangChain integrations, JS port, hosted by AgentCore and other runtimes ^{vr559}
Reliability and maturity	4	1.0 GA October 2025, Production/Stable, frequent releases, stable API commitment ^{vr560}

⁵⁵⁵ Verified fact — A4-S118: LangGraph README (GitHub) (LangChain) <https://raw.githubusercontent.com/langchain-ai/langgraph/main/README.md> accessed 2026-10-07; A4-S039: LangGraph Persistence / Durable execution / Interrupts - Docs by LangChain <https://docs.langchain.com/oss/python/langgraph/durable-execution> accessed 2026-10-07 [search-tool extract]

⁵⁵⁶ Verified fact — A4-S038: User management - Docs by LangChain (with self-hosted changelog) <https://docs.langchain.com/langsmith/user-management> accessed 2026-10-07 [search-tool extract]

⁵⁵⁷ Verified fact — A4-S034: LangSmith Regions FAQ (LangChain) <https://docs.langchain.com/langsmith/regions-faq> accessed 2026-10-07 [search-tool extract]; B-L3-S007: LangGraph security advisories (GitHub) and NVD entries (LangChain / GitHub / NIST NVD) <https://github.com/langchain-ai/langgraph/security> accessed 2026-10-08 [search-tool extract]

⁵⁵⁸ Verified fact — A4-S036: LangSmith Plans and Pricing (LangChain) <https://www.langchain.com/pricing> accessed 2026-10-07 [search-tool extract]; A4-S037: LangSmith Deployment - Docs by LangChain <https://docs.langchain.com/langsmith/deployment> accessed 2026-10-07 [search-tool extract]

⁵⁵⁹ Verified fact — A4-S118: LangGraph README (GitHub) (LangChain) <https://raw.githubusercontent.com/langchain-ai/langgraph/main/README.md> accessed 2026-10-07; A4-S116: Bedrock AgentCore SDK README (GitHub) (AWS) <https://raw.githubusercontent.com/aws/bedrock-agentcore-sdk-python/main/README.md> accessed 2026-10-07

⁵⁶⁰ Verified fact — A4-S001: PyPI JSON metadata: langgraph (Python Package Index) <https://pypi.org/pypi/langgraph/json> accessed 2026-10-07; A4-S033: LangChain and LangGraph Agent Frameworks Reach v1.0 Milestones <https://www.langchain.com/blog/langchain-langgraph-1dot0> accessed 2026-10-07 [search-tool extract]

Criterion	Score	Rationale
Cost / TCO	4	OSS free; LangSmith seat and LSU pricing published ^{vr561}
Lock-in / portability	3	MIT, self-hostable, but graph/checkpoint APIs proprietary to LangGraph and deployment is proprietary ^{vr562}
Total (generic / FS)	4.40 / 4.20	

Capabilities. Low-level stateful graph runtime (MIT) combining deterministic code and LLM decisions in one graph, with checkpointed durable execution, interrupts for human-in-the-loop, time travel, memory and streaming; managed runtime is LangSmith Deployment (formerly LangGraph Platform) ^{vr563}.

Strengths

- Covers every L3 research question in one GA (1.x) framework with a no-breaking-changes commitment until 2.0 ^{vr564 AJ}
- Deployment range from library to SaaS, BYOC, self-hosted and on-prem Kubernetes ^{vr565}

Limitations and risks

- Graph and checkpoint APIs are LangGraph-specific ^{vr566}
- Three checkpoint deserialisation advisories in 2025-26, all patched; the checkpoint store is a security boundary ^{vr567}
- LangSmith ISO 27001 claimed but scope unconfirmed ^{vr568}
- Full self-hosted platform needs the Enterprise plan ^{vr569}

Choose when

⁵⁶¹ Verified fact — A4-S036: LangSmith Plans and Pricing (LangChain) <https://www.langchain.com/pricing> accessed 2026-10-07 [search-tool extract]

⁵⁶² Verified fact — A4-S001: PyPI JSON metadata: langgraph (Python Package Index) <https://pypi.org/pypi/langgraph/json> accessed 2026-10-07; A4-S039: LangGraph Persistence / Durable execution / Interrupts - Docs by LangChain <https://docs.langchain.com/oss/python/langgraph/durable-execution> accessed 2026-10-07 [search-tool extract]

⁵⁶³ Verified fact — A4-S118: LangGraph README (GitHub) (LangChain) <https://raw.githubusercontent.com/langchain-ai/langgraph/main/README.md> accessed 2026-10-07; A4-S039: LangGraph Persistence / Durable execution / Interrupts - Docs by LangChain <https://docs.langchain.com/oss/python/langgraph/durable-execution> accessed 2026-10-07 [search-tool extract]; A4-S031: Product naming changes: LangSmith Deployment and LangSmith Studio (LangChain) <https://changelog.langchain.com/announcements/product-naming-changes-langsmith-deployment-and-langsmith-studio> accessed 2026-10-07 [search-tool extract]

⁵⁶⁴ Verified fact — A4-S033: LangChain and LangGraph Agent Frameworks Reach v1.0 Milestones <https://www.langchain.com/blog/langchain-langgraph-1dot0> accessed 2026-10-07 [search-tool extract]

⁵⁶⁵ Verified fact — A4-S036: LangSmith Plans and Pricing (LangChain) <https://www.langchain.com/pricing> accessed 2026-10-07 [search-tool extract]; A4-S037: LangSmith Deployment - Docs by LangChain <https://docs.langchain.com/langsmith/deployment> accessed 2026-10-07 [search-tool extract]

⁵⁶⁶ Verified fact — A4-S039: LangGraph Persistence / Durable execution / Interrupts - Docs by LangChain <https://docs.langchain.com/oss/python/langgraph/durable-execution> accessed 2026-10-07 [search-tool extract]

⁵⁶⁷ Verified fact — B-L3-S007: LangGraph security advisories (GitHub) and NVD entries (LangChain / GitHub / NIST NVD) <https://github.com/langchain-ai/langgraph/security> accessed 2026-10-08 [search-tool extract]

⁵⁶⁸ Verified fact — A4-S034: LangSmith Regions FAQ (LangChain) <https://docs.langchain.com/langsmith/regions-faq> accessed 2026-10-07 [search-tool extract]; A4-S035: LangChain Enterprise <https://www.langchain.com/enterprise> accessed 2026-10-07 [search-tool extract]; A4-S149: LangSmith Engine security - Docs by LangChain (with Trust Center, enterprise and shared-responsibility pages) <https://docs.langchain.com/langsmith/engine-security> accessed 2026-10-07 [search-tool extract]

⁵⁶⁹ Verified fact — A4-S037: LangSmith Deployment - Docs by LangChain <https://docs.langchain.com/langsmith/deployment> accessed 2026-10-07 [search-tool extract]

- You need deterministic workflow graphs with one or more LLM steps, durable checkpoints and approval interrupts ^{AJ}
- Python or TypeScript teams standardising on one orchestration framework ^{AJ}

Avoid when

- The workflow is a single model call that needs no framework ^{AJ}
- You cannot restrict write access to the checkpoint database ^{AJ}

Nearest competitors: L3-microsoft-agent-framework, L3-pydantic-ai, L3-google-adk, L3-temporal

Regulated-FS note. Pin langgraph and langgraph-checkpoint, keep checkpoints in a firm-controlled Postgres with write access limited to the runtime, and export traces via OTel as well as LangSmith ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	LangChain, Inc.	Verified fact	A4-S001, A4-S032
Category	Low-level stateful agent and workflow orchestration framework (graph runtime) with a commercial managed deployment service	Verified fact	A4-S118, A4-S037
Version / lineup	langgraph 1.2.14 on PyPI (released 6 October 2026); @langchain/langgraph 1.4.21 on npm (7 October 2026); 1.0.0 released 17 October 2025 (Python) / 18 October 2025 (JS)	Verified fact	A4-S001, A4-S026
Licence	Open source, MIT (framework). LangSmith (Observability, Evaluation, Deployment) is proprietary SaaS / self-hosted enterprise software	Verified fact	A4-S001, A4-S026, A4-S036
Status events	14 October 2025: LangGraph Platform renamed LangSmith Deployment; LangGraph Studio renamed LangSmith Studio; 17 October 2025: LangGraph 1.0.0 GA on PyPI; 1 October 2026: existing LangSmith Deployment customers move from per-run/uptime pricing to usage-based LSU pricing	Verified fact	A4-S031, A4-S001, A4-S036, V1-S056
Strategic direction	LangGraph 1.0 positioned as 'the first stable major release in the durable agent framework space' with no breaking changes until 2.0; LangChain 1.0 create_agent runs on the LangGraph runtime; Deep Agents harness built on LangGraph. Interrupt 2026 (13-14 May 2026) launched LangSmith Engine, Managed Deep Agents, SmithDB, Context Hub, an LLM Gateway and Sandboxes GA; Agent Builder rebranded LangSmith Fleet (March 2026)	Verified fact	A4-S033, A4-S118, A4-S040

Field	Value	Label	Sources
What it does	Provides low-level infrastructure for long-running, stateful workflows or agents: durable execution that resumes from where it left off, human-in-the-loop via interrupts, short- and long-term memory, streaming, and deployment through LangSmith. Deterministic code and LLM-driven decisions can be combined in one graph; a Functional API (@entrypoint/@task) offers a workflow style.	Verified fact	A4-S118, A4-S039
Stack position	Agent orchestration runtime (L3). LangSmith Deployment is a separate agent-hosting runtime; LangSmith Observability/Evaluation sits in L9. The graphic's label 'workflow' captures only part of its role: LangGraph covers both deterministic workflows and agent loops.	Verified fact	A4-S039, A4-S031
Integration	Python and JS/TS libraries; LangGraph SDK / Agent Server API for deployed graphs; LangChain model and tool integrations; tracing to LangSmith	Verified fact	A4-S118, A4-S037
Dependencies	Python >=3.10 (or JS/TS); checkpointeer backend for persistence (durable, database-backed checkpointeer recommended in production, e.g. Postgres via langgraph-checkpoint-postgres); LangSmith for deployment/observability (optional); any LLM provider via LangChain integrations	Verified fact	A4-S001, A4-S039
Certifications	LangSmith: SOC 2 Type II and HIPAA (BAA on Enterprise) consistently stated; ISO 27001 stated only on the enterprise marketing page and LangSmith Engine security docs ('certified to ISO 27001'), while LangSmith enterprise/shared-responsibility docs and the EU-residency announcement omit it and the trust-centre listing seen does not name it. Treat ISO 27001 as claimed, scope unconfirmed. Not applicable to the OSS library.	Verified fact	A4-S034, A4-S035, A4-S149
GDPR / residency	LangSmith regional instances: GCP EU (eu.smith.langchain.com), GCP US, GCP APAC, AWS US; available on all plans; region migration not supported; no EU legal entity for contracting; DPA on request	Verified fact	A4-S034
Security features	Not publicly verified	Not publicly verified	
Access controls	LangSmith Enterprise: SAML 2.0 / OIDC SSO with JIT provisioning, SCIM 2.0, workspace RBAC with custom roles; self-hosted v0.14: ABAC and audit logs (OCSF 1.7.0) enabled by default	Verified fact	A4-S038
Enterprise support	LangSmith Enterprise plan (custom pricing, invoiced annually upfront)	Verified fact	A4-S036
Pricing	LangGraph OSS free (MIT). LangSmith Plus US\$39 per seat per month (10k base traces/month then pay-as-you-go; 1 free small serverless deployment); Enterprise custom. Deployment billed on resources consumed in LangChain Standard Units (LSU), replacing per-run/uptime pricing (pricing page shows 1 LSU = US\$1 in the Engine section)	Verified fact	A4-S036
Infrastructure cost	Self-hosting requires a durable checkpointeer database; self-hosted LangSmith Deployment runs control plane and Agent Servers on customer Kubernetes	Verified fact	A4-S039, A4-S037
Ecosystem	LangChain integrations; Deep Agents; LangSmith; JS/TS port (LangGraph.js); NVIDIA Dynamo publishes a LangChain integration	Verified fact	A4-S118, A4-S026,

Field	Value	Label	Sources
			A4-S098
Adoption signals	README names Klarna, Replit and Elastic; LangChain raised US\$125M at US\$1.25B valuation (IVP lead) alongside the 1.0 releases (October 2025)	Verified fact	A4-S118, A4-S032
Maturity	First PyPI release 8 January 2024; 1.0 GA October 2025 (Development Status: Production/Stable); frequent releases (1.2.14 on 6 October 2026)	Verified fact	A4-S001
Deployment	saas: Yes (LangSmith Deployment cloud; Plus plan or above); managed_cloud: Yes (LangSmith Deployment managed cloud); vpc_byoc: Yes (BYOC and hybrid: data plane in customer VPC; Enterprise plan); private_cloud: Yes (self-hosted LangSmith with Deployment control plane on Kubernetes; Enterprise plan); self_hosted: Yes (open-source library runs anywhere; full self-hosted platform requires Enterprise plan); on_prem: Yes (self-hosted Kubernetes deployment of LangSmith; Enterprise)		

Temporal (durable execution platform; Python SDK temporalio) (L3-temporal)

Tier: Strategic · **Flags:** none · **Original graphic label:** not in graphic

Rationale: Strategic as the durable-execution substrate under deterministic workflows; independent of model vendors.

Criterion	Score	Rationale
Technical	4	Leading durability, retries and deterministic workflow semantics; no agent loop itself ^{vr570}
Enterprise readiness	4	SAML SSO, SCIM, roles and custom roles, audit logs, Cloud Ops API; SLA not verified ^{vr571}
Security and compliance	3	SOC 2 Type 2, HIPAA BAA, GDPR DPA, client-side encryption; no ISO 27001 found ^{vr572}
Deployment flexibility	5	Temporal Cloud on 14 AWS and 6 GCP regions incl. Europe; MIT self-host anywhere ^{vr573}
Ecosystem	4	Built into Pydantic AI, OpenAI Agents SDK, Mistral Workflows ^{vr574}

⁵⁷⁰ Verified fact — A4-S019: PyPI JSON metadata: temporalio (Python Package Index) <https://pypi.org/pypi/temporalio/json> accessed 2026-10-07; A4-S045: Pydantic AI Durable Execution (overview, Temporal, DBOS, Prefect) https://pydantic.dev/docs/ai/capabilities/durable_execution/overview/ accessed 2026-10-07 [search-tool extract]

⁵⁷¹ Verified fact — B-L3-S003: Temporal Cloud account access / SCIM / audit logs (Temporal Technologies Inc.) <https://docs.temporal.io/cloud/manage-access> accessed 2026-10-08 [search-tool extract]

⁵⁷² Verified fact — B-L3-S001: Temporal security / Trust Center / Cloud security model (Temporal Technologies Inc.) <https://trust.temporal.io/> accessed 2026-10-08 [search-tool extract]

⁵⁷³ Verified fact — B-L3-S002: Temporal server repository; production deployment and Cloud overview docs (Temporal Technologies Inc.) <https://github.com/temporalio/temporal> accessed 2026-10-08 [search-tool extract]

⁵⁷⁴ Verified fact — A4-S045: Pydantic AI Durable Execution (overview, Temporal, DBOS, Prefect) https://pydantic.dev/docs/ai/capabilities/durable_execution/overview/ accessed 2026-10-07 [search-tool extract]; A4-S052: OpenAI Agents SDK documentation (agents, sessions, handoffs, guardrails, tracing, running agents) <https://openai.github.io/openai-agents-python/> accessed 2026-10-07 [search-tool extract]; A4-S058: Workflows for work that runs the business (Mistral Workflows) (Mistral AI) <https://mistral.ai/news/workflows/> accessed 2026-10-07 [search-tool extract]

Criterion	Score	Rationale
Reliability and maturity	4	Python SDK since March 2022; long-established ^{vf575}
Cost / TCO	3	Cloud pricing not verified; self-host operations heavy ^{NPV vf576}
Lock-in / portability	4	MIT server; portable between self-host and Cloud; code is Temporal-specific ^{vf577}
Total (generic / FS)	3.90 / 3.90	

Capabilities. Durable execution engine: workflows whose activities are recorded so execution resumes after failure; used beneath Pydantic AI, the OpenAI Agents SDK and Mistral Workflows. MIT server, self-hosted or Temporal Cloud on AWS and GCP ^{vf578}.

Strengths

- Separates reliability (retries, resume, timers, signals) from agent logic ^{AJ}
- Same server in self-hosted and Cloud; moves without code changes ^{vf579}
- Cloud: SAML SSO, SCIM, roles, exportable audit logs ^{vf580}

Limitations and risks

- Not an agent framework ^{AJ}
- Self-hosting needs Kubernetes and persistence stores, sequential upgrades ^{vf581}
- No ISO 27001 found; certification evidence from search extracts ^{vf582}
- Workflow code becomes a hard dependency ^{AJ}
- Cloud pricing not verified ^{NPV}

Choose when

- Any regulated agent workflow that must resume without re-calling tools ^{AJ}

Avoid when

⁵⁷⁵ Verified fact — A4-S019: PyPI JSON metadata: temporalio (Python Package Index) <https://pypi.org/pypi/temporalio/json> accessed 2026-10-07

⁵⁷⁶ Verified fact — B-L3-S002: Temporal server repository; production deployment and Cloud overview docs (Temporal Technologies Inc.) <https://github.com/temporalio/temporal> accessed 2026-10-08 [search-tool extract]

⁵⁷⁷ Verified fact — B-L3-S002: Temporal server repository; production deployment and Cloud overview docs (Temporal Technologies Inc.) <https://github.com/temporalio/temporal> accessed 2026-10-08 [search-tool extract]

⁵⁷⁸ Verified fact — A4-S019: PyPI JSON metadata: temporalio (Python Package Index) <https://pypi.org/pypi/temporalio/json> accessed 2026-10-07; A4-S045: Pydantic AI Durable Execution (overview, Temporal, DBOS, Prefect) https://pydantic.dev/docs/ai/capabilities/durable_execution/overview/ accessed 2026-10-07 [search-tool extract]; A4-S052: OpenAI Agents SDK documentation (agents, sessions, handoffs, guardrails, tracing, running agents) <https://openai.github.io/openai-agents-python/> accessed 2026-10-07 [search-tool extract]; A4-S058: Workflows for work that runs the business (Mistral Workflows) (Mistral AI) <https://mistral.ai/news/workflows/> accessed 2026-10-07 [search-tool extract]; B-L3-S002: Temporal server repository; production deployment and Cloud overview docs (Temporal Technologies Inc.) <https://github.com/temporalio/temporal> accessed 2026-10-08 [search-tool extract]

⁵⁷⁹ Verified fact — B-L3-S002: Temporal server repository; production deployment and Cloud overview docs (Temporal Technologies Inc.) <https://github.com/temporalio/temporal> accessed 2026-10-08 [search-tool extract]

⁵⁸⁰ Verified fact — B-L3-S003: Temporal Cloud account access / SCIM / audit logs (Temporal Technologies Inc.) <https://docs.temporal.io/cloud/manage-access> accessed 2026-10-08 [search-tool extract]

⁵⁸¹ Verified fact — B-L3-S002: Temporal server repository; production deployment and Cloud overview docs (Temporal Technologies Inc.) <https://github.com/temporalio/temporal> accessed 2026-10-08 [search-tool extract]

⁵⁸² Verified fact — B-L3-S001: Temporal security / Trust Center / Cloud security model (Temporal Technologies Inc.) <https://trust.temporal.io/> accessed 2026-10-08 [search-tool extract]

- Short, stateless single-call tasks ^{AJ}

Nearest competitors: DBOS, Restate, L3-langgraph, L3-microsoft-agent-framework

Regulated-FS note. Encrypt payloads client-side so Temporal Cloud stores ciphertext, or self-host in-region ⁵⁸³
VF Rec.

Facts (as of 2026-10-07)

Field	Value	Label	Sources
Company	Temporal Technologies Inc.	Verified fact	A4-S019
Category	Durable execution / workflow orchestration engine used as the reliability layer under agent frameworks	Verified fact	A4-S019, A4-S045, A4-S058
Version / lineup	temporalio (Python SDK) 1.34.0 (30 September 2026)	Verified fact	A4-S019
Licence	Python SDK MIT; server licence and Temporal Cloud terms not verified in this run	Verified fact	A4-S019
Status events	Not publicly verified	Not publicly verified	
Strategic direction	Not publicly verified	Not publicly verified	
What it does	Runs workflows whose steps (activities) are recorded so execution resumes after failures. Agent frameworks run model calls, tool calls and	Verified fact	A4-S045, A4-S052,

⁵⁸³ Verified fact — B-L3-S001: Temporal security / Trust Center / Cloud security model (Temporal Technologies Inc.) <https://trust.temporal.io/> accessed 2026-10-08 [search-tool extract]

Field	Value	Label	Sources
	MCP traffic as Temporal activities (Pydantic AI), offer Temporal integrations for long-running human-in-the-loop agents (OpenAI Agents SDK), or build their workflow product on it (Mistral Workflows).		A4-S058
Stack position	Cross-cutting reliability substrate beneath L3 (deterministic workflow layer in H2), not an agent framework itself	Architectural judgment	
Integration	Not publicly verified	Not publicly verified	
Dependencies	Not publicly verified	Not publicly verified	
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Not publicly verified	Not publicly verified	
Infrastructure	Not publicly verified	Not	

Field	Value	Label	Sources
cost		publicly verified	
Ecosystem	Integrated by Pydantic AI, OpenAI Agents SDK and Mistral Workflows; alternatives DBOS (dbos 3.2.0, MIT) and Restate (restate-sdk 1.0.5)	Verified fact	A4-S045, A4-S052, A4-S058, A4-S020, A4-S104
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Python SDK since March 2022	Verified fact	A4-S019
Deployment	saas: Not publicly verified; managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Not publicly verified; on_prem: Not publicly verified		

Microsoft Agent Framework (MAF) (L3-microsoft-agent-framework)

Tier: Strategic · **Flags:** none · **Original graphic label:** Agent Framework – Microsoft

Rationale: Strategic, conditional: where Microsoft/Azure or .NET is the primary platform; supersedes Semantic Kernel and AutoGen.

Criterion	Score	Rationale
Technical	4	Agents plus graph workflows, checkpointing, HITL, time travel, A2A/MCP; durability beta ^{vr584}
Enterprise readiness	4	Library (rule 2) with Microsoft long-term support commitment; Foundry Hosted Agents GA ^{vr585}

⁵⁸⁴ Verified fact — A4-S066: Microsoft Agent Framework README (GitHub) <https://raw.githubusercontent.com/microsoft/agent-framework/main/README.md> accessed 2026-10-07; A4-S101: PyPI JSON metadata: agent-framework-durabletask (Python Package Index) <https://pypi.org/pypi/agent-framework-durabletask/json> accessed 2026-10-07

⁵⁸⁵ Verified fact — A4-S012: PyPI JSON metadata: semantic-kernel (Python Package Index) <https://pypi.org/pypi/semantic-kernel/json> accessed 2026-10-07; B-L3-S006: Foundry Hosted Agents (Agent Framework hosting docs and Foundry blog) (Microsoft) <https://learn.microsoft.com/en-us/agent-framework/hosting/foundry-hosted-agent> accessed 2026-10-08 [search-tool extract]

Criterion	Score	Rationale
Security and compliance	3	Library (rule 2); hygiene not assessed; Azure/Foundry certifications out of record scope ^{AJ}
Deployment flexibility	4	Library anywhere; Foundry Hosted Agents and Azure Functions managed ^{VF⁵⁸⁶}
Ecosystem	4	Foundry, Azure OpenAI, OpenAI, Copilot Studio, A2A, MCP, OTel ^{VF⁵⁸⁷}
Reliability and maturity	4	GA April 2026 with LTS commitment; successor to two mature frameworks ^{VF⁵⁸⁸}
Cost / TCO	4	Free MIT; durable hosting extra ^{VF⁵⁸⁹}
Lock-in / portability	3	MIT, multi-provider, but Azure-first integrations ^{VF⁵⁹⁰}
Total (generic / FS)	3.80	
	/	
	3.65	

Capabilities. MIT framework (Python, .NET, Go) that separates Agents from graph-based Workflows (sequential, concurrent, handoff, group collaboration) with checkpointing, streaming, HITL and time travel; durable agents via Durable Task or Azure Durable Functions (beta); A2A, MCP, OTel; declared successor to Semantic Kernel and AutoGen; Foundry Hosted Agents for managed hosting ^{VF⁵⁹¹}.

Strengths

- Explicit agents-versus-workflows split is the clearest H2 implementation ^{VF⁵⁹² AJ}
- 1.0 with stable APIs and a long-term support commitment ^{VF⁵⁹³}
- .NET first-class for Microsoft estates ^{VF⁵⁹⁴}

⁵⁸⁶ Verified fact — A4-S008: PyPI JSON metadata: agent-framework (Python Package Index) <https://pypi.org/pypi/agent-framework/json> accessed 2026-10-07; A4-S066: Microsoft Agent Framework README (GitHub) <https://raw.githubusercontent.com/microsoft/agent-framework/main/README.md> accessed 2026-10-07

⁵⁸⁷ Verified fact — A4-S066: Microsoft Agent Framework README (GitHub) <https://raw.githubusercontent.com/microsoft/agent-framework/main/README.md> accessed 2026-10-07; A4-S008: PyPI JSON metadata: agent-framework (Python Package Index) <https://pypi.org/pypi/agent-framework/json> accessed 2026-10-07

⁵⁸⁸ Verified fact — A4-S008: PyPI JSON metadata: agent-framework (Python Package Index) <https://pypi.org/pypi/agent-framework/json> accessed 2026-10-07; A4-S012: PyPI JSON metadata: semantic-kernel (Python Package Index) <https://pypi.org/pypi/semantic-kernel/json> accessed 2026-10-07

⁵⁸⁹ Verified fact — A4-S067: Microsoft Agent Framework LICENSE <https://raw.githubusercontent.com/microsoft/agent-framework/main/LICENSE> accessed 2026-10-07

⁵⁹⁰ Verified fact — A4-S066: Microsoft Agent Framework README (GitHub) <https://raw.githubusercontent.com/microsoft/agent-framework/main/README.md> accessed 2026-10-07

⁵⁹¹ Verified fact — A4-S066: Microsoft Agent Framework README (GitHub) <https://raw.githubusercontent.com/microsoft/agent-framework/main/README.md> accessed 2026-10-07; A4-S012: PyPI JSON metadata: semantic-kernel (Python Package Index) <https://pypi.org/pypi/semantic-kernel/json> accessed 2026-10-07; A4-S101: PyPI JSON metadata: agent-framework-durabletask (Python Package Index) <https://pypi.org/pypi/agent-framework-durabletask/json> accessed 2026-10-07; A4-S102: PyPI JSON metadata: agent-framework-azurefunctions (Python Package Index) <https://pypi.org/pypi/agent-framework-azurefunctions/json> accessed 2026-10-07; B-L3-S006: Foundry Hosted Agents (Agent Framework hosting docs and Foundry blog) (Microsoft) <https://learn.microsoft.com/en-us/agent-framework/hosting/foundry-hosted-agent> accessed 2026-10-08 [search-tool extract]

⁵⁹² Verified fact — A4-S066: Microsoft Agent Framework README (GitHub) <https://raw.githubusercontent.com/microsoft/agent-framework/main/README.md> accessed 2026-10-07

⁵⁹³ Verified fact — A4-S012: PyPI JSON metadata: semantic-kernel (Python Package Index) <https://pypi.org/pypi/semantic-kernel/json> accessed 2026-10-07; A4-S068: AutoGen README (GitHub) (Microsoft) <https://raw.githubusercontent.com/microsoft/autogen/main/README.md> accessed 2026-10-07

⁵⁹⁴ Verified fact — A4-S066: Microsoft Agent Framework README (GitHub) <https://raw.githubusercontent.com/microsoft/agent-framework/main/README.md> accessed 2026-10-07

Limitations and risks

- Durable extensions still beta ^{vf595}
- Foundry hosting integration package for Python prerelease ^{vf596}
- First-class integrations are Azure/Foundry ^{vf597}
- Young GA (April 2026) ^{vf598}

Choose when

- Microsoft/.NET estate, or Semantic Kernel/AutoGen code to migrate ^{AI}

Avoid when

- You need GA durable execution today without adding Temporal or Durable Task ^{AI}

Nearest competitors: L3-langgraph, L3-google-adk, L3-aws-strands-agentcore

Regulated-FS note. Migrate Semantic Kernel and AutoGen workloads here; until durable packages are GA, back long-running workflows with a GA durable engine ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Microsoft Corporation	Verified fact	A4-S067
Category	Multi-language agent and multi-agent workflow framework (Python, .NET, Go)	Verified fact	A4-S066
Version / lineup	agent-framework (Python) 1.20.0 (2 October 2026); 1.0.0 GA 2 April 2026 (Development Status: Production/Stable); durable extensions agent-framework-durabletask and agent-framework-azurefunctions still beta (1.0.0b260922)	Verified fact	A4-S008, A4-S101, A4-S102
Licence	Open source, MIT	Verified fact	A4-S067, A4-S008
Status events	1 October 2025: first public preview on PyPI; 2 April 2026: 1.0.0 GA; Semantic Kernel README: 'Semantic Kernel is now Microsoft Agent Framework'; AutoGen placed in maintenance mode (no new features; community managed); last autogen-agentchat release 0.7.5 on 30 September 2025	Verified fact	A4-S008, A4-S012, A4-S068, A4-S021, V1-S050,

⁵⁹⁵ Verified fact — A4-S101: PyPI JSON metadata: agent-framework-durabletask (Python Package Index) <https://pypi.org/pypi/agent-framework-durabletask/json> accessed 2026-10-07; A4-S102: PyPI JSON metadata: agent-framework-azurefunctions (Python Package Index) <https://pypi.org/pypi/agent-framework-azurefunctions/json> accessed 2026-10-07

⁵⁹⁶ Verified fact — B-L3-S006: Foundry Hosted Agents (Agent Framework hosting docs and Foundry blog) (Microsoft) <https://learn.microsoft.com/en-us/agent-framework/hosting/foundry-hosted-agent> accessed 2026-10-08 [search-tool extract]

⁵⁹⁷ Verified fact — A4-S066: Microsoft Agent Framework README (GitHub) <https://raw.githubusercontent.com/microsoft/agent-framework/main/README.md> accessed 2026-10-07

⁵⁹⁸ Verified fact — A4-S008: PyPI JSON metadata: agent-framework (Python Package Index) <https://pypi.org/pypi/agent-framework/json> accessed 2026-10-07

Field	Value	Label	Sources
			V1-S052
Strategic direction	Declared 'enterprise-ready successor' to both Semantic Kernel and AutoGen; graph-based workflows (sequential, concurrent, handoff, group collaboration) with checkpointing, streaming, human-in-the-loop and time-travel; Foundry Hosted Agents; A2A and MCP interoperability; declarative YAML agents; Agent Skills	Verified fact	A4-S066, A4-S012, A4-S068
What it does	Builds agents (chat-client based, multi-provider) and multi-agent workflows; middleware; OpenTelemetry observability; DevUI; durable agents via Durable Task or Azure Durable Functions that persist state and recover from failures.	Verified fact	A4-S066, A4-S101, A4-S102
Stack position	L3; explicitly separates Agents from graph-based Workflows within one framework	Verified fact	A4-S066
Integration	Python packages (agent-framework-core, -foundry, preview - copilotstudio), NuGet for .NET; A2A and MCP; OpenTelemetry	Verified fact	A4-S008, A4-S066, A4-S012
Dependencies	Python >=3.10 or .NET (Go SDK separate repo); providers incl. Microsoft Foundry, Azure OpenAI, OpenAI, GitHub Copilot SDK; optional Durable Task / Azure Functions	Verified fact	A4-S008, A4-S066, A4-S101
Certifications	Not applicable to the MIT library; Azure/Foundry certifications out of scope for this record	Architectural judgment	
GDPR / residency	Inherits the chosen model provider/hosting (e.g. Azure region)	Architectural judgment	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Microsoft states 1.0 comes with stable APIs and a commitment to long-term support	Verified fact	A4-S012, A4-S068
Pricing	Free (MIT); hosting and models billed separately	Verified fact	A4-S067
Infrastructure cost	Durable agents require a Durable Task scheduler or Azure Functions	Architectural judgment	
Ecosystem	Microsoft Foundry, Azure OpenAI, OpenAI, Copilot Studio, GitHub Copilot SDK; A2A; MCP	Verified fact	A4-S066, A4-S008
Adoption	Not publicly verified	Not	

Field	Value	Label	Sources
signals		publicly verified	
Maturity	GA since April 2026; weekly-to-fortnightly minor releases (1.13 on 30 July 2026 to 1.20 on 2 October 2026)	Verified fact	A4-S008
Deployment	saas: No for the framework; Foundry Hosted Agents provides managed hosting; managed_cloud: Yes (Foundry Hosted Agents; Azure Durable Functions); vpc_byoc: Yes (library runs inside the customer's own environment); private_cloud: Yes (library runs inside the customer's own environment); self_hosted: Yes; on_prem: Yes (library)		

Pydantic AI (L3-pydantic-ai)

Tier: Strategic · **Flags:** none · **Original graphic label:** Pydantic AI – type-safe

Rationale: Strategic, conditional: for Python teams wanting type-safe agents, typically as the typed agent step inside a Temporal or DBOS workflow; pin the major version (breaking v2 ten months after v1) and note that commercial support is not verified. Upgraded from Tactical by the reader at Checkpoint 4 (CP4-4); scores unchanged (FS 3.55) ^{AJ}.

Criterion	Score	Rationale
Technical	4	Typed agent loop, graph control flow, durable capabilities via Temporal/DBOS/Prefect, MCP ^{VF599}
Enterprise readiness	3	Library (rule 2): enables controls in your estate; support not verified ^{NPV}
Security and compliance	3	Library hygiene: v1 security fixes promised for six months after v2 ^{VF600} ; inherits host controls
Deployment flexibility	4	Library anywhere; no managed runtime ^{VF601}
Ecosystem	4	Temporal, DBOS, Prefect integrations co-maintained; Logfire OTel ^{VF602}
Reliability and maturity	3	Production/Stable, but breaking v2 ten months after v1 ^{VF603}
Cost / TCO	4	Free MIT; durability backend adds cost ^{VF604 AJ}

⁵⁹⁹ Verified fact — A4-S119: Pydantic AI README (GitHub) <https://raw.githubusercontent.com/pydantic/pydantic-ai/main/README.md> accessed 2026-10-07; A4-S045: Pydantic AI Durable Execution (overview, Temporal, DBOS, Prefect) https://pydantic.dev/docs/ai/capabilities/durable_execution/overview/ accessed 2026-10-07 [search-tool extract]

⁶⁰⁰ Verified fact — V1-S074: Pydantic AI Upgrade Guide; Pydantic AI v2 article; Version Policy <https://pydantic.dev/docs/ai/project/changelog/> accessed 2026-10-08 [search-tool extract]

⁶⁰¹ Verified fact — A4-S003: PyPI JSON metadata: pydantic-ai (Python Package Index) <https://pypi.org/pypi/pydantic-ai/json> accessed 2026-10-07

⁶⁰² Verified fact — A4-S045: Pydantic AI Durable Execution (overview, Temporal, DBOS, Prefect) https://pydantic.dev/docs/ai/capabilities/durable_execution/overview/ accessed 2026-10-07 [search-tool extract]

⁶⁰³ Verified fact — A4-S003: PyPI JSON metadata: pydantic-ai (Python Package Index) <https://pypi.org/pypi/pydantic-ai/json> accessed 2026-10-07; A4-S044: Pydantic AI v2: capabilities, a leaner core, and the Harness (with upgrade guide) <https://pydantic.dev/articles/pydantic-ai-v2> accessed 2026-10-07 [search-tool extract]

⁶⁰⁴ Verified fact — A4-S003: PyPI JSON metadata: pydantic-ai (Python Package Index) <https://pypi.org/pypi/pydantic-ai/json> accessed 2026-10-07

Criterion	Score	Rationale
Lock-in / portability	4	MIT, model-agnostic, durability on standard engines ^{VF605}
Total (generic / FS)	3.60 / 3.55	

Capabilities. Typed, model-agnostic Python agent loop with structured outputs and tools, pydantic_graph for graph control flow, and durable execution attached as a capability via Temporal, DBOS or Prefect ^{VF606}.

Strengths

- Type-checked inputs and outputs suit numeric and structured tasks ^{AJ}
- Delegates durability to established engines instead of inventing its own ^{VF607 AJ}

Limitations and risks

- v2 (June 2026) removed graph persistence; two majors in about ten months ^{VF608}
- Commercial support not publicly verified ^{NPV}
- Small vendor: US\$12.5M Series A is the only round found ^{VF609}

Choose when

- Python teams wanting a typed agent step inside a Temporal or DBOS workflow ^{AJ}

Avoid when

- You need a vendor-supported platform with SLAs ^{AJ}

Nearest competitors: L3-langgraph, L3-openai-agents-sdk, L3-temporal

Regulated-FS note. Pin the major version; v1 receives security fixes for at least six months after v2 ^{VF610 Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Pydantic Services Inc.	Verifie d fact	A4-S003, A4-S046
Category	Typed Python agent framework (agent loop) with optional graph library	Verifie	A4-S119,

⁶⁰⁵ Verified fact — A4-S003: PyPI JSON metadata: pydantic-ai (Python Package Index) <https://pypi.org/pypi/pydantic-ai/json> accessed 2026-10-07; A4-S119: Pydantic AI README (GitHub) <https://raw.githubusercontent.com/pydantic/pydantic-ai/main/README.md> accessed 2026-10-07

⁶⁰⁶ Verified fact — A4-S119: Pydantic AI README (GitHub) <https://raw.githubusercontent.com/pydantic/pydantic-ai/main/README.md> accessed 2026-10-07; A4-S044: Pydantic AI v2: capabilities, a leaner core, and the Harness (with upgrade guide) <https://pydantic.dev/articles/pydantic-ai-v2> accessed 2026-10-07 [search-tool extract]; A4-S045: Pydantic AI Durable Execution (overview, Temporal, DBOS, Prefect) https://pydantic.dev/docs/ai/capabilities/durable_execution/overview/ accessed 2026-10-07 [search-tool extract]

⁶⁰⁷ Verified fact — A4-S045: Pydantic AI Durable Execution (overview, Temporal, DBOS, Prefect) https://pydantic.dev/docs/ai/capabilities/durable_execution/overview/ accessed 2026-10-07 [search-tool extract]

⁶⁰⁸ Verified fact — A4-S003: PyPI JSON metadata: pydantic-ai (Python Package Index) <https://pypi.org/pypi/pydantic-ai/json> accessed 2026-10-07; A4-S044: Pydantic AI v2: capabilities, a leaner core, and the Harness (with upgrade guide) <https://pydantic.dev/articles/pydantic-ai-v2> accessed 2026-10-07 [search-tool extract]

⁶⁰⁹ Verified fact — A4-S046: Announcement: Pydantic Logfire launch & Series A Funding (and AI Gateway moving to Logfire) <https://pydantic.dev/articles/logfire-announcement> accessed 2026-10-07 [search-tool extract]

⁶¹⁰ Verified fact — V1-S074: Pydantic AI Upgrade Guide; Pydantic AI v2 article; Version Policy <https://pydantic.dev/docs/ai/project/changelog/> accessed 2026-10-08 [search-tool extract]

Field	Value	Label	Sources
	(pydantic_graph) and durable-execution integrations	d fact	A4-S044
Version / lineup	pydantic-ai 2.54.0 (3 October 2026); v2.0.0 released 23 June 2026; v1 line still patched (1.107.7 on 30 September 2026)	Verified fact	A4-S003
Licence	Open source, MIT	Verified fact	A4-S003, A4-S119
Status events	5 September 2025: v1.0.0; 23 June 2026: v2.0.0 (breaking: pydantic_graph.persistence removed); Pydantic AI Gateway (gateway.pydantic.dev) deprecated and moved into Logfire	Verified fact	A4-S003, A4-S044, A4-S046, V1-S074
Strategic direction	v2 (first beta 20 May 2026) introduced a capability layer, a leaner core and 'the Harness'; durability is moving onto the capability layer (runtime extension point tracked for after v2). AI Gateway merged into Pydantic Logfire (adds failover, load balancing, DLP, spend caps)	Verified fact	A4-S044, A4-S046
What it does	A typed, extensible agent loop for Python with model-agnostic providers, structured outputs and tools; pydantic_graph provides graph-based control flow; durable execution is attached as a capability via Temporal, DBOS or Prefect (plus Harness packages such as AWS Lambda and Step Persistence).	Verified fact	A4-S119, A4-S044, A4-S045
Stack position	L3 agent framework; observability via Logfire (L9) and gateway via Logfire (C1) are separate commercial products	Verified fact	A4-S046
Integration	Python library; MCP support; OpenTelemetry via Logfire; durable-execution capabilities (TemporalDurability, DBOSDurability, PrefectDurability)	Verified fact	A4-S045, A4-S119
Dependencies	Python >=3.10; Pydantic; model provider SDKs; optional Temporal/DBOS/Prefect for durability	Verified fact	A4-S003, A4-S045
Certifications	Not applicable to the open-source library; Logfire certifications out of scope for this record	Architectural judgment	
GDPR / residency	Not applicable (library; data flows go to the customer's chosen model provider)	Architectural judgment	
Security features	Logfire AI Gateway (separate product) offers DLP scanning of prompts/completions for secrets and PII, and spend caps	Verified fact	A4-S046
Access controls	Not applicable (library)	Architectural judgment	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Free (MIT)	Verified fact	A4-S003
Infrastructure	Costs are model tokens plus any durable-execution backend (Temporal	Architectural judgment	

Field	Value	Label	Sources
cost	cluster, database for DBOS)	ctural judge ment	
Ecosystem	Temporal, DBOS and Prefect integrations co-maintained with those vendors; Logfire; GitHub Agentic Workflows harness	Verifie d fact	A4-S045, A4-S119
Adoption signals	Pydantic raised a US\$12.5M Series A led by Sequoia (announced with Logfire GA, c. late 2024); no later round found	Verifie d fact	A4-S046
Maturity	First release May 2024; 1.0 September 2025; 2.0 June 2026; Development Status: Production/Stable; 344 releases on PyPI	Verifie d fact	A4-S003
Deployment	saas: No; managed_cloud: No; vpc_byoc: Yes (library runs inside the customer's own environment); private_cloud: Yes (library runs inside the customer's own environment); self_hosted: Yes (library); on_prem: Yes (library)		

Agent Development Kit (ADK) (L3-google-adk)

Tier: Strategic · Flags: none · Original graphic label: not in graphic

Rationale: Strategic, conditional: where Google Cloud is your primary cloud (rule 10: ADK on Agent Engine is Google's lead agent framework and runtime, matching Strands/AgentCore on AWS and Microsoft Agent Framework on Azure); no criterion at 1; scored as a library under rule 2, so enterprise readiness and security stay at 3. Tier changed from Tactical by CP4 reviewer C for rule-10 parity. Confirmed by the reader at Checkpoint 4 (CP4-4).

Criterion	Score	Rationale
Technical	4	Workflow Runtime, multi-agent, A2A Task API, HITL, evaluation ^{vf611}
Enterprise readiness	3	Library (rule 2); no support commitment recorded ^{NPV}
Security and compliance	3	Library (rule 2); managed route on Agent Platform in ISO 42001 and SOC 2 scope ^{vf^} [Verified fact — A5-S028: ISO/IEC 42001 - Compliance
Deployment flexibility	4	Library anywhere; Cloud Run and Agent Engine ^{vf612}
Ecosystem	3	MCP, OpenAPI, A2A; broader ecosystem not verified ^{vf613 NPV}
Reliability and maturity	3	1.0 May 2025, 2.0 May 2026 ^{vf614}
Cost / TCO	4	Free Apache-2.0 ^{vf615}
Lock-in / portability	3	Apache-2.0 but Gemini-optimised and GCP-targeted ^{vf616}

⁶¹¹ Verified fact — A4-S114: Agent Development Kit (ADK) 2.0 README (GitHub) (Google) <https://raw.githubusercontent.com/google/adk-python/main/README.md> accessed 2026-10-07

⁶¹² Verified fact — A4-S114: Agent Development Kit (ADK) 2.0 README (GitHub) (Google) <https://raw.githubusercontent.com/google/adk-python/main/README.md> accessed 2026-10-07

⁶¹³ Verified fact — A4-S114: Agent Development Kit (ADK) 2.0 README (GitHub) (Google) <https://raw.githubusercontent.com/google/adk-python/main/README.md> accessed 2026-10-07

⁶¹⁴ Verified fact — A4-S017: PyPI JSON metadata: google-adk (Python Package Index) <https://pypi.org/pypi/google-adk/json> accessed 2026-10-07

⁶¹⁵ Verified fact — A4-S114: Agent Development Kit (ADK) 2.0 README (GitHub) (Google) <https://raw.githubusercontent.com/google/adk-python/main/README.md> accessed 2026-10-07

Criterion	Score	Rationale
Total (generic / FS)	3.45 / 3.35	

Capabilities. Apache-2.0 code-first agent framework (Python, Java, Kotlin, Go, TS); ADK 2.0 adds a graph-based Workflow Runtime (routing, fan-out/fan-in, loops, retry, state, HITL, nested workflows) and a Task API for agent-to-agent delegation; deploys to Cloud Run or Vertex AI Agent Engine ^{VF617}.

Strengths

- Explicit deterministic Workflow Runtime alongside agents ^{VF618 AJ}
- Tool-confirmation HITL and evaluation built in ^{VF619}

Limitations and risks

- Two major versions in twelve months ^{VF620}
- Gemini-optimised; Google Cloud deployment targets ^{VF621}
- Security features and access controls not verified for the library ^{NPV}

Choose when

- Google Cloud estates, especially with Agent Engine ^{AJ}

Avoid when

- You need cross-cloud hosting parity ^{AJ}

Nearest competitors: L3-microsoft-agent-framework, L3-langgraph, L3-aws-strands-agentcore

Regulated-FS note. Treat Agent Engine as a hyperscaler service under CP2 Q1 and confirm its controls per service ^{Rec.}.

Facts (as of 2026-10-07)

Field	Value	Label	Sources
Company	Google	Verified fact	A4-S114
Category	Open-source, code-first agent framework with a graph-based deterministic Workflow Runtime	Verified fact	A4-S114

⁶¹⁶ Verified fact — A4-S114: Agent Development Kit (ADK) 2.0 README (GitHub) (Google) <https://raw.githubusercontent.com/google/adk-python/main/README.md> accessed 2026-10-07

⁶¹⁷ Verified fact — A4-S114: Agent Development Kit (ADK) 2.0 README (GitHub) (Google) <https://raw.githubusercontent.com/google/adk-python/main/README.md> accessed 2026-10-07; A4-S017: PyPI JSON metadata: google-adk (Python Package Index) <https://pypi.org/pypi/google-adk/json> accessed 2026-10-07

⁶¹⁸ Verified fact — A4-S114: Agent Development Kit (ADK) 2.0 README (GitHub) (Google) <https://raw.githubusercontent.com/google/adk-python/main/README.md> accessed 2026-10-07

⁶¹⁹ Verified fact — A4-S114: Agent Development Kit (ADK) 2.0 README (GitHub) (Google) <https://raw.githubusercontent.com/google/adk-python/main/README.md> accessed 2026-10-07

⁶²⁰ Verified fact — A4-S017: PyPI JSON metadata: google-adk (Python Package Index) <https://pypi.org/pypi/google-adk/json> accessed 2026-10-07

⁶²¹ Verified fact — A4-S114: Agent Development Kit (ADK) 2.0 README (GitHub) (Google) <https://raw.githubusercontent.com/google/adk-python/main/README.md> accessed 2026-10-07

Field	Value	Label	Sources
Version / lineup	google-adk 2.11.0 (2 October 2026); 2.0.0 on 19 May 2026; 1.0.0 on 20 May 2025; also Java, Kotlin, Go and TypeScript ports	Verified fact	A4-S017, A4-S114
Licence	Open source, Apache-2.0	Verified fact	A4-S114
Status events	19 May 2026: ADK 2.0.0	Verified fact	A4-S017
Strategic direction	ADK 2.0 adds a Workflow Runtime (routing, fan-out/fan-in, loops, retry, state, human-in-the-loop, nested workflows) and a Task API for agent-to-agent delegation; optimised for Gemini but model-agnostic	Verified fact	A4-S114
What it does	Builds, evaluates and deploys agents and multi-agent hierarchies with tools (OpenAPI, MCP), tool-confirmation HITL, and deterministic graph workflows; deploys to Cloud Run or Vertex AI Agent Engine.	Verified fact	A4-S114
Stack position	L3 framework; Vertex AI Agent Engine is the managed runtime	Verified fact	A4-S114
Integration	Libraries; MCP and OpenAPI tools; A2A delegation via Task API	Verified fact	A4-S114
Dependencies	Python (or Java/Kotlin/Go/TS); Gemini or other models	Verified fact	A4-S114
Certifications	Not applicable to the library	Architectural judgement	
GDPR / residency	Inherits hosting/model choice	Architectural judgement	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Free (Apache-2.0)	Verified fact	A4-S114
Infrastructure cost	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	1.0 May 2025; 2.0 May 2026	Verified fact	A4-S017
Deployment	saas: No; managed_cloud: Yes (Vertex AI Agent Engine, Cloud Run); vpc_byoc: Yes (library runs inside the customer's own environment); private_cloud: Yes (library runs inside the customer's own environment); self_hosted: Yes; on_prem: Yes (library)		

OpenAI Agents SDK (Python: `openai-agents`; JS/TS: `@openai/agents`) (L3-`openai-agents-sdk`)

Tier: Tactical · **Flags:** none · **Original graphic label:** Agent SDK (OpenAI logo)

Rationale: Good agent-step SDK, but pre-1.0 and surrounded by product churn.

Criterion	Score	Rationale
Technical	4	Agent loop, handoffs, guardrails, sessions, approvals, tracing, sandboxes, Temporal/DBOS durability ^{vi.622}
Enterprise readiness	3	Library (rule 2); enterprise support not verified ^{NPV}
Security and compliance	3	Library (rule 2); harness keeps credentials away from model-generated code ^{vi.623} ; hygiene not assessed
Deployment flexibility	4	Library anywhere; model calls to chosen provider ^{vi.624}
Ecosystem	4	MCP, sandbox partners, Temporal, DBOS, Bedrock Managed Agents ^{vi.625}
Reliability and maturity	2	Pre-1.0 versioning; neighbouring products sunset in 2026 ^{vi.626}

⁶²² Verified fact — A4-S120: OpenAI Agents SDK README (GitHub) <https://raw.githubusercontent.com/openai/openai-agents-python/main/README.md> accessed 2026-10-07; A4-S052: OpenAI Agents SDK documentation (agents, sessions, handoffs, guardrails, tracing, running agents) <https://openai.github.io/openai-agents-python/> accessed 2026-10-07 [search-tool extract]; A4-S053: The next evolution of the Agents SDK (OpenAI) <https://openai.com/index/the-next-evolution-of-the-agents-sdk/> accessed 2026-10-07 [search-tool extract]

⁶²³ Verified fact — A4-S053: The next evolution of the Agents SDK (OpenAI) <https://openai.com/index/the-next-evolution-of-the-agents-sdk/> accessed 2026-10-07 [search-tool extract]

⁶²⁴ Verified fact — A4-S005: PyPI JSON metadata: openai-agents (Python Package Index) <https://pypi.org/pypi/openai-agents/json> accessed 2026-10-07

⁶²⁵ Verified fact — A4-S053: The next evolution of the Agents SDK (OpenAI) <https://openai.com/index/the-next-evolution-of-the-agents-sdk/> accessed 2026-10-07 [search-tool extract]; A4-S055: OpenAI API Changelog (Agents API public beta, Sep 2026) and Introducing the Agents API <https://developers.openai.com/api/docs/changelog> accessed 2026-10-07 [search-tool extract]

Criterion	Score	Rationale
Cost / TCO	4	Free MIT ^{vf627}
Lock-in / portability	3	MIT and provider-agnostic, but defaults favour OpenAI services ^{vf628}
Total (generic / FS)	3.45	
	/	
	3.30	

Capabilities. MIT, provider-agnostic multi-agent SDK: agents with tools, handoffs or agents-as-tools, input/output guardrails, sessions with resumable approvals, tracing, sandboxed agents; Temporal and DBOS integrations for durable runs. Distinct from the hosted Agents API (beta) and Agent Builder (shutting down 30 November 2026) ^{vf629}.

Strengths

- Compact primitives with guardrails and approvals built in ^{vf630 AJ}
- Works with 100+ non-OpenAI models ^{vf631}

Limitations and risks

- Pre-1.0 (0.23.1) ^{vf632}
- Tracing defaults to OpenAI's dashboard ^{vf633}
- Adjacent OpenAI products are retired frequently ^{vf634}
- Hosted Agents API is US-only with no ZDR at launch ^{vf635}

⁶²⁶ Verified fact — A4-S005: PyPI JSON metadata: openai-agents (Python Package Index) <https://pypi.org/pypi/openai-agents/json> accessed 2026-10-07; A4-S054: OpenAI Deprecations (Agent Builder) and Introducing AgentKit update <https://developers.openai.com/api/docs/deprecations> accessed 2026-10-07 [search-tool extract]; A4-S056: Assistants API beta deprecation - August 26, 2026 sunset (OpenAI) <https://community.openai.com/t/assistants-api-beta-deprecation-august-26-2026-sunset/1354666> accessed 2026-10-07 [search-tool extract]

⁶²⁷ Verified fact — A4-S005: PyPI JSON metadata: openai-agents (Python Package Index) <https://pypi.org/pypi/openai-agents/json> accessed 2026-10-07

⁶²⁸ Verified fact — A4-S120: OpenAI Agents SDK README (GitHub) <https://raw.githubusercontent.com/openai/openai-agents-python/main/README.md> accessed 2026-10-07; A4-S052: OpenAI Agents SDK documentation (agents, sessions, handoffs, guardrails, tracing, running agents) <https://openai.github.io/openai-agents-python/> accessed 2026-10-07 [search-tool extract]

⁶²⁹ Verified fact — A4-S120: OpenAI Agents SDK README (GitHub) <https://raw.githubusercontent.com/openai/openai-agents-python/main/README.md> accessed 2026-10-07; A4-S052: OpenAI Agents SDK documentation (agents, sessions, handoffs, guardrails, tracing, running agents) <https://openai.github.io/openai-agents-python/> accessed 2026-10-07 [search-tool extract]; A4-S053: The next evolution of the Agents SDK (OpenAI) <https://openai.com/index/the-next-evolution-of-the-agents-sdk/> accessed 2026-10-07 [search-tool extract]; A4-S055: OpenAI API Changelog (Agents API public beta, Sep 2026) and Introducing the Agents API <https://developers.openai.com/api/docs/changelog> accessed 2026-10-07 [search-tool extract]; A4-S054: OpenAI Deprecations (Agent Builder) and Introducing AgentKit update <https://developers.openai.com/api/docs/deprecations> accessed 2026-10-07 [search-tool extract]

⁶³⁰ Verified fact — A4-S120: OpenAI Agents SDK README (GitHub) <https://raw.githubusercontent.com/openai/openai-agents-python/main/README.md> accessed 2026-10-07

⁶³¹ Verified fact — A4-S120: OpenAI Agents SDK README (GitHub) <https://raw.githubusercontent.com/openai/openai-agents-python/main/README.md> accessed 2026-10-07

⁶³² Verified fact — A4-S005: PyPI JSON metadata: openai-agents (Python Package Index) <https://pypi.org/pypi/openai-agents/json> accessed 2026-10-07

⁶³³ Verified fact — A4-S052: OpenAI Agents SDK documentation (agents, sessions, handoffs, guardrails, tracing, running agents) <https://openai.github.io/openai-agents-python/> accessed 2026-10-07 [search-tool extract]

⁶³⁴ Verified fact — A4-S056: Assistants API beta deprecation - August 26, 2026 sunset (OpenAI) <https://community.openai.com/t/assistants-api-beta-deprecation-august-26-2026-sunset/1354666> accessed 2026-10-07 [search-tool extract]; A4-S054: OpenAI Deprecations (Agent Builder) and Introducing AgentKit update <https://developers.openai.com/api/docs/deprecations> accessed 2026-10-07 [search-tool extract]

⁶³⁵ Verified fact — A4-S055: OpenAI API Changelog (Agents API public beta, Sep 2026) and Introducing the Agents API <https://developers.openai.com/api/docs/changelog> accessed 2026-10-07 [search-tool extract]

Choose when

- An agent step inside a durable workflow, with a custom trace processor ^{AJ}

Avoid when

- You would use the hosted Agents API for UK/EU client data ^{AJ}

Nearest competitors: L3-pydantic-ai, L3-langgraph, L3-claude-agent-sdk

Regulated-FS note. Replace the default trace exporter with a firm-owned OTel processor before any client data flows ^{SEC}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	OpenAI	Verified fact	A4-S005
Category	Open-source, provider-agnostic multi-agent SDK (agent loop, handoffs, guardrails, sessions, tracing, sandbox harness)	Verified fact	A4-S120, A4-S052
Version / lineup	openai-agents 0.23.1 (2 October 2026); @openai/agents 0.19.0 (5 October 2026)	Verified fact	A4-S005, A4-S024
Licence	Open source, MIT	Verified fact	A4-S005, A4-S024
Status events	4 March 2025: first PyPI release; 3 June 2026: Agent Builder (AgentKit) deprecation announced; shutdown 30 November 2026; Agents SDK is the code-based migration path; 26 August 2026: Assistants API sunset (Responses API is the replacement); 10 September 2026: separate hosted Agents API in public beta	Verified fact	A4-S005, A4-S054, A4-S056, A4-S055, V1-S051, V1-S083
Strategic direction	15 April 2026 'next evolution': model-native harness (configurable memory, sandbox-aware orchestration, Codex-like filesystem tools, MCP/skills/AGENTS.md), native sandbox execution with Blaxel, Cloudflare, Daytona, E2B, Modal, Runloop, Vercel, snapshot/rehydration; TypeScript parity May 2026. OpenAI also launched a hosted Agents API (public beta 10 September 2026) running a managed Codex harness, and is retiring Agent Builder in favour of the Agents SDK	Verified fact	A4-S053, A4-S055, A4-S054

Field	Value	Label	Sources
What it does	Lightweight framework for multi-agent workflows: agents with tools, handoffs (delegated agent takes over the conversation) or agents-as-tools, input/output guardrails, sessions (persistent memory, resumable approvals), built-in tracing, voice/realtime agents and sandboxed agents. Supports OpenAI Responses and Chat Completions APIs and 100+ other LLMs.	Verified fact	A4-S120, A4-S052
Stack position	L3 agent framework. The hosted Agents API is a managed agent runtime (distinct product); tracing defaults to OpenAI's dashboard (L9 overlap)	Verified fact	A4-S055, A4-S052
Integration	Python/TS libraries; MCP; custom trace processors; Temporal and DBOS integrations for durable long-running workflows incl. human-in-the-loop	Verified fact	A4-S052, A4-S053
Dependencies	Python >=3.10 or Node; an LLM provider (OpenAI by default); optional sandbox provider; optional Temporal/DBOS for durability	Verified fact	A4-S005, A4-S053, A4-S052
Certifications	Not applicable to the MIT library; OpenAI platform certifications covered by the L1 stream	Architectural judgment	
GDPR / residency	SDK: inherits the configured model provider. Hosted Agents API (separate product): US data residency only and no Zero Data Retention support at launch	Verified fact	A4-S055
Security features	Harness separated from compute so credentials stay away from model-generated code; sandbox snapshot and rehydration	Verified fact	A4-S053
Access controls	Not applicable (library)	Architectural judgment	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	SDK free (MIT); model usage billed by provider. Hosted Agents API: sandboxes at standard container rates plus model usage	Verified fact	A4-S005, A4-S055
Infrastructure cost	Sandboxed agents add container/sandbox-provider cost on top of tokens	Architectural judgment	
Ecosystem	Sandbox partners (Blaxel, Cloudflare, Daytona, E2B, Modal, Runloop, Vercel); Temporal, DBOS; Amazon Bedrock Managed Agents built with OpenAI on the Agents API	Verified fact	A4-S053, A4-S052, A4-S055
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	First release March 2025; still 0.x versioning (0.23.1); frequent releases (123 on PyPI)	Verified fact	A4-S005

Field	Value	Label	Sources
Deployment	saas: No for the SDK (library); hosted alternative is the separate Agents API (beta); managed_cloud: No for the SDK; Agents API runs in OpenAI-hosted or customer-connected sandboxes; vpc_byoc: Yes (library runs inside the customer's own environment); private_cloud: Yes (library runs inside the customer's own environment); self_hosted: Yes (library); on_prem: Yes (library; model calls still go to the chosen provider)		

Strands Agents (open-source SDK) and Amazon Bedrock AgentCore (managed agent runtime and services) (L3-aws-strands-agentcore)

Tier: Strategic · **Flags:** none · **Original graphic label:** not in graphic

Rationale: Strategic, conditional: where AWS is your primary cloud (rule 10, lead AWS agent runtime); deployment and lock-in at 2 accepted under rule 11 as an existing platform commitment.

Criterion	Score	Rationale
Technical	4	Isolated runtime, memory, gateway, identity, any framework; no deterministic durable workflow engine ^{vf636}
Enterprise readiness	4	Hyperscaler presumption (rule 6); platform controls presumed (CP2 Q1); confirm per service
Security and compliance	4	AgentCore in AWS SOC 1/2/3 and ISO 27001 scope, HIPAA eligible; SOC 2 type not stated, so 4 (rule 8) ^{vf637}
Deployment flexibility	2	AgentCore AWS-only; Strands portable ^{vf638}
Ecosystem	4	Any framework, MCP gateway, AG-UI ^{vf639}

⁶³⁶ Verified fact — A4-S116: Bedrock AgentCore SDK README (GitHub) (AWS) <https://raw.githubusercontent.com/aws/bedrock-agentcore-sdk-python/main/README.md> accessed 2026-10-07; B-L3-S004: AgentCore Runtime sessions / microVMs / quotas / security best practices (Amazon Web Services) <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/runtime-sessions.html> accessed 2026-10-08 [search-tool extract]

⁶³⁷ Verified fact — B-L5-S003: Compliance validation for Amazon Bedrock AgentCore (and AWS Services in Scope pages) (Amazon Web Services) <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/compliance-validation.html> accessed 2026-10-08 [search-tool extract]

⁶³⁸ Verified fact — A4-S116: Bedrock AgentCore SDK README (GitHub) (AWS) <https://raw.githubusercontent.com/aws/bedrock-agentcore-sdk-python/main/README.md> accessed 2026-10-07; A4-S018: PyPI JSON metadata: strands-agents (Python Package Index) <https://pypi.org/pypi/strands-agents/json> accessed 2026-10-07

⁶³⁹ Verified fact — A4-S116: Bedrock AgentCore SDK README (GitHub) (AWS) <https://raw.githubusercontent.com/aws/bedrock-agentcore-sdk-python/main/README.md> accessed 2026-10-07

Criterion	Score	Rationale
Reliability and maturity	3	AgentCore GA 13 October 2025; Strands 1.0 July 2025; SDK Alpha ^{VF640}
Cost / TCO	3	Runtime pricing not retrieved ^{NPV} ; consumption model assumed ^{AJ}
Lock-in / portability	2	Proprietary AWS-only service APIs ^{VF641}
Total (generic / FS)	3.40	
	/	
	3.25	

Capabilities. Strands Agents (Apache-2.0 model-driven agent SDK) plus Amazon Bedrock AgentCore, a framework-agnostic managed agent platform: Runtime with per-session microVM isolation (up to 8 hours per lifecycle), Memory, Gateway, Identity, Observability; hosts Strands, LangGraph, CrewAI, AutoGen or custom agents ^{VF642}.

Strengths

- Per-session microVM isolation and VPC/PrivateLink ^{VF643}
- Runs any framework, so it separates runtime choice from framework choice ^{VF644 AJ}

Limitations and risks

- AWS-only, proprietary APIs ^{VF645}
- Session state ephemeral by default; Runtime is not a durable workflow engine ^{VF646}
- AgentCore SDK classified Alpha ^{VF647}
- Execution-role credentials reachable from code in the microVM ^{VF648}
- Runtime pricing not verified ^{NPV}

⁶⁴⁰ Verified fact — V1-S087: Amazon Bedrock AgentCore is now generally available; AgentCore FAQs; release notes (Amazon Web Services) <https://aws.amazon.com/about-aws/whats-new/2025/10/amazon-bedrock-agentcore-available/> accessed 2026-10-08 [search-tool extract]; A4-S018: PyPI JSON metadata: strands-agents (Python Package Index) <https://pypi.org/pypi/strands-agents/json> accessed 2026-10-07; A4-S121: PyPI JSON metadata: bedrock-agentcore (Python Package Index) <https://pypi.org/pypi/bedrock-agentcore/json> accessed 2026-10-07

⁶⁴¹ Verified fact — A4-S116: Bedrock AgentCore SDK README (GitHub) (AWS) <https://raw.githubusercontent.com/aws/bedrock-agentcore-sdk-python/main/README.md> accessed 2026-10-07

⁶⁴² Verified fact — A4-S115: Strands Agents README (GitHub) (AWS (Strands Agents)) <https://raw.githubusercontent.com/strands-agents/sdk-python/main/README.md> accessed 2026-10-07; A4-S116: Bedrock AgentCore SDK README (GitHub) (AWS) <https://raw.githubusercontent.com/aws/bedrock-agentcore-sdk-python/main/README.md> accessed 2026-10-07; B-L3-S004: AgentCore Runtime sessions / microVMs / quotas / security best practices (Amazon Web Services) <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/runtime-sessions.html> accessed 2026-10-08 [search-tool extract]

⁶⁴³ Verified fact — B-L3-S004: AgentCore Runtime sessions / microVMs / quotas / security best practices (Amazon Web Services) <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/runtime-sessions.html> accessed 2026-10-08 [search-tool extract]

⁶⁴⁴ Verified fact — A4-S116: Bedrock AgentCore SDK README (GitHub) (AWS) <https://raw.githubusercontent.com/aws/bedrock-agentcore-sdk-python/main/README.md> accessed 2026-10-07

⁶⁴⁵ Verified fact — A4-S116: Bedrock AgentCore SDK README (GitHub) (AWS) <https://raw.githubusercontent.com/aws/bedrock-agentcore-sdk-python/main/README.md> accessed 2026-10-07

⁶⁴⁶ Verified fact — B-L3-S004: AgentCore Runtime sessions / microVMs / quotas / security best practices (Amazon Web Services) <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/runtime-sessions.html> accessed 2026-10-08 [search-tool extract]

⁶⁴⁷ Verified fact — A4-S121: PyPI JSON metadata: bedrock-agentcore (Python Package Index) <https://pypi.org/pypi/bedrock-agentcore/json> accessed 2026-10-07

⁶⁴⁸ Verified fact — B-L3-S004: AgentCore Runtime sessions / microVMs / quotas / security best practices (Amazon Web Services) <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/runtime-sessions.html> accessed 2026-10-08 [search-tool extract]

Choose when

- AWS is the primary cloud and you need a managed, isolated runtime for framework-built agents ^{AJ}

Avoid when

- Multi-cloud portability of the runtime is a requirement ^{AJ}

Nearest competitors: L3-microsoft-agent-framework, L3-google-adk, L3-langgraph

Regulated-FS note. Keep the workflow definition in a portable framework; scope execution roles tightly; platform controls presumed (CP2 Q1); confirm per service ^{REC}.

Facts (as of 2026-10-07)

Field	Value	Label	Sources
Company	Amazon Web Services	Verified fact	A4-S116
Category	Model-driven agent SDK plus framework-agnostic managed agent platform (Runtime, Memory, Gateway, Identity)	Verified fact	A4-S115, A4-S116
Version / lineup	strands-agents 1.58.1 (6 October 2026; 1.0.0 15 July 2025); bedrock-agentcore SDK 1.24.1 (7 October 2026; classifier Alpha)	Verified fact	A4-S018, A4-S121
Licence	SDKs Apache-2.0; AgentCore is a proprietary AWS managed service	Verified fact	A4-S018, A4-S121, A4-S116
Status events	Not publicly verified	Not publicly verified	
Strategic direction	Not publicly verified	Not publicly verified	
What it does	AgentCore deploys and operates agents 'using any framework and model' (Strands, LangGraph, CrewAI, AutoGen or custom) with session-isolated Runtime, Memory, Gateway (APIs to MCP tools) and Identity; supports AG-UI protocol.	Verified fact	A4-S116
Stack position	L3 framework (Strands) and managed agent runtime (AgentCore);	Verified	A4-S116

Field	Value	Label	Sources
	AgentCore Gateway/Identity overlap L4/C4	d fact	
Integration	Not publicly verified	Not publicly verified	
Dependencies	Not publicly verified	Not publicly verified	
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Not publicly verified	Not publicly verified	
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
Adoption signals	Not publicly verified	y verifie d Not publicl y verifie d	
Maturity	Strands 1.0 July 2025; AgentCore SDK still Alpha-classified	Verifie d fact	A4-S018, A4-S121
Deployment	saas: No; managed_cloud: Yes (AgentCore); vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes (Strands SDK); on_prem: Not publicly verified		

CrewAI (open-source framework: Crews and Flows); commercial platform CrewAI AMP (Agent Management Platform; earlier materials say CrewAI Enterprise; explicit rename notice not found) (L3-crewai)

Tier: Tactical · **Flags:** none · **Original graphic label:** CrewAI – multi-agent

Rationale: Both modes available, but abstractions are proprietary and durability is unproven.

Criterion	Score	Rationale
Technical	3	Crews plus Flows cover multi-agent and workflow modes; HITL/durability not evidenced ^{vf649} _{npv}
Enterprise readiness	3	AMP Enterprise SSO and RBAC verified; no SCIM, SLA or audit evidence, so 3 (rule 7) ^{vf650}
Security and compliance	3	SOC 2 Type 2 report (June 2026) listed; HIPAA Type 1 only ^{vf651}
Deployment flexibility	4	SaaS, VPC on AWS/Azure/GCP, on-prem, OSS self-host; air-gap not verified ^{vf652}
Ecosystem	4	crewai-tools, AMP, HPE partnership; hosted by AgentCore ^{vf653}
Reliability and maturity	3	1.x since October 2025; very high release frequency ^{vf654}

⁶⁴⁹ Verified fact — A4-S050: CrewAI Introduction / Flows / Evaluating Use Cases - CrewAI docs <https://docs.crewai.com/en/introduction> accessed 2026-10-07 [search-tool extract]

⁶⁵⁰ Verified fact — A4-S047: CrewAI Pricing <https://crewai.com/pricing> accessed 2026-10-07 [search-tool extract]

⁶⁵¹ Verified fact — A4-S048: Trust Center - CrewAI <https://trust.crewai.com/> accessed 2026-10-07 [search-tool extract]

⁶⁵² Verified fact — A4-S047: CrewAI Pricing <https://crewai.com/pricing> accessed 2026-10-07 [search-tool extract]; A4-S049: CrewAI AMP - Agent Management Platform (product page, docs and blog) <https://crewai.com/agent-management-platform> accessed 2026-10-07 [search-tool extract]

⁶⁵³ Verified fact — A4-S004: PyPI JSON metadata: crewai (Python Package Index) <https://pypi.org/pypi/crewai/json> accessed 2026-10-07; A4-S049: CrewAI AMP - Agent Management Platform (product page, docs and blog) <https://crewai.com/agent-management-platform> accessed 2026-10-07 [search-tool extract]; A4-S116: Bedrock AgentCore SDK README (GitHub) (AWS) <https://raw.githubusercontent.com/aws/bedrock-agentcore-sdk-python/main/README.md> accessed 2026-10-07

⁶⁵⁴ Verified fact — A4-S004: PyPI JSON metadata: crewai (Python Package Index) <https://pypi.org/pypi/crewai/json> accessed 2026-10-07

Criterion	Score	Rationale
Cost / TCO	3	Free tier 50 executions/month; Enterprise custom-priced ^{VF655}
Lock-in / portability	3	MIT framework, proprietary abstractions and AMP ^{VF656}
Total (generic / FS)	3.25 / 3.20	

Capabilities. MIT multi-agent framework with two modes: Crews (autonomous role-based agents) and Flows (event-driven, stateful process definitions); commercial platform CrewAI AMP (earlier materials call it CrewAI Enterprise) ^{VF657}.

Strengths

- Vendor itself recommends Flows as the outer process with Crews inside, which matches the H2 split ^{VF658 AJ}
- AMP deploys to cloud, customer VPC or on-premises ^{VF659}

Limitations and risks

- Crew and Flow abstractions are CrewAI-specific ^{AJ}
- Durable execution and HITL not evidenced ^{NPV}
- Funding confirmed only to about US\$20M ^{VF660}
- Multi-agent Crews multiply LLM calls per task ^{AJ}

Choose when

- Teams that want a packaged multi-agent platform with on-prem option ^{AJ}

Avoid when

- The task is deterministic and does not need multiple autonomous agents ^{AJ}

Nearest competitors: L3-langgraph, L3-microsoft-agent-framework, L3-google-adk

Regulated-FS note. Use Flows for any regulated process and confine Crews to drafting or research sub-steps behind an approval gate ^{REC}.

Facts (as of 2026-10-08)

⁶⁵⁵ Verified fact — A4-S047: CrewAI Pricing <https://crewai.com/pricing> accessed 2026-10-07 [search-tool extract]

⁶⁵⁶ Verified fact — A4-S126: crewAI LICENSE (crewAI, Inc.) <https://raw.githubusercontent.com/crewAIInc/crewAI/main/LICENSE> accessed 2026-10-07; A4-S047: CrewAI Pricing <https://crewai.com/pricing> accessed 2026-10-07 [search-tool extract]

⁶⁵⁷ Verified fact — A4-S004: PyPI JSON metadata: crewai (Python Package Index) <https://pypi.org/pypi/crewai/json> accessed 2026-10-07; A4-S050: CrewAI Introduction / Flows / Evaluating Use Cases - CrewAI docs <https://docs.crewai.com/en/introduction> accessed 2026-10-07 [search-tool extract]; A4-S049: CrewAI AMP - Agent Management Platform (product page, docs and blog) <https://crewai.com/agent-management-platform> accessed 2026-10-07 [search-tool extract]; V1-S075: CrewAI AMP - CrewAI Platform (docs); CrewAI AMP - The Agent Management Platform (blog) <https://docs.crewai.com/en/enterprise/introduction> accessed 2026-10-08 [search-tool extract]

⁶⁵⁸ Verified fact — A4-S050: CrewAI Introduction / Flows / Evaluating Use Cases - CrewAI docs <https://docs.crewai.com/en/introduction> accessed 2026-10-07 [search-tool extract]

⁶⁵⁹ Verified fact — A4-S047: CrewAI Pricing <https://crewai.com/pricing> accessed 2026-10-07 [search-tool extract]; A4-S049: CrewAI AMP - Agent Management Platform (product page, docs and blog) <https://crewai.com/agent-management-platform> accessed 2026-10-07 [search-tool extract]

⁶⁶⁰ Verified fact — A4-S148: CrewAI OSS 1.0 - We are going GA (and CrewAI blog posts) <https://blog.crewai.com/crewai-oss-1-0-we-are-going-ga/> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
Company	crewAI, Inc.	Verified fact	A4-S126
Category	Multi-agent orchestration framework (role-based Crews) with event-driven Flows, plus a commercial agent management platform	Verified fact	A4-S004, A4-S050
Version / lineup	crewai 1.15.24 (7 October 2026); 1.0.0 released 20 October 2025	Verified fact	A4-S004
Licence	Open source, MIT (framework); CrewAI AMP proprietary	Verified fact	A4-S004, A4-S126, A4-S047
Status events	20 October 2025: CrewAI 1.0.0; CrewAI AMP is the commercial platform; earlier materials call it CrewAI Enterprise (rename inferred from /enterprise/ doc paths and older naming; no explicit rename notice found; one blog used 'AOP')	Verified fact	A4-S004, A4-S049, V1-S075
Strategic direction	Vendor recommends Flows as the outer 'process definition' with Crews invoked inside for autonomous collaborative steps; AMP adds deployment, SSO/RBAC, PII redaction and policies	Verified fact	A4-S050, A4-S047
What it does	Crews: teams of role-playing agents that collaborate autonomously. Flows: event-driven, stateful process definitions (@start, @listen, @router, and _/or_) giving granular control; Flows can trigger Crews. AMP deploys and manages these in CrewAI cloud, customer VPC or on-premises.	Verified fact	A4-S004, A4-S050, A4-S049
Stack position	L3 multi-agent framework; AMP also acts as an agent runtime/management platform	Verified fact	A4-S049
Integration	Python library and CLI; crewai-tools package; AMP deployment	Verified fact	A4-S004, A4-S049
Dependencies	Python >=3.10,<3.14; standalone (not built on LangChain); LLM providers via integrations	Verified fact	A4-S004
Certifications	SOC 2 Type 2 (report dated June 2026 listed in trust centre); HIPAA audit report dated February 2026 listed (described as 'Type 1'; scope/BAA not shown)	Verified fact	A4-S048
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Enterprise plan: PII redaction, policies, workload identity	Verified fact	A4-S047
Access controls	Enterprise plan: SSO, RBAC	Verified fact	A4-S047
Enterprise support	Enterprise plan with 45-day onboarding; forward-deployed engineering and training sold separately	Verified fact	A4-S047
Pricing	Free tier: 50 workflow executions/month; Enterprise custom-priced (pricing page metadata about a year old)	Verified fact	A4-S047
Infrastructure cost	Multi-agent Crews multiply LLM calls per task; token cost dominates	Architectural judge	

Field	Value	Label	Sources
		ment	
Ecosystem	crewai-tools; AMP; HPE on-prem partnership	Verified fact	A4-S004, A4-S049
Adoption signals	Vendor and TechCrunch (Oct 2024): US\$18M seed+Series A (Boldstart, Craft, Earl Grey, Insight Partners); CrewAI 2025 blog: ~US\$20M raised. A ~US\$20M Series B (15 April 2026) appears only in PitchBook/Forge; no vendor or major-outlet announcement found	Verified fact	A4-S148, A4-S051
Maturity	First release November 2023; 1.0 in October 2025; very high release frequency (461 PyPI releases incl. nightly dev builds)	Verified fact	A4-S004
Deployment	saas: Yes (CrewAI cloud); managed_cloud: Yes; vpc_byoc: Yes (customer VPC on AWS, Azure or GCP); private_cloud: Yes (private VPC); self_hosted: Yes (OSS framework); on_prem: Yes (AMP on customer infrastructure; HPE pre-installed hardware offer)		

LlamaIndex (open-source framework, incl. Workflows); commercial platform now named LlamaParse (formerly LlamaCloud), which includes LlamaAgents (L3-llamaindex)

Tier: Tactical · Flags: Renamed · Original graphic label: LlamaIndex – document agents

Rationale: Capable but 0.x and no longer the vendor’s focus; Renamed refers to LlamaCloud -> LlamaParse.

Criterion	Score	Rationale
Technical	3	Workflows plus agents and RAG; HITL and durability not evidenced ^{vf661} _{hgv}
Enterprise readiness	3	Library (rule 2); LlamaParse Enterprise SSO is one verified control ^{vf662}
Security and compliance	3	Library hygiene not assessed; LlamaParse SOC 2 Type II and HIPAA BAA ^{vf663}
Deployment	4	Library anywhere; LlamaAgents managed or bring-your-own infrastructure ^{vf664}

⁶⁶¹ Verified fact — A4-S041: Announcing Workflows 1.0: a lightweight framework for agentic systems (LlamaIndex) <https://www.llamaindex.ai/blog/announcing-workflows-1-0-a-lightweight-framework-for-agentic-systems> accessed 2026-10-07 [search-tool extract]

⁶⁶² Verified fact — A1-S115: Compliance - LlamaParse documentation (Enterprise Readiness) (LlamaIndex, Inc.) <https://developers.llamaindex.ai/llamaparse/general/enterprise-readiness/compliance/> accessed 2026-10-07 [search-tool extract]; A1-S116: Pricing - LlamaParse documentation (LlamaIndex, Inc.) <https://developers.llamaindex.ai/llamaparse/general/pricing/> accessed 2026-10-07 [search-tool extract]

⁶⁶³ Verified fact — A1-S115: Compliance - LlamaParse documentation (Enterprise Readiness) (LlamaIndex, Inc.) <https://developers.llamaindex.ai/llamaparse/general/enterprise-readiness/compliance/> accessed 2026-10-07 [search-tool extract]

⁶⁶⁴ Verified fact — A4-S042: LlamaAgents Open Preview: Build Document Agents (and LlamaIndex newsletters Jan-Feb 2026) <https://www.llamaindex.ai/blog/llamaagents-build-serve-and-deploy-document-agents> accessed 2026-10-07 [search-tool extract]

Criterion	Score	Rationale
flexibility		
Ecosystem	4	Broad LLM and vector-store integrations ^{vf665}
Reliability and maturity	2	0.x versioning after three years; vendor focus moved to LlamaParse ^{vf666}
Cost / TCO	4	Framework free (MIT) ^{vf667}
Lock-in / portability	3	MIT but Workflows API is LlamaIndex-specific ^{vf668}
Total (generic / FS)	3.25 / 3.15	

Capabilities. MIT framework for RAG and agentic applications with Workflows, an event-driven, step-based engine with typed state and OpenTelemetry instrumentation; LlamaAgents deploys document agents on the proprietary LlamaParse platform (formerly LlamaCloud) ^{vf669}.

Strengths

- Workflows is a clean deterministic step engine usable outside the LlamaIndex ecosystem ^{vf670 aj}
- Large integration catalogue for retrieval ^{vf671}

Limitations and risks

- Vendor says its primary focus has shifted to LlamaParse ^{vf672}
- Framework still 0.x after three years ^{vf673}
- Human-in-the-loop and durable execution not evidenced in the fact base ^{npv}

Choose when

- Retrieval-heavy agents already built on LlamaIndex ^{aj}

⁶⁶⁵ Verified fact — A4-S117: LlamaIndex README (GitHub) https://raw.githubusercontent.com/run-llama/llama_index/main/README.md accessed 2026-10-07

⁶⁶⁶ Verified fact — A4-S002: PyPI JSON metadata: llama-index-core (Python Package Index) <https://pypi.org/pypi/llama-index-core/json> accessed 2026-10-07; A4-S117: LlamaIndex README (GitHub) https://raw.githubusercontent.com/run-llama/llama_index/main/README.md accessed 2026-10-07

⁶⁶⁷ Verified fact — A4-S002: PyPI JSON metadata: llama-index-core (Python Package Index) <https://pypi.org/pypi/llama-index-core/json> accessed 2026-10-07

⁶⁶⁸ Verified fact — A4-S041: Announcing Workflows 1.0: a lightweight framework for agentic systems (LlamaIndex) <https://www.llamaindex.ai/blog/announcing-workflows-1-0-a-lightweight-framework-for-agentic-systems> accessed 2026-10-07 [search-tool extract]

⁶⁶⁹ Verified fact — A4-S117: LlamaIndex README (GitHub) https://raw.githubusercontent.com/run-llama/llama_index/main/README.md accessed 2026-10-07; A4-S041: Announcing Workflows 1.0: a lightweight framework for agentic systems (LlamaIndex) <https://www.llamaindex.ai/blog/announcing-workflows-1-0-a-lightweight-framework-for-agentic-systems> accessed 2026-10-07 [search-tool extract]; A4-S042: LlamaAgents Open Preview: Build Document Agents (and LlamaIndex newsletters Jan-Feb 2026) <https://www.llamaindex.ai/blog/llamaagents-build-serve-and-deploy-document-agents> accessed 2026-10-07 [search-tool extract]

⁶⁷⁰ Verified fact — A4-S041: Announcing Workflows 1.0: a lightweight framework for agentic systems (LlamaIndex) <https://www.llamaindex.ai/blog/announcing-workflows-1-0-a-lightweight-framework-for-agentic-systems> accessed 2026-10-07 [search-tool extract]

⁶⁷¹ Verified fact — A4-S117: LlamaIndex README (GitHub) https://raw.githubusercontent.com/run-llama/llama_index/main/README.md accessed 2026-10-07

⁶⁷² Verified fact — A4-S117: LlamaIndex README (GitHub) https://raw.githubusercontent.com/run-llama/llama_index/main/README.md accessed 2026-10-07

⁶⁷³ Verified fact — A4-S002: PyPI JSON metadata: llama-index-core (Python Package Index) <https://pypi.org/pypi/llama-index-core/json> accessed 2026-10-07

- Document agents where LlamaParse is the chosen L8 parser ^{AJ}

Avoid when

- You need a long-term orchestration standard with roadmap commitment ^{AJ}

Nearest competitors: L3-langgraph, L3-pydantic-ai, L8-llamaparse

Regulated-FS note. Treat as a retrieval toolkit inside a governed workflow rather than as the orchestration standard ^{Rec}.

Facts (as of 2026-10-07)

Field	Value	Label	Sources
Company	LlamaIndex, Inc. (run-llama)	Verified fact	A4-S117
Category	RAG / agentic application framework with an event-driven workflow engine; vendor's commercial focus is document parsing and document agents	Verified fact	A4-S117, A4-S041
Version / lineup	llama-index / llama-index-core 0.14.25 (21 September 2026); llama-index-workflows 2.25.0 (25 September 2026)	Verified fact	A4-S002, A4-S013
Licence	Open source, MIT (framework and workflows); LlamaParse platform is proprietary SaaS	Verified fact	A4-S002, A4-S013, A4-S127, A4-S117
Status events	30 June 2025: Workflows 1.0 released as a standalone package (llama-index-workflows); November 2025: LlamaAgents open preview; 2026: LlamaCloud renamed LlamaParse; 2026: vendor states primary focus shifted from OSS framework to LlamaParse	Verified fact	A4-S041, A4-S042, A4-S117
Strategic direction	README (2026): 'our primary focus has shifted towards LlamaParse, along with liteparse and our benchmarking efforts'; the OSS framework remains available as an open toolkit. LlamaAgents (document agents built on Workflows) in open preview from November 2025; LlamaAgents Builder (natural-language to Workflow code) February 2026	Verified fact	A4-S117, A4-S042
What it does	Open-source framework for building agentic and RAG applications over data, with Workflows: a lightweight, event-driven, step-based engine for multi-step agentic applications in Python and TypeScript (typed state, resource injection, OpenTelemetry instrumentation). LlamaParse adds Parse, Extract, Index and LlamaAgents (deployed document agents).	Verified fact	A4-S117, A4-S041, A4-S013
Stack position	L3 for the framework/Workflows; the commercial product is mainly L8	Verified	A4-S117

Field	Value	Label	Sources
	(document extraction, LlamaParse) and the graphic's 'document agents' descriptor reflects that. The framework is no longer the vendor's primary focus.	d fact	
Integration	Python/TS libraries; llamactl CLI to deploy Agent Workflows to LlamaCloud/LlamaParse or as a headless API; OpenTelemetry instrumentation	Verified fact	A4-S042, A4-S041
Dependencies	Python >=3.9/3.10 or TypeScript; LLM and embedding providers; vector stores via integrations; LlamaParse API for managed parsing (optional)	Verified fact	A4-S002
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Framework free (MIT). LlamaParse metered in credits; third-party pages report ~US\$1.25 per 1,000 credits in 2026 and ~10,000 free credits per month; vendor pricing page not retrieved	Reported	A4-S043
Infrastructure cost	Self-hosted framework cost is driven by LLM/embedding tokens and the chosen vector store, not the library	Architectural judgment	
Ecosystem	Large integration catalogue (LLMs, vector stores); Workflows usable outside the LlamaIndex ecosystem; LlamaParse usable with or without the framework	Verified fact	A4-S117, A4-S041
Adoption signals	Funding reported as US\$27.5M (Greylock, Norwest; Series A May 2025) by Tracxn; another source reports US\$19M (conflict)	Reported	A4-S043
Maturity	Framework first released February 2023; still 0.x versioning (0.14.25); Workflows 1.0 since June 2025 and now 2.x	Verified fact	A4-S002, A4-S013,

Field	Value	Label	Sources
			A4-S041
Deployment	saas: Yes (LlamaParse / LlamaAgents managed cloud); managed_cloud: Yes (LlamaAgents managed cloud); vpc_byoc: Yes (LlamaAgents: 'bring-your-own infrastructure' option); private_cloud: Not publicly verified; self_hosted: Yes (OSS framework); on_prem: Not publicly verified		

AI SDK (by Vercel); npm package 'ai' (L3-vercel-ai-sdk)

Tier: Tactical · **Flags:** none · **Original graphic label:** AI SDK – AI SDK (triangle logo)

Rationale: Good TS application toolkit; fast-moving majors; not an orchestration standard.

Criterion	Score	Rationale
Technical	3	Unified model API, tools, ToolLoopAgent, UI streaming; durability via separate Workflow SDK ^{vf674}
Enterprise readiness	3	Library (rule 2); enterprise support not verified ^{NPV}
Security and compliance	3	Library (rule 2); Apache-2.0 licence clear ^{vf675} ; hygiene not assessed
Deployment flexibility	4	Any Node runtime ^{vf676}
Ecosystem	4	Major UI frameworks and model providers ^{vf677}
Reliability and maturity	2	Major version roughly every six months ^{vf678}
Cost / TCO	4	Free; gateway priced separately ^{vf679}
Lock-in / portability	3	Apache-2.0; optional gateway default ^{vf680}

⁶⁷⁴ Verified fact — A4-S069: AI SDK package README (GitHub) (Vercel) <https://raw.githubusercontent.com/vercel/ai/main/packages/ai/README.md> accessed 2026-10-07; A4-S071: Workflow SDK README (GitHub) (Vercel) <https://raw.githubusercontent.com/vercel/workflow/main/packages/workflow/README.md> accessed 2026-10-07

⁶⁷⁵ Verified fact — A4-S070: AI SDK LICENSE (Vercel) <https://raw.githubusercontent.com/vercel/ai/main/LICENSE> accessed 2026-10-07

⁶⁷⁶ Verified fact — A4-S069: AI SDK package README (GitHub) (Vercel) <https://raw.githubusercontent.com/vercel/ai/main/packages/ai/README.md> accessed 2026-10-07

⁶⁷⁷ Verified fact — A4-S069: AI SDK package README (GitHub) (Vercel) <https://raw.githubusercontent.com/vercel/ai/main/packages/ai/README.md> accessed 2026-10-07

⁶⁷⁸ Verified fact — A4-S022: npm registry metadata: ai <https://registry.npmjs.org/ai> accessed 2026-10-07

⁶⁷⁹ Verified fact — A4-S070: AI SDK LICENSE (Vercel) <https://raw.githubusercontent.com/vercel/ai/main/LICENSE> accessed 2026-10-07

⁶⁸⁰ Verified fact — A4-S069: AI SDK package README (GitHub) (Vercel) <https://raw.githubusercontent.com/vercel/ai/main/packages/ai/README.md> accessed 2026-10-07

Criterion	Score	Rationale
Total (generic / FS)	3.25 / 3.15	

Capabilities. Apache-2.0 TypeScript toolkit (npm ‘ai’) for provider-agnostic generation, structured output, tool calling and ToolLoopAgent agents, with UI hooks for React, Svelte, Vue and Angular; companion Workflow SDK makes TS/JS functions durable ^{vr681}.

Strengths

- Best fit for TypeScript front ends that stream agent output to the browser ^{AI}
- Workflow SDK can be self-hosted with Postgres ^{vr682}

Limitations and risks

- Three major versions in about eleven months ^{vr683}
- Defaults to Vercel AI Gateway for model access ^{vr684}
- Vercel certifications not verified ^{NPV}

Choose when

- The agent is part of a Next.js or TypeScript application tier ^{AI}

Avoid when

- Back-office regulated workflows owned by a Python data team ^{AI}

Nearest competitors: L3-openai-agents-sdk, L3-langgraph, L3-microsoft-agent-framework

Regulated-FS note. Use direct provider packages or the firm’s gateway, not the Vercel AI Gateway default, for client data ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Vercel, Inc.	Verified fact	A4-S070, A4-S069
Category	Provider-agnostic TypeScript toolkit for AI applications and agents (generation, structured output, tool-loop agents, UI hooks)	Verified fact	A4-S069
Version / lineup	ai 7.0.131 (7 October 2026); 7.0.0 released 25 June 2026 (6.0.0 22 December 2025; 5.0.0 31 July 2025)	Verified fact	A4-S022, V1-S082
Licence	Open source, Apache-2.0	Verified fact	A4-S022,

⁶⁸¹ Verified fact — A4-S069: AI SDK package README (GitHub) (Vercel) <https://raw.githubusercontent.com/vercel/ai/main/packages/ai/README.md> accessed 2026-10-07; A4-S071: Workflow SDK README (GitHub) (Vercel) <https://raw.githubusercontent.com/vercel/workflow/main/packages/workflow/README.md> accessed 2026-10-07; A4-S106: npm registry metadata: workflow (Vercel Workflow SDK) <https://registry.npmjs.org/workflow> accessed 2026-10-07

⁶⁸² Verified fact — A4-S071: Workflow SDK README (GitHub) (Vercel) <https://raw.githubusercontent.com/vercel/workflow/main/packages/workflow/README.md> accessed 2026-10-07

⁶⁸³ Verified fact — A4-S022: npm registry metadata: ai <https://registry.npmjs.org/ai> accessed 2026-10-07

⁶⁸⁴ Verified fact — A4-S069: AI SDK package README (GitHub) (Vercel) <https://raw.githubusercontent.com/vercel/ai/main/packages/ai/README.md> accessed 2026-10-07

Field	Value	Label	Sources
		d fact	A4-S070
Status events	25 June 2026: AI SDK 7.0.0; 30 September 2026: Workflow SDK 5.0.0	Verified fact	A4-S022, A4-S106
Strategic direction	Defaults to the Vercel AI Gateway for model access (direct provider packages also supported); ToolLoopAgent abstraction; companion Workflow SDK (npm 'workflow' 5.1.0) makes TS/JS functions durable	Verified fact	A4-S069, A4-S071, A4-S106
What it does	Unified API across model providers for text and structured generation, tool calling and agents (ToolLoopAgent), plus AI SDK UI hooks for React, Svelte, Vue and Angular and streaming agent responses to the browser.	Verified fact	A4-S069
Stack position	L3 (TypeScript agent/app framework) with an application/UI layer component; model routing via Vercel AI Gateway overlaps L2/C1	Verified fact	A4-S069
Integration	TypeScript library; provider adapters; AI Gateway model strings; UI framework hooks	Verified fact	A4-S069
Dependencies	Node.js 22+; provider packages (@ai-sdk/openai, @ai-sdk/anthropic, @ai-sdk/google, ...) or Vercel AI Gateway	Verified fact	A4-S069
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not applicable (library)	Architectural judgment	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Free (Apache-2.0); AI Gateway usage priced separately (not retrieved)	Verified fact	A4-S070
Infrastructure cost	Not material for the library	Architectural judgment	
Ecosystem	Next.js, React, Svelte, Vue, Angular; providers OpenAI, Anthropic, Google and others; Vercel Sandbox; Workflow SDK	Verified fact	A4-S069, A4-S071

Field	Value	Label	Sources
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Package name 'ai' reused; AI SDK 3.0 February 2024; major version roughly every 6 months (5.0 July 2025, 6.0 December 2025, 7.0 June 2026)	Verified fact	A4-S022
Deployment	saas: No; managed_cloud: No; vpc_byoc: Yes (library runs inside the customer's own environment); private_cloud: Yes (library runs inside the customer's own environment); self_hosted: Yes (library runs in any Node runtime); on_prem: Yes (library)		

Claude Agent SDK (Python: `claude-agent-sdk`; TypeScript: `@anthropic-ai/claude-agent-sdk`); formerly Claude Code SDK (`L3-claude-agent-sdk`)

Tier: Experimental · **Flags:** Renamed · **Original graphic label:** Agent SDK (Anthropic logo)

Rationale: Alpha (the vendor's own PyPI classifier), single-model and subprocess-bound; stays Experimental. Maturity set to 2 by the reader at Checkpoint 4 (CP4-9), in line with the OpenAI Agents SDK; technical 3 and deployment 3 are the peer-consistent values (CP4 review C). Conflict of interest disclosed.

Criterion	Score	Rationale
Technical	3	Rich autonomous harness, but no workflow engine or built-in durable execution ^{vf685} ; borderline 3/4 resolved down
Enterprise readiness	3	Library (rule 2): permission modes and hooks; API-key auth; enterprise support not verified ^{vf686 NPV}
Security and compliance	3	Library (rule 2); inherits Claude API terms where ZDR is available, but Managed Agents excluded from ZDR ^{vf687}
Deployment flexibility	3	Self-hosted in containers/Kubernetes; model inference remains a cloud API; subprocess-per-session hosting ^{vf688} ; borderline 3/4 resolved down
Ecosystem	3	MCP, plugins and skills; Claude models only ^{vf689}
Reliability and maturity	2	Alpha development status (PyPI 'Development Status: 3 - Alpha'), pre-1.0, renamed September 2025 ^{vf690} . Scored 2, the same as the pre-1.0 OpenAI Agents

⁶⁸⁵ Verified fact — A4-S027: Agent SDK overview - Claude Code Docs (Anthropic) <https://code.claude.com/docs/en/agent-sdk/overview> accessed 2026-10-07; A4-S122: Hosting the Agent SDK - Claude Code Docs (Anthropic) <https://code.claude.com/docs/en/agent-sdk/hosting> accessed 2026-10-07

⁶⁸⁶ Verified fact — A4-S027: Agent SDK overview - Claude Code Docs (Anthropic) <https://code.claude.com/docs/en/agent-sdk/overview> accessed 2026-10-07

⁶⁸⁷ Verified fact — A4-S123: API and data retention - Claude Platform Docs (Anthropic) <https://platform.claude.com/docs/en/build-with-claude/zero-data-retention> accessed 2026-10-07

⁶⁸⁸ Verified fact — A4-S122: Hosting the Agent SDK - Claude Code Docs (Anthropic) <https://code.claude.com/docs/en/agent-sdk/hosting> accessed 2026-10-07

⁶⁸⁹ Verified fact — A4-S027: Agent SDK overview - Claude Code Docs (Anthropic) <https://code.claude.com/docs/en/agent-sdk/overview> accessed 2026-10-07

⁶⁹⁰ Verified fact — A4-S006: PyPI JSON metadata: claude-agent-sdk (Python Package Index) <https://pypi.org/pypi/claude-agent-sdk/json> accessed 2026-10-07; A4-S029: Building agents with the Claude Agent SDK (Anthropic) <https://www.anthropic.com/engineering/building-agents-with-the-claude-agent-sdk> accessed 2026-10-07; B-REVC-S001: PyPI JSON API: claude-agent-sdk (with openai-agents; google-adk; sglang; pydantic-ai; langgraph) (Python

Criterion	Score	Rationale
		SDK, by the reader at Checkpoint 4 (CP4-9; previously 1 under a strict reading of the anchor). The Alpha label is recorded as a limitation and reflected in the Experimental tier.
Cost / TCO	3	No SDK charge; compute scales with concurrent sessions ^{vi.691} ^{AJ}
Lock-in / portability	2	Claude-only, bundled CLI binary, Commercial Terms ^{vi.692}
Total (generic / FS)	2.85	
	/	
	2.75	

Capabilities. Anthropic’s agent harness SDK (formerly Claude Code SDK): embeds Claude Code’s agent loop, built-in file/shell/web tools, hooks, subagents, MCP, permission modes and resumable/forkable sessions as a Python or TypeScript library; spawns one Claude Code CLI subprocess per session ^{vi.693}. Conflict of interest: Anthropic product scored by an Anthropic model.

Strengths

- Permission modes and hooks give fine-grained tool approval ^{vi.694}
- Strong fit for code- and file-centric autonomous tasks ^{AJ}

Limitations and risks

- Python package classified Alpha; 0.x versioning ^{vi.695}
- Claude models only, via Claude API, Bedrock or Google Cloud ^{vi.696}
- Use governed by Anthropic Commercial Terms despite the MIT repository licence ^{vi.697}
- One long-lived subprocess per session; transcripts on local disk unless a SessionStore is configured ^{vi.698}
- Claude Managed Agents is excluded from ZDR ^{vi.699}

Package Index) <https://pypi.org/pypi/claude-agent-sdk/json> accessed 2026-10-08 [search-tool extract]

⁶⁹¹ Verified fact — A4-S122: Hosting the Agent SDK - Claude Code Docs (Anthropic) <https://code.claude.com/docs/en/agent-sdk/hosting> accessed 2026-10-07

⁶⁹² Verified fact — A4-S027: Agent SDK overview - Claude Code Docs (Anthropic) <https://code.claude.com/docs/en/agent-sdk/overview> accessed 2026-10-07; A4-S092: Claude Agent SDK (Python) LICENSE (Anthropic) <https://raw.githubusercontent.com/anthropics/claude-agent-sdk-python/main/LICENSE> accessed 2026-10-07

⁶⁹³ Verified fact — A4-S027: Agent SDK overview - Claude Code Docs (Anthropic) <https://code.claude.com/docs/en/agent-sdk/overview> accessed 2026-10-07; A4-S122: Hosting the Agent SDK - Claude Code Docs (Anthropic) <https://code.claude.com/docs/en/agent-sdk/hosting> accessed 2026-10-07; A4-S028: Migrate to Claude Agent SDK - Claude Code Docs (Anthropic) <https://platform.claude.com/docs/en/agent-sdk/migration-guide> accessed 2026-10-07

⁶⁹⁴ Verified fact — A4-S027: Agent SDK overview - Claude Code Docs (Anthropic) <https://code.claude.com/docs/en/agent-sdk/overview> accessed 2026-10-07

⁶⁹⁵ Verified fact — A4-S006: PyPI JSON metadata: claude-agent-sdk (Python Package Index) <https://pypi.org/pypi/claude-agent-sdk/json> accessed 2026-10-07

⁶⁹⁶ Verified fact — A4-S006: PyPI JSON metadata: claude-agent-sdk (Python Package Index) <https://pypi.org/pypi/claude-agent-sdk/json> accessed 2026-10-07; A4-S122: Hosting the Agent SDK - Claude Code Docs (Anthropic) <https://code.claude.com/docs/en/agent-sdk/hosting> accessed 2026-10-07

⁶⁹⁷ Verified fact — A4-S092: Claude Agent SDK (Python) LICENSE (Anthropic) <https://raw.githubusercontent.com/anthropics/claude-agent-sdk-python/main/LICENSE> accessed 2026-10-07; V1-S048: Agent SDK overview (License and terms section) (Anthropic) <https://docs.claude.com/en/docs/agent-sdk/overview> accessed 2026-10-08

⁶⁹⁸ Verified fact — A4-S122: Hosting the Agent SDK - Claude Code Docs (Anthropic) <https://code.claude.com/docs/en/agent-sdk/hosting> accessed 2026-10-07

- No deterministic workflow engine ^{VF700}

Choose when

- A sandboxed autonomous sub-task (e.g. code or document manipulation) inside a firm-owned workflow, where Claude is already an approved model ^{AJ}

Avoid when

- You need model portability or a deterministic workflow engine ^{AJ}
- A regulated process would run on an Alpha-classified dependency ^{AJ}

Nearest competitors: L3-openai-agents-sdk, L3-langgraph, L3-pydantic-ai

Regulated-FS note. Run only in sandboxed containers with network controls, under a workflow engine that owns state and approvals; independent alternatives are LangGraph and Pydantic AI ^{ROC}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Anthropic, PBC	Verified fact	A4-S092, A4-S027
Category	Agent harness SDK that embeds Claude Code's agent loop, built-in tools and context management as a library	Verified fact	A4-S027
Version / lineup	claude-agent-sdk 0.2.164 (6 October 2026; PyPI classifier 'Development Status: 3 - Alpha'); @anthropic-ai/claude-agent-sdk 0.3.293 (7 October 2026)	Verified fact	A4-S006, A4-S023
Licence	Python repository LICENSE is MIT; npm package licence field reads 'SEE LICENSE IN README.md'; Anthropic docs state use of the SDK is governed by Anthropic's Commercial Terms of Service (except components under their own licence)	Verified fact	A4-S092, A4-S006, A4-S023, A4-S027, V1-S048
Status events	29 September 2025: Claude Code SDK renamed Claude Agent SDK (claude-agent-sdk 0.1.0 first published to PyPI 28 September 2025 UTC) (packages claude-code-sdk -> claude-agent-sdk, @anthropic-ai/claude-code -> @anthropic-ai/claude-agent-sdk); claude-code-sdk marked deprecated and no longer maintained (last release 0.0.25, 29 September 2025)	Verified fact	A4-S029, A4-S028, A4-S011, V1-S049
Strategic direction	Positioned for agents beyond coding; Anthropic also offers Claude Managed Agents (beta), a hosted agent harness with Anthropic-managed or self-hosted sandboxes, as the hosted alternative	Verified fact	A4-S028, A4-S027, A4-S124
What it does	Gives developers the same tools, agent loop and context management that power Claude Code, programmable in Python and TypeScript: built-in file/shell/web tools, hooks, subagents, MCP, permissions, sessions (resume/fork), skills, commands, memory and plugins.	Verified fact	A4-S027
Stack position	L3 agent harness (autonomous agent loop). It is not a deterministic workflow engine; it runs as a supervised CLI subprocess per session	Verified fact	A4-S027, A4-S122

⁶⁹⁹ Verified fact — A4-S123: API and data retention - Claude Platform Docs (Anthropic) <https://platform.claude.com/docs/en/build-with-claude/zero-data-retention> accessed 2026-10-07

⁷⁰⁰ Verified fact — A4-S027: Agent SDK overview - Claude Code Docs (Anthropic) <https://code.claude.com/docs/en/agent-sdk/overview> accessed 2026-10-07

Field	Value	Label	Sources
Integration	Python/TS libraries; MCP for tools/data; hooks; CLI subprocess with JSON output for other languages	Verified fact	A4-S027
Dependencies	Bundled Claude Code CLI binary (spawned as a subprocess per session over stdio); Python >=3.10 or Node; Claude models via the Claude API, or provider endpoints on Amazon Bedrock or Google Cloud's Agent Platform; durable store needed to persist session transcripts beyond local disk	Verified fact	A4-S006, A4-S122
Certifications	Not applicable to the SDK; Anthropic platform certifications are covered in the L1 Anthropic record (stream A5)	Architectural judgement	
GDPR / residency	SDK inherits the API arrangement: ZDR is available per organisation for eligible Claude API features and for Claude Code used with Commercial-org API keys; Claude Managed Agents is excluded from ZDR (session transcripts persist until deleted); on Bedrock / Google Cloud the cloud provider is the data processor	Verified fact	A4-S123
Security features	Permission modes controlling which tools run automatically or need approval; hooks; recommended sandboxed containers for process isolation and network control	Verified fact	A4-S027, A4-S122
Access controls	Third-party products may not offer claude.ai login or rate limits unless previously approved; API-key authentication required	Verified fact	A4-S027
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	SDK has no separate charge identified; usage billed as Claude API tokens (pricing covered by L1 stream)	Architectural judgement	
Infrastructure cost	One long-lived subprocess per concurrent session (own shell, working directory and transcript), so compute scales with concurrent sessions rather than requests	Verified fact	A4-S122
Ecosystem	MCP, Claude Code plugins/skills; runs on Claude API, Amazon Bedrock or Google Cloud; OpenAI publishes a migration cookbook from the Claude Agent SDK to its Agents SDK and Anthropic publishes the reverse	Verified fact	A4-S027, A4-S122, A4-S028
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Renamed and relaunched September 2025; 0.x versioning; Python package classified Alpha; very frequent releases (149 on PyPI)	Verified fact	A4-S006, A4-S029
Deployment	saas: No for the SDK; hosted alternative is Claude Managed Agents (beta); managed_cloud: No for the SDK; Managed Agents runs in Anthropic-managed cloud sandboxes or self-hosted sandboxes; vpc_byoc: Yes (self-hosted in customer containers/Kubernetes); private_cloud: Yes (self-hosted); self_hosted: Yes; on_prem: Yes		

Field	Value	Label	Sources
	(process runs on customer infrastructure; model inference remains a cloud API call)		

Mistral Agents API (exposed through the mistralai Python and TypeScript client SDKs, with an ‘agents’ extra) and Mistral Workflows (mistralai-workflows SDK); no separately branded ‘Mistral Agents SDK’ product found (L3-mistral-agents)

Tier: Experimental · **Flags:** none · **Original graphic label:** Mistral – Agents SDK

Rationale: Real products under a different name; beta status and proprietary hosted state. Tile relabelled ‘Mistral Agents API (+ Workflows)’.

Criterion	Score	Rationale
Technical	3	Hosted agents plus durable Workflows, but beta ^{VF701}
Enterprise readiness	2	SSO/RBAC/audit not publicly verified; capped at 2 ^{NPV}
Security and compliance	3	Company-level SOC 2 Type II, ISO 27001, ISO 27701; product scope not stated, so one below (rule 8) ^{VF702}
Deployment flexibility	3	SaaS; Workflows workers in customer Kubernetes, control plane Mistral-hosted ^{VF703}
Ecosystem	3	MCP connectors; Temporal underneath ^{VF704}
Reliability and maturity	2	Beta / Public Preview components; breaking SDK majors ^{VF705}
Cost / TCO	2	Pricing not publicly verified; customer-run worker infrastructure ^{NPV VF706}
Lock-in / portability	2	Proprietary hosted API, Mistral-model-centric ^{VF707}
Total (generic / FS)	2.60 / 2.55	

⁷⁰¹ Verified fact — A4-S057: Build AI agents with the Mistral Agents API (and Agents & Conversations / Handoffs docs) (Mistral AI) <https://mistral.ai/news/agents-api/> accessed 2026-10-07 [search-tool extract]; A4-S058: Workflows for work that runs the business (Mistral Workflows) (Mistral AI) <https://mistral.ai/news/workflows/> accessed 2026-10-07 [search-tool extract]

⁷⁰² Verified fact — A4-S060: Do you have SOC 2 or ISO 27001 certification? (Mistral Help Center / Trust Center) (Mistral AI) <https://help.mistral.ai/en/articles/347638-do-you-have-soc-2-or-iso-27001-certification> accessed 2026-10-07 [search-tool extract]

⁷⁰³ Verified fact — A4-S058: Workflows for work that runs the business (Mistral Workflows) (Mistral AI) <https://mistral.ai/news/workflows/> accessed 2026-10-07 [search-tool extract]

⁷⁰⁴ Verified fact — A4-S057: Build AI agents with the Mistral Agents API (and Agents & Conversations / Handoffs docs) (Mistral AI) <https://mistral.ai/news/agents-api/> accessed 2026-10-07 [search-tool extract]; A4-S058: Workflows for work that runs the business (Mistral Workflows) (Mistral AI) <https://mistral.ai/news/workflows/> accessed 2026-10-07 [search-tool extract]

⁷⁰⁵ Verified fact — A4-S061: PyPI JSON metadata: mistralai-workflows (Python Package Index) <https://pypi.org/pypi/mistralai-workflows/json> accessed 2026-10-07; A4-S007: PyPI JSON metadata: mistralai (Python Package Index) <https://pypi.org/pypi/mistralai/json> accessed 2026-10-07

⁷⁰⁶ Verified fact — A4-S058: Workflows for work that runs the business (Mistral Workflows) (Mistral AI) <https://mistral.ai/news/workflows/> accessed 2026-10-07 [search-tool extract]

⁷⁰⁷ Verified fact — A4-S057: Build AI agents with the Mistral Agents API (and Agents & Conversations / Handoffs docs) (Mistral AI) <https://mistral.ai/news/agents-api/> accessed 2026-10-07 [search-tool extract]

Evidence rules applied: enterprise_readiness capped at 2: access controls NPV

Capabilities. No distinct ‘Mistral Agents SDK’ exists. Mistral offers the Agents API (persistent agents, connectors incl. MCP, branchable stateful conversations, handoffs) through the general mistralai SDKs, and Mistral Workflows, a Temporal-based durable workflow SDK with Mistral-hosted control plane and customer-run workers^{vf708}.

Strengths

- EU hosting by default for the company's services^{vf709}
- Workflows brings Temporal-grade durability with HITL decorators^{vf710}

Limitations and risks

- Agents API in beta namespace; Workflows Beta/Public Preview; breaking SDK majors March and September 2026^{vf711}
- State lives in Mistral cloud unless store=False^{vf712}
- Access controls not publicly verified^{NPV}
- Pricing not publicly verified^{NPV}

Choose when

- An EU-hosted, Mistral-model estate wanting managed agents^{AJ}

Avoid when

- You need a portable framework or GA components^{AJ}

Nearest competitors: L3-openai-agents-sdk, L3-temporal, L3-langgraph

Regulated-FS note. Run conversations with store=False and keep the workflow definition in the firm’s own Temporal or framework code^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Mistral AI	Verified fact	A4-S057, A4-S125

⁷⁰⁸ Verified fact — A4-S057: Build AI agents with the Mistral Agents API (and Agents & Conversations / Handoffs docs) (Mistral AI) <https://mistral.ai/news/agents-api/> accessed 2026-10-07 [search-tool extract]; A4-S058: Workflows for work that runs the business (Mistral Workflows) (Mistral AI) <https://mistral.ai/news/workflows/> accessed 2026-10-07 [search-tool extract]; A4-S061: PyPI JSON metadata: mistralai-workflows (Python Package Index) <https://pypi.org/pypi/mistralai-workflows/json> accessed 2026-10-07

⁷⁰⁹ Verified fact — A5-S075: Mistral Help Centre: SOC 2 / ISO 27001; data storage location; ZDR; Regional inference docs; Trust Center (Mistral AI) <https://help.mistral.ai/en/articles/347638-do-you-have-soc-2-or-iso-27001-certification> accessed 2026-10-07 [search-tool extract]

⁷¹⁰ Verified fact — A4-S058: Workflows for work that runs the business (Mistral Workflows) (Mistral AI) <https://mistral.ai/news/workflows/> accessed 2026-10-07 [search-tool extract]

⁷¹¹ Verified fact — A4-S057: Build AI agents with the Mistral Agents API (and Agents & Conversations / Handoffs docs) (Mistral AI) <https://mistral.ai/news/agents-api/> accessed 2026-10-07 [search-tool extract]; A4-S061: PyPI JSON metadata: mistralai-workflows (Python Package Index) <https://pypi.org/pypi/mistralai-workflows/json> accessed 2026-10-07; A4-S007: PyPI JSON metadata: mistralai (Python Package Index) <https://pypi.org/pypi/mistralai/json> accessed 2026-10-07; V1-S065: Workflows for work that runs the business; Mistral Workflows docs (Mistral AI) <https://mistral.ai/news/workflows/> accessed 2026-10-08 [search-tool extract]

⁷¹² Verified fact — A4-S057: Build AI agents with the Mistral Agents API (and Agents & Conversations / Handoffs docs) (Mistral AI) <https://mistral.ai/news/agents-api/> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
Category	Hosted agent API (stateful conversations, connectors, handoffs) plus a Temporal-based durable workflow platform	Verified fact	A4-S057, A4-S058
Version / lineup	mistralai (Python) 3.1.0 (6 October 2026; 3.0.0 on 28 September 2026); @mistralai/mistralai 2.7.0 (9 September 2026); mistralai-workflows 3.15.0 (14 September 2026, Beta)	Verified fact	A4-S007, A4-S025, A4-S061
Licence	Client SDKs Apache-2.0; mistralai-workflows Apache-2.0; Agents API and Workflows control plane are proprietary hosted services	Verified fact	A4-S125, A4-S025, A4-S061
Status events	SDK 3.0.0 (28 September 2026): HTTPX2, code_interpreter/web_search tools removed from chat/agents completions (use Conversations API or an agent); Le Chat rebranded (Work/Vibe naming in docs); Le Chat agents cannot be used programmatically	Verified fact	A4-S059, A4-S007, A4-S057
Strategic direction	Agents API launched as a dedicated agentic API alongside chat completions; Studio adds built-in and custom MCP connectors with human-in-the-loop approvals; Mistral Workflows (v3.0 public, c. September 2026) built on Temporal with Mistral-hosted control plane and customer-run workers	Verified fact	A4-S057, A4-S058
What it does	Agents API: create persistent agents with built-in connectors (code execution, web search, image generation, document library/RAG, MCP), stateful conversations that can be branched, and handoffs between agents executed server-side or client-side. Workflows: Python SDK with decorators for retries, timeouts, tracing, rate limiting and human-in-the-loop, with event-history durability.	Verified fact	A4-S057, A4-S058, A4-S061
Stack position	L3, but as a vendor-hosted agent runtime rather than a portable framework; Workflows is a durable workflow engine (H2)	Verified fact	A4-S057, A4-S058
Integration	REST API and client SDKs (Python/TS); MCP connectors; Workflows Python SDK	Verified fact	A4-S057, A4-S061
Dependencies	Mistral cloud (La Plateforme/Studio); Mistral models; Workflows workers on customer Kubernetes via Helm connecting to Mistral-hosted Temporal cluster	Verified fact	A4-S058
Certifications	Mistral AI: SOC 2 Type II, ISO/IEC 27001:2022, ISO/IEC 27701:2019 (company-level; scope/dates not seen)	Verified fact	A4-S060
GDPR / residency	Third-party summaries say EU hosting by default with optional US endpoint and ZDR negotiable via enterprise sales; Agents API conversations can be run with store=False (history not stored on Mistral cloud)	Reported	A4-S060, A4-S057
Security features	Connector tool allow/deny lists; human-in-the-loop approval for sensitive actions	Verified fact	A4-S057
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
Pricing	Not publicly verified	Not publicly verified	
Infrastructure cost	Workflows requires customer-operated worker infrastructure (Kubernetes)	Verified fact	A4-S058
Ecosystem	MCP connectors (built-in and custom); Temporal (Workflows engine)	Verified fact	A4-S057, A4-S058
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Agents API under client.beta namespace in SDK examples; Workflows SDK classified Beta; durable agents in Public Preview	Verified fact	A4-S057, A4-S061, V1-S065
Deployment	saas: Yes; managed_cloud: Yes; vpc_byoc: Yes (partial: Workflows workers run in customer Kubernetes; control plane Mistral-hosted); private_cloud: Not publicly verified; self_hosted: Yes (partial: Workflows LocalSession for experimental or on-premises use); on_prem: Not publicly verified		

L2: Inference, serving and model access

vLLM (L2-vllm)

Tier: Strategic · **Flags:** none · **Original graphic label:** vLLM – high-throughput

Rationale: Default serving engine for any self-hosted route; neutral governance; conditional on patch and model-provenance discipline

Criterion	Score	Rationale
Technical	5	Leading depth: quantisation, speculative decoding, prefix caching, disaggregation, multi-hardware ^{vr713}
Enterprise readiness	4	Rule 2: enables in-estate serving; inherits host controls; commercial support via Red Hat AI Inference lifts the 4 cap ^{vr714}
Security and	3	Rule 2 hygiene: published advisory process with fixed versions ^{vr715} , but several

⁷¹³ Verified fact — A4-S063: vLLM README (GitHub) (vLLM project) <https://raw.githubusercontent.com/vllm-project/vllm/main/README.md> accessed 2026-10-07

⁷¹⁴ Verified fact — B-L2-S001: Red Hat AI Inference 3.4 release notes https://docs.redhat.com/en/documentation/red_hat_ai_inference/3.4/html/release_notes/rhaii-340-ga-release-notes_release-notes accessed 2026-10-08 [search-tool extract]

⁷¹⁵ Verified fact — A4-S063: vLLM README (GitHub) (vLLM project) <https://raw.githubusercontent.com/vllm-project/vllm/main/README.md> accessed 2026-10-07; B-L2-S004: Security Advisories - vllm-project/vllm (vLLM project (GitHub)) <https://github.com/vllm-project/vllm/security/advisories> accessed 2026-10-08 [search-tool extract]

Criterion	Score	Rationale
compliance		2026 RCE advisories, one critical
Deployment flexibility	5	Self-host, private cloud, on-prem, and managed via third parties ^{VF716}
Ecosystem	5	De-facto standard with OpenAI-compatible API; 2,000+ contributors ^{VF717}
Reliability and maturity	4	Foundation-hosted since May 2025 ^{VF718} ; held at 4 for pre-1.0 versioning and fast cadence ^{VF719}
Cost / TCO	4	Free; cost is GPU hours and operations ^{AJ}
Lock-in / portability	5	Apache-2.0, open APIs, neutral governance ^{VF720}
Total (generic / FS)	4.35	
	/	
	4.30	

Capabilities. Apache-2.0 LLM inference and serving engine (0.31.0, 5 October 2026), PyTorch Foundation-hosted since 7 May 2025, with PagedAttention, continuous batching, prefix caching, quantisation (FP8, MXFP4, NVFP4, INT4/8, GPTQ/AWQ, GGUF), speculative decoding (n-gram, suffix, EAGLE, DFlash), disaggregated prefill/decode/encode, OpenAI-compatible and Anthropic Messages APIs, 200+ architectures and multi-vendor hardware ^{VF721}. A commercially supported distribution exists (Red Hat AI Inference) ^{VF722}.

Strengths

- Broadest optimisation feature set inside one engine ^{VF723 AJ}
- Neutral foundation governance and Apache-2.0 ^{VF724}
- De-facto substrate: under Hugging Face Endpoints, ILM-d and Dynamo ^{VF725}

⁷¹⁶ Verified fact — A4-S063: vLLM README (GitHub) (vLLM project) <https://raw.githubusercontent.com/vllm-project/vllm/main/README.md> accessed 2026-10-07; A4-S081: About Inference Endpoints (Hugging Face) <https://raw.githubusercontent.com/huggingface/hf-endpoints-documentation/main/docs/source/about.md> accessed 2026-10-07

⁷¹⁷ Verified fact — A4-S063: vLLM README (GitHub) (vLLM project) <https://raw.githubusercontent.com/vllm-project/vllm/main/README.md> accessed 2026-10-07

⁷¹⁸ Verified fact — A4-S146: PyTorch Foundation Welcomes vLLM as a Hosted Project (and umbrella-foundation press release, 7 May 2025) (PyTorch Foundation (Linux Foundation)) <https://pytorch.org/blog/pytorch-foundation-welcomes-vllm/> accessed 2026-10-07 [search-tool extract]

⁷¹⁹ Verified fact — A4-S009: PyPI JSON metadata: vllm (Python Package Index) <https://pypi.org/pypi/vllm/json> accessed 2026-10-07

⁷²⁰ Verified fact — A4-S009: PyPI JSON metadata: vllm (Python Package Index) <https://pypi.org/pypi/vllm/json> accessed 2026-10-07; A4-S146: PyTorch Foundation Welcomes vLLM as a Hosted Project (and umbrella-foundation press release, 7 May 2025) (PyTorch Foundation (Linux Foundation)) <https://pytorch.org/blog/pytorch-foundation-welcomes-vllm/> accessed 2026-10-07 [search-tool extract]

⁷²¹ Verified fact — A4-S009: PyPI JSON metadata: vllm (Python Package Index) <https://pypi.org/pypi/vllm/json> accessed 2026-10-07; A4-S063: vLLM README (GitHub) (vLLM project) <https://raw.githubusercontent.com/vllm-project/vllm/main/README.md> accessed 2026-10-07; A4-S146: PyTorch Foundation Welcomes vLLM as a Hosted Project (and umbrella-foundation press release, 7 May 2025) (PyTorch Foundation (Linux Foundation)) <https://pytorch.org/blog/pytorch-foundation-welcomes-vllm/> accessed 2026-10-07 [search-tool extract]

⁷²² Verified fact — B-L2-S001: Red Hat AI Inference 3.4 release notes https://docs.redhat.com/en/documentation/red_hat_ai_inference/3.4/html/release_notes/rhaii-340-ga-release-notes_release-notes accessed 2026-10-08 [search-tool extract]

⁷²³ Verified fact — A4-S063: vLLM README (GitHub) (vLLM project) <https://raw.githubusercontent.com/vllm-project/vllm/main/README.md> accessed 2026-10-07

⁷²⁴ Verified fact — A4-S146: PyTorch Foundation Welcomes vLLM as a Hosted Project (and umbrella-foundation press release, 7 May 2025) (PyTorch Foundation (Linux Foundation)) <https://pytorch.org/blog/pytorch-foundation-welcomes-vllm/> accessed 2026-10-07 [search-tool extract]; A4-S009: PyPI JSON metadata: vllm (Python Package Index) <https://pypi.org/pypi/vllm/json> accessed 2026-10-07

⁷²⁵ Verified fact — A4-S081: About Inference Endpoints (Hugging Face) <https://raw.githubusercontent.com/huggingface/hf-endpoints-documentation/main/docs/source/about.md> accessed 2026-10-07; A4-S089: ILM-d README (GitHub) (ILM-d project) <https://raw.githubusercontent.com/ilm-d/ilm-d/main/README.md> accessed 2026-10-07; A4-S090: NVIDIA Dynamo README (GitHub)

Limitations and risks

- Pre-1.0 with roughly fortnightly releases (28 in 2026) ⁷²⁶
- A run of 2026 code-execution advisories, including critical CVE-2026-22778, mostly via untrusted model files or media inputs ⁷²⁷
- No identity, audit or tenancy of its own; inherits host and gateway controls ^{AJ}

Choose when

- You self-host open-weight models on your own or private-cloud GPUs ^{AJ}
- You want the default engine with the widest model and hardware coverage ^{AJ}

Avoid when

- You cannot run a monthly patch cadence and model-provenance controls ^{AJ}
- You have no GPU capacity or SRE capability; use managed access instead ^{AJ}

Nearest competitors: L2-sglang, TensorRT-LLM, L2-hugging-face

Regulated-FS note. Run behind the gateway (C1), load only allow-listed, scanned weights with trust_remote_code off, pin versions and track advisories ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	vLLM project, a PyTorch Foundation-hosted project (Linux Foundation) since May 2025; originated at UC Berkeley Sky Computing Lab; lead maintainers include Woosuk Kwon, Zhuohan Li, Simon Mo, Kaichao You and Robert Shaw	Verified fact	A4-S146, A4-S063, A4-S064
Category	Open-source LLM inference and serving engine	Verified fact	A4-S063
Version / lineup	vllm 0.31.0 (5 October 2026); 28 releases in 2026 (0.14.0 in January to 0.31.0)	Verified fact	A4-S009
Licence	Open source, Apache-2.0	Verified fact	A4-S009
Status events	7 May 2025: accepted as a PyTorch Foundation-hosted project when the foundation became an umbrella foundation	Verified fact	A4-S146, V1-S064
Strategic direction	PagedAttention, continuous batching, prefix caching, quantisation (FP8, MXFP4, NVFP4, INT4/8, GPTQ/AWQ, GGUF), speculative decoding (n-gram, suffix, EAGLE, DFlash), disaggregated prefill/decode/encode; OpenAI-compatible server plus Anthropic Messages API and gRPC; 200+	Verified fact	A4-S063

<https://raw.githubusercontent.com/ai-dynamo/dynamo/main/README.md> accessed 2026-10-07

⁷²⁶ Verified fact — A4-S009: PyPI JSON metadata: vllm (Python Package Index) <https://pypi.org/pypi/vllm/json> accessed 2026-10-07

⁷²⁷ Verified fact — B-L2-S004: Security Advisories - vllm-project/vllm (vLLM project (GitHub)) <https://github.com/vllm-project/vllm/security/advisories> accessed 2026-10-08 [search-tool extract]

Field	Value	Label	Sources
	architectures; NVIDIA, AMD, Intel GPUs, CPUs and plugins (TPU, Gaudi, Ascend, Apple Silicon)		
What it does	High-throughput, memory-efficient inference and serving library and server for LLMs, multimodal, embedding and reward models.	Verified fact	A4-S063
Stack position	Serving engine (L2) underneath managed services (e.g. Hugging Face Endpoints) and distributed orchestration layers (llm-d, NVIDIA Dynamo)	Verified fact	A4-S081, A4-S089, A4-S090
Integration	OpenAI-compatible API, Anthropic Messages API, gRPC; Python library	Verified fact	A4-S063
Dependencies	Python <3.15,>=3.10; PyTorch; accelerators (NVIDIA/AMD/Intel GPUs, CPUs, hardware plugins)	Verified fact	A4-S009, A4-S063
Certifications	Not applicable (open-source software)	Architectural judgement	
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Free (Apache-2.0)	Verified fact	A4-S009
Infrastructure cost	Dominated by accelerator hours and memory; throughput features (batching, quantisation, prefix caching) drive cost per token	Architectural judgement	
Ecosystem	Hugging Face models; llm-d builds on vLLM; Dynamo supports vLLM backend; sponsors listed on vllm.ai	Verified fact	A4-S063, A4-S089, A4-S090
Adoption signals	README: 'over 2000 contributors' from 'many dozens of academic institutions and companies'	Verified fact	A4-S063
Maturity	First PyPI release June 2023; still 0.x; roughly fortnightly releases	Verified	A4-S009

Field	Value	Label	Sources
		d fact	
Deployment	saas: No; managed_cloud: Yes (via third parties, e.g. Hugging Face Inference Endpoints); vpc_byoc: Yes (self-deployed); private_cloud: Yes; self_hosted: Yes; on_prem: Yes		

Fireworks AI (L2-fireworks-ai)

Tier: Strategic · **Flags:** none · **Original graphic label:** Fireworks AI – fast inference

Rationale: Strategic, conditional: for managed open-model inference, once its ISO 27001, 27701 and 42001 certificates are confirmed (trust-centre and docs pages conflict); the strongest controls among the independent inference clouds. For client data use EU dedicated or BYOC deployments, because the self-serve residency setting is US-only. Upgraded from Tactical ('candidate for Strategic after due diligence') by the reader at Checkpoint 4 (CP4-4); scores unchanged (FS 3.65) ^{AJ}.

Criterion	Score	Rationale
Technical	4	Serverless, dedicated, batch, fine-tuning/RFT, BYOC, FireRouter ^{VF728}
Enterprise readiness	4	SSO, RBAC and audit logs with CLI (rule 7 to 4); audit logs are recent ^{VF729}
Security and compliance	3	SOC 2 Type II, CMK, ZDR default; ISO claims not relied on (V1 section 4), so 3 ^{VF730}
Deployment flexibility	4	SaaS, dedicated, reserved, BYOC; no on-prem ^{VF731}
Ecosystem	4	OpenAI-compatible; Microsoft Foundry; Hugging Face partner ^{VF732}
Reliability and maturity	3	GA, well funded; revenue concentration reported ^{VF733}
Cost / TCO	3	Transparent but GPU rates higher than peers; batch at 50% ^{VF734}
Lock-in / portability	4	Open models portable; BYOC reduces data-plane lock-in ^{VF735}
Total (generic / FS)	3.65	

⁷²⁸ Verified fact — A4-S135: Fireworks - Pricing (and Serverless Pricing docs) (Fireworks AI) <https://fireworks.ai/pricing> accessed 2026-10-07 [search-tool extract]; A4-S152: Custom SSO - Fireworks AI Docs (with Enterprise, Virtual Cloud and changelog pages) <https://docs.fireworks.ai/accounts/sso> accessed 2026-10-07 [search-tool extract]

⁷²⁹ Verified fact — A4-S152: Custom SSO - Fireworks AI Docs (with Enterprise, Virtual Cloud and changelog pages) <https://docs.fireworks.ai/accounts/sso> accessed 2026-10-07 [search-tool extract]

⁷³⁰ Verified fact — A4-S133: Trust Center - Fireworks AI (with docs Certifications page and certification blogs) <https://trust.fireworks.ai/> accessed 2026-10-07 [search-tool extract]; V1-S067: Fireworks data security / Security & Compliance FAQ / triple ISO blog (Fireworks AI) https://docs.fireworks.ai/guides/security_compliance/data_security accessed 2026-10-08 [search-tool extract]

⁷³¹ Verified fact — A4-S152: Custom SSO - Fireworks AI Docs (with Enterprise, Virtual Cloud and changelog pages) <https://docs.fireworks.ai/accounts/sso> accessed 2026-10-07 [search-tool extract]

⁷³² Verified fact — A4-S134: Zero Data Retention / Data residency / Regions - Fireworks AI Docs https://docs.fireworks.ai/guides/security_compliance/data_handling accessed 2026-10-07 [search-tool extract]; A4-S075: Hugging Face Inference Providers <https://raw.githubusercontent.com/huggingface/hub-docs/main/docs/inference-providers/index.md> accessed 2026-10-07

⁷³³ Verified fact — A4-S141: Fireworks Raises a \$1.5 Billion Series D (press release; CNBC 16 Jul 2026) (Fireworks AI (Business Wire)) [https://www.businesswire.com/news/home/20260716264405/en/Fireworks-Raises-a-\\$1.5-Billion-Series-D-to-Lead-the-Specialized-Intelligence-Revolution](https://www.businesswire.com/news/home/20260716264405/en/Fireworks-Raises-a-$1.5-Billion-Series-D-to-Lead-the-Specialized-Intelligence-Revolution) accessed 2026-10-07 [search-tool extract]

⁷³⁴ Verified fact — A4-S135: Fireworks - Pricing (and Serverless Pricing docs) (Fireworks AI) <https://fireworks.ai/pricing> accessed 2026-10-07 [search-tool extract]

Criterion	Score	Rationale
	/	
	3.65	

Capabilities. Managed inference and fine-tuning for open models (serverless, on-demand and reserved dedicated, batch, RFT), hybrid BYOC in the customer VPC, customer-managed keys, SSO (OIDC/SAML), RBAC and audit logs; ZDR by default for open models except the Response API (30-day storage by default); SOC 2 Type II, while ISO 27001/27701/42001 claims conflict across its own pages ^{vf736}.

Strengths

- ZDR by default and BYOC data plane ^{vf737}
- Customer-managed keys logged in the customer's cloud audit log ^{vf738}
- Microsoft Foundry partnership ^{vf739}

Limitations and risks

- ISO claims conflict ('achieved' vs 'in progress') ^{vf740}
- Self-serve data-residency setting lists US only; EU via sales ^{vf741}
- Revenue concentration reported ^{vf742}
- Response API exception to ZDR ^{vf743}

Choose when

- Dedicated or BYOC open-model serving where you want a managed engine without running GPUs ^{AI}

Avoid when

⁷³⁵ Verified fact — A4-S152: Custom SSO - Fireworks AI Docs (with Enterprise, Virtual Cloud and changelog pages) <https://docs.fireworks.ai/accounts/sso> accessed 2026-10-07 [search-tool extract]

⁷³⁶ Verified fact — A4-S135: Fireworks - Pricing (and Serverless Pricing docs) (Fireworks AI) <https://fireworks.ai/pricing> accessed 2026-10-07 [search-tool extract]; A4-S152: Custom SSO - Fireworks AI Docs (with Enterprise, Virtual Cloud and changelog pages) <https://docs.fireworks.ai/accounts/sso> accessed 2026-10-07 [search-tool extract]; A4-S134: Zero Data Retention / Data residency / Regions - Fireworks AI Docs https://docs.fireworks.ai/guides/security_compliance/data_handling accessed 2026-10-07 [search-tool extract]; A4-S133: Trust Center - Fireworks AI (with docs Certifications page and certification blogs) <https://trust.fireworks.ai/> accessed 2026-10-07 [search-tool extract]; V1-S067: Fireworks data security / Security & Compliance FAQ / triple ISO blog (Fireworks AI) https://docs.fireworks.ai/guides/security_compliance/data_security accessed 2026-10-08 [search-tool extract]

⁷³⁷ Verified fact — A4-S134: Zero Data Retention / Data residency / Regions - Fireworks AI Docs https://docs.fireworks.ai/guides/security_compliance/data_handling accessed 2026-10-07 [search-tool extract]; A4-S152: Custom SSO - Fireworks AI Docs (with Enterprise, Virtual Cloud and changelog pages) <https://docs.fireworks.ai/accounts/sso> accessed 2026-10-07 [search-tool extract]

⁷³⁸ Verified fact — A4-S152: Custom SSO - Fireworks AI Docs (with Enterprise, Virtual Cloud and changelog pages) <https://docs.fireworks.ai/accounts/sso> accessed 2026-10-07 [search-tool extract]

⁷³⁹ Verified fact — A4-S134: Zero Data Retention / Data residency / Regions - Fireworks AI Docs https://docs.fireworks.ai/guides/security_compliance/data_handling accessed 2026-10-07 [search-tool extract]

⁷⁴⁰ Verified fact — V1-S067: Fireworks data security / Security & Compliance FAQ / triple ISO blog (Fireworks AI) https://docs.fireworks.ai/guides/security_compliance/data_security accessed 2026-10-08 [search-tool extract]

⁷⁴¹ Verified fact — A4-S134: Zero Data Retention / Data residency / Regions - Fireworks AI Docs https://docs.fireworks.ai/guides/security_compliance/data_handling accessed 2026-10-07 [search-tool extract]

⁷⁴² Verified fact — A4-S141: Fireworks Raises a \$1.5 Billion Series D (press release; CNBC 16 Jul 2026) (Fireworks AI (Business Wire)) [https://www.businesswire.com/news/home/20260716264405/en/Fireworks-Raises-a-\\$1.5-Billion-Series-D-to-Lead-the-Specialized-Intelligence-Revolution](https://www.businesswire.com/news/home/20260716264405/en/Fireworks-Raises-a-$1.5-Billion-Series-D-to-Lead-the-Specialized-Intelligence-Revolution) accessed 2026-10-07 [search-tool extract]

⁷⁴³ Verified fact — A4-S134: Zero Data Retention / Data residency / Regions - Fireworks AI Docs https://docs.fireworks.ai/guides/security_compliance/data_handling accessed 2026-10-07 [search-tool extract]

- The Response API for client data without store=False ^{744 AJ}
- ISO certificates are a hard gate before you have seen them ^{AJ}

Nearest competitors: L2-together-ai, L2-hugging-face, Hyperscaler model catalogues

Regulated-FS note. Request ISO certificates and BAA terms from the trust centre; prefer BYOC or EU dedicated deployments; disable Response API storage ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Fireworks AI, Inc.; privately held	Verified fact	A4-S136, A4-S141
Category	Managed inference and fine-tuning platform for open models (serverless, on-demand dedicated, reserved, BYOC)	Verified fact	A4-S135, A4-S152
Version / lineup	Official SDK fireworks-ai 1.2.20 (6 October 2026); serverless, on-demand deployments, batch inference, fine-tuning/RFT, Virtual Cloud (GA), FireRouter	Verified fact	A4-S094, A4-S134, A4-S152
Licence	Proprietary cloud service; official Python SDK Apache-2.0	Verified fact	A4-S094
Status events	28 October 2025: Series C US\$250M at US\$4B; 15-16 July 2026: Series D US\$1.505B at US\$17.5B (Atreides, Index, TCV)	Verified fact	A4-S136, A4-S141, V1-S058
Strategic direction	Series D (July 2026) to expand compute and engineering; positioning on 'specialised intelligence' (customers fine-tuning open models on proprietary data); reports >US\$1B annualised revenue and >40 trillion tokens/day	Verified fact	A4-S141
What it does	Serves open models via serverless per-token APIs and per-GPU-second dedicated deployments, batch inference at 50% of serverless price, fine-tuning and reinforcement fine-tuning, with enterprise reserved capacity and hybrid BYOC.	Verified fact	A4-S135, A4-S152
Stack position	Managed inference cloud (L2); FireRouter adds routing	Architectural judgement	
Integration	Official Python SDK; OpenAI-compatible APIs; Microsoft Foundry	Verified fact	A4-S094,

⁷⁴⁴ Verified fact — A4-S134: Zero Data Retention / Data residency / Regions - Fireworks AI Docs
https://docs.fireworks.ai/guides/security_compliance/data_handling accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
	partnership; Hugging Face Inference Providers partner	d fact	A4-S134, A4-S075
Dependencies	Not publicly verified	Not publicly verified	
Certifications	SOC 2 Type II; HIPAA (report of compliance listed; 'HIPAA-ready' on inference page); ISO 27001:2022, ISO 27701, ISO/IEC 42001:2023; GDPR (trust centre). BAA availability not seen; one Fireworks docs page still lists the three ISO certifications as "(in progress)" - request certificates from trust.fireworks.ai	Verified fact	A4-S133, V1-S067
GDPR / residency	ZDR by default for open models (prompts/outputs only in volatile memory); exceptions: Response API stores conversations by default for 30 days (store=False avoids), training/fine-tuning/agent features. Enterprise data-residency setting rejects out-of-region requests, but documented options are None and US only (others via sales); EU regions (Frankfurt, Iceland) exist for deployments	Verified fact	A4-S134
Security features	Customer-managed encryption keys (usage logged in customer cloud audit log); enterprise-enforceable ZDR policy; service accounts	Verified fact	A4-S152, A4-S134
Access controls	Custom SSO via OIDC or SAML 2.0 (enterprise); RBAC; Audit Logs page and CLI	Verified fact	A4-S152
Enterprise support	Enterprise reserved tier (multi-region deployments, custom optimisations, BYOC)	Verified fact	A4-S152
Pricing	On-demand GPUs: H100/H200 US\$8.00/h, B200 US\$13.00/h, B300 US\$15.00/h, billed per GPU-second (page ~167 days old; a secondary review cites US\$7.00/h). Serverless per 1M tokens: <4B US\$0.10, 4-16B US\$0.20, >16B US\$0.90; Kimi K3 US\$3 in/US\$15 out; batch 50% of serverless	Verified fact	A4-S135
Infrastructure cost	Dedicated deployments scale to zero; billed per GPU-second	Verified fact	A4-S135
Ecosystem	Microsoft Foundry; Hugging Face Inference Providers; Anthropic as a ZDR sub-processor for select services	Verified fact	A4-S134, A4-S075
Adoption signals	Series D US\$1.505B at US\$17.5B (July 2026); >US\$1B annualised revenue; CNBC: about half of revenue from Cursor as of 2025	Verified fact	A4-S141
Maturity	SDK since October 2022; Series D-stage company	Verified fact	A4-S094, A4-S141
Deployment	saas: Yes (serverless); managed_cloud: Yes (on-demand and reserved dedicated deployments); vpc_byoc: Yes (hybrid BYOC: Fireworks inference engine in customer VPC; enterprise reserved tier); private_cloud: Yes (reserved capacity); self_hosted: Yes (partial: BYOC in customer cloud account); on_prem: Not publicly verified		

SGLang (L2-sglang)

Tier: Strategic · **Flags:** none · **Original graphic label:** SGLang – efficient engine

Rationale: Strategic, conditional: as the qualified backup engine to vLLM, once CVE-2026-3059 is confirmed fixed in the deployed version (NVD and OSV reference a fix in 0.5.10; the GitHub advisory lists none), with internal ports isolated; not the default engine. Upgraded from Tactical by the reader at Checkpoint 4 (CP4-4); scores unchanged (FS 3.65), with security 2 carried as the tier condition ^{AJ}.

Criterion	Score	Rationale
Technical	5	Leading depth: RadixAttention, PD disaggregation, speculative decoding, quantisation ^{VF745}
Enterprise readiness	3	Rule 2: inherits host controls; no verified commercial support, so the 4 cap applies; scored 3 ^{AJ}
Security and compliance	2	Rule 2 hygiene: critical unauthenticated RCE (fix evidenced only by NVD/OSV references to v0.5.10; GitHub advisory lists none), disclosure-response concerns and a reported bypass of an earlier fix ^{VF746} ; kept at 2 at Checkpoint 4 (CP4-4 changed the tier only; a confirmed fix is the tier condition)
Deployment flexibility	5	Self-host, VPC, on-prem; managed via Hugging Face Endpoints ^{VF747}
Ecosystem	4	OpenAI-compatible; Dynamo backend; RL frameworks ^{VF748}
Reliability and maturity	3	0.x since January 2024; new commercial steward in May 2026 ^{VF749}
Cost / TCO	4	Free; GPU and operations cost ^{AJ}
Lock-in / portability	4	Apache-2.0, but governance is a non-profit plus VC-backed steward rather than a neutral foundation ^{VF750}

⁷⁴⁵ Verified fact — A4-S010: PyPI JSON metadata: sglang (Python Package Index) <https://pypi.org/pypi/sglang/json> accessed 2026-10-07

⁷⁴⁶ Verified fact — B-L2-S009: SGLang multimodal generation module unauthenticated RCE via ZMQ broker (CVE-2026-3059) - GitHub Advisory Database <https://github.com/advisories/GHSA-rgq9-fqf5-fv58> accessed 2026-10-08 [search-tool extract]; B-REVC-S002: NVD CVE-2026-3059 (with OSV and GitHub Advisory Database GHSA-rgq9-fqf5-fv58) (NIST NVD) <https://nvd.nist.gov/vuln/detail/CVE-2026-3059> accessed 2026-10-08 [search-tool extract]

⁷⁴⁷ Verified fact — A4-S065: SGLang README (GitHub) (SGLang project (LMSYS)) <https://raw.githubusercontent.com/sgl-project/sglang/main/README.md> accessed 2026-10-07; A4-S081: About Inference Endpoints (Hugging Face) <https://raw.githubusercontent.com/huggingface/hf-endpoints-documentation/main/docs/source/about.md> accessed 2026-10-07

⁷⁴⁸ Verified fact — A4-S010: PyPI JSON metadata: sglang (Python Package Index) <https://pypi.org/pypi/sglang/json> accessed 2026-10-07; A4-S090: NVIDIA Dynamo README (GitHub) <https://raw.githubusercontent.com/ai-dynamo/dynamo/main/README.md> accessed 2026-10-07; A4-S065: SGLang README (GitHub) (SGLang project (LMSYS)) <https://raw.githubusercontent.com/sgl-project/sglang/main/README.md> accessed 2026-10-07

⁷⁴⁹ Verified fact — A4-S010: PyPI JSON metadata: sglang (Python Package Index) <https://pypi.org/pypi/sglang/json> accessed 2026-10-07; A4-S147: RadixArk Launches with \$100 Million in Seed Funding Led by Accel to Grow SGLang (Business Wire, May 2026) (RadixArk (Business Wire)) <https://www.morningstar.com/news/business-wire/20260505077157/radixark-launches-with-100-million-in-seed-funding-led-by-accel-to-grow-sglang-and-democratize-frontier-ai-infrastructure> accessed 2026-10-07 [search-tool extract]

⁷⁵⁰ Verified fact — A4-S128: SGLang LICENSE (SGLang project) <https://raw.githubusercontent.com/sgl-project/sglang/main/LICENSE> accessed 2026-10-07; A4-S147: RadixArk Launches with \$100 Million in Seed Funding Led by Accel to Grow SGLang (Business Wire, May 2026) (RadixArk (Business Wire)) <https://www.morningstar.com/news/business-wire/20260505077157/radixark-launches-with-100-million-in-seed-funding-led-by-accel-to-grow-sglang-and-democratize-frontier-ai-infrastructure> accessed 2026-10-07 [search-tool extract]

Criterion	Score	Rationale
Total (generic / FS)	3.80	
	/	
	3.65	

Evidence rules applied: enterprise_readiness capped at 4 (rule 2, no verified commercial support); not binding at 3

Capabilities. Apache-2.0 serving framework (0.5.21, 1 October 2026) hosted by the LMSYS non-profit, with RadixAttention prefix caching, PD disaggregation, speculative decoding, FP4/FP8 quantisation, TPU and multi-vendor hardware support and day-0 support for new open models; RadixArk (US\$100M seed, May 2026) is its commercial steward ^{vf751}.

Strengths

- Prefix-cache-heavy and agentic workloads (RadixAttention) ^{vf752} ^{AI}
- Fast day-0 model support ^{vf753}
- Apache-2.0, OpenAI-compatible ^{vf754}

Limitations and risks

- CVE-2026-3059 (critical, unauthenticated RCE via ZMQ broker, <=0.5.9): the GitHub advisory lists no patched version, while NVD references a fix in v0.5.10 and OSV a fixed commit; an earlier fix (CVE-2025-10164) has an unanswered bypass report ^{vf755}
- Pre-1.0; governance with a non-profit plus a VC-backed steward, no formal transfer found ^{vf756}
- No verified commercial support offer ^{NPV}

Choose when

- Second engine where prefix reuse or a model's day-0 support matters ^{AI}
- TPU serving ^{vf757}

Avoid when

⁷⁵¹ Verified fact — A4-S010: PyPI JSON metadata: sglang (Python Package Index) <https://pypi.org/pypi/sglang/json> accessed 2026-10-07; A4-S065: SGLang README (GitHub) (SGLang project (LMSYS)) <https://raw.githubusercontent.com/sgl-project/sglang/main/README.md> accessed 2026-10-07; A4-S147: RadixArk Launches with \$100 Million in Seed Funding Led by Accel to Grow SGLang (Business Wire, May 2026) (RadixArk (Business Wire)) <https://www.morningstar.com/news/business-wire/20260505077157/radixark-launches-with-100-million-in-seed-funding-led-by-accel-to-grow-sglang-and-democratize-frontier-ai-infrastructure> accessed 2026-10-07 [search-tool extract]; V1-S063: RadixArk Launches with \$100 Million in Seed Funding Led by Accel (RadixArk (Business Wire)) <https://www.morningstar.com/news/business-wire/20260505077157/radixark-launches-with-100-million-in-seed-funding-led-by-accel-to-grow-sglang-and-democratize-frontier-ai-infrastructure> accessed 2026-10-08 [search-tool extract]

⁷⁵² Verified fact — A4-S010: PyPI JSON metadata: sglang (Python Package Index) <https://pypi.org/pypi/sglang/json> accessed 2026-10-07

⁷⁵³ Verified fact — A4-S065: SGLang README (GitHub) (SGLang project (LMSYS)) <https://raw.githubusercontent.com/sgl-project/sglang/main/README.md> accessed 2026-10-07

⁷⁵⁴ Verified fact — A4-S128: SGLang LICENSE (SGLang project) <https://raw.githubusercontent.com/sgl-project/sglang/main/LICENSE> accessed 2026-10-07; A4-S010: PyPI JSON metadata: sglang (Python Package Index) <https://pypi.org/pypi/sglang/json> accessed 2026-10-07

⁷⁵⁵ Verified fact — B-L2-S009: SGLang multimodal generation module unauthenticated RCE via ZMQ broker (CVE-2026-3059) - GitHub Advisory Database <https://github.com/advisories/GHSA-rgq9-fqf5-fv58> accessed 2026-10-08 [search-tool extract]; B-REVC-S002: NVD CVE-2026-3059 (with OSV and GitHub Advisory Database GHSA-rgq9-fqf5-fv58) (NIST NVD) <https://nvd.nist.gov/vuln/detail/CVE-2026-3059> accessed 2026-10-08 [search-tool extract]

⁷⁵⁶ Verified fact — A4-S010: PyPI JSON metadata: sglang (Python Package Index) <https://pypi.org/pypi/sglang/json> accessed 2026-10-07; A4-S147: RadixArk Launches with \$100 Million in Seed Funding Led by Accel to Grow SGLang (Business Wire, May 2026) (RadixArk (Business Wire)) <https://www.morningstar.com/news/business-wire/20260505077157/radixark-launches-with-100-million-in-seed-funding-led-by-accel-to-grow-sglang-and-democratize-frontier-ai-infrastructure> accessed 2026-10-07 [search-tool extract]

⁷⁵⁷ Verified fact — A4-S010: PyPI JSON metadata: sglang (Python Package Index) <https://pypi.org/pypi/sglang/json> accessed 2026-10-07

- The engine port could be reachable beyond a locked-down serving namespace ^{AJ}
- You need a vendor-supported distribution today ^{AJ}

Nearest competitors: L2-vllm, TensorRT-LLM, L2-nvidia-dynamo

Regulated-FS note. Qualify as the alternative engine in the exit plan; isolate internal ports and confirm CVE-2026-3059 status for the deployed version before use ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	SGLang open-source project hosted by LMSYS (non-profit); commercial steward RadixArk (founded by SGLang creators Ying Sheng and Banghua Zhu; US\$100M seed, May 2026)	Verified fact	A4-S065, A4-S147
Category	Open-source inference/serving framework for LLMs, VLMs and diffusion models; RL rollout backend	Verified fact	A4-S065
Version / lineup	sclang 0.5.21 (1 October 2026); 21 releases in 2026	Verified fact	A4-S010
Licence	Open source, Apache-2.0	Verified fact	A4-S010, A4-S128
Status events	5 May 2026: RadixArk launches with US\$100M seed (Accel lead, Spark co-lead; NVIDIA, AMD participate) at US\$400M post-money to steward SGLang and build a commercial platform	Verified fact	A4-S147, V1-S063
Strategic direction	Optimised for agentic workloads, RL rollouts and large-scale serving: RadixAttention prefix caching, PD disaggregation, speculative decoding (DFlash, Spec V2, June 2026), FP4/FP8 quantisation; TPU support with Google and RadixArk (July 2026); day-0 support for new open models (e.g. DeepSeek-V4, Kimi K3)	Verified fact	A4-S010, A4-S065
What it does	High-performance serving framework delivering low-latency, high-throughput inference from a single GPU to large clusters, OpenAI-API compatible, with broad model and hardware support (NVIDIA, AMD, Intel Xeon, Google TPU, Ascend).	Verified fact	A4-S010
Stack position	Serving engine (L2); natively supported by Hugging Face Inference Endpoints and as a Dynamo backend	Verified fact	A4-S081, A4-S090
Integration	OpenAI-compatible API; Python runtime; Docker	Verified	A4-S010,

Field	Value	Label	Sources
		d fact	A4-S065
Dependencies	Python >=3.10; accelerators; Docker image lmsysorg/sglang	Verified fact	A4-S010, A4-S065
Certifications	Not applicable (open-source software)	Architectural judgment	
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Free (Apache-2.0)	Verified fact	A4-S010
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	RL frameworks Miles (RadixArk), slime, AReaL, Tunix, verl; Dynamo; Hugging Face Endpoints	Verified fact	A4-S065, A4-S090, A4-S081
Adoption signals	Project states it powers over 400,000 GPUs worldwide	Verified fact	A4-S010
Maturity	First release January 2024; still 0.x; frequent releases	Verified fact	A4-S010
Deployment	saas: No; managed_cloud: Yes (via third parties, e.g. Hugging Face Inference Endpoints); vpc_byoc: Yes; private_cloud: Yes; self_hosted: Yes; on_prem: Yes		

Hugging Face: Hub (models/datasets/Spaces/buckets/Jobs), Inference Providers (routed serverless inference), Inference Endpoints (managed dedicated deployments); Text Generation Inference (TGI) in maintenance mode (L2-hugging-face)

Tier: Strategic · **Flags:** Duplicated, Deprecated · **Original graphic label:** Hugging Face – models & APIs

Rationale: Strategic, conditional: as the governed open-weight supply source (Enterprise plan); Endpoints Tactical; Inference Providers not for client data

Criterion	Score	Rationale
Technical	4	Hub, router and dedicated serving with engine choice ^{vf758}
Enterprise readiness	4	SSO (SAML/OIDC), SCIM, RBAC and audit logs on Team/Enterprise (rule 7) ^{vf759}
Security and compliance	3	SOC 2 Type 2 covering Hub and Endpoints; no ISO 27001 found ^{vf760}
Deployment flexibility	3	SaaS plus managed endpoints on three clouds with PrivateLink; not BYOC ^{vf761}
Ecosystem	5	De-facto distribution standard; OpenAI-compatible router; many partners ^{vf762}
Reliability and maturity	3	Hub mature since 2020, but TGI archived and Endpoints instance deprecations ^{vf763}
Cost / TCO	4	Pass-through pricing; per-minute instance billing ^{vf764}
Lock-in / portability	4	Models portable, open engines, OpenAI-compatible routing ^{vf765}

⁷⁵⁸ Verified fact — A4-S075: Hugging Face Inference Providers <https://raw.githubusercontent.com/huggingface/hub-docs/main/docs/inference-providers/index.md> accessed 2026-10-07; A4-S081: About Inference Endpoints (Hugging Face) <https://raw.githubusercontent.com/huggingface/hf-endpoints-documentation/main/docs/source/about.md> accessed 2026-10-07

⁷⁵⁹ Verified fact — A4-S080: Hugging Face Single Sign-On (SSO) <https://raw.githubusercontent.com/huggingface/hub-docs/main/docs/hub/security-sso.md> accessed 2026-10-07; A4-S079: Hugging Face Audit Logs <https://raw.githubusercontent.com/huggingface/hub-docs/main/docs/hub/audit-logs.md> accessed 2026-10-07

⁷⁶⁰ Verified fact — A4-S074: Hugging Face Hub Security <https://raw.githubusercontent.com/huggingface/hub-docs/main/docs/hub/security.md> accessed 2026-10-07; A4-S082: Inference Endpoints Security & Compliance (Hugging Face) <https://raw.githubusercontent.com/huggingface/hf-endpoints-documentation/main/docs/source/guides/security.md> accessed 2026-10-07

⁷⁶¹ Verified fact — A4-S083: Inference Endpoints Pricing (Hugging Face) <https://raw.githubusercontent.com/huggingface/hf-endpoints-documentation/main/docs/source/support/pricing.md> accessed 2026-10-07; A4-S082: Inference Endpoints Security & Compliance (Hugging Face) <https://raw.githubusercontent.com/huggingface/hf-endpoints-documentation/main/docs/source/guides/security.md> accessed 2026-10-07

⁷⁶² Verified fact — A4-S075: Hugging Face Inference Providers <https://raw.githubusercontent.com/huggingface/hub-docs/main/docs/inference-providers/index.md> accessed 2026-10-07

⁷⁶³ Verified fact — A4-S014: PyPI JSON metadata: huggingface-hub (Python Package Index) <https://pypi.org/pypi/huggingface-hub/json> accessed 2026-10-07; V1-S054: GitHub repository metadata: huggingface/text-generation-inference (archived) (GitHub (via GitHub MCP search_repositories)) <https://api.github.com/repos/huggingface/text-generation-inference> accessed 2026-10-08 [search-tool extract]; A4-S083: Inference Endpoints Pricing (Hugging Face) <https://raw.githubusercontent.com/huggingface/hf-endpoints-documentation/main/docs/source/support/pricing.md> accessed 2026-10-07

⁷⁶⁴ Verified fact — A4-S076: Inference Providers Pricing and Billing (Hugging Face) <https://raw.githubusercontent.com/huggingface/hub-docs/main/docs/inference-providers/pricing.md> accessed 2026-10-07; A4-S083: Inference Endpoints Pricing (Hugging Face) <https://raw.githubusercontent.com/huggingface/hf-endpoints-documentation/main/docs/source/support/pricing.md> accessed 2026-10-07

⁷⁶⁵ Verified fact — A4-S076: Inference Providers Pricing and Billing (Hugging Face) <https://raw.githubusercontent.com/huggingface/hub-docs/main/docs/inference-providers/pricing.md> accessed 2026-10-07; A4-S075: Hugging Face Inference Providers <https://raw.githubusercontent.com/huggingface/hub-docs/main/docs/inference-providers/index.md> accessed 2026-10-07

Criterion	Score	Rationale
Total (generic / FS)	3.70 / 3.60	

Capabilities. Four products under one tile: the Hub (model and dataset distribution with SSO, SCIM, RBAC, audit logs and EU storage on Team/Enterprise), Inference Providers (OpenAI-compatible router to partner clouds with pass-through pricing), Inference Endpoints (managed dedicated serving on AWS, Azure or GCP with vLLM, SGLang, TGI, llama.cpp or TEI, SOC 2 Type 2, AWS PrivateLink) and TGI (maintenance mode; repository archived 21 March 2026) ^{vf766}.

Strengths

- The de-facto open-weight distribution point, with enterprise identity controls ^{vf767} ^{AJ}
- Endpoints let you choose the engine and the cloud region ^{vf768}
- No markup on routed provider calls ^{vf769}

Limitations and risks

- TGI archived; Endpoints customers on TGI must migrate ^{vf770}
- Inference Providers inherits each partner's security and retention ^{vf771}
- BAA not explicitly tied to Inference Endpoints; no ISO 27001 found ^{vf772}

Choose when

- You need a governed source of open weights and a managed dedicated endpoint in a chosen cloud region ^{AJ}

Avoid when

⁷⁶⁶ Verified fact — A4-S074: Hugging Face Hub Security <https://raw.githubusercontent.com/huggingface/hub-docs/main/docs/hub/security.md> accessed 2026-10-07; A4-S075: Hugging Face Inference Providers <https://raw.githubusercontent.com/huggingface/hub-docs/main/docs/inference-providers/index.md> accessed 2026-10-07; A4-S080: Hugging Face Single Sign-On (SSO) <https://raw.githubusercontent.com/huggingface/hub-docs/main/docs/hub/security-ss0.md> accessed 2026-10-07; A4-S081: About Inference Endpoints (Hugging Face) <https://raw.githubusercontent.com/huggingface/hf-endpoints-documentation/main/docs/source/about.md> accessed 2026-10-07; A4-S082: Inference Endpoints Security & Compliance (Hugging Face) <https://raw.githubusercontent.com/huggingface/hf-endpoints-documentation/main/docs/source/guides/security.md> accessed 2026-10-07; V1-S054: GitHub repository metadata: huggingface/text-generation-inference (archived) (GitHub (via GitHub MCP search_repositories)) <https://api.github.com/repos/huggingface/text-generation-inference> accessed 2026-10-08 [search-tool extract]

⁷⁶⁷ Verified fact — A4-S075: Hugging Face Inference Providers <https://raw.githubusercontent.com/huggingface/hub-docs/main/docs/inference-providers/index.md> accessed 2026-10-07; A4-S080: Hugging Face Single Sign-On (SSO) <https://raw.githubusercontent.com/huggingface/hub-docs/main/docs/hub/security-ss0.md> accessed 2026-10-07

⁷⁶⁸ Verified fact — A4-S081: About Inference Endpoints (Hugging Face) <https://raw.githubusercontent.com/huggingface/hf-endpoints-documentation/main/docs/source/about.md> accessed 2026-10-07; A4-S083: Inference Endpoints Pricing (Hugging Face) <https://raw.githubusercontent.com/huggingface/hf-endpoints-documentation/main/docs/source/support/pricing.md> accessed 2026-10-07

⁷⁶⁹ Verified fact — A4-S076: Inference Providers Pricing and Billing (Hugging Face) <https://raw.githubusercontent.com/huggingface/hub-docs/main/docs/inference-providers/pricing.md> accessed 2026-10-07

⁷⁷⁰ Verified fact — V1-S054: GitHub repository metadata: huggingface/text-generation-inference (archived) (GitHub (via GitHub MCP search_repositories)) <https://api.github.com/repos/huggingface/text-generation-inference> accessed 2026-10-08 [search-tool extract]

⁷⁷¹ Verified fact — A4-S077: Inference Providers Security & Compliance (Hugging Face) <https://raw.githubusercontent.com/huggingface/hub-docs/main/docs/inference-providers/security.md> accessed 2026-10-07

⁷⁷² Verified fact — V1-S097: Inference Endpoints Security & Compliance; SOC 2/ISO 27001 compliance guide; Hub security (Hugging Face) <https://huggingface.co/docs/inference-endpoints/en/security> accessed 2026-10-08 [search-tool extract]

- Routing client data through Inference Providers to partners you have not assessed ^{AJ}

Nearest competitors: L2-together-ai, L2-fireworks-ai, Amazon SageMaker / Azure ML model catalogues

Regulated-FS note. Use the Hub (Enterprise, EU storage) as the governed model supply with scanning and an internal mirror; use Endpoints only in an approved region with PrivateLink ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Hugging Face	Verified fact	A4-S074
Category	Model hub plus inference router and managed dedicated inference service	Verified fact	A4-S075, A4-S081
Version / lineup	huggingface_hub client 2.1.1 (1 October 2026); Inference Endpoints natively supports vLLM, TGI, SGLang, llama.cpp and TEI engines	Verified fact	A4-S014, A4-S081
Licence	Hub client and TGI Apache-2.0; Hub, Inference Providers and Inference Endpoints are proprietary hosted services	Verified fact	A4-S014, A4-S072, A4-S081
Status events	TGI in maintenance mode 'as of 12/11/2025' (only minor bug fixes, docs and lightweight maintenance); Endpoints: AWS H100 instances marked 'Deprecated from December 2025'; 21 March 2026: TGI GitHub repository archived (read-only); latest release v3.3.7; Hugging Face recommends vLLM or SGLang	Verified fact	A4-S073, A4-S072, A4-S083, V1-S054, V1-S055
Strategic direction	TGI moved to maintenance mode; Hugging Face recommends vLLM and SGLang (and llama.cpp/MLX locally). Inference Providers routes to partner clouds (e.g. Cerebras, Groq, Together, Fireworks, Baseten, Nscale, OVHcloud, Scaleway, Z.ai) with pass-through pricing	Verified fact	A4-S072, A4-S073, A4-S075, A4-S076
What it does	Hub hosts and versions models/datasets with private repos, tokens, resource groups and malware/pickle/secrets scanning. Inference Providers gives one token and API across many third-party providers	Verified fact	A4-S074, A4-S075, A4-S081,

Field	Value	Label	Sources
	(OpenAI-compatible, provider selectable by suffix). Inference Endpoints deploys a chosen Hub model on a chosen engine to managed AWS/Azure/GCP instances billed per minute.		A4-S083
Stack position	Spans model distribution (L1 adjacency), routing/aggregation (gateway-like, L2/C1) and managed serving (L2). The single tile conflates three products	Verified fact	A4-S075, A4-S081
Integration	huggingface_hub (Python) and JS clients, hf CLI; OpenAI-compatible router; Endpoints REST	Verified fact	A4-S014, A4-S075
Dependencies	Endpoints run on AWS, Azure or GCP instances; Inference Providers depend on partner providers' own security and retention	Verified fact	A4-S083, A4-S077
Certifications	SOC 2 Type 2 (Hub and Inference Endpoints); GDPR compliant; BAA and GDPR DPA via Enterprise plan	Verified fact	A4-S074, A4-S082, V1-S097
GDPR / residency	Storage regions US and EU for Team/Enterprise orgs (APAC, GCC coming); Inference Providers does not store request/response bodies, logs kept up to 30 days; Endpoints do not store payloads, logs kept 30 days	Verified fact	A4-S078, A4-S077, A4-S082
Security features	TLS in transit; AWS PrivateLink for Endpoints; malware, pickle and secrets scanning; resource groups	Verified fact	A4-S082, A4-S074
Access controls	SSO (SAML 2.0/OIDC) on Team & Enterprise, Managed SSO on Enterprise Plus, SCIM, RBAC, audit logs (Team & Enterprise)	Verified fact	A4-S080, A4-S079, A4-S082
Enterprise support	Team, Enterprise and Enterprise Plus plans	Verified fact	A4-S080
Pricing	Inference Providers: provider rates passed through with no HF markup; PRO and Team/Enterprise seats include US\$2.00/month compute credits. Endpoints: hourly instance rates billed per minute, e.g. AWS L4 x1 US\$0.80/h, AWS H200 x1 US\$5/h	Verified fact	A4-S076, A4-S083
Infrastructure cost	Endpoint cost = instance-hours (GPU type and count) billed per minute	Verified fact	A4-S083
Ecosystem	Partners: Baseten, Cerebras, Cohere, DeepInfra, Fal, Featherless, Fireworks, Groq, Novita, Nscale, OVHcloud, Public AI, Replicate, Scaleway, Together, WaveSpeed, Z.ai; engines vLLM, SGLang, llama.cpp, TEI	Verified fact	A4-S075, A4-S081
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Hub client since 2020; TGI superseded by vLLM/SGLang	Verified fact	A4-S014, A4-S073
Deployment	saas: Yes (Hub, Inference Providers); managed_cloud: Yes (Inference Endpoints on AWS/Azure/GCP); vpc_byoc: No (AWS PrivateLink private connectivity offered; not BYOC); private_cloud: Not publicly verified; self_hosted: Yes for open-source engines (TGI) and client libraries only; on_prem: Not publicly verified		

Together AI (l2-together-ai)

Tier: Tactical · **Flags:** none · **Original graphic label:** Together AI – open-source cloud

Rationale: Capable open-model cloud; unsafe defaults (ZDR off) and partial RBAC keep it Tactical

Criterion	Score	Rationale
Technical	4	Serverless, dedicated, PTU, fine-tuning, GPU clusters ^{vf773}
Enterprise readiness	3	SSO on Scale/Enterprise; partial RBAC; audit trails for GPU clusters only (rule 7) ^{vf774}
Security and compliance	3	SOC 2 Type 2 and ISO 27001 vendor-stated without product scope: one below anchor (rule 8) ^{vf775}
Deployment flexibility	4	SaaS, dedicated, VPC deployments incl. EU, reserved capacity; no self-host ^{vf776}
Ecosystem	3	Hugging Face partner; NVIDIA partner ^{vf777}
Reliability and maturity	3	GA services; PTU SLA but serverless none; RBAC still rolling out ^{vf778}
Cost / TCO	4	Transparent per-token and per-hour pricing; one H100 price conflict ^{vf779}
Lock-in / portability	4	Open-weight models and standard usage; contractual capacity ^{vf780}
Total (generic / FS)	3.50 / 3.50	

Capabilities. AI-native cloud: serverless and dedicated inference for open-weight models, Provisioned Throughput with SLA, fine-tuning and GPU clusters; SOC 2 Type 2 and ISO 27001:2022 (vendor-stated); ZDR available but off by default; EU dedicated endpoints only on Scale and Enterprise; US\$800M Series C on 1 July 2026 ^{vf781}.

⁷⁷³ Verified fact — A4-S131: Pricing - Together AI docs (serverless, dedicated endpoints, GPU clusters) <https://docs.together.ai/docs/inference/pricing> accessed 2026-10-07 [search-tool extract]

⁷⁷⁴ Verified fact — A4-S151: Single sign-on (SSO) and Roles & permissions (RBAC) - Together AI docs <https://docs.together.ai/docs/sso> accessed 2026-10-07 [search-tool extract]

⁷⁷⁵ Verified fact — A4-S129: Powering Secure AI: Together AI Achieves SOC 2 Type 2 Compliance (and product pages) <https://www.together.ai/blog/soc-2-compliance> accessed 2026-10-07 [search-tool extract]; V1-S079: Together AI Trust Center; ISO 27001:2022 certification blog; SOC 2 Type 2 blog <https://trust.together.ai/> accessed 2026-10-08 [search-tool extract]

⁷⁷⁶ Verified fact — A4-S130: Zero data retention / Privacy and security - Together AI docs (and EU data centers support article) <https://docs.together.ai/docs/zero-data-retention> accessed 2026-10-07 [search-tool extract]; A4-S131: Pricing - Together AI docs (serverless, dedicated endpoints, GPU clusters) <https://docs.together.ai/docs/inference/pricing> accessed 2026-10-07 [search-tool extract]

⁷⁷⁷ Verified fact — A4-S075: Hugging Face Inference Providers <https://raw.githubusercontent.com/huggingface/hub-docs/main/docs/inference-providers/index.md> accessed 2026-10-07; A4-S132: Together AI Raises \$800 Million at \$8.3 Billion Valuation (press release, 1 Jul 2026; TechCrunch coverage) (Together AI (Business Wire)) <https://www.businesswire.com/news/home/20260701243402/en/Together-AI-Raises-00-Million-at-.3-Billion-Valuation-to-Make-Frontier-AI-Accessible-to-All> accessed 2026-10-07 [search-tool extract]

⁷⁷⁸ Verified fact — A4-S131: Pricing - Together AI docs (serverless, dedicated endpoints, GPU clusters) <https://docs.together.ai/docs/inference/pricing> accessed 2026-10-07 [search-tool extract]; A4-S151: Single sign-on (SSO) and Roles & permissions (RBAC) - Together AI docs <https://docs.together.ai/docs/sso> accessed 2026-10-07 [search-tool extract]

⁷⁷⁹ Verified fact — A4-S131: Pricing - Together AI docs (serverless, dedicated endpoints, GPU clusters) <https://docs.together.ai/docs/inference/pricing> accessed 2026-10-07 [search-tool extract]

⁷⁸⁰ Verified fact — A4-S131: Pricing - Together AI docs (serverless, dedicated endpoints, GPU clusters) <https://docs.together.ai/docs/inference/pricing> accessed 2026-10-07 [search-tool extract]

Strengths

- Broad open-model catalogue plus dedicated and reserved capacity ^{VF782}
- EU and US cluster sites ^{VF783}

Limitations and risks

- Default storage of prompts and responses, with possible product-improvement use, unless ZDR is enabled ^{VF784}
- Serverless has no region selection ^{VF785}
- Product-level RBAC still being rolled out ^{VF786}
- HIPAA wording inconsistent; customer BAA not confirmed ^{VF787}

Choose when

- Open-weight models on dedicated EU endpoints with ZDR on, under a Scale or Enterprise contract ^{AJ}

Avoid when

- Serverless use with client data, or any use with ZDR left at default ^{AJ}

Nearest competitors: L2-fireworks-ai, L2-hugging-face, Hyperscaler model catalogues (Bedrock, Foundry, Vertex)

Regulated-FS note. Contract ZDR on, EU dedicated endpoints and the PTU SLA; confirm certification scope from the trust centre ^{ROC}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Together Computer, Inc. (trading as Together AI); privately held	Verified fact	A4-S132
Category	AI-native cloud: serverless and dedicated inference for open-weight	Verified	A4-S131,

⁷⁸¹ Verified fact — A4-S131: Pricing - Together AI docs (serverless, dedicated endpoints, GPU clusters) <https://docs.together.ai/docs/inference/pricing> accessed 2026-10-07 [search-tool extract]; A4-S129: Powering Secure AI: Together AI Achieves SOC 2 Type 2 Compliance (and product pages) <https://www.together.ai/blog/soc-2-compliance> accessed 2026-10-07 [search-tool extract]; A4-S130: Zero data retention / Privacy and security - Together AI docs (and EU data centers support article) <https://docs.together.ai/docs/zero-data-retention> accessed 2026-10-07 [search-tool extract]; V1-S079: Together AI Trust Center; ISO 27001:2022 certification blog; SOC 2 Type 2 blog <https://trust.together.ai/> accessed 2026-10-08 [search-tool extract]; V1-S057: Together AI Raises \$800 Million at \$8.3 Billion Valuation (Together AI (Business Wire)) [https://www.businesswire.com/news/home/20260701243402/en/Together-AI-Raises-\\$800-Million-at-\\$8.3-Billion-Valuation-to-Make-Frontier-AI-Accessible-to-All](https://www.businesswire.com/news/home/20260701243402/en/Together-AI-Raises-$800-Million-at-$8.3-Billion-Valuation-to-Make-Frontier-AI-Accessible-to-All) accessed 2026-10-08 [search-tool extract]

⁷⁸² Verified fact — A4-S131: Pricing - Together AI docs (serverless, dedicated endpoints, GPU clusters) <https://docs.together.ai/docs/inference/pricing> accessed 2026-10-07 [search-tool extract]

⁷⁸³ Verified fact — A4-S130: Zero data retention / Privacy and security - Together AI docs (and EU data centers support article) <https://docs.together.ai/docs/zero-data-retention> accessed 2026-10-07 [search-tool extract]

⁷⁸⁴ Verified fact — A4-S130: Zero data retention / Privacy and security - Together AI docs (and EU data centers support article) <https://docs.together.ai/docs/zero-data-retention> accessed 2026-10-07 [search-tool extract]

⁷⁸⁵ Verified fact — A4-S130: Zero data retention / Privacy and security - Together AI docs (and EU data centers support article) <https://docs.together.ai/docs/zero-data-retention> accessed 2026-10-07 [search-tool extract]

⁷⁸⁶ Verified fact — A4-S151: Single sign-on (SSO) and Roles & permissions (RBAC) - Together AI docs <https://docs.together.ai/docs/ssu> accessed 2026-10-07 [search-tool extract]

⁷⁸⁷ Verified fact — V1-S079: Together AI Trust Center; ISO 27001:2022 certification blog; SOC 2 Type 2 blog <https://trust.together.ai/> accessed 2026-10-08 [search-tool extract]

Field	Value	Label	Sources
	models, fine-tuning, GPU clusters, provisioned throughput	d fact	A4-S093
Version / lineup	Official SDK together 2.40.0 (7 October 2026); products: serverless inference, dedicated endpoints, Provisioned Throughput (PTU), fine-tuning, GPU clusters (on-demand, reserved, Instant Clusters), container deployments ('Jig', beta)	Verified fact	A4-S093, A4-S131
Licence	Proprietary cloud service; official Python SDK Apache-2.0	Verified fact	A4-S093
Status events	February 2025: Series B US\$305M at US\$3.3B; 1 July 2026: Series C US\$800M at US\$8.3B post-money (Aramco Ventures lead)	Verified fact	A4-S132, V1-S057
Strategic direction	Raised US\$800M Series C (1 July 2026) to scale its 'neocloud'; reports annual bookings above US\$1.15B; adding Provisioned Throughput with SLA and EU dedicated capacity	Verified fact	A4-S132, A4-S131, A4-S130
What it does	Hosts open-weight models behind serverless per-token APIs and per-minute dedicated endpoints, offers reserved token capacity (PTU), fine-tuning, and on-demand or reserved NVIDIA GPU clusters (H100 to GB300).	Verified fact	A4-S131, A4-S093
Stack position	Managed inference cloud plus GPU IaaS (L2 and below); routers such as OpenRouter and Hugging Face Inference Providers sit in front of it	Architectural judgement	
Integration	Official Python SDK and CLI; Hugging Face Inference Providers partner	Verified fact	A4-S093, A4-S075
Dependencies	Together-operated data centres (cluster sites listed in US and Europe: France, Netherlands, Sweden, Romania)	Verified fact	A4-S130
Certifications	SOC 2 Type 2 and ISO 27001:2022 (vendor blog and product pages); HIPAA described inconsistently ('compliant', 'HIPAA-aligned', 'HIPAA-ready') with BAAs mentioned; report scope/dates not seen	Verified fact	A4-S129, V1-S079
GDPR / residency	ZDR available at organisation level but OFF by default: by default prompts and responses are stored and may be used for product improvement unless ZDR is enabled; training on customer data is opt-in (off by default); metadata retained for billing. Serverless has no region selection; EU dedicated endpoints only on Scale and Enterprise plans	Verified fact	A4-S130
Security features	Encryption in transit and at rest and audit logging cited for HIPAA programme; private networking/VPC deployments	Verified fact	A4-S129, A4-S130
Access controls	SSO (SAML/OIDC; Okta, Entra, Google Workspace, JumpCloud) on Scale and Enterprise plans; org/project RBAC (admin/developer) but product-level RBAC for fine-tuning, endpoints and serverless 'still being rolled out'; per-user audit trails for GPU clusters	Verified fact	A4-S151
Enterprise support	Scale and Enterprise plans; PTUs carry an uptime SLA, serverless does not	Verified fact	A4-S130, A4-S131
Pricing	Serverless per 1M tokens (in/out), e.g. GPT-OSS 120B US\$0.15/US\$0.60, Llama 3.3 70B Turbo US\$1.04/US\$1.04, Kimi K3 US\$3.00/US\$15.00; dedicated H100 US\$3.99/h in docs vs US\$5.49/h on pricing page (conflict); GPU clusters on-demand H100 US\$3.99, B200 US\$8.19 per GPU-hour; PTU US\$0.05 per PTU-minute	Verified fact	A4-S131
Infrastructure cost	Dedicated endpoints bill per running replica-minute regardless of traffic; serverless cheaper for bursty load	Verified fact	A4-S131

Field	Value	Label	Sources
Ecosystem	Hugging Face Inference Providers partner; NVIDIA partner	Verified fact	A4-S075, A4-S132
Adoption signals	US\$800M Series C at US\$8.3B (July 2026); annual bookings reported above US\$1.15B	Verified fact	A4-S132, V1-S057
Maturity	SDK first released April 2023; company funded through Series C	Verified fact	A4-S093, A4-S132
Deployment	saas: Yes (serverless); managed_cloud: Yes (dedicated endpoints, GPU clusters); vpc_byoc: Yes (private networking and VPC-based deployments, incl. EU regions); private_cloud: Yes (dedicated / reserved capacity); self_hosted: No; on_prem: Not publicly verified		

llm-d (L2-llm-d)

Tier: Experimental · **Flags:** none · **Original graphic label:** not in graphic

Rationale: The neutral-governance option for the optimisation sub-layer, but pre-1.0 and only Technology Preview support

Criterion	Score	Rationale
Technical	4	Prefix-cache-aware routing, tiered KV cache, PD disaggregation, wide EP ^{vf788}
Enterprise readiness	3	Rule 2: inherits Kubernetes controls; Red Hat offer is Technology Preview, so the 4 cap applies; 3 ^{vf789}
Security and compliance	2	Rule 2 hygiene: security policy, CVE handling and signing not verified ^{NPV}
Deployment flexibility	4	Self-host, VPC, private cloud, on-prem on Kubernetes; no SaaS ^{vf790}
Ecosystem	4	vLLM, SGLang, KServe, Gateway API Inference Extension; broad founding coalition ^{vf791}
Reliability and	2	CNCF sandbox, v0.7; Technology Preview support ^{vf792}

⁷⁸⁸ Verified fact — A4-S089: llm-d README (GitHub) (llm-d project) <https://raw.githubusercontent.com/llm-d/llm-d/main/README.md> accessed 2026-10-07

⁷⁸⁹ Verified fact — B-L2-S002: Supported platforms for Distributed Inference with llm-d (Red Hat AI 3) https://docs.redhat.com/en/documentation/red_hat_ai/3/html/supported_product_and_hardware_configurations/llmd-supported-platforms_supported-configurations accessed 2026-10-08 [search-tool extract]

⁷⁹⁰ Verified fact — A4-S089: llm-d README (GitHub) (llm-d project) <https://raw.githubusercontent.com/llm-d/llm-d/main/README.md> accessed 2026-10-07

⁷⁹¹ Verified fact — A4-S089: llm-d README (GitHub) (llm-d project) <https://raw.githubusercontent.com/llm-d/llm-d/main/README.md> accessed 2026-10-07; A4-S062: Introducing the next generation of AI inference, powered by llm-d (Google Cloud) <https://cloud.google.com/blog/products/ai-machine-learning/enhancing-vllm-for-distributed-inference-with-llm-d> accessed 2026-10-07

⁷⁹² Verified fact — A4-S089: llm-d README (GitHub) (llm-d project) <https://raw.githubusercontent.com/llm-d/llm-d/main/README.md> accessed 2026-10-07; B-L2-S002: Supported platforms for Distributed Inference with llm-d (Red Hat AI 3) https://docs.redhat.com/en/documentation/red_hat_ai/3/html/supported_product_and_hardware_configurations/llmd-supported-platforms_supported-configurations accessed 2026-10-08 [search-tool extract]

Criterion	Score	Rationale
maturity		
Cost / TCO	3	Free; heavy Kubernetes and GPU operations ^{AJ}
Lock-in / portability	5	Apache-2.0 under CNCF neutral governance ^{VF793}
Total (generic / FS)	3.30	
	/	
	3.35	

Evidence rules applied: enterprise_readiness capped at 4 (rule 2, support is Technology Preview only); not binding at 3

Capabilities. Apache-2.0 Kubernetes-native distributed inference stack above vLLM and SGLang: prefix-cache- and load-aware routing, tiered KV-cache management, PD disaggregation, wide expert parallelism and SLO-aware autoscaling; CNCF sandbox project (March 2026) founded by Red Hat, Google Cloud, IBM Research, CoreWeave and NVIDIA; v0.7 in May 2026 ^{VF794}. Red Hat supports it only as a Technology Preview ^{VF795}.

Strengths

- Neutral governance (CNCF) and multi-vendor founders ^{VF796}
- Builds on the Kubernetes Gateway API Inference Extension rather than a private router ^{VF797}

Limitations and risks

- Pre-1.0 (v0.7) and sandbox stage ^{VF798}
- Vendor support is Technology Preview, not covered by production SLAs ^{VF799}
- Needs Kubernetes and data-centre accelerators ^{VF800}

Choose when

- Kubernetes is your platform standard and you are building a multi-model internal inference service ^{AJ}

⁷⁹³ Verified fact — A4-S089: llm-d README (GitHub) (llm-d project) <https://raw.githubusercontent.com/llm-d/llm-d/main/README.md> accessed 2026-10-07

⁷⁹⁴ Verified fact — A4-S089: llm-d README (GitHub) (llm-d project) <https://raw.githubusercontent.com/llm-d/llm-d/main/README.md> accessed 2026-10-07; A4-S062: Introducing the next generation of AI inference, powered by llm-d (Google Cloud) <https://cloud.google.com/blog/products/ai-machine-learning/enhancing-vllm-for-distributed-inference-with-llm-d> accessed 2026-10-07

⁷⁹⁵ Verified fact — B-L2-S002: Supported platforms for Distributed Inference with llm-d (Red Hat AI 3) https://docs.redhat.com/en/documentation/red_hat_ai/3/html/supported_product_and_hardware_configurations/llmd-supported-platforms_supported-configurations accessed 2026-10-08 [search-tool extract]

⁷⁹⁶ Verified fact — A4-S089: llm-d README (GitHub) (llm-d project) <https://raw.githubusercontent.com/llm-d/llm-d/main/README.md> accessed 2026-10-07

⁷⁹⁷ Verified fact — A4-S062: Introducing the next generation of AI inference, powered by llm-d (Google Cloud) <https://cloud.google.com/blog/products/ai-machine-learning/enhancing-vllm-for-distributed-inference-with-llm-d> accessed 2026-10-07

⁷⁹⁸ Verified fact — A4-S089: llm-d README (GitHub) (llm-d project) <https://raw.githubusercontent.com/llm-d/llm-d/main/README.md> accessed 2026-10-07

⁷⁹⁹ Verified fact — B-L2-S002: Supported platforms for Distributed Inference with llm-d (Red Hat AI 3) https://docs.redhat.com/en/documentation/red_hat_ai/3/html/supported_product_and_hardware_configurations/llmd-supported-platforms_supported-configurations accessed 2026-10-08 [search-tool extract]

⁸⁰⁰ Verified fact — A4-S089: llm-d README (GitHub) (llm-d project) <https://raw.githubusercontent.com/llm-d/llm-d/main/README.md> accessed 2026-10-07; B-L2-S002: Supported platforms for Distributed Inference with llm-d (Red Hat AI 3) https://docs.redhat.com/en/documentation/red_hat_ai/3/html/supported_product_and_hardware_configurations/llmd-supported-platforms_supported-configurations accessed 2026-10-08 [search-tool extract]

Avoid when

- You need production-supported software now ⁸⁰¹ _{AJ}
- Volumes do not justify multi-replica routing _{AJ}

Nearest competitors: L2-nvidia-dynamo, L2-vllm, KServe

Regulated-FS note. Pilot in non-production with Red Hat or GKE; revisit at 1.0 and at Red Hat GA support ^{Rec}.

Facts (as of 2026-10-07)

Field	Value	Label	Sources
Company	llm-d project (CNCF sandbox; founded by Red Hat, Google Cloud, IBM Research, CoreWeave and NVIDIA)	Verified fact	A4-S089, A4-S062
Category	Kubernetes-native distributed inference serving stack above model servers	Verified fact	A4-S089
Version / lineup	Not publicly verified	Not publicly verified	
Licence	Not publicly verified	Not publicly verified	
Status events	Not publicly verified	Not publicly verified	

⁸⁰¹ Verified fact — B-L2-S002: Supported platforms for Distributed Inference with llm-d (Red Hat AI 3)
https://docs.redhat.com/en/documentation/red_hat_ai/3/html/supported_product_and_hardware_configurations/llmd-supported-platforms_supported-configurations accessed 2026-10-08 [search-tool extract]

Field	Value	Label	Sources
Strategic direction	Launched 20 May 2025 (Google Cloud as founding contributor) building on vLLM and the Gateway API Inference Extension (GKE Inference Gateway)	Verified fact	A4-S062
What it does	Intelligent prefix-cache- and load-aware routing, tiered KV-cache management, PD disaggregation and wide expert parallelism, SLO-aware autoscaling and batch APIs, on top of vLLM/SGLang.	Verified fact	A4-S089
Stack position	Inference optimisation / routing sub-layer above serving engines (H1)	Verified fact	A4-S089
Integration	Not publicly verified	Not publicly verified	
Dependencies	Not publicly verified	Not publicly verified	
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	vLLM, SGLang, Kubernetes, KServe; partners incl. AMD, Cisco, Hugging Face, Intel, Lambda, Mistral AI	Verified fact	A4-S089, A4-S062
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Not publicly verified	Not publicly verified	
Deployment	saas: No; managed_cloud: Not publicly verified; vpc_byoc: Yes; private_cloud: Yes; self_hosted: Yes; on_prem: Yes		

OpenRouter (L2-openrouter)

Tier: Tactical • **Flags:** Acquired • **Original graphic label:** OpenRouter – multi-provider

Rationale: Useful router for evaluation; third-party SaaS with billing intermediation and a pending change of owner

Criterion	Score	Rationale
Technical	4	Routing, fallback, provider selection, budgets, allowlists, ZDR, regions ⁸⁰²
Enterprise readiness	4	SSO, SCIM, org roles on Enterprise; contractual SLA on Enterprise (rule 7 to 4) ⁸⁰³

⁸⁰² Verified fact — A4-S111: OpenRouter Docs: Guardrails <https://raw.githubusercontent.com/OpenRouterTeam/docs/main/guides/features/guardrails.md> accessed 2026-10-07; A4-S113: OpenRouter Docs: FAQ (pricing and fees) <https://raw.githubusercontent.com/OpenRouterTeam/docs/main/faq.md> accessed 2026-10-07

⁸⁰³ Verified fact — A4-S110: OpenRouter Docs: Single Sign-On <https://raw.githubusercontent.com/OpenRouterTeam/docs/main/guides/features/sso.md> accessed 2026-10-07; A4-S112: OpenRouter Docs: Enterprise Quickstart <https://raw.githubusercontent.com/OpenRouterTeam/docs/main/cookbook/get-started/enterprise-quickstart.md> accessed 2026-10-07; A4-S150: In-Region Routing: keep your data in the US or EU (OpenRouter blog) and OpenRouter Pricing/Enterprise pages <https://openrouter.ai/blog/announcements/us-in-region-routing/> accessed 2026-10-07 [search-tool extract]

Criterion	Score	Rationale
Security and compliance	3	SOC 2 Type 2 (report on request); no BAA; no ISO 27001 found ^{vr804}
Deployment flexibility	2	SaaS only, with EU/US processing regions on higher plans ^{vr805}
Ecosystem	4	OpenAI-compatible; many providers ^{vr806}
Reliability and maturity	3	Operating at scale but no published SLA figure; Stripe acquisition pending (rule 3 mention) ^{vr807}
Cost / TCO	3	Provider prices passed through plus 5.5-8% platform fee ^{vr808}
Lock-in / portability	2	OpenAI-compatible (3) but proprietary guardrails and billing; reduced by 1 for ownership change (rule 3) ^{vr809}
Total (generic / FS)	3.25 / 3.05	

Capabilities. Hosted multi-provider model router with budgets, model and provider allowlists, ZDR enforcement, EU/US in-region routing (Business and Enterprise), SSO and SCIM (Enterprise); SOC 2 Type 2, no HIPAA BAA; 5.5% fee on credit purchases; Stripe agreed to acquire it (announced 19 August 2026, pending as of 8 October 2026) ^{vr810}.

Strengths

- Fastest way to evaluate many models behind one OpenAI-compatible API ^{vr811 AJ}

⁸⁰⁴ Verified fact — A4-S142: OpenRouter Trust Center (and OpenRouter blog / Enterprise Access Agreement) <https://trust.openrouter.ai/> accessed 2026-10-07 [search-tool extract]

⁸⁰⁵ Verified fact — A4-S107: OpenRouter Docs: Data Collection (source of openrouter.ai/docs) <https://raw.githubusercontent.com/OpenRouterTeam/docs/main/guides/privacy/data-collection.md> accessed 2026-10-07; A4-S109: OpenRouter Docs: In-Region Routing (EU/US) <https://raw.githubusercontent.com/OpenRouterTeam/docs/main/guides/features/in-region-routing.md> accessed 2026-10-07

⁸⁰⁶ Verified fact — A4-S109: OpenRouter Docs: In-Region Routing (EU/US) <https://raw.githubusercontent.com/OpenRouterTeam/docs/main/guides/features/in-region-routing.md> accessed 2026-10-07; A4-S112: OpenRouter Docs: Enterprise Quickstart <https://raw.githubusercontent.com/OpenRouterTeam/docs/main/cookbook/get-started/enterprise-quickstart.md> accessed 2026-10-07

⁸⁰⁷ Verified fact — A4-S150: In-Region Routing: keep your data in the US or EU (OpenRouter blog) and OpenRouter Pricing/Enterprise pages <https://openrouter.ai/blog/announcements/us-in-region-routing/> accessed 2026-10-07 [search-tool extract]; V1-S059: Stripe agrees to acquire OpenRouter to help businesses optimize token routing and usage <https://stripe.com/newsroom/news/stripe-agrees-to-acquire-openrouter> accessed 2026-10-08 [search-tool extract]

⁸⁰⁸ Verified fact — V1-S062: Simplifying Our Platform Fee (OpenRouter) <https://openrouter.ai/blog/announcements/simplifying-our-platform-fee/> accessed 2026-10-08 [search-tool extract]; A4-S144: OpenRouter Business / plans (and pricing blog posts) <https://openrouter.ai/business> accessed 2026-10-07 [search-tool extract]

⁸⁰⁹ Verified fact — A4-S111: OpenRouter Docs: Guardrails <https://raw.githubusercontent.com/OpenRouterTeam/docs/main/guides/features/guardrails.md> accessed 2026-10-07; V1-S059: Stripe agrees to acquire OpenRouter to help businesses optimize token routing and usage <https://stripe.com/newsroom/news/stripe-agrees-to-acquire-openrouter> accessed 2026-10-08 [search-tool extract]

⁸¹⁰ Verified fact — A4-S111: OpenRouter Docs: Guardrails <https://raw.githubusercontent.com/OpenRouterTeam/docs/main/guides/features/guardrails.md> accessed 2026-10-07; A4-S109: OpenRouter Docs: In-Region Routing (EU/US) <https://raw.githubusercontent.com/OpenRouterTeam/docs/main/guides/features/in-region-routing.md> accessed 2026-10-07; A4-S110: OpenRouter Docs: Single Sign-On <https://raw.githubusercontent.com/OpenRouterTeam/docs/main/guides/features/sso.md> accessed 2026-10-07; A4-S142: OpenRouter Trust Center (and OpenRouter blog / Enterprise Access Agreement) <https://trust.openrouter.ai/> accessed 2026-10-07 [search-tool extract]; A4-S144: OpenRouter Business / plans (and pricing blog posts) <https://openrouter.ai/business> accessed 2026-10-07 [search-tool extract]; V1-S062: Simplifying Our Platform Fee (OpenRouter) <https://openrouter.ai/blog/announcements/simplifying-our-platform-fee/> accessed 2026-10-08 [search-tool extract]; V1-S059: Stripe agrees to acquire OpenRouter to help businesses optimize token routing and usage <https://stripe.com/newsroom/news/stripe-agrees-to-acquire-openrouter> accessed 2026-10-08 [search-tool extract]

- In-region routing fails rather than falling back out of region ^{vr⁸¹²}

Limitations and risks

- Third-party SaaS in the data path; Batch API, web search and fetch not region-resident ^{vr⁸¹³}
- Billing intermediation (5.5% Standard, 8% Business) ^{vr⁸¹⁴}
- Ownership change pending; deal value is press-reported only ^{vr⁸¹⁵}
- No numeric uptime target published ^{vr⁸¹⁶}

Choose when

- Model exploration and non-confidential workloads, behind the firm's gateway ^{aj}

Avoid when

- Client or personal data without ZDR and in-region routing contractually enforced ^{aj}
- As the firm's gateway of record (see C1) ^{aj}

Nearest competitors: L2-hugging-face, C1-litellm, C1-portkey

Regulated-FS note. Treat as a model-access source behind C1, not as C1; reassess terms and data-processing roles when the Stripe acquisition closes ^{rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	OpenRouter, Inc.; subject to a pending acquisition by Stripe (agreement announced 19 August 2026; not confirmed closed as of 8 October 2026)	Verified fact	A4-S143, V1-S059
Category	Hosted multi-provider model router / aggregator with gateway-style governance (budgets, allowlists, ZDR, regional routing)	Verified fact	A4-S111, A4-S109
Version / lineup	Official Python SDK openrouter 1.3.32 (7 October 2026; first release 13 November 2025)	Verified fact	A4-S096
Licence	Proprietary SaaS; SDK Apache-2.0	Verified fact	A4-S096
Status events	26 May 2026: US\$113M Series B led by CapitalG (company press release); 19 August 2026: Stripe signed an agreement to acquire OpenRouter (announced by both companies); closing expected "in the coming weeks", subject to customary conditions; no completion notice	Verified fact	V1-S061, V1-S059, V1-S060

⁸¹¹ Verified fact — A4-S113: OpenRouter Docs: FAQ (pricing and fees) <https://raw.githubusercontent.com/OpenRouterTeam/docs/main/faq.mdx> accessed 2026-10-07

⁸¹² Verified fact — A4-S109: OpenRouter Docs: In-Region Routing (EU/US) <https://raw.githubusercontent.com/OpenRouterTeam/docs/main/guides/features/in-region-routing.mdx> accessed 2026-10-07; A4-S150: In-Region Routing: keep your data in the US or EU (OpenRouter blog) and OpenRouter Pricing/Enterprise pages <https://openrouter.ai/blog/announcements/us-in-region-routing/> accessed 2026-10-07 [search-tool extract]

⁸¹³ Verified fact — A4-S150: In-Region Routing: keep your data in the US or EU (OpenRouter blog) and OpenRouter Pricing/Enterprise pages <https://openrouter.ai/blog/announcements/us-in-region-routing/> accessed 2026-10-07 [search-tool extract]

⁸¹⁴ Verified fact — V1-S062: Simplifying Our Platform Fee (OpenRouter) <https://openrouter.ai/blog/announcements/simplifying-our-platform-fee/> accessed 2026-10-08 [search-tool extract]; A4-S144: OpenRouter Business / plans (and pricing blog posts) <https://openrouter.ai/business> accessed 2026-10-07 [search-tool extract]

⁸¹⁵ Verified fact — V1-S059: Stripe agrees to acquire OpenRouter to help businesses optimize token routing and usage <https://stripe.com/newsroom/news/stripe-agrees-to-acquire-openrouter> accessed 2026-10-08 [search-tool extract]

⁸¹⁶ Verified fact — A4-S150: In-Region Routing: keep your data in the US or EU (OpenRouter blog) and OpenRouter Pricing/Enterprise pages <https://openrouter.ai/blog/announcements/us-in-region-routing/> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
	found as of 8 October 2026; price undisclosed (press reports range from over US\$7B to over US\$8B); OpenRouter says name, product and roadmap are unchanged		
Strategic direction	Enterprise features: workspaces, workspace budgets, guardrails (budget limits, model/provider allowlists, ZDR, data regions, prompt-injection and sensitive-info guardrails), SSO and SCIM group mappings, EU/US in-region routing	Verified fact	A4-S112, A4-S111, A4-S109
What it does	Single API to many models and providers with routing, fallbacks and provider selection; passes through provider pricing; lets organisations enforce ZDR, regions, budgets and model/provider allowlists per workspace, key or member.	Verified fact	A4-S113, A4-S108, A4-S111
Stack position	Model access router (L2) with control-plane features overlapping C1 (AI gateway) but as a third-party SaaS in the data path	Verified fact	A4-S111
Integration	OpenAI-compatible API, @openrouter/sdk and Python SDK; BYOK; Datadog/Braintrust broadcast	Verified fact	A4-S109, A4-S113, A4-S096
Dependencies	Underlying model providers; their data policies (OpenRouter tracks per-endpoint retention/training policy)	Verified fact	A4-S108
Certifications	SOC 2 Type 2 (trust centre; report on request); no HIPAA BAA offered	Verified fact	A4-S142
GDPR / residency	Prompts/responses not stored unless the customer opts in; metadata stored. ZDR enforceable globally, per model group, guardrail or request. EU/US in-region routing on Business and Enterprise plans; fails rather than falling back out-of-region; Batch API, web search and web fetch not region-resident	Verified fact	A4-S107, A4-S108, A4-S109, A4-S150
Security features	Guardrails for prompt injection, sensitive information and secret formats; ZDR does not cover plugins/tools such as web search	Verified fact	A4-S111, A4-S108
Access controls	SSO (Okta, Entra ID, Google Workspace, custom SAML) on Enterprise plans; SCIM group mappings; org admin roles	Verified fact	A4-S110, A4-S112
Enterprise support	Business plan (self-serve, no minimum) and Enterprise plan (contractual SLAs, support SLA, dedicated Slack, CSM, forward-deployed engineer); no numeric uptime target published; ~50-minute full outage in August 2025 noted by vendor	Verified fact	A4-S109, A4-S150
Pricing	Provider prices passed through without markup; platform fee on credit purchases: Standard 5.5% (US\$0.80 minimum), Business 8%, Enterprise custom; BYOK 5% of model cost after plan allowance	Verified fact	A4-S144, A4-S113, V1-S062
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Many providers; coding-agent launchers (ori) for Claude Code, Codex, OpenCode	Verified fact	A4-S112
Adoption signals	Series B US\$113M led by CapitalG (26 May 2026, company release), about US\$1.3B post-money (New York Times, reported); Series A US\$40M (June 2025); 25 trillion tokens per week (May 2026, company); Stripe acquisition agreement August 2026 (value reported, not	Verified fact	A4-S143, V1-S061, V1-S060

Field	Value	Label	Sources
Maturity	disclosed) Not publicly verified	Not publicly verified	
Deployment	saas: Yes; managed_cloud: Yes; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: No; on_prem: No		

NVIDIA Dynamo (L2-nvidia-dynamo)

Tier: Experimental · Flags: none · Original graphic label: not in graphic

Rationale: Clear H1 evidence and real capability, but beta packaging and month-long support windows make it unsuitable as a critical dependency today

Criterion	Score	Rationale
Technical	4	Disaggregation, KV-aware routing, KV offload, SLA planner ⁸¹⁷
Enterprise readiness	3	Rule 2: commercial support exists (NVAIE feature branch) ⁸¹⁸ ; inherits host controls; 3
Security and compliance	2	Rule 2 hygiene: security policy and CVE handling not verified ^{NPV} ; fixes only in new releases ⁸¹⁹
Deployment flexibility	4	Self-host on Kubernetes, VPC, on-prem; guides for EKS, GKE, AKS ⁸²⁰
Ecosystem	3	Three engine backends, GAIE plugin, LMCache ⁸²¹

⁸¹⁷ Verified fact — A4-S090: NVIDIA Dynamo README (GitHub) <https://raw.githubusercontent.com/ai-dynamo/dynamo/main/README.md> accessed 2026-10-07

⁸¹⁸ Verified fact — B-L2-S003: Dynamo Support Coverage (NVIDIA) <https://docs.nvidia.com/dynamo/dev/reference/enterprise/support-coverage> accessed 2026-10-08 [search-tool extract]

⁸¹⁹ Verified fact — B-L2-S003: Dynamo Support Coverage (NVIDIA) <https://docs.nvidia.com/dynamo/dev/reference/enterprise/support-coverage> accessed 2026-10-08 [search-tool extract]

⁸²⁰ Verified fact — A4-S090: NVIDIA Dynamo README (GitHub) <https://raw.githubusercontent.com/ai-dynamo/dynamo/main/README.md> accessed 2026-10-07

⁸²¹ Verified fact — A4-S090: NVIDIA Dynamo README (GitHub) <https://raw.githubusercontent.com/ai-dynamo/dynamo/main/README.md> accessed 2026-10-07; A4-S091: LMCache README (GitHub) (LMCache project) <https://raw.githubusercontent.com/LMCache/LMCache/dev/README.md> accessed 2026-10-07

Criterion	Score	Rationale
Reliability and maturity	2	Beta classifier; 1.0 claim conflicts; one-month support windows ^{vr822}
Cost / TCO	3	Free; heavy Kubernetes and GPU operations, subscription for support ^{vr823} ^{aj}
Lock-in / portability	3	Apache-2.0 but NVIDIA-led and NVIDIA-optimised ^{vr824}
Total (generic / FS)	3.10 / 3.00	

Capabilities. Apache-2.0 distributed inference orchestration layer above vLLM, SGLang and TensorRT-LLM: disaggregated serving, KV-aware routing, multi-tier KV block manager, SLA-based planner and autoscaling, a Kubernetes Gateway API Inference Extension plugin; ai-dynamo 1.5.1 (7 October 2026, PyPI classifier Beta) ^{vr825}. Enterprise support is an NVIDIA AI Enterprise feature branch ^{vr826}.

Strengths

- Explicit optimisation sub-layer that does not replace the engine ^{vr827}
- Engine-agnostic across three backends; integrates LMCACHE ^{vr828}

Limitations and risks

- Beta classifier on PyPI while the README calls 1.0 production-ready (conflict) ^{vr829}
- Support is a one-month feature branch with no backports and needs a subscription ^{vr830}
- NVIDIA-led and NVIDIA-GPU-optimised ^{vr831}

⁸²² Verified fact — A4-S098: PyPI JSON metadata: ai-dynamo (Python Package Index) <https://pypi.org/pypi/ai-dynamo/json> accessed 2026-10-07; A4-S090: NVIDIA Dynamo README (GitHub) <https://raw.githubusercontent.com/ai-dynamo/dynamo/main/README.md> accessed 2026-10-07; B-L2-S003: Dynamo Support Coverage (NVIDIA) <https://docs.nvidia.com/dynamo/dev/reference/enterprise/support-coverage> accessed 2026-10-08 [search-tool extract]

⁸²³ Verified fact — B-L2-S003: Dynamo Support Coverage (NVIDIA) <https://docs.nvidia.com/dynamo/dev/reference/enterprise/support-coverage> accessed 2026-10-08 [search-tool extract]

⁸²⁴ Verified fact — A4-S090: NVIDIA Dynamo README (GitHub) <https://raw.githubusercontent.com/ai-dynamo/dynamo/main/README.md> accessed 2026-10-07

⁸²⁵ Verified fact — A4-S090: NVIDIA Dynamo README (GitHub) <https://raw.githubusercontent.com/ai-dynamo/dynamo/main/README.md> accessed 2026-10-07; A4-S098: PyPI JSON metadata: ai-dynamo (Python Package Index) <https://pypi.org/pypi/ai-dynamo/json> accessed 2026-10-07

⁸²⁶ Verified fact — B-L2-S003: Dynamo Support Coverage (NVIDIA) <https://docs.nvidia.com/dynamo/dev/reference/enterprise/support-coverage> accessed 2026-10-08 [search-tool extract]

⁸²⁷ Verified fact — A4-S090: NVIDIA Dynamo README (GitHub) <https://raw.githubusercontent.com/ai-dynamo/dynamo/main/README.md> accessed 2026-10-07

⁸²⁸ Verified fact — A4-S090: NVIDIA Dynamo README (GitHub) <https://raw.githubusercontent.com/ai-dynamo/dynamo/main/README.md> accessed 2026-10-07; A4-S091: LMCACHE README (GitHub) (LMCACHE project) <https://raw.githubusercontent.com/LMCACHE/LMCACHE/dev/README.md> accessed 2026-10-07

⁸²⁹ Verified fact — A4-S098: PyPI JSON metadata: ai-dynamo (Python Package Index) <https://pypi.org/pypi/ai-dynamo/json> accessed 2026-10-07; A4-S090: NVIDIA Dynamo README (GitHub) <https://raw.githubusercontent.com/ai-dynamo/dynamo/main/README.md> accessed 2026-10-07

⁸³⁰ Verified fact — B-L2-S003: Dynamo Support Coverage (NVIDIA) <https://docs.nvidia.com/dynamo/dev/reference/enterprise/support-coverage> accessed 2026-10-08 [search-tool extract]

⁸³¹ Verified fact — A4-S090: NVIDIA Dynamo README (GitHub) <https://raw.githubusercontent.com/ai-dynamo/dynamo/main/README.md> accessed 2026-10-07

Choose when

- You run multi-node NVIDIA GPU serving at a scale where disaggregation pays ^{AJ}

Avoid when

- Single-node or modest volumes; vLLM alone is enough ^{AJ}
- You need long-term-support releases ^{V⁸³² AJ}

Nearest competitors: L2-llm-d, L2-vllm, L2-sglang

Regulated-FS note. Treat as an optimisation layer you can remove: keep the engine’s OpenAI-compatible API as the contract to the gateway ^{Rec.}

Facts (as of 2026-10-07)

Field	Value	Label	Sources
Company	NVIDIA	Verified fact	A4-S098
Category	Distributed inference orchestration / optimisation layer above inference engines	Verified fact	A4-S090
Version / lineup	ai-dynamo 1.5.1 (7 October 2026; PyPI classifier Beta); first release March 2025	Verified fact	A4-S098
Licence	Open source, Apache-2.0	Verified fact	A4-S098, A4-S090
Status events	Not publicly verified	Not publicly verified	
Strategic direction	Not publicly verified	Not publicly verified	
What it does	‘The orchestration layer above inference engines — it doesn’t replace SGLang, TensorRT-LLM, or vLLM’: disaggregated serving, KV-aware routing, multi-tier KV caching (KVBM), SLA-based planner and autoscaling; 1.0 added zero-config deploy, agentic-inference hints and	Verified fact	A4-S090

⁸³² Verified fact — B-L2-S003: Dynamo Support Coverage (NVIDIA) <https://docs.nvidia.com/dynamo/dev/reference/enterprise/support-coverage> accessed 2026-10-08 [search-tool extract]

Field	Value	Label	Sources
	multimodal E/P/D.		
Stack position	Inference optimisation sub-layer between serving engines and gateways (H1)	Verified fact	A4-S090
Integration	Not publicly verified	Not publicly verified	
Dependencies	Not publicly verified	Not publicly verified	
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Free (Apache-2.0)	Verified fact	A4-S098
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Backends SGLang, TensorRT-LLM, vLLM; integrates LMCache (September 2025); LangChain and NeMo Agent Toolkit integrations	Verified fact	A4-S090, A4-S091

Field	Value	Label	Sources
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Not publicly verified	Not publicly verified	
Deployment	saas: No; managed_cloud: Not publicly verified; vpc_byoc: Yes; private_cloud: Yes; self_hosted: Yes; on_prem: Yes		

Ollama (local runtime) with Ollama Cloud (hosted cloud models) (L2-ollama)

Tier: Tactical · Flags: none · Original graphic label: Ollama – run locally

Rationale: Useful developer runtime; Ollama Cloud is an unverified third-party processor

Criterion	Score	Rationale
Technical	3	Covers local serving competently; not a multi-tenant production engine ^{vr833} _{AJ}
Enterprise readiness	2	No SSO, RBAC or audit verified for Cloud; NPV cap ^{NPV}
Security and compliance	2	Cloud certifications NPV (cap); local runtime has no documented authentication ^{vr834} _{NPV}
Deployment flexibility	4	Local, on-prem and SaaS; no VPC ^{vr835}
Ecosystem	4	Large community integrations; OpenAI/Anthropic-compatible APIs ^{vr836}

⁸³³ Verified fact — A4-S084: Ollama Cloud docs <https://raw.githubusercontent.com/ollama/ollama/main/docs/cloud.mdx> accessed 2026-10-07

⁸³⁴ Verified fact — B-L2-S007: FAQ - Ollama <https://docs.ollama.com/faq> accessed 2026-10-08 [search-tool extract]

⁸³⁵ Verified fact — A4-S084: Ollama Cloud docs <https://raw.githubusercontent.com/ollama/ollama/main/docs/cloud.mdx> accessed 2026-10-07; A4-S085: Ollama FAQ <https://raw.githubusercontent.com/ollama/ollama/main/docs/faq.mdx> accessed 2026-10-07

⁸³⁶ Verified fact — A4-S084: Ollama Cloud docs <https://raw.githubusercontent.com/ollama/ollama/main/docs/cloud.mdx> accessed 2026-10-07

Criterion	Score	Rationale
Reliability and maturity	2	Server version and maturity not verified; cloud pricing changed 31 August 2026 ^{vr⁸³⁷ NPV}
Cost / TCO	4	Runtime free; Cloud plans published ^{vr⁸³⁸}
Lock-in / portability	4	MIT runtime and portable models; Cloud proprietary ^{vr⁸³⁹}
Total (generic / FS)	3.00 / 2.95	

Evidence rules applied: enterprise_readiness capped at 2: SSO/RBAC/audit NPV; security_compliance capped at 2: Ollama Cloud certifications NPV

Capabilities. MIT-licensed local LLM runtime and model packager, plus Ollama Cloud, a hosted service for larger models with OpenAI- and Anthropic-compatible APIs; cloud features can be disabled; usage-based cloud plans from 31 August 2026 ^{vr⁸⁴⁰}. Binds 127.0.0.1:11434 by default ^{vr⁸⁴¹}.

Strengths

- Simplest local runtime for laptops and disconnected prototyping ^{AJ}
- MIT runtime, portable models ^{vr⁸⁴²}
- Cloud can be switched off (OLLAMA_NO_CLOUD) ^{vr⁸⁴³}

Limitations and risks

- No longer local-only: cloud models change the data-flow assumption ^{vr⁸⁴⁴}
- Ollama Cloud certifications, SSO and audit not publicly verified ^{NPV}
- Built-in API authentication not documented; exposure relies on a proxy ^{vr⁸⁴⁵ AJ}

Choose when

- Developer workstations and offline evaluation with cloud disabled ^{AJ}

Avoid when

- Any shared or production serving ^{AJ}

⁸³⁷ Verified fact — V1-S066: Pricing - Ollama; Ollama's transparent pricing (31 Aug 2026) <https://ollama.com/pricing> accessed 2026-10-08 [search-tool extract]

⁸³⁸ Verified fact — V1-S066: Pricing - Ollama; Ollama's transparent pricing (31 Aug 2026) <https://ollama.com/pricing> accessed 2026-10-08 [search-tool extract]

⁸³⁹ Verified fact — A4-S086: Ollama LICENSE <https://raw.githubusercontent.com/ollama/ollama/main/LICENSE> accessed 2026-10-07

⁸⁴⁰ Verified fact — A4-S084: Ollama Cloud docs <https://raw.githubusercontent.com/ollama/ollama/main/docs/cloud.mdx> accessed 2026-10-07; A4-S085: Ollama FAQ <https://raw.githubusercontent.com/ollama/ollama/main/docs/faq.mdx> accessed 2026-10-07; A4-S086: Ollama LICENSE <https://raw.githubusercontent.com/ollama/ollama/main/LICENSE> accessed 2026-10-07; V1-S066: Pricing - Ollama; Ollama's transparent pricing (31 Aug 2026) <https://ollama.com/pricing> accessed 2026-10-08 [search-tool extract]

⁸⁴¹ Verified fact — B-L2-S007: FAQ - Ollama <https://docs.ollama.com/faq> accessed 2026-10-08 [search-tool extract]

⁸⁴² Verified fact — A4-S086: Ollama LICENSE <https://raw.githubusercontent.com/ollama/ollama/main/LICENSE> accessed 2026-10-07

⁸⁴³ Verified fact — A4-S085: Ollama FAQ <https://raw.githubusercontent.com/ollama/ollama/main/docs/faq.mdx> accessed 2026-10-07

⁸⁴⁴ Verified fact — A4-S084: Ollama Cloud docs <https://raw.githubusercontent.com/ollama/ollama/main/docs/cloud.mdx> accessed 2026-10-07

⁸⁴⁵ Verified fact — B-L2-S007: FAQ - Ollama <https://docs.ollama.com/faq> accessed 2026-10-08 [search-tool extract]

- Client data on Ollama Cloud ^{AI}

Nearest competitors: L2-lm-studio, llama.cpp, L2-vllm

Regulated-FS note. Allow on managed desktops only with OLLAMA_NO_CLOUD enforced and localhost binding; never as a production endpoint ^{Rec.}

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Ollama	Verified fact	A4-S086
Category	Local LLM runtime and model packager, plus an optional hosted cloud-model service	Verified fact	A4-S084
Version / lineup	Official Python client ollama 0.6.3 (29 September 2026); server/app version not retrieved	Verified fact	A4-S015
Licence	Open source, MIT (runtime); Ollama Cloud is a hosted service with plans	Verified fact	A4-S086, A4-S084
Status events	Cloud models introduced (e.g. *-cloud tags; gemma4:cloud); cloud features can be disabled (OLLAMA_NO_CLOUD=1)	Verified fact	A4-S015, A4-S085
Strategic direction	Cloud models run in Ollama's cloud from apps, CLI or API with an API key; launchers for coding agents (Claude Code, Codex, OpenCode); OpenAI- and Anthropic-compatible APIs	Verified fact	A4-S084
What it does	Runs open-weight models locally (no prompt visibility to Ollama when local) and optionally offloads larger models to Ollama's cloud; embeddings; OpenAI/Anthropic-compatible endpoints.	Verified fact	A4-S085, A4-S084, A4-S015
Stack position	Developer/edge local runtime (L2); not a multi-tenant production serving engine. Ollama Cloud adds a hosted inference option	Verified fact	A4-S084
Integration	Not publicly verified	Not	

Field	Value	Label	Sources
		publicly verified	
Dependencies	Not publicly verified	Not publicly verified	
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	Local: Ollama does not see prompts. Cloud: prompts/responses processed to serve requests but not stored or logged and never used for training; basic account info and usage metadata collected	Verified fact	A4-S085, A4-S084
Security features	Cloud features can be disabled for local-only mode (disable_ollama_cloud / OLLAMA_NO_CLOUD)	Verified fact	A4-S085
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Local runtime free (MIT). Ollama Cloud usage-based plans from 31 August 2026: Free (small monthly usage on starter models, pay-as-you-go credits), Pro US\$20/month (US\$60 usage included), Max US\$100/month (US\$300 included), Team US\$500/month (US\$1,000 shared usage, unlimited users); per-token rates by model; no hourly or weekly caps; off-peak discounts	Verified fact	A4-S084, V1-S066
Infrastructure cost	Local cost is the user's own hardware (GPU/Apple silicon RAM)	Architectural judgement	
Ecosystem	Large community integration list; Google Cloud Run tutorial; Claude Code/Codex/OpenCode launchers	Verified fact	A4-S084
Adoption signals	US\$65M funding round reported July 2026 (SiliconANGLE, 9 July 2026)	Reported	V1-S071
Maturity	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
Deployment	saas: Yes (Ollama Cloud); managed_cloud: Yes (Ollama Cloud); vpc_byoc: No; private_cloud: Not publicly verified; self_hosted: Yes; on_prem: Yes		

Cerebras Systems (Cerebras Inference / Cerebras Cloud; CS-3 systems)
(L2 - cerebras)

Tier: Tactical · Flags: none · Original graphic label: Cerebras – ultra-scale cloud

Rationale: Distinctive speed but thin enterprise controls and no EU processing yet

Criterion	Score	Rationale
Technical	3	Fast inference on a smaller catalogue; on-prem systems ^{vf846}
Enterprise readiness	3	Console roles verified (rule 7 lifts the cap to 3); no SSO or audit documented ^{vf847}
Security and compliance	2	SOC 2 Type 2 listed in the trust centre without product scope (rule 8) ^{vf848}
Deployment	3	SaaS, dedicated, on-prem systems; no EU region yet ^{vf849}

⁸⁴⁶ Verified fact — A4-S095: PyPI JSON metadata: cerebras-cloud-sdk (Python Package Index) <https://pypi.org/pypi/cerebras-cloud-sdk/json> accessed 2026-10-07; A4-S139: Pricing - Build with Cerebras Inference (Cerebras Systems) <https://inference-docs.cerebras.ai/support/pricing> accessed 2026-10-07 [search-tool extract]

⁸⁴⁷ Verified fact — A4-S153: Cloud Console / Projects - Cerebras Inference docs (with Cerebras Privacy Policy) (Cerebras Systems) <https://inference-docs.cerebras.ai/console/overview> accessed 2026-10-07 [search-tool extract]

⁸⁴⁸ Verified fact — A4-S138: Cerebras Systems Trust Center (and Cerebras Inference blog posts on data retention) <https://trust.cerebras.ai/> accessed 2026-10-07 [search-tool extract]; V1-S095: Cerebras Systems Trust Center; six new datacenters press release (March 2025) <https://trust.cerebras.ai/> accessed 2026-10-08 [search-tool extract]

⁸⁴⁹ Verified fact — A4-S139: Pricing - Build with Cerebras Inference (Cerebras Systems) <https://inference-docs.cerebras.ai/support/pricing> accessed 2026-10-07 [search-tool extract]; A4-S095: PyPI JSON metadata: cerebras-cloud-sdk (Python Package Index) <https://pypi.org/pypi/cerebras-cloud-sdk/json> accessed 2026-10-07; A4-S140: Cerebras Systems Accelerates European Expansion with 200MW of AI Compute Capacity by End of 2027 (8 Jul 2026) and Mikkeli, Finland release (31 Aug 2026) <https://investors.cerebras.ai/news-releases/news-release-details/cerebras-systems-accelerates-european-expansion-200mw-ai-compute> accessed 2026-10-07 [search-tool extract]

Criterion	Score	Rationale
flexibility		
Ecosystem	3	OpenAI-compatible; AWS Marketplace; Hugging Face partner ^{vf850}
Reliability and maturity	3	Public company; inference cloud since 2024; customer concentration ^{vf851}
Cost / TCO	3	Transparent per-token pricing; enterprise flat pricing ^{vf852}
Lock-in / portability	3	OpenAI-compatible API, proprietary hardware ^{vf853}
Total (generic / FS)	2.85 / 2.80	

Capabilities. Chip and system vendor (WSE-3, CS-3) that also runs an OpenAI-compatible inference cloud and sells on-premise systems; IPO priced 13 May 2026 (Nasdaq: CBR5); 750MW OpenAI agreement; SOC 2 Type 2, GDPR and CCPA listed, HIPAA not; EU capacity targeted for end-2026 ^{vf854}.

Strengths

- Very low-latency inference as a design point ^{vf855 AJ}
- On-premise CS-3 option ^{vf856}

Limitations and risks

- No SSO documented; console roles only ^{vf857}
- Data may be processed outside the US unless otherwise agreed; no EU capacity yet ^{vf858}

⁸⁵⁰ Verified fact — A4-S139: Pricing - Build with Cerebras Inference (Cerebras Systems) <https://inference-docs.cerebras.ai/support/pricing> accessed 2026-10-07 [search-tool extract]; A4-S075: Hugging Face Inference Providers <https://raw.githubusercontent.com/huggingface/hub-docs/main/docs/inference-providers/index.md> accessed 2026-10-07

⁸⁵¹ Verified fact — A4-S137: Cerebras Systems Announces Pricing of Initial Public Offering <https://www.cerebras.ai/press-release/cerebras-systems-announces-pricing-of-initial-public-offering> accessed 2026-10-07 [search-tool extract]; A4-S154: Cerebras Systems Inc. Form 10-Q (quarter ended 30 June 2026) and related S-1/8-K filings (Cerebras Systems Inc. (SEC EDGAR)) <https://www.sec.gov/Archives/edgar/data/0002021728/000162828026056357/cbrs-20260630.htm> accessed 2026-10-07 [search-tool extract]

⁸⁵² Verified fact — A4-S139: Pricing - Build with Cerebras Inference (Cerebras Systems) <https://inference-docs.cerebras.ai/support/pricing> accessed 2026-10-07 [search-tool extract]

⁸⁵³ Verified fact — A4-S139: Pricing - Build with Cerebras Inference (Cerebras Systems) <https://inference-docs.cerebras.ai/support/pricing> accessed 2026-10-07 [search-tool extract]; A4-S095: PyPI JSON metadata: cerebras-cloud-sdk (Python Package Index) <https://pypi.org/pypi/cerebras-cloud-sdk/json> accessed 2026-10-07

⁸⁵⁴ Verified fact — A4-S095: PyPI JSON metadata: cerebras-cloud-sdk (Python Package Index) <https://pypi.org/pypi/cerebras-cloud-sdk/json> accessed 2026-10-07; A4-S137: Cerebras Systems Announces Pricing of Initial Public Offering <https://www.cerebras.ai/press-release/cerebras-systems-announces-pricing-of-initial-public-offering> accessed 2026-10-07 [search-tool extract]; A4-S154: Cerebras Systems Inc. Form 10-Q (quarter ended 30 June 2026) and related S-1/8-K filings (Cerebras Systems Inc. (SEC EDGAR)) <https://www.sec.gov/Archives/edgar/data/0002021728/000162828026056357/cbrs-20260630.htm> accessed 2026-10-07 [search-tool extract]; A4-S138: Cerebras Systems Trust Center (and Cerebras Inference blog posts on data retention) <https://trust.cerebras.ai/> accessed 2026-10-07 [search-tool extract]; A4-S140: Cerebras Systems Accelerates European Expansion with 200MW of AI Compute Capacity by End of 2027 (8 Jul 2026) and Mikkeli, Finland release (31 Aug 2026) <https://investors.cerebras.ai/news-releases/news-release-details/cerebras-systems-accelerates-european-expansion-200mw-ai-compute> accessed 2026-10-07 [search-tool extract]; V1-S095: Cerebras Systems Trust Center; six new datacenters press release (March 2025) <https://trust.cerebras.ai/> accessed 2026-10-08 [search-tool extract]

⁸⁵⁵ Verified fact — A4-S095: PyPI JSON metadata: cerebras-cloud-sdk (Python Package Index) <https://pypi.org/pypi/cerebras-cloud-sdk/json> accessed 2026-10-07

⁸⁵⁶ Verified fact — A4-S095: PyPI JSON metadata: cerebras-cloud-sdk (Python Package Index) <https://pypi.org/pypi/cerebras-cloud-sdk/json> accessed 2026-10-07

⁸⁵⁷ Verified fact — A4-S153: Cloud Console / Projects - Cerebras Inference docs (with Cerebras Privacy Policy) (Cerebras Systems) <https://inference-docs.cerebras.ai/console/overview> accessed 2026-10-07 [search-tool extract]

- ZDR evidence is privacy-policy and blog level ^{VF859}
- Customer concentration in the OpenAI agreement ^{VF860}

Choose when

- Latency-critical, non-confidential workloads on supported open models ^{AJ}

Avoid when

- Client data until EU capacity and contractual ZDR exist ^{AJ}

Nearest competitors: L2-together-ai, L2-fireworks-ai, Groq

Regulated-FS note. Re-assess when EU capacity is live and SSO and contractual ZDR are documented ^{Rec.}

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Cerebras Systems Inc., listed on Nasdaq (CBRS) since 14 May 2026	Verified fact	A4-S137
Category	AI chip and system vendor (WSE-3, CS-3) that also operates an inference cloud and sells on-premise systems	Verified fact	A4-S095, A4-S154
Version / lineup	cerebras-cloud-sdk 1.91.0 (16 July 2026); hardware WSE-3 / CS-3; Inference tiers: free, pay-per-token (Exploration), Enterprise; Cerebras Code subscriptions	Verified fact	A4-S095, A4-S139
Licence	Proprietary hardware and cloud service; SDK Apache-2.0	Verified fact	A4-S095
Status events	December 2025: OpenAI master relationship agreement (750MW); 13 May 2026: IPO priced at US\$185/share (30M Class A shares, ~US\$5.55B); Nasdaq: CBRS from 14 May 2026; 8 July 2026: European expansion announced	Verified fact	A4-S154, A4-S137, A4-S140, V1-S053
Strategic direction	Large inference supply agreement with OpenAI (750MW, Dec 2025; option for further 1.25GW by 2030); European expansion: first EU capacity by end-2026 (France, Nordics), 200MW by end-2027, 165MW Mikkeli (Finland) site	Verified fact	A4-S154, A4-S140
What it does	Provides high-speed inference through an OpenAI-compatible REST API	Verified fact	A4-S095,

⁸⁵⁸ Verified fact — A4-S153: Cloud Console / Projects - Cerebras Inference docs (with Cerebras Privacy Policy) (Cerebras Systems) <https://inference-docs.cerebras.ai/console/overview> accessed 2026-10-07 [search-tool extract]; A4-S140: Cerebras Systems Accelerates European Expansion with 200MW of AI Compute Capacity by End of 2027 (8 Jul 2026) and Mikkeli, Finland release (31 Aug 2026) <https://investors.cerebras.ai/news-releases/news-release-details/cerebras-systems-accelerates-european-expansion-200mw-ai-compute> accessed 2026-10-07 [search-tool extract]

⁸⁵⁹ Verified fact — A4-S153: Cloud Console / Projects - Cerebras Inference docs (with Cerebras Privacy Policy) (Cerebras Systems) <https://inference-docs.cerebras.ai/console/overview> accessed 2026-10-07 [search-tool extract]; A4-S138: Cerebras Systems Trust Center (and Cerebras Inference blog posts on data retention) <https://trust.cerebras.ai/> accessed 2026-10-07 [search-tool extract]

⁸⁶⁰ Verified fact — A4-S154: Cerebras Systems Inc. Form 10-Q (quarter ended 30 June 2026) and related S-1/8-K filings (Cerebras Systems Inc. (SEC EDGAR)) <https://www.sec.gov/Archives/edgar/data/0002021728/000162828026056357/cbrs-20260630.htm> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
	on CS-3 systems in Cerebras data centres, and sells CS-3 systems for on-premise deployment.	d fact	A4-S139
Stack position	Both hardware vendor (below L2) and inference provider (L2). The graphic's 'ultra-scale cloud' omits the hardware business	Verified fact	A4-S095
Integration	Official SDK; OpenAI-compatible API; AWS Marketplace billing; Hugging Face Inference Providers partner	Verified fact	A4-S095, A4-S139, A4-S075
Dependencies	Not publicly verified	Not publicly verified	
Certifications	SOC 2 Type 2, GDPR, CCPA listed in trust centre; HIPAA not listed	Verified fact	A4-S138, V1-S095
GDPR / residency	Privacy policy: inputs and outputs of inference services not retained; logs deleted when no longer needed; data may be processed outside the US unless otherwise agreed. Blog claims (Jan 2025): inference runs in US data centres with zero data retention. EU capacity announced but not yet online (target end-2026)	Verified fact	A4-S153, A4-S138, A4-S140
Security features	Encryption at rest and in transit (trust centre sections)	Verified fact	A4-S138
Access controls	Console roles: Organization Admin, Project Admin, Project Member; projects group keys and rate limits; API-key auth; no SSO documentation found	Verified fact	A4-S153
Enterprise support	Enterprise tier: custom SLAs, dedicated support, fine-tuned models, 3/6/12-month contracts	Verified fact	A4-S139
Pricing	Pay-per-token per 1M tokens (in/out): Llama 3.1 8B US\$0.10/US\$0.10, Llama 3.3 70B US\$0.85/US\$1.20, Qwen 3 32B US\$0.40/US\$0.80; free tier limited to 8,192-token context; Enterprise flat monthly pricing by tokens-per-minute; Cerebras Code Pro US\$50/month, Max US\$200/month	Verified fact	A4-S139
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	IPO raised ~US\$5.55B (May 2026); OpenAI 750MW commitment; 600MW of data-centre capacity under contract (Q2 2026 results)	Verified fact	A4-S137, A4-S154
Maturity	Public company since May 2026; inference cloud since 2024	Verified fact	A4-S137, A4-S139
Deployment	saas: Yes; managed_cloud: Yes (Enterprise dedicated capacity); vpc_byoc: Not publicly verified; private_cloud: Not publicly verified;		

Field	Value	Label	Sources
	self_hosted: No; on_prem: Yes (on-premise CS-3 systems)		

LM Studio (desktop app) with lms CLI and Python/JS SDKs (L2-lm-studio)

Tier: Tactical · **Flags:** none · **Original graphic label:** LM Studio – desktop app

Rationale: Desktop-only tool; not part of the serving architecture

Criterion	Score	Rationale
Technical	2	Desktop runtime only; missing production-serving functions ^{vr⁸⁶¹ AJ}
Enterprise readiness	2	SSO on Enterprise verified at low confidence (rule 7 allows up to 3); no audit or support evidence, so 2 ^{vr⁸⁶²}
Security and compliance	2	Desktop software, proprietary; security features NPV ^{NPV}
Deployment flexibility	2	Desktop on-prem only; no server, VPC or SaaS ^{vr⁸⁶³}
Ecosystem	2	Integrations largely NPV beyond CLI and SDKs ^{vr⁸⁶⁴}
Reliability and maturity	2	Maturity NPV; SDK last release August 2025 ^{vr⁸⁶⁵}
Cost / TCO	4	Free for internal business use ^{vr⁸⁶⁶}
Lock-in / portability	3	Proprietary app, but models and the OpenAI-style local API are portable ^{AJ}
Total (generic / FS)	2.25	

⁸⁶¹ Verified fact — A4-S087: lms CLI README (LM Studio) <https://raw.githubusercontent.com/lmstudio-ai/lms/main/README.md> accessed 2026-10-07

⁸⁶² Verified fact — A4-S145: LM Studio is free for use at work (blog) and LM Studio App Terms of Service <https://lmstudio.ai/blog/free-for-work> accessed 2026-10-07 [search-tool extract]

⁸⁶³ Verified fact — A4-S087: lms CLI README (LM Studio) <https://raw.githubusercontent.com/lmstudio-ai/lms/main/README.md> accessed 2026-10-07

⁸⁶⁴ Verified fact — A4-S087: lms CLI README (LM Studio) <https://raw.githubusercontent.com/lmstudio-ai/lms/main/README.md> accessed 2026-10-07; A4-S016: PyPI JSON metadata: lmstudio (Python Package Index) <https://pypi.org/pypi/lmstudio/json> accessed 2026-10-07

⁸⁶⁵ Verified fact — A4-S016: PyPI JSON metadata: lmstudio (Python Package Index) <https://pypi.org/pypi/lmstudio/json> accessed 2026-10-07

⁸⁶⁶ Verified fact — A4-S145: LM Studio is free for use at work (blog) and LM Studio App Terms of Service <https://lmstudio.ai/blog/free-for-work> accessed 2026-10-07 [search-tool extract]

Criterion	Score	Rationale
	/	
	2.25	

Evidence rules applied: security_compliance capped at 2: security features and certifications NPV

Capabilities. Proprietary desktop app for downloading and running local models with a local API server, MIT lms CLI and SDKs; free for personal and internal business use since about July 2025; may not be redistributed or offered as a service; a paid Enterprise plan adds SSO and model/MCP gating ^{vr867}.

Strengths

- Low-friction desktop evaluation of open models ^{aj}
- Enterprise plan gating of models and MCP servers ^{vr868}

Limitations and risks

- Proprietary app terms forbid offering it as a service ^{vr869}
- Security features, support, versioning and maturity largely not publicly verified ^{npv}
- Python SDK last released August 2025 ^{vr870}

Choose when

- Managed developer desktops where a GUI and model gating help adoption ^{aj}

Avoid when

- Any server or shared use ^{vr871 aj}

Nearest competitors: L2-ollama, llama.cpp

Regulated-FS note. Permit only via the Enterprise plan on managed devices with an approved-model list; exclude client data ^{rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	LM Studio	Verified fact	A4-S088
Category	Desktop application for running local models, with a local API server	Verified fact	A4-S087

⁸⁶⁷ Verified fact — A4-S087: lms CLI README (LM Studio) <https://raw.githubusercontent.com/lmstudio-ai/lms/main/README.md> accessed 2026-10-07; A4-S088: lms LICENSE (LM Studio) <https://raw.githubusercontent.com/lmstudio-ai/lms/main/LICENSE> accessed 2026-10-07; A4-S145: LM Studio is free for use at work (blog) and LM Studio App Terms of Service <https://lmstudio.ai/blog/free-for-work> accessed 2026-10-07 [search-tool extract]; V1-S076: LM Studio is free for use at work; LM Studio App Terms of Service (LM Studio (Element Labs)) <https://lmstudio.ai/blog/free-for-work> accessed 2026-10-08 [search-tool extract]

⁸⁶⁸ Verified fact — A4-S145: LM Studio is free for use at work (blog) and LM Studio App Terms of Service <https://lmstudio.ai/blog/free-for-work> accessed 2026-10-07 [search-tool extract]

⁸⁶⁹ Verified fact — A4-S145: LM Studio is free for use at work (blog) and LM Studio App Terms of Service <https://lmstudio.ai/blog/free-for-work> accessed 2026-10-07 [search-tool extract]

⁸⁷⁰ Verified fact — A4-S016: PyPI JSON metadata: lmstudio (Python Package Index) <https://pypi.org/pypi/lmstudio/json> accessed 2026-10-07

⁸⁷¹ Verified fact — A4-S145: LM Studio is free for use at work (blog) and LM Studio App Terms of Service <https://lmstudio.ai/blog/free-for-work> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
Version / lineup	lmstudio Python SDK 1.5.0 (22 August 2025); lms CLI ships with LM Studio 0.2.22 and newer; desktop app version not retrieved	Verified fact	A4-S016, A4-S087
Licence	Desktop app proprietary (App Terms of Service): licensed for personal and/or internal business use, free at work without a separate commercial licence; prohibits modification, redistribution, sublicensing, offering as a service, reverse engineering. lms CLI MIT	Verified fact	A4-S145, A4-S088, V1-S076
Status events	Not publicly verified	Not publicly verified	
Strategic direction	Not publicly verified	Not publicly verified	
What it does	Downloads and runs models locally; lms CLI starts/stops the local API server and lists models; SDKs allow programmatic access.	Verified fact	A4-S087, A4-S016
Stack position	Developer-desktop runtime (L2); not a production serving tier	Architectural judgement	
Integration	Not publicly verified	Not publicly verified	
Dependencies	Not publicly verified	Not publicly verified	
Certifications	Not applicable (desktop software)	Architectural judgement	
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	SSO and model/MCP gating on paid Enterprise plan	Verified fact	A4-S145

Field	Value	Label	Sources
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Free for personal and internal work use; paid Enterprise plan for SSO and model/MCP gating; LM Link free during preview	Verified fact	A4-S145
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Not publicly verified	Not publicly verified	
Deployment	saas: No; managed_cloud: No; vpc_byoc: No; private_cloud: Not publicly verified; self_hosted: Yes; on_prem: Yes		

L1: Foundation models

OpenAI GPT model family (GPT-6: Astra, Sol, Luna; GPT-6.1 Sol), plus GPT-5.6 Sol/Terra/Luna and open-weight gpt-oss (L1-openai)

Tier: Strategic • **Flags:** none • **Original graphic label:** OpenAI – GPT-6

Rationale: Broadest evidenced lineup, two hyperscaler routes plus first-party EU processing, product-scoped SOC 2 and ISO 27001; concentration is managed by pairing with a second vendor, not by avoiding OpenAI ^{AJ}.

Criterion	Score	Rationale
Technical	5	Covers every plan \$5 L1 question on primary evidence: frontier, mid and small tiers, reasoning, image input, coding, a specialist cyber line and Apache-2.0 open weights (A5-S002, A5-S005, A5-S008). Benchmarks not used.

Criterion	Score	Rationale
Enterprise readiness	4	First-party SSO/RBAC/audit and support terms are NPV; GPT-6 is consumed in-tenant through Bedrock and Foundry, so the hyperscaler presumption gives 4 (rule 6). Platform controls presumed (CP2 Q1); confirm per service.
Security and compliance	4	SOC 2 Type 2 and ISO 27001 within API scope, ZDR and first-party EU residency (A5-S006, A5-S007, V2-S062); not 5 because ISO 42001 scope does not name the API and CMK is not verified (rule 8).
Deployment flexibility	4	First-party SaaS (US, EU, UK-storage endpoints), Bedrock, Foundry, and self-hosted/air-gapped gpt-oss; 4 not 5 because the frontier tiers cannot be self-hosted and gpt-oss dates from August 2025.
Ecosystem	5	The OpenAI API shape is the de facto interface: other vendors publish OpenAI-compatible endpoints (A5-S010, A5-S077, A5-S038); official SDKs; two hyperscalers.
Reliability and maturity	4	Years of production use and a published deprecation policy (B-L1-S001); fast tier churn (GPT-5.6 in July, GPT-6 in September 2026) and the Astra GA conflict keep it at 4.
Cost / TCO	4	Transparent per-tier pricing (Luna US\$0.10/US\$0.50, Sol US\$2/US\$10, Astra US\$10/US\$50 per 1M) with 50% batch discount (A5-S004); competitive, not uniquely efficient.
Lock-in / portability	3	Proprietary frontier API on a de facto standard interface, on two non-OpenAI clouds, with Apache-2.0 gpt-oss as a portability option.
Total (generic / FS)	4.25	

Criterion	Score	Rationale
	/	
	4.05	

Capabilities. Tiered proprietary family: GPT-6 Astra (gated flagship, introduced 3 September 2026), GPT-6 Sol and Luna (22 September 2026), GPT-6.1 Sol (29 September 2026), with GPT-5.6 Sol/Terra/Luna still offered; all GPT-6 tiers are reasoning models with text and image input; Daybreak cyber models in limited availability on Bedrock; open-weight gpt-oss-120b/20b under Apache 2.0 (August 2025) ^{vr872}.

Strengths

- Complete tier ladder (frontier, mid, small) plus Apache-2.0 open weights, so one vendor covers every portfolio tier except sovereign self-hosted frontier ^{vr873}
- Available on AWS Bedrock and Microsoft Foundry as well as the first-party API, including EU Data Zone deployments on Foundry ^{vr874}
- First-party EU residency: eu.api.openai.com stores and processes in region (new projects, with Modified Abuse Monitoring or ZDR) ^{vr875}
- SOC 2 Type 2 for the API Platform and ISO/IEC 27001, 27017, 27018, 27701 covering the API ^{vr876}
- Published deprecation notice policy: at least 6 months for GA models, at least 3 months for specialised variants ^{vr877}

Limitations and risks

- Frontier tiers are closed-weight and API-only; fine-tuning is not supported for Astra ^{vr878}

⁸⁷² Verified fact — A5-S001: Previewing GPT-5.6 Sol / GPT-5.6 Sol, Terra and Luna (with help-centre preview article and API changelog) (OpenAI) <https://openai.com/index/previewing-gpt-5-6-sol/> accessed 2026-10-07 [search-tool extract]; A5-S002: GPT-6 Astra: A new generation of intelligence (plus release notes, Introducing GPT-6 Sol and Luna, GPT-6.1 Sol, GPT-6 for everyone) (OpenAI) <https://openai.com/index/gpt-6-astra/> accessed 2026-10-07 [search-tool extract]; A5-S003: OpenAI announces rollout of GPT-6 Astra model (with Axios and OpenRouter results) (CNBC) <https://www.cnbc.com/2026/09/03/open-ai-astra-gpt-6-cyber.html> accessed 2026-10-07 [search-tool extract]; A5-S005: Introducing gpt-oss (and gpt-oss-safeguard; help-centre open-weight article) (OpenAI) <https://openai.com/index/introducing-gpt-oss/> accessed 2026-10-07 [search-tool extract]; A5-S008: OpenAI GPT-6 Astra / GPT-6 Sol and Luna generally available on Amazon Bedrock (AWS) <https://aws.amazon.com/about-aws/whats-new/2026/09/openai-gpt-6-astra-on-amazon-bedrock/> accessed 2026-10-07 [search-tool extract]; A5-S009: GPT-6 Astra, Sol, and Luna for production AI agents in Microsoft Foundry <https://azure.microsoft.com/en-us/blog/gpt-6-astra-sol-and-luna-for-production-agents-in-microsoft-foundry/> accessed 2026-10-07 [search-tool extract]; A5-S083: ChatGPT Release Notes ('Introducing GPT-6 Astra', 3 Sept 2026); Path to Astra (1 Sept 2026); Safety overview: GPT-6 Astra (OpenAI) <https://help.openai.com/en/articles/6825453-chatgpt-release-notes> accessed 2026-10-07 [search-tool extract]

⁸⁷³ Verified fact — A5-S002: GPT-6 Astra: A new generation of intelligence (plus release notes, Introducing GPT-6 Sol and Luna, GPT-6.1 Sol, GPT-6 for everyone) (OpenAI) <https://openai.com/index/gpt-6-astra/> accessed 2026-10-07 [search-tool extract]; A5-S005: Introducing gpt-oss (and gpt-oss-safeguard; help-centre open-weight article) (OpenAI) <https://openai.com/index/introducing-gpt-oss/> accessed 2026-10-07 [search-tool extract]

⁸⁷⁴ Verified fact — A5-S008: OpenAI GPT-6 Astra / GPT-6 Sol and Luna generally available on Amazon Bedrock (AWS) <https://aws.amazon.com/about-aws/whats-new/2026/09/openai-gpt-6-astra-on-amazon-bedrock/> accessed 2026-10-07 [search-tool extract]; A5-S009: GPT-6 Astra, Sol, and Luna for production AI agents in Microsoft Foundry <https://azure.microsoft.com/en-us/blog/gpt-6-astra-sol-and-luna-for-production-agents-in-microsoft-foundry/> accessed 2026-10-07 [search-tool extract]

⁸⁷⁵ Verified fact — A5-S006: Data controls in the OpenAI platform (with Enterprise privacy, EU/UK data residency announcements) <https://developers.openai.com/api/docs/guides/your-data> accessed 2026-10-07 [search-tool extract]; V2-S079: Data controls in the OpenAI platform (with help article 'Data residency for the OpenAI API'; 'Expanding data residency access') <https://developers.openai.com/api/docs/guides/your-data> accessed 2026-10-08 [search-tool extract]

⁸⁷⁶ Verified fact — A5-S007: OpenAI Trust Portal (with Security and privacy at OpenAI; Enterprise privacy FAQ) <https://trust.openai.com/> accessed 2026-10-07 [search-tool extract]; V2-S062: OpenAI Trust Portal (with openai.com/security-and-privacy) <https://trust.openai.com/> accessed 2026-10-08 [search-tool extract]

⁸⁷⁷ Verified fact — B-L1-S001: Deprecations (OpenAI API docs) <https://developers.openai.com/api/docs/deprecations> accessed 2026-10-08 [search-tool extract]

⁸⁷⁸ Verified fact — A5-S003: OpenAI announces rollout of GPT-6 Astra model (with Axios and OpenRouter results) (CNBC) <https://www.cnbc.com/2026/09/03/open-ai-astra-gpt-6-cyber.html> accessed 2026-10-07 [search-tool extract]

- Astra is gated (rated ‘Critical’ for cybersecurity; enterprise admins must enable it), and OpenAI said ‘not yet generally available’ while AWS lists it GA on Bedrock ^{VF879}
- UK endpoint stores data in the UK but does not process there ^{VF880}
- Bedrock Astra is in-region only in us-east-1 and us-west-2 ^{VF881}
- ISO/IEC 42001 scope wording does not name the API Platform; CMK and first-party SSO/SCIM/RBAC were not verified ^{VF882 NPV}
- Preview models may be retired on about 2 weeks’ notice ^{VF883}
- Not on Google Cloud except gpt-oss ^{VF884}

Choose when

- You need a primary or fallback frontier vendor available in-tenant on Azure or AWS with a first-party EU residency option ^{AJ}
- You want one vendor relationship that also supplies a small classification tier (Luna) and a self-hostable open-weight model (gpt-oss) ^{AJ}

Avoid when

- Google Cloud is the only permitted cloud and you need GPT-6 tiers ^{AJ}
- You need UK in-country processing ^{AJ}
- You would use Promptfoo, now announced as part of OpenAI, as the only independent test of an OpenAI-based system ^{AJ}

Nearest competitors: L1-anthropic, L1-google-gemini, L1-mistral

Regulated-FS note. Consume via Microsoft Foundry EU Data Zone or the EU first-party endpoint with ZDR or Modified Abuse Monitoring; pin dated snapshots, track the deprecations page, and keep a second vendor qualified as the SS2/21 exit route ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	OpenAI	Verified fact	A5-S047, A5-S002

⁸⁷⁹ Verified fact — A5-S002: GPT-6 Astra: A new generation of intelligence (plus release notes, Introducing GPT-6 Sol and Luna, GPT-6.1 Sol, GPT-6 for everyone) (OpenAI) <https://openai.com/index/gpt-6-astra/> accessed 2026-10-07 [search-tool extract]; A5-S003: OpenAI announces rollout of GPT-6 Astra model (with Axios and OpenRouter results) (CNBC) <https://www.cnn.com/2026/09/03/open-ai-astra-gpt-6-cyber.html> accessed 2026-10-07 [search-tool extract]; A5-S008: OpenAI GPT-6 Astra / GPT-6 Sol and Luna generally available on Amazon Bedrock (AWS) <https://aws.amazon.com/about-aws/whats-new/2026/09/openai-gpt-6-astra-on-amazon-bedrock/> accessed 2026-10-07 [search-tool extract]

⁸⁸⁰ Verified fact — A5-S006: Data controls in the OpenAI platform (with Enterprise privacy, EU/UK data residency announcements) <https://developers.openai.com/api/docs/guides/your-data> accessed 2026-10-07 [search-tool extract]; V2-S079: Data controls in the OpenAI platform (with help article ‘Data residency for the OpenAI API’; ‘Expanding data residency access’) <https://developers.openai.com/api/docs/guides/your-data> accessed 2026-10-08 [search-tool extract]

⁸⁸¹ Verified fact — A5-S008: OpenAI GPT-6 Astra / GPT-6 Sol and Luna generally available on Amazon Bedrock (AWS) <https://aws.amazon.com/about-aws/whats-new/2026/09/openai-gpt-6-astra-on-amazon-bedrock/> accessed 2026-10-07 [search-tool extract]

⁸⁸² Verified fact — V2-S062: OpenAI Trust Portal (with openai.com/security-and-privacy) <https://trust.openai.com/> accessed 2026-10-08 [search-tool extract]

⁸⁸³ Verified fact — B-L1-S001: Deprecations (OpenAI API docs) <https://developers.openai.com/api/docs/deprecations> accessed 2026-10-08 [search-tool extract]

⁸⁸⁴ Verified fact — A5-S027: Generative AI pricing - Google Cloud (Gemini Enterprise Agent Platform, formerly Vertex AI) <https://cloud.google.com/gemini-enterprise-agent-platform/generative-ai/pricing> accessed 2026-10-07

Field	Value	Label	Sources
Category	Frontier model vendor: proprietary API models (reasoning, multimodal input, coding) plus Apache-2.0 open-weight models	Verified fact	A5-S002, A5-S005
Version / lineup	flagship: GPT-6 Astra, introduced 3 September 2026 per ChatGPT release notes (limited organisations first; “Path to Astra” post of 1 September explains a several-week hold for cyber safeguards); OpenAI forum post says API and Pro/Enterprise access on 4 September; Bedrock model card lists 8 September (AWS GA date). Gated: enterprise admins must enable; first OpenAI model rated “Critical” for cybersecurity. 1,050,000-token context, 128K output, knowledge cut-off 30 April 2026. [A5-S083, A5-S002, A5-S003, A5-S008]; mid: GPT-6 Sol (22 September 2026) and GPT-6.1 Sol (29 September 2026; now the def	Verified fact	A5-S001, A5-S002, A5-S003, A5-S004, A5-S005, A5-S008, A5-S009, A5-S083, V2-S008, V2-S024
Licence	Proprietary, API-only for GPT-6 and GPT-5.6 (closed weights; fine-tuning not supported for Astra). Open weights for gpt-oss under Apache 2.0 plus gpt-oss usage policy.	Verified fact	A5-S003, A5-S005
Status events	9 July 2026: GPT-5.6 Sol/Terra/Luna GA; 30 July 2026: GPT-5.6 Luna -80% and Terra -20% price cuts; 3 September 2026: GPT-6 Astra introduced (limited organisations); API from 4 September; Bedrock GA 8 September; 22 September 2026: GPT-6 Sol and Luna; 29 September 2026: GPT-6.1 Sol; 7 October 2026: GPT-6 rolled out to ChatGPT Chat	Verified fact	A5-S001, A5-S002, A5-S083, V2-S024
Strategic direction	Generation number plus capability-tier naming (Sol/Terra/Luna introduced with GPT-5.6, July 2026) so tiers can advance on separate cadences; staged, gated releases for cyber-capable models (Daybreak trusted-access programme); rapid price cuts (GPT-6 Sol/Luna 50% below GPT-5.6 promotional pricing); distribution through AWS Bedrock and Microsoft Foundry.	Verified fact	A5-S001, A5-S002, A5-S008, A5-S009
What it does	General-purpose large language models served through the OpenAI API (Responses and Chat Completions), ChatGPT, Codex and partner clouds; tiers trade capability against cost and latency.	Verified fact	A5-S002, A5-S008
Stack position	L1 foundation model. Placement in the graphic is correct; OpenAI also appears at L3 (Agents SDK) and L7 (embeddings).	Architectural judgement	
Integration	REST API (Responses, Chat Completions, Batch), official SDKs (openai Python 3.26.0, Apache-2.0, 6 October 2026); also via Bedrock (bedrock-mantle and bedrock-runtime endpoints) and Microsoft Foundry deployments.	Verified fact	A5-S047, A5-S008, A5-S009
Dependencies	Hosted service on OpenAI infrastructure (Astra trained on >100,000 GPUs at Stargate, Texas) or partner clouds (AWS Bedrock, Microsoft Foundry). gpt-oss runs on own GPUs (120b fits one H100) via vLLM, Ollama, llama.cpp.	Verified fact	A5-S003, A5-S005, A5-S008, A5-S009
Certifications	SOC 2 Type 2 (API Platform; latest report 1 July 2025 to 30 June 2026); ISO/IEC 27001:2022, 27017, 27018, 27701 (Schellman; API, ChatGPT Enterprise, Edu; not ChatGPT Business); ISO/IEC 42001:2023 AI management system (scope wording covers consumer and business AI products and models; the API Platform is not named explicitly); SOC 3; CSA STAR Level 1 self-assessment.	Verified fact	A5-S007, V2-S062
GDPR /	DPA available. API: no training on API data by default; inputs/outputs	Verified fact	A5-S006,

Field	Value	Label	Sources
residency	may be retained up to 30 days unless Zero Data Retention (approval required; some endpoints not ZDR-eligible). EU residency: eu.api.openai.com, storage and processing in region, new projects only, requires Modified Abuse Monitoring or ZDR. UK: gb.api.openai.com storage only, processing not in UK. Microsoft Foundry offers EU Data Zone deployments; Bedrock Astra in-region only in us-east-1/us-west-2 (Mantle).	d fact	A5-S009, A5-S008, V2-S079
Security features	Private Safety Processing option (customer-controlled storage under ZDR) in phased rollout; other controls (CMK, private networking) not verified in this run.	Verified fact	A5-S006
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	gpt-6-astra: US\$10 input / US\$50 output per 1M tokens (<=272K input); US\$20 / US\$75 above 272K; batch 50% off; gpt-6.1-sol / gpt-6-sol: US\$2 input / US\$10 output per 1M; gpt-6-luna: US\$0.10 input / US\$0.50 output per 1M; gpt-5.6-sol (promotional to at least 21 November 2026): US\$4 / US\$20 per 1M; Microsoft Foundry GPT-6 Sol Global Standard: US\$2 / US\$10 per 1M (Data Zone slightly higher)	Verified fact	A5-S004, A5-S001, A5-S009, V2-S008, V2-S024
Infrastructure cost	gpt-oss-120b (117B total, 5.1B active parameters) fits a single H100; gpt-oss-20b targets local use.	Verified fact	A5-S005
Ecosystem	AWS Bedrock (GPT-6 Astra/Sol/Luna, GPT-5.6, gpt-oss), Microsoft Foundry (GPT-6 family, 28 Global regions, US/EU Data Zones), Google Cloud (gpt-oss-120b/20b only).	Verified fact	A5-S008, A5-S009, A5-S027
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	GPT-6 Sol/Luna GA in API and partner clouds; GPT-6 Astra gated, not generally available per OpenAI release notes, though AWS lists it GA on Bedrock (conflict recorded).	Verified fact	A5-S002, A5-S008
Deployment	saas: Yes; managed_cloud: Yes (AWS Bedrock; Microsoft Foundry); vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes, gpt-oss only; on_prem: Yes, gpt-oss only		

Mistral AI model family (Mistral Large 4 preview; Mistral Medium 3.5; Mistral Large 3; Mistral Small 4.0; Ministral 3; Magistral; Devstral 2; Codestral; Mistral OCR; Voxtral) (L1-mistral)

Tier: Strategic • **Flags:** Superseded • **Original graphic label:** Mistral – Medium 3.1

Rationale: The strongest combination of EU residency, open weights and multi-cloud availability in the layer; Strategic as the EU and open-weight leg of the portfolio, not as the sole frontier model ⁸. 'Superseded' refers to the graphic's Medium 3.1 label.

Criterion	Score	Rationale
Technical	4	Broad lineup across general, reasoning, coding, OCR and speech, with open weights (A5-S074, A5-S038); frontier tier only in preview.
Enterprise readiness	4	First-party access controls NPV; hyperscaler routes give the presumption (rule 6). Platform controls presumed (CP2 Q1); confirm per service.
Security and compliance	3	SOC 2 Type II and ISO 27001/27701 plus EU residency (A5-S075) would anchor at 4; product scope not stated, so one point lower (rule 8).
Deployment flexibility	5	EU SaaS, three hyperscalers, Scaleway, self-host, on-premises and air-gap with open weights (A5-S074, A5-S038).
Ecosystem	4	Three hyperscalers and mainstream open-serving support (A5-S039).
Reliability and maturity	3	GA mid tier with a published retirement table (B-L1-S002); frontier preview days old; fast retirements (Medium 3.1 within about a year).
Cost / TCO	4	Medium 3.5 at US\$1.50/US\$7.50 per 1M (A5-S074) and free open weights; EU endpoint carries 10% surcharge (A5-S075).
Lock-in / portability	4	Open weights under Apache 2.0 or Modified MIT; revenue thresholds on Medium

Criterion	Score	Rationale
		3.5 (rule 4) keep it from 5.
Total (generic / FS)	3.90	
	/	
	3.85	

Capabilities. Mistral AI family: Mistral Large 4 (public preview 6 October 2026, API-only, weights targeted for 27 October 2026, licence unpublished), Mistral Medium 3.5 (28 April 2026, 128B dense, 256K context, Modified MIT v26.04), Large 3 (Apache 2.0), Small 4.0, Ministral 3, Magistral reasoning, Devstral 2 and Codestral coding, Mistral OCR and Voxtral speech ^{vr885} ^{vr886}.

Strengths

- EU hosting by default, with an EU regional endpoint (api.eu.mistral.ai) committing inference location ^{vr887}
- Open weights across tiers (Large 3 Apache 2.0; Medium 3.5 Modified MIT; Ministral 3 Apache 2.0) ^{vr888}
- Offered on Google Cloud, Bedrock and Microsoft Foundry (different subsets) ^{vr889}
- Published deprecation and retirement table with replacements ^{vr890}
- Signatory of the GPAI Code of Practice ^{vr891}

Limitations and risks

- Frontier tier is a days-old preview; Large 4 licence and list price unpublished; parameter counts differ between launch post and docs ^{vr892}
- Medium 3.5's Modified MIT licence has revenue-based exceptions ^{vr893}

⁸⁸⁵ Verified fact — A5-S074: Mistral Medium 3.5 model card; Introducing Mistral 3 (Large 3, Ministral 3); Mistral-Medium-3.5-128B (Hugging Face) (Mistral AI) <https://docs.mistral.ai/models/model-cards/mistral-medium-3-5-26-04> accessed 2026-10-07 [search-tool extract]; A5-S076: Introducing Mistral Large 4 (and Mistral changelog) (Mistral AI) <https://mistral.ai/news/mistral-large-4/> accessed 2026-10-07 [search-tool extract]; V2-S017: Mistral launches Mistral Large 4 (with VentureBeat, AI News, The Tribune) (SiliconANGLE) <https://siliconangle.com/2026/10/06/mistral-launches-open-source-mistral-large-4-details-ai-roadmap/> accessed 2026-10-08 [search-tool extract]; V2-S018: Mistral Medium 3.5 model card (with mistral.ai Vibe remote agents post, changelog) (Mistral AI) <https://docs.mistral.ai/models/model-cards/mistral-medium-3-5-26-04> accessed 2026-10-08 [search-tool extract]

⁸⁸⁶ Reported — A5-S038: LiteLLM 1.104.1 model_prices_and_context_window map (PyPI wheel) (BerriAI / LiteLLM (community-maintained)) <https://pypi.org/project/litellm/1.104.1/> accessed 2026-10-07 [search-tool extract]

⁸⁸⁷ Verified fact — A5-S075: Mistral Help Centre: SOC 2 / ISO 27001; data storage location; ZDR; Regional inference docs; Trust Center (Mistral AI) <https://help.mistral.ai/en/articles/347638-do-you-have-soc-2-or-iso-27001-certification> accessed 2026-10-07 [search-tool extract]

⁸⁸⁸ Verified fact — A5-S074: Mistral Medium 3.5 model card; Introducing Mistral 3 (Large 3, Ministral 3); Mistral-Medium-3.5-128B (Hugging Face) (Mistral AI) <https://docs.mistral.ai/models/model-cards/mistral-medium-3-5-26-04> accessed 2026-10-07 [search-tool extract]

⁸⁸⁹ Reported — A5-S027: Generative AI pricing - Google Cloud (Gemini Enterprise Agent Platform, formerly Vertex AI) <https://cloud.google.com/gemini-enterprise-agent-platform/generative-ai/pricing> accessed 2026-10-07; A5-S038: LiteLLM 1.104.1 model_prices_and_context_window map (PyPI wheel) (BerriAI / LiteLLM (community-maintained)) <https://pypi.org/project/litellm/1.104.1/> accessed 2026-10-07 [search-tool extract]

⁸⁹⁰ Verified fact — B-L1-S002: Models overview: deprecated and retired models (Mistral Docs) (Mistral AI) <https://docs.mistral.ai/models> accessed 2026-10-08 [search-tool extract]

⁸⁹¹ Verified fact — R-EU-GPAI-COP: General-Purpose AI Code of Practice (AI Act Articles 53 and 55) (see regulatory_facts.json); A8-S015: The General-Purpose AI Code of Practice (contents and signatory list) (European Commission) <https://digital-strategy.ec.europa.eu/en/policies/contents-code-gpai> accessed 2026-10-07 [search-tool extract]

⁸⁹² Verified fact — A5-S076: Introducing Mistral Large 4 (and Mistral changelog) (Mistral AI) <https://mistral.ai/news/mistral-large-4/> accessed 2026-10-07 [search-tool extract]; V2-S017: Mistral launches Mistral Large 4 (with VentureBeat, AI News, The Tribune) (SiliconANGLE) <https://siliconangle.com/2026/10/06/mistral-launches-open-source-mistral-large-4-details-ai-roadmap/> accessed 2026-10-08 [search-tool extract]

⁸⁹³ Verified fact — A5-S074: Mistral Medium 3.5 model card; Introducing Mistral 3 (Large 3, Ministral 3); Mistral-Medium-3.5-128B (Hugging Face) (Mistral AI) <https://docs.mistral.ai/models/model-cards/mistral-medium-3-5-26-04> accessed 2026-10-07 [search-tool extract]; V2-S018: Mistral Medium 3.5 model card (with mistral.ai Vibe remote agents post, changelog) (Mistral AI) <https://docs.mistral.ai/models/model-cards/mistral-medium-3-5-26-04> accessed 2026-10-08 [search-tool extract]

- Certifications stated on the help centre without product scope; ZDR only for pay-as-you-go stateless calls at Mistral's discretion; default 30-day retention ⁸⁹⁴
- Mistral Medium 3.1, the graphic's label, retired on 31 August 2026 ⁸⁹⁵
- Resells Z.ai GLM models on its platform, so a Mistral contract does not by itself exclude Chinese-origin models ⁸⁹⁶

Choose when

- You need an EU-domiciled vendor with an EU inference endpoint as primary or fallback ^{AJ}
- You want open weights from a non-Chinese vendor for a self-hosted sensitive-data tier ^{AJ}

Avoid when

- You need a GA frontier-class model today ^{AJ}
- Your licence policy cannot accept revenue-threshold licences for Medium 3.5 ^{AJ}

Nearest competitors: L1-openai, L1-google-gemma, L1-meta

Regulated-FS note. Use api.eu.mistral.ai or self-hosted weights; obtain the SOC 2 report and ISO certificate scope from the Trust Center; exclude resold third-party models via model allow-lists at the gateway ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Mistral AI (official mistralai SDK author "Mistral")	Verified fact	A5-S047
Category	Model vendor with API and open-weight models (general, reasoning, coding, OCR, speech)	Reported	A5-S038
Version / lineup	frontier_preview: Mistral Large 4, public preview 6 October 2026; ~1T total / 49-52B active MoE, multimodal, 1M context; open weights promised by end of October 2026 (27 October per VentureBeat); preview API-only in Mistral Studio. [A5-S076]; flagship_mid: Mistral Medium 3.5, 128B dense, 256K context, released 28 April 2026 (changelog lists 26 April), open weights under Modified MIT v26.04; API US\$1.5 / US\$7.5 per 1M. [A5-S074]; large_open: Mistral Large 3 (December 2025), 675B total / 41B active MoE, Apache 2.0. [A5-S074]; small: Mistral Small 4.0 (26.03); Ministral 3 3B/8B/14B (25.12). [A5-S	Verified fact	A5-S038, A5-S027, A5-S039, A5-S074, A5-S076, V2-S017, V2-S018
Licence	Mixed: Large 3 and Ministral 3 Apache 2.0; Medium 3.5 Modified MIT (revenue-based exceptions); Large 4 licence not yet published; some models API-only.	Verified fact	A5-S074, A5-S076, V2-S018
Status events	December 2025: Mistral Large 3, Ministral 3; March 2026: Mistral Small 4.0; April 2026: Mistral Medium 3.5 supersedes Medium 3.1 (dates inferred from model-card version codes); 6 October 2026: Mistral Large 4 public preview	Reported	A5-S038, A5-S076
Strategic	Not publicly verified	Not	

⁸⁹⁴ Verified fact — A5-S075: Mistral Help Centre: SOC 2 / ISO 27001; data storage location; ZDR; Regional inference docs; Trust Center (Mistral AI) <https://help.mistral.ai/en/articles/347638-do-you-have-soc-2-or-iso-27001-certification> accessed 2026-10-07 [search-tool extract]

⁸⁹⁵ Verified fact — B-L1-S002: Models overview: deprecated and retired models (Mistral Docs) (Mistral AI) <https://docs.mistral.ai/models> accessed 2026-10-08 [search-tool extract]

⁸⁹⁶ Reported — A5-S038: LiteLLM 1.104.1 model_prices_and_context_window map (PyPI wheel) (BerriAI / LiteLLM (community-maintained)) <https://pypi.org/project/litellm/1.104.1/> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
direction		publicly verified	
What it does	General, reasoning, coding, OCR and speech models via Mistral's API (La Plateforme) and partner clouds, with some models released as open weights.	Reported	A5-S038
Stack position	L1 foundation model; Mistral also appears at L3 (Agents) and L8 (Mistral OCR).	Architectural judgement	
Integration	Mistral API; mistralai Python SDK 3.1.0 (6 October 2026).	Verified fact	A5-S047
Dependencies	Not publicly verified	Not publicly verified	
Certifications	SOC 2 Type II and ISO 27001/27701 (per Mistral help centre; reports via Trust Center on request).	Verified fact	A5-S075
GDPR / residency	Data hosted in the EU by default; optional US endpoint; EU regional endpoint api.eu.mistral.ai (10% surcharge) commits inference location; some features may transfer data outside EU (listed on Trust Center). API data not used for training; default 30-day retention for abuse monitoring; ZDR only for pay-as-you-go stateless calls, at Mistral's discretion.	Verified fact	A5-S075
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Mistral Medium 3.5: US\$1.50 input / US\$7.50 output per 1M; Mistral Large 3: US\$0.50 / US\$1.50 per 1M (aggregator); Mistral Small 4.0: US\$0.15 / US\$0.60 per 1M (aggregator); Mistral Large 4 preview: 50% launch discount for 2 weeks; list price not verified	Verified fact	A5-S074, A5-S038, A5-S076
Infrastructure cost	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
Ecosystem	Google Cloud, Amazon Bedrock, Microsoft Foundry, Scaleway (Medium 3.5 128B), OpenRouter; SGLang day-0 support for Large 3.	Reported	A5-S027, A5-S038, A5-S039
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Not publicly verified	Not publicly verified	
Deployment	saas: Yes; managed_cloud: Yes (Google Cloud: Medium 3, Small 3.1, Codestral 2, OCR; Bedrock: Large 3, Devstral 2, Ministral 3, Magistral Small, Pixtral Large incl. eu., Voxtral; Microsoft Foundry: Large 3, Medium 3.5, OCR 4); vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes (open-weight models: Large 3, Medium 3.5, Ministral 3); on_prem: Yes (open-weight models)		

Anthropic Claude model family (Claude Fable 5.1, Claude Opus 5.5, Claude Sonnet 5.5, Claude Haiku 5.5; restricted Claude Mythos 5.1) (L1-anthropic)

Tier: Strategic · **Flags:** none · **Original graphic label:** Claude – Opus 5.5

Rationale: Strategic, conditional: consumed only through a hyperscaler EU or UK region (Bedrock regional endpoint or Google Cloud EU), as one of two frontier and mid-tier vendors with a non-Anthropic fallback qualified on the same evaluation suite, not as the sole frontier model; independent alternatives are OpenAI GPT-6.1 Sol, Gemini 3.8 Flash and Mistral Medium 3.5. FS 3.80 with no criterion below 3. Tier and neutral scoring set by the reader at Checkpoint 4 (CP4-1); the author’s pipeline had resolved three borderline scores against Anthropic. Security 5 and cost 4 are the neutral rubric values identified in CP4 review C (previously 4 and 3). The first-party residency gap and the June 2026 Fable 5 suspension remain factual weaknesses and are stated as conditions. Conflict of interest: the author is an Anthropic model ^{AI}.

Criterion	Score	Rationale
Technical	4	Four GA tiers with 1M context, reasoning, image input and coding (A5-S010); no open-weight, audio or image-output models and no GA specialist line, so fewer plan \$5 questions covered than OpenAI or Google; borderline 4/5 resolved against Anthropic (conflict rule). CP4 review C found 4 is also the neutral value on peers; unchanged at CP4.
Enterprise readiness	4	Workspaces, Admin API, Compliance API / Activity Feed and WIF verified (A5-S013, A5-S018) plus in-tenant hyperscaler routes (rule 6); not 5 because SSO/SCIM detail and SLA are NPV. Platform controls presumed (CP2 Q1); confirm per service.

Criterion	Score	Rationale
Security and compliance	5	Product-scoped SOC 2 Type II, ISO 27001 and ISO 42001 covering the API and Claude in Bedrock and Google Cloud, plus CMEK in the admin console (A5-S021, V2-S063, A5-S013), meet the rule-8 test for 5. Set to 5, the neutral rubric value, by the reader at Checkpoint 4 (CP4-1; previously held at 4 against Anthropic). Residency and ZDR are optional extras under the anchor, so the absence of a first-party EU/UK inference geography (V2-S075) and the top tier's conflicting ZDR position (A5-S023, A5-S016) are carried as tier conditions, not score deductions; the Azure-hosted Foundry deployment is still 'In-Process Q4 2026' for certification (V2-S063).
Deployment flexibility	3	First-party SaaS plus Bedrock, Google Cloud, Foundry and Claude Platform on AWS; no self-hosting or on-premises (A5-S010).
Ecosystem	4	Three hyperscalers and an OpenAI SDK compatibility endpoint (A5-S010); the API itself is not the de facto standard; borderline 4/5 resolved against Anthropic. CP4 review C found 4 is also the neutral value on peers; unchanged at CP4.
Reliability and maturity	3	Published retirement commitments (A5-S010) are positive, but top-tier access was suspended 12 June to 1 July 2026 (V2-S004), four models shipped in five weeks, and the D.C. Circuit upheld a federal supply-chain-risk designation (A5-S084).
Cost / TCO	4	Transparent; list prices equal OpenAI's at every tier (Table 5.1 = Astra US\$10/US\$50; Sonnet 5.5 = GPT-6.1 Sol US\$2/US\$10; Haiku 5.5 = Luna US\$0.10/US\$0.50 per 1M) with a 50% batch discount (A5-S011, A5-S004), and OpenAI scores 4. The 10% regional premium does not hold peers at 3 (Mistral and Grok score 4), and the dearer recommended default (Opus 5.5, US\$4/US\$20) is a deployment choice, not a vendor price. Set to 4, the neutral rubric value, by the reader at Checkpoint 4 (CP4-1; previously 3).
Lock-in / portability	3	Proprietary API with no open weights, but on three clouds and with an OpenAI-compatible endpoint, so a gateway can switch it out.
Total (generic / FS)	3.85 / 3.80	

Capabilities. Proprietary Claude family: Claude Fable 5.1 (top GA tier, 1 September 2026), Opus 5.5 (22 September 2026; Anthropic's recommended starting point), Sonnet 5.5 (28 September 2026) and Haiku 5.5 (7 October 2026); Mythos 5.1 is the same model as Fable 5.1 with looser safeguards for trusted-access programmes only; all four GA models have adaptive thinking, text and image input, text output, 1M-token context and 128K output; no open weights ⁸⁹⁷. Conflict of interest: the author is an Anthropic model.

Strengths

- Available on all three hyperscalers (Bedrock, Google Cloud, Microsoft Foundry) plus Claude Platform on AWS and the first-party API ⁸⁹⁸

⁸⁹⁷ Verified fact — A5-S010: Models overview - Claude API Docs (Anthropic) <https://platform.claude.com/docs/en/models/overview> accessed 2026-10-07; A5-S012: Anthropic Newsroom <https://www.anthropic.com/news> accessed 2026-10-07; A5-S016: Introducing Claude Fable 5.1 and Claude Mythos 5.1 (Anthropic) <https://www.anthropic.com/claude-fable-and-mythos-5-1> accessed 2026-10-07; A5-S019: Introducing Claude Opus 5.5 (Anthropic) <https://www.anthropic.com/claude-opus-5-5> accessed 2026-10-07; A5-S020: Introducing Claude Haiku 5.5 (Anthropic) <https://www.anthropic.com/claude-haiku-5-5> accessed 2026-10-07; V2-S066: Introducing Claude Fable 5.1 and Claude Mythos 5.1 (Anthropic) <https://www.anthropic.com/claude-fable-and-mythos-5-1> accessed 2026-10-08

⁸⁹⁸ Verified fact — A5-S010: Models overview - Claude API Docs (Anthropic) <https://platform.claude.com/docs/en/models/overview> accessed 2026-10-07

- SOC 2 Type II, ISO 27001:2022 and ISO/IEC 42001:2023 with stated scope covering the API and Claude in Bedrock and Google Cloud ^{vf899}
- Documented admin controls: Workspaces, Admin API, Workload Identity Federation, Compliance API / Activity Feed with 6-year retention, CMEK listed in the admin console ^{vf900}
- Published retirement commitments ('not sooner than' September/October 2027) for current models ^{vf901}

Limitations and risks

- No first-party EU or UK inference option: inference_geo is 'global' or 'us' only and workspace geo is 'us' only; EU processing requires Google Cloud EU or Bedrock regional endpoints at a 10% premium ^{vf902}
- Claude Fable 5 access was suspended on 12 June 2026 and restored from 1 July 2026 ^{vf903}
- Fable 5.1 is a 'Covered Model' requiring 30-day retention unless expressly authorised, which conflicts with the launch post on ZDR eligibility ^{vf904}
- On 25 September 2026 the D.C. Circuit (No. 26-1049) upheld, 2-1, the US Department of Defense's FASCSA supply-chain-risk designation of Anthropic, with effect stayed pending a rehearing petition; a separate N.D. Cal. ruling of 27 August 2026 held the other (s.3252) designation unlawful ^{vf905} ^{vf906}
- Bartz v. Anthropic US\$1.5bn class settlement over pirated books received final approval on 20 July 2026 ^{vf907} ^{vf908}
- No open weights; no audio or image output ^{vf909}
- Claude in Microsoft Foundry hosted on Azure is listed as certification 'In-Process Q4 2026' ^{vf910}
- Enterprise support tiers and SLA not publicly verified ^{NPV}

⁸⁹⁹ Verified fact — A5-S021: Trust Center - Anthropic (with help-centre article 'What Certifications has Anthropic obtained?') <https://trust.anthropic.com/> accessed 2026-10-07 [search-tool extract]; V2-S063: Trust Center - Anthropic (with 'What Certifications has Anthropic obtained?' help article; ISO 42001 announcement) <https://trust.anthropic.com/> accessed 2026-10-08 [search-tool extract]

⁹⁰⁰ Verified fact — A5-S013: Data residency - Claude Platform Docs (Anthropic) <https://platform.claude.com/docs/en/manage-claude/data-residency> accessed 2026-10-07; A5-S018: API and data retention - Claude Platform Docs (Anthropic) <https://platform.claude.com/docs/en/manage-claude/api-and-data-retention> accessed 2026-10-07

⁹⁰¹ Verified fact — A5-S010: Models overview - Claude API Docs (Anthropic) <https://platform.claude.com/docs/en/models/overview> accessed 2026-10-07

⁹⁰² Verified fact — A5-S013: Data residency - Claude Platform Docs (Anthropic) <https://platform.claude.com/docs/en/manage-claude/data-residency> accessed 2026-10-07; V2-S075: Data residency - Claude Platform Docs (Anthropic) <https://platform.claude.com/docs/en/build-with-claude/data-residency> accessed 2026-10-08; A5-S011: Pricing - Claude API Docs (Anthropic) <https://platform.claude.com/docs/en/about-claude/pricing> accessed 2026-10-07; A5-S027: Generative AI pricing - Google Cloud (Gemini Enterprise Agent Platform, formerly Vertex AI) <https://cloud.google.com/gemini-enterprise-agent-platform/generative-ai/pricing> accessed 2026-10-07

⁹⁰³ Verified fact — V2-S004: Claude Fable - Anthropic <https://www.anthropic.com/claude/fable> accessed 2026-10-08

⁹⁰⁴ Verified fact — A5-S023: Updates to Consumer Terms and Privacy Policy (Anthropic) <https://www.anthropic.com/news/updates-to-our-consumer-terms> accessed 2026-10-07; A5-S016: Introducing Claude Fable 5.1 and Claude Mythos 5.1 (Anthropic) <https://www.anthropic.com/claude-fable-and-mythos-5-1> accessed 2026-10-07

⁹⁰⁵ Verified fact — A5-S084: Anthropic PBC v. United States Department of War, No. 26-1049 (D.C. Cir., decided 25 Sept 2026) - opinion; with CNBC and Bloomberg coverage (US Court of Appeals for the D.C. Circuit) <https://media.cadc.uscourts.gov/opinions/docs/2026/09/26-1049-2194984.pdf> accessed 2026-10-07 [search-tool extract]; V2-S068: Judge blocks Pentagon blacklist of Anthropic as supply chain risk (with Justia order, Doc. 134, N.D. Cal. 3:26-cv-01996; Taft law bulletin; TechCrunch) (CNBC) <https://www.cnbc.com/2026/08/28/judge-blocks-pentagon-blacklist-anthropic.html> accessed 2026-10-08 [search-tool extract]

⁹⁰⁶ Reported — V2-S005: D.C. Circuit Upholds Exclusion of Anthropic from DOW Supply Chain Under FASCSA (with Miller & Chevalier, CCIA, ktslaw) (Holland & Knight) <https://www.hklaw.com/en/insights/publications/2026/09/dc-circuit-upholds-exclusion-of-anthropic-from-dow-supply-chain> accessed 2026-10-08 [search-tool extract]

⁹⁰⁷ Verified fact — A5-S085: Anthropic Copyright Settlement website (Bartz v. Anthropic PBC, N.D. Cal. 4:24-cv-05417); Washington Post 21 July 2026; NPR (Court-appointed settlement administrator / Washington Post /) <https://www.anthropiccopyrightsettlement.com/> accessed 2026-10-07 [search-tool extract]

⁹⁰⁸ Reported — V2-S007: Bartz v. Anthropic Settlement Receives Final Approval (with Authors Guild, Jurist) (Authors Alliance) <https://www.authorsalliance.org/2026/07/21/bartz-v-anthropic-settlement-receives-final-approval/> accessed 2026-10-08 [search-tool extract]

⁹⁰⁹ Verified fact — A5-S010: Models overview - Claude API Docs (Anthropic) <https://platform.claude.com/docs/en/models/overview> accessed 2026-10-07

Choose when

- You need a second frontier vendor that is available in-tenant on whichever hyperscaler is primary, including Google Cloud where OpenAI's GPT-6 is not offered ^{AJ}
- Long-context (1M tokens on every tier) drafting or agentic coding is the main workload ^{AJ}

Avoid when

- EU or UK processing must be contracted with the model vendor itself rather than through a hyperscaler ^{AJ}
- The firm has US defence-contract exposure and its legal team has not assessed the supply-chain-risk designation ^{AJ}
- You need a self-hostable model from the same vendor ^{AJ}

Nearest competitors: L1-openai, L1-google-gemini, L1-mistral

Regulated-FS note. Consume only through a hyperscaler EU region (Bedrock regional endpoint or Google Cloud EU), with a non-Anthropic fallback qualified on the same evaluation suite; independent alternatives are OpenAI GPT-6.1 Sol and Gemini 3.8 Flash ^{REC}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Anthropic, PBC (public benefit corporation; confidentially submitted draft S-1 for proposed IPO on 1 June 2026)	Verified fact	A5-S024
Category	Frontier model vendor: proprietary API models (reasoning, vision input, coding/agentic)	Verified fact	A5-S010
Version / lineup	flagship: Claude Fable 5.1, GA 1 September 2026, “for demanding reasoning and long-horizon agentic work”; Claude Mythos 5.1 is the same model with more permissive cyber/biology safeguards, available only via trusted-access programmes (limited US organisations). [A5-S016, A5-S022, A5-S010]; default_recommended: Claude Opus 5.5, 22 September 2026, first model of the Claude 5.5 family; Anthropic docs recommend it as the starting point for most workloads. [A5-S019, A5-S010]; mid: Claude Sonnet 5.5, 28 September 2026. [A5-S012, A5-S010]; small: Claude Haiku 5.5, 7 October 2026 (Haiku 4.5 still list	Verified fact	A5-S010, A5-S012, A5-S016, A5-S019, A5-S020, A5-S022, V2-S001, V2-S003, V2-S004, V2-S066
Licence	Proprietary; API and partner-cloud access only; no open weights.	Verified fact	A5-S010, V2-S001
Status events	4 September 2025: sales barred to entities controlled from unsupported	Verified fact	A5-S045,

⁹¹⁰ Verified fact — V2-S063: Trust Center - Anthropic (with ‘What Certifications has Anthropic obtained?’ help article; ISO 42001 announcement) <https://trust.anthropic.com/> accessed 2026-10-08 [search-tool extract]

Field	Value	Label	Sources
	regions such as China; US Department of Defense/War “supply chain risk” designation: Anthropic sued 9 March 2026; the DoD relied on two designations litigated separately - N.D. California (Judge Rita F. Lin, No. 3:26-cv-01996) granted Anthropic summary judgment on 27 August 2026 that the 10 U.S.C. s.3252 designation was unlawful (First Amendment retaliation, due process, arbitrary and capricious); D.C. Circuit (No. 26-1049, Katsas and Rao; Henderson dissenting) upheld the other under FASCSA on 25 September 2026, effect stay	d fact	A5-S025, A5-S024, A5-S026, A5-S016, A5-S019, A5-S012, A5-S020, A5-S084, A5-S085, V2-S004, V2-S005, V2-S006, V2-S007, V2-S068
Strategic direction	Tiered safeguards (Fable vs Mythos; Cyber and Life Sciences Verification Programmes); Enterprise Frontier Safeguards (customer-controlled data storage) phased from autumn 2026; price cuts (Opus 5.5 inputs/outputs 20% below Opus 5; cache reads cheaper); multi-cloud distribution; stated “pacing the frontier” policy; IPO preparation.	Verified fact	A5-S016, A5-S019, A5-S020, A5-S024
What it does	General-purpose LLMs for coding, agentic work and knowledge work, served via the Claude API, Claude apps and partner clouds.	Verified fact	A5-S010, A5-S019
Stack position	L1 foundation model. Placement correct; Anthropic also appears at L3 (Claude Agent SDK) and originated L4 standards (MCP, Agent Skills).	Architectural judgement	
Integration	Messages API, Batch API, Models API; OpenAI SDK compatibility endpoint; official SDKs (anthropic Python 1.12.0, MIT, 7 October 2026); model IDs published for Bedrock, Google Cloud, Microsoft Foundry and Claude Platform on AWS.	Verified fact	A5-S010, A5-S013, A5-S047
Dependencies	Hosted service only (Anthropic first-party, Claude Platform on AWS, Amazon Bedrock, Google Cloud, Microsoft Foundry).	Verified fact	A5-S010, A5-S018
Certifications	SOC 2 Type I and Type II, ISO 27001:2022, ISO/IEC 42001:2023 (Schellman), HIPAA-ready configuration, CSA STAR Level 2; scope covers API, Claude Enterprise/Team, Claude in Bedrock and Google Cloud; Claude in Microsoft Foundry hosted on Anthropic is in scope; the Foundry deployment hosted on Azure is listed “In-Process Q4 2026”; Claude for Government listed N/A. SOC 2 bridge letters dated August and October 2026.	Verified fact	A5-S021, A5-S017, V2-S063
GDPR / residency	Commercial/API data not used for training without express permission; consumer plans (Free/Pro/Max) opt-in training with 5-year retention since 28 August 2025 (commercial and API excluded). ZDR on request per organisation; Fable 5.1/Mythos 5.1/Fable 5/Mythos 5 are “Covered Models” requiring 30-day retention and not ZDR unless expressly authorised (conflicts with launch post saying eligible customers may use Fable 5.1 with ZDR until EFS ships). First-party inference_geo supports only “global” or “us”; workspace (storage) geo only “us”; no first-party EU/UK inference option documented. EU proces	Verified fact	A5-S018, A5-S023, A5-S013, A5-S011, A5-S027, A5-S016, V2-S075, V2-S066
Security features	Admin console lists Encryption keys (CMEK), Access Transparency, Inference hooks, Compliance API, Workload Identity Federation	Verified fact	A5-S013

Field	Value	Label	Sources
	(documentation sections).		
Access controls	Workspaces, Admin API, user management, Workload Identity Federation, Compliance API / Activity Feed (6-year retention) documented.	Verified fact	A5-S013, A5-S018
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Claude Fable 5.1: US\$10 input / US\$50 output per 1M (cache read US\$0.25); Claude Opus 5.5: US\$4 / US\$20 per 1M (cache read US\$0.20); Claude Sonnet 5.5: US\$2 / US\$10 per 1M; Claude Haiku 5.5: US\$0.10 / US\$0.50 per 1M for prompts up to 100K tokens; US\$0.50 / US\$2.50 above; modifiers: Batch API 50% off; US-only inference_geo 1.1x; full 1M context at standard price except Haiku 5.5	Verified fact	A5-S011, A5-S010, V2-S002
Infrastructure cost	Not applicable (no self-hosting).	Verified fact	A5-S010
Ecosystem	Amazon Bedrock, Google Cloud (Vertex AI / Gemini Enterprise Agent Platform), Microsoft Foundry, Claude Platform on AWS; Claude Code, Cowork.	Verified fact	A5-S010, A5-S027
Adoption signals	Anthropic-announced customers include Barclays (1 October 2026); Millennium cited in Fable 5.1 launch (vendor-selected).	Verified fact	A5-S012, A5-S016
Maturity	Four GA tiers released 1 September to 7 October 2026; published retirement commitments ("not sooner than" September/October 2027).	Verified fact	A5-S010
Deployment	saas: Yes; managed_cloud: Yes (Bedrock, Google Cloud, Microsoft Foundry, Claude Platform on AWS); vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: No; on_prem: No		

Google Gemma 4 (open-weight family) (L1-google-gemma)

Tier: Strategic · **Flags:** none · **Original graphic label:** Gemma – 2.9

Rationale: Strategic, conditional: as the small, self-hosted open-weight tier of the portfolio; not a substitute for a frontier model ^{AJ}.

Criterion	Score	Rationale
Technical	3	Competent small and medium multimodal models; no frontier tier (A5-S034).
Enterprise readiness	4	Managed Gemma 4 on Google Cloud and Bedrock gives the hyperscaler presumption (rule 6); self-hosted weights inherit host controls. Platform controls presumed (CP2 Q1); confirm per service.
Security and compliance	3	Scored as self-hosted weights (rule 2): clear Apache 2.0 licence (V2-S020); signed releases and CVE handling not evidenced; inherits host controls.
Deployment	5	Managed, self-hosted, on-premises, edge and air-gap (A5-S034).

Criterion	Score	Rationale
flexibility		
Ecosystem	4	Mainstream open-weight tooling and three hosting routes (A5-S034, A5-S027).
Reliability and maturity	4	Fourth generation, GA open release from Google (A5-S034).
Cost / TCO	4	Free weights; the larger sizes need a data-centre GPU, so operations are not trivial.
Lock-in / portability	4	Permissive open weights; governance by one vendor, not neutral, so not 5.
Total (generic / FS)	3.80	
	/	
	3.80	

Capabilities. Gemma 4 open-weight family: E2B, E4B, 26B MoE and 31B dense (31 March / 2 April 2026) and a 12B unified multimodal model (3 June 2026); image and video input on all sizes, audio on E2B/E4B; up to 256K context; Apache 2.0 ^{vf911}.

Strengths

- Apache 2.0 weights from a major vendor, runnable from phones and laptops to a single data-centre GPU ^{vf912} ^{rs913}
- Managed hosting on Google Cloud and Bedrock as well as self-hosting ^{vf914} ^{rs915}
- Broad tool support (Hugging Face, Kaggle, Ollama, Unsloth) ^{vf916}

Limitations and risks

- Small and medium sizes only; no frontier-class model ^{vf917}
- No vendor support for weights verified; security depends on the host ^{aj}
- One README links a separate Gemma 4 licence page that was not read ^{vf918}

⁹¹¹ Verified fact — A5-S034: Gemma 4: Byte for byte, the most capable open models (with Gemma releases page, Gemma 4 12B post) (Google) <https://blog.google/innovation-and-ai/technology/developers-tools/gemma-4/> accessed 2026-10-07 [search-tool extract]; A5-S035: google/gemma-4-31B-it and related Gemma 4 model cards - Hugging Face (Google (Hugging Face)) <https://huggingface.co/google/gemma-4-31B-it> accessed 2026-10-07 [search-tool extract]; V2-S020: Gemma 4: Byte for byte, the most capable open models (with Gemma releases page, Gemma 4 model card, Introducing Gemma 4 12B) (Google) <https://blog.google/innovation-and-ai/technology/developers-tools/gemma-4/> accessed 2026-10-08 [search-tool extract]

⁹¹² Verified fact — A5-S034: Gemma 4: Byte for byte, the most capable open models (with Gemma releases page, Gemma 4 12B post) (Google) <https://blog.google/innovation-and-ai/technology/developers-tools/gemma-4/> accessed 2026-10-07 [search-tool extract]; V2-S020: Gemma 4: Byte for byte, the most capable open models (with Gemma releases page, Gemma 4 model card, Introducing Gemma 4 12B) (Google) <https://blog.google/innovation-and-ai/technology/developers-tools/gemma-4/> accessed 2026-10-08 [search-tool extract]

⁹¹³ Reported — A5-S035: google/gemma-4-31B-it and related Gemma 4 model cards - Hugging Face (Google (Hugging Face)) <https://huggingface.co/google/gemma-4-31B-it> accessed 2026-10-07 [search-tool extract]

⁹¹⁴ Verified fact — A5-S027: Generative AI pricing - Google Cloud (Gemini Enterprise Agent Platform, formerly Vertex AI) <https://cloud.google.com/gemini-enterprise-agent-platform/generative-ai/pricing> accessed 2026-10-07

⁹¹⁵ Reported — A5-S038: LiteLLM 1.104.1 model_prices_and_context_window map (PyPI wheel) (BerriAI / LiteLLM (community-maintained)) <https://pypi.org/project/litellm/1.104.1/> accessed 2026-10-07 [search-tool extract]

⁹¹⁶ Verified fact — A5-S034: Gemma 4: Byte for byte, the most capable open models (with Gemma releases page, Gemma 4 12B post) (Google) <https://blog.google/innovation-and-ai/technology/developers-tools/gemma-4/> accessed 2026-10-07 [search-tool extract]

⁹¹⁷ Verified fact — A5-S034: Gemma 4: Byte for byte, the most capable open models (with Gemma releases page, Gemma 4 12B post) (Google) <https://blog.google/innovation-and-ai/technology/developers-tools/gemma-4/> accessed 2026-10-07 [search-tool extract]

⁹¹⁸ Verified fact — A5-S034: Gemma 4: Byte for byte, the most capable open models (with Gemma releases page, Gemma 4 12B post) (Google) <https://blog.google/innovation-and-ai/technology/developers-tools/gemma-4/> accessed 2026-10-07 [search-tool extract]

Choose when

- You need a self-hosted classification, extraction or redaction model inside the estate ^{AJ}
- Data must never leave the firm's infrastructure ^{AJ}

Avoid when

- The task needs frontier reasoning or long-form drafting quality ^{AJ}

Nearest competitors: L1-mistral, L1-meta, L1-alibaba-qwen

Regulated-FS note. Self-host behind the gateway with model files scanned and pinned by hash (C7); validate as a model in its own right ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Google (Google DeepMind)	Verified fact	A5-S034
Category	Open-weight model family (small/medium, multimodal, on-device)	Verified fact	A5-S034
Version / lineup	current: Gemma 4: E2B, E4B, 26B MoE (A4B), 31B Dense released 31 March 2026 (releases page) / 2 April 2026 (blog); Gemma 4 12B unified encoder-free multimodal added 3 June 2026; MTP drafters April 2026. [A5-S034]; context: 128K (E2B/E4B), up to 256K (larger models); LiteLLM lists 262,144 for 26B/31B. [A5-S034, A5-S038]; multimodal: All models take image and video input; E2B/E4B also audio. [A5-S034]; previous: Gemma 3 (1B/4B/12B/27B) still self-deployable on Google Cloud and on Bedrock. [A5-S027, A5-S038]; graphic_label_check: "2.9" is WRONG: no Gemma 2.9 exists in any source found; current ge	Verified fact	A5-S034, A5-S035, A5-S027, A5-S038, V2-S020
Licence	Open weights under Apache 2.0 (Hugging Face model cards and Google blog); one README also links a separate Gemma 4 licence page on ai.google.dev (not read).	Verified fact	A5-S034, A5-S035, V2-S020
Status events	31 March / 2 April 2026: Gemma 4 launch; 3 June 2026: Gemma 4 12B	Verified fact	A5-S034, V2-S020
Strategic direction	Edge and on-device focus (Android AICore, AI Edge Gallery), agentic features (function calling, structured JSON), Google reports 150 million Gemma 4 downloads.	Verified fact	A5-S034
What it does	Open-weight multimodal models for local, edge and self-hosted deployment, also served by Google and third parties.	Verified fact	A5-S034
Stack position	L1 open-weight model; the graphic shows it beside Gemini, but its natural deployment is self-hosted via L2 serving engines.	Architectural judgment	
Integration	Weights on Hugging Face/Kaggle/Ollama; hosted Gemma 4 on Gemini API and Google Cloud (26B), Bedrock Mantle (31B, 26B, E2B incl. US GovCloud), Cloudflare, Together, OpenRouter.	Verified fact	A5-S034, A5-S027, A5-S038
Dependencies	Runs on own hardware (laptop to GPU server) via Hugging Face, Kaggle, Ollama; Google AI Edge for devices.	Verified fact	A5-S034
Certifications	Not applicable to weights; Google Cloud hosting inherits Google Cloud certifications.	Architectural	

Field	Value	Label	Sources
GDPR / residency	Self-hosting keeps data wherever the operator runs the model.	judgement Architectural judgement	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	No vendor support for open weights verified; support via hosting cloud.	Architectural judgement	
Pricing	weights: Free (Apache 2.0); Google Cloud Gemma 4 26B managed: US\$0.15 input / US\$0.60 output per 1M	Verified fact	A5-S034, A5-S027
Infrastructure cost	E2B/E4B target phones and laptops; 26B MoE and 31B dense need a single data-centre GPU class (quantisation-aware-training GGUF releases available).	Reported	A5-S034, A5-S035
Ecosystem	Hugging Face, Kaggle, Ollama, Unsloth (QAT, MTP, GGUF, MLX), Google Cloud, Bedrock, Cloudflare Workers AI, Together, DeepInfra.	Verified fact	A5-S034, A5-S043, A5-S038
Adoption signals	Google reports 150 million Gemma 4 downloads.	Verified fact	A5-S034
Maturity	Fourth generation; GA open release.	Verified fact	A5-S034
Deployment	saas: Yes (Gemini API / AI Studio hosting); managed_cloud: Yes (Google Cloud; Amazon Bedrock); vpc_byoc: Yes (self-deploy in own cloud account); private_cloud: Yes (open weights); self_hosted: Yes; on_prem: Yes		

Google Gemini model family (Gemini 3.x: 3.1 Pro, 3.8 Flash, 3.5 Flash-Lite; Gemini 4 Argon in limited release) (L1-google-gemini)

Tier: Strategic · **Flags:** none · **Original graphic label:** Gemini – Gemini

Rationale: Strategic, conditional: where Google Cloud is your primary cloud (rule 10, the lead model service on that cloud); deployment and lock-in score 2, allowed under rule 11 as an existing platform commitment; FS 3.35 ^{AJ}.

Criterion	Score	Rationale
Technical	5	Broadest evidenced modality range plus reasoning, small, coding-oriented and cyber variants, with Gemma as Google's open-weight line (A5-S030, A5-S027); benchmarks not used.
Enterprise readiness	4	Consumed through Google Cloud: hyperscaler presumption (rule 6). Platform controls presumed (CP2 Q1); confirm per service.
Security and compliance	4	SOC 2 and ISO 42001 for Google Cloud generative AI services, UK/EU data-at-rest residency, ZDR configuration (A5-S028, A5-S029, A5-S033); ISO 27001 scope not in the fact base; stays at 4 (rules 8 and 10).
Deployment flexibility	2	Google Cloud (regional endpoints) and the Gemini API only; no other cloud, no self-hosting (A5-S027, A5-S032).
Ecosystem	3	Deep inside Google Cloud (Model Garden, agent runtime) but isolated from AWS and Azure estates (A5-S027).
Reliability and maturity	3	Flash tiers GA; Pro preview for eight months; Flash versions retire within about five to six months of release (B-L1-S003).
Cost / TCO	3	Transparent, but 3.8 Flash doubles on 1 January 2027 and regional endpoints carry a 10% premium (A5-S027, V2-S010).
Lock-in / portability	2	Proprietary API on one cloud only (A5-S027).
Total (generic / FS)	3.50	
	/	
	3.35	

Capabilities. Proprietary Gemini 3.x production lineup: 3.1 Pro (still preview since 19 February 2026), 3.8 Flash (GA 2 September 2026), 3.5 Flash-Lite, 3.8 Live and TTS models, image models and cyber-specialised Flash variants; Gemini 4 Argon announced 30 September 2026 with access restricted to trusted cyber defenders (Fairwind) and not on Google Cloud at launch; natively multimodal input (text, image, video, audio)

⁹¹⁹

Strengths

- Widest native modality coverage in the layer: text, image, video and audio input, live audio, TTS and image generation ⁹²⁰
- Google Cloud generative AI services in scope for SOC 2 and ISO/IEC 42001 ⁹²¹
- Data-at-rest residency for generative AI in the UK and several EU countries; ZDR achievable on Google Cloud by disabling logging, caching and session resumption ⁹²²

⁹¹⁹ Verified fact — A5-S027: Generative AI pricing - Google Cloud (Gemini Enterprise Agent Platform, formerly Vertex AI)

<https://cloud.google.com/gemini-enterprise-agent-platform/generative-ai/pricing> accessed 2026-10-07; A5-S030: Gemini API Release notes (with Models, Deprecations, 'What's new in Gemini 3.8 Flash' pages) (Google) <https://ai.google.dev/gemini-api/docs/changelog> accessed 2026-10-07 [search-tool extract]; A5-S031: Gemini 4 Argon: our next era of frontier intelligence (Google) <https://blog.google/innovation-and-ai/models-and-research/gemini-models/gemini-4-argon/> accessed 2026-10-07 [search-tool extract]; A5-S032: Gemini Developer API pricing (and Gemini 3.1 Pro preview model page) (Google) <https://ai.google.dev/gemini-api/docs/pricing> accessed 2026-10-07 [search-tool extract]; V2-S009: Gemini 4 Argon: our next era of frontier intelligence (with deepmind.google cyber page, Fairwind Program page) (Google) <https://blog.google/innovation-and-ai/models-and-research/gemini-models/gemini-4-argon/> accessed 2026-10-08 [search-tool extract]; V2-S010: Gemini API release notes (with Gemini 3.8 Flash model page, Gemini 3.1 Pro preview model page, Gemini Enterprise release notes) (Google) <https://ai.google.dev/gemini-api/docs/changelog> accessed 2026-10-08 [search-tool extract]

⁹²⁰ Verified fact — A5-S030: Gemini API Release notes (with Models, Deprecations, 'What's new in Gemini 3.8 Flash' pages) (Google)

<https://ai.google.dev/gemini-api/docs/changelog> accessed 2026-10-07 [search-tool extract]; A5-S027: Generative AI pricing - Google Cloud (Gemini Enterprise Agent Platform, formerly Vertex AI) <https://cloud.google.com/gemini-enterprise-agent-platform/generative-ai/pricing> accessed 2026-10-07

⁹²¹ Verified fact — A5-S028: ISO/IEC 42001 - Compliance | Google Cloud <https://cloud.google.com/security/compliance/iso-42001> accessed 2026-10-07;

A5-S037: Certifications and security for Gemini products in Google Cloud (with SOC 2 compliance page) <https://docs.cloud.google.com/gemini/docs/discover/certifications> accessed 2026-10-07 [search-tool extract]

- Per-model retirement dates published on Google Cloud model pages ^{VF}⁹²³

Limitations and risks

- Pro tier has been preview-only since February 2026; Gemini 3.5 Pro was reportedly cancelled ^{VF}⁹²⁴ ^R⁹²⁵
- Short Flash lifetimes: 3.7 Flash released 13 August 2026 retires 28 January 2027; 3.6 Flash retires 19 November 2026 ^{VF}⁹²⁶
- Google Cloud only; no AWS or Azure listing found; no self-hosting ^{VF}⁹²⁷
- Paid Gemini API (non-Google Cloud) may cache content in any country ^{VF}⁹²⁸
- 3.8 Flash introductory price rises from US\$0.75/US\$3.75 to US\$1.50/US\$7.50 per 1M on 1 January 2027; non-global endpoints cost 10% more ^{VF}⁹²⁹
- Frontier tier (Argon) not available for general enterprise use; pricing not published ^{VF}⁹³⁰ ^{NPV}

Choose when

- Google Cloud is the primary cloud ^{AJ}
- The workload needs native audio, video or live multimodal input ^{AJ}

Avoid when

- You need a GA Pro-class model with a long support window today ^{AJ}
- Your primary cloud is AWS or Azure and you do not want a cross-cloud data flow ^{AJ}
- You would use the consumer Gemini API rather than Google Cloud for client data ^{AJ}

Nearest competitors: L1-openai, L1-anthropic, L1-mistral

Regulated-FS note. Use only through Google Cloud regional (EU or UK) endpoints with logging and caching disabled, pin dated model IDs, and plan re-validation around the short Flash retirement dates ^{Rec}.

⁹²² Verified fact — A5-S029: Google Cloud generative AI data residency guarantees for data stored at-rest <https://cloud.google.com/blog/products/ai-machine-learning/google-cloud-generative-ai-data-residency-guarantees-for-data-stored-at-rest> accessed 2026-10-07; A5-S033: Gemini API Additional Terms of Service (with ZDR and logs-policy pages; Vertex AI zero data retention page) (Google) <https://ai.google.dev/gemini-api/terms> accessed 2026-10-07 [search-tool extract]

⁹²³ Verified fact — B-L1-S003: Gemini 3.7 Flash model page and related lifecycle pages (Gemini Enterprise Agent Platform) (Google Cloud) <https://docs.cloud.google.com/gemini-enterprise-agent-platform/models/gemini/3-7-flash> accessed 2026-10-08 [search-tool extract]

⁹²⁴ Verified fact — A5-S032: Gemini Developer API pricing (and Gemini 3.1 Pro preview model page) (Google) <https://ai.google.dev/gemini-api/docs/pricing> accessed 2026-10-07 [search-tool extract]

⁹²⁵ Reported — A5-S036: Google announces Gemini 4 Argon as its new frontier model (9to5Google) <https://9to5google.com/2026/09/30/gemini-4-argon-announcement/> accessed 2026-10-07 [search-tool extract]

⁹²⁶ Verified fact — B-L1-S003: Gemini 3.7 Flash model page and related lifecycle pages (Gemini Enterprise Agent Platform) (Google Cloud) <https://docs.cloud.google.com/gemini-enterprise-agent-platform/models/gemini/3-7-flash> accessed 2026-10-08 [search-tool extract]

⁹²⁷ Verified fact — A5-S027: Generative AI pricing - Google Cloud (Gemini Enterprise Agent Platform, formerly Vertex AI) <https://cloud.google.com/gemini-enterprise-agent-platform/generative-ai/pricing> accessed 2026-10-07; A5-S032: Gemini Developer API pricing (and Gemini 3.1 Pro preview model page) (Google) <https://ai.google.dev/gemini-api/docs/pricing> accessed 2026-10-07 [search-tool extract]

⁹²⁸ Verified fact — A5-S033: Gemini API Additional Terms of Service (with ZDR and logs-policy pages; Vertex AI zero data retention page) (Google) <https://ai.google.dev/gemini-api/terms> accessed 2026-10-07 [search-tool extract]

⁹²⁹ Verified fact — A5-S027: Generative AI pricing - Google Cloud (Gemini Enterprise Agent Platform, formerly Vertex AI) <https://cloud.google.com/gemini-enterprise-agent-platform/generative-ai/pricing> accessed 2026-10-07; V2-S010: Gemini API release notes (with Gemini 3.8 Flash model page, Gemini 3.1 Pro preview model page, Gemini Enterprise release notes) (Google) <https://ai.google.dev/gemini-api/docs/changelog> accessed 2026-10-08 [search-tool extract]

⁹³⁰ Verified fact — A5-S031: Gemini 4 Argon: our next era of frontier intelligence (Google) <https://blog.google/innovation-and-ai/models-and-research/gemini-models/gemini-4-argon/> accessed 2026-10-07 [search-tool extract]

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Google (Google DeepMind / Google LLC)	Verified fact	A5-S047, A5-S031
Category	Frontier model vendor: proprietary API models (reasoning, native multimodal input, live audio, TTS, image)	Verified fact	A5-S030, A5-S027
Version / lineup	frontier_restricted: Gemini 4 Argon, announced 30 September 2026; rolling out first to trusted cyber defenders (Fairwind Program); paid API and AI Ultra “next”, no date; 1M-token output limit claimed. Not on Vertex AI at launch. [A5-S031, A5-S036]; pro: Gemini 3.1 Pro, preview since 19 February 2026, still labelled preview (gemini-3.1-pro-preview); 1,048,576 input / 65,536 output tokens. Gemini 3 Pro retired 9 March 2026. Gemini 3.5 Pro never released (reported cancelled). [A5-S030, A5-S032, A5-S036]; mid_flash: Gemini 3.8 Flash GA 2 September 2026 (“most intelligent Flash”, long-horizon coding)	Verified fact	A5-S027, A5-S030, A5-S031, A5-S032, A5-S036, V2-S009, V2-S010
Licence	Proprietary, API-only.	Verified fact	A5-S032
Status events	9 March 2026: Gemini 3 Pro shut down (replaced by 3.1 Pro); 18 September 2026: Gemini 2.5 access limited to recent users; 30 September 2026: Gemini 4 Argon announced (limited); April 2026: Vertex AI renamed Gemini Enterprise Agent Platform (renames listed in 22 April 2026 release notes)	Verified fact	A5-S030, A5-S031, A5-S028, V2-S009, V2-S037
Strategic direction	Fast Flash cadence (3.5 to 3.8 between May and September 2026) with introductory pricing; Pro tier stalled at 3.1 preview; new Gemini 4 naming with Argon; cyber-specialised variants (3.5/3.8 Flash Cyber);	Verified fact	A5-S030, A5-S031, A5-S027,

Field	Value	Label	Sources
	Vertex AI renamed Gemini Enterprise Agent Platform.		A5-S028
What it does	Natively multimodal LLMs (text, image, video, audio input) via the Gemini Developer API / AI Studio and Google Cloud (Gemini Enterprise Agent Platform, formerly Vertex AI).	Verified fact	A5-S027, A5-S033
Stack position	L1 foundation model; Google also appears at L7 (Gemini Embedding) and owns the Gemma open-weight family.	Architectural judgement	
Integration	Gemini API (google-genai SDK 2.29.0, Apache-2.0, 7 October 2026), Interactions API, Live API, Google Cloud endpoints (global, multi-region, regional).	Verified fact	A5-S047, A5-S030, A5-S027
Dependencies	Hosted on Google infrastructure only.	Verified fact	A5-S027
Certifications	Google Cloud generative AI services (Gemini Enterprise Agent Platform) in scope for ISO/IEC 42001:2023 and SOC 2; Gemini in Workspace and Gemini app FedRAMP High.	Verified fact	A5-S028, A5-S037
GDPR / residency	Paid Gemini API: prompts/responses not used to improve products, processed under Google DPA, logged for a limited period for abuse detection and may be cached in any country. Free tier content used for improvement (except EEA/UK/Switzerland, where paid terms apply). Vertex/Agent Platform: no training without permission; ZDR achievable by disabling logging, session resumption and caching and avoiding Search grounding. Data-at-rest residency for generative AI in UK, Netherlands, France, Germany, Belgium etc. (2023 commitment); non-global endpoints priced 10% higher from 1 July 2026.	Verified fact	A5-S033, A5-S032, A5-S029, A5-S027
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Gemini 3.1 Pro Preview: US\$2 input / US\$12 output per 1M (<=200K); US\$4 / US\$18 above; Gemini 3.8 Flash: US\$0.75 / US\$3.75 per 1M introductory to 31 December 2026; US\$1.50 / US\$7.50 from 1 January 2027; Gemini 3.5 Flash-Lite: US\$0.30 / US\$2.50 per 1M; Gemini 4 Argon: Not publicly verified	Verified fact	A5-S027, A5-S032, A5-S030, V2-S010
Infrastructure cost	Not applicable (no self-hosting).	Architectural judgement	

Field	Value	Label	Sources
Ecosystem	Google Cloud Gemini Enterprise Agent Platform (200+ models in Model Garden), AI Studio, Antigravity agent runtime defaults to 3.8 Flash.	Verified fact	A5-S027, A5-S030
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Flash tiers GA; Pro tier preview only (3.1 Pro); frontier (Argon) restricted.	Verified fact	A5-S030, A5-S032, A5-S031
Deployment	saas: Yes; managed_cloud: Yes (Google Cloud only); vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: No; on_prem: Not publicly verified		

xAI Grok model family (Grok 4.7 flagship; 4.6, 4.5, 4.3, 4.20; Grok 4.1 Fast; grok-code-fast-1 / Grok Build) (L1-xai-grok)

Tier: Tactical • **Flags:** Acquired • **Original graphic label:** Grok

Rationale: Capable, cheap and on every hyperscaler, but the change of control, naming churn, missing ISO certification and derived-data terms keep it out of the foundation tier ^{AJ}.

Criterion	Score	Rationale
Technical	4	Flagship, reasoning, multi-agent, fast, coding and generative-media variants (A5-S079, A5-S027); no open weights verified.
Enterprise readiness	4	Admin settings and audit logs verified (A5-S080) plus hyperscaler routes (rule 6). Platform controls presumed (CP2 Q1); confirm per service.
Security and compliance	3	SOC 2 Type II, ZDR, HIPAA BAA (A5-S080); no ISO 27001 and no EU residency for the API, so the anchor-3 baseline only.
Deployment flexibility	3	First-party SaaS plus all three hyperscalers; no self-hosting verified.
Ecosystem	4	Google Cloud, Bedrock (incl. GovCloud) and Foundry listings; official SDK (A5-

Criterion	Score	Rationale
		S027, A5-S047).
Reliability and maturity	3	GA with a steady release cadence, but acquired on 2 February 2026 with naming churn since (V2-S011) and open regulatory investigations into the consumer product (V2-S070).
Cost / TCO	4	Transparent and low list prices (A5-S079); 1.1x for the US regional endpoint.
Lock-in / portability	2	Proprietary API on standard routes would be 3; reduced by 1 for the 2026 change of ownership (rule 3).
Total (generic / FS)	3.50 / 3.25	

Capabilities. Proprietary Grok family from SpaceXAI (formerly xAI): Grok 4.7 flagship (21 September 2026, 500K context, configurable reasoning effort), Grok 4.6, 4.5, 4.3 and 4.20 (reasoning, non-reasoning and multi-agent), Grok 4.1 Fast, grok-code-fast-1, image and video generation ^{VF931, R932}.

Strengths

- Available on all three hyperscalers, including AWS GovCloud for Grok 4.6 ^{VF933, R934}
- SOC 2 Type II, HIPAA-eligible deployments, team-level ZDR and admin audit logs ^{VF935}
- Low flagship list price: US\$2/US\$6 per 1M up to 200K tokens ^{VF936}

Limitations and risks

- SpaceX acquired xAI in an all-stock deal on 2 February 2026; the unit now trades as SpaceXAI, and a further rename was announced on 4 October 2026 but had not taken effect ^{VF937, R938}
- No ISO certifications found; Trust Center under NDA ^{VF939}
- Default endpoint gives no region guarantee; only a US regional endpoint is documented for the API ^{VF940}

⁹³¹ Verified fact — A5-S079: Introducing Grok 4.7 (21 September 2026) and Grok Models & Pricing docs (xAI (SpaceXAI)) <https://x.ai/news/grok-4-7> accessed 2026-10-07 [search-tool extract]; V2-S012: Grok 4.7 model page (with x.ai/news/grok-4-7, docs.x.ai pricing and release notes) (SpaceXAI (xAI)) <https://docs.x.ai/developers/grok-4-7> accessed 2026-10-08 [search-tool extract]

⁹³² Reported — A5-S038: LiteLLM 1.104.1 model_prices_and_context_window map (PyPI wheel) (BerriAI / LiteLLM (community-maintained)) <https://pypi.org/project/litellm/1.104.1/> accessed 2026-10-07 [search-tool extract]

⁹³³ Verified fact — A5-S027: Generative AI pricing - Google Cloud (Gemini Enterprise Agent Platform, formerly Vertex AI) <https://cloud.google.com/gemini-enterprise-agent-platform/generative-ai/pricing> accessed 2026-10-07

⁹³⁴ Reported — A5-S038: LiteLLM 1.104.1 model_prices_and_context_window map (PyPI wheel) (BerriAI / LiteLLM (community-maintained)) <https://pypi.org/project/litellm/1.104.1/> accessed 2026-10-07 [search-tool extract]

⁹³⁵ Verified fact — A5-S080: FAQ - API Security; Regional Endpoints; Enterprise Terms of Service (xAI/SpaceXAI) (xAI (SpaceXAI)) <https://docs.x.ai/developers/faq/security> accessed 2026-10-07 [search-tool extract]

⁹³⁶ Verified fact — A5-S079: Introducing Grok 4.7 (21 September 2026) and Grok Models & Pricing docs (xAI (SpaceXAI)) <https://x.ai/news/grok-4-7> accessed 2026-10-07 [search-tool extract]; V2-S012: Grok 4.7 model page (with x.ai/news/grok-4-7, docs.x.ai pricing and release notes) (SpaceXAI (xAI)) <https://docs.x.ai/developers/grok-4-7> accessed 2026-10-08 [search-tool extract]

⁹³⁷ Verified fact — V2-S011: xAI joins SpaceX (with TechCrunch 2 Feb 2026; Forbes/Euronews 4-5 Oct 2026 on SpaceXSI) (xAI / SpaceX) <https://x.ai/news/xai-joins-spacex> accessed 2026-10-08 [search-tool extract]

⁹³⁸ Reported — A5-S081: Musk's xAI completes public shift to SpaceXAI branding (TechBriefly, 7 July 2026) and related coverage (TechBriefly / Teslarati / Digital Applied) <https://techbriefly.com/2026/07/07/spacexai-rebrand/> accessed 2026-10-07 [search-tool extract]

⁹³⁹ Verified fact — A5-S080: FAQ - API Security; Regional Endpoints; Enterprise Terms of Service (xAI/SpaceXAI) (xAI (SpaceXAI)) <https://docs.x.ai/developers/faq/security> accessed 2026-10-07 [search-tool extract]

- Enterprise terms let the vendor create and own de-identified derived data except under ZDR ^{VF941}
- EU Commission DSA proceedings (26 January 2026) and Ofcom, ICO and Irish DPC investigations into Grok-generated images on X; no outcomes found ^{VF942} ^{R943}
- Signed only the Safety and Security chapter of the GPAI Code of Practice ^{VF944}

Choose when

- A low-cost reasoning model already offered in-tenant on your hyperscaler is wanted as a secondary or evaluation option ^{AJ}

Avoid when

- Client data would go to the first-party API without ZDR ^{AJ}
- Ownership stability or ISO 27001 is a gating criterion ^{AJ}

Nearest competitors: L1-openai, L1-anthropic, L1-google-gemini

Regulated-FS note. If used, use only through a hyperscaler region with platform data terms, and refresh third-party due diligence after the change of control ^{REC}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	xAI, acquired by SpaceX in an all-stock deal announced and closed 2 February 2026 (wholly owned subsidiary); the AI unit was rebranded SpaceXAI in mid-2026 (July per Wikipedia, 7 May per another reference; no primary source for the date); on 4 October 2026 Elon Musk said SpaceXAI will be renamed SpaceXSI, which had not taken effect as of 5 October 2026 reports	Reported	A5-S081, A5-S079, V2-S011
Category	Frontier model vendor: proprietary API models (reasoning and non-reasoning variants, vision, coding, image/video generation)	Reported	A5-S038
Version / lineup	flagship: Grok 4.7, launched 21 September 2026; 500K context; configurable reasoning effort (low to xhigh). [A5-S079]; previous_flagships: Grok 4.6 (Bedrock incl. US GovCloud, Azure, Google Cloud; xAI SDK README example), Grok 4.5. [A5-S038, A5-S027, A5-S047]; mid_reasoning: Grok 4.20 Reasoning / Non-Reasoning / Multi-agent (model ID suffix 0309), Grok 4.3; 1M context. [A5-S038, A5-S027]; small_fast: Grok 4.1 Fast Reasoning / Non-Reasoning. [A5-S027]; coding: grok-code-fast-1; grok-build. [A5-S038]; multimodal: Grok Imagine image 2.0 and video 1.5; voice transcription. [A5-S038]; open_weight:	Verified fact	A5-S038, A5-S027, A5-S047, A5-S079, V2-S012
Licence	Proprietary API (open-weight status of older Grok models not verified).	Report	A5-S038

⁹⁴⁰ Verified fact — A5-S080: FAQ - API Security; Regional Endpoints; Enterprise Terms of Service (xAI/SpaceXAI) (xAI (SpaceXAI)) <https://docs.x.ai/developers/faq/security> accessed 2026-10-07 [search-tool extract]

⁹⁴¹ Verified fact — A5-S080: FAQ - API Security; Regional Endpoints; Enterprise Terms of Service (xAI/SpaceXAI) (xAI (SpaceXAI)) <https://docs.x.ai/developers/faq/security> accessed 2026-10-07 [search-tool extract]

⁹⁴² Verified fact — V2-S070: EU opens formal investigation into X over Grok sexually explicit images (The Record; with Anadolu) <https://therecord.media/grok-sexually-explicit-images-eu-formal-investigation> accessed 2026-10-08 [search-tool extract]

⁹⁴³ Reported — A5-S082: Elon Musk’s X probed by EU over sexually explicit images on Grok (CNBC, 26 Jan 2026) with CNN, Deadline (Ofcom) and Irish DPC coverage <https://www.cnn.com/2026/01/26/elon-musk-grok-eu-explicit-images-investigation.html> accessed 2026-10-07 [search-tool extract]

⁹⁴⁴ Verified fact — R-EU-GPAI-COP: General-Purpose AI Code of Practice (AI Act Articles 53 and 55) (see regulatory_facts.json); A8-S015: The General-Purpose AI Code of Practice (contents and signatory list) (European Commission) <https://digital-strategy.ec.europa.eu/en/policies/contents-code-gpai> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
Status events	2 February 2026: SpaceX acquired xAI (all-stock; xAI became a wholly owned SpaceX subsidiary); mid-2026 (July per Wikipedia; 7 May per another reference): rebranded SpaceXAI (Grok names unchanged); 21 September 2026: Grok 4.7; January-February 2026: EU Commission DSA proceedings (26 January), Ofcom (12 January), ICO (3 February) and Irish DPC (17 February) investigations into X over Grok-generated sexualised images; no outcomes found as of spring 2026; 4 October 2026: Musk announced SpaceXAI will be renamed SpaceXSI (following a 29 September 2026 US executive order on “super intelligence” term	ed Reported	A5-S081, A5-S079, A5-S082, V2-S011, V2-S070
Strategic direction	Not publicly verified	Not publicly verified	
What it does	General-purpose reasoning LLMs and multi-agent variants via the xAI API and partner clouds.	Reported	A5-S038
Stack position	L1 foundation model; placement correct.	Architectural judgment	
Integration	xAI API; official xai-sdk (Python 1.20.0, Apache-2.0, 24 September 2026).	Verified fact	A5-S047
Dependencies	Not publicly verified	Not publicly verified	
Certifications	SOC 2 Type II (per xAI API FAQ and business pages; Trust Center under NDA); HIPAA-eligible deployments with BAA. ISO certifications not found.	Verified fact	A5-S080
GDPR / residency	API inputs/outputs not used for training without explicit permission; default 30-day encrypted retention for abuse auditing; team-level Zero Data Retention (deleted within one hour; disables stateful APIs). Enterprise terms let xAI create and own de-identified/aggregated derived data except under ZDR. US regional endpoint (us.api.x.ai) available; default endpoint gives no region guarantee; EU residency mentioned only on voice/imagine product pages. Grok on Google Cloud follows Google Cloud data terms.	Verified fact	A5-S080
Security features	Not publicly verified	Not publicly verified	
Access controls	Team-level admin settings and audit logs of administrative events (content excluded under ZDR).	Verified fact	A5-S080
Enterprise support	Not publicly verified	Not publicl	

Field	Value	Label	Sources
		y verifie d	
Pricing	Grok 4.7 (xAI API): US\$2 input / US\$6 output per 1M (<=200K); US\$4 / US\$12 above 200K; cached US\$0.50; US regional endpoint 1.1x; Grok 4.20 / 4.3 (Google Cloud): US\$1.25 / US\$2.50 per 1M; Grok 4.1 Fast (Google Cloud): US\$0.20 / US\$0.50 per 1M	Verifie d fact	A5-S079, A5-S027, V2-S012
Infrastructure cost	Not publicly verified	Not publicl y verifie d	
Ecosystem	Google Cloud (Grok 4.7, 4.6, 4.20, 4.3, 4.1 Fast), Amazon Bedrock (Grok 4.6 global/US/US-Gov), Microsoft Foundry (Grok 4.6, 4.3, 4.20, 4.1 Fast, code-fast-1).	Verifie d fact	A5-S027, A5-S038
Adoption signals	Not publicly verified	Not publicl y verifie d	
Maturity	Not publicly verified	Not publicl y verifie d	
Deployment	saas: Yes; managed_cloud: Yes (Google Cloud; Amazon Bedrock incl. US GovCloud; Microsoft Foundry); vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Not publicly verified; on_prem: Not publicly verified		

DeepSeek V4 family (DeepSeek-V4-Pro; DeepSeek-V4.1-Flash, which replaced V4-Flash in the API) (L1-deepseek)

Tier: Tactical · **Flags:** Not recommended · **Original graphic label:** DeepSeek – V4

Rationale: Tactical, conditional: self-hosted MIT weights or Microsoft Foundry in-tenant only; the ‘Not recommended’ flag applies to DeepSeek’s hosted API (PRC storage, unremediated Garante limitation) ^Λ.

Criterion	Score	Rationale
Technical	4	Large MoE reasoning models with 1M context and vision in Flash (A5-S063, A5-S064); no capability claim taken from vendor benchmarks.
Enterprise readiness	4	4 only for the Microsoft Foundry route, where Azure sells DeepSeek V4 directly and processes it in-tenant in the customer’s geography (rule 6; A5-S087); DeepSeek’s own API has no verified SSO/RBAC/audit and would be capped at 2.

Criterion	Score	Rationale
		Hosting caveat (CP4-3): on Azure, Microsoft Foundry's Kimi K3 and GLM-5.x run on Fireworks outside the customer tenant, while DeepSeek V4 is sold directly and processed in-tenant (A5-S087). Kept at 4 by the reader at Checkpoint 4 (CP4-3). Platform controls presumed (CP2 Q1); confirm per service.
Security and compliance	1	Hosted API meets anchor 1: no verified certification, PRC storage (A5-S062) and an unremediated Garante limitation (A5-S048, V2-S023). Self-hosted weights would score about 3 under rule 2 (inherits host controls), with CAISI's hijacking findings (A5-S049) as a model-behaviour risk.
Deployment flexibility	4	Self-host and air-gap possible (MIT), Foundry in-tenant, own SaaS; 4 not 5 because the vendor SaaS is PRC-only and V4 is not on Bedrock or Google Cloud (A5-S086, A5-S027).
Ecosystem	4	Broad open-serving support and third-party hosts (A5-S039, A5-S044, A5-S087).
Reliability and maturity	2	Phase-out of V4-Pro announced 10 September and reversed 17 September 2026 (V2-S013); certifications and support NPV.
Cost / TCO	3	Weights free (MIT) but V4-Pro needs a large multi-GPU cluster; hosted prices are aggregator-level and not used (V2 §4 item 3).
Lock-in / portability	4	MIT weights mean no model lock-in once self-hosted; jurisdictional exit risk is captured in security and tier.
Total (generic / FS)	3.25 / 3.15	

Evidence rules applied: security_compliance at 1 (below the NPV cap of 2): certifications NPV plus unremediated regulatory finding on the hosted API

Capabilities. DeepSeek V4 family: V4-Pro (1.6T total / 49B active, GA 13 August 2026, 1M context; a planned phase-out was reversed on 17 September 2026) and V4.1-Flash (10 September 2026, replaced V4-Flash in the API, native vision); thinking and non-thinking modes; open weights under MIT ⁹⁴⁵.

Strengths

- MIT-licensed open weights: the most permissive licence among the large Chinese-origin families ⁹⁴⁶
- Microsoft Foundry sells V4-Pro and V4-Flash directly, processed in the customer's geography or Data Zone ⁹⁴⁷
- Day-0 support in common open serving engines (SGLang, LMDeploy, KTransformers) ⁹⁴⁸

⁹⁴⁵ Verified fact — A5-S063: DeepSeek V4 Preview Release (24 April 2026) and deepseek-ai/DeepSeek-V4-Pro model card (MIT) <https://api-docs.deepseek.com/news/news260424/> accessed 2026-10-07 [search-tool extract]; A5-S064: DeepSeek-V4.1-Flash: Smarter, Faster, More Efficient (10 September 2026) and API Change Log <https://api-docs.deepseek.com/news/news260910/> accessed 2026-10-07 [search-tool extract]; A5-S065: DeepSeek Models & Pricing https://api-docs.deepseek.com/quick_start/pricing/ accessed 2026-10-07 [search-tool extract]; V2-S013: DeepSeek-V4.1-Flash: Smarter, Faster, More Efficient (with SiliconANGLE 10 Sep 2026, digitalapplied) <https://api-docs.deepseek.com/news/news260910/> accessed 2026-10-08 [search-tool extract]

⁹⁴⁶ Verified fact — A5-S063: DeepSeek V4 Preview Release (24 April 2026) and deepseek-ai/DeepSeek-V4-Pro model card (MIT) <https://api-docs.deepseek.com/news/news260424/> accessed 2026-10-07 [search-tool extract]

⁹⁴⁷ Verified fact — A5-S087: Introducing DeepSeek V4 Flash and V4 Pro in Microsoft Foundry; Foundry Models sold by Azure; Data, privacy, and security for Foundry Models sold by Azure; Bring <https://techcommunity.microsoft.com/blog/azure-ai-foundry-blog/introducing-deepseek-v4-flash-and-v4-pro-in-microsoft-foundry/4515174> accessed 2026-10-07 [search-tool extract]

⁹⁴⁸ Reported — A5-S039: SGLang README 'News' section (PyPI) (SGLang project (LMSYS)) <https://pypi.org/project/sglang/> accessed 2026-10-07 [search-tool extract]; A5-S044: LMDeploy README supported models (PyPI) (InternLM / Shanghai AI Lab (LMDeploy project)) <https://pypi.org/project/lmdeloy/> accessed 2026-10-07 [search-tool extract]; A5-S042: KTransformers README updates (PyPI) (kvcache-ai (KTransformers project)) <https://pypi.org/project/ktransformers/> accessed 2026-10-07 [search-tool extract]

Limitations and risks

- Hosted API: DeepSeek stores data in the PRC under PRC-law terms; API training position not stated; no EU/UK residency ⁹⁴⁹
- Italy's Garante imposed an urgent limitation on DeepSeek's processing of Italian users' data on 30 January 2025; no lifting or fine found ⁹⁵⁰
- Korea PIPC found unconsented transfers of prompts and device data (24 April 2025); Berlin DPA reported the app under DSA Art. 16 (27 June 2025) ⁹⁵¹
- Government-device bans: Australia, Taiwan agencies, US DoD (FY2026 NDAA s.1532) and intelligence community, several US states and Commerce bureaus, UK DWP; no US private-sector ban as of October 2026 ⁹⁵²
- NIST CAISI (September 2025) found DeepSeek R1/V3.1 more susceptible to agent hijacking and jailbreaks, with censorship risk ⁹⁵³
- Certifications, access controls and enterprise support not publicly verified ^{NPV}
- API prices are aggregator-sourced for decision purposes and not used as inputs ⁹⁵⁴

Choose when

- A firm's sovereignty policy permits Chinese-origin weights and a self-hosted or Foundry in-tenant model is needed for low-risk, non-agentic internal tasks ^{AJ}

Avoid when

- Any route sends data to DeepSeek's own API ^{AJ}
- The model would act agentially with tool access, given CAISI's hijacking findings ^{AJ}

⁹⁴⁹ Verified fact — A5-S062: DeepSeek Privacy Policy; DeepSeek Open Platform Privacy Policy and Terms of Service <https://cdn.deepseek.com/policies/en-US/deepseek-privacy-policy.html> accessed 2026-10-07 [search-tool extract]

⁹⁵⁰ Verified fact — A5-S048: Provvedimento del 30 gennaio 2025 (10098477) (and press releases 10097450, 10096856) (Garante per la protezione dei dati personali (Italy)) <https://www.garanteprivacy.it/web/guest/home/docweb/-/docweb-display/docweb/10098477> accessed 2026-10-07 [search-tool extract]; V2-S023: Generative AI and GDPR Enforcement in Europe: A Lot of Noise, One Fine, Zero Survivors (Cross-Border Data Forum) <https://www.crossborderdataforum.org/generative-ai-and-gdpr-enforcement-in-europe-a-lot-of-noise-one-fine-zero-survivors/> accessed 2026-10-08 [search-tool extract]

⁹⁵¹ Verified fact — A5-S053: Gov't confirms DeepSeek's unauthorized data transfer abroad (Korea Times, 24 April 2025) and PIPC notice (The Korea Times / Personal Information Protection Commission) <https://www.koreatimes.co.kr/amp/southkorea/globalcommunity/20250424/deepseek-transferred-korean-users-data-abroad-without-permission-govt> accessed 2026-10-07 [search-tool extract]; A5-S058: Germany tells Apple, Google to remove DeepSeek from the country's app stores (with BleepingComputer coverage) (TechCrunch) <https://techcrunch.com/2025/06/27/germany-tells-apple-google-to-remove-deepseek-from-the-countrys-app-stores> accessed 2026-10-07 [search-tool extract]

⁹⁵² Verified fact — A5-S054: Direction 001-2025 on DeepSeek Products, Applications and Web Services (PSPF) (Australian Government, Department of Home Affairs (Protectiv) <https://www.protectivesecurity.gov.au/publications-library/direction-001-2025-deepseek-products-applications-and-web-services> accessed 2026-10-07 [search-tool extract]; A5-S059: Taiwan bans government agencies from using DeepSeek (AFP via Korea Times, 1 Feb 2025) and NSB inspection of five PRC AI models (Nov 2025) (The Korea Times (AFP) / Taiwan National Security Bureau repo) <https://www.koreatimes.co.kr/amp/world/20250201/taiwan-bans-government-agencies-from-using-deepseek> accessed 2026-10-07 [search-tool extract]; A5-S057: Cyber and Artificial Intelligence Provisions in the FY2026 National Defense Authorization Act (CRS IF13197, 7 April 2026) (Congressional Research Service) <https://www.congress.gov/crs-product/IF13197> accessed 2026-10-07 [search-tool extract]; A5-S056: New York, Texas, and Virginia Ban DeepSeek AI Over Security Concerns (with Fordham IPLJ and StateTech coverage) (National Law Review) <https://natlawreview.com/article/three-states-ban-deepseek-use-state-devices-and-networks> accessed 2026-10-07 [search-tool extract]; A5-S061: UK Parliament written question HL4479 (DeepSeek) and DWP Acceptable Use Policy (gov.uk) (UK Parliament / UK Government) <https://questions-statements.parliament.uk/written-questions/detail/2025-01-28/HL4479/> accessed 2026-10-07 [search-tool extract]; A5-S060: Lawmakers probe growing use of Chinese AI models in U.S. companies (CNBC, 8 July 2026); Axios 20 July 2026; CNBC 27 August 2026 <https://www.cnbc.com/2026/07/08/chinese-ai-models-probe-us-lawmakers.html> accessed 2026-10-07 [search-tool extract]

⁹⁵³ Verified fact — A5-S049: CAISI Evaluation of DeepSeek AI Models Finds Shortcomings and Risks (30 Sept 2025; report PDF) (NIST Center for AI Standards and Innovation (US Department o) <https://www.nist.gov/news-events/news/2025/09/caisi-evaluation-deepseek-ai-models-finds-shortcomings-and-risks> accessed 2026-10-07 [search-tool extract]

⁹⁵⁴ Reported — A5-S038: LiteLLM 1.104.1 model_prices_and_context_window map (PyPI wheel) (BerriAI / LiteLLM (community-maintained)) <https://pypi.org/project/litellm/1.104.1/> accessed 2026-10-07 [search-tool extract]

- The firm has US DoD contract exposure (s.1532) ^{AJ}

Nearest competitors: L1-mistral, L1-google-gemma, L1-alibaba-qwen

Regulated-FS note. Hosted API not recommended for a regulated firm; if used at all, self-host the MIT weights or use Foundry in-tenant, behind the gateway and guardrails, with a documented sovereignty assessment ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	DeepSeek (Hangzhou DeepSeek AI; PRC). Open Platform terms governed by laws of mainland PRC; US bills name High Flyer as owner.	Reported	A5-S062, A5-S055, A5-S049
Category	Open-weight model vendor with hosted API (reasoning, coding; Flash vision experimental)	Reported	A5-S038, A5-S041
Version / lineup	flagship: DeepSeek-V4-Pro: preview 24 April 2026 (1.6T total / 49B active); GA 13 August 2026 (V4-Pro-0813); 1M context, 384K output. The planned phase-out (routing V4-Pro calls to V4.1-Flash from 14 September 2026) was reversed: the pricing-page footnote updated 17 September 2026 says the V4-Pro API continues with billing unchanged (the 10 September news post still carries the phase-out wording). [A5-S063, A5-S064, A5-S065]; small_fast: DeepSeek-V4.1-Flash: released 10 September 2026; 552B MoE, 8B/16B active; native vision; API name deepseek-flash; replaces V4-Flash (preview 24 April; official)	Verified fact	A5-S063, A5-S064, A5-S065, A5-S027, A5-S086, V2-S013
Licence	Open weights under MIT licence; also hosted API.	Verified fact	A5-S063
Status events	24 April 2026: V4 preview (Pro, Flash); 31 July 2026: V4-Flash-0731 official; 13 August 2026: V4-Pro GA; peak/off-peak pricing from 16 August; 10 September 2026: V4.1-Flash; price cuts; V4-Pro phase-out announced, then reversed on 17 September 2026 (V4-Pro API continues)	Verified fact	A5-S063, A5-S064, V2-S013

Field	Value	Label	Sources
Strategic direction	Not publicly verified	Not publicly verified	
What it does	Large mixture-of-experts LLMs released as open weights and served via DeepSeek's own API and third-party clouds.	Reported	A5-S044, A5-S038
Stack position	L1 foundation model; open weights also make it an L2 self-hosting candidate.	Architectural judgment	
Integration	DeepSeek API (OpenAI-compatible model IDs per aggregator); open weights served by SGLang, vLLM-class engines, KTransformers, LMDeploy.	Reported	A5-S038, A5-S039, A5-S042, A5-S044
Dependencies	Self-hosting requires large multi-GPU clusters (V4-Pro ~1.6T parameters); V4-Flash demonstrated on single Ascend NPU with CPU expert offload (KTransformers).	Reported	A5-S044, A5-S042
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	DeepSeek states it collects, processes and stores personal data in the PRC (consumer and Open Platform policies); Open Platform terms under mainland PRC law; consumer data may be used for training with opt-out; API training position not stated. No EU/UK residency option. Non-China processing only via third-party hosting (e.g. Microsoft Foundry) or self-hosted weights.	Verified fact	A5-S062, A5-S087
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	deepseek-v4-pro (peak): US\$1.32 input (cache miss) / US\$3.96 output per 1M; off-peak half; deepseek-flash = V4.1-Flash (peak): US\$0.30 / US\$1.20 per 1M; off-peak US\$0.15 / US\$0.60; Microsoft Foundry V4-Pro: US\$1.74 / US\$3.48 per 1M (aggregator)	Verified fact	A5-S065, A5-S038

Field	Value	Label	Sources
Infrastructure cost	V4-Pro ~1.6T and V4-Flash ~284B total parameters (LMDeploy); quantised V4-Flash runs on a single Ascend NPU with CPU offload (KTransformers).	Reported	A5-S044, A5-S042
Ecosystem	Day-0 support in SGLang; supported by LMDeploy, KTransformers, Xinference, Unsloth; hosted by Microsoft Foundry, Alibaba Model Studio, Together, Fireworks, Nebius, Databricks, OpenRouter.	Reported	A5-S039, A5-S044, A5-S042, A5-S041, A5-S043, A5-S038
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Not publicly verified	Not publicly verified	
Deployment	saas: Yes (DeepSeek API, data stored in PRC); managed_cloud: Yes: Microsoft Foundry sells V4-Pro/V4-Flash directly by Azure (customer geography, Global or DataZone processing); Bedrock and Google Cloud offer only V3.x/R1 (Bedrock V3.2 includes EU profile); vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes (open weights); on_prem: Yes (open weights)		

Z.ai (Zhipu) GLM model family (GLM-5.3, GLM-5.3-Flash, GLM-5.2, GLM-5.1, GLM-5; GLM-4.7) (L1-zai-glm)

Tier: Tactical · **Flags:** none · **Original graphic label:** QI4 (Z logo) – Q4

Rationale: Tactical, conditional: self-hosted GLM-5.3-Flash (MIT) only, after sanctions review; Entity List status and unverified data terms rule out the hosted API ⁴¹.

Criterion	Score	Rationale
Technical	4	Large agentic/coding model, a multimodal MIT Flash tier and an image model; CAISI's independent assessment places GLM-5.3 near the frontier on cyber tasks (A5-S052).
Enterprise readiness	4	The Bedrock (AWS-operated) route gives the hyperscaler presumption (rule 6): GLM 5.3 cross-Region only from 5 October 2026, GLM 5 in-Region in London (A5-S086). Hosting caveat (CP4-3): on Azure, Microsoft Foundry's Kimi K3 and GLM-5.x run on Fireworks outside the customer tenant, while DeepSeek V4 is sold directly and processed in-tenant (A5-S087). The Azure route therefore does not

Criterion	Score	Rationale
		support this score, and Z.ai API controls are NPV (would be 2). Kept at 4 by the reader at Checkpoint 4 (CP4-3). Platform controls presumed (CP2 Q1); confirm per service.
Security and compliance	2	Certifications NPV (cap 2); Singapore processing and DPA conflict (A5-S073).
Deployment flexibility	4	Own API, three hyperscalers (London in-Region for GLM 5), Mistral platform, self-host and on-premises.
Ecosystem	3	Present on many platforms, but the current GLM-5.3 reaches them via cross-Region or pass-through routes; Mistral retires GLM 5.2 on its platform on 31 October 2026 (B-L1-S002).
Reliability and maturity	2	Entity List exposure is a continuity risk for any commercial relationship (A5-S072); maturity and support NPV.
Cost / TCO	3	MIT Flash weights are cheap to run; GLM-5.3 needs multi-GPU serving; hosted prices not decision inputs (V2 \$4 item 3).
Lock-in / portability	3	MIT for Flash; MIT-style grant with a MaaS condition for GLM-5.3 (rule 4).
Total (generic / FS)	3.25 / 3.15	

Evidence rules applied: security_compliance capped at 2: certifications NPV

Capabilities. Z.ai (Zhipu) GLM-5.x family: GLM-5.3 (753B, API mid-August 2026 (14 August per CAISI, 18 August per press), weights about 28 August under a bespoke licence), GLM-5.3-Flash (26 August 2026, 320B / 18B active, natively multimodal, MIT), GLM-5.2 open weights, GLM-Image; the graphic's 'QI4' label matches no model⁹⁵⁵ ⁹⁵⁶.

Strengths

- GLM-5.3-Flash is MIT-licensed and targets consumer GPUs⁹⁵⁷ ⁹⁵⁸
- Available on Google Cloud, Bedrock (GLM 5 in-Region in London), Microsoft Foundry and Mistral's platform⁹⁵⁹ ⁹⁶⁰

⁹⁵⁵ Verified fact — A5-S071: zai-org/GLM-5.3 and GLM-5.3-Flash (Hugging Face, licences) and Z.ai pricing <https://huggingface.co/zai-org/GLM-5.3> accessed 2026-10-07 [search-tool extract]; V2-S021: CAISI's Assessment of Z.ai's GLM-5.3 Cyber Capabilities (with UK AISI / CAISI Preliminary Assessment of Kimi K3, July 2026) (NIST CAISI) <https://www.nist.gov/news-events/news/2026/09/caisis-assessment-zais-glm-53-cyber-capabilities> accessed 2026-10-08 [search-tool extract]

⁹⁵⁶ Reported — V2-S016: Z.ai Releases GLM-5.3-Flash (with testingcatalog, digitalapplied 'GLM-5.3's Weights Are Out. The Licence Is Not MIT') (MarkTechPost) <https://www.marktechpost.com/2026/08/26/z-ai-releases-glm-5-3-flash-a-320b-a18b-natively-multimodal-moe-with-a-1m-token-context/> accessed 2026-10-08 [search-tool extract]; A5-S038: LiteLLM 1.104.1 model_prices_and_context_window map (PyPI wheel) (BerriAI / LiteLLM (community-maintained)) <https://pypi.org/project/litellm/1.104.1/> accessed 2026-10-07 [search-tool extract]; A5-S042: KTransformers README updates (PyPI) (kvcache-ai (KTransformers project)) <https://pypi.org/project/ktransformers/> accessed 2026-10-07 [search-tool extract]

⁹⁵⁷ Verified fact — A5-S071: zai-org/GLM-5.3 and GLM-5.3-Flash (Hugging Face, licences) and Z.ai pricing <https://huggingface.co/zai-org/GLM-5.3> accessed 2026-10-07 [search-tool extract]

⁹⁵⁸ Reported — A5-S042: KTransformers README updates (PyPI) (kvcache-ai (KTransformers project)) <https://pypi.org/project/ktransformers/> accessed 2026-10-07 [search-tool extract]

⁹⁵⁹ Verified fact — A5-S086: Introducing Kimi K3 on Amazon Bedrock; GLM 5.3 GA on Amazon Bedrock (Oct 2026); Bedrock model cards (GLM 5, DeepSeek, Qwen3) (AWS) <https://aws.amazon.com/blogs/machine-learning/introducing-kimi-k3-on-amazon-bedrock/> accessed 2026-10-07 [search-tool extract]; A5-S027: Generative AI pricing - Google Cloud (Gemini Enterprise Agent Platform, formerly Vertex AI) <https://cloud.google.com/gemini-enterprise-agent-platform/generative-ai/pricing> accessed 2026-10-07

⁹⁶⁰ Reported — A5-S038: LiteLLM 1.104.1 model_prices_and_context_window map (PyPI wheel) (BerriAI / LiteLLM (community-maintained)) <https://pypi.org/project/litellm/1.104.1/> accessed 2026-10-07 [search-tool extract]

- NIST CAISI (September 2026) assessed GLM-5.3 as the most cyber-capable open-weight model to date, about four months behind the US frontier ^{VF961}

Limitations and risks

- Zhipu AI has been on the US Entity List since 16 January 2025, with a presumption of denial for EAR items ^{VF962}
- Data ‘generally processed in Singapore’; DPA versions conflict on storage; privacy policy cites training under legitimate interest ^{VF963}
- GLM-5.3 on Bedrock is US/global cross-Region only; Foundry’s GLM is Fireworks-operated outside the tenant ^{VF964}
- CAISI found GLM-5.2 safeguards allow agentic exploit assistance ^{VF965}
- Certifications, access controls and support not publicly verified ^{NIPV}

Choose when

- Sovereignty policy and sanctions counsel permit it, and a small MIT model (GLM-5.3-Flash) is needed self-hosted ^{AJ}

Avoid when

- Any direct commercial relationship with Z.ai would be required, until legal review of the Entity List exposure ^{AJ}
- Client data would go to the Z.ai API ^{AJ}

Nearest competitors: L1-deepseek, L1-alibaba-qwen, L1-mistral

Regulated-FS note. Self-hosted MIT weights only, after sanctions and licence review; no Z.ai API for client data ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Z.ai (Zhipu AI; listed in US Entity List as Beijing Zhipu Huazhang Technology Co., Ltd. a.k.a. Zhipu AI). International services provided by Jingsheng Hengxing Technology Pte. Ltd. (Singapore).	Verified fact	A5-S072, A5-S073
Category	Open-weight and API model vendor (agentic/coding LLMs, multimodal,	Report	A5-S038,

⁹⁶¹ Verified fact — A5-S052: CAISI Assessment of Z.ai’s GLM-5.2 (July 2026) and CAISI’s Assessment of Z.ai’s GLM-5.3 Cyber Capabilities (September 2026) (NIST CAISI) <https://www.nist.gov/news-events/news/2026/07/caisi-assessment-zais-glm-52> accessed 2026-10-07 [search-tool extract]; V2-S021: CAISI’s Assessment of Z.ai’s GLM-5.3 Cyber Capabilities (with UK AISI / CAISI Preliminary Assessment of Kimi K3, July 2026) (NIST CAISI) <https://www.nist.gov/news-events/news/2026/09/caisis-assessment-zais-glm-53-cyber-capabilities> accessed 2026-10-08 [search-tool extract]

⁹⁶² Verified fact — A5-S072: Addition of Entities to and Revision of Entry on the Entity List (90 FR 4619, 16 January 2025) (US Department of Commerce, Bureau of Industry and Security ()) <https://www.federalregister.gov/documents/2025/01/16/2025-00704/addition-of-entities-to-and-revision-of-entry-on-the-entity-list> accessed 2026-10-07 [search-tool extract]

⁹⁶³ Verified fact — A5-S073: Z.ai Privacy Policy and API Data Processing Addendum (docs.z.ai) (Z.ai (Jingsheng Hengxing Technology Pte. Ltd.)) <https://docs.z.ai/legal-agreement/privacy-policy> accessed 2026-10-07 [search-tool extract]

⁹⁶⁴ Verified fact — A5-S086: Introducing Kimi K3 on Amazon Bedrock; GLM 5.3 GA on Amazon Bedrock (Oct 2026); Bedrock model cards (GLM 5, DeepSeek, Qwen3) (AWS) <https://aws.amazon.com/blogs/machine-learning/introducing-kimi-k3-on-amazon-bedrock/> accessed 2026-10-07 [search-tool extract]; A5-S087: Introducing DeepSeek V4 Flash and V4 Pro in Microsoft Foundry; Foundry Models sold by Azure; Data, privacy, and security for Foundry Models sold by Azure; Bring <https://techcommunity.microsoft.com/blog/azure-ai-foundry-blog/introducing-deepseek-v4-flash-and-v4-pro-in-microsoft-foundry/4515174> accessed 2026-10-07 [search-tool extract]

⁹⁶⁵ Verified fact — A5-S052: CAISI Assessment of Z.ai’s GLM-5.2 (July 2026) and CAISI’s Assessment of Z.ai’s GLM-5.3 Cyber Capabilities (September 2026) (NIST CAISI) <https://www.nist.gov/news-events/news/2026/07/caisi-assessment-zais-glm-52> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
	image)	ed	A5-S041
Version / lineup	flagship: GLM-5.3 (753B; same base as GLM-5.2, post-trained; API launch mid-August 2026 (14 August per NIST CAISI, 18 August per press); weights about 28 August 2026 under a bespoke glm-5.3 licence; on Bedrock from 5 October 2026), GLM-5.2 (open weights, 1M context, June 2026). [A5-S071, A5-S086, A5-S042]; small_fast: GLM-5.3-Flash (320B total / 18B active, first natively multimodal GLM-5, MIT); GLM-5.3-FlashX. [A5-S071]; previous: GLM-5 (day-0 support 12 February 2026), GLM-5.1, GLM-5-Code, GLM-4.7 / 4.7-Flash. [A5-S042, A5-S038]; multimodal: GLM-Image. [A5-S041]; graphic_label_check: "QI4" /	Report ed	A5-S038, A5-S041, A5-S042, A5-S043, A5-S044, A5-S071, A5-S086, V2-S016, V2-S021
Licence	GLM-5.3-Flash: MIT. GLM-5.3: MIT-style grant with "Model as a Service" condition (MaaS businesses >US\$10bn revenue need Z.ai security review).	Verifie d fact	A5-S071
Status events	12 February 2026: GLM-5; 17 June 2026: GLM-5.2; 26 August 2026: GLM-5.3-Flash open weights (MIT); 16 January 2025: Zhipu AI added to US Entity List (presumption of denial for EAR items); 5 October 2026: GLM 5.3 GA on Amazon Bedrock (eligible enterprise customers); 14 August 2026 (CAISI) / 18 August 2026 (press): GLM-5.3 released; weights about 28 August 2026; September 2026: NIST CAISI assessment finds GLM-5.3 the most cyber-capable open-weight model to date, about four months behind the US frontier	Report ed	A5-S042, A5-S072, A5-S086, V2-S016, V2-S021
Strategic direction	Not publicly verified	Not publicl y verifie d	
What it does	Large MoE LLMs for agentic and coding workloads released as open weights and via the Z.ai API.	Report ed	A5-S043, A5-S038
Stack position	L1 foundation model.	Archite ctural judge ment	
Integration	Z.ai API (zai-sdk 0.2.3, 16 June 2026); open weights via SGLang, KTransformers, Unsloth, LMDeploy, Xinference; also resold on Mistral's platform (zai-glm-5-3).	Verifie d fact	A5-S047, A5-S039, A5-S038
Dependencies	GLM-5.2 ~744-754B parameters requires multi-GPU serving; NVFP4 serving reported at 500 TPS with SGLang.	Report ed	A5-S043, A5-S044, A5-S039
Certifications	Not publicly verified	Not publicl y verifie d	
GDPR / residency	Personal data "generally processed in Singapore"; may transfer to Z.ai Group affiliates overseas. Current DPA says API content is not stored (older version said temporary storage - conflict). General privacy policy cites model training under legitimate interest; API terms do not expressly exclude API inputs from training. No EU residency option	Verifie d fact	A5-S073, A5-S086

Field	Value	Label	Sources
	found. Bedrock GLM 5 in-Region in eu-west-2 (London); GLM 5.3 via US/global cross-Region only.		
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	GLM-5.3 (Z.ai): US\$1.40 input / US\$4.40 output per 1M; cached US\$0.26; GLM-5.3-Flash: US\$0.15 / US\$0.50 per 1M; GLM-5.2 (Google Cloud): US\$1.40 / US\$4.40 per 1M	Verified fact	A5-S071, A5-S027
Infrastructure cost	GLM-5.2 runnable locally only with heavy quantisation (Unsloth dynamic GGUFs); GLM-5.3-Flash targets consumer GPUs.	Reported	A5-S043, A5-S042
Ecosystem	Google Cloud, Bedrock, Microsoft Foundry, Mistral platform; SGLang, KTransformers, Unsloth, LMDeploy, XInference.	Reported	A5-S027, A5-S038, A5-S039, A5-S042, A5-S043, A5-S044, A5-S041
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Not publicly verified	Not publicly verified	
Deployment	saas: Yes (Z.ai API); managed_cloud: Yes (Google Cloud GLM-4.7/5/5.2; Amazon Bedrock GLM-5/4.7 US regions; Microsoft Foundry Fireworks-operated GLM-5 to 5.3; Mistral platform GLM-5.x); vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes (open weights); on_prem: Yes (open weights)		

Meta Muse family (Muse Spark 1.1-1.3 via Meta Model API; Muse Glimmer 30B open weights; Muse Code; Muse Image) plus Llama 4 (Scout, Maverick) (L1-meta)

Tier: Tactical · **Flags:** Renamed · **Original graphic label:** Meta – Llama (new: Muse)

Rationale: Brand transition from Llama to Muse is under way, the API is weeks past GA and its data terms are unverified; useful for open-weight continuity only ⁴¹.

Criterion	Score	Rationale
Technical	3	API flagship, small open model, code and image models (A5-S077, A5-S078); capability evidence beyond names is thin and Llama has not been refreshed since April 2025.
Enterprise readiness	4	Muse Spark 1.3 on Foundry and Llama 4 on all three hyperscalers give the presumption (rule 6); Meta Model API controls NPV. Platform controls presumed (CP2 Q1); confirm per service.
Security and compliance	2	Certifications and data terms NPV (cap 2); contributor tier trains on data (V2-S019).
Deployment flexibility	4	Meta API, hyperscalers and self-hosted open weights.
Ecosystem	4	Llama’s broad hosting footprint plus Foundry, OCI and Google Cloud preview for Muse (A5-S077, A5-S027).
Reliability and maturity	2	Brand transition in 2026, API GA only since late September 2026, stale SDK (A5-S047).
Cost / TCO	3	Muse prices are aggregator-level; Llama 4 on Google Cloud is cheap (A5-S027); contributor pricing is excluded as a data-use condition.
Lock-in / portability	3	Mixed: Apache 2.0 (Glimmer, secondary), community licence with MAU threshold (Llama 4, rule 4), API-only flagship.

Criterion	Score	Rationale
Total (generic / FS)	3.15 / 3.05	

Evidence rules applied: security_compliance capped at 2: certifications NPV

Capabilities. Meta Muse family from Meta Superintelligence Labs: Muse Spark (announced April 2026; 1.1 to 1.3 via the Meta Model API, GA at Meta Connect 23-24 September 2026; 1M context), Muse Glimmer 30B open weights (August 2026, Apache 2.0 per secondary sources), Muse Code and Muse Image; plus Llama 4 Scout and Maverick under the Llama 4 Community License ^{vr⁹⁶⁶ r⁹⁶⁷}.

Strengths

- Llama 4 is available on Google Cloud, Bedrock and Foundry; Muse Spark 1.3 is on Foundry ^{vr⁹⁶⁸ r⁹⁶⁹}
- An open small model (Glimmer 30B) alongside the API flagship ^{vr⁹⁷⁰ r⁹⁷¹}

Limitations and risks

- Meta Model API data terms, certifications and EU residency not publicly verified ^{NPV}
- The 'contributor' tier is discounted in exchange for Meta training on prompts and completions: a data-use condition, not a cheaper price ^{vr⁹⁷²}
- Llama 4 Community License requires a separate licence above 700M MAU; no Llama release since Llama 4 ^{vr⁹⁷³ r⁹⁷⁴}
- Official llama-api-client SDK last released 18 December 2025 ^{vr⁹⁷⁵}

⁹⁶⁶ Verified fact — A5-S077: Introducing Muse Spark (Meta, April 2026); Introducing Muse Spark 1.1 / Meta Model API (ai.meta.com); Meta Connect 2026 recap <https://about.fb.com/news/2026/04/introducing-muse-spark-meta-superintelligence-labs/> accessed 2026-10-07 [search-tool extract]; A5-S078: meta-models/Muse-Glimmer-30B model card (Apache 2.0) and Llama 4 Community License Agreement (llama.com) (Meta) <https://huggingface.co/meta-models/Muse-Glimmer-30B> accessed 2026-10-07 [search-tool extract]; V2-S019: Meta Connect 2026: The end-to-end recap (with research.meta.ai 'Introducing Muse Spark 1.3') <https://developers.meta.com/blog/meta-connect-recap/> accessed 2026-10-08 [search-tool extract]; A5-S027: Generative AI pricing - Google Cloud (Gemini Enterprise Agent Platform, formerly Vertex AI) <https://cloud.google.com/gemini-enterprise-agent-platform/generative-ai/pricing> accessed 2026-10-07

⁹⁶⁷ Reported — A5-S038: LiteLLM 1.104.1 model_prices_and_context_window map (PyPI wheel) (BerriAI / LiteLLM (community-maintained)) <https://pypi.org/project/litellm/1.104.1/> accessed 2026-10-07 [search-tool extract]; A5-S040: ms-swift README news (PyPI) (ModelScope / Alibaba (ms-swift project)) <https://pypi.org/project/ms-swift/> accessed 2026-10-07 [search-tool extract]

⁹⁶⁸ Verified fact — A5-S027: Generative AI pricing - Google Cloud (Gemini Enterprise Agent Platform, formerly Vertex AI) <https://cloud.google.com/gemini-enterprise-agent-platform/generative-ai/pricing> accessed 2026-10-07

⁹⁶⁹ Reported — A5-S038: LiteLLM 1.104.1 model_prices_and_context_window map (PyPI wheel) (BerriAI / LiteLLM (community-maintained)) <https://pypi.org/project/litellm/1.104.1/> accessed 2026-10-07 [search-tool extract]

⁹⁷⁰ Verified fact — A5-S078: meta-models/Muse-Glimmer-30B model card (Apache 2.0) and Llama 4 Community License Agreement (llama.com) (Meta) <https://huggingface.co/meta-models/Muse-Glimmer-30B> accessed 2026-10-07 [search-tool extract]

⁹⁷¹ Reported — A5-S040: ms-swift README news (PyPI) (ModelScope / Alibaba (ms-swift project)) <https://pypi.org/project/ms-swift/> accessed 2026-10-07 [search-tool extract]

⁹⁷² Verified fact — V2-S019: Meta Connect 2026: The end-to-end recap (with research.meta.ai 'Introducing Muse Spark 1.3') <https://developers.meta.com/blog/meta-connect-recap/> accessed 2026-10-08 [search-tool extract]

⁹⁷³ Verified fact — A5-S077: Introducing Muse Spark (Meta, April 2026); Introducing Muse Spark 1.1 / Meta Model API (ai.meta.com); Meta Connect 2026 recap <https://about.fb.com/news/2026/04/introducing-muse-spark-meta-superintelligence-labs/> accessed 2026-10-07 [search-tool extract]

⁹⁷⁴ Reported — A5-S038: LiteLLM 1.104.1 model_prices_and_context_window map (PyPI wheel) (BerriAI / LiteLLM (community-maintained)) <https://pypi.org/project/litellm/1.104.1/> accessed 2026-10-07 [search-tool extract]

⁹⁷⁵ Verified fact — A5-S047: Official vendor Python SDK metadata (openai, anthropic, google-genai, xai-sdk, mistralai, dashscope, zai-sdk, llama-api-client) via PyPI JSON API (PyPI (vendor-published package metadata)) <https://pypi.org/> accessed 2026-10-07 [search-tool extract]

- Glimmer's Apache 2.0 licence rests on secondary sources ⁹⁷⁶

Choose when

- You already self-host Llama 4 and want continuity, or you want Glimmer 30B as an additional open-weight option ^{AJ}

Avoid when

- Client data would go to the Meta Model API before its data terms are verified ^{AJ}
- Any route would use the contributor tier ^{AJ}

Nearest competitors: L1-google-gemma, L1-mistral, L1-openai

Regulated-FS note. Block the contributor tier at the gateway; use Llama 4 or Glimmer only self-hosted or via a hyperscaler ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Meta (models built by Meta Superintelligence Labs)	Verified fact	A5-S077, A5-S078
Category	Model vendor: hosted API models (Muse Spark) and open-weight models (Muse Glimmer, Llama 4)	Reported	A5-S038, A5-S040
Version / lineup	flagship_api: Muse Spark announced April 2026 (Meta Superintelligence Labs); Muse Spark 1.1 with public Meta Model API preview 9 July 2026; 1.2 on 5 August 2026; 1.3 on 2 September 2026; Meta Model API GA globally (announced at Meta Connect, 23-24 September 2026; pre-Connect developer page still said public preview); 1,048,576 context; OpenAI/Anthropic-SDK compatible; on Oracle Cloud, Microsoft Foundry, Google Cloud private preview. [A5-S077, A5-S038]; open_weight: Muse Glimmer 30B, released August 2026 under Apache 2.0 (distilled from Muse Spark, agentic, consumer hardware). [A5-S078, A5-S040]	Verified fact	A5-S038, A5-S040, A5-S027, A5-S047, A5-S077, A5-S078, V2-S019
Licence	Muse Spark: hosted API only. Muse Glimmer 30B: Apache 2.0. Llama 4: Llama 4 Community License (attribution; >700M MAU need separate licence; California law).	Verified fact	A5-S077, A5-S078
Status events	April 2026: Muse Spark announced; 9 July 2026: Meta Model API public preview (Muse Spark 1.1); 5 August 2026: Muse Spark 1.2, Muse Code; August 2026: Muse Glimmer 30B (Apache 2.0); 2 September 2026: Muse Spark 1.3; Meta Connect 2026: Meta Model API GA	Verified fact	A5-S077, A5-S078, V2-S019
Strategic	Frontier effort moved to Meta Superintelligence Labs' Muse brand:	Verified	A5-S077,

⁹⁷⁶ Reported — V2-S019: Meta Connect 2026: The end-to-end recap (with research.meta.ai 'Introducing Muse Spark 1.3') <https://developers.meta.com/blog/meta-connect-recap/> accessed 2026-10-08 [search-tool extract]

Field	Value	Label	Sources
direction	hosted API flagship (Muse Spark) plus a permissively licensed small open model (Muse Glimmer), distribution via Oracle, Microsoft and Google clouds.	d fact	A5-S078
What it does	Multimodal reasoning LLMs via Meta's developer API and partner clouds; open-weight models for self-hosting.	Report ed	A5-S038, A5-S040
Stack position	L1 foundation model.	Archite ctural judge ment	
Integration	Meta developer API (Llama API client SDK); Microsoft Foundry; open weights via ModelScope/Hugging Face and inference providers.	Report ed	A5-S047, A5-S038, A5-S040
Dependencies	Not publicly verified	Not publicl y verifie d	
Certifications	Not publicly verified	Not publicl y verifie d	
GDPR / residency	Not publicly verified	Not publicl y verifie d	
Security features	Not publicly verified	Not publicl y verifie d	
Access controls	Not publicly verified	Not publicl y verifie d	
Enterprise support	Not publicly verified	Not publicl y verifie d	
Pricing	Muse Spark 1.3 (Meta API): US\$1.25 input / US\$4.25 output per 1M (aggregator); contributor variant US\$0.10 / US\$0.20; Muse Glimmer 30B (Together): US\$0.35 / US\$1.50 per 1M; Llama 4 Maverick (Google Cloud): US\$0.35 / US\$1.15 per 1M; Llama 4 Scout (Google Cloud): US\$0.25 / US\$0.70 per 1M	Report ed	A5-S038, A5-S027
Infrastructure	Not publicly verified	Not	

Field	Value	Label	Sources
cost		publicly verified	
Ecosystem	Microsoft Foundry, Google Cloud, Amazon Bedrock, Together, Fireworks, DeepInfra, OpenRouter, ms-swift.	Reported	A5-S038, A5-S027, A5-S040
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Not publicly verified	Not publicly verified	
Deployment	saas: Yes (Meta API); managed_cloud: Yes (Microsoft Foundry Muse Spark 1.3; Llama 4 on Google Cloud, Bedrock, Foundry); vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes (Muse Glimmer 30B, Llama 4); on_prem: Yes (open-weight models)		

Alibaba Qwen model family (Qwen3.8: Max, Flash, Omni-Flash API; open-weight Qwen3.8-27B, Qwen3.8-2.4T-A95B, Qwen3.8-Flash-Next) (L1-alibaba-qwen)

Tier: Tactical · Flags: none · Original graphic label: Qwen – 3.8

Rationale: Tactical, conditional: self-hosted Apache-2.0 sizes only, subject to sovereignty policy; hosted Max and Model Studio lack verified certifications and controls ^N.

Criterion	Score	Rationale
Technical	4	Dense and MoE sizes, omni-modal, coding and embedding variants (A5-S068, A5-S038); evidence is partly secondary.
Enterprise readiness	2	Qwen3.8 is on no hyperscaler, so rule 6 does not apply; Model Studio SSO/RBAC/audit NPV (cap 2).
Security and compliance	2	Certifications NPV (cap 2); Frankfurt region and no-training statement are positives (A5-S067).
Deployment	4	Model Studio (EU scope available) plus self-host and on-premises for open sizes;

Criterion	Score	Rationale
flexibility		no current-generation hyperscaler route.
Ecosystem	4	Broad open-serving and third-party hosting support (A5-S038, A5-S041, A5-S043).
Reliability and maturity	3	Frequent generations (3.5 to 3.8 within 2026); maturity evidence NPV; 3 on observed cadence.
Cost / TCO	3	Apache-2.0 small sizes are cheap to run; hosted prices are not decision inputs (V2 \$4 item 3); flagship needs multi-node serving.
Lock-in / portability	3	Apache-2.0 for small sizes; custom licence with a revenue threshold for the flagship (rule 4).
Total (generic / FS)	3.15 / 3.00	

Evidence rules applied: enterprise_readiness capped at 2: SSO/RBAC/audit NPV and no current-generation hyperscaler route; security_compliance capped at 2: certifications NPV

Capabilities. Qwen3.8 family: hosted qwen3.8-max (launched 3 August 2026), qwen3.8-flash and omni-flash through Alibaba Cloud Model Studio; open weights Qwen3.8-27B (Apache 2.0), Qwen3.8-2.4T-A95B (12 August 2026, custom Qwen3.8-Max License) and Qwen3.8-Flash-Next; coding and embedding lines ^{vf977} ^{r978}.

Strengths

- Widest open-weight size range in the layer (27B to 2.4T total) ^{vf979} ^{r980}
- Model Studio is region-bound, with a Frankfurt EU scope, and says it never trains on customer data ^{vf981}
- Qwen3.8-27B under Apache 2.0 ^{vf982}

Limitations and risks

- Qwen3.8 is not offered on any hyperscaler; Bedrock and Google Cloud carry only Qwen3 ^{vf983}

⁹⁷⁷ Verified fact — A5-S066: Alibaba Unveils Qwen3.8-27B and Releases Weights of Qwen3.8 Flagship Model (with Qwen/Qwen3.8-27B model card) (Alibaba Cloud) https://www.alibabacloud.com/blog/alibaba-unveils-qwen3-8-27b-and-releases-weights-of-qwen3-8-flagship-model_603463 accessed 2026-10-07 [search-tool extract]; A5-S068: Alibaba Cloud Model Studio model pricing (qwen3.8-max, qwen3.8-flash) and Alizila Qwen3.8-Flash release <https://www.alibabacloud.com/help/en/model-studio/model-pricing> accessed 2026-10-07 [search-tool extract]

⁹⁷⁸ Reported — V2-S014: Alibaba's AI model Qwen3.8-Max made widely accessible ahead of open-weights release (with Quartz, MarkTechPost 3 Aug 2026, The Decoder, DeepLearning.AI The Batc (South China Morning Post) <https://www.scmp.com/tech/article/3362738/alibabas-ai-model-qwen3-8-max-made-widely-accessible-ahead-open-weights-release> accessed 2026-10-08 [search-tool extract]; A5-S038: LiteLLM 1.104.1 model_prices_and_context_window map (PyPI wheel) (BerriAI / LiteLLM (community-maintained)) <https://pypi.org/project/litellm/1.104.1/> accessed 2026-10-07 [search-tool extract]; A5-S040: ms-swift README news (PyPI) (ModelScope / Alibaba (ms-swift project)) <https://pypi.org/project/ms-swift/> accessed 2026-10-07 [search-tool extract]

⁹⁷⁹ Verified fact — A5-S066: Alibaba Unveils Qwen3.8-27B and Releases Weights of Qwen3.8 Flagship Model (with Qwen/Qwen3.8-27B model card) (Alibaba Cloud) https://www.alibabacloud.com/blog/alibaba-unveils-qwen3-8-27b-and-releases-weights-of-qwen3-8-flagship-model_603463 accessed 2026-10-07 [search-tool extract]

⁹⁸⁰ Reported — A5-S041: Xinference README latest-release notes (PyPI) (Xorbits (xinference project)) <https://pypi.org/project/xinference/> accessed 2026-10-07 [search-tool extract]

⁹⁸¹ Verified fact — A5-S067: Regions and endpoints; FAQ - Alibaba Cloud Model Studio <https://www.alibabacloud.com/help/en/model-studio/regions> accessed 2026-10-07 [search-tool extract]

⁹⁸² Verified fact — A5-S066: Alibaba Unveils Qwen3.8-27B and Releases Weights of Qwen3.8 Flagship Model (with Qwen/Qwen3.8-27B model card) (Alibaba Cloud) https://www.alibabacloud.com/blog/alibaba-unveils-qwen3-8-27b-and-releases-weights-of-qwen3-8-flagship-model_603463 accessed 2026-10-07 [search-tool extract]

⁹⁸³ Verified fact — A5-S086: Introducing Kimi K3 on Amazon Bedrock; GLM 5.3 GA on Amazon Bedrock (Oct 2026); Bedrock model cards (GLM 5, DeepSeek, Qwen3) (AWS) <https://aws.amazon.com/blogs/machine-learning/introducing-kimi-k3-on-amazon-bedrock/> accessed 2026-10-07 [search-tool extract]; A5-S027: Generative AI pricing - Google Cloud (Gemini Enterprise Agent Platform, formerly Vertex AI) <https://cloud.google.com/gemini->

- Flagship weights use a custom licence: Model-as-a-Service businesses above US\$50m trailing revenue need a separate licence (licence file not read) ⁹⁸⁴
- Taiwan's NSB (November 2025) found security and bias failings in Tongyi/Qwen among five PRC models; no CAISI evaluation of Qwen found ⁹⁸⁵
- Reported July 2026 consideration in Beijing of restricting overseas access to leading Chinese models including open-weight Qwen (unenacted) ⁹⁸⁶
- Certifications, access controls and support for Model Studio not publicly verified ^{NPV}

Choose when

- A small Apache-2.0 model (Qwen3.8-27B) is needed self-hosted and the firm's sovereignty policy permits Chinese-origin weights ^{AJ}

Avoid when

- The firm needs a hyperscaler-hosted current-generation route ^{AJ}
- You would deploy the flagship weights without legal review of the custom licence ^{AJ}

Nearest competitors: L1-google-gemma, L1-mistral, L1-deepseek

Regulated-FS note. Self-host the Apache-2.0 sizes only; treat Model Studio Frankfurt as unassessed until certifications are verified ^{Rec.}

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Alibaba Cloud (Qwen team); official dashscope SDK published by Alibaba Cloud	Verified fact	A5-S047
Category	Open-weight and API model vendor (general, reasoning, coding, omni-modal, embeddings)	Reported	A5-S038, A5-S040
Version / lineup	flagship_api: qwen3.8-max (snapshot qwen3.8-max-0902), 1M context, 128K output; US\$2 / US\$6 per 1M (International scope). [A5-S068, A5-S038]; small_api: qwen3.8-flash and qwen3.8-omni-flash, ~992K context. [A5-S038]; open_weight: Qwen3.8-2.4T-A95B flagship weights (hosted Qwen3.8-Max launched 3 August 2026; weights 12 August 2026; ~95B active; open weights reportedly text-only and without the full 1M context; custom Qwen3.8-Max License), Qwen3.8-27B (Apache 2.0, 262K native context), Qwen3.8-Flash (open-weight multimodal MoE, ~late August 2026), Qwen3.8-Flash-Next. [A5-S066, A5-S068, A5-	Verified fact	A5-S038, A5-S040, A5-S041, A5-S044, A5-S027, A5-S066, A5-S068, V2-S014

[enterprise-agent-platform/generative-ai/pricing](#) accessed 2026-10-07

⁹⁸⁴ Reported — V2-S014: Alibaba's AI model Qwen3.8-Max made widely accessible ahead of open-weights release (with Quartz, MarkTechPost 3 Aug 2026, The Decoder, DeepLearning.AI The Batc (South China Morning Post) <https://www.scmp.com/tech/article/3362738/alibabas-ai-model-qwen38-max-made-widely-accessible-ahead-open-weights-release> accessed 2026-10-08 [search-tool extract]

⁹⁸⁵ Verified fact — A5-S059: Taiwan bans government agencies from using DeepSeek (AFP via Korea Times, 1 Feb 2025) and NSB inspection of five PRC AI models (Nov 2025) (The Korea Times (AFP) / Taiwan National Security Bureau repo) <https://www.koreatimes.co.kr/amp/world/20250201/taiwan-bans-government-agencies-from-using-deepseek> accessed 2026-10-07 [search-tool extract]; A5-S051: UK AISI / CAISI Preliminary Assessment of Kimi K3's Cyber Capabilities (July 2026); CAISI Evaluation of Kimi K2 Thinking (Dec 2025) (NIST CAISI and UK AI Security Institute) <https://www.nist.gov/news-events/news/2026/07/uk-aisi-caisi-preliminary-assessment-kimi-k3s-cyber-capabilities> accessed 2026-10-07 [search-tool extract]

⁹⁸⁶ Reported — A5-S060: Lawmakers probe growing use of Chinese AI models in U.S. companies (CNBC, 8 July 2026); Axios 20 July 2026; CNBC 27 August 2026 <https://www.cnbc.com/2026/07/08/chinese-ai-models-probe-us-lawmakers.html> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
Licence	S040, Mixed: Qwen3.8-27B Apache 2.0; Qwen3.8-2.4T-A95B (open weights 12 August 2026) under a custom “Qwen3.8-Max License”: Model-as-a-Service or AI-assistant businesses with over US\$50m trailing revenue need a separate licence, with attribution duties for very large products (Reported; licence file not read); Max/Plus API tiers proprietary; Qwen-Image-2.1 reportedly research-only.	Reported	A5-S066, V2-S014
Status events	February 2026: Qwen3.5 (LMDeploy support); 26 August 2026: Qwen3.8-Flash-Next (ms-swift day-0); 3 August 2026: Qwen3.8-Max hosted launch (Reported); 12 August 2026: Qwen3.8-2.4T-A95B open weights under Qwen3.8-Max License (Reported)	Reported	A5-S044, A5-S040, V2-S014
Strategic direction	Not publicly verified	Not publicly verified	
What it does	Family of dense and MoE LLMs, multimodal and coding variants, offered as open weights and via Alibaba Cloud Model Studio (DashScope) API.	Reported	A5-S038, A5-S041
Stack position	L1 foundation model; Qwen3 embeddings also appear at L7.	Architectural judgement	
Integration	DashScope API (dashscope SDK 1.27.7, Apache 2.0, 24 September 2026); open weights via Hugging Face/ModelScope served by vLLM, SGLang, LMDeploy, Xinference.	Verified fact	A5-S047, A5-S041
Dependencies	Self-hosting: model sizes range from 27B to 2.4T total parameters (Qwen3.8).	Reported	A5-S041
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	Alibaba says it never uses customer data for model training (Model Studio FAQ). Region-bound storage: Frankfurt region supports an EU deployment scope (workspace-dedicated endpoint), Singapore “International” scope, US (Virginia) US scope, Beijing mainland scope; static data remains in the selected region. Bedrock-hosted Qwen3 available in London, Ireland, Stockholm, Milan regions.	Verified fact	A5-S067, A5-S086
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	qwen3.8-max (International): US\$2 input / US\$6 output per 1M; cached US\$0.25; qwen3.8-flash (Singapore): US\$0.15 / US\$0.47 per 1M; Global/US US\$0.113 / US\$0.382; Google Cloud Qwen3-235B-A22B-2507: US\$0.22 / US\$0.88 per 1M	Verified fact	A5-S068, A5-S027
Infrastructure cost	Qwen3.8 open sizes 27B (single-GPU class with quantisation) to 2.4T-A95B (multi-node).	Reported	A5-S041, A5-S043
Ecosystem	Google Cloud, Amazon Bedrock (Qwen3 incl. eu-west-2 London), Together, DeepInfra, Groq, OpenRouter; Unsloth, ms-swift, LLaMA-Factory, Xinference support.	Reported	A5-S027, A5-S038, A5-S040, A5-S041, A5-S043
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Not publicly verified	Not publicly verified	
Deployment	saas: Yes (Alibaba Cloud Model Studio); managed_cloud: Yes (Google Cloud Qwen3 models; Amazon Bedrock Qwen3 family incl. EU/UK regions); Qwen3.8 not listed on hyperscalers; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes (open-weight sizes); on_prem: Yes (open-weight sizes)		

Moonshot AI Kimi model family (Kimi K3; K2.7-Code, K2.6, K2.5, K2-Thinking) (L1-moonshot-kimi)

Tier: Experimental · **Flags:** none · **Original graphic label:** Kimi – K3

Rationale: Large and capable on paper, but evidence is largely secondary, the weights are expensive to host, the licence is custom and no in-region hyperscaler route exists ⁴⁰.

Criterion	Score	Rationale
Technical	3	One flagship plus a coding variant; CAISI places K3 below frontier cyber capability (A5-S051); narrower lineup than peers.
Enterprise readiness	4	The Bedrock (AWS-operated) route, through US, India and global cross-Region profiles only (A5-S086), gives the hyperscaler presumption (rule 6). Hosting caveat (CP4-3): on Azure, Microsoft Foundry's Kimi K3 and GLM-5.x run on Fireworks outside the customer tenant, while DeepSeek V4 is sold directly and processed in-tenant (A5-S087). The Azure route therefore does not support this

Criterion	Score	Rationale
		score, and the Kimi API's controls are NPV (would be 2). Kept at 4 by the reader at Checkpoint 4 (CP4-3). Platform controls presumed (CP2 Q1); confirm per service.
Security and compliance	2	Certifications NPV (cap 2); Singapore storage; training-policy conflict (A5-S070).
Deployment flexibility	3	Own API, Bedrock cross-Region only, Foundry outside tenant, self-host of 2.8T weights; no in-region managed route.
Ecosystem	3	SGLang, Xinference and two hyperscalers, but the current model reaches them only through cross-Region or pass-through routes.
Reliability and maturity	2	Maturity, support and adoption NPV; facts rest on secondary sources (V2-S015).
Cost / TCO	2	Self-hosting a 2.8T MoE is a heavy operations burden; hosted prices are not decision inputs (V2 \$4 item 3).
Lock-in / portability	3	Open weights, but under a custom licence with thresholds (rule 4).
Total (generic / FS)	2.80	
	/	
	2.80	

Evidence rules applied: security_compliance capped at 2: certifications NPV

Capabilities. Kimi K3 (launched 16 July 2026; weights by 27 July 2026; 2.8T-parameter MoE, about 104B active reported; native multimodal; 1M context) plus K2.7-Code and earlier K2.x models ^{vf987 r988}.

Strengths

- Open weights for a very large multimodal model ^{vf989}
- Offered on Amazon Bedrock and Microsoft Foundry ^{vf990}

Limitations and risks

- International API stores data in Singapore; mainland platform in the PRC; API page says no training but the privacy policy lists training as a purpose ^{vf991}
- Bedrock offers K3 only through cross-Region profiles (US, India, global); Foundry's K3 is Fireworks-operated with inference outside the customer's tenant ^{vf992}
- Custom Kimi K3 License with revenue and MAU thresholds, not MIT ^{vf993 r994}

⁹⁸⁷ Verified fact — A5-S069: Kimi K3 Tech Blog; Kimi K3 API quickstart and pricing; moonshotai/Kimi-K3 LICENSE (Hugging Face) (Moonshot AI) <https://www.kimi.com/en/blog/kimi-k3> accessed 2026-10-07 [search-tool extract]

⁹⁸⁸ Reported — V2-S015: Kimi K3 coverage (aiweekly, morphllm, Hugging Face community blog, cryptobriefing, ayautomate) (Various (secondary)) <https://huggingface.co/moonshotai/Kimi-K3> accessed 2026-10-08 [search-tool extract]; A5-S038: LiteLLM 1.104.1 model_prices_and_context_window map (PyPI wheel) (BerriAI / LiteLLM (community-maintained)) <https://pypi.org/project/litellm/1.104.1/> accessed 2026-10-07 [search-tool extract]

⁹⁸⁹ Verified fact — A5-S069: Kimi K3 Tech Blog; Kimi K3 API quickstart and pricing; moonshotai/Kimi-K3 LICENSE (Hugging Face) (Moonshot AI) <https://www.kimi.com/en/blog/kimi-k3> accessed 2026-10-07 [search-tool extract]

⁹⁹⁰ Verified fact — A5-S086: Introducing Kimi K3 on Amazon Bedrock; GLM 5.3 GA on Amazon Bedrock (Oct 2026); Bedrock model cards (GLM 5, DeepSeek, Qwen3) (AWS) <https://aws.amazon.com/blogs/machine-learning/introducing-kimi-k3-on-amazon-bedrock/> accessed 2026-10-07 [search-tool extract]; A5-S087: Introducing DeepSeek V4 Flash and V4 Pro in Microsoft Foundry; Foundry Models sold by Azure; Data, privacy, and security for Foundry Models sold by Azure; Bring <https://techcommunity.microsoft.com/blog/azure-ai-foundry-blog/introducing-deepseek-v4-flash-and-v4-pro-in-microsoft-foundry/4515174> accessed 2026-10-07 [search-tool extract]

⁹⁹¹ Verified fact — A5-S070: Kimi OpenPlatform Privacy Policy; Kimi API Data Security and Privacy Protection (Moonshot AI) <https://platform.kimi.ai/docs/agreement/userprivacy> accessed 2026-10-07 [search-tool extract]

- CAISI with UK AISI (July 2026): below frontier cyber capability; safeguards did not stop attempted exploit development; K2 Thinking heavily censored in Chinese ^{VF 995}
- Context only, not a scored limitation: Anthropic alleges that Moonshot distilled Claude; this is an unadjudicated competitor allegation and the author is an Anthropic model, so it is not used in any score ^{VF 996}
- Certifications, access controls and support not publicly verified ^{NIPV}

Choose when

- Research or evaluation of large open-weight multimodal models in an isolated environment ^{AI}

Avoid when

- Any production use with client data in a regulated firm today ^{AI}

Nearest competitors: L1-deepseek, L1-zai-glm, L1-alibaba-qwen

Regulated-FS note. Do not use for client data; if evaluated, self-host in an isolated sandbox after licence review ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Moonshot AI: international API operated by Moonshot AI PTE. LTD. (Singapore); mainland platform by Beijing Moonshot Technology (PRC)	Verified fact	A5-S070
Category	Open-weight and API model vendor (agentic, reasoning, multimodal)	Reported	A5-S038, A5-S040
Version / lineup	flagship: Kimi K3, launched 16 July 2026; weights released by 27 July 2026 on Hugging Face; 2.8T-parameter MoE (about 104B active, reported); native multimodal; 1M context. [A5-S069, A5-S039, A5-S040]; coding: Kimi K2.7-Code (262K context). [A5-S038]; previous: Kimi K2.6, K2.5 (January 2026 per KTransformers), K2-Thinking (on Google Cloud and Bedrock). [A5-S038, A5-S042, A5-S027]; graphic_label_check: "K3" is CORRECT (16 July 2026), verified from Moonshot pages.	Verified fact	A5-S038, A5-S039, A5-S040, A5-S041, A5-S042, A5-S027, A5-S069, V2-S015, V2-S021

⁹⁹² Verified fact — A5-S086: Introducing Kimi K3 on Amazon Bedrock; GLM 5.3 GA on Amazon Bedrock (Oct 2026); Bedrock model cards (GLM 5, DeepSeek, Qwen3) (AWS) <https://aws.amazon.com/blogs/machine-learning/introducing-kimi-k3-on-amazon-bedrock/> accessed 2026-10-07 [search-tool extract]; A5-S087: Introducing DeepSeek V4 Flash and V4 Pro in Microsoft Foundry; Foundry Models sold by Azure; Data, privacy, and security for Foundry Models sold by Azure; Bring <https://techcommunity.microsoft.com/blog/azure-ai-foundry-blog/introducing-deepseek-v4-flash-and-v4-pro-in-microsoft-foundry/4515174> accessed 2026-10-07 [search-tool extract]

⁹⁹³ Verified fact — A5-S069: Kimi K3 Tech Blog; Kimi K3 API quickstart and pricing; moonshotai/Kimi-K3 LICENSE (Hugging Face) (Moonshot AI) <https://www.kimi.com/en/blog/kimi-k3> accessed 2026-10-07 [search-tool extract]

⁹⁹⁴ Reported — V2-S015: Kimi K3 coverage (aiweekly, morphllm, Hugging Face community blog, cryptobriefing, ayautomate) (Various (secondary)) <https://huggingface.co/moonshotai/Kimi-K3> accessed 2026-10-08 [search-tool extract]

⁹⁹⁵ Verified fact — A5-S051: UK AISI / CAISI Preliminary Assessment of Kimi K3's Cyber Capabilities (July 2026); CAISI Evaluation of Kimi K2 Thinking (Dec 2025) (NIST CAISI and UK AI Security Institute) <https://www.nist.gov/news-events/news/2026/07/uk-aisi-caisi-preliminary-assessment-kimi-k3s-cyber-capabilities> accessed 2026-10-07 [search-tool extract]; V2-S021: CAISI's Assessment of Z.ai's GLM-5.3 Cyber Capabilities (with UK AISI / CAISI Preliminary Assessment of Kimi K3, July 2026) (NIST CAISI) <https://www.nist.gov/news-events/news/2026/09/caisi-assessment-zais-glm-53-cyber-capabilities> accessed 2026-10-08 [search-tool extract]

⁹⁹⁶ Verified fact — A5-S046: Detecting and preventing distillation attacks (Anthropic) <https://www.anthropic.com/news/detecting-and-preventing-distillation-attacks> accessed 2026-10-07; V2-S022: Detecting and preventing distillation attacks (Anthropic) <https://www.anthropic.com/news/detecting-and-preventing-distillation-attacks> accessed 2026-10-08 [search-tool extract]

Field	Value	Label	Sources
Licence	Open weights under the custom “Kimi K3 License” (MIT-style grant; hosted-service providers above a revenue threshold need a separate agreement; attribution for products >100M MAU or >US\$20M monthly revenue). Kimi K2.7-Code uses a “Modified MIT” licence.	Verified fact	A5-S069, V2-S015
Status events	January 2026: Kimi K2.5; July 2026: Kimi K3 open release	Reported	A5-S042, A5-S039, A5-S040
Strategic direction	Not publicly verified	Not publicly verified	
What it does	Large multimodal agentic LLMs offered as open weights and via Moonshot’s Kimi platform API.	Reported	A5-S040, A5-S038
Stack position	L1 foundation model.	Architectural judgment	
Integration	Kimi platform API (platform.kimi.ai per aggregator source links); open weights via SGLang, Xinference, ms-swift, Unsloth.	Reported	A5-S038, A5-S039, A5-S041
Dependencies	Not publicly verified	Not publicly verified	
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	International API platform stores data on servers in Singapore; mainland platform keeps data in the PRC. Kimi API help page says API inputs/outputs are not used for training; the international privacy policy (30 April 2025) lists model training as a purpose (conflict). No EU residency option found. Bedrock offers Kimi K3 via US, India and global cross-Region profiles (no in-Region); Microsoft Foundry offers K3 via Fireworks pass-through (US Data Zone).	Verified fact	A5-S070, A5-S086, A5-S087
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Kimi K3 (Kimi API): US\$3 input / US\$15 output per 1M; cached input US\$0.30; flat across 1M context; Kimi K3 (Bedrock global): US\$3 / US\$15 per 1M (aggregator)	Verified fact	A5-S069, A5-S038
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Amazon Bedrock, Microsoft Foundry, Google Cloud (K2-Thinking), SGLang, Xinference, ms-swift, Unsloth.	Reported	A5-S038, A5-S027, A5-S039, A5-S041, A5-S040, A5-S043
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Not publicly verified	Not publicly verified	
Deployment	saas: Yes (Kimi platform API); managed_cloud: Yes (Bedrock Kimi K3 cross-Region profiles; Microsoft Foundry via Fireworks - inference on Fireworks GPUs outside customer tenant; Google Cloud K2-Thinking); vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes (open weights); on_prem: Yes (open weights)		

C1: AI / LLM gateway

LiteLLM (Python SDK and LiteLLM Proxy / ‘AI Gateway’) (c1-litellm)

Tier: Strategic · **Flags:** none · **Original graphic label:** not in graphic

Rationale: Strategic, conditional: only as a hardened, pinned, Enterprise-licensed internal service, because the March 2026 compromise and fail-open defaults put security and maturity at 3 ^{AJ}.

Criterion	Score	Rationale
Technical	5	Leading coverage of the C1 research questions: routing, fallback, budgets, rate limits, semantic cache, guardrail hooks, logging, LLM+MCP+A2A ^{VF997}

Criterion	Score	Rationale
Enterprise readiness	4	SSO, RBAC, SCIM and audit logs on the Enterprise tier, custom SLAs and named support channels ^{vf998} ; rule 2 lets an OSS product with commercial support exceed 4, but the licence gating keeps it at 4
Security and compliance	3	SOC 2 Type 2 refreshed September 2026; ISO 27001 unconfirmed ^{vf999} . Serious supply-chain incident in March 2026, remediated (CI/CD v2, cosign, Mandiant review) ^{vf1000} ; held at 3
Deployment flexibility	4	Self-hosted, customer-account (VPC) and hosted options verified ^{vf1001} ; on-prem and air-gap not verified ^{NPV}
Ecosystem	5	De facto open interface (OpenAI format), 100+ providers, MCP, A2A, logging callbacks; used by AWS's Guidance ^{vf1002}
Reliability and maturity	3	Since July 2023 with very high cadence ^{vf1003} , but a 2026 supply-chain compromise and fail-open defaults ^{vf1004}
Cost / TCO	4	Free to self-host; Enterprise licence priced on capacity (pricing description conflicts); database and Redis to operate ^{vf1005}
Lock-in / portability	4	MIT core and OpenAI-format API; enterprise features proprietary ^{vf1006} ; no ownership change found
Total (generic / FS)	4.05 / 3.90	

⁹⁹⁷ Verified fact — A6-S015: LiteLLM docs: MCP Overview, Agent Gateway (A2A), Caching, Budgets/Rate Limits, v1.89.0 release notes (BerriAI (LiteLLM)) <https://docs.litellm.ai/docs/mcp> accessed 2026-10-07 [search-tool extract]

⁹⁹⁸ Verified fact — A6-S007: LiteLLM pricing / Enterprise / Audit Logs docs (BerriAI (LiteLLM)) <https://www.litellm.ai/pricing> accessed 2026-10-07 [search-tool extract]; A6-S051: LiteLLM README (GitHub) (BerriAI (LiteLLM)) <https://raw.githubusercontent.com/BerriAI/litellm/main/README.md> accessed 2026-10-07

⁹⁹⁹ Verified fact — A6-S108: Our Updated SOC 2 Type 2 Report Is Available; LiteLLM + Vanta: SOC 2 Type 2 and ISO 27001 Recertification; Data Privacy and Security; Trust Center (BerriAI (LiteLLM)) <https://docs.litellm.ai/blog/soc2-type-2-report> accessed 2026-10-07 [search-tool extract]

¹⁰⁰⁰ Verified fact — A6-S008: Security Update: Suspected Supply Chain Incident (and Security Townhall Updates) (BerriAI (LiteLLM)) <https://docs.litellm.ai/blog/security-update-march-2026> accessed 2026-10-07 [search-tool extract]; A6-S009: v1.83.0 - Official Release (Post Supply Chain Incident) (BerriAI (LiteLLM)) https://docs.litellm.ai/release_notes/v1.83.0/v1-83-0 accessed 2026-10-07 [search-tool extract]

¹⁰⁰¹ Verified fact — A6-S007: LiteLLM pricing / Enterprise / Audit Logs docs (BerriAI (LiteLLM)) <https://www.litellm.ai/pricing> accessed 2026-10-07 [search-tool extract]; A6-S022: Guidance for Multi-Provider Generative AI Gateway on AWS; Implementing resilience patterns with Amazon Bedrock and LLM gateway (Amazon Web Services) <https://aws.amazon.com/solutions/guidance/multi-provider-generative-ai-gateway-on-aws/> accessed 2026-10-07 [search-tool extract]; A6-S051: LiteLLM README (GitHub) (BerriAI (LiteLLM)) <https://raw.githubusercontent.com/BerriAI/litellm/main/README.md> accessed 2026-10-07

¹⁰⁰² Verified fact — A6-S015: LiteLLM docs: MCP Overview, Agent Gateway (A2A), Caching, Budgets/Rate Limits, v1.89.0 release notes (BerriAI (LiteLLM)) <https://docs.litellm.ai/docs/mcp> accessed 2026-10-07 [search-tool extract]; A6-S022: Guidance for Multi-Provider Generative AI Gateway on AWS; Implementing resilience patterns with Amazon Bedrock and LLM gateway (Amazon Web Services) <https://aws.amazon.com/solutions/guidance/multi-provider-generative-ai-gateway-on-aws/> accessed 2026-10-07 [search-tool extract]

¹⁰⁰³ Verified fact — A6-S001: litellm - PyPI JSON metadata (Python Package Index) <https://pypi.org/pypi/litellm/json> accessed 2026-10-07

¹⁰⁰⁴ Verified fact — A6-S008: Security Update: Suspected Supply Chain Incident (and Security Townhall Updates) (BerriAI (LiteLLM)) <https://docs.litellm.ai/blog/security-update-march-2026> accessed 2026-10-07 [search-tool extract]; A6-S015: LiteLLM docs: MCP Overview, Agent Gateway (A2A), Caching, Budgets/Rate Limits, v1.89.0 release notes (BerriAI (LiteLLM)) <https://docs.litellm.ai/docs/mcp> accessed 2026-10-07 [search-tool extract]

¹⁰⁰⁵ Verified fact — A6-S007: LiteLLM pricing / Enterprise / Audit Logs docs (BerriAI (LiteLLM)) <https://www.litellm.ai/pricing> accessed 2026-10-07 [search-tool extract]; A6-S015: LiteLLM docs: MCP Overview, Agent Gateway (A2A), Caching, Budgets/Rate Limits, v1.89.0 release notes (BerriAI (LiteLLM)) <https://docs.litellm.ai/docs/mcp> accessed 2026-10-07 [search-tool extract]

¹⁰⁰⁶ Verified fact — A6-S001: litellm - PyPI JSON metadata (Python Package Index) <https://pypi.org/pypi/litellm/json> accessed 2026-10-07; A6-S002: litellm-enterprise - PyPI JSON metadata (Python Package Index) <https://pypi.org/pypi/litellm-enterprise/json> accessed 2026-10-07

Evidence rules applied: Rule 2 basis: self-hosted open-core software with commercial support; NPV cap not applied

Capabilities. Open-core Python SDK and proxy exposing 100+ providers through an OpenAI-format API, with virtual keys, budgets per key/user/team/customer, TPM/RPM limits, fallbacks, exact and semantic caching, pre/during/post-call guardrail hooks and logging callbacks; the same proxy fronts MCP servers and A2A agents ^{vf-1007}.

Strengths

- Widest provider coverage and the de facto open-source gateway; basis of AWS's own multi-provider Guidance ^{vf-1008 AJ}
- One control plane for LLM, MCP and A2A traffic with shared auth, limits and spend tracking ^{vf-1009}
- MIT core and OpenAI-format API keep the application exit cheap ^{vf-1010 AJ}

Limitations and risks

- PyPI supply-chain compromise on 24 March 2026 (1.82.7 and 1.82.8) via CI credentials; clean v1.83.0 released 30 March 2026 (US time) from a rebuilt pipeline ^{vf-1011}
- Several controls fail open or are off by default: max_budget without a database, prompt-injection guardrails off, admin key bypasses budgets; A2A agents open to all callers until an allowlist is set ^{vf-1012}
- SSO beyond 5 users, RBAC, SCIM and audit logs need the commercial licence; pricing model described inconsistently ^{vf-1013}
- ISO 27001 recertification not confirmed ^{vf-1014}

Choose when

- You want a self-hosted, provider-neutral gateway for LLM, MCP and A2A traffic and will run it as a hardened internal service ^{AJ}

¹⁰⁰⁷ Verified fact — A6-S015: LiteLLM docs: MCP Overview, Agent Gateway (A2A), Caching, Budgets/Rate Limits, v1.89.0 release notes (BerriAI (LiteLLM)) <https://docs.litellm.ai/docs/mcp> accessed 2026-10-07 [search-tool extract]; A6-S051: LiteLLM README (GitHub) (BerriAI (LiteLLM)) <https://raw.githubusercontent.com/BerriAI/litellm/main/README.md> accessed 2026-10-07

¹⁰⁰⁸ Verified fact — A6-S015: LiteLLM docs: MCP Overview, Agent Gateway (A2A), Caching, Budgets/Rate Limits, v1.89.0 release notes (BerriAI (LiteLLM)) <https://docs.litellm.ai/docs/mcp> accessed 2026-10-07 [search-tool extract]; A6-S022: Guidance for Multi-Provider Generative AI Gateway on AWS; Implementing resilience patterns with Amazon Bedrock and LLM gateway (Amazon Web Services) <https://aws.amazon.com/solutions/guidance/multi-provider-generative-ai-gateway-on-aws/> accessed 2026-10-07 [search-tool extract]

¹⁰⁰⁹ Verified fact — A6-S015: LiteLLM docs: MCP Overview, Agent Gateway (A2A), Caching, Budgets/Rate Limits, v1.89.0 release notes (BerriAI (LiteLLM)) <https://docs.litellm.ai/docs/mcp> accessed 2026-10-07 [search-tool extract]

¹⁰¹⁰ Verified fact — A6-S001: litellm - PyPI JSON metadata (Python Package Index) <https://pypi.org/pypi/litellm/json> accessed 2026-10-07; A6-S015: LiteLLM docs: MCP Overview, Agent Gateway (A2A), Caching, Budgets/Rate Limits, v1.89.0 release notes (BerriAI (LiteLLM)) <https://docs.litellm.ai/docs/mcp> accessed 2026-10-07 [search-tool extract]

¹⁰¹¹ Verified fact — A6-S008: Security Update: Suspected Supply Chain Incident (and Security Townhall Updates) (BerriAI (LiteLLM)) <https://docs.litellm.ai/blog/security-update-march-2026> accessed 2026-10-07 [search-tool extract]; A6-S009: v1.83.0 - Official Release (Post Supply Chain Incident) (BerriAI (LiteLLM)) https://docs.litellm.ai/release_notes/v1.83.0/v1-83-0 accessed 2026-10-07 [search-tool extract]; V2-S027: LiteLLM PyPI supply chain attack (with NetSPI, Comet, Snyk/Datadog reporting) (FutureSearch) <https://futuresearch.ai/blog/litellm-pypi-supply-chain-attack/> accessed 2026-10-08 [search-tool extract]; V2-S028: PyPI JSON API: litellm, guardrails-ai, presidio-analyzer, nemoguardrails, llamafirewall, langfuse, promptfoo, modelscan, picklescan, fickling, safetensors, mode <https://pypi.org/pypi/litellm/json> accessed 2026-10-08 [search-tool extract]

¹⁰¹² Verified fact — A6-S015: LiteLLM docs: MCP Overview, Agent Gateway (A2A), Caching, Budgets/Rate Limits, v1.89.0 release notes (BerriAI (LiteLLM)) <https://docs.litellm.ai/docs/mcp> accessed 2026-10-07 [search-tool extract]

¹⁰¹³ Verified fact — A6-S007: LiteLLM pricing / Enterprise / Audit Logs docs (BerriAI (LiteLLM)) <https://www.litellm.ai/pricing> accessed 2026-10-07 [search-tool extract]

¹⁰¹⁴ Verified fact — A6-S108: Our Updated SOC 2 Type 2 Report Is Available; LiteLLM + Vanta: SOC 2 Type 2 and ISO 27001 Recertification; Data Privacy and Security; Trust Center (BerriAI (LiteLLM)) <https://docs.litellm.ai/blog/soc2-type-2-report> accessed 2026-10-07 [search-tool extract]

- You will buy the Enterprise licence for SSO, RBAC and audit logs ^{AJ}

Avoid when

- You cannot pin versions, mirror packages and verify signed images ^{AJ}
- You would run the open edition without a database and accept fail-open budgets ^{AJ}

Nearest competitors: C1-kong-ai-gateway, C1-portkey, C1-agentgateway

Regulated-FS note. Treat as a Tier-1 internal service: install only from a private mirror with pinned hashes, deploy the cosign-signed images, run fail-closed configuration and buy Enterprise for audit evidence ^{Rec.}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	BerriAI (PyPI author of litellm); no acquisition found in sources consulted	Verified fact	A6-S001
Category	AI/LLM gateway (open-source proxy) that also acts as MCP gateway and A2A agent gateway	Verified fact	A6-S015, A6-S051
Version / lineup	litellm 1.104.1 (PyPI, 7 October 2026; 1.105.0rc2 pre-release the same day); proprietary litellm-enterprise 0.1.74 (7 October 2026)	Verified fact	A6-S001, A6-S002, V2-S028
Licence	Open core: core package MIT; enterprise features under the LiteLLM Commercial License, shipped as litellm-enterprise (LicenseRef-Proprietary)	Verified fact	A6-S001, A6-S002, A6-S051
Status events	24 March 2026: malicious litellm 1.82.7 and 1.82.8 published to PyPI using release credentials stolen via a compromised Trivy scanner in CI; quarantined after about 40 minutes per LiteLLM (Snyk reports an exposure window of about three hours); both versions are no longer listed on PyPI; 30 March 2026 (US time; PyPI upload 31 March 2026 05:08 UTC): v1.83.0 released as first build from rebuilt CI/CD v2 pipeline after forensic review with Mandiant and Veria Labs	Verified fact	A6-S008, A6-S009, A6-S010, V2-S027, V2-S028
Strategic direction	Positions the proxy as one control plane for LLMs, MCP servers and A2A agents with shared auth, rate limiting and usage dashboard; v1.89.0 added A2A agent providers and per-MCP-server controls; after March 2026 incident moved to CI/CD v2 with ephemeral credentials and cosign-signed images	Verified fact	A6-S015, A6-S009, A6-S051
What it does	Python SDK and proxy server that expose 100+ LLM providers through	Verified fact	A6-S015,

Field	Value	Label	Sources
	an OpenAI-format API, with virtual keys, budgets (key/user/team/customer), TPM/RPM limits, fallbacks, exact and semantic caching, guardrail hooks (pre-call, during-call, post-call) and logging callbacks; the same proxy fronts registered MCP servers and A2A agents with access control, cost tracking and guardrails.	d fact	A6-S051
Stack position	Gateway/control plane between applications or agents and model providers, MCP tool servers and A2A agents. Absent from the graphic (L2 shows OpenRouter, a hosted router, not an enterprise gateway).	Verified fact	A6-S015
Integration	OpenAI-compatible REST API; MCP endpoint (fixed endpoint for all registered MCP tools, OAuth passthrough with issuer-scoped JWT auth); A2A message/send and message/stream; logging callbacks (e.g. per-team Langfuse projects), log export to S3/GCS/Azure Blob; JWT/OIDC request auth	Verified fact	A6-S015, A6-S007
Dependencies	Python >=3.10,<3.15; a database is required for global budgets and spend tracking; Redis, Valkey or Qdrant back semantic caching; official Docker images on GHCR	Verified fact	A6-S001, A6-S015, A6-S051
Certifications	SOC 2 Type 2: updated report announced September 2026 (recertification with Vanta, independent auditor), available via trust.litellm.ai on request. ISO 27001: recertification announced March 2026; completion not confirmed (Data Privacy page lists only SOC 2 Type II)	Verified fact	A6-S108
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Docker images on GHCR signed with cosign from v1.83.0; virtual keys; guardrail framework applied across chat, embeddings, MCP and A2A routes (prompt-injection guardrails off by default); ISMS, regular vulnerability scans and a vulnerability disclosure policy	Verified fact	A6-S051, A6-S009, A6-S015, A6-S108
Access controls	Admin UI SSO via Okta, Azure AD, Google Workspace or any OIDC/SAML IdP (free up to 5 users, then enterprise licence); RBAC, SCIM and audit logs of key/team/user/model changes in Enterprise tier	Verified fact	A6-S007
Enterprise support	Enterprise tier: professional support via dedicated Discord/Slack, custom SLAs, feature prioritisation, custom integrations	Verified fact	A6-S051
Pricing	Open-source proxy free to self-host. Enterprise: annual licence priced on gateway request capacity, deployment architecture and support ('never per token'); 30-day trial key by email. Docs Enterprise page instead describes usage-based pricing (conflict).	Verified fact	A6-S007
Infrastructure cost	Self-hosted cost is container compute plus the database needed for budgets/spend and Redis/Valkey for caching and multi-instance rate limits	Verified fact	A6-S015
Ecosystem	100+ LLM providers; MCP and A2A; Langfuse and other logging callbacks; basis of AWS 'Guidance for Multi-Provider Generative AI Gateway on AWS'	Verified fact	A6-S015, A6-S022
Adoption signals	About 3 to 3.4 million PyPI downloads per day (InfoQ and Trend Micro, March 2026)	Reported	A6-S010

Field	Value	Label	Sources
Maturity	First PyPI release 27 July 2023; very high release cadence (several releases on 7 October 2026 alone)	Verified fact	A6-S001
Deployment	saas: Yes (hosted LiteLLM proxy referenced in README and Enterprise docs); managed_cloud: Not publicly verified; vpc_byoc: Yes (customer-account deployment pattern, e.g. AWS Guidance deploys LiteLLM on ECS/EKS); private_cloud: Not publicly verified; self_hosted: Yes (free to self-host; Docker images); on_prem: Not publicly verified		

Kong AI Gateway (AI Gateway 2.x runtime in Kong Konnect; AI plugins in Kong Gateway 3.x) (c1-kong-ai-gateway)

Tier: Strategic · **Flags:** none · **Original graphic label:** not in graphic

Rationale: Strategic, conditional: where Kong is the API standard; cost (2) is a stated condition because pricing is not public ^{AJ}.

Criterion	Score	Rationale
Technical	5	Leading coverage across LLM, MCP and A2A, including guardrails, PII sanitisation and cost governance ^{VF1015}
Enterprise readiness	4	Konnect SSO (SAML or OIDC) with teams, predefined roles and IdP group mapping (B-REVA-S004), audit logs, Kong Identity principals and a 99.99% Konnect SLA (A6-S109, A6-S016, A6-S018): all three controls plus SLA reach 4 under rule 7
Security and compliance	4	ISO 27001:2022 and SOC 2 Type II both naming AI Gateway; PCI DSS for Konnect ^{VF1016} ; no CMK, ISO 42001 or FedRAMP evidence, so not 5
Deployment flexibility	4	Konnect SaaS, Dedicated Cloud Gateways, hybrid and self-managed ^{VF1017} ; on-prem and air-gap not verified ^{NPV}
Ecosystem	4	Large API-gateway ecosystem; AWS SigV4, Foundry, SageMaker and NeMo integrations ^{VF1018}
Reliability and maturity	3	AI plugins since 3.6 (2024) but AI Gateway 2.0 GA only on 1 September 2026, with a configuration-model change ^{VF1019}
Cost / TCO	2	Pricing 'contact sales'; advanced plugins licence-gated ^{VF1020}

¹⁰¹⁵ Verified fact — A6-S016: Kong AI Gateway 2.0 Is Now GA — And It's Already Moving Faster (Kong Inc.) <https://konghq.com/blog/product-releases/kong-ai-gateway-2-0-ga> accessed 2026-10-07 [search-tool extract]; A6-S017: Kong AI Gateway changelog; Kong Gateway changelog; AI Proxy Advanced plugin docs (Kong Inc.) <https://developer.konghq.com/ai-gateway/changelog/> accessed 2026-10-07 [search-tool extract]

¹⁰¹⁶ Verified fact — A6-S109: Kong Trust Center - Resources; Trust and Compliance at Kong (Kong Inc.) <https://trust.konghq.com/resources> accessed 2026-10-07 [search-tool extract]

¹⁰¹⁷ Verified fact — A6-S018: Kong startup program page (compliance claims, deployment modes) (Kong Inc.) <https://konghq.com/solutions/startup-program> accessed 2026-10-07 [search-tool extract]

¹⁰¹⁸ Verified fact — A6-S017: Kong AI Gateway changelog; Kong Gateway changelog; AI Proxy Advanced plugin docs (Kong Inc.) <https://developer.konghq.com/ai-gateway/changelog/> accessed 2026-10-07 [search-tool extract]; A6-S016: Kong AI Gateway 2.0 Is Now GA — And It's Already Moving Faster (Kong Inc.) <https://konghq.com/blog/product-releases/kong-ai-gateway-2-0-ga> accessed 2026-10-07 [search-tool extract]

¹⁰¹⁹ Verified fact — A6-S016: Kong AI Gateway 2.0 Is Now GA — And It's Already Moving Faster (Kong Inc.) <https://konghq.com/blog/product-releases/kong-ai-gateway-2-0-ga> accessed 2026-10-07 [search-tool extract]; A6-S017: Kong AI Gateway changelog; Kong Gateway changelog; AI Proxy Advanced plugin docs (Kong Inc.) <https://developer.konghq.com/ai-gateway/changelog/> accessed 2026-10-07 [search-tool extract]; V2-S034: Kong AI Gateway 2.0 Is Now GA (with Kong press release 30 Sep 2026 on AI Gateway 2.2 GA) <https://konghq.com/blog/product-releases/kong-ai-gateway-2-0-ga> accessed 2026-10-08 [search-tool extract]

Criterion	Score	Rationale
Lock-in / portability	3	Apache-2.0 OSS core, but 2.x runtime and advanced plugins licence-gated ^{vf-1021}
Total (generic / FS)	3.85 / 3.80	

Capabilities. AI gateway for LLM, MCP and A2A traffic on Kong's API platform: multi-provider routing and load balancing, token budgets and rate limits, semantic caching, semantic prompt/response guards, PII sanitisation, cloud guardrail and NeMo Guardrails integrations, MCP access control and tool filtering; AI Gateway 2.x is a dedicated runtime in Konnect ^{vf-1022}.

Strengths

- Broad AI policy set on an enterprise API-gateway estate many firms already run ^{vf-1023 AJ}
- Product-scoped SOC 2 Type II and ISO 27001:2022 covering AI Gateway ^{vf-1024}
- Tracks the MCP 2026-07-28 revision and offers per-caller MCP tool exposure ^{vf-1025}
- Konnect organisation SSO (SAML or OIDC), teams and roles with IdP group mapping ^{vf-1026}

Limitations and risks

- Configuration model changed from 3.x plugins to 2.x entities; migration needed before 3.18 ^{vf-1027}
- Advanced AI plugins and AI Gateway 2.x are licence-gated or Konnect-delivered ^{vf-1028}
- Pricing not published ^{vf-1029}

¹⁰²⁰ Verified fact — A6-S017: Kong AI Gateway changelog; Kong Gateway changelog; AI Proxy Advanced plugin docs (Kong Inc.) <https://developer.konghq.com/ai-gateway/changelog/> accessed 2026-10-07 [search-tool extract]; A6-S018: Kong startup program page (compliance claims, deployment modes) (Kong Inc.) <https://konghq.com/solutions/startup-program> accessed 2026-10-07 [search-tool extract]

¹⁰²¹ Verified fact — A6-S017: Kong AI Gateway changelog; Kong Gateway changelog; AI Proxy Advanced plugin docs (Kong Inc.) <https://developer.konghq.com/ai-gateway/changelog/> accessed 2026-10-07 [search-tool extract]

¹⁰²² Verified fact — A6-S016: Kong AI Gateway 2.0 Is Now GA — And It's Already Moving Faster (Kong Inc.) <https://konghq.com/blog/product-releases/kong-ai-gateway-2-0-ga> accessed 2026-10-07 [search-tool extract]; A6-S017: Kong AI Gateway changelog; Kong Gateway changelog; AI Proxy Advanced plugin docs (Kong Inc.) <https://developer.konghq.com/ai-gateway/changelog/> accessed 2026-10-07 [search-tool extract]

¹⁰²³ Verified fact — A6-S016: Kong AI Gateway 2.0 Is Now GA — And It's Already Moving Faster (Kong Inc.) <https://konghq.com/blog/product-releases/kong-ai-gateway-2-0-ga> accessed 2026-10-07 [search-tool extract]

¹⁰²⁴ Verified fact — A6-S109: Kong Trust Center - Resources; Trust and Compliance at Kong (Kong Inc.) <https://trust.konghq.com/resources> accessed 2026-10-07 [search-tool extract]

¹⁰²⁵ Verified fact — A6-S017: Kong AI Gateway changelog; Kong Gateway changelog; AI Proxy Advanced plugin docs (Kong Inc.) <https://developer.konghq.com/ai-gateway/changelog/> accessed 2026-10-07 [search-tool extract]; A6-S016: Kong AI Gateway 2.0 Is Now GA — And It's Already Moving Faster (Kong Inc.) <https://konghq.com/blog/product-releases/kong-ai-gateway-2-0-ga> accessed 2026-10-07 [search-tool extract]

¹⁰²⁶ Verified fact — B-REVA-S004: Konnect platform authentication and authorization (Kong Inc.) <https://developer.konghq.com/konnect-platform/authentication/> accessed 2026-10-08 [search-tool extract]

¹⁰²⁷ Verified fact — A6-S016: Kong AI Gateway 2.0 Is Now GA — And It's Already Moving Faster (Kong Inc.) <https://konghq.com/blog/product-releases/kong-ai-gateway-2-0-ga> accessed 2026-10-07 [search-tool extract]; A6-S017: Kong AI Gateway changelog; Kong Gateway changelog; AI Proxy Advanced plugin docs (Kong Inc.) <https://developer.konghq.com/ai-gateway/changelog/> accessed 2026-10-07 [search-tool extract]

¹⁰²⁸ Verified fact — A6-S017: Kong AI Gateway changelog; Kong Gateway changelog; AI Proxy Advanced plugin docs (Kong Inc.) <https://developer.konghq.com/ai-gateway/changelog/> accessed 2026-10-07 [search-tool extract]

¹⁰²⁹ Verified fact — A6-S017: Kong AI Gateway changelog; Kong Gateway changelog; AI Proxy Advanced plugin docs (Kong Inc.) <https://developer.konghq.com/ai-gateway/changelog/> accessed 2026-10-07 [search-tool extract]; A6-S018: Kong startup program page (compliance claims, deployment modes) (Kong Inc.) <https://konghq.com/solutions/startup-program> accessed 2026-10-07 [search-tool extract]

Choose when

- Kong is already your API gateway standard ^{AJ}
- You need product-scoped certifications for the gateway itself ^{AJ}

Avoid when

- You want a free, self-hosted gateway with no licence dependency ^{AJ}
- You cannot plan the 3.x-to-2.x migration in the next year ^{AJ}

Nearest competitors: C1-litellm, C1-google-apigee-ai-gateway, C1-azure-apim-ai-gateway

Regulated-FS note. Prefer the hybrid mode (customer-run data plane) so prompts and logs stay in region; confirm audit-log export on Konnect ^{Rec.}

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Kong Inc.	Verified fact	A6-S016
Category	AI gateway (LLM, MCP and A2A traffic) built on an API gateway vendor's platform	Verified fact	A6-S016, A6-S017
Version / lineup	AI Gateway 2.0 GA on 1 September 2026; 2.1.0 (22 September 2026) and 2.2.0 (30 September 2026); AI plugins supported on Kong Gateway 3.14 LTS (3.14.0.0, 7 April 2026)	Verified fact	A6-S016, A6-S017, V2-S034
Licence	Open core: Kong Gateway OSS (Apache-2.0) with basic AI plugins released as open source in 3.6; advanced AI plugins (e.g. AI Proxy Advanced) require an AI/Enterprise licence; AI Gateway 2.x delivered via Konnect	Verified fact	A6-S017, A6-S018
Status events	1 September 2026: Kong AI Gateway 2.0 GA as a dedicated runtime (configuration moves from plugins to AI Model Provider/AI Model entities)	Verified fact	A6-S016, A6-S017, V2-S034
Strategic direction	Separate AI runtime with its own control plane, admin API and release cadence for LLM, MCP and agent workloads; MCP Server Bundling, principal-aware AI traffic policies via Kong Identity, modality-aware cost governance; migration from 3.x plugins recommended before 3.18	Verified fact	A6-S016, A6-S017
What it does	Governs LLM, MCP and A2A traffic: multi-provider routing and load balancing, token budgeting and rate limiting, semantic caching, semantic prompt/response guards, PII sanitisation, integrations with cloud guardrail services and NVIDIA NeMo Guardrails, MCP access control and tool filtering, A2A traffic management.	Verified fact	A6-S016, A6-S017

Field	Value	Label	Sources
Stack position	Gateway/control plane; extends an existing enterprise API gateway estate to AI traffic	Verified fact	A6-S016
Integration	Gateway plugins/entities; supports MCP protocol revision 2026-07-28 (2.1.0); AWS SigV4 upstream auth for Bedrock AgentCore; providers incl. Microsoft Foundry, Amazon SageMaker, Kimi	Verified fact	A6-S016, A6-S017
Dependencies	Not publicly verified	Not publicly verified	
Certifications	ISO/IEC 27001:2022 certificate LC-516-21 (Linford & Company, issued 1 September 2026, valid to 31 August 2029) covering the API and AI Connectivity Platform; SOC 2 Type II (security, availability, confidentiality) covering Konnect, Dedicated Cloud Gateways, Kong Gateway Enterprise, Kong AI Gateway, Kong Mesh, Insomnia; PCI DSS v4.0.1 (Konnect, Dedicated Cloud Gateways); CSA STAR Level 1 self-assessment	Verified fact	A6-S109
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Konnect and Dev Portal audit logs; Kong Identity principals usable in AI Gateway policies	Verified fact	A6-S109, A6-S016
Enterprise support	99.99% SLA advertised for Konnect cloud	Verified fact	A6-S018
Pricing	Not published in sources consulted beyond 'Contact sales'/consumption model; AI Gateway paid plugins listed as add-on or included depending on tier	Verified fact	A6-S017, A6-S018
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	Series E of US\$175m at US\$2bn valuation (date not verified); total equity raised estimated at US\$345m (Sacra)	Reported	A6-S018
Maturity	AI plugins since Kong Gateway 3.6 (2024); AI Gateway 2.x GA since 1	Verified	A6-S016,

Field	Value	Label	Sources
	September 2026	d fact	A6-S017
Deployment	saas: Yes (Konnect managed control plane); managed_cloud: Yes (Dedicated Cloud Gateways on AWS, Azure or GCP); vpc_byoc: Yes (hybrid: Kong-run control plane, customer-run data plane); private_cloud: Not publicly verified; self_hosted: Yes (fully self-managed gateways); on_prem: Not publicly verified		

Apigee (API management) used as an AI gateway (c1-google-apigee-ai-gateway)

Tier: Strategic · **Flags:** none · **Original graphic label:** not in graphic

Rationale: Strategic, conditional: where Google Cloud is your primary cloud or Apigee is already the API standard (CP3 Q2, rubric rule 10); lock-in at 2 is the stated condition ¹⁰³⁰.

Criterion	Score	Rationale
Technical	4	Token limits, semantic cache, routing with circuit breaking, auditing, Model Armor, MCP, A2A ^{vr1030}
Enterprise readiness	4	Hyperscaler presumption (CP2 Q1); access-control detail not in the fact base ^{NPV} ; confirm per service
Security and compliance	4	Apigee named in Google Cloud SOC 2 and ISO/IEC 27001 scope ^{vr1031} ; CMK not verified, so not 5
Deployment flexibility	4	Apigee X SaaS and Apigee hybrid self-managed runtime ^{vr1032}
Ecosystem	4	OAuth/JWT, MCP, A2A, AP2, multicloud routing ^{vr1033}
Reliability and maturity	4	Mature API management; AI and MCP features added 2025–26 ^{vr1034}
Cost / TCO	3	Transparent pay-as-you-go per call, with paid analytics and security add-ons ^{vr1035}
Lock-in / portability	2	Proprietary policy model ^{vr1036}
Total (generic / FS)	3.80	/

¹⁰³⁰ Verified fact — A6-S024: Apigee AI solutions: gateway to enterprise-ready AI (Google Cloud) <https://cloud.google.com/solutions/apigee-ai> accessed 2026-10-07; A6-S023: Apigee docs: AI gateway capabilities; semantic caching tutorial; MCP support for Apigee (Google Cloud) <https://docs.cloud.google.com/apigee/docs/api-platform/get-started/ai-capabilities> accessed 2026-10-07 [search-tool extract]

¹⁰³¹ Verified fact — A6-S070: SOC 2 compliance (services in scope) (Google Cloud) <https://cloud.google.com/security/compliance/soc-2> accessed 2026-10-07; A6-S071: ISO/IEC 27001 compliance (services in scope) (Google Cloud) <https://cloud.google.com/security/compliance/iso-27001> accessed 2026-10-07

¹⁰³² Verified fact — A6-S069: Apigee Pricing (Google Cloud) <https://cloud.google.com/apigee/pricing> accessed 2026-10-07; A6-S025: Apigee release notes (all; hybrid) (Google Cloud) <https://docs.cloud.google.com/apigee/docs/release-notes> accessed 2026-10-07 [search-tool extract]

¹⁰³³ Verified fact — A6-S024: Apigee AI solutions: gateway to enterprise-ready AI (Google Cloud) <https://cloud.google.com/solutions/apigee-ai> accessed 2026-10-07

¹⁰³⁴ Verified fact — A6-S023: Apigee docs: AI gateway capabilities; semantic caching tutorial; MCP support for Apigee (Google Cloud) <https://docs.cloud.google.com/apigee/docs/api-platform/get-started/ai-capabilities> accessed 2026-10-07 [search-tool extract]; A6-S025: Apigee release notes (all; hybrid) (Google Cloud) <https://docs.cloud.google.com/apigee/docs/release-notes> accessed 2026-10-07 [search-tool extract]

¹⁰³⁵ Verified fact — A6-S069: Apigee Pricing (Google Cloud) <https://cloud.google.com/apigee/pricing> accessed 2026-10-07

¹⁰³⁶ Verified fact — A6-S024: Apigee AI solutions: gateway to enterprise-ready AI (Google Cloud) <https://cloud.google.com/solutions/apigee-ai> accessed 2026-10-07

Criterion	Score	Rationale
	3.65	

Evidence rules applied: enterprise_readiness set to 4 under the hyperscaler presumption (CP2 Q1); platform controls presumed; confirm per service

Capabilities. Apigee API management marketed as an AI gateway: token limits and monitoring, semantic caching, multicloud model routing with circuit breaking, LLM auditing and logging, Model Armor screening, MCP (GA 31 March 2026) and A2A support; SaaS or hybrid self-managed runtime ^{vf1037}.

Strengths

- Mature API-management product with product-scoped SOC 2 and ISO 27001 ^{vf1038}
- Hybrid runtime gives a customer-run data plane ^{vf1039}
- Inline Model Armor screening ^{vf1040}

Limitations and risks

- Proprietary policy model ^{vf1041}
- Per-call pricing plus add-ons for analytics and security ^{vf1042}
- Semantic cache needed an SSRF fix on 30 September 2026 ^{vf1043}

Choose when

- Google Cloud is primary, or Apigee is already your API standard ^{aj}
- You want the gateway data plane in your own Kubernetes (hybrid) ^{aj}

Avoid when

- You want a light, low-cost model gateway with no API-management estate ^{aj}

Nearest competitors: C1-azure-apim-ai-gateway, C1-kong-ai-gateway, C1-litellm

Regulated-FS note. Run hybrid in a UK or EU region and confirm residency of LLM audit logs; keep semantic caching off for client-specific routes ^{rec}.

Facts (as of 2026-10-07)

¹⁰³⁷ Verified fact — A6-S024: Apigee AI solutions: gateway to enterprise-ready AI (Google Cloud) <https://cloud.google.com/solutions/apigee-ai> accessed 2026-10-07; A6-S023: Apigee docs: AI gateway capabilities; semantic caching tutorial; MCP support for Apigee (Google Cloud) <https://docs.cloud.google.com/apigee/docs/api-platform/get-started/ai-capabilities> accessed 2026-10-07 [search-tool extract]; A6-S025: Apigee release notes (all; hybrid) (Google Cloud) <https://docs.cloud.google.com/apigee/docs/release-notes> accessed 2026-10-07 [search-tool extract]

¹⁰³⁸ Verified fact — A6-S070: SOC 2 compliance (services in scope) (Google Cloud) <https://cloud.google.com/security/compliance/soc-2> accessed 2026-10-07; A6-S071: ISO/IEC 27001 compliance (services in scope) (Google Cloud) <https://cloud.google.com/security/compliance/iso-27001> accessed 2026-10-07

¹⁰³⁹ Verified fact — A6-S025: Apigee release notes (all; hybrid) (Google Cloud) <https://docs.cloud.google.com/apigee/docs/release-notes> accessed 2026-10-07 [search-tool extract]

¹⁰⁴⁰ Verified fact — A6-S024: Apigee AI solutions: gateway to enterprise-ready AI (Google Cloud) <https://cloud.google.com/solutions/apigee-ai> accessed 2026-10-07

¹⁰⁴¹ Verified fact — A6-S024: Apigee AI solutions: gateway to enterprise-ready AI (Google Cloud) <https://cloud.google.com/solutions/apigee-ai> accessed 2026-10-07

¹⁰⁴² Verified fact — A6-S069: Apigee Pricing (Google Cloud) <https://cloud.google.com/apigee/pricing> accessed 2026-10-07

¹⁰⁴³ Verified fact — A6-S025: Apigee release notes (all; hybrid) (Google Cloud) <https://docs.cloud.google.com/apigee/docs/release-notes> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
Company	Google Cloud	Verified fact	A6-S024
Category	Cloud API management platform with AI gateway policies (LLM, MCP, A2A)	Verified fact	A6-S024
Version / lineup	Apigee X (SaaS) and Apigee hybrid 1.17 (1.17.1 on 30 September 2026); Apigee MCP support GA 31 March 2026; API hub MCP server GA 24 July 2026	Verified fact	A6-S023, A6-S025
Licence	Proprietary managed service (hybrid runtime self-managed)	Verified fact	A6-S069
Strategic direction	Apigee marketed as the AI gateway for agentic AI: supports SSE and JSON-RPC for MCP, A2A and AP2; multicloud model routing; MCP server and MCP transcoding; Model Armor integration	Verified fact	A6-S024
What it does	Token limit enforcement and token consumption monitoring, semantic caching, multicloud model routing with circuit breaking, LLM auditing and logging, API keys/OAuth 2.0/JWT, quotas, Model Armor prompt/response screening, exposing existing APIs as MCP tools.	Verified fact	A6-S024, A6-S023
Stack position	Gateway/control plane for AI traffic in Google Cloud and multicloud estates	Verified fact	A6-S024
Integration	Apigee proxies and policies; extension processor applies policies to traffic that bypasses a proxy; MCP endpoints (hybrid 1.17)	Verified fact	A6-S023, A6-S025
Dependencies	Model Armor for prompt/response screening; vector search index and embeddings for semantic cache	Verified fact	A6-S023, A6-S024
Certifications	Apigee listed in Google Cloud SOC 2 and ISO/IEC 27001 scope	Verified fact	A6-S070, A6-S071
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Pay-as-you-go: Standard proxy US\$20 per 1M calls (up to 50M), Extensible proxy US\$100 per 1M calls (up to 50M), tiered down at volume; API Analytics add-on US\$20 per 1M calls; Advanced API Security	Verified fact	A6-S069

Field	Value	Label	Sources
	US\$350 per 1M calls; subscription tiers also offered		
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Mature API management product; AI-specific policies and MCP support added 2025-2026	Verified fact	A6-S023, A6-S025
Deployment	saas: Yes (Apigee on Google Cloud); managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes (Apigee hybrid runtime); on_prem: Not publicly verified		

Portkey AI Gateway, now branded ‘Prisma AIRS AI Gateway’ (Palo Alto Networks) (c1-portkey)

Tier: Tactical • **Flags:** Acquired, Renamed • **Original graphic label:** not in graphic

Rationale: Tactical: capable, but ownership and branding changed four months ago and the post-acquisition commercial terms are not public ^{A1}.

Criterion	Score	Rationale
Technical	4	Routing, fallback, budgets, guardrails, MCP gateway ^{vf1044} ; A2A not evidenced ^{NPV}
Enterprise readiness	4	SSO/SAML 2.0, SCIM, organisation and workspace roles and organisation-wide audit logs (B-C1-S001, B-C1-S002), with IdP group mapping in the docs changelog (B-C1-S003): all three controls plus SCIM reach 4 under rule 7 (CP3 review A, consistent with Firecrawl); condition: evidence is mainly vendor pages and post-acquisition terms are not public

¹⁰⁴⁴ Verified fact — A6-S049: Portkey AI Gateway README (GitHub) (Portkey (Palo Alto Networks)) <https://raw.githubusercontent.com/Portkey-AI/gateway/main/README.md> accessed 2026-10-07; A6-S014: What is Portkey? - Portkey (Prisma AIRS AI Gateway) Docs (Portkey (Palo Alto Networks)) <https://portkey.ai/docs/introduction/what-is-portkey> accessed 2026-10-07 [search-tool extract]

Criterion	Score	Rationale
Security and compliance	3	SOC 2 Type 2, ISO 27001, GDPR and HIPAA claimed, tier coverage unclear ^{vf1045} , scope rule (CP2 Q4) applied
Deployment flexibility	4	SaaS, VPC, private cloud and self-hosted OSS; air-gap no longer offered ^{vf1046}
Ecosystem	4	OpenAI-compatible API and SDKs, 1,600+ models, MCP, Cloudflare Workers ^{vf1047}
Reliability and maturity	3	Acquired May 2026; Prisma AIRS AI Gateway GA 16 July 2026 ^{vf1048} ; OSS Gateway 2.0 pre-release ^{vf1049}
Cost / TCO	3	Published tiers (free, US\$49/month Production, custom Enterprise) may be stale; post-acquisition pricing not verified ^{vf1050}
Lock-in / portability	3	MIT core reduces lock-in, but governance features are proprietary; ownership change minus 1 (rule 3) ^{vf1051}
Total (generic / FS)	3.60 / 3.50	

Evidence rules applied: enterprise_readiness: NPV cap lifted by verified SSO, roles, audit logs and SCIM (rule 7)

Capabilities. AI gateway routing to 1,600+ models with fallbacks, retries, load balancing, caching, budgets, guardrails and logging, plus an MCP Gateway with a single auth layer; now Palo Alto Networks' Prisma AIRS AI Gateway ^{vf1052}.

Strengths

- Mature routing and fallback configuration as JSON Configs, with guardrails in the same config ^{vf1053}

¹⁰⁴⁵ Verified fact — A6-S013: Portkey pricing and Feature Comparison docs (Portkey (Palo Alto Networks)) <https://portkey.ai/pricing> accessed 2026-10-07 [search-tool extract]

¹⁰⁴⁶ Verified fact — A6-S013: Portkey pricing and Feature Comparison docs (Portkey (Palo Alto Networks)) <https://portkey.ai/pricing> accessed 2026-10-07 [search-tool extract]; A6-S049: Portkey AI Gateway README (GitHub) (Portkey (Palo Alto Networks)) <https://raw.githubusercontent.com/Portkey-AI/gateway/main/README.md> accessed 2026-10-07

¹⁰⁴⁷ Verified fact — A6-S049: Portkey AI Gateway README (GitHub) (Portkey (Palo Alto Networks)) <https://raw.githubusercontent.com/Portkey-AI/gateway/main/README.md> accessed 2026-10-07

¹⁰⁴⁸ Verified fact — V2-S048: Announcing the General Availability of Prisma AIRS AI Gateway (with Calclist/SDxCentral on Koi completion) (Palo Alto Networks) <https://www.paloaltonetworks.com/blog/2026/07/announcing-general-availability-of-prisma-air-ai-gateway/> accessed 2026-10-08 [search-tool extract]

¹⁰⁴⁹ Verified fact — A6-S049: Portkey AI Gateway README (GitHub) (Portkey (Palo Alto Networks)) <https://raw.githubusercontent.com/Portkey-AI/gateway/main/README.md> accessed 2026-10-07

¹⁰⁵⁰ Verified fact — A6-S013: Portkey pricing and Feature Comparison docs (Portkey (Palo Alto Networks)) <https://portkey.ai/pricing> accessed 2026-10-07 [search-tool extract]

¹⁰⁵¹ Verified fact — A6-S050: Portkey AI Gateway LICENSE (Portkey (Palo Alto Networks)) <https://raw.githubusercontent.com/Portkey-AI/gateway/main/LICENSE> accessed 2026-10-07; A6-S011: Palo Alto Networks Completes Acquisition of Portkey to Secure AI Agents (and 30 Apr 2026 announcement) <https://www.paloaltonetworks.com/company/press/2026/palo-alto-networks-completes-acquisition-of-portkey-to-secure-ai-agents> accessed 2026-10-07 [search-tool extract]

¹⁰⁵² Verified fact — A6-S049: Portkey AI Gateway README (GitHub) (Portkey (Palo Alto Networks)) <https://raw.githubusercontent.com/Portkey-AI/gateway/main/README.md> accessed 2026-10-07; A6-S011: Palo Alto Networks Completes Acquisition of Portkey to Secure AI Agents (and 30 Apr 2026 announcement) <https://www.paloaltonetworks.com/company/press/2026/palo-alto-networks-completes-acquisition-of-portkey-to-secure-ai-agents> accessed 2026-10-07 [search-tool extract]; A6-S014: What is Portkey? - Portkey (Prisma AIRS AI Gateway) Docs (Portkey (Palo Alto Networks)) <https://portkey.ai/docs/introduction/what-is-portkey> accessed 2026-10-07 [search-tool extract]

- MIT open-source gateway with a pre-release Gateway 2.0 that merges enterprise gateway code into open source ^{vf-1054}
- Now sits inside a security vendor's runtime-inspection platform ^{vf-1055}

Limitations and risks

- Acquired by Palo Alto Networks, completed 29 May 2026; US\$117m per the 10-K ^{vf-1056}; roadmap now tied to Prisma AIRS ^{vf-1057}
- Certifications claimed on pricing pages with unclear tier coverage; pricing page possibly stale ^{vf-1058}
- Gateway 2.0 is pre-release; fully air-gapped deployment no longer offered ^{vf-1059}
- Access-control evidence comes largely from marketing pages ^{vf-1060}

Choose when

- You already run Prisma AIRS and want gateway and runtime inspection from one security vendor ^{aj}
- You want a managed gateway with VPC or private-cloud options ^{vf-1061 aj}

Avoid when

- You need vendor-neutral governance of the traffic path, or air-gapped deployment ^{aj}
- You cannot absorb a contract re-papering after the change of control ^{aj}

Nearest competitors: C1-litellm, C1-kong-ai-gateway, C1-cloudflare-ai-gateway

Regulated-FS note. Refresh third-party due diligence and the DORA register entry after the change of control; confirm which certifications cover the Prisma AIRS-branded service ^{rec}.

¹⁰⁵³ Verified fact — A6-S049: Portkey AI Gateway README (GitHub) (Portkey (Palo Alto Networks)) <https://raw.githubusercontent.com/Portkey-AI/gateway/main/README.md> accessed 2026-10-07

¹⁰⁵⁴ Verified fact — A6-S049: Portkey AI Gateway README (GitHub) (Portkey (Palo Alto Networks)) <https://raw.githubusercontent.com/Portkey-AI/gateway/main/README.md> accessed 2026-10-07; A6-S050: Portkey AI Gateway LICENSE (Portkey (Palo Alto Networks)) <https://raw.githubusercontent.com/Portkey-AI/gateway/main/LICENSE> accessed 2026-10-07

¹⁰⁵⁵ Verified fact — A6-S011: Palo Alto Networks Completes Acquisition of Portkey to Secure AI Agents (and 30 Apr 2026 announcement) <https://www.paloaltonetworks.com/company/press/2026/palo-alto-networks-completes-acquisition-of-portkey-to-secure-ai-agents> accessed 2026-10-07 [search-tool extract]

¹⁰⁵⁶ Verified fact — A6-S012: Palo Alto Networks Inc - Form 10-K - FY2026 (fiscal year ended 31 July 2026) (US SEC EDGAR / Palo Alto Networks) <https://www.sec.gov/Archives/edgar/data/0001327567/000132756726000023/panw-20260731.htm> accessed 2026-10-07 [search-tool extract]; V2-S025: Palo Alto Networks Completes Acquisition of Portkey to Secure AI Agents (with PANW FY2026 10-K and 10-Q, SDxCentral) <https://www.paloaltonetworks.com/company/press/2026/palo-alto-networks-completes-acquisition-of-portkey-to-secure-ai-agents> accessed 2026-10-08 [search-tool extract]

¹⁰⁵⁷ Verified fact — A6-S011: Palo Alto Networks Completes Acquisition of Portkey to Secure AI Agents (and 30 Apr 2026 announcement) <https://www.paloaltonetworks.com/company/press/2026/palo-alto-networks-completes-acquisition-of-portkey-to-secure-ai-agents> accessed 2026-10-07 [search-tool extract]

¹⁰⁵⁸ Verified fact — A6-S013: Portkey pricing and Feature Comparison docs (Portkey (Palo Alto Networks)) <https://portkey.ai/pricing> accessed 2026-10-07 [search-tool extract]

¹⁰⁵⁹ Verified fact — A6-S049: Portkey AI Gateway README (GitHub) (Portkey (Palo Alto Networks)) <https://raw.githubusercontent.com/Portkey-AI/gateway/main/README.md> accessed 2026-10-07; A6-S013: Portkey pricing and Feature Comparison docs (Portkey (Palo Alto Networks)) <https://portkey.ai/pricing> accessed 2026-10-07 [search-tool extract]

¹⁰⁶⁰ Verified fact — B-C1-S001: Precise RBAC for AI innovation - Role-based access control (plus SSO docs) (Portkey (Palo Alto Networks)) <https://portkey.ai/for/rbac> accessed 2026-10-08 [search-tool extract]; B-C1-S002: Organisation-wide Audit logs (Portkey (Palo Alto Networks)) <https://portkey.ai/for/org-wide-audit-logs> accessed 2026-10-08 [search-tool extract]

¹⁰⁶¹ Verified fact — A6-S013: Portkey pricing and Feature Comparison docs (Portkey (Palo Alto Networks)) <https://portkey.ai/pricing> accessed 2026-10-07 [search-tool extract]

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Palo Alto Networks (acquired Portkey, Inc.; completed 29 May 2026)	Verified fact	A6-S011, A6-S012, V2-S025
Category	AI gateway (LLM gateway plus MCP gateway) with guardrails, observability and governance	Verified fact	A6-S049, A6-S014
Version / lineup	Open-source gateway on GitHub with 'Gateway 2.0 (Pre-Release)' that merges Portkey's core enterprise gateway into open source; Prisma AIRS AI Gateway described by PANW as generally available	Verified fact	A6-S049, A6-S011
Licence	Open core: open-source gateway MIT (Copyright 2024 Portkey, Inc.); hosted and enterprise platform proprietary	Verified fact	A6-S050, A6-S013
Status events	30 April 2026: Palo Alto Networks announced intent to acquire Portkey; 29 May 2026: acquisition completed; FY2026 Form 10-K reports total purchase consideration of US\$117m (10-Q had stated US\$140m incl. replacement awards); 2026: product documentation retitled 'Portkey (Prisma AIRS AI Gateway)'	Verified fact	A6-S011, A6-S012, A6-S014, V2-S025, V2-S048
Strategic direction	Becoming the AI gateway inside PANW's Prisma AIRS platform, inspecting AI traffic at runtime and enforcing security and governance policy for agents; core enterprise gateway being merged into open source with Gateway 2.0	Verified fact	A6-S011, A6-S049
What it does	Routes requests to 1,600+ models through one API with fallbacks, retries, load balancing, timeouts, caching, budgets/quotas, guardrails and logging; MCP Gateway gives a single auth layer, access control and observability for agent-to-tool connections.	Verified fact	A6-S049, A6-S011, A6-S014
Stack position	Gateway/control plane in the traffic path between applications or agents and model providers and MCP servers; now also a security enforcement point for PANW	Verified fact	A6-S011
Integration	OpenAI-compatible API and SDKs; JSON 'Configs' for routing rules, retries and input/output guardrails; MCP Gateway	Verified fact	A6-S049
Dependencies	Open-source gateway runs on Node.js (npm), Docker or Cloudflare Workers	Verified fact	A6-S049
Certifications	SOC 2 Type 2, ISO 27001, GDPR and HIPAA claimed on Portkey enterprise/pricing pages; tier coverage unclear	Verified fact	A6-S013
GDPR /	GDPR compliance claimed; custom BAAs and data isolation on	Verified fact	A6-S013

Field	Value	Label	Sources
residency	Enterprise; EU processing region not verified	d fact	
Security features	Data isolation and export to customer data lakes (Enterprise); runtime inspection of AI traffic under Prisma AIRS	Verified fact	A6-S013, A6-S011
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Developer: free (10k recorded logs/month, 3-day retention). Production: US\$49/month (100k requests; US\$9 per extra 100k up to 3M). Enterprise: custom. Post-acquisition pricing under Prisma AIRS not verified.	Verified fact	A6-S013
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	1,600+ language, vision, audio and image models; MCP servers; deployable on Cloudflare Workers	Verified fact	A6-S049
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Prisma AIRS AI Gateway described as generally available; open-source Gateway 2.0 still pre-release	Verified fact	A6-S011, A6-S049
Deployment	saas: Yes (hosted gateway / managed SaaS); managed_cloud: Not publicly verified; vpc_byoc: Yes (VPC hosting on Enterprise plan); private_cloud: Yes (private cloud deployment on Enterprise plan); self_hosted: Yes (open-source gateway); on_prem: No for fully air-gapped deployment ('No Longer Offered' in feature comparison)		

agentgateway (c1-agentgateway)

Tier: Tactical · **Flags:** none · **Original graphic label:** not in graphic

Rationale: Tactical: the most neutral governance in the layer, but young and with unverified hygiene ^{A1}.

Criterion	Score	Rationale
Technical	4	LLM, MCP and A2A gateway with budgets, failover and OAuth ^{vi1062} , caching and guardrail depth not verified ^{NPV}

¹⁰⁶² Verified fact — A6-S061: agentgateway README (GitHub) (agentgateway project (Linux Foundation))
<https://raw.githubusercontent.com/agentgateway/agentgateway/main/README.md> accessed 2026-10-07

Criterion	Score	Rationale
Enterprise readiness	3	Rule 2: OAuth on MCP and a commercial enterprise distribution with audit tooling ^{vf1063}
Security and compliance	2	Rule 2 hygiene (security policy, CVE handling, signed releases) not verified ^{NPV}
Deployment flexibility	4	Self-hosted ^{vf1064} ; enterprise distribution from Solo ^{vf1065}
Ecosystem	3	OpenAI-compatible, MCP, A2A, OpenAPI; multi-vendor contributors ^{vf1066}
Reliability and maturity	3	1.x line, v1.6.0 on 2 October 2026; foundation-hosted since August 2025 ^{vf1067}
Cost / TCO	4	Free open source; operations cost is your own ^{vf1068}
Lock-in / portability	5	Apache-2.0 with neutral governance ^{vf1069}
Total (generic / FS)	3.40	
	/	
	3.45	

Evidence rules applied: Rule 2 basis: self-hosted open-source software; NPV cap not applied; commercial support exists (B-C1-S010)

Capabilities. Open-source proxy for agent-to-LLM, agent-to-tool and agent-to-agent traffic: OpenAI-compatible routing with budget and spend controls, load balancing and failover, MCP federation with OAuth, and an A2A gateway ^{vf1070}.

Strengths

¹⁰⁶³ Verified fact — A6-S061: agentgateway README (GitHub) (agentgateway project (Linux Foundation)) <https://raw.githubusercontent.com/agentgateway/agentgateway/main/README.md> accessed 2026-10-07; B-C1-S010: Solo.io Launches Solo Enterprise for agentgateway and Solo Labs for MCP <https://www.solo.io/press-releases/enterprise-agentgateway-mcp-labs> accessed 2026-10-08 [search-tool extract]

¹⁰⁶⁴ Verified fact — A6-S061: agentgateway README (GitHub) (agentgateway project (Linux Foundation)) <https://raw.githubusercontent.com/agentgateway/agentgateway/main/README.md> accessed 2026-10-07

¹⁰⁶⁵ Verified fact — B-C1-S010: Solo.io Launches Solo Enterprise for agentgateway and Solo Labs for MCP <https://www.solo.io/press-releases/enterprise-agentgateway-mcp-labs> accessed 2026-10-08 [search-tool extract]

¹⁰⁶⁶ Verified fact — A6-S061: agentgateway README (GitHub) (agentgateway project (Linux Foundation)) <https://raw.githubusercontent.com/agentgateway/agentgateway/main/README.md> accessed 2026-10-07; B-C1-S009: Linux Foundation Welcomes Agentgateway Project (The Linux Foundation) <https://www.linuxfoundation.org/press/linux-foundation-welcomes-agentgateway-project-to-accelerate-ai-agent-adoption-while-maintaining-security-observability-and-governance> accessed 2026-10-08 [search-tool extract]

¹⁰⁶⁷ Verified fact — A6-S062: Go module proxy: github.com/agentgateway/agentgateway @latest (Go module proxy (proxy.golang.org)) <https://proxy.golang.org/github.com/agentgateway/agentgateway/@latest> accessed 2026-10-07; B-C1-S009: Linux Foundation Welcomes Agentgateway Project (The Linux Foundation) <https://www.linuxfoundation.org/press/linux-foundation-welcomes-agentgateway-project-to-accelerate-ai-agent-adoption-while-maintaining-security-observability-and-governance> accessed 2026-10-08 [search-tool extract]

¹⁰⁶⁸ Verified fact — A6-S061: agentgateway README (GitHub) (agentgateway project (Linux Foundation)) <https://raw.githubusercontent.com/agentgateway/agentgateway/main/README.md> accessed 2026-10-07

¹⁰⁶⁹ Verified fact — A6-S061: agentgateway README (GitHub) (agentgateway project (Linux Foundation)) <https://raw.githubusercontent.com/agentgateway/agentgateway/main/README.md> accessed 2026-10-07; B-C1-S009: Linux Foundation Welcomes Agentgateway Project (The Linux Foundation) <https://www.linuxfoundation.org/press/linux-foundation-welcomes-agentgateway-project-to-accelerate-ai-agent-adoption-while-maintaining-security-observability-and-governance> accessed 2026-10-08 [search-tool extract]

¹⁰⁷⁰ Verified fact — A6-S061: agentgateway README (GitHub) (agentgateway project (Linux Foundation)) <https://raw.githubusercontent.com/agentgateway/agentgateway/main/README.md> accessed 2026-10-07

- Apache-2.0 under vendor-neutral foundation governance ^{VF-1071}
- Covers LLM, MCP and A2A in one data plane ^{VF-1072}
- Commercial distribution and support exist (Solo Enterprise for agentgateway) ^{VF-1073}

Limitations and risks

- Security hygiene (policy, signed releases) and access-control detail not verified ^{NPV}
- Current foundation home reported differently (Linux Foundation vs Agentic AI Foundation) ^{VF-1074}
- Adoption not verified ^{NPV}

Choose when

- You want a neutral, open-source gateway for MCP and A2A traffic in Kubernetes estates ^{AJ}

Avoid when

- You need certified SaaS or turnkey cost attribution today ^{AJ}

Nearest competitors: C1-litellm, C1-envoy-ai-gateway, C1-kong-ai-gateway

Regulated-FS note. Pilot for MCP and A2A traffic with the enterprise distribution, and verify its release-signing and CVE process before production ^{REC}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	agentgateway project, a Linux Foundation project	Verified fact	A6-S061
Category	Open-source AI-native proxy: LLM gateway, MCP gateway and A2A gateway	Verified fact	A6-S061
Version / lineup	v1.6.0 (2 October 2026)	Verified fact	A6-S062, V2-S061
Licence	Open source (Apache-2.0)	Verified fact	A6-S061
Strategic direction	Not publicly verified	Not publicly verified	
What it does	Proxy providing security, observability and governance for agent-to-LLM, agent-to-tool and agent-to-agent traffic: OpenAI-compatible LLM	Verified fact	A6-S061

¹⁰⁷¹ Verified fact — A6-S061: agentgateway README (GitHub) (agentgateway project (Linux Foundation)) <https://raw.githubusercontent.com/agentgateway/agentgateway/main/README.md> accessed 2026-10-07; B-C1-S009: Linux Foundation Welcomes Agentgateway Project (The Linux Foundation) <https://www.linuxfoundation.org/press/linux-foundation-welcomes-agentgateway-project-to-accelerate-ai-agent-adoption-while-maintaining-security-observability-and-governance> accessed 2026-10-08 [search-tool extract]

¹⁰⁷² Verified fact — A6-S061: agentgateway README (GitHub) (agentgateway project (Linux Foundation)) <https://raw.githubusercontent.com/agentgateway/agentgateway/main/README.md> accessed 2026-10-07

¹⁰⁷³ Verified fact — B-C1-S010: Solo.io Launches Solo Enterprise for agentgateway and Solo Labs for MCP <https://www.solo.io/press-releases/enterprise-agentgateway-mcp-labs> accessed 2026-10-08 [search-tool extract]

¹⁰⁷⁴ Verified fact — B-C1-S009: Linux Foundation Welcomes Agentgateway Project (The Linux Foundation) <https://www.linuxfoundation.org/press/linux-foundation-welcomes-agentgateway-project-to-accelerate-ai-agent-adoption-while-maintaining-security-observability-and-governance> accessed 2026-10-08 [search-tool extract]

Field	Value	Label	Sources
	routing with budget and spend controls, load balancing and failover; MCP tool federation over stdio/HTTP/SSE/Streamable HTTP with OAuth; A2A gateway.		
Stack position	Gateway/control plane spanning model, tool and agent traffic	Verified fact	A6-S061
Integration	OpenAI-compatible API, MCP, A2A, OpenAPI integration	Verified fact	A6-S061
Dependencies	Not publicly verified	Not publicly verified	
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Free open source	Verified fact	A6-S061
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
Adoption signals	Not publicly verified	d Not publicly verified	
Maturity	1.x release line; latest v1.6.0 on 2 October 2026	Verified fact	A6-S062
Deployment	saas: Not publicly verified; managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes; on_prem: Not publicly verified		

AI gateway in Azure API Management (plus the AI Gateway tier, preview) (c1-azure-apim-ai-gateway)

Tier: Strategic · **Flags:** none · **Original graphic label:** not in graphic

Rationale: Strategic, conditional: where Azure is your primary cloud (CP3 Q2, rubric rule 10), using the GA AI gateway policies in existing APIM tiers, not the preview AI Gateway tier (no SLA); lock-in 2 is accepted under rule 11 because the condition is an existing platform commitment ^{A1}.

Criterion	Score	Rationale
Technical	4	Token limits, metrics, semantic cache, load balancing, circuit breaking, content safety on LLM/MCP/A2A ^{vs1075}
Enterprise readiness	4	Hyperscaler presumption (CP2 Q1): platform controls presumed; Entra sign-in and caller-keyed limits verified ^{vs1076} ; confirm per service
Security and compliance	4	FedRAMP High audit scope names API Management ^{vs1077} ; Azure ISO 27001 and SOC 2 are platform-level without naming the service ^{vs1078} ; scope rule applied, so 4
Deployment	2	Managed Azure service only in the fact base; preview tier in two regions ^{vs1079} ;

¹⁰⁷⁵ Verified fact — A6-S053: AI gateway capabilities in Azure API Management (Microsoft Learn source, ms.date 29 May 2026) <https://raw.githubusercontent.com/MicrosoftDocs/azure-docs/main/articles/api-management/genai-gateway-capabilities.md> accessed 2026-10-07; A6-S020: Azure API Management AI gateway docs and Tech Community announcements (AI Gateway tier preview; Build 2026 updates) (Microsoft) <https://learn.microsoft.com/en-us/azure/api-management/ai-gateway-overview> accessed 2026-10-07 [search-tool extract]

¹⁰⁷⁶ Verified fact — A6-S020: Azure API Management AI gateway docs and Tech Community announcements (AI Gateway tier preview; Build 2026 updates) (Microsoft) <https://learn.microsoft.com/en-us/azure/api-management/ai-gateway-overview> accessed 2026-10-07 [search-tool extract]; A6-S053: AI gateway capabilities in Azure API Management (Microsoft Learn source, ms.date 29 May 2026) <https://raw.githubusercontent.com/MicrosoftDocs/azure-docs/main/articles/api-management/genai-gateway-capabilities.md> accessed 2026-10-07

¹⁰⁷⁷ Verified fact — A6-S056: Azure services in FedRAMP and DoD audit scope (Microsoft Learn source) <https://raw.githubusercontent.com/MicrosoftDocs/azure-docs/main/articles/azure-government/compliance/azure-services-in-fedramp-auditscope.md> accessed 2026-10-07

¹⁰⁷⁸ Verified fact — B-C1-S007: ISO/IEC 27001 - Azure Compliance (Microsoft) <https://learn.microsoft.com/en-us/azure/compliance/offerings/offering-iso-27001> accessed 2026-10-08 [search-tool extract]; B-C1-S008: System and Organization Controls (SOC) 2 Type 2 - Azure Compliance (Microsoft) <https://learn.microsoft.com/en-us/azure/compliance/offerings/offering-soc-2> accessed 2026-10-08 [search-tool extract]

¹⁰⁷⁹ Verified fact — A6-S053: AI gateway capabilities in Azure API Management (Microsoft Learn source, ms.date 29 May 2026) <https://raw.githubusercontent.com/MicrosoftDocs/azure-docs/main/articles/api-management/genai-gateway-capabilities.md> accessed 2026-10-07; A6-S020: Azure API Management AI gateway docs and Tech Community announcements (AI Gateway tier preview; Build 2026 updates) (Microsoft) <https://learn.microsoft.com/en-us/azure/api-management/ai-gateway-overview> accessed 2026-10-07 [search-tool extract]

Criterion	Score	Rationale
flexibility		self-hosted gateway not verified ^{NPV}
Ecosystem	4	OpenAI-compatible unified API (preview), Anthropic Messages API, Foundry, MCP and A2A import ^{VF1080}
Reliability and maturity	3	Core LLM policies GA; tier and unified API preview without SLA ^{VF1081}
Cost / TCO	3	Preview tier free; classic tier prices not verified ^{VF1082 NPV}
Lock-in / portability	2	APIM-specific policy XML and Azure dependency ^{VF1083}
Total (generic / FS)	3.40	
	/	
	3.25	

Evidence rules applied: enterprise_readiness set to 4 under the hyperscaler presumption (CP2 Q1): platform controls presumed; confirm per service

Capabilities. AI gateway policies in Azure API Management: token limits and quotas per consumer, token metrics, semantic caching, load balancing and circuit breaking, Content Safety (incl. Prompt Shields) on LLM, MCP and A2A traffic, MCP exposure and governance, OAuth via credential manager; plus a dedicated AI Gateway tier in preview ^{VF1084}.

Strengths

- Extends an API gateway many Azure estates already operate; Microsoft calls it ‘not a separate offering’ ^{VF1085}
- Content safety applies to MCP tool-call arguments and A2A payloads, not only prompts ^{VF1086}
- API Management is in Azure FedRAMP High audit scope ^{VF1087}

Limitations and risks

¹⁰⁸⁰ Verified fact — A6-S053: AI gateway capabilities in Azure API Management (Microsoft Learn source, ms.date 29 May 2026) <https://raw.githubusercontent.com/MicrosoftDocs/azure-docs/main/articles/api-management/genai-gateway-capabilities.md> accessed 2026-10-07

¹⁰⁸¹ Verified fact — A6-S020: Azure API Management AI gateway docs and Tech Community announcements (AI Gateway tier preview; Build 2026 updates) (Microsoft) <https://learn.microsoft.com/en-us/azure/api-management/ai-gateway-overview> accessed 2026-10-07 [search-tool extract]; A6-S053: AI gateway capabilities in Azure API Management (Microsoft Learn source, ms.date 29 May 2026) <https://raw.githubusercontent.com/MicrosoftDocs/azure-docs/main/articles/api-management/genai-gateway-capabilities.md> accessed 2026-10-07

¹⁰⁸² Verified fact — A6-S020: Azure API Management AI gateway docs and Tech Community announcements (AI Gateway tier preview; Build 2026 updates) (Microsoft) <https://learn.microsoft.com/en-us/azure/api-management/ai-gateway-overview> accessed 2026-10-07 [search-tool extract]

¹⁰⁸³ Verified fact — A6-S053: AI gateway capabilities in Azure API Management (Microsoft Learn source, ms.date 29 May 2026) <https://raw.githubusercontent.com/MicrosoftDocs/azure-docs/main/articles/api-management/genai-gateway-capabilities.md> accessed 2026-10-07

¹⁰⁸⁴ Verified fact — A6-S053: AI gateway capabilities in Azure API Management (Microsoft Learn source, ms.date 29 May 2026) <https://raw.githubusercontent.com/MicrosoftDocs/azure-docs/main/articles/api-management/genai-gateway-capabilities.md> accessed 2026-10-07; A6-S020: Azure API Management AI gateway docs and Tech Community announcements (AI Gateway tier preview; Build 2026 updates) (Microsoft) <https://learn.microsoft.com/en-us/azure/api-management/ai-gateway-overview> accessed 2026-10-07 [search-tool extract]

¹⁰⁸⁵ Verified fact — A6-S053: AI gateway capabilities in Azure API Management (Microsoft Learn source, ms.date 29 May 2026) <https://raw.githubusercontent.com/MicrosoftDocs/azure-docs/main/articles/api-management/genai-gateway-capabilities.md> accessed 2026-10-07

¹⁰⁸⁶ Verified fact — A6-S020: Azure API Management AI gateway docs and Tech Community announcements (AI Gateway tier preview; Build 2026 updates) (Microsoft) <https://learn.microsoft.com/en-us/azure/api-management/ai-gateway-overview> accessed 2026-10-07 [search-tool extract]

¹⁰⁸⁷ Verified fact — A6-S056: Azure services in FedRAMP and DoD audit scope (Microsoft Learn source) <https://raw.githubusercontent.com/MicrosoftDocs/azure-docs/main/articles/azure-government/compliance/azure-services-in-fedramp-auditscope.md> accessed 2026-10-07

- AI Gateway tier is preview: East US 2 and Sweden Central only, free, no SLA ^{VF¹⁰⁸⁸}
- Policies are APIM-specific XML ^{VF¹⁰⁸⁹}
- Classic tier pricing and self-hosted options not verified ^{NPV}

Choose when

- Azure is your primary cloud and APIM your API standard ^{AJ}
- You want content safety enforced on tool and agent traffic at the gateway ^{AJ}

Avoid when

- You need the AI Gateway tier in production now (no SLA) ^{AJ}
- Your multi-cloud exit plan needs a gateway outside any one hyperscaler ^{AJ}

Nearest competitors: C1-google-apigee-ai-gateway, C1-kong-ai-gateway, C1-litellm

Regulated-FS note. Use the GA policies in existing tiers for production; do not place regulated traffic on the preview tier; keep policy logic documented so it can be re-expressed elsewhere ^{REC}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Microsoft	Verified fact	A6-S053
Category	Cloud-native API management platform with AI gateway policies (LLM, MCP and A2A)	Verified fact	A6-S053
Version / lineup	Policy-based AI gateway available across APIM tiers (doc dated 29 May 2026); AI Gateway tier in public preview since about late July 2026; unified model API (preview)	Verified fact	A6-S053, A6-S020, V2-S073
Licence	Proprietary managed Azure service	Verified fact	A6-S053
Strategic direction	Extends APIM to govern models from Foundry, OpenAI-compatible third parties, Anthropic and Google Vertex AI, MCP servers and A2A agents; integration into Microsoft Foundry as its AI gateway control plane (preview)	Verified fact	A6-S053, A6-S020
What it does	Token limit and quota policies per consumer key, token metrics, semantic caching (Azure Managed Redis), load balancing and circuit breaking across AI backends, content safety checks via Azure AI Content	Verified fact	A6-S053, A6-S020

¹⁰⁸⁸ Verified fact — A6-S020: Azure API Management AI gateway docs and Tech Community announcements (AI Gateway tier preview; Build 2026 updates) (Microsoft) <https://learn.microsoft.com/en-us/azure/api-management/ai-gateway-overview> accessed 2026-10-07 [search-tool extract]; V2-S073: AI Gateway tier (preview) overview - Azure API Management (with Tech Community 'AI Gateway tier of API Management now in public preview') (Microsoft) <https://learn.microsoft.com/en-us/azure/api-management/ai-gateway-overview> accessed 2026-10-08 [search-tool extract]

¹⁰⁸⁹ Verified fact — A6-S053: AI gateway capabilities in Azure API Management (Microsoft Learn source, ms.date 29 May 2026) <https://raw.githubusercontent.com/MicrosoftDocs/azure-docs/main/articles/api-management/genai-gateway-capabilities.md> accessed 2026-10-07

Field	Value	Label	Sources
	Safety (incl. Prompt Shields) on LLM, MCP and A2A traffic, exposing REST APIs as MCP servers and governing existing MCP servers, OAuth via credential manager.		
Stack position	Extension of APIM's existing API gateway, 'not a separate offering'; control plane for AI traffic in Azure estates	Verified fact	A6-S053
Integration	Policy XML (e.g. llm-token-limit), OpenAI-compatible unified model API (preview), Anthropic Messages API (v2 tiers), MCP and A2A API import	Verified fact	A6-S053
Dependencies	Azure API Management instance; Azure Managed Redis (or RedisSearch-compatible cache) and an embeddings deployment for semantic caching; Azure AI Content Safety for moderation; Application Insights for metrics	Verified fact	A6-S053, A6-S020
Certifications	API Management is in Azure FedRAMP High and DoD IL2 audit scope (Azure public, list updated February 2026)	Verified fact	A6-S056
GDPR / residency	AI Gateway tier preview limited to East US 2 and Sweden Central regions	Verified fact	A6-S020
Security features	Not publicly verified	Not publicly verified	
Access controls	Microsoft Entra ID sign-in required for AI Gateway tier quickstart; token limits keyed on subscription, IP or caller identity; OAuth for MCP via credential manager	Verified fact	A6-S020, A6-S053
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	AI Gateway tier free during public preview, pricing to be announced; classic APIM tier pricing not verified	Verified fact	A6-S020
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Core LLM policies GA; AI Gateway tier and unified model API in preview with no SLA	Verified fact	A6-S020, A6-S053

Field	Value	Label	Sources
Deployment	saas: Yes (managed Azure service); managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Not publicly verified; on_prem: Not publicly verified		

Amazon Bedrock AgentCore Gateway (plus ‘Guidance for Multi-Provider Generative AI Gateway on AWS’) (c1-aws-agentcore-gateway)

Tier: Strategic · Flags: none · Original graphic label: not in graphic

Rationale: Strategic, conditional: where AWS is your primary cloud (CP3 Q2, rubric rule 10), as the gateway for MCP and tool traffic. Maturity 2 is a stated condition: keep model routing on a gateway with GA inference routing until AWS states GA and Regions for inference targets. Same tier as L4-aws-agentcore-gateway-identity ^{AI}.

Criterion	Score	Rationale
Technical	3	Strong MCP gateway and policy; LLM routing new with no GA wording; no semantic cache evidenced ^{vr1090} ^{NPV}
Enterprise readiness	4	Hyperscaler presumption (CP2 Q1); IAM/JWT auth and principal rules verified ^{vr1091} ; confirm per service
Security and compliance	4	AgentCore in SOC 1/2/3 scope (July 2026) and assessed for ISO/IEC 27001 (February 2026) ^{vr1092} ; CMK for the token vault ^{vr1093} . Held at 4, not 5: the ISO evidence is a partly garbled search extract and FedRAMP status conflicts
Deployment	2	Managed AWS service only; Gateway regional list not verified ^{vr1094}

¹⁰⁹⁰ Verified fact — A6-S074: AWS API model: Amazon Bedrock AgentCore control plane (botocore service-2.json) (Amazon Web Services (via boto/botocore)) <https://raw.githubusercontent.com/boto/botocore/develop/botocore/data/bedrock-agentcore-control/2023-06-05/service-2.json> accessed 2026-10-07; A6-S105: AgentCore Gateway: Inference targets; Inference connector/provider targets; Add rate limits to a gateway (Amazon Web Services) <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/gateway-targets-inference.html> accessed 2026-10-07 [search-tool extract]

¹⁰⁹¹ Verified fact — A6-S074: AWS API model: Amazon Bedrock AgentCore control plane (botocore service-2.json) (Amazon Web Services (via boto/botocore)) <https://raw.githubusercontent.com/boto/botocore/develop/botocore/data/bedrock-agentcore-control/2023-06-05/service-2.json> accessed 2026-10-07

¹⁰⁹² Verified fact — B-C1-S004: Release notes for Amazon Bedrock AgentCore (Amazon Web Services) <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/release-notes.html> accessed 2026-10-08 [search-tool extract]; B-C1-S005: AWS Services in Scope by Compliance Program - SOC (Amazon Web Services) <https://aws.amazon.com/compliance/services-in-scope/SOC/> accessed 2026-10-08 [search-tool extract]; B-C1-S006: Amazon Bedrock AgentCore FAQs (Amazon Web Services) <https://aws.amazon.com/bedrock/agentcore/faqs/> accessed 2026-10-08 [search-tool extract]

¹⁰⁹³ Verified fact — A6-S074: AWS API model: Amazon Bedrock AgentCore control plane (botocore service-2.json) (Amazon Web Services (via boto/botocore)) <https://raw.githubusercontent.com/boto/botocore/develop/botocore/data/bedrock-agentcore-control/2023-06-05/service-2.json> accessed 2026-10-07

¹⁰⁹⁴ Verified fact — A6-S021: Amazon Bedrock AgentCore Pricing; AgentCore GA announcement (Oct 2025); AgentCore Gateway docs (Amazon Web Services) <https://aws.amazon.com/bedrock/agentcore/pricing/> accessed 2026-10-07 [search-tool extract]; A6-S026: Policy in Amazon Bedrock AgentCore is now generally available; AgentCore Policy docs; Why Policy in AgentCore chose Cedar (Amazon Web Services) <https://aws.amazon.com/about-aws/whats-new/2026/03/policy-amazon-bedrock-agentcore-generally-available/> accessed 2026-10-07 [search-tool extract]

Criterion	Score	Rationale
flexibility		
Ecosystem	3	MCP-native; Kong SigV4 and Entra Agent ID integrations ^{vr1095}
Reliability and maturity	2	AgentCore GA October 2025; inference targets and token limits announced 6 August 2026 without GA wording ^{vr1096}
Cost / TCO	4	US\$0.005 per 1,000 invocations and similar unit prices ^{vr1097}
Lock-in / portability	2	AWS-specific control plane and IAM ^{vr1098}
Total (generic / FS)	3.10	
	/	
	3.00	

Evidence rules applied: enterprise_readiness set to 4 under the hyperscaler presumption (CP2 Q1): platform controls presumed; confirm per service

Capabilities. Managed gateway that turns APIs, Lambda functions and MCP servers into MCP tools behind one endpoint, with IAM or JWT inbound auth, request/response interceptors, Cedar policy enforcement and, newly, inference targets that route to LLM providers with token-aware rate limits; AWS has no product named 'AI gateway' ^{vr1099}.

Strengths

- Strongest policy model for tool calls: deny-by-default Cedar policies with decisions logged to CloudWatch ^{vr1100}
- Tracks MCP 2026-07-28 ^{vr1101}

¹⁰⁹⁵ Verified fact — A6-S016: Kong AI Gateway 2.0 Is Now GA — And It's Already Moving Faster (Kong Inc.) <https://konghq.com/blog/product-releases/kong-ai-gateway-2-0-ga> accessed 2026-10-07 [search-tool extract]; A6-S058: What's new in Microsoft Entra Agent ID (Microsoft Learn source) <https://raw.githubusercontent.com/MicrosoftDocs/entra-docs/main/docs/agent-id/whats-new-agent-id.md> accessed 2026-10-07

¹⁰⁹⁶ Verified fact — A6-S021: Amazon Bedrock AgentCore Pricing; AgentCore GA announcement (Oct 2025); AgentCore Gateway docs (Amazon Web Services) <https://aws.amazon.com/bedrock/agentcore/pricing/> accessed 2026-10-07 [search-tool extract]; A6-S106: Announcing temporal policies and rate limiting in Amazon Bedrock AgentCore (Amazon Web Services) <https://aws.amazon.com/about-aws/whats-new/2026/08/temporal-policies-agentcore/> accessed 2026-10-07 [search-tool extract]

¹⁰⁹⁷ Verified fact — A6-S021: Amazon Bedrock AgentCore Pricing; AgentCore GA announcement (Oct 2025); AgentCore Gateway docs (Amazon Web Services) <https://aws.amazon.com/bedrock/agentcore/pricing/> accessed 2026-10-07 [search-tool extract]

¹⁰⁹⁸ Verified fact — A6-S074: AWS API model: Amazon Bedrock AgentCore control plane (botocore service-2.json) (Amazon Web Services (via boto/botocore)) <https://raw.githubusercontent.com/boto/botocore/develop/botocore/data/bedrock-agentcore-control/2023-06-05/service-2.json> accessed 2026-10-07

¹⁰⁹⁹ Verified fact — A6-S021: Amazon Bedrock AgentCore Pricing; AgentCore GA announcement (Oct 2025); AgentCore Gateway docs (Amazon Web Services) <https://aws.amazon.com/bedrock/agentcore/pricing/> accessed 2026-10-07 [search-tool extract]; A6-S074: AWS API model: Amazon Bedrock AgentCore control plane (botocore service-2.json) (Amazon Web Services (via boto/botocore)) <https://raw.githubusercontent.com/boto/botocore/develop/botocore/data/bedrock-agentcore-control/2023-06-05/service-2.json> accessed 2026-10-07; A6-S105: AgentCore Gateway: Inference targets; Inference connector/provider targets; Add rate limits to a gateway (Amazon Web Services) <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/gateway-targets-inference.html> accessed 2026-10-07 [search-tool extract]; A6-S026: Policy in Amazon Bedrock AgentCore is now generally available; AgentCore Policy docs; Why Policy in AgentCore chose Cedar (Amazon Web Services) <https://aws.amazon.com/about-aws/whats-new/2026/03/policy-amazon-bedrock-agentcore-generally-available/> accessed 2026-10-07 [search-tool extract]

¹¹⁰⁰ Verified fact — A6-S026: Policy in Amazon Bedrock AgentCore is now generally available; AgentCore Policy docs; Why Policy in AgentCore chose Cedar (Amazon Web Services) <https://aws.amazon.com/about-aws/whats-new/2026/03/policy-amazon-bedrock-agentcore-generally-available/> accessed 2026-10-07 [search-tool extract]

¹¹⁰¹ Verified fact — A6-S021: Amazon Bedrock AgentCore Pricing; AgentCore GA announcement (Oct 2025); AgentCore Gateway docs (Amazon Web Services) <https://aws.amazon.com/bedrock/agentcore/pricing/> accessed 2026-10-07 [search-tool extract]

- Low, transparent per-invocation pricing ^{VF1102}

Limitations and risks

- No explicit GA statement or regional list for inference targets; documentation conflicts on multi-target selection ^{VF1103}
- Fast-moving API surface (171 control-plane operations) ^{VF1104}
- AWS-specific control plane ^{VF1105}

Choose when

- You run agents on AWS and need governed MCP tool access with policy enforcement ^{AJ}

Avoid when

- You need a mature multi-provider LLM gateway today; use AWS’s LiteLLM-based Guidance or another C1 product for model traffic ^{VF1106 AJ}

Nearest competitors: C1-litellm, C1-kong-ai-gateway, C1-agentgateway

Regulated-FS note. Use it for tool traffic first; keep model routing on a gateway with GA inference routing until AWS states GA and regions for inference targets ^{Rec}.

Facts (as of 2026-10-07)

Field	Value	Label	Sources
Company	Amazon Web Services	Verified fact	A6-S021
Category	Managed agent/MCP gateway with emerging LLM inference routing; plus a LiteLLM-based reference pattern	Verified fact	A6-S021, A6-S074, A6-S022
Version / lineup	AgentCore GA October 2025; Gateway supports MCP versions 2026-07-28, 2025-11-25, 2025-06-18, 2025-03-26; inference targets documented (connectors for bedrock-mantle, openai, anthropic; provider configs for OpenAI-compatible endpoints); token-based rate limiting announced 6 August 2026	Verified fact	A6-S021, A6-S105, A6-S106, A6-S074
Licence	Proprietary managed service (Guidance pattern uses open-source LiteLLM, MIT)	Verified fact	A6-S021, A6-S022

¹¹⁰² Verified fact — A6-S021: Amazon Bedrock AgentCore Pricing; AgentCore GA announcement (Oct 2025); AgentCore Gateway docs (Amazon Web Services) <https://aws.amazon.com/bedrock/agentcore/pricing/> accessed 2026-10-07 [search-tool extract]

¹¹⁰³ Verified fact — A6-S105: AgentCore Gateway: Inference targets; Inference connector/provider targets; Add rate limits to a gateway (Amazon Web Services) <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/gateway-targets-inference.html> accessed 2026-10-07 [search-tool extract]; A6-S106: Announcing temporal policies and rate limiting in Amazon Bedrock AgentCore (Amazon Web Services) <https://aws.amazon.com/about-aws/whats-new/2026/08/temporal-policies-agentcore/> accessed 2026-10-07 [search-tool extract]

¹¹⁰⁴ Verified fact — A6-S074: AWS API model: Amazon Bedrock AgentCore control plane (botocore service-2.json) (Amazon Web Services (via boto/botocore)) <https://raw.githubusercontent.com/boto/botocore/develop/botocore/data/bedrock-agentcore-control/2023-06-05/service-2.json> accessed 2026-10-07

¹¹⁰⁵ Verified fact — A6-S074: AWS API model: Amazon Bedrock AgentCore control plane (botocore service-2.json) (Amazon Web Services (via boto/botocore)) <https://raw.githubusercontent.com/boto/botocore/develop/botocore/data/bedrock-agentcore-control/2023-06-05/service-2.json> accessed 2026-10-07

¹¹⁰⁶ Verified fact — A6-S022: Guidance for Multi-Provider Generative AI Gateway on AWS; Implementing resilience patterns with Amazon Bedrock and LLM gateway (Amazon Web Services) <https://aws.amazon.com/solutions/guidance/multi-provider-generative-ai-gateway-on-aws/> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
Strategic direction	AgentCore Gateway is growing from an MCP tool gateway into a combined tool and model gateway: inference targets route to LLM providers; rate limits cover request rates, token consumption and concurrency; rules give principal-based access control and routing; Policy engines (Cedar) authorise every tool call	Verified fact	A6-S074, A6-S026
What it does	Turns APIs (OpenAPI, Smithy), Lambda functions and existing MCP servers into MCP tools behind one endpoint with inbound auth (custom JWT, IAM) and outbound credentials via AgentCore Identity, semantic tool search, request/response interceptors and policy enforcement; can route to LLM providers via inference targets. Inference targets expose an /inference path usable with the OpenAI or Anthropic SDK (route by model field; 'target/model' pins a provider); rate limits cover requests, tokens per minute (estimated up front, reconciled with provider usage) and connections, scoped by JWT claims, IAM	Verified fact	A6-S021, A6-S074, A6-S026, A6-S105
Stack position	Combined tool (MCP) and model (inference) gateway in the AWS agent stack; the LiteLLM-based Guidance remains AWS's published multi-provider gateway pattern	Verified fact	A6-S105, A6-S022
Integration	MCP (protocol type MCP); authoriser types CUSTOM_JWT, AWS_IAM, NONE, AUTHENTICATE_ONLY; interception points REQUEST/RESPONSE; MCP target types OpenAPI schema, Smithy model, Lambda, MCP server, API Gateway, connector	Verified fact	A6-S074
Dependencies	AWS account; AgentCore Identity for credentials; optional AgentCore Policy engine; Guidance pattern uses ECS/EKS, ALB, WAF, CloudFront/Route 53	Verified fact	A6-S021, A6-S022, A6-S074
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	AgentCore Policy GA in 13 regions incl. Europe (Ireland); Gateway regional list not verified	Verified fact	A6-S026
Security features	Customer-managed KMS key for AgentCore token vault (SetTokenVaultCMK API); policy decisions logged to CloudWatch	Verified fact	A6-S074, A6-S026
Access controls	IAM or custom JWT inbound auth; gateway rules keyed on IAM principals; Cedar policy engine with default deny on tool calls	Verified fact	A6-S074, A6-S026
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Gateway: US\$0.005 per 1,000 API invocations (ListTools, InvokeTool, Ping); US\$0.025 per 1,000 Search API calls; US\$0.02 per 100 tools indexed per month; data egress to customer VPC US\$0.006/GB	Verified fact	A6-S021
Infrastructure cost	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
Ecosystem	Kong AI Gateway 2.0 adds SigV4 auth for Bedrock AgentCore endpoints; Microsoft Entra Agent ID documents securing Bedrock agents	Verified fact	A6-S016, A6-S058
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	AgentCore GA October 2025; inference targets documented and token rate limiting announced 6 August 2026, but no explicit GA statement or regional list for inference targets was found; docs conflict on multi-target selection (round-robin vs random)	Verified fact	A6-S021, A6-S105, A6-S106
Deployment	saas: Yes (managed AWS service); managed_cloud: Not publicly verified; vpc_byoc: Guidance pattern deploys into the customer's own AWS account (ECS/EKS); private_cloud: Not publicly verified; self_hosted: Not publicly verified; on_prem: Not publicly verified		

Agent Router (formerly Envoy AI Gateway) (c1-envoy-ai-gateway)

Tier: Tactical · **Flags:** Renamed · **Original graphic label:** not in graphic

Rationale: Tactical: sound plumbing for Envoy estates, but narrower than the leaders and with unverified hygiene ^{AJ}.

Criterion	Score	Rationale
Technical	3	Routing, quotas, failover, credentials and attribution for LLM and MCP ^{vf¹¹⁰⁷} ; guardrails and caching not verified ^{NPV}
Enterprise readiness	2	Rule 2: Tier One gateway centralises auth and rate limits ^{vf¹¹⁰⁸} ; RBAC and audit not verified; no commercial support offer verified beyond a listed 'Tetrat Agent Router Service' provider ^{vf¹¹⁰⁹}
Security and compliance	2	Rule 2 hygiene not verified ^{NPV}
Deployment flexibility	3	Self-hosted on Kubernetes only ^{vf¹¹¹⁰}
Ecosystem	3	Envoy ecosystem; OpenAI-compatible API ^{vf¹¹¹¹}
Reliability and	3	1.x line, v1.2.0 on 6 October 2026; renamed and moved in 2026 ^{vf¹¹¹²}

¹¹⁰⁷ Verified fact — A6-S063: Envoy AI Gateway / Agent Router README (GitHub) (Envoy AI Gateway project (Agentic AI Foundation)) <https://raw.githubusercontent.com/envoyproxy/ai-gateway/main/README.md> accessed 2026-10-07

¹¹⁰⁸ Verified fact — A6-S063: Envoy AI Gateway / Agent Router README (GitHub) (Envoy AI Gateway project (Agentic AI Foundation)) <https://raw.githubusercontent.com/envoyproxy/ai-gateway/main/README.md> accessed 2026-10-07

¹¹⁰⁹ Verified fact — A6-S063: Envoy AI Gateway / Agent Router README (GitHub) (Envoy AI Gateway project (Agentic AI Foundation)) <https://raw.githubusercontent.com/envoyproxy/ai-gateway/main/README.md> accessed 2026-10-07

¹¹¹⁰ Verified fact — A6-S063: Envoy AI Gateway / Agent Router README (GitHub) (Envoy AI Gateway project (Agentic AI Foundation)) <https://raw.githubusercontent.com/envoyproxy/ai-gateway/main/README.md> accessed 2026-10-07

¹¹¹¹ Verified fact — A6-S063: Envoy AI Gateway / Agent Router README (GitHub) (Envoy AI Gateway project (Agentic AI Foundation)) <https://raw.githubusercontent.com/envoyproxy/ai-gateway/main/README.md> accessed 2026-10-07

¹¹¹² Verified fact — A6-S064: Go module proxy: github.com/envoyproxy/ai-gateway @latest (Go module proxy (proxy.golang.org)) <https://proxy.golang.org/github.com/envoyproxy/ai-gateway/@latest> accessed 2026-10-07; V2-S029: Agent Router (formerly Envoy AI Gateway)

Criterion	Score	Rationale
maturity		
Cost / TCO	4	Free open source; Kubernetes operations ^{VF-1113}
Lock-in / portability	5	Apache-2.0, neutral foundation ^{VF-1114}
Total (generic / FS)	2.90 / 3.00	

Evidence rules applied: Rule 2 basis: self-hosted open-source software; NPV cap not applied

Capabilities. Agent Router (formerly Envoy AI Gateway): one OpenAI-compatible API for hosted and self-hosted models and MCP servers, with centralised credentials, routing, quotas, failover and usage attribution enforced by Envoy and Envoy Gateway, in a two-tier design ^{VF-1115}.

Strengths

- Built on Envoy, which many Kubernetes platforms already run ^{VF-1116 AI}
- Apache-2.0 under the Agentic AI Foundation ^{VF-1117}
- Usage attribution and quotas suit chargeback ^{VF-1118}

Limitations and risks

- Renamed and moved foundation in 2026; repository moved ^{VF-1119}
- Access control, security features and hygiene not verified ^{NPV}
- Kubernetes and Envoy Gateway required ^{VF-1120}

Choose when

- Your platform team runs Envoy Gateway and wants AI routing in the same mesh ^{AI}

README (Agent Router project (Agentic AI Foundation)) <https://raw.githubusercontent.com/envoyproxy/ai-gateway/main/README.md> accessed 2026-10-08

¹¹¹³ Verified fact — A6-S063: Envoy AI Gateway / Agent Router README (GitHub) (Envoy AI Gateway project (Agentic AI Foundation)) <https://raw.githubusercontent.com/envoyproxy/ai-gateway/main/README.md> accessed 2026-10-07

¹¹¹⁴ Verified fact — A6-S063: Envoy AI Gateway / Agent Router README (GitHub) (Envoy AI Gateway project (Agentic AI Foundation)) <https://raw.githubusercontent.com/envoyproxy/ai-gateway/main/README.md> accessed 2026-10-07

¹¹¹⁵ Verified fact — A6-S063: Envoy AI Gateway / Agent Router README (GitHub) (Envoy AI Gateway project (Agentic AI Foundation)) <https://raw.githubusercontent.com/envoyproxy/ai-gateway/main/README.md> accessed 2026-10-07

¹¹¹⁶ Verified fact — A6-S063: Envoy AI Gateway / Agent Router README (GitHub) (Envoy AI Gateway project (Agentic AI Foundation)) <https://raw.githubusercontent.com/envoyproxy/ai-gateway/main/README.md> accessed 2026-10-07

¹¹¹⁷ Verified fact — A6-S063: Envoy AI Gateway / Agent Router README (GitHub) (Envoy AI Gateway project (Agentic AI Foundation)) <https://raw.githubusercontent.com/envoyproxy/ai-gateway/main/README.md> accessed 2026-10-07

¹¹¹⁸ Verified fact — A6-S063: Envoy AI Gateway / Agent Router README (GitHub) (Envoy AI Gateway project (Agentic AI Foundation)) <https://raw.githubusercontent.com/envoyproxy/ai-gateway/main/README.md> accessed 2026-10-07

¹¹¹⁹ Verified fact — A6-S063: Envoy AI Gateway / Agent Router README (GitHub) (Envoy AI Gateway project (Agentic AI Foundation)) <https://raw.githubusercontent.com/envoyproxy/ai-gateway/main/README.md> accessed 2026-10-07; V2-S029: Agent Router (formerly Envoy AI Gateway) README (Agent Router project (Agentic AI Foundation)) <https://raw.githubusercontent.com/envoyproxy/ai-gateway/main/README.md> accessed 2026-10-08

¹¹²⁰ Verified fact — A6-S063: Envoy AI Gateway / Agent Router README (GitHub) (Envoy AI Gateway project (Agentic AI Foundation)) <https://raw.githubusercontent.com/envoyproxy/ai-gateway/main/README.md> accessed 2026-10-07

Avoid when

- You do not run Kubernetes, or need guardrail and caching features in the gateway itself ^{A1}

Nearest competitors: C1-agentgateway, C1-litellm, C1-kong-ai-gateway

Regulated-FS note. Treat as platform plumbing: pair with a separate guardrail service and log store, and track the rename in the software inventory ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Agentic AI Foundation project (formerly under the Envoy project)	Verified fact	A6-S063
Category	Open-source AI gateway on Envoy Proxy / Envoy Gateway (Kubernetes)	Verified fact	A6-S063
Version / lineup	v1.2.0 (6 October 2026)	Verified fact	A6-S064, V2-S061
Licence	Open source (Apache-2.0)	Verified fact	A6-S063
Status events	2026: renamed from Envoy AI Gateway to Agent Router and moved to the Agentic AI Foundation; deployed resources not renamed	Verified fact	A6-S063, V2-S029
Strategic direction	Not publicly verified	Not publicly verified	
What it does	One OpenAI-compatible API for hosted and self-hosted models and MCP servers; platform teams centralise credentials, routing, quotas, failover and usage attribution, enforced by Envoy and Envoy Gateway; two-tier design with a Tier One gateway for auth, top-level routing and global rate limiting.	Verified fact	A6-S063
Stack position	Gateway/control plane in Kubernetes-based platforms	Verified fact	A6-S063
Integration	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
		y verifie d	
Dependencies	Envoy Proxy and Envoy Gateway on Kubernetes (CLI also available)	Verifie d fact	A6-S063
Certifications	Not publicly verified	Not publicl y verifie d	
GDPR / residency	Not publicly verified	Not publicl y verifie d	
Security features	Not publicly verified	Not publicl y verifie d	
Access controls	Not publicly verified	Not publicl y verifie d	
Enterprise support	Not publicly verified	Not publicl y verifie d	
Pricing	Free open source; a commercial 'Tetrade Agent Router Service' is listed among providers	Verifie d fact	A6-S063
Infrastructure cost	Not publicly verified	Not publicl y verifie d	
Ecosystem	Not publicly verified	Not publicl y verifie d	
Adoption signals	Not publicly verified	Not publicl y verifie d	
Maturity	1.x release line; v1.2.0 on 6 October 2026	Verifie	A6-S064

Field	Value	Label	Sources
Deployment	saas: Not publicly verified; managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes; on_prem: Not publicly verified	d fact	

Cloudflare AI Gateway (c1-cloudflare-ai-gateway)

Tier: Tactical · **Flags:** none · **Original graphic label:** not in graphic

Rationale: Tactical: useful edge controls, but managed-only with unverified log localisation and no MCP evidence ^{AJ}.

Criterion	Score	Rationale
Technical	3	Covers routing, fallback, caching, rate limits, spend limits, DLP and guardrails ^{vr1121} ; MCP/A2A and policy depth not evidenced ^{NPV}
Enterprise readiness	3	Identity-based budgets via Cloudflare Access verified ^{vr1122} ; one control verified lifts the cap to a maximum of 3 (rule 7); enterprise support NPV
Security and compliance	3	SOC 2 Type II names AI Gateway; ISO 27001:2022 is platform-wide without naming it ^{vr1123} ; Trust Hub pages flagged as possibly old, so 3
Deployment flexibility	2	SaaS only; no self-host or on-prem ^{vr1124}
Ecosystem	4	Many providers via BYOK or Unified Billing, Workers AI, Logpush, Cloudflare One DLP ^{vr1125}
Reliability and maturity	3	GA with frequent 2026 changes, including two log-pricing changes ^{vr1126}
Cost / TCO	4	Core features and DLP free; guardrails billed as Workers AI inference; 5% Unified Billing fee; per-GB log pricing from 1 December 2026 ^{vr1127}

¹¹²¹ Verified fact — A6-S019: Cloudflare AI Gateway docs: Pricing, Features, Changelog; Cloudflare blog spend limits (Cloudflare, Inc.) <https://developers.cloudflare.com/ai-gateway/reference/pricing/> accessed 2026-10-07 [search-tool extract]; A6-S052: Cloudflare AI Gateway docs: Pricing (source mdx) (Cloudflare, Inc.) <https://raw.githubusercontent.com/cloudflare/cloudflare-docs/production/src/content/docs/ai-gateway/reference/pricing.mdx> accessed 2026-10-07

¹¹²² Verified fact — A6-S019: Cloudflare AI Gateway docs: Pricing, Features, Changelog; Cloudflare blog spend limits (Cloudflare, Inc.) <https://developers.cloudflare.com/ai-gateway/reference/pricing/> accessed 2026-10-07 [search-tool extract]

¹¹²³ Verified fact — A6-S110: Cloudflare Trust Hub: SOC 2 FAQs; ISO Certifications FAQs; GDPR FAQs (Cloudflare, Inc.) <https://www.cloudflare.com/trust-hub/compliance-resources/soc-2/> accessed 2026-10-07 [search-tool extract]

¹¹²⁴ Verified fact — A6-S052: Cloudflare AI Gateway docs: Pricing (source mdx) (Cloudflare, Inc.) <https://raw.githubusercontent.com/cloudflare/cloudflare-docs/production/src/content/docs/ai-gateway/reference/pricing.mdx> accessed 2026-10-07

¹¹²⁵ Verified fact — A6-S052: Cloudflare AI Gateway docs: Pricing (source mdx) (Cloudflare, Inc.) <https://raw.githubusercontent.com/cloudflare/cloudflare-docs/production/src/content/docs/ai-gateway/reference/pricing.mdx> accessed 2026-10-07; A6-S019: Cloudflare AI Gateway docs: Pricing, Features, Changelog; Cloudflare blog spend limits (Cloudflare, Inc.) <https://developers.cloudflare.com/ai-gateway/reference/pricing/> accessed 2026-10-07 [search-tool extract]

¹¹²⁶ Verified fact — A6-S019: Cloudflare AI Gateway docs: Pricing, Features, Changelog; Cloudflare blog spend limits (Cloudflare, Inc.) <https://developers.cloudflare.com/ai-gateway/reference/pricing/> accessed 2026-10-07 [search-tool extract]; V2-S074: Pricing - Cloudflare AI Gateway docs (with AI Gateway changelog 2026-09-01; Cloudflare observability blog) <https://developers.cloudflare.com/ai-gateway/reference/pricing/> accessed 2026-10-08 [search-tool extract]

¹¹²⁷ Verified fact — A6-S052: Cloudflare AI Gateway docs: Pricing (source mdx) (Cloudflare, Inc.) <https://raw.githubusercontent.com/cloudflare/cloudflare-docs/production/src/content/docs/ai-gateway/reference/pricing.mdx> accessed 2026-10-07; V2-S074: Pricing - Cloudflare AI Gateway docs (with AI Gateway changelog 2026-09-01; Cloudflare observability blog) <https://developers.cloudflare.com/ai-gateway/reference/pricing/> accessed 2026-10-08 [search-tool extract]

Criterion	Score	Rationale
Lock-in / portability	2	Proprietary managed service in the data path; Unified Billing deepens dependency ^{VF1128}
Total (generic / FS)	3.00 / 2.80	

Evidence rules applied: enterprise_readiness: NPV cap lifted to max 3 by one verified control (identity-based controls), rule 7

Capabilities. Managed edge proxy for AI provider calls with analytics, caching, rate limiting, logging, Llama Guard guardrails, DLP scanning, dynamic routing and fallback, spend limits, identity-based budgets and Unified Billing ^{VF1129}.

Strengths

- Lowest-friction adoption: a one-line endpoint change, core features free ^{VF1130}
- DLP and guardrails available at the edge ^{VF1131}
- SOC 2 Type II with AI Gateway named in scope ^{VF1132}

Limitations and risks

- Managed only: no self-hosting or on-prem ^{VF1133}
- AI Gateway-specific data localisation not verified ^{VF1134}
- Unified Billing (5% fee) ties provider spend to Cloudflare; future features may be premium; log pricing changed in September 2026 and again from 1 December 2026 ^{VF1135}
- MCP and A2A governance not evidenced ^{NPV}

Choose when

¹¹²⁸ Verified fact — A6-S052: Cloudflare AI Gateway docs: Pricing (source mdx) (Cloudflare, Inc.) <https://raw.githubusercontent.com/cloudflare/cloudflare-docs/production/src/content/docs/ai-gateway/reference/pricing.mdx> accessed 2026-10-07

¹¹²⁹ Verified fact — A6-S019: Cloudflare AI Gateway docs: Pricing, Features, Changelog; Cloudflare blog spend limits (Cloudflare, Inc.) <https://developers.cloudflare.com/ai-gateway/reference/pricing/> accessed 2026-10-07 [search-tool extract]; A6-S052: Cloudflare AI Gateway docs: Pricing (source mdx) (Cloudflare, Inc.) <https://raw.githubusercontent.com/cloudflare/cloudflare-docs/production/src/content/docs/ai-gateway/reference/pricing.mdx> accessed 2026-10-07

¹¹³⁰ Verified fact — A6-S052: Cloudflare AI Gateway docs: Pricing (source mdx) (Cloudflare, Inc.) <https://raw.githubusercontent.com/cloudflare/cloudflare-docs/production/src/content/docs/ai-gateway/reference/pricing.mdx> accessed 2026-10-07

¹¹³¹ Verified fact — A6-S052: Cloudflare AI Gateway docs: Pricing (source mdx) (Cloudflare, Inc.) <https://raw.githubusercontent.com/cloudflare/cloudflare-docs/production/src/content/docs/ai-gateway/reference/pricing.mdx> accessed 2026-10-07

¹¹³² Verified fact — A6-S110: Cloudflare Trust Hub: SOC 2 FAQs; ISO Certifications FAQs; GDPR FAQs (Cloudflare, Inc.) <https://www.cloudflare.com/trust-hub/compliance-resources/soc-2/> accessed 2026-10-07 [search-tool extract]

¹¹³³ Verified fact — A6-S052: Cloudflare AI Gateway docs: Pricing (source mdx) (Cloudflare, Inc.) <https://raw.githubusercontent.com/cloudflare/cloudflare-docs/production/src/content/docs/ai-gateway/reference/pricing.mdx> accessed 2026-10-07

¹¹³⁴ Verified fact — A6-S110: Cloudflare Trust Hub: SOC 2 FAQs; ISO Certifications FAQs; GDPR FAQs (Cloudflare, Inc.) <https://www.cloudflare.com/trust-hub/compliance-resources/soc-2/> accessed 2026-10-07 [search-tool extract]

¹¹³⁵ Verified fact — A6-S052: Cloudflare AI Gateway docs: Pricing (source mdx) (Cloudflare, Inc.) <https://raw.githubusercontent.com/cloudflare/cloudflare-docs/production/src/content/docs/ai-gateway/reference/pricing.mdx> accessed 2026-10-07; V2-S074: Pricing - Cloudflare AI Gateway docs (with AI Gateway changelog 2026-09-01; Cloudflare observability blog) <https://developers.cloudflare.com/ai-gateway/reference/pricing/> accessed 2026-10-08 [search-tool extract]

- You already run Cloudflare Zero Trust and want quick visibility, caching and spend limits for non-confidential workloads ^{AJ}

Avoid when

- Prompts contain client data that must stay in-estate or in a verified region ^{AJ}
- You want the gateway to be the exit route from a provider rather than another dependency ^{AJ}

Nearest competitors: C1-portkey, C1-litellm, L2-openrouter

Regulated-FS note. Use BYOK rather than Unified Billing, confirm AI Gateway log localisation in writing, and keep regulated client data off this path until it is confirmed ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Cloudflare, Inc.	Verified fact	A6-S052
Category	Managed SaaS AI gateway at the network edge	Verified fact	A6-S019, A6-S052
Version / lineup	Continuously delivered SaaS (no version numbers); 2026 additions include spend limits (June), identity-based controls (August), Unified Billing invoice changes (1 September) and new log pricing (24 September)	Verified fact	A6-S019, A6-S052
Licence	Proprietary managed service	Verified fact	A6-S052
Strategic direction	Positioned as an 'AI Application Control Plane': unified billing across providers and Workers AI, cost-based spend limits, identity-based budgets via Cloudflare Access, Dynamic Routing and Auto Router	Verified fact	A6-S019
What it does	Proxy for AI provider calls with analytics, caching, rate limiting, logging, guardrails (Llama Guard on Workers AI), DLP scanning, dynamic routing/fallback across providers, spend limits and unified billing.	Verified fact	A6-S019, A6-S052
Stack position	Gateway/control plane delivered from Cloudflare's network; complements rather than replaces a self-hosted gateway	Verified fact	A6-S019
Integration	One-line endpoint change to route provider calls; BYOK or Unified Billing; Logpush	Verified fact	A6-S052, A6-S019
Dependencies	Cloudflare account; Workers AI for guardrail inference; Zero Trust subscription for full DLP profiles	Verified fact	A6-S052
Certifications	SOC 2 Type II (security, confidentiality, availability) with AI Gateway	Verified	A6-S110

Field	Value	Label	Sources
	named in scope; ISO 27001:2022 for the global cloud platform (AI Gateway not named individually); ISO 27701, ISO 27018 and PCI DSS Level 1 also maintained	d fact	
GDPR / residency	EU transfers rely on EU-U.S. Data Privacy Framework certifications with SCCs as fallback; Data Localization Suite (in SOC 2 scope) sets where data is stored and protected; AI Gateway-specific localisation behaviour not verified	Verified fact	A6-S110
Security features	DLP scanning of prompts/responses (two predefined profiles free; full profiles with Zero Trust DLP); guardrails on prompts and responses	Verified fact	A6-S052
Access controls	Identity-based controls using Cloudflare Access identity for per-user budgets (August 2026)	Verified fact	A6-S019
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Core features (analytics, caching, rate limiting) free on all plans; DLP free; guardrails billed as Workers AI token inference; 5% fee on Unified Billing credits (no markup on provider tokens); Logpush 10M requests/month then US\$0.05 per million on Workers Paid; logs follow Workers Logs pricing for gateways created on/after 24 September 2026. From 1 December 2026 (announced): unified per-GB log pricing covering AI Gateway logs, US\$0.25 per GB ingested and US\$0.10 per GB-month stored after 50 GB / 10 GB-month included (blog announcement; terms may change).	Verified fact	A6-S052, V2-S074
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Third-party model providers via BYOK or Unified Billing; Workers AI models; Cloudflare One DLP	Verified fact	A6-S052, A6-S019
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Generally available managed service with frequent changelog entries in 2026	Verified fact	A6-S019
Deployment	saas: Yes; managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: No (managed service only); on_prem: No		

C2: Guardrails

NVIDIA NeMo Guardrails (open-source library and Guardrails server) (C2-nemo-guardrails)

Tier: Tactical · **Flags:** none · **Original graphic label:** not in graphic

Rationale: Tactical: the most capable in-estate framework, but 0.x Beta with breaking changes; a Strategic candidate at 1.0 ^{AJ}.

Criterion	Score	Rationale
Technical	4	Input, output, dialogue, retrieval and tool rails; jailbreak, content and topic safety; self-check rails; IORails ^{vr1136}
Enterprise readiness	3	Rule 2: server mode, OTel-style spans; identity inherits host; commercial support via NVIDIA AI Enterprise ^{vr1137}
Security and compliance	3	Rule 2 hygiene: fail-closed streaming rails and explicit block-vs-failure outcomes ^{vr1138} ; security policy and signed releases NPV; NVIDIA certifications for the microservice NPV
Deployment flexibility	4	Library, server and Helm microservice on customer Kubernetes ^{vr1139}
Ecosystem	4	Kong NeMo plugin, F5 AI Guardrails integration, NIM safety models ^{vr1140}
Reliability and maturity	2	First release April 2023 but still 0.x Beta, breaking changes in minors (0.24.0) ^{vr1141}
Cost / TCO	4	Free; optional AI Enterprise at US\$4,500 per GPU per year; model rails add inference cost ^{vr1142}
Lock-in / portability	4	Apache-2.0 but NVIDIA-governed; model rails can use any LLM endpoint ^{vr1143}

¹¹³⁶ Verified fact — A6-S027: NeMo Guardrails Release Notes; IORails engine feature support; Guardrail Catalog (NVIDIA) <https://docs.nvidia.com/nemo/guardrails/latest/about/release-notes.html> accessed 2026-10-07 [search-tool extract]; A6-S036: NeMo Guardrails CHANGELOG.md (NVIDIA) <https://raw.githubusercontent.com/NVIDIA-NeMo/Guardrails/develop/CHANGELOG.md> accessed 2026-10-07

¹¹³⁷ Verified fact — A6-S036: NeMo Guardrails CHANGELOG.md (NVIDIA) <https://raw.githubusercontent.com/NVIDIA-NeMo/Guardrails/develop/CHANGELOG.md> accessed 2026-10-07; A6-S111: About Guardrails - NVIDIA NeMo Microservices (25.11.0); NVIDIA Releases NIM Microservices to Safeguard Applications for Agentic AI; NGC Helm chart <https://docs.nvidia.com/nemo/microservices/25.11.0/guardrails/index.html> accessed 2026-10-07 [search-tool extract]

¹¹³⁸ Verified fact — A6-S036: NeMo Guardrails CHANGELOG.md (NVIDIA) <https://raw.githubusercontent.com/NVIDIA-NeMo/Guardrails/develop/CHANGELOG.md> accessed 2026-10-07

¹¹³⁹ Verified fact — A6-S036: NeMo Guardrails CHANGELOG.md (NVIDIA) <https://raw.githubusercontent.com/NVIDIA-NeMo/Guardrails/develop/CHANGELOG.md> accessed 2026-10-07; A6-S111: About Guardrails - NVIDIA NeMo Microservices (25.11.0); NVIDIA Releases NIM Microservices to Safeguard Applications for Agentic AI; NGC Helm chart <https://docs.nvidia.com/nemo/microservices/25.11.0/guardrails/index.html> accessed 2026-10-07 [search-tool extract]

¹¹⁴⁰ Verified fact — A6-S017: Kong AI Gateway changelog; Kong Gateway changelog; AI Proxy Advanced plugin docs (Kong Inc.) <https://developer.konghq.com/ai-gateway/changelog/> accessed 2026-10-07 [search-tool extract]; A6-S036: NeMo Guardrails CHANGELOG.md (NVIDIA) <https://raw.githubusercontent.com/NVIDIA-NeMo/Guardrails/develop/CHANGELOG.md> accessed 2026-10-07

¹¹⁴¹ Verified fact — A6-S003: nemoguardrails - PyPI JSON metadata (Python Package Index) <https://pypi.org/pypi/nemoguardrails/json> accessed 2026-10-07; A6-S036: NeMo Guardrails CHANGELOG.md (NVIDIA) <https://raw.githubusercontent.com/NVIDIA-NeMo/Guardrails/develop/CHANGELOG.md> accessed 2026-10-07

¹¹⁴² Verified fact — A6-S112: Pricing - NVIDIA Enterprise Licensing Guide <https://docs.nvidia.com/ai-enterprise/planning-resource/licensing-guide/latest/pricing.html> accessed 2026-10-07 [search-tool extract]; A6-S003: nemoguardrails - PyPI JSON metadata (Python Package Index) <https://pypi.org/pypi/nemoguardrails/json> accessed 2026-10-07

Criterion	Score	Rationale
Total (generic / FS)	3.50 / 3.45	

Evidence rules applied: Rule 2 basis: self-hosted open-source library; NPV cap not applied; commercial support exists (NVIDIA AI Enterprise)

Capabilities. Apache-2.0 toolkit adding programmable rails around LLM applications: content safety, topic safety, jailbreak detection (including a NIM-based model), self-check rails, Colang dialogue rails and third-party integrations; runs as a library or an OpenAI-compatible Guardrails server with a /v1/checks endpoint; new IORails engine for input, output and tool rails ^{vr1144}.

Strengths

- Widest rail types in one in-estate framework: input, output, dialogue, retrieval and tool rails ^{vr1145}
- Engine-neutral allow/block/transform contract and fail-closed streaming output rails ^{vr1146}
- Supported microservice via NVIDIA AI Enterprise ^{vr1147}

Limitations and risks

- Still 0.x and classified Beta, with breaking changes in minor releases ^{vr1148}
- Model-based rails depend on NIM/Nemotron or other LLM endpoints ^{vr1149}
- The supported microservice accepts Colang 1.0 configurations only and lacks some toolkit server features ^{vr1150}

Choose when

- You want an in-estate guardrail orchestration layer combining classifiers, rules and model-based checks ^{ai}

¹¹⁴³ Verified fact — A6-S003: nemoguardrails - PyPI JSON metadata (Python Package Index) <https://pypi.org/pypi/nemoguardrails/json> accessed 2026-10-07; A6-S036: NeMo Guardrails CHANGELOG.md (NVIDIA) <https://raw.githubusercontent.com/NVIDIA-NeMo/Guardrails/develop/CHANGELOG.md> accessed 2026-10-07

¹¹⁴⁴ Verified fact — A6-S036: NeMo Guardrails CHANGELOG.md (NVIDIA) <https://raw.githubusercontent.com/NVIDIA-NeMo/Guardrails/develop/CHANGELOG.md> accessed 2026-10-07; A6-S027: NeMo Guardrails Release Notes; IORails engine feature support; Guardrail Catalog (NVIDIA) <https://docs.nvidia.com/nemo/guardrails/latest/about/release-notes.html> accessed 2026-10-07 [search-tool extract]

¹¹⁴⁵ Verified fact — A6-S027: NeMo Guardrails Release Notes; IORails engine feature support; Guardrail Catalog (NVIDIA) <https://docs.nvidia.com/nemo/guardrails/latest/about/release-notes.html> accessed 2026-10-07 [search-tool extract]; A6-S036: NeMo Guardrails CHANGELOG.md (NVIDIA) <https://raw.githubusercontent.com/NVIDIA-NeMo/Guardrails/develop/CHANGELOG.md> accessed 2026-10-07

¹¹⁴⁶ Verified fact — A6-S036: NeMo Guardrails CHANGELOG.md (NVIDIA) <https://raw.githubusercontent.com/NVIDIA-NeMo/Guardrails/develop/CHANGELOG.md> accessed 2026-10-07

¹¹⁴⁷ Verified fact — A6-S111: About Guardrails - NVIDIA NeMo Microservices (25.11.0); NVIDIA Releases NIM Microservices to Safeguard Applications for Agentic AI; NGC Helm chart <https://docs.nvidia.com/nemo/microservices/25.11.0/guardrails/index.html> accessed 2026-10-07 [search-tool extract]; A6-S112: Pricing - NVIDIA Enterprise Licensing Guide <https://docs.nvidia.com/ai-enterprise/planning-resource/licensing-guide/latest/pricing.html> accessed 2026-10-07 [search-tool extract]

¹¹⁴⁸ Verified fact — A6-S003: nemoguardrails - PyPI JSON metadata (Python Package Index) <https://pypi.org/pypi/nemoguardrails/json> accessed 2026-10-07; A6-S036: NeMo Guardrails CHANGELOG.md (NVIDIA) <https://raw.githubusercontent.com/NVIDIA-NeMo/Guardrails/develop/CHANGELOG.md> accessed 2026-10-07

¹¹⁴⁹ Verified fact — A6-S027: NeMo Guardrails Release Notes; IORails engine feature support; Guardrail Catalog (NVIDIA) <https://docs.nvidia.com/nemo/guardrails/latest/about/release-notes.html> accessed 2026-10-07 [search-tool extract]; A6-S036: NeMo Guardrails CHANGELOG.md (NVIDIA) <https://raw.githubusercontent.com/NVIDIA-NeMo/Guardrails/develop/CHANGELOG.md> accessed 2026-10-07

¹¹⁵⁰ Verified fact — A6-S111: About Guardrails - NVIDIA NeMo Microservices (25.11.0); NVIDIA Releases NIM Microservices to Safeguard Applications for Agentic AI; NGC Helm chart <https://docs.nvidia.com/nemo/microservices/25.11.0/guardrails/index.html> accessed 2026-10-07 [search-tool extract]

- You run NVIDIA AI Enterprise and want supported safety NIMs ^{AJ}

Avoid when

- You cannot absorb breaking changes in a pre-1.0 dependency ^{AJ}

Nearest competitors: C2-guardrails-ai, C2-meta-llama-protections, C7-lakera

Regulated-FS note. Pin versions, run it as a sidecar or server called from the gateway, and keep rail definitions in Git with the L9 red-team suite as the regression gate ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	NVIDIA (repository NVIDIA-NeMo/Guardrails)	Verified fact	A6-S003
Category	Open-source programmable guardrails toolkit (input, output, dialogue, retrieval and tool rails)	Verified fact	A6-S003, A6-S027
Version / lineup	0.24.1 (16 September 2026); 0.24.0 (25 August 2026)	Verified fact	A6-S003, A6-S036, V2-S028
Licence	Open source (Apache-2.0)	Verified fact	A6-S003
Strategic direction	New optimised IORails engine for input/output/tool rails without Colang runtime dependency; engine-neutral RailOutcome allow/block/transform contract; typed rail manifests; server health endpoints and output-rail check mode; third-party integrations such as F5 AI Guardrails	Verified fact	A6-S036, A6-S027
What it does	Adds programmable rails around LLM applications: content safety, topic safety and jailbreak detection (incl. NIM-based jailbreak model), self-check rails, dialogue rails in Colang, and third-party guardrail integrations; can run as a library or as an OpenAI-compatible Guardrails server with a /v1/checks endpoint.	Verified fact	A6-S036, A6-S027
Stack position	Guardrail layer wrapping model calls inside the application or as a sidecar/server; can also be called from gateways (Kong added an NVIDIA NeMo Guardrails plugin in AI Gateway 2.0.1)	Verified fact	A6-S027, A6-S017

Field	Value	Label	Sources
Integration	Python library; OpenAI-compatible server with /v1/chat/completions-style and /v1/checks endpoints; OpenTelemetry-style tracing spans	Verified fact	A6-S036, A6-S027
Dependencies	Python >=3.10,<3.14; LLM or NIM endpoints for model-based rails; transformers and torch for the local classifier; microservice places NemoGuard safety NIMs (content safety, topic control, jailbreak detection) between the application and its LLM	Verified fact	A6-S003, A6-S036, A6-S111
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Streaming output rails fail closed when actions fail (0.24.0); IORails distinguishes policy blocks from rail execution failures	Verified fact	A6-S036
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	NeMo Guardrails microservice (NeMo Microservices, Helm chart on NGC) is tagged 'NVIDIA AI Enterprise Supported'; AI Enterprise subscriptions include Business Standard support, Business Critical at extra cost	Verified fact	A6-S111, A6-S112
Pricing	Open-source toolkit free (Apache-2.0). Supported microservice via NVIDIA AI Enterprise: list US\$4,500 per GPU per year (self-managed, support included) or US\$1 per GPU-hour on cloud marketplaces	Verified fact	A6-S003, A6-S112
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	First PyPI release 25 April 2023; still 0.x and classified Beta; roughly monthly-to-bimonthly minor releases in 2026 (0.21 March, 0.22 May, 0.23 July, 0.24 August)	Verified fact	A6-S003

Field	Value	Label	Sources
Deployment	saas: Not publicly verified; managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Yes (NeMo Guardrails microservice deployed via Helm on customer Kubernetes); self_hosted: Yes (library or server); on_prem: Not publicly verified		

Amazon Bedrock Guardrails (c2-bedrock-guardrails)

Tier: Strategic · **Flags:** none · **Original graphic label:** not in graphic

Rationale: Strategic, conditional: where AWS is your primary cloud (CP3 Q2, rubric rule 10). The strongest managed policy set and AWS's lead guardrail service; use the Classic tier or region-constrained profiles for client data, because the Standard tier uses cross-region inference ^{AJ}.

Criterion	Score	Rationale
Technical	5	Content and prompt-attack filters, denied topics, PII (incl. UK identifiers), contextual grounding, Automated Reasoning, standalone API ^{vf1151}
Enterprise readiness	4	Hyperscaler presumption (CP2 Q1); confirm per service
Security and compliance	4	Bedrock in AWS SOC 1/2/3 scope since 15 August 2023 and on the ISO 27001 list ^{vf1152} ; Guardrails not named separately, so the scope rule applies; customer KMS key supported ^{vf1153}
Deployment flexibility	2	Managed AWS only; cross-region profiles ^{vf1154}
Ecosystem	3	Referenced by Kong and AWS's gateway Guidance ^{vf1155}
Reliability and	3	Priced since at least December 2024; tiers added June 2025 (A6-S101, A6-S072);

¹¹⁵¹ Verified fact — A6-S072: AWS API model: Amazon Bedrock control plane (botocore service-2.json) - guardrail shapes (Amazon Web Services (via boto/botocore)) <https://raw.githubusercontent.com/boto/botocore/develop/botocore/data/bedrock/2023-04-20/service-2.json> accessed 2026-10-07; A6-S073: AWS API model: Amazon Bedrock runtime (botocore service-2.json) - ApplyGuardrail (Amazon Web Services (via boto/botocore)) <https://raw.githubusercontent.com/boto/botocore/develop/botocore/data/bedrock-runtime/2023-09-30/service-2.json> accessed 2026-10-07

¹¹⁵² Verified fact — B-C2-S001: AWS Services in Scope by Compliance Program - SOC (Amazon Bedrock) (Amazon Web Services) <https://aws.amazon.com/compliance/services-in-scope/SOC/> accessed 2026-10-08 [search-tool extract]; B-C2-S002: ISO Certified - AWS (Amazon Web Services) <https://aws.amazon.com/compliance/iso-certified/> accessed 2026-10-08 [search-tool extract]

¹¹⁵³ Verified fact — A6-S072: AWS API model: Amazon Bedrock control plane (botocore service-2.json) - guardrail shapes (Amazon Web Services (via boto/botocore)) <https://raw.githubusercontent.com/boto/botocore/develop/botocore/data/bedrock/2023-04-20/service-2.json> accessed 2026-10-07

¹¹⁵⁴ Verified fact — A6-S072: AWS API model: Amazon Bedrock control plane (botocore service-2.json) - guardrail shapes (Amazon Web Services (via boto/botocore)) <https://raw.githubusercontent.com/boto/botocore/develop/botocore/data/bedrock/2023-04-20/service-2.json> accessed 2026-10-07

¹¹⁵⁵ Verified fact — A6-S017: Kong AI Gateway changelog; Kong Gateway changelog; AI Proxy Advanced plugin docs (Kong Inc.) <https://developer.konghq.com/ai-gateway/changelog/> accessed 2026-10-07 [search-tool extract]; A6-S022: Guidance for Multi-Provider Generative AI Gateway on AWS; Implementing resilience patterns with Amazon Bedrock and LLM gateway (Amazon Web Services) <https://aws.amazon.com/solutions/guidance/multi-provider-generative-ai-gateway-on-aws/> accessed 2026-10-07 [search-tool extract]

Criterion	Score	Rationale
maturity		maturity fact cell NPV and InvokeGuardrailChecks GA not verified; 3, consistent with Azure AI Content Safety (GA August 2024) and Model Armor (CP3 review A)
Cost / TCO	3	Transparent per-policy unit pricing, e.g. US\$0.15 per 1,000 text units for content filters ^{vf-1156}
Lock-in / portability	2	AWS-specific API ^{vf-1157}
Total (generic / FS)	3.50	
	/	
	3.35	

Evidence rules applied: enterprise_readiness set to 4 under the hyperscaler presumption (CP2 Q1): platform controls presumed; confirm per service

Capabilities. Managed guardrails for prompts and responses: content filters (including PROMPT_ATTACK) with Classic and Standard tiers, denied topics, word filters, PII detection with block or anonymise actions (31 entity types including UK NHS and NI numbers, IBAN, SWIFT), contextual grounding and Automated Reasoning checks; usable with any model through the standalone ApplyGuardrail API ^{vf-1158}.

Strengths

- Broadest managed policy set, including grounding and Automated Reasoning ^{vf-1159 AJ}
- Model-independent via ApplyGuardrail ^{vf-1160}
- Customer KMS key encryption ^{vf-1161}

Limitations and risks

- Standard tier uses cross-region inference, a residency consideration ^{vf-1162}
- Per-policy pricing stacks ^{vf-1163}
- AWS-specific API ^{vf-1164}

¹¹⁵⁶ Verified fact — A6-S101: Amazon Bedrock Pricing (Guardrails); Guardrails reduces pricing by up to 85% (Dec 2024); Safeguard tiers (June 2025) (Amazon Web Services) <https://aws.amazon.com/bedrock/pricing/> accessed 2026-10-07 [search-tool extract]

¹¹⁵⁷ Verified fact — A6-S072: AWS API model: Amazon Bedrock control plane (botocore service-2.json) - guardrail shapes (Amazon Web Services (via boto/botocore)) <https://raw.githubusercontent.com/boto/botocore/develop/botocore/data/bedrock/2023-04-20/service-2.json> accessed 2026-10-07

¹¹⁵⁸ Verified fact — A6-S072: AWS API model: Amazon Bedrock control plane (botocore service-2.json) - guardrail shapes (Amazon Web Services (via boto/botocore)) <https://raw.githubusercontent.com/boto/botocore/develop/botocore/data/bedrock/2023-04-20/service-2.json> accessed 2026-10-07; A6-S073: AWS API model: Amazon Bedrock runtime (botocore service-2.json) - ApplyGuardrail (Amazon Web Services (via boto/botocore)) <https://raw.githubusercontent.com/boto/botocore/develop/botocore/data/bedrock-runtime/2023-09-30/service-2.json> accessed 2026-10-07

¹¹⁵⁹ Verified fact — A6-S072: AWS API model: Amazon Bedrock control plane (botocore service-2.json) - guardrail shapes (Amazon Web Services (via boto/botocore)) <https://raw.githubusercontent.com/boto/botocore/develop/botocore/data/bedrock/2023-04-20/service-2.json> accessed 2026-10-07

¹¹⁶⁰ Verified fact — A6-S073: AWS API model: Amazon Bedrock runtime (botocore service-2.json) - ApplyGuardrail (Amazon Web Services (via boto/botocore)) <https://raw.githubusercontent.com/boto/botocore/develop/botocore/data/bedrock-runtime/2023-09-30/service-2.json> accessed 2026-10-07

¹¹⁶¹ Verified fact — A6-S072: AWS API model: Amazon Bedrock control plane (botocore service-2.json) - guardrail shapes (Amazon Web Services (via boto/botocore)) <https://raw.githubusercontent.com/boto/botocore/develop/botocore/data/bedrock/2023-04-20/service-2.json> accessed 2026-10-07

¹¹⁶² Verified fact — A6-S072: AWS API model: Amazon Bedrock control plane (botocore service-2.json) - guardrail shapes (Amazon Web Services (via boto/botocore)) <https://raw.githubusercontent.com/boto/botocore/develop/botocore/data/bedrock/2023-04-20/service-2.json> accessed 2026-10-07; A6-S101: Amazon Bedrock Pricing (Guardrails); Guardrails reduces pricing by up to 85% (Dec 2024); Safeguard tiers (June 2025) (Amazon Web Services) <https://aws.amazon.com/bedrock/pricing/> accessed 2026-10-07 [search-tool extract]

¹¹⁶³ Verified fact — A6-S101: Amazon Bedrock Pricing (Guardrails); Guardrails reduces pricing by up to 85% (Dec 2024); Safeguard tiers (June 2025) (Amazon Web Services) <https://aws.amazon.com/bedrock/pricing/> accessed 2026-10-07 [search-tool extract]

Choose when

- You run on AWS and want managed input/output guardrails, PII handling and grounding checks ^{AJ}
- You want guardrails callable for non-Bedrock models ^{AJ}

Avoid when

- You need the guardrail to run outside AWS, or cannot accept cross-region inference on the Standard tier ^{AJ}

Nearest competitors: C2-azure-ai-content-safety, C2-google-model-armor, C2-nemo-guardrails

Regulated-FS note. Pin guardrail versions in C5, use the Classic tier or region-constrained profiles for client data, and evaluate false-positive rates per policy on your own data ^{Rec}.

Facts (as of 2026-10-07)

Field	Value	Label	Sources
Company	Amazon Web Services	Verified fact	A6-S072
Category	Managed guardrail service for generative AI inputs and outputs	Verified fact	A6-S072
Version / lineup	Managed service; API model includes content filters with CLASSIC and STANDARD tiers, denied topics, word filters, sensitive-information (PII) filters, contextual grounding, Automated Reasoning policies and cross-region guardrail profiles (botocore 1.43.109, 7 October 2026); Standard/Classic safeguard tiers for content filters and denied topics introduced June 2025 (Standard uses cross-region inference)	Verified fact	A6-S072, A6-S075, A6-S101
Licence	Proprietary managed service	Verified fact	A6-S072
Strategic direction	Not publicly verified	Not publicly verified	
What it does	Configurable guardrails applied to prompts and model responses: content filters (sexual, violence, hate, insults, misconduct, prompt attack), denied topics, word filters, PII detection with block or anonymise actions (31 entity types incl. UK NHS and NI numbers, IBAN, SWIFT), contextual grounding checks and Automated Reasoning checks; usable with Bedrock models or standalone via the ApplyGuardrail API.	Verified fact	A6-S072, A6-S073

¹¹⁶⁴ Verified fact — A6-S072: AWS API model: Amazon Bedrock control plane (botocore service-2.json) - guardrail shapes (Amazon Web Services (via boto/botocore)) <https://raw.githubusercontent.com/boto/botocore/develop/botocore/data/bedrock/2023-04-20/service-2.json> accessed 2026-10-07

Field	Value	Label	Sources
Stack position	Guardrail service callable inline with Bedrock inference or independently of any model (ApplyGuardrail); also used in the AWS multi-provider gateway Guidance	Verified fact	A6-S073, A6-S022
Integration	AWS API/SDK: CreateGuardrail (control plane), ApplyGuardrail and InvokeGuardrailChecks (runtime)	Verified fact	A6-S072, A6-S073
Dependencies	Not publicly verified	Not publicly verified	
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	Cross-region guardrail profiles route guardrail inference to defined destination regions (crossRegionConfig)	Verified fact	A6-S072
Security features	Guardrails can be encrypted with a customer KMS key (kmsKeyId)	Verified fact	A6-S072
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Per 1,000 text units (1 unit = 1,000 characters, billed per configured policy): content filters US\$0.15 and denied topics US\$0.15 (since 1 December 2024); sensitive-information filters US\$0.10 and Automated Reasoning checks US\$0.17 (from AWS worked pricing examples); no separate Standard-tier rate found	Verified fact	A6-S101
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Referenced as a guardrail option in Kong AI Gateway ('AWS, Azure and GCP guardrail services')	Verified fact	A6-S017
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
		y verifie d	
Deployment	saas: Yes (managed AWS service); managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: No; on_prem: Not publicly verified		

Model Armor (c2-google-model-armor)

Tier: Strategic · Flags: none · Original graphic label: not in graphic

Rationale: Strategic, conditional: where Google Cloud is your primary cloud (CP3 Q2, rubric rule 10). Google’s lead guardrail service, efficient and residency-aware; narrower than Bedrock (grounding not evidenced), and the London residency position must be confirmed in writing where UK residency is required ^{AJ}.

Criterion	Score	Rationale
Technical	4	Prompt injection and jailbreak, malicious URLs and files, harm, sensitive data ^{VF1165} ; grounding not evidenced ^{NPV}
Enterprise readiness	4	Hyperscaler presumption (CP2 Q1); confirm per service
Security and compliance	4	Named in Google Cloud SOC 2 and ISO/IEC 27001 scope ^{VF1166} ; CMK not verified, so not 5
Deployment flexibility	2	Managed only, with regional and multi-regional endpoints ^{VF1167}
Ecosystem	4	Apigee, Agent Platform, Google MCP servers, Service Extensions, Firebase, LangChain ^{VF1168}
Reliability and	3	Active release notes; GA date not verified ^{VF1169} ^{NPV}

¹¹⁶⁵ Verified fact — A6-S067: Model Armor product page (incl. pricing) (Google Cloud) <https://cloud.google.com/security/products/model-armor> accessed 2026-10-07

¹¹⁶⁶ Verified fact — A6-S070: SOC 2 compliance (services in scope) (Google Cloud) <https://cloud.google.com/security/compliance/soc-2> accessed 2026-10-07; A6-S071: ISO/IEC 27001 compliance (services in scope) (Google Cloud) <https://cloud.google.com/security/compliance/iso-27001> accessed 2026-10-07

¹¹⁶⁷ Verified fact — B-C2-S003: Model Armor: Data residency and endpoints (Google Cloud) <https://docs.cloud.google.com/model-armor/data-residency> accessed 2026-10-08 [search-tool extract]; B-C2-S004: Model Armor locations (Google Cloud) <https://docs.cloud.google.com/model-armor/locations> accessed 2026-10-08 [search-tool extract]

¹¹⁶⁸ Verified fact — A6-S067: Model Armor product page (incl. pricing) (Google Cloud) <https://cloud.google.com/security/products/model-armor> accessed 2026-10-07

¹¹⁶⁹ Verified fact — B-C2-S005: Model Armor release notes (Google Cloud) <https://docs.cloud.google.com/model-armor/release-notes> accessed 2026-10-08 [search-tool extract]

Criterion	Score	Rationale
maturity		
Cost / TCO	5	Free to 2M tokens/month, then US\$0.10 per 1M tokens ^{vf¹¹⁷⁰}
Lock-in / portability	2	Google-specific API ^{vf¹¹⁷¹}
Total (generic / FS)	3.60	
	/	
	3.35	

Evidence rules applied: enterprise_readiness set to 4 under the hyperscaler presumption (CP2 Q1): platform controls presumed; confirm per service

Capabilities. Managed screening of prompts, responses and agent interactions for prompt injection and jailbreaks, malicious URLs and files, harmful content with adjustable thresholds, and sensitive-data leaks (built on Sensitive Data Protection); invoked inline from Apigee, Agent Platform, Google MCP servers, Service Extensions, Firebase or LangChain ^{vf¹¹⁷²}.

Strengths

- Strict data residency by default at template level, with six European regions plus an eu multi-region ^{vf¹¹⁷³}
- Low unit price: free to 2M tokens/month, then US\$0.10 per 1M tokens ^{vf¹¹⁷⁴}
- Product-scoped SOC 2 and ISO 27001 ^{vf¹¹⁷⁵}

Limitations and risks

- No grounding or denied-topic checks in the fact base ^{NPV}
- London (europe-west2) has reduced features and conflicting in-use residency statements ^{vf¹¹⁷⁶}
- Google-specific API ^{vf¹¹⁷⁷}

Choose when

- You run on Google Cloud or Apigee and want low-cost, region-pinned screening ^{AJ}

Avoid when

¹¹⁷⁰ Verified fact — A6-S067: Model Armor product page (incl. pricing) (Google Cloud) <https://cloud.google.com/security/products/model-armor> accessed 2026-10-07

¹¹⁷¹ Verified fact — A6-S067: Model Armor product page (incl. pricing) (Google Cloud) <https://cloud.google.com/security/products/model-armor> accessed 2026-10-07

¹¹⁷² Verified fact — A6-S067: Model Armor product page (incl. pricing) (Google Cloud) <https://cloud.google.com/security/products/model-armor> accessed 2026-10-07

¹¹⁷³ Verified fact — B-C2-S003: Model Armor: Data residency and endpoints (Google Cloud) <https://docs.cloud.google.com/model-armor/data-residency> accessed 2026-10-08 [search-tool extract]; B-C2-S004: Model Armor locations (Google Cloud) <https://docs.cloud.google.com/model-armor/locations> accessed 2026-10-08 [search-tool extract]

¹¹⁷⁴ Verified fact — A6-S067: Model Armor product page (incl. pricing) (Google Cloud) <https://cloud.google.com/security/products/model-armor> accessed 2026-10-07

¹¹⁷⁵ Verified fact — A6-S070: SOC 2 compliance (services in scope) (Google Cloud) <https://cloud.google.com/security/compliance/soc-2> accessed 2026-10-07; A6-S071: ISO/IEC 27001 compliance (services in scope) (Google Cloud) <https://cloud.google.com/security/compliance/iso-27001> accessed 2026-10-07

¹¹⁷⁶ Verified fact — B-C2-S003: Model Armor: Data residency and endpoints (Google Cloud) <https://docs.cloud.google.com/model-armor/data-residency> accessed 2026-10-08 [search-tool extract]; B-C2-S005: Model Armor release notes (Google Cloud) <https://docs.cloud.google.com/model-armor/release-notes> accessed 2026-10-08 [search-tool extract]

¹¹⁷⁷ Verified fact — A6-S067: Model Armor product page (incl. pricing) (Google Cloud) <https://cloud.google.com/security/products/model-armor> accessed 2026-10-07

- You need grounding checks, or the full feature set in London ^{AJ}

Nearest competitors: C2-bedrock-guardrails, C2-azure-ai-content-safety, C7-lakera

Regulated-FS note. Pin templates to an EU region with strict residency left on; if UK residency is required, confirm the London feature and in-use residency status in writing ^{Rec.}

Facts (as of 2026-10-07)

Field	Value	Label	Sources
Company	Google Cloud	Verified fact	A6-S067
Category	Managed runtime security/guardrail service for prompts, responses and agent interactions	Verified fact	A6-S067
Version / lineup	Not publicly verified	Not publicly verified	
Licence	Not publicly verified	Not publicly verified	
Strategic direction	Not publicly verified	Not publicly verified	
What it does	Screens prompts, responses and agent interactions for prompt injection and jailbreaks, malicious URLs and files, harmful content with adjustable confidence thresholds, and sensitive data leaks (built on Sensitive Data Protection).	Verified fact	A6-S067
Stack position	Guardrail service invoked inline from Apigee, Gemini Enterprise Agent Platform, Google MCP servers, Service Extensions, Firebase or LangChain	Verified fact	A6-S067, A6-S023

Field	Value	Label	Sources
Integration	API and inline integrations (Apigee, Agent Platform, Google MCP servers, Service Extensions, Firebase, LangChain)	Verified fact	A6-S067
Dependencies	Not publicly verified	Not publicly verified	
Certifications	Listed in Google Cloud SOC 2 and ISO/IEC 27001 scope	Verified fact	A6-S070, A6-S071
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Free up to 2 million tokens/month, then US\$0.10 per additional 1 million tokens (pay-as-you-go and SCC Premium)	Verified fact	A6-S067
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Not publicly verified	Not publicl	

Field	Value	Label	Sources
		y verifie d	
Deployment	saas: Yes; managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Not publicly verified; on_prem: Not publicly verified		

Azure AI Content Safety (Prompt Shields); Azure product and pricing pages now titled ‘Content Safety in Foundry Control Plane’ (c2-azure-ai-content-safety)

Tier: Strategic · **Flags:** none · **Original graphic label:** not in graphic

Rationale: Strategic, conditional: where Azure is your primary cloud (CP3 Q2, rubric rule 10). Azure’s lead guardrail service; GA features only (Prompt Shields, protected material) on regulated routes, because the agent-oriented and groundedness features are preview ^{AJ}.

Criterion	Score	Rationale
Technical	4	Harm categories, Prompt Shields (direct and document attacks), protected material; groundedness and Task Adherence preview ^{vf1178}
Enterprise readiness	4	Hyperscaler presumption (CP2 Q1); confirm per service
Security and compliance	4	Azure ISO 27001 and SOC 2 are platform-level without naming the service ^{vf1179} ; BYOK supported ^{vf1180} ; scope rule applied, so 4

¹¹⁷⁸ Verified fact — A6-S054: What is Azure AI Content Safety? (Microsoft Learn source) <https://raw.githubusercontent.com/MicrosoftDocs/azure-ai-docs/main/articles/ai-services/content-safety/overview.md> accessed 2026-10-07; A6-S055: What’s new in Azure AI Content Safety? (Microsoft Learn source) <https://raw.githubusercontent.com/MicrosoftDocs/azure-ai-docs/main/articles/ai-services/content-safety/whats-new.md> accessed 2026-10-07

¹¹⁷⁹ Verified fact — B-C1-S007: ISO/IEC 27001 - Azure Compliance (Microsoft) <https://learn.microsoft.com/en-us/azure/compliance/offerings/offering-iso-27001> accessed 2026-10-08 [search-tool extract]; B-C1-S008: System and Organization Controls (SOC) 2 Type 2 - Azure Compliance (Microsoft) <https://learn.microsoft.com/en-us/azure/compliance/offerings/offering-soc-2> accessed 2026-10-08 [search-tool extract]

¹¹⁸⁰ Verified fact — A6-S054: What is Azure AI Content Safety? (Microsoft Learn source) <https://raw.githubusercontent.com/MicrosoftDocs/azure-ai-docs/main/articles/ai-services/content-safety/overview.md> accessed 2026-10-07

Criterion	Score	Rationale
Deployment flexibility	2	Azure resource in supported regions only ^{vf-1181}
Ecosystem	4	APIM llm-content-safety policy, Foundry guardrails, SDKs ^{vf-1182}
Reliability and maturity	3	Core GA since 2024; agent features preview; 90-day deprecation policy ^{vf-1183}
Cost / TCO	2	Free tier 5,000 records/month; \$0 billed per 1,000 records per model, unit prices not verified ^{vf-1184}
Lock-in / portability	2	Azure-specific API ^{vf-1185}
Total (generic / FS)	3.30 / 3.20	

Evidence rules applied: enterprise_readiness set to 4 under the hyperscaler presumption (CP2 Q1): platform controls presumed; confirm per service

Capabilities. Managed APIs for harm categories with severity levels, Prompt Shields for user-input and document attacks, protected material detection, groundedness detection (preview), custom categories (preview) and Task Adherence for agent tool use (preview); callable from APIM on LLM, MCP and A2A traffic ^{vf-1186}.

Strengths

- Prompt Shields GA since August 2024, covering indirect (document) attacks ^{vf-1187}
- Enforceable at the APIM gateway on tool and agent payloads ^{vf-1188}
- BYOK encryption ^{vf-1189}

Limitations and risks

¹¹⁸¹ Verified fact — A6-S054: What is Azure AI Content Safety? (Microsoft Learn source) <https://raw.githubusercontent.com/MicrosoftDocs/azure-ai-docs/main/articles/ai-services/content-safety/overview.md> accessed 2026-10-07

¹¹⁸² Verified fact — A6-S053: AI gateway capabilities in Azure API Management (Microsoft Learn source, ms.date 29 May 2026) <https://raw.githubusercontent.com/MicrosoftDocs/azure-docs/main/articles/api-management/genai-gateway-capabilities.md> accessed 2026-10-07; A6-S103: Guardrails and controls overview in Microsoft Foundry <https://learn.microsoft.com/en-us/azure/foundry/guardrails/guardrails-overview> accessed 2026-10-07 [search-tool extract]

¹¹⁸³ Verified fact — A6-S054: What is Azure AI Content Safety? (Microsoft Learn source) <https://raw.githubusercontent.com/MicrosoftDocs/azure-ai-docs/main/articles/ai-services/content-safety/overview.md> accessed 2026-10-07; A6-S055: What's new in Azure AI Content Safety? (Microsoft Learn source) <https://raw.githubusercontent.com/MicrosoftDocs/azure-ai-docs/main/articles/ai-services/content-safety/whats-new.md> accessed 2026-10-07

¹¹⁸⁴ Verified fact — A6-S102: Content Safety in Foundry Control Plane - Pricing; Azure AI Content Safety FAQ (Microsoft) <https://azure.microsoft.com/en-us/pricing/details/content-safety/> accessed 2026-10-07 [search-tool extract]

¹¹⁸⁵ Verified fact — A6-S054: What is Azure AI Content Safety? (Microsoft Learn source) <https://raw.githubusercontent.com/MicrosoftDocs/azure-ai-docs/main/articles/ai-services/content-safety/overview.md> accessed 2026-10-07

¹¹⁸⁶ Verified fact — A6-S054: What is Azure AI Content Safety? (Microsoft Learn source) <https://raw.githubusercontent.com/MicrosoftDocs/azure-ai-docs/main/articles/ai-services/content-safety/overview.md> accessed 2026-10-07; A6-S055: What's new in Azure AI Content Safety? (Microsoft Learn source) <https://raw.githubusercontent.com/MicrosoftDocs/azure-ai-docs/main/articles/ai-services/content-safety/whats-new.md> accessed 2026-10-07; A6-S020: Azure API Management AI gateway docs and Tech Community announcements (AI Gateway tier preview; Build 2026 updates) (Microsoft) <https://learn.microsoft.com/en-us/azure/api-management/ai-gateway-overview> accessed 2026-10-07 [search-tool extract]

¹¹⁸⁷ Verified fact — A6-S055: What's new in Azure AI Content Safety? (Microsoft Learn source) <https://raw.githubusercontent.com/MicrosoftDocs/azure-ai-docs/main/articles/ai-services/content-safety/whats-new.md> accessed 2026-10-07; A6-S054: What is Azure AI Content Safety? (Microsoft Learn source) <https://raw.githubusercontent.com/MicrosoftDocs/azure-ai-docs/main/articles/ai-services/content-safety/overview.md> accessed 2026-10-07

¹¹⁸⁸ Verified fact — A6-S020: Azure API Management AI gateway docs and Tech Community announcements (AI Gateway tier preview; Build 2026 updates) (Microsoft) <https://learn.microsoft.com/en-us/azure/api-management/ai-gateway-overview> accessed 2026-10-07 [search-tool extract]

- Groundedness, custom categories and Task Adherence are preview ^{vr1190}
- 50 unit prices not verified ^{vr1191}
- Foundry tool-call inspection is preview and only for Foundry Agent Service agents ^{vr1192}

Choose when

- You run on Azure and want prompt-attack and harm screening at the APIM gateway ^{AI}

Avoid when

- You need GA groundedness checks today, or a guardrail outside Azure ^{AI}

Nearest competitors: C2-bedrock-guardrails, C2-google-model-armor, C7-lakera

Regulated-FS note. Use Prompt Shields on retrieved documents as well as user input; treat preview features as non-production; confirm product scope in the Service Trust Portal ^{Rec}.

Facts (as of 2026-10-07)

Field	Value	Label	Sources
Company	Microsoft	Verified fact	A6-S054
Category	Managed content-safety and prompt-attack detection APIs	Verified fact	A6-S054
Version / lineup	Prompt Shields and Protected Material (text) GA August 2024; Task Adherence public preview November 2025; groundedness detection and custom categories in preview; Python SDK azure-ai-content-safety 1.0.0 (12 December 2023)	Verified fact	A6-S055, A6-S054, A6-S084
Licence	Proprietary managed service (SDK MIT)	Verified fact	A6-S054, A6-S084
Strategic direction	Moving from content moderation to agent safety: Task Adherence API (preview) for agent tool use; Prompt Shields for direct and indirect attacks; Microsoft Foundry 'guardrails' (named collections of controls, default Microsoft.DefaultV2) build on Content Safety models and inspect user input, output, tool calls and tool responses (tool inspection in preview), currently only for Foundry Agent Service agents	Verified fact	A6-S055, A6-S054, A6-S103
What it does	APIs to analyse text and images for sexual, violence, hate and self-harm content with severity levels; Prompt Shields for user-input and document attacks; protected material detection; groundedness detection (preview); custom categories (preview); Task Adherence for agent tool use (preview).	Verified fact	A6-S054, A6-S055
Stack position	Guardrail service called by applications, Microsoft Foundry and Azure	Verified fact	A6-S053,

¹¹⁸⁹ Verified fact — A6-S054: What is Azure AI Content Safety? (Microsoft Learn source) <https://raw.githubusercontent.com/MicrosoftDocs/azure-ai-docs/main/articles/ai-services/content-safety/overview.md> accessed 2026-10-07

¹¹⁹⁰ Verified fact — A6-S054: What is Azure AI Content Safety? (Microsoft Learn source) <https://raw.githubusercontent.com/MicrosoftDocs/azure-ai-docs/main/articles/ai-services/content-safety/overview.md> accessed 2026-10-07; A6-S055: What's new in Azure AI Content Safety? (Microsoft Learn source) <https://raw.githubusercontent.com/MicrosoftDocs/azure-ai-docs/main/articles/ai-services/content-safety/whats-new.md> accessed 2026-10-07

¹¹⁹¹ Verified fact — A6-S102: Content Safety in Foundry Control Plane - Pricing; Azure AI Content Safety FAQ (Microsoft) <https://azure.microsoft.com/en-us/pricing/details/content-safety/> accessed 2026-10-07 [search-tool extract]

¹¹⁹² Verified fact — A6-S103: Guardrails and controls overview in Microsoft Foundry <https://learn.microsoft.com/en-us/azure/foundry/guardrails/guardrails-overview> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
Integration	API Management (llm-content-safety policy)	d fact	A6-S020
	REST API and SDKs; APIM llm-content-safety policy; Content Safety Studio	Verifie d fact	A6-S054, A6-S053
	Dependencies	Not publicl y verifie d	
Certifications	Not publicly verified	Not publicl y verifie d	
GDPR / residency	Resource must be created in a supported region; region list not reproduced in sources consulted	Verifie d fact	A6-S054
Security features	Encryption at rest with customer-managed keys (BYOK) supported	Verifie d fact	A6-S054
Access controls	Not publicly verified	Not publicl y verifie d	
Enterprise support	Not publicly verified	Not publicl y verifie d	
Pricing	Free tier: 5,000 text records/month (F0, 5 RPS). Standard (S0): billed per 1,000 text records, each model (incl. Prompt Shields) with its own rate; a text record is up to 1,000 characters. Published S0 unit prices did not render; a community figure of US\$0.38 per 1,000 records is unverified	Verifie d fact	A6-S102, A6-S054
Infrastructure cost	Not publicly verified	Not publicl y verifie d	
Ecosystem	Not publicly verified	Not publicl y verifie d	
Adoption signals	Not publicly verified	Not publicl y verifie d	
Maturity	Core moderation and Prompt Shields GA; several agent-oriented features in preview; 90-day deprecation policy for superseded API	Verifie	A6-S054,

Field	Value	Label	Sources
	versions	d fact	A6-S055
Deployment	saas: Yes (Azure resource in supported regions); managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Not publicly verified; on_prem: Not publicly verified		

Llama Protections: Llama Guard 4 (12B), Llama Prompt Guard 2 (86M/22M), LlamaFirewall, Code Shield (c2-meta-llama-protections)

Tier: Tactical · **Flags:** none · **Original graphic label:** not in graphic

Rationale: Tactical: useful self-hosted detectors, but no release for 17 months is a maintenance risk for a security control ^{AJ}.

Criterion	Score	Rationale
Technical	3	Safety taxonomy classifier, prompt-attack classifier, agent alignment check and code scanner ^{vf¹¹⁹³} ; 512-token window and reported multilingual weakness ^{vf¹¹⁹⁴} _{i¹¹⁹⁵}
Enterprise readiness	2	Rule 2: components only; no commercial support from Meta verified ^{NPV}
Security and compliance	2	Rule 2: custom licence; LlamaFirewall licence not stated on PyPI ^{vf¹¹⁹⁶} ; CVE handling NPV
Deployment flexibility	5	Open weights run anywhere, including air-gapped; also hosted by third parties ^{vf¹¹⁹⁷}
Ecosystem	4	Embedded in Cloudflare and Guardrails Hub; hosted on NVIDIA build and Vertex Model Garden ^{vf¹¹⁹⁸}
Reliability and maturity	2	Mature lineage but no release after May 2025 ^{vf¹¹⁹⁹}

¹¹⁹³ Verified fact — A6-S030: meta-llama/Llama-Guard-4-12B model card and LICENSE (Hugging Face); Llama Prompt Guard 2 model cards (Meta (via Hugging Face)) <https://huggingface.co/meta-llama/Llama-Guard-4-12B> accessed 2026-10-07 [search-tool extract]; A6-S039: LlamaFirewall README (GitHub) (Meta) <https://raw.githubusercontent.com/meta-llama/PurpleLlama/main/LlamaFirewall/README.md> accessed 2026-10-07

¹¹⁹⁴ Verified fact — A6-S030: meta-llama/Llama-Guard-4-12B model card and LICENSE (Hugging Face); Llama Prompt Guard 2 model cards (Meta (via Hugging Face)) <https://huggingface.co/meta-llama/Llama-Guard-4-12B> accessed 2026-10-07 [search-tool extract]

¹¹⁹⁵ Reported — A6-S031: Sharing new open source protection tools and advancements in AI privacy and security; Llama Protections page (Meta) <https://ai.meta.com/blog/ai-defenders-program-llama-protection-tools/> accessed 2026-10-07 [search-tool extract]

¹¹⁹⁶ Verified fact — A6-S030: meta-llama/Llama-Guard-4-12B model card and LICENSE (Hugging Face); Llama Prompt Guard 2 model cards (Meta (via Hugging Face)) <https://huggingface.co/meta-llama/Llama-Guard-4-12B> accessed 2026-10-07 [search-tool extract]; A6-S006: llamafirewall - PyPI JSON metadata (Python Package Index) <https://pypi.org/pypi/llamafirewall/json> accessed 2026-10-07

¹¹⁹⁷ Verified fact — A6-S030: meta-llama/Llama-Guard-4-12B model card and LICENSE (Hugging Face); Llama Prompt Guard 2 model cards (Meta (via Hugging Face)) <https://huggingface.co/meta-llama/Llama-Guard-4-12B> accessed 2026-10-07 [search-tool extract]

¹¹⁹⁸ Verified fact — A6-S052: Cloudflare AI Gateway docs: Pricing (source mdx) (Cloudflare, Inc.) <https://raw.githubusercontent.com/cloudflare/cloudflare-docs/production/src/content/docs/ai-gateway/reference/pricing.mdx> accessed 2026-10-07; A6-S029: Guardrails AI docs: Validators; Remote Validation Inference; Deploying Guardrails (Guardrails AI (Harvey)) <https://guardrailsai.com/guardrails/docs/concepts/validators> accessed 2026-10-07 [search-tool extract]; A6-S030: meta-llama/Llama-Guard-4-12B model card and LICENSE (Hugging Face); Llama Prompt Guard 2 model cards (Meta (via Hugging Face)) <https://huggingface.co/meta-llama/Llama-Guard-4-12B> accessed 2026-10-07 [search-tool extract]

¹¹⁹⁹ Verified fact — A6-S006: llamafirewall - PyPI JSON metadata (Python Package Index) <https://pypi.org/pypi/llamafirewall/json> accessed 2026-10-07; A6-S107: Llama Guard 4 model card; Llama Prompt Guard 2 model card; Llama Protections (Meta) <https://developer.meta.com/ai/docs/model-cards-and-prompt-formats/llama-guard-4/> accessed 2026-10-07 [search-tool extract]

Criterion	Score	Rationale
Cost / TCO	4	Free; 12B model needs GPU-class serving, 22M model is cheap ^{vf1200}
Lock-in / portability	3	Portable weights under a custom community licence ^{vf1201}
Total (generic / FS)	3.10 / 2.95	

Evidence rules applied: Rule 2 basis: self-hosted open weights and open-source framework; NPV cap not applied

Capabilities. Meta's open-weight safety components: Llama Guard 4 (12B) classifies prompts and responses (text and images) against a safety taxonomy; Prompt Guard 2 (86M/22M) detects prompt injection and jailbreaks within a 512-token window; LlamaFirewall orchestrates PromptGuard, AlignmentCheck and CodeShield across agent inputs, reasoning and outputs ^{vf1202}.

Strengths

- Small, cheap prompt-attack classifier (22M) suited to inline use ^{vf1203}
- AlignmentCheck targets agent goal hijacking ^{vf1204}
- Widely embedded: Cloudflare runs Llama Guard; Guardrails Hub has a validator ^{vf1205}

Limitations and risks

- No release found since Llama Guard 4 / Prompt Guard 2 (29 April 2025) and Llamafirewall 1.0.3 (29 May 2025) ^{vf1206}
- Llama community licence with a 700M MAU clause and Acceptable Use Policy, not OSI ^{vf1207}

¹²⁰⁰ Verified fact — A6-S030: meta-llama/Llama-Guard-4-12B model card and LICENSE (Hugging Face); Llama Prompt Guard 2 model cards (Meta (via Hugging Face)) <https://huggingface.co/meta-llama/Llama-Guard-4-12B> accessed 2026-10-07 [search-tool extract]

¹²⁰¹ Verified fact — A6-S030: meta-llama/Llama-Guard-4-12B model card and LICENSE (Hugging Face); Llama Prompt Guard 2 model cards (Meta (via Hugging Face)) <https://huggingface.co/meta-llama/Llama-Guard-4-12B> accessed 2026-10-07 [search-tool extract]; A6-S038: PurpleLlama README (GitHub) (Meta) <https://raw.githubusercontent.com/meta-llama/PurpleLlama/main/README.md> accessed 2026-10-07

¹²⁰² Verified fact — A6-S030: meta-llama/Llama-Guard-4-12B model card and LICENSE (Hugging Face); Llama Prompt Guard 2 model cards (Meta (via Hugging Face)) <https://huggingface.co/meta-llama/Llama-Guard-4-12B> accessed 2026-10-07 [search-tool extract]; A6-S039: LlamaFirewall README (GitHub) (Meta) <https://raw.githubusercontent.com/meta-llama/PurpleLlama/main/LlamaFirewall/README.md> accessed 2026-10-07; A6-S038: PurpleLlama README (GitHub) (Meta) <https://raw.githubusercontent.com/meta-llama/PurpleLlama/main/README.md> accessed 2026-10-07

¹²⁰³ Verified fact — A6-S030: meta-llama/Llama-Guard-4-12B model card and LICENSE (Hugging Face); Llama Prompt Guard 2 model cards (Meta (via Hugging Face)) <https://huggingface.co/meta-llama/Llama-Guard-4-12B> accessed 2026-10-07 [search-tool extract]

¹²⁰⁴ Verified fact — A6-S039: LlamaFirewall README (GitHub) (Meta) <https://raw.githubusercontent.com/meta-llama/PurpleLlama/main/LlamaFirewall/README.md> accessed 2026-10-07

¹²⁰⁵ Verified fact — A6-S052: Cloudflare AI Gateway docs: Pricing (source mdx) (Cloudflare, Inc.) <https://raw.githubusercontent.com/cloudflare/cloudflare-docs/production/src/content/docs/ai-gateway/reference/pricing.mdx> accessed 2026-10-07; A6-S029: Guardrails AI docs: Validators; Remote Validation Inference; Deploying Guardrails (Guardrails AI (Harvey)) <https://guardrailsai.com/guardrails/docs/concepts/validators> accessed 2026-10-07 [search-tool extract]

¹²⁰⁶ Verified fact — A6-S006: llamafirewall - PyPI JSON metadata (Python Package Index) <https://pypi.org/pypi/llamafirewall/json> accessed 2026-10-07; A6-S030: meta-llama/Llama-Guard-4-12B model card and LICENSE (Hugging Face); Llama Prompt Guard 2 model cards (Meta (via Hugging Face)) <https://huggingface.co/meta-llama/Llama-Guard-4-12B> accessed 2026-10-07 [search-tool extract]; A6-S107: Llama Guard 4 model card; Llama Prompt Guard 2 model card; Llama Protections (Meta) <https://developer.meta.com/ai/docs/model-cards-and-prompt-formats/llama-guard-4/> accessed 2026-10-07 [search-tool extract]

¹²⁰⁷ Verified fact — A6-S030: meta-llama/Llama-Guard-4-12B model card and LICENSE (Hugging Face); Llama Prompt Guard 2 model cards (Meta (via Hugging Face)) <https://huggingface.co/meta-llama/Llama-Guard-4-12B> accessed 2026-10-07 [search-tool extract]; A6-S038: PurpleLlama README (GitHub) (Meta) <https://raw.githubusercontent.com/meta-llama/PurpleLlama/main/README.md> accessed 2026-10-07

- An independent tester reported Prompt Guard weak against non-English and obfuscated prompts ¹²⁰⁸

Choose when

- You want an in-estate, low-cost first-pass classifier inside a broader guardrail framework ^{AJ}

Avoid when

- It would be your only prompt-attack defence, or you need a maintained component with a release cadence ^{AJ}

Nearest competitors: C2-azure-ai-content-safety, C2-google-model-armor, C7-lakera

Regulated-FS note. Use only as one detector among several, with your own evaluation set; record the licence terms and the maintenance risk in the inventory ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Meta	Verified fact	A6-S006, A6-S031
Category	Open-weight safety classifiers plus an open-source agent guardrail framework	Verified fact	A6-S030, A6-S039
Version / lineup	Llama Guard 4 12B and Prompt Guard 2 86M/22M (checkpoints 29 April 2025); llamafirewall 1.0.3 (PyPI, 29 May 2025)	Verified fact	A6-S030, A6-S006, V2-S028
Licence	Models: Llama Community licences (Llama Guard 4 under Llama 4 Community License, incl. 700M MAU clause and Acceptable Use Policy); evals/benchmarks MIT; LlamaFirewall licence not stated on PyPI	Verified fact	A6-S030, A6-S038, A6-S006
Strategic direction	Agent-focused layered defence: LlamaFirewall orchestrates PromptGuard, AlignmentCheck (chain-of-thought auditing for goal hijacking) and CodeShield scanners; no newer release found since mid-2025	Verified fact	A6-S039, A6-S031
What it does	Llama Guard 4 classifies prompts and responses (text and images) against a safety taxonomy; Prompt Guard 2 detects prompt injection and jailbreaks (512-token window); LlamaFirewall combines scanners across agent inputs, reasoning and outputs; Code Shield filters insecure generated code.	Verified fact	A6-S030, A6-S039, A6-S038
Stack position	Model-level guardrail components called by applications, gateways or guardrail frameworks (e.g. Guardrails Hub has a Llama Guard validator;	Verified fact	A6-S029, A6-S052

¹²⁰⁸ Reported — A6-S031: Sharing new open source protection tools and advancements in AI privacy and security; Llama Protections page (Meta) <https://ai.meta.com/blog/ai-defenders-program-llama-protection-tools/> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
	Cloudflare AI Gateway uses Llama Guard)		
Integration	Hugging Face model weights; Llama API moderations endpoint; hosted on third parties (e.g. NVIDIA build, Vertex Model Garden); pip install llamafirewall	Verified fact	A6-S030, A6-S039
Dependencies	GPU/CPU inference for 12B (Llama Guard 4) or 22M/86M (Prompt Guard 2) models; LlamaFirewall Python package	Verified fact	A6-S030, A6-S006
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Free to download under licence terms	Verified fact	A6-S030
Infrastructure cost	Llama Guard 4 is a 12B-parameter model (GPU class serving); Prompt Guard 2 22M designed for low-latency, low-compute use	Verified fact	A6-S030
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Mature model lineage (Llama Guard 1 to 4); Meta's current model cards still present Llama Guard 4 (12B) as 'the latest safeguard model'; no 2026 release found in a vendor-domain search	Verified fact	A6-S038, A6-S030, A6-S006,

Field	Value	Label	Sources
			A6-S107
Deployment	saas: Not publicly verified; managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes (open weights and open-source framework); on_prem: Not publicly verified		

Guardrails AI (open-source ‘guardrails’ framework and Guardrails Hub validators) (c2-guardrails-ai)

Tier: Experimental • **Flags:** Acquired • **Original graphic label:** not in graphic

Rationale: Experimental: the owner is now an application company, the distribution model changed weeks before the deal, and the roadmap is not stated ^{AJ}.

Criterion	Score	Rationale
Technical	3	Validators and structured-output validation; no native grounding or prompt-attack depth evidenced beyond validators ^{VF1209}
Enterprise readiness	2	Rule 2: library plus REST server; enterprise platform ‘talk to sales’ per a secondary source only ^{A1210}
Security and compliance	2	Rule 2 hygiene not verified ^{NPV} ; distribution change mid-2026 ^{VF1211}
Deployment flexibility	3	Self-hosted only now that remote inference is retired ^{VF1212}
Ecosystem	3	Hub validators (including Llama Guard) and 250,000+ monthly downloads (vendor claim) ^{VF1213}
Reliability and	2	0.x since 2023; acquired; distribution changed ^{VF1214}

¹²⁰⁹ Verified fact — A6-S037: Guardrails AI README (GitHub) (Guardrails AI (Harvey)) <https://raw.githubusercontent.com/guardrails-ai/guardrails/main/README.md> accessed 2026-10-07; A6-S029: Guardrails AI docs: Validators; Remote Validation Inference; Deploying Guardrails (Guardrails AI (Harvey)) <https://guardrailsai.com/guardrails/docs/concepts/validators> accessed 2026-10-07 [search-tool extract]

¹²¹⁰ Reported — A6-S028: Guardrails AI Joins Harvey <https://www.harvey.ai/blog/guardrails-ai-joins-harvey> accessed 2026-10-07 [search-tool extract]

¹²¹¹ Verified fact — A6-S037: Guardrails AI README (GitHub) (Guardrails AI (Harvey)) <https://raw.githubusercontent.com/guardrails-ai/guardrails/main/README.md> accessed 2026-10-07

¹²¹² Verified fact — A6-S037: Guardrails AI README (GitHub) (Guardrails AI (Harvey)) <https://raw.githubusercontent.com/guardrails-ai/guardrails/main/README.md> accessed 2026-10-07

¹²¹³ Verified fact — A6-S029: Guardrails AI docs: Validators; Remote Validation Inference; Deploying Guardrails (Guardrails AI (Harvey)) <https://guardrailsai.com/guardrails/docs/concepts/validators> accessed 2026-10-07 [search-tool extract]; A6-S028: Guardrails AI Joins Harvey <https://www.harvey.ai/blog/guardrails-ai-joins-harvey> accessed 2026-10-07 [search-tool extract]

¹²¹⁴ Verified fact — A6-S004: guardrails-ai - PyPI JSON metadata (Python Package Index) <https://pypi.org/pypi/guardrails-ai/json> accessed 2026-10-07; A6-S028: Guardrails AI Joins Harvey <https://www.harvey.ai/blog/guardrails-ai-joins-harvey> accessed 2026-10-07 [search-tool extract]; A6-S037: Guardrails AI README (GitHub) (Guardrails AI (Harvey)) <https://raw.githubusercontent.com/guardrails-ai/guardrails/main/README.md> accessed 2026-10-07

Criterion	Score	Rationale
maturity		
Cost / TCO	4	Free open source ^{vr1215}
Lock-in / portability	3	Apache-2.0, but acquired and not neutrally governed: minus 1 (rule 3) ^{vr1216}
Total (generic / FS)	2.70	
	/	
	2.60	

Evidence rules applied: Rule 2 basis: self-hosted open-source library; NPV cap not applied

Capabilities. Apache-2.0 Python framework running input and output Guards with composable validators and structured-output validation, plus a Guardrails Server REST API; validators now ship as PyPI packages after the hosted hub and remote inference were retired ^{vr1217}.

Strengths

- Simple validator model for structured-output and format checks ^{vr1218} ^{AI}
- Apache-2.0 ^{vr1219}

Limitations and risks

- Acquired by Harvey, a legal AI application company, on 9 September 2026; open-source roadmap not stated ^{vr1220}
- Hosted hub install and remote inference retired; cutoff given as 25 August (repository) or 6 August 2026 (docs) ^{vr1221}
- 0.x versioning ^{vr1222}

Choose when

- You already use it for structured-output validation and can vendor the validators you need ^{AI}

Avoid when

¹²¹⁵ Verified fact — A6-S004: guardrails-ai - PyPI JSON metadata (Python Package Index) <https://pypi.org/pypi/guardrails-ai/json> accessed 2026-10-07

¹²¹⁶ Verified fact — A6-S004: guardrails-ai - PyPI JSON metadata (Python Package Index) <https://pypi.org/pypi/guardrails-ai/json> accessed 2026-10-07; A6-S028: Guardrails AI Joins Harvey <https://www.harvey.ai/blog/guardrails-ai-joins-harvey> accessed 2026-10-07 [search-tool extract]

¹²¹⁷ Verified fact — A6-S037: Guardrails AI README (GitHub) (Guardrails AI (Harvey)) <https://raw.githubusercontent.com/guardrails-ai/guardrails/main/README.md> accessed 2026-10-07; A6-S029: Guardrails AI docs: Validators; Remote Validation Inference; Deploying Guardrails (Guardrails AI (Harvey)) <https://guardrailsai.com/guardrails/docs/concepts/validators> accessed 2026-10-07 [search-tool extract]; A6-S004: guardrails-ai - PyPI JSON metadata (Python Package Index) <https://pypi.org/pypi/guardrails-ai/json> accessed 2026-10-07

¹²¹⁸ Verified fact — A6-S037: Guardrails AI README (GitHub) (Guardrails AI (Harvey)) <https://raw.githubusercontent.com/guardrails-ai/guardrails/main/README.md> accessed 2026-10-07

¹²¹⁹ Verified fact — A6-S004: guardrails-ai - PyPI JSON metadata (Python Package Index) <https://pypi.org/pypi/guardrails-ai/json> accessed 2026-10-07

¹²²⁰ Verified fact — A6-S028: Guardrails AI Joins Harvey <https://www.harvey.ai/blog/guardrails-ai-joins-harvey> accessed 2026-10-07 [search-tool extract]; V2-S026: Guardrails AI Joins Harvey (with Artificial Lawyer 9 Sep 2026, The Next Web, unite.ai) <https://www.harvey.ai/blog/guardrails-ai-joins-harvey> accessed 2026-10-08 [search-tool extract]

¹²²¹ Verified fact — V2-S071: Guardrails HUB_UPDATE.md (Guardrails AI (Harvey)) https://raw.githubusercontent.com/guardrails-ai/guardrails/main/HUB_UPDATE.md accessed 2026-10-08; V2-S072: Guardrails 0.11 migration guide (with beri.net article) (Guardrails AI (Harvey)) <https://guardrailsai.com/guardrails/docs/migration-guides/0-11-migration> accessed 2026-10-08 [search-tool extract]

¹²²² Verified fact — A6-S004: guardrails-ai - PyPI JSON metadata (Python Package Index) <https://pypi.org/pypi/guardrails-ai/json> accessed 2026-10-07

- You are choosing a new strategic guardrail dependency ^{AJ}

Nearest competitors: C2-nemo-guardrails, C2-meta-llama-protections, L9-opik

Regulated-FS note. Freeze and vendor the validators in use, plan a migration path, and do not rely on any hosted Guardrails service ^{Rec.}

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Harvey (acquired Guardrails AI, announced 9 September 2026)	Verified fact	A6-S028
Category	Open-source input/output validation framework for LLM applications	Verified fact	A6-S037
Version / lineup	guardrails-ai 0.11.0 (14 August 2026)	Verified fact	A6-S004, V2-S028
Licence	Open source (Apache-2.0)	Verified fact	A6-S004, A6-S037
Status events	6 July 2026: announced move of validators to PyPI and discontinuation of hosted remote inference (hard cutoff 25 August 2026 per the repository HUB_UPDATE.md; the docs-site 0.11 migration guide gives 6 August 2026); 9 September 2026: acquired by Harvey; co-founders Shreya Rajpal and Zayd Simjee joined Harvey; terms undisclosed	Verified fact	A6-S037, A6-S028, V2-S026, V2-S071, V2-S072
Strategic direction	Validators moving to standard PyPI packages; hosted remote inferencing discontinued (cutoff 25 August 2026); team joined Harvey's product and engineering organisation to work on agent reliability	Verified fact	A6-S037, A6-S028
What it does	Runs input and output Guards in the application that detect, quantify and mitigate specific risks using composable validators (from Guardrails Hub), and supports structured output validation; also offers a Guardrails Server REST API.	Verified fact	A6-S037, A6-S029
Stack position	Application-level guardrail library around model calls	Verified fact	A6-S037
Integration	Python SDK; validators installed with pip as guardrails-ai-; Guardrails Server REST API	Verified fact	A6-S029, A6-S037
Dependencies	Python >=3.10,<3.14; individual validators may need their own models	Verified	A6-S004,

Field	Value	Label	Sources
Certifications	Not publicly verified	d fact Not publicl y verifie d	A6-S029
GDPR / residency	Not publicly verified	Not publicl y verifie d	
Security features	Not publicly verified	Not publicl y verifie d	
Access controls	Not publicly verified	Not publicl y verifie d	
Enterprise support	Not publicly verified	Not publicl y verifie d	
Pricing	Open-source framework free; enterprise platform and Snowglobe sold via sales (last known)	Report ed	A6-S028
Infrastructure cost	Not publicly verified	Not publicl y verifie d	
Ecosystem	Not publicly verified	Not publicl y verifie d	
Adoption signals	Harvey states the framework is downloaded more than 250,000 times a month (vendor claim)	Verifie d fact	A6-S028
Maturity	First PyPI release 13 March 2023; 0.x versioning	Verifie d fact	A6-S004
Deployment	saas: No longer for free remote inference (retired 25 August 2026); managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes; on_prem: Not publicly verified		

C3: DLP and PII protection

Presidio (Data Privacy Stack) (c3-presidio)

Tier: Strategic · **Flags:** Renamed · **Original graphic label:** not in graphic

Rationale: Strategic, conditional: as the in-estate engine behind a firm-owned privacy service with recall testing and a sampling second detector; enterprise readiness is what the firm builds around it ^W.

Criterion	Score	Rationale
Technical	3	Covers detection and masking across text, images and structured data; no vault tokenisation, discovery or classification of stores.
Enterprise readiness	3	Rule 2 library: enables in-estate detection; no commercial support; REST services unauthenticated by default (NPV).
Security and compliance	3	Rule 2 hygiene: security policy with private reporting verified (B-C3-S001); MIT clear; signed releases NPV; inherits host controls.
Deployment flexibility	5	Self-hosted library or containers, in-estate and offline-capable with local NER models.
Ecosystem	3	Python and REST interfaces; gateway and guardrail integrations not verified in the fact base.
Reliability and maturity	3	Releases since March 2020 with several a year; governance moved to volunteers in 2026.
Cost / TCO	4	Free; operating cost is recogniser tuning and testing.
Lock-in / portability	5	MIT with neutral community governance; rule 3 reduction not applied (permissive OSS, neutral governance).
Total (generic / FS)	3.50 /	

Criterion	Score	Rationale
	3.65	

Evidence rules applied: Rule 2 (self-hosted library): enterprise_readiness and security_compliance capped at 4; NPV cap not applied; inherits host controls

Capabilities. Open-source PII detection and de-identification SDK (analyzer, anonymizer, image redactor incl. DICOM, structured data) using NER, regex, rules and checksums with context; LLM-based recognisers added; Python packages and Docker REST services ^{vf¹²²³}. 2.2.364 (22 July 2026), MIT ^{vf¹²²⁴}.

Strengths

- Runs entirely in-estate, so 'must not leave' text is inspected before any external call ^{AJ}
- MIT licence and community governance with a Technical Steering Committee ^{vf¹²²⁵}
- Custom recognisers are code the firm can own, version and test ^{AJ}
- Published security policy with private vulnerability reporting and 48-hour acknowledgement ^{vf¹²²⁶}

Limitations and risks

- README warns detection is not guaranteed to find all sensitive data ^{vf¹²²⁷}
- Maintained by volunteers; not owned or operated by a commercial entity; no commercial support ^{vf¹²²⁸}
- Ownership moved from Microsoft in 2026; images moved from MCR to GHCR ^{vf¹²²⁹}
- No vault, discovery or built-in authentication on REST services documented ^{NPV}
- Signed releases not verified ^{NPV}

Choose when

- You need in-estate detection behind a firm-owned privacy service ^{AJ}
- You need firm-specific recognisers (client codes, account numbers) ^{AJ}

Avoid when

¹²²³ Verified fact — A6-S068: Presidio README (GitHub, data-privacy-stack) (Presidio / Data Privacy Stack) <https://raw.githubusercontent.com/data-privacy-stack/presidio/main/README.MD> accessed 2026-10-07; A6-S040: Presidio has transitioned to a community-owned project (project_transition.md) (Presidio / Data Privacy Stack) https://raw.githubusercontent.com/data-privacy-stack/presidio/main/docs/project_transition.md accessed 2026-10-07; A6-S041: Presidio CHANGELOG.md (Presidio / Data Privacy Stack) <https://raw.githubusercontent.com/data-privacy-stack/presidio/main/CHANGELOG.md> accessed 2026-10-07

¹²²⁴ Verified fact — A6-S005: presidio-analyzer - PyPI JSON metadata (Python Package Index) <https://pypi.org/pypi/presidio-analyzer/json> accessed 2026-10-07; V2-S028: PyPI JSON API: litellm, guardrails-ai, presidio-analyzer, nemoguardrails, llamafirewall, langfuse, promptfoo, modelscan, picklescan, fickleling, safetensors, mode <https://pypi.org/pypi/litellm/json> accessed 2026-10-08 [search-tool extract]

¹²²⁵ Verified fact — A6-S040: Presidio has transitioned to a community-owned project (project_transition.md) (Presidio / Data Privacy Stack) https://raw.githubusercontent.com/data-privacy-stack/presidio/main/docs/project_transition.md accessed 2026-10-07; V2-S030: Presidio has transitioned to a community-owned project (Presidio / Data Privacy Stack) https://raw.githubusercontent.com/data-privacy-stack/presidio/main/docs/project_transition.md accessed 2026-10-08

¹²²⁶ Verified fact — B-C3-S001: Presidio Security Policy (SECURITY.md) (Data Privacy Stack (Presidio project)) <https://raw.githubusercontent.com/data-privacy-stack/presidio/main/SECURITY.md> accessed 2026-10-08

¹²²⁷ Verified fact — A6-S068: Presidio README (GitHub, data-privacy-stack) (Presidio / Data Privacy Stack) <https://raw.githubusercontent.com/data-privacy-stack/presidio/main/README.MD> accessed 2026-10-07

¹²²⁸ Verified fact — A6-S040: Presidio has transitioned to a community-owned project (project_transition.md) (Presidio / Data Privacy Stack) https://raw.githubusercontent.com/data-privacy-stack/presidio/main/docs/project_transition.md accessed 2026-10-07

¹²²⁹ Verified fact — A6-S040: Presidio has transitioned to a community-owned project (project_transition.md) (Presidio / Data Privacy Stack) https://raw.githubusercontent.com/data-privacy-stack/presidio/main/docs/project_transition.md accessed 2026-10-07; A6-S041: Presidio CHANGELOG.md (Presidio / Data Privacy Stack) <https://raw.githubusercontent.com/data-privacy-stack/presidio/main/CHANGELOG.md> accessed 2026-10-07

- You need a supported product with an SLA ^{AJ}
- You need reversible tokenisation at scale or estate-wide discovery ^{AJ}

Nearest competitors: C3-google-sdp, C2-bedrock-guardrails, C3-protegrity

Regulated-FS note. Pin versions, update image references after the GHCR move, and gate upgrades on recall per entity class measured on the firm's test corpus ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Community-governed open-source project under the Data Privacy Stack organisation (formerly Microsoft); Technical Steering Committee	Verified fact	A6-S040, V2-S030
Category	Open-source PII detection and de-identification SDK (text, images, structured data)	Verified fact	A6-S068
Version / lineup	presidio-analyzer and presidio-anonymizer 2.2.364 (22 July 2026)	Verified fact	A6-S005, V2-S028
Licence	Open source (MIT)	Verified fact	A6-S005, A6-S040
Status events	2026: transitioned from a Microsoft-owned project to community governance under Data Privacy Stack (rebrand in 2.2.363, 28 June 2026); container images moved from MCR to GHCR	Verified fact	A6-S040, A6-S041
Strategic direction	Independent, vendor-neutral governance; continued recognisers for new jurisdictions (e.g. South African, Philippine IDs) and LLM-based recognisers (LangExtract)	Verified fact	A6-S040, A6-S041
What it does	Identifies PII in text and images using NER, regular expressions, rule-based logic and checksums with context, and de-identifies it; separate modules for analyzer, anonymizer, image redactor (incl. DICOM) and structured data. README warns detection is not guaranteed to find all sensitive data.	Verified fact	A6-S068
Stack position	Library or microservice used in ingestion pipelines, gateways and guardrail hooks to detect and mask PII	Verified fact	A6-S068
Integration	Python packages, REST services in Docker images (ghcr.io/data-privacy-stack/presidio-*)	Verified fact	A6-S040, A6-S068
Dependencies	Python >=3.10,<3.15; spaCy/Stanza or Hugging Face NER models	Verified	A6-S005,

Field	Value	Label	Sources
	(NoOpNlpEngine available)	d fact	A6-S041
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Free open source	Verified fact	A6-S040
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	First PyPI release 10 March 2020; 2.2.x line with several releases per year	Verified fact	A6-S005
Deployment	saas: No (not operated by a commercial entity); managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes; on_prem: Not publicly verified		

Sensitive Data Protection (including Cloud Data Loss Prevention / DLP API) (c3-google-sdp)

Tier: Strategic · **Flags:** none · **Original graphic label:** not in graphic

Rationale: Strategic, conditional: where Google Cloud is your primary cloud (CP3 Q2, rubric rule 10), because deployment flexibility and lock-in score 2 ^{AJ}.

Criterion	Score	Rationale
Technical	5	Discovery, profiling, 200+ detectors, masking, tokenisation and bucketing, runtime prompt/response use via Model Armor.
Enterprise readiness	4	Hyperscaler presumption (CP2 Q1): platform controls presumed (CP2 Q1); confirm per service.
Security and compliance	4	SOC 2 and ISO 27001 product-scoped; KMS-wrapped de-identification keys; CMEK for stored profiles and EU locations not verified, so 4 not 5.
Deployment flexibility	2	SaaS only on Google Cloud; regional processing possible; no self-host.
Ecosystem	4	Built into Google Cloud storage services, Model Armor and Apigee; hybrid jobs for external data.
Reliability and maturity	5	Long-established service (client library since February 2018).
Cost / TCO	3	Transparent but volume-priced: US\$3.00/GiB after 1 GiB free; discovery subscription US\$2,500/unit/month.
Lock-in / portability	2	Google Cloud-specific API; outputs portable.
Total (generic / FS)	3.80 / 3.60	

Capabilities. Managed discovery, classification and de-identification (formerly Cloud DLP): profiling of BigQuery, Cloud SQL, Cloud Storage and Agent Platform; 200+ detectors; masking, tokenisation, bucketing; hybrid jobs for external sources; underpins Model Armor ^{vr1230}.

Strengths

- Broadest managed detection and transformation set in the control ^{AJ}
- Product-scoped SOC 2 and ISO 27001 ^{vr1231}
- Tokenisation keys wrapped by Cloud KMS, global or in the request region ^{vr1232}
- Transparent per-GiB pricing ^{vr1233}

¹²³⁰ Verified fact — A6-S066: Sensitive Data Protection product page (Google Cloud) <https://cloud.google.com/security/products/sensitive-data-protection> accessed 2026-10-07; A6-S065: Sensitive Data Protection pricing (Google Cloud) <https://cloud.google.com/sensitive-data-protection/pricing> accessed 2026-10-07; A6-S067: Model Armor product page (incl. pricing) (Google Cloud) <https://cloud.google.com/security/products/model-armor> accessed 2026-10-07

¹²³¹ Verified fact — A6-S070: SOC 2 compliance (services in scope) (Google Cloud) <https://cloud.google.com/security/compliance/soc-2> accessed 2026-10-07; A6-S071: ISO/IEC 27001 compliance (services in scope) (Google Cloud) <https://cloud.google.com/security/compliance/iso-27001> accessed 2026-10-07

¹²³² Verified fact — B-C3-S002: De-identify and re-identify sensitive data (Sensitive Data Protection) (Google Cloud) <https://docs.cloud.google.com/sensitive-data-protection/docs/inspect-sensitive-text-de-identify> accessed 2026-10-08 [search-tool extract]

¹²³³ Verified fact — A6-S065: Sensitive Data Protection pricing (Google Cloud) <https://cloud.google.com/sensitive-data-protection/pricing> accessed 2026-10-07

Limitations and risks

- No self-hosted option ^{VF}1234
- Google Cloud-specific API ^{VF}1235
- Supported EU locations not retrieved ^{NPV}
- Raw text must be sent to the service for inspection ^{AJ}

Choose when

- Google Cloud is the data platform ^{AJ}
- Model Armor or Apigee are already in the path ^{AJ}

Avoid when

- Policy forbids sending raw client text to a cloud service before tokenisation ^{AJ}
- You need a multi-cloud engine with one API ^{AJ}

Nearest competitors: C3-presidio, C3-microsoft-purview-dspm-ai, C2-bedrock-guardrails

Regulated-FS note. Pin processing and key location to an approved EU/UK region and confirm against the locations page; platform controls presumed (CP2 Q1); confirm per service ^{REC}.

Facts (as of 2026-10-07)

Field	Value	Label	Sources
Company	Google Cloud	Verified fact	A6-S066
Category	Managed sensitive-data discovery, classification and de-identification service	Verified fact	A6-S066
Version / lineup	Managed service; Python client google-cloud-dlp 3.40.0 (1 October 2026)	Verified fact	A6-S083
Licence	Proprietary managed service (client libraries Apache-2.0)	Verified fact	A6-S066, A6-S083
Strategic direction	Positioned for AI workloads: de-identify training/tuning data and protect generative AI prompts and responses at run time; underpins Model Armor’s sensitive-data screening	Verified fact	A6-S066, A6-S067
What it does	Discovery and profiling of sensitive data across BigQuery, Cloud SQL, Cloud Storage and Agent Platform; inspection with 200+ predefined detectors; de-identification by masking, tokenisation, bucketing; content methods for any data source (hybrid jobs).	Verified fact	A6-S066, A6-S065
Stack position	DLP service used in ingestion pipelines and at run time on prompts/responses (directly or through Model Armor)	Verified fact	A6-S066, A6-S067

¹²³⁴ Verified fact — A6-S066: Sensitive Data Protection product page (Google Cloud) <https://cloud.google.com/security/products/sensitive-data-protection> accessed 2026-10-07

¹²³⁵ Verified fact — A6-S066: Sensitive Data Protection product page (Google Cloud) <https://cloud.google.com/security/products/sensitive-data-protection> accessed 2026-10-07

Field	Value	Label	Sources
Integration	DLP API and client libraries; built-in integrations with Google Cloud storage services; hybrid jobs for external sources	Verified fact	A6-S066, A6-S065
Dependencies	Not publicly verified	Not publicly verified	
Certifications	Listed in Google Cloud SOC 2 and ISO/IEC 27001 scope	Verified fact	A6-S070, A6-S071
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Content/hybrid inspection: first 1 GiB/month free, then US\$3.00/GiB to 1 TiB and US\$2.00/GiB above; storage inspection from US\$1.00/GiB tiered to US\$0.60/GiB; discovery in consumption mode (bytes profiled) or subscription at US\$2,500 per subscription unit/month	Verified fact	A6-S065
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
Maturity	Long-established service (client library first released February 2018)	Verified fact	A6-S083
Deployment	saas: Yes; managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: No; on_prem: Not publicly verified		

Microsoft Purview Data Security Posture Management (new unified DSPM, GA May 2026), absorbing the earlier 'DSPM for AI' experience (C3-microsoft-purview-dspm-ai)

Tier: Strategic · **Flags:** Renamed · **Original graphic label:** not in graphic

Rationale: Strategic, conditional: where Azure and Microsoft 365 are your primary cloud and E5 or the Purview Suite is already licensed (CP3 Q2, rubric rule 10). Microsoft's lead AI data-security posture service, for posture and Copilot evidence only: it is not the prompt-path DLP control for custom agents ^{AJ}.

Criterion	Score	Rationale
Technical	3	Posture, discovery and risk assessment of AI interactions; no verified inline transformation for custom apps.
Enterprise readiness	4	Hyperscaler presumption (CP2 Q1): platform controls presumed (CP2 Q1); confirm per service.
Security and compliance	4	Purview portal ISO 27001 and Purview FedRAMP High; DSPM not named and commercial SOC 2 scope does not list the portal; scope rule applied (4).
Deployment flexibility	2	Microsoft cloud service only; GCC coverage reduced.
Ecosystem	4	Microsoft 365, Entra-registered apps, ChatGPT Enterprise and Foundry; partner sources preview.
Reliability and maturity	3	Unified DSPM GA May 2026 after consolidation; preview components remain.
Cost / TCO	2	Requires E5 or Purview Suite; licensing for Business Premium conflicting.
Lock-in / portability	2	Microsoft-specific posture model and labels.
Total (generic / FS)	3.10	

Criterion	Score	Rationale
	/	
	3.05	

Capabilities. Unified Purview Data Security Posture Management (GA May 2026) absorbing DSPM for AI: discovers and assesses data-security risk in AI use, including prompts and responses for Microsoft 365 Copilot, Entra-registered AI apps and ChatGPT Enterprise; covers Foundry workloads; partner sources and the posture agent in preview ^{VF 1236}.

Strengths

- Posture over the AI apps employees already use, inside the Microsoft compliance estate ^{AJ}
- Purview portal in ISO 27001 scope; Purview in FedRAMP High scope ^{VF 1237}

Limitations and risks

- Posture and governance, not an inline tokenisation engine for custom agents ^{AJ}
- Requires Microsoft 365 E5 or Purview Suite; Business Premium coverage conflicting ^{VF 1238} ^{ij 1239}
- Partner data sources and the posture agent still preview ^{VF 1240}
- Feature list of the new DSPM beyond AI coverage not verified ^{NPV}

Choose when

- Microsoft 365 Copilot or ChatGPT Enterprise is in use and E5/Purview Suite is licensed ^{AJ}

Avoid when

- You need run-time masking in a custom agent's prompt path ^{AJ}
- You are not a Microsoft estate ^{AJ}

Nearest competitors: C3-google-sdp, C3-protegrity, C3-skyflow

Regulated-FS note. Use for SaaS AI and Copilot posture evidence; do not count it as the prompt-path DLP control for custom agents; platform controls presumed (CP2 Q1); confirm per service ^{REC}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Microsoft	Verifie	A6-S056

¹²³⁶ Verified fact — A6-S090: What's new in Microsoft Purview <https://learn.microsoft.com/en-us/purview/whats-new> accessed 2026-10-07 [search-tool extract]; A6-S091: Considerations for deploying Microsoft Purview DSPM for AI; Get started with DSPM (classic); Purview for Microsoft Foundry <https://learn.microsoft.com/en-us/purview/dspm-for-ai-considerations> accessed 2026-10-07 [search-tool extract]; V2-S036: What's new in Microsoft Purview (with DSPM task mapping page; Tech Community 'Beyond Visibility: the new Microsoft Purview DSPM experience') <https://learn.microsoft.com/en-us/purview/whats-new> accessed 2026-10-08 [search-tool extract]

¹²³⁷ Verified fact — B-C4-S008: ISO/IEC 27001 and SOC 2 Type 2 offering pages (Microsoft Compliance) <https://learn.microsoft.com/en-us/compliance/regulatory/offering-iso-27001> accessed 2026-10-08 [search-tool extract]; A6-S056: Azure services in FedRAMP and DoD audit scope (Microsoft Learn source) <https://raw.githubusercontent.com/MicrosoftDocs/azure-docs/main/articles/azure-government/compliance/azure-services-in-fedramp-auditscope.md> accessed 2026-10-07

¹²³⁸ Verified fact — A6-S091: Considerations for deploying Microsoft Purview DSPM for AI; Get started with DSPM (classic); Purview for Microsoft Foundry <https://learn.microsoft.com/en-us/purview/dspm-for-ai-considerations> accessed 2026-10-07 [search-tool extract]

¹²³⁹ Reported — A6-S092: DSPM for AI with Business Premium Add-On - Microsoft Q&A (Microsoft Q&A (community forum)) <https://learn.microsoft.com/en-us/answers/questions/5813926/dspm-for-ai-with-business-premium-add-on> accessed 2026-10-07 [search-tool extract]

¹²⁴⁰ Verified fact — V2-S036: What's new in Microsoft Purview (with DSPM task mapping page; Tech Community 'Beyond Visibility: the new Microsoft Purview DSPM experience') <https://learn.microsoft.com/en-us/purview/whats-new> accessed 2026-10-08 [search-tool extract]

Field	Value	Label	Sources
		d fact	
Category	Data security posture management for AI and data estates within the Microsoft Purview suite	Verified fact	A6-S090, A6-S091
Version / lineup	New DSPM generally available May 2026; classic DSPM and DSPM for AI remained until June 2026; partner (non-Microsoft) data sources and the Data Security Posture Agent still preview	Verified fact	A6-S090, V2-S036
Licence	Proprietary SaaS; requires Microsoft 365 E5 or Microsoft Purview Suite (formerly Microsoft 365 E5 Compliance)	Verified fact	A6-S091
Status events	May-June 2026: DSPM for AI consolidated into the new unified Purview DSPM (GA May 2026)	Verified fact	A6-S090, V2-S036
Strategic direction	Not publicly verified	Not publicly verified	
What it does	Discovers and assesses data-security risk in AI use, including interactions, prompts and responses for Microsoft 365 Copilot, Entra-registered AI apps and ChatGPT Enterprise (E5 required for prompt/response coverage); Purview also covers Microsoft Foundry workloads	Verified fact	A6-S091
Stack position	Not publicly verified	Not publicly verified	
Integration	Not publicly verified	Not publicly verified	
Dependencies	Not publicly verified	Not publicly verified	
Certifications	Microsoft Purview (incl. Data Map, Data Estate Insights and governance portal) in Azure FedRAMP High and DoD IL2 scope	Verified fact	A6-S056
GDPR / residency	In US Government GCC only Microsoft 365 Copilot and supported AI sites are available in DSPM for AI	Verified fact	A6-S091
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
Enterprise support	Not publicly verified	verified Not publicly verified	
Pricing	Included in Microsoft 365 E5 / Purview Suite; community answers state the Purview Suite for Business Premium add-on (US\$10/user/month) includes DSPM for AI - conflicting, unverified by an official licensing matrix	Reported	A6-S091, A6-S092
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Agent 365 is available as an add-on to Microsoft E5/A5/Business Premium or Microsoft Defender Suite + Microsoft Purview Suite	Verified fact	A6-S059
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Not publicly verified	Not publicly verified	
Deployment	saas: Yes (Microsoft cloud service); managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Not publicly verified; on_prem: Not publicly verified		

Skyflow (Data Privacy Vault; Detect) (c3-skyflow)

Tier: Tactical · **Flags:** none · **Original graphic label:** not in graphic

Rationale: Purpose-built for reversible LLM tokenisation, but a vendor-held vault is a lock-in and outsourcing decision, not a default ^{AJ}.

Criterion	Score	Rationale
Technical	4	Vault tokenisation, de-identification and authorised rehydration across the AI flow.
Enterprise readiness	3	CP2 Q2: role-scoped credentials (RBAC) verified; SSO and audit NPV; max 3.

Criterion	Score	Rationale
Security and compliance	4	ISO 27001:2022, SOC 2 Type II and PCI DSS L1 listed on the trust page; no CMK/ISO 42001/FedRAMP verified; reports not public.
Deployment flexibility	2	Hosted vaults in many regions; self-host not verified.
Ecosystem	3	REST APIs and SDKs; mainstream.
Reliability and maturity	3	SDK since October 2021; v1 EOL forces migration; funding last verified 2024.
Cost / TCO	2	Pricing not published.
Lock-in / portability	2	Vendor-held vault and token map; exit requires bulk detokenisation.
Total (generic / FS)	3.05 / 3.00	

Evidence rules applied: enterprise_readiness: CP2 Q2 partial evidence (RBAC only verified), max 3

Capabilities. Data privacy vault: stores sensitive data and returns tokens; detokenises under role-scoped credentials; Detect APIs de-identify text and files; LLM Privacy Vault tokenises or masks data before models, prompts, RAG, tools, traces and agent workflows with authorised rehydration; EU vaults ^{vf-1241}.

Strengths

- Built around reversible tokenisation of LLM traffic ^{vf-1242}
- Residency by vault region, including EU vaults ^{vf-1243}
- Security page lists ISO 27001:2022, SOC 2 Type II and PCI DSS Level 1 ^{vf-1244}

Limitations and risks

- Vendor holds the sensitive values and sits on the critical path of re-identification ^{AI}
- Self-hosting, SSO, audit logs and pricing not verified ^{NPV}
- SDK v1 end of life 31 October 2026 ^{vf-1245}
- No funding round verified after March 2024 ^{vf-1246}
- Certification reports not public ^{vf-1247}

¹²⁴¹ Verified fact — A6-S081: skyflow - PyPI JSON metadata (Python Package Index) <https://pypi.org/pypi/skyflow/json> accessed 2026-10-07; A6-S095: Skyflow GDPR Compliance; Data Privacy Vault; LLM Privacy Vault; SOC 2 solution pages; Why Skyflow (docs) <https://www.skyflow.com/gdpr-compliance> accessed 2026-10-07 [search-tool extract]

¹²⁴² Verified fact — A6-S095: Skyflow GDPR Compliance; Data Privacy Vault; LLM Privacy Vault; SOC 2 solution pages; Why Skyflow (docs) <https://www.skyflow.com/gdpr-compliance> accessed 2026-10-07 [search-tool extract]

¹²⁴³ Verified fact — A6-S095: Skyflow GDPR Compliance; Data Privacy Vault; LLM Privacy Vault; SOC 2 solution pages; Why Skyflow (docs) <https://www.skyflow.com/gdpr-compliance> accessed 2026-10-07 [search-tool extract]

¹²⁴⁴ Verified fact — B-C3-S005: Skyflow Security (trust page) and certification posts <https://www.skyflow.com/security> accessed 2026-10-08 [search-tool extract]

¹²⁴⁵ Verified fact — A6-S081: skyflow - PyPI JSON metadata (Python Package Index) <https://pypi.org/pypi/skyflow/json> accessed 2026-10-07

¹²⁴⁶ Verified fact — A6-S096: Skyflow Raises \$30 Million led by Khosla Ventures for Data Privacy in AI and Global Data Residency (Business Wire press release) (Skyflow (via Business Wire)) [https://www.businesswire.com/news/home/20240328765469/en/Skyflow-Raises-\\$30-Million-led-by-Khosla-Ventures-for-Data-Privacy-in-AI-and-Global-Data-Residency](https://www.businesswire.com/news/home/20240328765469/en/Skyflow-Raises-$30-Million-led-by-Khosla-Ventures-for-Data-Privacy-in-AI-and-Global-Data-Residency) accessed 2026-10-07 [search-tool extract]

¹²⁴⁷ Verified fact — B-C3-S005: Skyflow Security (trust page) and certification posts <https://www.skyflow.com/security> accessed 2026-10-08 [search-tool extract]

Choose when

- You need reversible pseudonymisation across many applications with per-region vaults ^{AJ}

Avoid when

- Policy requires identifiers to stay in the firm's estate ^{AJ}
- You cannot test bulk detokenisation for exit ^{AJ}

Nearest competitors: C3-protegrity, C3-google-sdp, firm-built token service

Regulated-FS note. Migrate off SDK v1 before 31 October 2026; test bulk detokenisation as the exit plan; assess financial resilience as a material outsourcing ^{Rec}.

Facts (as of 2026-10-07)

Field	Value	Label	Sources
Company	Skyflow	Verified fact	A6-S081
Category	Data privacy vault with tokenisation and de-identification (Detect) APIs	Verified fact	A6-S081
Version / lineup	Python SDK skyflow 2.1.3 (4 August 2026); SDK v1 in maintenance, end of life 31 October 2026	Verified fact	A6-S081
Licence	Proprietary SaaS; SDK published on PyPI	Verified fact	A6-S081
Strategic direction	LLM Privacy Vault: tokenises or masks sensitive data before it reaches models, prompts, RAG, tools, traces and agent workflows, with authorised rehydration; Runtime Data Control for Enterprise AI Search	Verified fact	A6-S095
What it does	Stores sensitive data in vaults and returns tokens; detokenises under role-scoped credentials; Detect APIs de-identify text and files (redaction types) for use in AI and analytics workflows.	Verified fact	A6-S081
Stack position	Vault/tokenisation service outside the AI stack; de-identification before prompts or ingestion	Verified fact	A6-S081
Integration	REST APIs and SDKs; vault URLs per cluster (https://{cluster_id}.vault.skyflowapis.com); environments PROD, SANDBOX, DEV, STAGE	Verified fact	A6-S081
Dependencies	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
Certifications	Vendor states compliance with ISO 27001, SOC 2 Type 1 and Type 2, PCI DSS Level 1, GDPR and HIPAA (vendor claim; reports not seen)	Verified fact	A6-S095
GDPR / residency	EU data privacy vaults offered; vault infrastructure 'available in over 100 countries'; regional deployments e.g. AWS Jakarta	Verified fact	A6-S095
Security features	Not publicly verified	Not publicly verified	
Access controls	Bearer tokens scoped to roles; API key and static bearer token authentication	Verified fact	A6-S081
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Not publicly verified	Not publicly verified	
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	Latest verified funding: US\$30m Series B extension led by Khosla Ventures (28 March 2024); total equity about US\$100m per Crunchbase (via Finovate); named customers GoodRx, Lenovo, Hippocratic AI	Verified fact	A6-S096
Maturity	SDK first released October 2021	Verified fact	A6-S081
Deployment	saas: Yes (hosted vaults); managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Not publicly verified; on_prem: Not publicly verified		

Protegrity (data protection platform; 'Protegrity AI Developer Edition') (c3-protegrity)

Tier: Tactical · **Flags:** none · **Original graphic label:** not in graphic

Rationale: Strong tokenisation and audit model for existing customers; AI editions not yet GA and certification currency unverified ^{AI}.

Criterion	Score	Rationale
Technical	4	Discovery, tokenisation, masking, reversible protect/unprotect and semantic guardrails.
Enterprise readiness	3	CP2 Q2: RBAC and audit verified for ESA (B-C3-S004); SSO NPV; max 3.
Security and compliance	2	Anchor-based 2: ISO 27001:2013 announced 2023 with renewal unverified; no SOC 2 found.
Deployment flexibility	3	Customer-deployed core platform and containerised Developer Edition; AI Team Edition AWS-only.
Ecosystem	3	Python module, APIs, protectors and gateway; mainstream rather than broad.
Reliability and maturity	3	Established core platform; AI Team Edition Tech Preview.
Cost / TCO	2	Pricing not published; enterprise licensing.
Lock-in / portability	2	Proprietary policies and tokens; reversal depends on Protegrity.
Total (generic / FS)	2.90 / 2.75	

Evidence rules applied: security_compliance 2 by anchor (certificate currency unverified, no SOC 2 found); not an NPV cap; enterprise_readiness: CP2 Q2 partial evidence (RBAC and audit verified, SSO NPV), max 3

Capabilities. Enterprise data-protection platform: Data Discovery for PII in unstructured text; Find and Redact, Protect and Unprotect under Protegrity protection policies (tokenisation, masking); Semantic Guardrail API; ESA central policy with role-per-data-element permissions and audit ^{vf1248}. AI Team Edition (17 November 2025) still Tech Preview, AWS-only ^{vf1249}.

Strengths

- Role-per-data-element Unprotect rights and audit at every protection point ^{vf1250}
- Same tokens and policies can span structured data and the AI path ^{AJ}

Limitations and risks

- AI Team Edition documented as Tech Preview; AWS-only ^{vf1251}
- Only certification verified is ISO 27001:2013 (2023); no SOC 2 report found ^{vf1252}
- SSO not verified; pricing not published ^{NPV}
- Tokenised data depends on Protegrity policies to reverse ^{AJ}

¹²⁴⁸ Verified fact — A6-S082: protegrity-developer-python - PyPI JSON metadata (Python Package Index) <https://pypi.org/pypi/protegrity-developer-python/json> accessed 2026-10-07; B-C3-S004: Policy Management and Enterprise Security Administrator (ESA) documentation 10.1 (Protegrity) https://docs.protegrity.com/10.1/docs/pmg/_print/ accessed 2026-10-08 [search-tool extract]

¹²⁴⁹ Verified fact — A6-S093: Protegrity AI Team Edition to Secure Agentic Workflows; AI Team Edition docs (Tech Preview); Protegrity Expands Team (April 2026) <https://www.protegrity.com/news/protegrity-ai-team-edition-to-secure-agentic-workflows> accessed 2026-10-07 [search-tool extract]

¹²⁵⁰ Verified fact — B-C3-S004: Policy Management and Enterprise Security Administrator (ESA) documentation 10.1 (Protegrity) https://docs.protegrity.com/10.1/docs/pmg/_print/ accessed 2026-10-08 [search-tool extract]

¹²⁵¹ Verified fact — A6-S093: Protegrity AI Team Edition to Secure Agentic Workflows; AI Team Edition docs (Tech Preview); Protegrity Expands Team (April 2026) <https://www.protegrity.com/news/protegrity-ai-team-edition-to-secure-agentic-workflows> accessed 2026-10-07 [search-tool extract]

¹²⁵² Verified fact — A6-S094: Protegrity Awarded ISO 27001:2013 Certification <https://www.protegrity.com/news/protegrity-awarded-iso-270012013-certification> accessed 2026-10-07 [search-tool extract]; B-C3-S003: Protegrity compliance pages (cheat sheet; news; AI Enterprise Edition) <https://www.protegrity.com/blog/compliance-cheat-sheet-gdpr-vs-cpra-vs-hipaa-vs-soc/> accessed 2026-10-08 [search-tool extract]

Choose when

- Protegrity already tokenises your structured data ^{AJ}

Avoid when

- You are not on AWS for the AI editions ^{AJ}
- A current SOC 2 Type II report is a procurement gate ^{AJ}

Nearest competitors: C3-skyflow, C3-google-sdp, C3-presidio

Regulated-FS note. Request the current ISO certificate and any SOC 2 report; treat AI Team Edition as a pilot until GA ^{Rec}.

Facts (as of 2026-10-07)

Field	Value	Label	Sources
Company	Protegrity	Verified fact	A6-S082
Category	Enterprise data protection: discovery, tokenisation, masking, plus semantic guardrails for GenAI	Verified fact	A6-S082
Version / lineup	Protegrity AI Team Edition launched 17 November 2025 (docs still say Tech Preview, not GA; April 2026 release says 'available now'); AI Enterprise Edition; protegrity-developer-python 1.1.1 (16 December 2025)	Verified fact	A6-S093, A6-S082
Licence	Proprietary platform; AI Developer Edition Python module MIT-licensed	Verified fact	A6-S082
Status events	17 November 2025: AI Team Edition launched for agentic workflows; April 2026: team expansion announcement (no funding or acquisition disclosed)	Verified fact	A6-S093
Strategic direction	Targets GenAI: protect sensitive data in prompts, outputs and training data; Semantic Guardrail API scans conversations for PII and risk	Verified fact	A6-S082
What it does	Data Discovery classifies PII in unstructured text; Find and Redact / Protect / Unprotect using Protegrity protection policies (tokenisation, masking); Application Protector for structured protect/unprotect; Semantic Guardrail scanning.	Verified fact	A6-S082
Stack position	DLP/tokenisation layer applied in data pipelines and on prompts/responses	Verified fact	A6-S082
Integration	Python module and APIs; containerised Developer Edition sandbox	Verified fact	A6-S082
Dependencies	AI Team Edition deployment currently AWS-specific	Verified fact	A6-S093

Field	Value	Label	Sources
Certifications	ISO 27001:2013 certification for its ISMS announced August 2023 (current renewal/2022 transition not verified); no Protegrity SOC 2 report found - SOC 2 references describe customer audit support	Verified fact	A6-S094, A6-S093
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Not publicly verified	Not publicly verified	
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
Deployment	saas: Not publicly verified; managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes (containerised Developer Edition); on_prem: Not publicly verified		

C4: Identity and access for agents

Open Policy Agent (OPA) (c4-opa)

Tier: Strategic • **Flags:** none • **Original graphic label:** not in graphic

Rationale: The neutral, mature policy decision point for agent tool governance across clouds ^{AJ}.

Criterion	Score	Rationale
Technical	4	General ABAC/RBAC policy engine; agent semantics are the firm's to model.
Enterprise readiness	3	Rule 2: enables central policy decisions in-estate; commercial support not verified.
Security and compliance	4	Rule 2 hygiene: security policy (B-C4-S001), CNCF graduated, Apache-2.0.
Deployment flexibility	5	Library, sidecar or daemon; self-hosted and on-premises.
Ecosystem	5	Used across services, APIs, Kubernetes and infrastructure.
Reliability and maturity	4	Graduated since 2021 with regular releases; maintainer funding not verified.
Cost / TCO	4	Free; Rego skills and policy testing are the cost.
Lock-in / portability	5	Open source, Apache-2.0, CNCF governance.
Total (generic / FS)	4.15 / 4.20	

Evidence rules applied: Rule 2 (self-hosted software / open specification): enterprise_readiness and security_compliance capped at 4; NPV cap not applied; inherits host controls

Capabilities. Open Policy Agent: general-purpose policy engine evaluating Rego policies against JSON input for allow/deny or richer decisions, as library or sidecar/daemon, across services, APIs, Kubernetes and infrastructure; v1.21.1 (29 September 2026), Apache-2.0, CNCF graduated (February 2021) ^{VF1253}.

Strengths

- One policy engine for gateway, tool server and infrastructure decisions ^{VF1254}

¹²⁵³ Verified fact — A6-S046: Open Policy Agent README (GitHub) (Open Policy Agent project (CNCF)) <https://raw.githubusercontent.com/open-policy-agent/opa/main/README.md> accessed 2026-10-07; A6-S048: Go module proxy: github.com/open-policy-agent/opa @latest (Go module proxy (proxy.golang.org)) <https://proxy.golang.org/github.com/open-policy-agent/opa/@latest> accessed 2026-10-07; A6-S088: Open Policy Agent LICENSE (Open Policy Agent project) <https://raw.githubusercontent.com/open-policy-agent/opa/main/LICENSE> accessed 2026-10-07

¹²⁵⁴ Verified fact — A6-S046: Open Policy Agent README (GitHub) (Open Policy Agent project (CNCF)) <https://raw.githubusercontent.com/open-policy-agent/opa/main/README.md> accessed 2026-10-07

- CNCF graduated with a published security policy ^{VF1255}
- Policies are code that can be unit-tested and versioned ^{AJ}

Limitations and risks

- No agent-specific features verified ^{NPV}
- Reported 2025 move of the original maintainers to Apple not verified ^{NPV}
- Rego is a learning curve ^{AJ}

Choose when

- You want one vendor-neutral PDP across clouds and runtimes ^{AJ}

Avoid when

- Your agent estate is AWS AgentCore and you want managed, automated-reasoning-checked policies ^{AJ}

Nearest competitors: C4-cedar, Gateway-native policy (Kong, AgentCore Policy)

Regulated-FS note. Evaluate every tool call at the gateway with principal, agent, tool and arguments as input; keep policies and their tests in Git under change control ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Open Policy Agent project, CNCF graduated (graduated February 2021)	Verified fact	A6-S046
Category	Open-source general-purpose policy engine (policy-as-code, Rego)	Verified fact	A6-S046
Version / lineup	v1.21.1 (29 September 2026)	Verified fact	A6-S048, A6-S047, V2-S061
Licence	Open source (Apache-2.0)	Verified fact	A6-S088
Strategic direction	Not publicly verified	Not publicly verified	
What it does	Evaluates context-aware policies written in Rego against JSON input to return allow/deny or richer decisions, enabling unified policy enforcement across services, APIs, Kubernetes and infrastructure.	Verified fact	A6-S046
Stack position	Policy decision point used by gateways, agent runtimes and tool servers	Verified	A6-S046

¹²⁵⁵ Verified fact — A6-S046: Open Policy Agent README (GitHub) (Open Policy Agent project (CNCF)) <https://raw.githubusercontent.com/open-policy-agent/opa/main/README.md> accessed 2026-10-07; B-C4-S001: OPA Security Policy (SECURITY.md) (Open Policy Agent project (CNCF)) <https://raw.githubusercontent.com/open-policy-agent/opa/main/SECURITY.md> accessed 2026-10-08

Field	Value	Label	Sources
	for ABAC/RBAC decisions	d fact	
Integration	Rego policies; library or sidecar/daemon; VS Code extension and Rego Playground	Verified fact	A6-S046
Dependencies	Not publicly verified	Not publicly verified	
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Free open source	Verified fact	A6-S088
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
		verified	
Maturity	CNCF graduated; 1.x line with regular releases (1.21.0 and 1.21.1 in 2026)	Verified fact	A6-S046, A6-S047
Deployment	saas: Not publicly verified; managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes; on_prem: Yes		

SPIFFE (specification) and SPIRE (SPIFFE Runtime Environment) (c4-spiffe-spire)

Tier: Strategic · **Flags:** none · **Original graphic label:** not in graphic

Rationale: The neutral workload-identity layer beneath agents and tools; mature, open and portable ^{AJ}.

Criterion	Score	Rationale
Technical	3	Workload identity and attestation; one core C4 function, not delegation or authorisation.
Enterprise readiness	3	Rule 2: enables estate-wide workload identity; commercial distributions not assessed.
Security and compliance	4	Rule 2 hygiene: security policy with supported versions (B-C4-S002), third-party audit, CNCF assessments.
Deployment flexibility	5	Self-hosted anywhere including on-premises.
Ecosystem	4	Envoy SDS, Kubernetes and cloud node attestors.
Reliability and maturity	5	CNCF graduated with regular releases.
Cost / TCO	3	Free, but server and attestor operations are material.
Lock-in / portability	5	Open standard, Apache-2.0, CNCF governance.
Total (generic / FS)	3.85 / 4.05	

Evidence rules applied: Rule 2 (self-hosted software / open specification): enterprise_readiness and security_compliance capped at 4; NPV cap not applied; inherits host controls

Capabilities. SPIFFE specification and SPIRE runtime: attests running workloads and issues SPIFFE IDs and X.509/JWT SVIDs through the Workload API for mTLS or signed JWTs; Envoy SDS; SPIRE v1.15.3 (21 August 2026), Apache-2.0, CNCF graduated ^{W1256}.

Strengths

- Removes static secrets for workload-to-workload trust ^{AJ}

¹²⁵⁶ Verified fact — A6-S087: SPIRE README (GitHub) (SPIFFE project (CNCF)) <https://raw.githubusercontent.com/spiffe/spire/main/README.md> accessed 2026-10-07; A6-S042: SPIRE CHANGELOG.md (SPIFFE project (CNCF)) <https://raw.githubusercontent.com/spiffe/spire/main/CHANGELOG.md> accessed 2026-10-07; A6-S043: Go module proxy: github.com/spiffe/spire @latest (Go module proxy (proxy.golang.org)) <https://proxy.golang.org/github.com/spiffe/spire/@latest> accessed 2026-10-07; A6-S089: SPIRE LICENSE (SPIFFE project) <https://raw.githubusercontent.com/spiffe/spire/main/LICENSE> accessed 2026-10-07

- CNCF graduated; Cure53 audit (2021) and CNCF security assessments ^{VF-1257}
- Security fixes for current and previous minor release series ^{VF-1258}

Limitations and risks

- Machine identity only; carries no delegated user authority ^{VF-1259}
- No agent-specific SPIFFE profile verified ^{NPV}
- Operating SPIRE servers and attestors is a platform-team commitment ^{AJ}

Choose when

- Agents, gateways and MCP servers run on Kubernetes or mixed infrastructure and need workload identity without long-lived secrets ^{AJ}

Avoid when

- A cloud-native workload identity already covers every runtime ^{AJ}

Nearest competitors: Cloud workload identity federation (Entra, AWS IAM), C7-hashicorp-vault

Regulated-FS note. Use SVIDs for agent-runtime-to-gateway and gateway-to-tool mTLS; never as a substitute for the user's delegated token ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	SPIFFE project, CNCF graduated	Verified fact	A6-S087
Category	Open specification and open-source implementation for workload identity	Verified fact	A6-S087
Version / lineup	SPIRE v1.15.3 (21 August 2026); 1.16.0 unreleased	Verified fact	A6-S043, A6-S042, V2-S061
Licence	Open source (Apache-2.0)	Verified fact	A6-S089
Strategic direction	Recent SPIRE releases add Slurm workload attestor, Azure Blob trust-bundle publishing, nested-agent bootstrap and Kubernetes attestation options	Verified fact	A6-S042
What it does	SPIRE exposes the SPIFFE Workload API, attests running software and issues SPIFFE IDs and SVIDs to it, so workloads can establish mutual trust (e.g. mTLS or signed JWTs) and authenticate to secret stores, databases or cloud services; also implements Envoy SDS.	Verified fact	A6-S087
Stack position	Workload identity layer beneath agent runtimes, gateways and MCP servers (machine identity rather than delegated user authority)	Verified fact	A6-S087
Integration	SPIFFE Workload API and SDS API; X.509 and JWT SVIDs	Verified fact	A6-S042

¹²⁵⁷ Verified fact — A6-S087: SPIRE README (GitHub) (SPIFFE project (CNCF)) <https://raw.githubusercontent.com/spiffe/spire/main/README.md> accessed 2026-10-07

¹²⁵⁸ Verified fact — B-C4-S002: SPIRE Security Policy (SECURITY.md) (SPIFFE project (CNCF)) <https://raw.githubusercontent.com/spiffe/spire/main/SECURITY.md> accessed 2026-10-08

¹²⁵⁹ Verified fact — A6-S087: SPIRE README (GitHub) (SPIFFE project (CNCF)) <https://raw.githubusercontent.com/spiffe/spire/main/README.md> accessed 2026-10-07

Field	Value	Label	Sources
Dependencies	SPIRE server and agents; node attestors (e.g. Kubernetes PSAT, AWS IID)	Verified fact	A6-S042
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Third-party security audit by Cure53 (February 2021); CNCF security assessments 2018 and 2020	Verified fact	A6-S087
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Free open source	Verified fact	A6-S089
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	CNCF graduated; regular releases (1.15.1 May, 1.15.2 July, 1.15.3 August 2026)	Verified fact	A6-S087, A6-S042
Deployment	saas: Not publicly verified; managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes; on_prem: Yes (self-managed software)		

Cedar policy language; Amazon Verified Permissions (managed Cedar); Policy in Amazon Bedrock AgentCore (Cedar-based) (c4-cedar)

Tier: Strategic · **Flags:** none · **Original graphic label:** not in graphic

Rationale: Strategic, conditional: where AWS is your primary cloud and agents run on AgentCore Gateway (CP3 Q2, rubric rule 10). AgentCore Policy is AWS's lead managed policy service for agent tool calls, and its enforcement point (AgentCore Gateway) is now Strategic on the same condition in C1 and L4. OPA remains the cloud-neutral default; keep Cedar source in Git so policies stay portable ^{AI}.

Criterion	Score	Rationale
Technical	4	Strong tool-call authorisation with partial evaluation and automated reasoning; no external lookups.
Enterprise readiness	4	Hyperscaler presumption (CP2 Q1) for AgentCore Policy and Verified Permissions: platform controls presumed (CP2 Q1); confirm per service; Cedar library under rule 2.
Security and compliance	4	Cedar library hygiene verified (B-C4-S003); Amazon Bedrock AgentCore listed in AWS SOC 1/2/3 scope (GA features in scope unless excluded) and in ISO/IEC 27001 programmes ^{vf1260} ; held at 4, not 5, for consistency with C1 and L4 (ISO wording partly garbled, FedRAMP unresolved).
Deployment flexibility	4	Cedar library anywhere; AgentCore Policy in 13 AWS regions incl. Ireland and GovCloud.
Ecosystem	3	AWS services and CNCF (per AWS); narrower than OPA.
Reliability and maturity	3	Cedar 4.x regular releases; AgentCore Policy GA March 2026; Dogwood new.

¹²⁶⁰ Verified fact — B-C1-S005: AWS Services in Scope by Compliance Program - SOC (Amazon Web Services) <https://aws.amazon.com/compliance/services-in-scope/SOC/> accessed 2026-10-08 [search-tool extract]; B-C1-S006: Amazon Bedrock AgentCore FAQs (Amazon Web Services) <https://aws.amazon.com/bedrock/agentcore/faqs/> accessed 2026-10-08 [search-tool extract]; B-L4-S002: Compliance validation for Amazon Bedrock AgentCore (Amazon Web Services) <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/compliance-validation.html> accessed 2026-10-08 [search-tool extract]

Criterion	Score	Rationale
Cost / TCO	4	Cedar free; managed pricing transparent.
Lock-in / portability	3	Cedar open source, but enforcement tied to AgentCore Gateway and authoring moving to Dogwood.
Total (generic / FS)	3.75 / 3.70	

Evidence rules applied: Rule 2 (self-hosted software / open specification): enterprise_readiness and security_compliance capped at 4; NPV cap not applied; inherits host controls (Cedar library part); CP3 review: AgentCore Policy security re-based on the AgentCore SOC and ISO evidence already logged by C1 and L4 (B-C1-S005, B-C1-S006, B-L4-S002); 3 to 4

Capabilities. Cedar policy language and engine (Apache-2.0; 4.13.0, 15 September 2026) for RBAC/ABAC with schema validation and automated-reasoning analysis; Amazon Verified Permissions (managed Cedar); Policy in Amazon Bedrock AgentCore (GA 3 March 2026, 13 Regions): deny-by-default evaluation of every gateway tool call on identity claims and arguments, denied tools filtered from tools/list, natural-language authoring, decisions logged to CloudWatch; policies now authored in Dogwood, an open-source Cedar superset ^{vf1261}.

Strengths

- Deny-by-default tool-call authorisation with partial evaluation of tools/list ^{vf1262}
- Designed for automated reasoning over policy sets ^{vf1263}
- Transparent per-request pricing: US\$0.000025 per authorisation request ^{vf1264}
- Security policy with 1-business-day acknowledgement and embargoed advisories ^{vf1265}

Limitations and risks

- Cedar cannot do external lookups at evaluation time; AWS recommends Lambda interceptors ^{vf1266}
- AgentCore Policy enforcement is tied to AgentCore Gateway ^{vf1267}

¹²⁶¹ Verified fact — A6-S044: Cedar CHANGELOG.md (cedar-policy crate) (Cedar project)

<https://raw.githubusercontent.com/cedar-policy/cedar/main/cedar-policy/CHANGELOG.md> accessed 2026-10-07; A6-S045: Cedar README (GitHub) (Cedar project) <https://raw.githubusercontent.com/cedar-policy/cedar/main/README.md> accessed 2026-10-07; A6-S026: Policy in Amazon Bedrock AgentCore is now generally available; AgentCore Policy docs; Why Policy in AgentCore chose Cedar (Amazon Web Services) <https://aws.amazon.com/about-aws/whats-new/2026/03/policy-amazon-bedrock-agentcore-generally-available/> accessed 2026-10-07 [search-tool extract]; A6-S104: Amazon Verified Permissions now supports policy store aliases and named policies and policy templates; Cedar 4 FAQ; differences from Cedar (Amazon Web Services) <https://aws.amazon.com/about-aws/whats-new/2026/04/amazon-verified-permissions-policy-store/> accessed 2026-10-07 [search-tool extract]; V2-S033: Policy in Amazon Bedrock AgentCore is now generally available (with AgentCore policy authoring guide and FAQ) (Amazon Web Services) <https://aws.amazon.com/about-aws/whats-new/2026/03/policy-amazon-bedrock-agentcore-generally-available/> accessed 2026-10-08 [search-tool extract]

¹²⁶² Verified fact — A6-S026: Policy in Amazon Bedrock AgentCore is now generally available; AgentCore Policy docs; Why Policy in AgentCore chose Cedar (Amazon Web Services) <https://aws.amazon.com/about-aws/whats-new/2026/03/policy-amazon-bedrock-agentcore-generally-available/> accessed 2026-10-07 [search-tool extract]

¹²⁶³ Verified fact — A6-S045: Cedar README (GitHub) (Cedar project) <https://raw.githubusercontent.com/cedar-policy/cedar/main/README.md> accessed 2026-10-07

¹²⁶⁴ Verified fact — A6-S021: Amazon Bedrock AgentCore Pricing; AgentCore GA announcement (Oct 2025); AgentCore Gateway docs (Amazon Web Services) <https://aws.amazon.com/bedrock/agentcore/pricing/> accessed 2026-10-07 [search-tool extract]

¹²⁶⁵ Verified fact — B-C4-S003: Cedar Security Policy (SECURITY.md) (Cedar project) <https://raw.githubusercontent.com/cedar-policy/cedar/main/SECURITY.md> accessed 2026-10-08

¹²⁶⁶ Verified fact — A6-S026: Policy in Amazon Bedrock AgentCore is now generally available; AgentCore Policy docs; Why Policy in AgentCore chose Cedar (Amazon Web Services) <https://aws.amazon.com/about-aws/whats-new/2026/03/policy-amazon-bedrock-agentcore-generally-available/> accessed 2026-10-07 [search-tool extract]

- Cedar 2 to 4 migration was breaking for Verified Permissions users ^{VF-1268}
- Dogwood governance and licence not checked ^{NIPV}
- AgentCore is in AWS SOC 1/2/3 scope and ISO/IEC 27001 programmes, but the ISO wording is partly garbled and FedRAMP status is unresolved ^{VF-1269}

Choose when

- The agent estate runs on AWS AgentCore Gateway ^{AJ}
- You want analysable policies with a smaller language than Rego ^{AJ}

Avoid when

- You need runtime external data lookups in policy ^{AJ}
- You are multi-cloud and want one PDP (prefer OPA) ^{AJ}

Nearest competitors: C4-opa, Gateway-native policy (Kong)

Regulated-FS note. Review AI-generated policies before activation, keep the Cedar source in Git, and avoid Dogwood-only constructs where portability matters; platform controls presumed (CP2 Q1); confirm per service ^{Rec.}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Cedar project (cedar-policy GitHub organisation; created by AWS); AgentCore Policy by Amazon Web Services	Verified fact	A6-S045, A6-S026
Category	Open-source authorisation policy language and engine; managed agent tool-call policy service (AgentCore Policy)	Verified fact	A6-S045, A6-S026
Version / lineup	cedar-policy 4.13.0 (15 September 2026; Cedar language version 4.5); Policy in Amazon Bedrock AgentCore GA 3 March 2026 in 13 Regions (preview December 2025)	Verified fact	A6-S044, A6-S085, A6-S026, V2-S033
Licence	Cedar: open source (Apache-2.0); AgentCore Policy: proprietary managed service	Verified fact	A6-S045, A6-S026
Status events	3 March 2026: Policy in Amazon Bedrock AgentCore generally available (Cedar-based); 6 April 2026: Amazon Verified Permissions adds policy store aliases and named policies/templates; from April 2026 authorisation APIs disabled for policy stores still on Cedar 2 (Cedar 4 required); May 2026: Verified Permissions aligned with Cedar, supports multiple namespaces; 6 August 2026: temporal policies and rate limiting announced for AgentCore	Verified fact	A6-S026, A6-S104, A6-S106, V2-S033

¹²⁶⁷ Verified fact — A6-S026: Policy in Amazon Bedrock AgentCore is now generally available; AgentCore Policy docs; Why Policy in AgentCore chose Cedar (Amazon Web Services) <https://aws.amazon.com/about-aws/whats-new/2026/03/policy-amazon-bedrock-agentcore-generally-available/> accessed 2026-10-07 [search-tool extract]

¹²⁶⁸ Verified fact — A6-S104: Amazon Verified Permissions now supports policy store aliases and named policies and policy templates; Cedar 4 FAQ; differences from Cedar (Amazon Web Services) <https://aws.amazon.com/about-aws/whats-new/2026/04/amazon-verified-permissions-policy-store/> accessed 2026-10-07 [search-tool extract]

¹²⁶⁹ Verified fact — B-C1-S005: AWS Services in Scope by Compliance Program - SOC (Amazon Web Services) <https://aws.amazon.com/compliance/services-in-scope/SOC/> accessed 2026-10-08 [search-tool extract]; B-C1-S006: Amazon Bedrock AgentCore FAQs (Amazon Web Services) <https://aws.amazon.com/bedrock/agentcore/faqs/> accessed 2026-10-08 [search-tool extract]; B-L4-S002: Compliance validation for Amazon Bedrock AgentCore (Amazon Web Services) <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/compliance-validation.html> accessed 2026-10-08 [search-tool extract]

Field	Value	Label	Sources
Strategic direction	AWS uses Cedar as the deny-by-default authorisation layer for agent tool calls through AgentCore Gateway, with natural-language policy authoring, schema generation from MCP tool descriptions and automated-reasoning analysis of policy sets; current AgentCore documentation has policies authored in Dogwood, described by AWS as an open-source, Cedar-compatible policy language for AI agents (every valid Cedar policy is valid Dogwood); AWS says Cedar has joined the CNCF	Verified fact	A6-S026, A6-S074, V2-S033
What it does	Cedar expresses RBAC and ABAC permissions (principal, action, resource, context) that are validated against a schema and designed for automated-reasoning analysis. AgentCore Policy attaches a Cedar policy engine to a Gateway; every agent-to-tool call is evaluated before execution using identity claims and tool arguments; denied tools are filtered from tools/list via partial evaluation; decisions are logged to CloudWatch.	Verified fact	A6-S045, A6-S026
Stack position	Policy decision point for agent tool calls (inside the gateway) and for application authorisation	Verified fact	A6-S026
Integration	Cedar Rust crate, WASM/TypeScript bindings, language server; AgentCore CreatePolicyEngine/CreatePolicy/StartPolicyGeneration APIs	Verified fact	A6-S045, A6-S085, A6-S074
Dependencies	Cedar: Rust crates (WASM bindings for JavaScript); AgentCore Policy: AgentCore Gateway	Verified fact	A6-S045, A6-S085, A6-S074
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	AgentCore Policy GA in 13 AWS regions incl. Europe (Ireland); also AWS GovCloud (US-West)	Verified fact	A6-S026
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Cedar free. AgentCore Policy: US\$0.000025 per authorisation request (first 100 temporal policies per engine no extra authorisation charge); US\$0.13 per 1,000 input tokens for natural-language policy processing	Verified fact	A6-S021

Field	Value	Label	Sources
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Cedar 4.x with regular releases (4.11.2 June, 4.12.0 July, 4.13.0 September 2026)	Verified fact	A6-S044
Deployment	saas: Yes (AgentCore Policy managed service); managed_cloud: Yes (Amazon Verified Permissions); vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes (Cedar library); on_prem: Not publicly verified		

Auth0 for AI Agents; Okta for AI Agents; Okta Agent SSO (Cross App Access, XAA) (c4-okta-auth0-ai-agents)

Tier: Strategic · **Flags:** none · **Original graphic label:** not in graphic

Rationale: Strategic, conditional: where Okta is the workforce IdP; FS 3.55 is below the usual 3.6 because deployment scores 2 (SaaS only), accepted because agent identities must live in the same directory as the delegating humans ^{AJ}.

Criterion	Score	Rationale
Technical	5	Delegation, token vaulting, async human approval, fine-grained RAG authorisation and enterprise-managed MCP access.
Enterprise readiness	4	CP2 Q2 (rule 7): SSO (Agent SSO, core Okta SSO), a dedicated AI agent administrator role plus agent ownership and certification campaigns (RBAC), and System Log events for AI agents in reference documentation (audit), plus the System Log management API ¹²⁷⁰ ; 4. Not 5: SLA and agent-specific SCIM not verified, and System Log retention is 90 days.
Security and compliance	4	ISO 27001:2022, SOC 2 Type II and FedRAMP at company level; product scope not stated (scope rule, 4).

¹²⁷⁰ Verified fact — B-C4-S007: Okta help: AI agents (add manually; configure; update) and access certifications for AI agents (Okta, Inc.) <https://help.okta.com/OIE/en-us/content/topics/ai-agents/ai-agent-add-manually.htm> accessed 2026-10-08 [search-tool extract]; B-REVB-S001: Okta for AI Agents (help home), AI agent administrators, View Agent Gateway activity, and System Log API docs (Okta, Inc.) <https://help.okta.com/oie/en-us/content/topics/ai-agents/ai-agents-home.htm> accessed 2026-10-08 [search-tool extract]

Criterion	Score	Rationale
Deployment flexibility	2	SaaS only (Okta and Auth0 clouds); private deployment not verified.
Ecosystem	4	XAA adopted as the MCP extension with named SaaS support; SDKs for Python and JavaScript frameworks.
Reliability and maturity	3	Core GA November 2025 to August 2026; SDKs changing fast.
Cost / TCO	3	XAA included in core SSO; Auth0 tiers published; Okta for AI Agents price not published.
Lock-in / portability	3	Proprietary platforms on open standards (OAuth, OIDC, CIBA, ID-JAG).
Total (generic / FS)	3.65 / 3.55	

Evidence rules applied: CP3 review: earlier CP2 Q2 cap (audit only on a training page) lifted; System Log coverage of AI agents and the AI agent administrator role are in Okta reference documentation (B-REVB-S001)

Capabilities. Auth0 for AI Agents (GA 19 November 2025: user authentication, Token Vault, asynchronous authorisation via CIBA, FGA for RAG; Auth for MCP and OBO Token Exchange GA May 2026); Okta for AI Agents (GA 30 April 2026: discover, onboard, protect, govern agent identities); Okta Agent SSO / Cross App Access (GA 24 August 2026), the MCP Enterprise-Managed Authorization extension ^{vf1271}.

Strengths

- Standards-based delegation: RFC 8693 token exchange, RFC 7523, ID-JAG, OpenID CIBA ^{vf1272}
- Agents in Universal Directory with human owners and access-certification campaigns ^{vf1273}
- XAA included in core Okta SSO; out-of-box support from Atlassian, Slack, Figma and others ^{vf1274}
- Company-level ISO 27001:2022, SOC 2 Type II and FedRAMP authorisations ^{vf1275}

Limitations and risks

- Auth0 AI SDKs flagged 'under heavy development'; @auth0/ai at major version 6 within about 13 months ^{vf1276}

¹²⁷¹ Verified fact — A6-S097: Auth0 for AI Agents is Now Generally Available (GA); Auth0 May 2026 product innovations (Auth0 (Okta)) <https://auth0.com/blog/auth0-for-ai-agents-generally-available/> accessed 2026-10-07 [search-tool extract]; A6-S100: Okta for AI Agents is Now Generally Available; Okta announces new blueprint for the secure agentic enterprise (Okta, Inc.) <https://www.okta.com/blog/ai/okta-for-ai-agents-general-availability/> accessed 2026-10-07 [search-tool extract]; A6-S099: Okta brings first-class identity to AI agents with Agent SSO; Okta Introduces Agent SSO for Cross App Access (XAA) (Okta, Inc.) <https://www.okta.com/newsroom/press-releases/okta-brings-first-class-identity-to-ai-agents-with-agent-sso/> accessed 2026-10-07 [search-tool extract]; V2-S035: Okta brings first-class identity to AI agents with Agent SSO (with Okta support article, OIE release notes, Oktane 2026 release) <https://www.okta.com/newsroom/press-releases/okta-brings-first-class-identity-to-ai-agents-with-agent-sso/> accessed 2026-10-08 [search-tool extract]

¹²⁷² Verified fact — A6-S079: MCP ext-auth: Enterprise-Managed Authorization specification (Stable) (Model Context Protocol project) <https://raw.githubusercontent.com/modelcontextprotocol/ext-auth/main/specification/stable/enterprise-managed-authorization.md> accessed 2026-10-07; A6-S076: auth0-ai - PyPI JSON metadata (Python Package Index) <https://pypi.org/pypi/auth0-ai/json> accessed 2026-10-07

¹²⁷³ Verified fact — B-C4-S007: Okta help: AI agents (add manually; configure; update) and access certifications for AI agents (Okta, Inc.) <https://help.okta.com/OIE/en-us/content/topics/ai-agents/ai-agent-add-manually.htm> accessed 2026-10-08 [search-tool extract]

¹²⁷⁴ Verified fact — A6-S099: Okta brings first-class identity to AI agents with Agent SSO; Okta Introduces Agent SSO for Cross App Access (XAA) (Okta, Inc.) <https://www.okta.com/newsroom/press-releases/okta-brings-first-class-identity-to-ai-agents-with-agent-sso/> accessed 2026-10-07 [search-tool extract]

¹²⁷⁵ Verified fact — B-C4-S006: Okta Security Trust Center (with Okta Sustainability Report 2025) (Okta, Inc.) <https://security.okta.com/> accessed 2026-10-08 [search-tool extract]

- Okta Agent Gateway and Shadow AI Agent Discovery announced, not confirmed shipped ^{VF1277}
- System Log records AI agent events and a dedicated AI agent administrator role exists ^{VF1278}; System Log retention is 90 days per Okta support, so export agent events to the firm's SIEM ^{VF1279}
- Okta for AI Agents price not published; EU residency not verified ^{NPV}

Choose when

- Okta is the workforce IdP ^{AJ}
- Customer-facing or SaaS-integrated agents need Token Vault and CIBA approvals ^{AJ}

Avoid when

- Entra is the workforce IdP ^{AJ}
- You need a self-hosted IdP ^{AJ}

Nearest competitors: C4-entra-agent-id, C7-hashicorp-vault, C4-mcp-authorization

Regulated-FS note. Use XAA/ID-JAG for enterprise-managed MCP access, pin SDK versions, and confirm product scope of the SOC 2 and ISO certificates under NDA ^{REC}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Okta, Inc. (Auth0 is an Okta product)	Verified fact	A6-S078, A6-S080
Category	Identity and authorisation for AI agents (delegated access, async approval, fine-grained authorisation, enterprise-managed app-to-app access)	Verified fact	A6-S078, A6-S080
Version / lineup	Auth0 for AI Agents GA 19 November 2025 (User Authentication, Token Vault, Asynchronous Authorization, FGA for RAG); May 2026 additions Auth for MCP and On-Behalf-Of Token Exchange GA; Okta for AI Agents GA 30 April 2026; Agent SSO / XAA GA 24 August 2026; SDKs auth0-ai 1.0.2 and @auth0/ai 6.0.2	Verified fact	A6-S097, A6-S100, A6-S099, A6-S076, A6-S077, V2-S035
Licence	Proprietary identity platforms; AI SDKs Apache-2.0	Verified fact	A6-S076, A6-S077
Status events	19 November 2025: Auth0 for AI Agents GA; 30 April 2026: Okta for AI Agents GA; 24 August 2026: Okta Agent SSO (Cross App Access) GA, included in core Okta SSO	Verified fact	A6-S097, A6-S100, A6-S099, V2-S035
Strategic direction	Cross App Access built on the IETF Identity Assertion Authorization Grant so the enterprise IdP mediates agent-to-app access; MCP	Verified fact	A6-S080, A6-S079

¹²⁷⁶ Verified fact — A6-S078: Auth0 AI for JavaScript README (GitHub) (Auth0 (Okta)) <https://raw.githubusercontent.com/auth0/auth0-ai-js/main/README.md> accessed 2026-10-07; A6-S077: @auth0/ai - npm registry metadata <https://registry.npmjs.org/@auth0/ai> accessed 2026-10-07

¹²⁷⁷ Verified fact — A6-S100: Okta for AI Agents is Now Generally Available; Okta announces new blueprint for the secure agentic enterprise (Okta, Inc.) <https://www.okta.com/blog/ai/okta-for-ai-agents-general-availability/> accessed 2026-10-07 [search-tool extract]

¹²⁷⁸ Verified fact — B-REVB-S001: Okta for AI Agents (help home), AI agent administrators, View Agent Gateway activity, and System Log API docs (Okta, Inc.) <https://help.okta.com/oie/en-us/content/topics/ai-agents/ai-agents-home.htm> accessed 2026-10-08 [search-tool extract]

¹²⁷⁹ Verified fact — B-REVB-S001: Okta for AI Agents (help home), AI agent administrators, View Agent Gateway activity, and System Log API docs (Okta, Inc.) <https://help.okta.com/oie/en-us/content/topics/ai-agents/ai-agents-home.htm> accessed 2026-10-08 [search-tool extract]

Field	Value	Label	Sources
	standardised this as the Enterprise-Managed Authorization extension		
What it does	Auth0 AI SDKs: user authentication for agents, calling third-party APIs on users' behalf, authorisation for RAG (fine-grained), and asynchronous human approval using OpenID CIBA. Cross App Access: MCP client exchanges an enterprise ID token for an ID-JAG at the IdP, then for an access token at the MCP server's authorisation server. Auth0 Token Vault stores and brokers third-party API tokens (e.g. Google Drive, Jira, Slack) for agents; Okta for AI Agents discovers, onboards, protects and governs agent identities across frameworks and clouds.	Verified fact	A6-S078, A6-S076, A6-S080, A6-S079, A6-S097, A6-S100
Stack position	Identity provider / authorisation server for agents and MCP servers	Verified fact	A6-S079
Integration	OAuth 2.0 / OIDC, OpenID CIBA, RFC 8693 token exchange and RFC 7523 JWT grant (ID-JAG); SDKs for Python and JavaScript agent frameworks	Verified fact	A6-S076, A6-S079, A6-S078
Dependencies	Not publicly verified	Not publicly verified	
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Okta Agent SSO (XAA) included in core Okta SSO at no additional cost; Okta for AI Agents is a separate subscription (price not published). Auth0: Token Vault connections by plan (Free 2; Essentials/Professional 3 plus add-on; Enterprise 4 plus add-on); Free US\$0 up to 25,000 MAU,	Verified fact	A6-S099, A6-S098

Field	Value	Label	Sources
	Professional US\$240/month (pricing page may be stale)		
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	XAA out-of-the-box support includes Anthropic (Claude), Asana, Atlassian, Canva, Datadog, Figma, Glean, Linear, Notion, Slack, Supabase; XAA incorporated as the MCP Enterprise-Managed Authorization extension	Verified fact	A6-S099, A6-S035
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Core products GA (November 2025 to August 2026); Auth0 AI SDKs still flagged 'under heavy development'; Okta Agent Gateway and Shadow AI Agent Discovery announced for Q3 2026 (not confirmed shipped); Auth0 Agent Gateway in beta	Verified fact	A6-S097, A6-S099, A6-S100, A6-S078, V2-S035
Deployment	saas: Yes (Okta and Auth0 cloud identity services); managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Not publicly verified; on_prem: Not publicly verified		

Microsoft Entra Agent ID (c4-entra-agent-id)

Tier: Strategic · **Flags:** none · **Original graphic label:** not in graphic

Rationale: Strategic, conditional: where Entra is the workforce IdP; FS 3.50 is below the usual 3.6 because deployment and cost score 2, accepted because agent identity must live in the same directory as the delegating humans ^{AI}.

Criterion	Score	Rationale
Technical	5	Blueprints, agent identities, OBO and autonomous flows, Conditional Access, governance, ID Protection and audit for agents.
Enterprise readiness	4	Hyperscaler presumption (CP2 Q1): platform controls presumed (CP2 Q1); confirm per service; access controls and audit logs for agents also verified (A6-S058).
Security and compliance	4	Entra ID in ISO 27001 (commercial) and FedRAMP High scope; commercial SOC 2 scope does not name it; Agent ID and Agent 365 not named separately (scope rule, 4).

Criterion	Score	Rationale
Deployment flexibility	2	Microsoft cloud service only.
Ecosystem	4	Guides for Bedrock and n8n agents, MCP and A2A, workload identity federation.
Reliability and maturity	3	GA April 2026; parts still preview.
Cost / TCO	2	Agent 365 licence needed for security features; prices not verified.
Lock-in / portability	3	Standard OAuth/OIDC tokens, but proprietary agent constructs and licensing.
Total (generic / FS)	3.55	
	/	
	3.50	

Capabilities. Agent identity platform in Microsoft Entra: agent identity blueprints, agent identities, agent users, owners/sponsors/managers; OAuth client_credentials for autonomous agents, jwt-bearer On-Behalf-Of, refresh_token for long-running delegated work, no interactive flows; Conditional Access, ID Governance access packages, ID Protection, network controls and sign-in/audit logs extended to agents; supports MCP, A2A and non-Microsoft agents via sidecar SDK or workload identity federation ^{vr1280}. GA April 2026 ^{vr1281}.

Strengths

- Agent identity in the same directory as the human principals, so OBO delegation and Conditional Access apply natively ^{aj}
- Explicit sponsor/owner model and access packages for agents ^{vr1282}
- Standard OAuth flows; no interactive flows for agent entities ^{vr1283}
- Entra ID in ISO 27001 and FedRAMP High scope ^{vr1284}

Limitations and risks

- Security features (Conditional Access, ID Protection, governance) for agents need Microsoft Agent 365 licences (included in E7; add-on to E5/A5/Business Premium) ^{vr1285}

¹²⁸⁰ Verified fact — A6-S057: What is Microsoft Entra Agent ID? (Microsoft Learn source) <https://raw.githubusercontent.com/MicrosoftDocs/entra-docs/main/docs/agent-id/what-is-microsoft-entra-agent-id.md> accessed 2026-10-07; A6-S058: What's new in Microsoft Entra Agent ID (Microsoft Learn source) <https://raw.githubusercontent.com/MicrosoftDocs/entra-docs/main/docs/agent-id/whats-new-agent-id.md> accessed 2026-10-07; A6-S060: Authentication protocols for agents (Microsoft Learn source) <https://raw.githubusercontent.com/MicrosoftDocs/entra-docs/main/docs/agent-id/agent-oauth-protocols.md> accessed 2026-10-07

¹²⁸¹ Verified fact — V2-S032: Microsoft Entra releases and announcements (with What's new in Entra Agent ID; Entra blog 'Build AI agents for production with secure identities from day one'; <https://learn.microsoft.com/en-us/entra/fundamentals/whats-new> accessed 2026-10-08 [search-tool extract]

¹²⁸² Verified fact — A6-S057: What is Microsoft Entra Agent ID? (Microsoft Learn source) <https://raw.githubusercontent.com/MicrosoftDocs/entra-docs/main/docs/agent-id/what-is-microsoft-entra-agent-id.md> accessed 2026-10-07; A6-S058: What's new in Microsoft Entra Agent ID (Microsoft Learn source) <https://raw.githubusercontent.com/MicrosoftDocs/entra-docs/main/docs/agent-id/whats-new-agent-id.md> accessed 2026-10-07

¹²⁸³ Verified fact — A6-S060: Authentication protocols for agents (Microsoft Learn source) <https://raw.githubusercontent.com/MicrosoftDocs/entra-docs/main/docs/agent-id/agent-oauth-protocols.md> accessed 2026-10-07

¹²⁸⁴ Verified fact — B-C4-S008: ISO/IEC 27001 and SOC 2 Type 2 offering pages (Microsoft Compliance) <https://learn.microsoft.com/en-us/compliance/regulatory/offering-iso-27001> accessed 2026-10-08 [search-tool extract]; A6-S056: Azure services in FedRAMP and DoD audit scope (Microsoft Learn source) <https://raw.githubusercontent.com/MicrosoftDocs/azure-docs/main/articles/azure-government/compliance/azure-services-in-fedramp-auditscope.md> accessed 2026-10-07

¹²⁸⁵ Verified fact — A6-S059: Microsoft Entra Agent ID licensing (include file) <https://raw.githubusercontent.com/MicrosoftDocs/entra-docs/main/docs/includes/licensing-agent-id.md> accessed 2026-10-07; V2-S032: Microsoft Entra releases and announcements (with What's new in Entra Agent ID; Entra blog 'Build AI agents for production with secure identities from day one'; <https://learn.microsoft.com/en-us/entra/fundamentals/whats-new> accessed 2026-10-08 [search-tool extract]

- Microsoft-specific constructs (blueprints, agent users) ¹²⁸⁶
- Some admin-centre wizards still preview; registry converging into Agent 365 ¹²⁸⁷
- Agent 365 list prices not verified ^{NPV}

Choose when

- Entra is the workforce IdP ^{AJ}
- Copilot Studio or Foundry agents are in the estate ^{AJ}

Avoid when

- Okta is the workforce IdP and Entra would be a second identity plane ^{AJ}

Nearest competitors: C4-okta-auth0-ai-agents, C7-hashicorp-vault, Amazon Bedrock AgentCore Identity

Regulated-FS note. Register every agent with a named sponsor, use OBO for user-initiated work, and budget Agent 365 licensing before relying on Conditional Access for agents; platform controls presumed (CP2 Q1); confirm per service ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Microsoft	Verified fact	A6-S057
Category	Agent identity platform within Microsoft Entra (identity provider for AI agents)	Verified fact	A6-S057
Version / lineup	Generally available (Entra release log lists GA under April 2026; What's new page dated 1 May 2026); some admin-centre creation wizards still preview	Verified fact	A6-S058, V2-S032
Licence	Proprietary; Agent ID available to all Microsoft Entra customers; Entra security features for agents require Microsoft Agent 365 licences	Verified fact	A6-S059
Status events	April 2026: Entra Agent ID generally available (Entra release log); agent registry converging under Microsoft Agent 365	Verified fact	A6-S058, V2-S032
Strategic direction	Identity foundation for Microsoft Agent 365 (agent registry converging there); extends Zero Trust controls to agents; supports non-Microsoft agents (AWS Bedrock, GCP, n8n) via Entra ID Auth SDK sidecar or workload identity federation	Verified fact	A6-S058, A6-S057
What it does	Creates and manages agent identities from agent identity blueprints (parent-child templates), agent service principals and agents' user accounts with owners, sponsors and managers; applies Conditional	Verified fact	A6-S057, A6-S058

¹²⁸⁶ Verified fact — A6-S057: What is Microsoft Entra Agent ID? (Microsoft Learn source) <https://raw.githubusercontent.com/MicrosoftDocs/entra-docs/main/docs/agent-id/what-is-microsoft-entra-agent-id.md> accessed 2026-10-07

¹²⁸⁷ Verified fact — A6-S058: What's new in Microsoft Entra Agent ID (Microsoft Learn source) <https://raw.githubusercontent.com/MicrosoftDocs/entra-docs/main/docs/agent-id/whats-new-agent-id.md> accessed 2026-10-07

Field	Value	Label	Sources
	Access, ID Governance (access packages for OBO and autonomous scenarios), ID Protection (risky agents), network controls and sign-in/audit logs to agents.		
Stack position	Identity provider and authorisation server for agents; sits beside the agent runtime and tool layer	Verified fact	A6-S057
Integration	OAuth 2.0 with Federated Identity Credentials: client_credentials for autonomous agents, jwt-bearer for On-Behalf-Of delegation, refresh_token for long-running user-delegated work; no interactive flows for agent entities; supports MCP and A2A	Verified fact	A6-S060, A6-S057
Dependencies	Microsoft Entra tenant; Microsoft Agent 365 licences for security features	Verified fact	A6-S059
Certifications	Microsoft Entra ID (P1/P2) and Entra ID Governance in Azure FedRAMP High scope (Agent ID not listed separately)	Verified fact	A6-S056
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Conditional Access for agents; access packages; sponsor lifecycle workflows; sign-in and audit logs for agents	Verified fact	A6-S058
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Agent ID included for Entra customers; Agent 365 per-user licence required for security extensions (included in Microsoft 365 E7); list prices not verified	Verified fact	A6-S059
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Guides for securing Amazon Bedrock and n8n agents; migration of Copilot Studio agents and custom app registrations	Verified fact	A6-S058
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	GA in April 2026 after preview (announced at Microsoft Build 2025); documentation marks some features as new or preview	Verified fact	A6-S058, V2-S032

Field	Value	Label	Sources
Deployment	saas: Yes (Microsoft Entra cloud service); managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Not publicly verified; on_prem: Not publicly verified		

MCP Authorization (MCP specification 2026-07-28) plus Enterprise-Managed Authorization extension (C4-mcp-authorization)

Tier: Strategic · **Flags:** none · **Original graphic label:** not in graphic

Rationale: Strategic, conditional: only behind a gateway with mandatory authorisation, an allow-list and pinned tool definitions. Tier set by the reader at Checkpoint 3 (CP3 Q1): MCP (L4-mcp) and its authorisation profile are one decision. FS 3.45 and reliability 2 (IETF drafts, four breaking revisions) are why the condition includes pinning the specification revision at the gateway. Conflict of interest: MCP originated at Anthropic and the author is an Anthropic model ^{AI}.

Criterion	Score	Rationale
Technical	4	Strong agent-to-tool contract (audience binding, PRM, issuer validation, EMA); optional and silent on fine-grained policy.
Enterprise readiness	3	Rule 2: EMA lets the corporate IdP govern access; enterprise features come from implementations.
Security and compliance	3	Rule 2 hygiene: private advisory reporting and documented trust model (B-C4-S004); optional authorisation and draft dependencies keep it at 3.
Deployment flexibility	4	Implementation-dependent; usable in self-hosted and managed gateways.
Ecosystem	4	Implemented by AWS AgentCore Gateway, Kong AI Gateway, Okta XAA and Azure API Management; MCP itself is the de facto tool protocol (L4 scores it 5), but authorisation is optional and adoption of the authorisation profile across servers is not publicly verified, so 4.
Reliability and maturity	2	Four breaking revisions in about 16 months; OAuth 2.1 still draft.
Cost / TCO	4	Free specification; implementation and re-certification on each revision cost effort.
Lock-in / portability	4	Open specification under Linux Foundation hosting; maintainer concentration at one vendor.
Total (generic / FS)	3.50 / 3.45	

Evidence rules applied: Rule 2 (self-hosted software / open specification): enterprise_readiness and security_compliance capped at 4; NPV cap not applied; inherits host controls

Capabilities. Authorisation profile of OAuth 2.1 (still an IETF draft) for MCP: server as resource server with Protected Resource Metadata (RFC 9728), PKCE, audience-bound tokens via Resource Indicators (RFC 8707), token passthrough forbidden; 2026-07-28 adds RFC 9207 issuer validation and issuer-bound clients and

deprecates Dynamic Client Registration; Enterprise-Managed Authorization (ID-JAG) extension Stable. Authorisation is OPTIONAL ^{vf1288}.

Strengths

- Audience binding and the ban on token passthrough address the confused-deputy problem ^{vf1289}
- Enterprise IdP decides which MCP servers an employee can use (EMA/ID-JAG) ^{vf1290}
- Implemented by AgentCore Gateway, Kong and Okta XAA ^{vf1291}
- Hosted by the Agentic AI Foundation (Linux Foundation) since 9 December 2025 ^{vf1292}

Limitations and risks

- Authorisation is OPTIONAL in MCP ^{vf1293}
- Frequent breaking revisions: 2025-03-26, 2025-06-18, 2025-11-25, 2026-07-28 ^{vf1294}
- Several dependencies are IETF drafts, including OAuth 2.1 ^{vf1295}
- Both Lead Maintainers are Anthropic staff ^{vf1296}; the author is an Anthropic model (conflict of interest)

¹²⁸⁸ Verified fact — A6-S033: MCP specification 2026-07-28: Authorization (Model Context Protocol project) <https://raw.githubusercontent.com/modelcontextprotocol/modelcontextprotocol/main/docs/specification/2026-07-28/basic/authorization/index.md> accessed 2026-10-07; A6-S034: MCP specification 2026-07-28: Authorization security considerations (Model Context Protocol project) <https://raw.githubusercontent.com/modelcontextprotocol/modelcontextprotocol/main/docs/specification/2026-07-28/basic/authorization/security-considerations.md> accessed 2026-10-07; A6-S032: MCP specification 2026-07-28: Key Changes (changelog) (Model Context Protocol project (Agentic AI Foundation)) <https://raw.githubusercontent.com/modelcontextprotocol/modelcontextprotocol/main/docs/specification/2026-07-28/changelog.md> accessed 2026-10-07; A6-S035: MCP extension: Enterprise-Managed Authorization (overview) (Model Context Protocol project) <https://raw.githubusercontent.com/modelcontextprotocol/modelcontextprotocol/main/docs/extensions/auth/enterprise-managed-authorization.md> accessed 2026-10-07; A6-S079: MCP ext-auth: Enterprise-Managed Authorization specification (Stable) (Model Context Protocol project) <https://raw.githubusercontent.com/modelcontextprotocol/ext-auth/main/specification/stable/enterprise-managed-authorization.md> accessed 2026-10-07

¹²⁸⁹ Verified fact — A6-S034: MCP specification 2026-07-28: Authorization security considerations (Model Context Protocol project) <https://raw.githubusercontent.com/modelcontextprotocol/modelcontextprotocol/main/docs/specification/2026-07-28/basic/authorization/security-considerations.md> accessed 2026-10-07

¹²⁹⁰ Verified fact — A6-S035: MCP extension: Enterprise-Managed Authorization (overview) (Model Context Protocol project) <https://raw.githubusercontent.com/modelcontextprotocol/modelcontextprotocol/main/docs/extensions/auth/enterprise-managed-authorization.md> accessed 2026-10-07; A6-S079: MCP ext-auth: Enterprise-Managed Authorization specification (Stable) (Model Context Protocol project) <https://raw.githubusercontent.com/modelcontextprotocol/ext-auth/main/specification/stable/enterprise-managed-authorization.md> accessed 2026-10-07

¹²⁹¹ Verified fact — A6-S021: Amazon Bedrock AgentCore Pricing; AgentCore GA announcement (Oct 2025); AgentCore Gateway docs (Amazon Web Services) <https://aws.amazon.com/bedrock/agentcore/pricing/> accessed 2026-10-07 [search-tool extract]; A6-S017: Kong AI Gateway changelog; Kong Gateway changelog; AI Proxy Advanced plugin docs (Kong Inc.) <https://developer.konghq.com/ai-gateway/changelog/> accessed 2026-10-07 [search-tool extract]; A6-S099: Okta brings first-class identity to AI agents with Agent SSO; Okta Introduces Agent SSO for Cross App Access (XAA) (Okta, Inc.) <https://www.okta.com/newsroom/press-releases/okta-brings-first-class-identity-to-ai-agents-with-agent-sso/> accessed 2026-10-07 [search-tool extract]

¹²⁹² Verified fact — A3-S018: MCP joins the Agentic AI Foundation (9 Dec 2025) (Model Context Protocol project (blog)) <https://blog.modelcontextprotocol.io/posts/2025-12-09-mcp-joins-agentic-ai-foundation/> accessed 2026-10-07

¹²⁹³ Verified fact — A6-S033: MCP specification 2026-07-28: Authorization (Model Context Protocol project) <https://raw.githubusercontent.com/modelcontextprotocol/modelcontextprotocol/main/docs/specification/2026-07-28/basic/authorization/index.md> accessed 2026-10-07

¹²⁹⁴ Verified fact — A6-S033: MCP specification 2026-07-28: Authorization (Model Context Protocol project) <https://raw.githubusercontent.com/modelcontextprotocol/modelcontextprotocol/main/docs/specification/2026-07-28/basic/authorization/index.md> accessed 2026-10-07; A6-S021: Amazon Bedrock AgentCore Pricing; AgentCore GA announcement (Oct 2025); AgentCore Gateway docs (Amazon Web Services) <https://aws.amazon.com/bedrock/agentcore/pricing/> accessed 2026-10-07 [search-tool extract]

¹²⁹⁵ Verified fact — A6-S033: MCP specification 2026-07-28: Authorization (Model Context Protocol project) <https://raw.githubusercontent.com/modelcontextprotocol/modelcontextprotocol/main/docs/specification/2026-07-28/basic/authorization/index.md> accessed 2026-10-07; A6-S079: MCP ext-auth: Enterprise-Managed Authorization specification (Stable) (Model Context Protocol project) <https://raw.githubusercontent.com/modelcontextprotocol/ext-auth/main/specification/stable/enterprise-managed-authorization.md> accessed 2026-10-07

¹²⁹⁶ Verified fact — A3-S082: Expanding the MCP Maintainer Team (8 Apr 2026) (Model Context Protocol project (blog)) <https://blog.modelcontextprotocol.io/posts/2026-04-08-maintainer-update/> accessed 2026-10-07

- Security policy states clients trust configured servers and that access control is the server developer's responsibility ^{VF 1297}

Choose when

- MCP is the firm's tool protocol ^{AJ}

Avoid when

- Tools are exposed as REST/OpenAPI behind a gateway, where plain OAuth 2.0 resource-server patterns suffice ^{AJ}

Nearest competitors: Gateway-native tool auth (Kong, AgentCore Gateway, Azure APIM credential manager), C4-okta-auth0-ai-agents, OAuth 2.0 resource-server pattern for REST tools

Regulated-FS note. Strategic, conditional, and mandatory wherever MCP is used (CP3 Q1): make authorisation mandatory by firm policy, pin the spec revision at the gateway, and require EMA so the corporate IdP mediates access; the durable controls are the IdP and the gateway PDP, which keep working if the tool protocol changes ^{REC}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Model Context Protocol project (specification and extensions published in modelcontextprotocol GitHub organisation)	Verified fact	A6-S032, A6-S079
Category	Open specification (authorisation profile of OAuth 2.1 for agent-to-tool access)	Verified fact	A6-S033
Version / lineup	Specification revision 2026-07-28 (previous 2025-11-25); Enterprise-Managed Authorization extension status 'Stable'; Python SDK mcp 2.3.0 (2 October 2026)	Verified fact	A6-S032, A6-S079, A6-S086, V2-S031
Licence	Open specification	Verified fact	A6-S033
Status events	28 July 2026: MCP specification revision 2026-07-28 published; OAuth Dynamic Client Registration deprecated as a registration mechanism	Verified fact	A6-S032, V2-S031
Strategic direction	Hardening for enterprise use: RFC 9207 issuer validation, issuer-bound client registration, Client ID Metadata Documents replacing Dynamic Client Registration, enterprise IdP control via ID-JAG; feature lifecycle policy with 12-month deprecation window; 2026-07-28 also makes the protocol stateless (no initialize handshake), removes SSE stream resumability, and deprecates the Roots, Sampling and Logging features	Verified fact	A6-S032, A6-S035, V2-S031
What it does	Defines how MCP clients obtain and present access tokens to protected MCP servers: MCP server is an OAuth 2.1 resource server; MUST publish Protected Resource Metadata (RFC 9728); clients MUST use PKCE and Resource Indicators (RFC 8707) so tokens are audience-bound; servers	Verified fact	A6-S033, A6-S034, A6-S035, A6-S079

¹²⁹⁷ Verified fact — B-C4-S004: Model Context Protocol Security Policy (SECURITY.md) (Model Context Protocol project) <https://raw.githubusercontent.com/modelcontextprotocol/modelcontextprotocol/main/SECURITY.md> accessed 2026-10-08

Field	Value	Label	Sources
	MUST reject tokens not issued for them (token passthrough forbidden). The Enterprise-Managed Authorization extension lets the corporate IdP decide which MCP servers an employee can use via an Identity Assertion JWT Authorization Grant.		
Stack position	Authorisation contract between agents (MCP clients), MCP servers and identity providers; the standards basis for an agent identity/authorisation sub-layer	Verified fact	A6-S033, A6-S035
Integration	HTTP Authorization bearer header on every request; WWW-Authenticate resource_metadata discovery; scope challenge handling	Verified fact	A6-S033
Dependencies	OAuth 2.1 (draft-ietf-oauth-v2-1-13), RFC 8414, RFC 7591 (deprecated path), RFC 8707, RFC 9728, RFC 9207, draft-ietf-oauth-client-id-metadata-document-00; extension uses draft-ietf-oauth-identity-assertion-authz-grant, RFC 8693, RFC 7523	Verified fact	A6-S033, A6-S079
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Free open specification	Verified fact	A6-S033
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Implemented by gateways: AWS AgentCore Gateway supports MCP 2026-07-28; Kong AI Gateway 2.1.0 supports 2026-07-28; Okta Cross App Access sample uses ID-JAG; Okta states XAA is 'formally	Verified fact	A6-S021, A6-S017, A6-S080,

Field	Value	Label	Sources
	incorporated as the official Enterprise-Managed Authorization extension' for MCP		A6-S099
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Authorisation is OPTIONAL in MCP; several underlying documents are IETF drafts; frequent breaking revisions (2025-03-26, 2025-06-18, 2025-11-25, 2026-07-28)	Verified fact	A6-S033, A6-S021
Deployment	saas: Not publicly verified; managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Not publicly verified; on_prem: Not publicly verified		

C5: Prompt and configuration management

LangSmith prompt management (Prompts, Prompt Hub) (c5-langsmith-prompts)

Tier: Strategic · **Flags:** none · **Original graphic label:** not in graphic

Rationale: Strategic only in a LangSmith/LangGraph estate, with GitHub sync enabled as the exit route; otherwise Tactical ^{AJ}.

Criterion	Score	Rationale
Technical	4	Commits, diffs, environments, promotion, rollback history, owners, webhooks and Git sync; no documented percentage rollout or user segmentation.
Enterprise readiness	4	Organisation and workspace roles, custom RBAC and per-prompt owners; Enterprise SSO, ABAC and support SLA from the platform record; SCIM not documented.
Security and compliance	4	SOC 2 Type II, ISO 27001:2022, HIPAA BAA on Enterprise; no CMK/BYOK documented, so not 5.
Deployment flexibility	5	SaaS in US, EU and APAC regions, BYOC on AWS, self-hosted and air-gapped self-host on Enterprise.
Ecosystem	4	LangChain ecosystem, langchain-prompty integration, webhooks and GitHub sync.
Reliability and maturity	4	GA since 2023, independent and funded (US\$125M Series B, October 2025).
Cost / TCO	3	Per-seat plus per-trace pricing; heavy self-host footprint.
Lock-in / portability	3	Proprietary service, but prompt text syncs to GitHub, so lock-in is lower than for LangSmith traces (L9 scores 2).
Total (generic / FS)	4.00 / 3.95	

Capabilities. Prompts stored as a commit history with diffs; reserved Staging and Production environments assigned by promotion; per-environment rollback history; 'owners only' mode restricting who may tag, promote or delete; webhooks on every commit; synchronisation of prompts with a GitHub repository; public prompt hub ^{vf¹²⁹⁸}.

Strengths

- The most complete change-control workflow among the registries profiled: commits, diffs, promotion, rollback history and prompt owners ^{vf¹²⁹⁹ AI}
- GitHub synchronisation gives a documented export path and lets Git remain the record ^{vf¹³⁰⁰}
- Webhooks on commit can trigger CI evaluation and change tickets ^{vf¹³⁰¹}
- SOC 2 Type II, ISO 27001:2022 and HIPAA; US, EU (Netherlands) and APAC regions; BYOC on AWS, self-host and air-gapped self-host on Enterprise ^{vf¹³⁰²}

Limitations and risks

- Proprietary service for prompt storage and promotion ^{vf¹³⁰³}
- No EU legal entity for contracting ^{vf¹³⁰⁴}
- Self-hosting is an Enterprise add-on with a heavy footprint (Kubernetes, PostgreSQL, Redis, ClickHouse) ^{vf¹³⁰⁵}
- Platform increasingly bundles the LangChain agent runtime ^{vf¹³⁰⁶}

Choose when

- LangSmith is already the L9 platform and LangGraph the orchestration standard ^{AI}
- Product owners need a UI promotion and rollback workflow while engineering keeps Git as the record via sync ^{AI}

¹²⁹⁸ Verified fact — A7-S076: LangSmith docs: Manage prompts (LangChain) <https://raw.githubusercontent.com/langchain-ai/docs/main/src/langsmith/manage-prompts.mdx> accessed 2026-10-07

¹²⁹⁹ Verified fact — A7-S076: LangSmith docs: Manage prompts (LangChain) <https://raw.githubusercontent.com/langchain-ai/docs/main/src/langsmith/manage-prompts.mdx> accessed 2026-10-07

¹³⁰⁰ Verified fact — A7-S076: LangSmith docs: Manage prompts (LangChain) <https://raw.githubusercontent.com/langchain-ai/docs/main/src/langsmith/manage-prompts.mdx> accessed 2026-10-07

¹³⁰¹ Verified fact — A7-S076: LangSmith docs: Manage prompts (LangChain) <https://raw.githubusercontent.com/langchain-ai/docs/main/src/langsmith/manage-prompts.mdx> accessed 2026-10-07

¹³⁰² Verified fact — A7-S077: LangSmith docs: Regions FAQ (LangChain) <https://raw.githubusercontent.com/langchain-ai/docs/main/src/langsmith/regions-faq.mdx> accessed 2026-10-07; A1-S036: LangSmith Regions FAQ - Docs by LangChain (LangChain, Inc.) <https://docs.langchain.com/langsmith/regions-faq> accessed 2026-10-07 [search-tool extract]; A1-S126: LangSmith Engine security - Docs by LangChain (LangChain, Inc.) <https://docs.langchain.com/langsmith/engine-security> accessed 2026-10-07 [search-tool extract]; A1-S034: Set up LangSmith (deployment options) - Docs by LangChain (LangChain, Inc.) <https://docs.langchain.com/langsmith/platform-setup> accessed 2026-10-07 [search-tool extract]

¹³⁰³ Verified fact — A7-S076: LangSmith docs: Manage prompts (LangChain) <https://raw.githubusercontent.com/langchain-ai/docs/main/src/langsmith/manage-prompts.mdx> accessed 2026-10-07; A7-S079: LangSmith docs: Self-hosted (LangChain) <https://raw.githubusercontent.com/langchain-ai/docs/main/src/langsmith/self-hosted.mdx> accessed 2026-10-07

¹³⁰⁴ Verified fact — A7-S077: LangSmith docs: Regions FAQ (LangChain) <https://raw.githubusercontent.com/langchain-ai/docs/main/src/langsmith/regions-faq.mdx> accessed 2026-10-07

¹³⁰⁵ Verified fact — A7-S079: LangSmith docs: Self-hosted (LangChain) <https://raw.githubusercontent.com/langchain-ai/docs/main/src/langsmith/self-hosted.mdx> accessed 2026-10-07; A1-S034: Set up LangSmith (deployment options) - Docs by LangChain (LangChain, Inc.) <https://docs.langchain.com/langsmith/platform-setup> accessed 2026-10-07 [search-tool extract]

¹³⁰⁶ Verified fact — A1-S039: Everything we shipped at Interrupt (LangChain, Inc.) <https://www.langchain.com/blog/interrupt-2026-overview> accessed 2026-10-07 [search-tool extract]

Avoid when

- You want the prompt registry separated from the agent runtime vendor ^{AJ}
- Contracting with an EU legal entity is a requirement ^{AJ}

Nearest competitors: C5-langfuse-prompts, C5-promptlayer, C5-prompts-as-code

Regulated-FS note. Enable owners-only mode on production prompts, route promotion through a service identity triggered after the eval gate, and use GitHub sync so that the approved prompt text also lives in the firm's repository ^{Rec.}.

Facts (as of 2026-10-07)

Field	Value	Label	Sources
Company	LangChain, Inc.	Verified fact	A7-S076, A7-S011
Category	Prompt management within an LLM observability, evaluation and deployment platform	Verified fact	A7-S076, A7-S078
Version / lineup	langsmith Python SDK 0.14.4 released 2 October 2026 (platform is a continuously deployed SaaS)	Verified fact	A7-S011
Licence	Proprietary platform (self-hosting is an Enterprise add-on); langsmith SDK MIT	Verified fact	A7-S079, A7-S011
Strategic direction	Prompt workflow modelled on commits: environments (Staging/Production) as reserved commit tags, promotion UI, prompt owners, webhooks on commit and synchronisation of prompts with a GitHub repository	Verified fact	A7-S076
What it does	Stores prompts as a commit history with diffs; commit tags and the reserved 'staging' and 'production' tags mark which version is live in each environment. 'Owners only' mode restricts who can tag, promote or delete a prompt. Webhooks fire on each commit, and a public prompt hub offers community prompts.	Verified fact	A7-S076
Stack position	Control-plane capability (C5) inside the L9 LangSmith platform; the graphic shows LangSmith only as 'trace & eval'	Verified fact	A7-S076
Integration	LangSmith UI, API and SDK; webhooks on prompt commit; GitHub synchronisation	Verified fact	A7-S076
Dependencies	Not publicly verified	Not publicly	

Field	Value	Label	Sources
Certifications	SOC 2 Type 2; HIPAA compliant; GDPR	verified Verified fact	A7-S077
GDPR / residency	SaaS regions: US (GCP us-central1), EU (GCP europe-west4, Netherlands), APAC (GCP australia-southeast1, from May 2026), US on AWS (us-east-2, from April 2026). No EU legal entity for contracting.	Verified fact	A7-S077, A7-S078
Security features	Not publicly verified	Not publicly verified	
Access controls	Organisation and workspace roles, custom RBAC roles, personal access tokens; per-prompt owners	Verified fact	A7-S080, A7-S076
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Not publicly verified	Not publicly verified	
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	langsmith SDK first published on PyPI 26 June 2023	Verified fact	A7-S011
Maturity	Not publicly verified	Not publicly verified	
Deployment	saas: Yes; managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes (Enterprise add-on, Kubernetes); on_prem: Not publicly verified		

Langfuse Prompt Management (a capability of the Langfuse platform)

(C5-langfuse-prompts)

Tier: Strategic · **Flags:** Acquired · **Original graphic label:** not in graphic

Rationale: Strategic only where Langfuse is the L9 platform of record and the Enterprise licence is bought for protected labels, RBAC and audit logs; otherwise Tactical ^{AJ}.

Criterion	Score	Rationale
Technical	4	Versions, environment/tenant/experiment labels, client caching, loud failure on missing labels and trace linkage; no documented approval workflow, Git sync or percentage rollout.
Enterprise readiness	4	Rule 7, aligned with the CP2 rework of L9 Langfuse (4): SSO, project-level RBAC, audit logs, SCIM and the Org Management API verified ^{ver1307} ; Enterprise-licensed when self-hosted, which is a stated condition of the tier; SLA terms not published.
Security and compliance	4	SOC 2 Type II and ISO 27001 with annual audits, DPA, EU region on Cloud; no CMK/BYOK documented, so not 5.
Deployment flexibility	4	SaaS (US, EU, JP, HIPAA regions), self-hosted in any region and fully offline; kept at 4 for consistency with L9 because vendor-managed BYOC is not documented (see scoring notes).
Ecosystem	4	Open-source SDKs in Python and JS/TS, OTel-based platform, wide adoption of the parent platform; prompt API is Langfuse-specific.
Reliability and maturity	4	Platform at v4, frequent SDK releases (4.17.0 on 5 October 2026); ownership changed to ClickHouse in January 2026 with no planned licence change.
Cost / TCO	4	MIT core free; the governance controls this control needs are in the paid

¹³⁰⁷ Verified fact — A7-S074: Langfuse docs: Enterprise License Key (self-hosted) (Langfuse (ClickHouse)) <https://raw.githubusercontent.com/langfuse/langfuse-docs/main/content/self-hosting/license-key.mdx> accessed 2026-10-07; A1-S033: Self-Hosted Pricing - Langfuse (Langfuse (ClickHouse)) <https://langfuse.com/pricing-self-host> accessed 2026-10-07 [search-tool extract]

Criterion	Score	Rationale
Lock-in / portability	3	Enterprise tier. Base 4 (MIT, self-hostable, prompts are text) reduced by 1 for the 2026 ownership change.
Total (generic / FS)	3.95 / 3.85	

Evidence rules applied: CP3 review: enterprise readiness raised from 3 to 4 to match the CP2-reworked L9 Langfuse score (rule 7: all three controls plus SCIM and an admin API)

Capabilities. Prompt registry inside the Langfuse platform: every prompt version gets an immutable version ID; labels (production, staging, tenant or experiment) select which version the SDK fetches; SDKs cache prompts client-side; prompt versions are linked to traces so quality and cost can be analysed by version; protected prompt labels restrict who may move a production label (Enterprise) ^{vf¹³⁰⁸}.

Strengths

- Version-to-trace linkage answers 'which prompt version produced this output' inside the same store that holds the evaluation evidence (see L9) ^{vf¹³⁰⁹}
- Client-side SDK caching keeps the registry off the hot path ^{vf¹³¹⁰}
- A missing label returns an error rather than silently serving another version, so a mislabelled environment fails loudly ^{vf¹³¹¹}
- MIT core; self-hosted or Cloud with SOC 2 Type II, ISO 27001 and an EU (Ireland) region ^{vf¹³¹²}

Limitations and risks

- Protected prompt labels, project-level RBAC, audit logs and SCIM need an Enterprise licence key when self-hosted, so segregation of duties on production prompts is a paid feature ^{vf¹³¹³}
- No native pull-request style approval or Git synchronisation documented in the fact base ^{NPV}
- Owned by ClickHouse (acquisition announced 16 January 2026) ^{vf¹³¹⁴}

¹³⁰⁸ Verified fact — A7-S071: Langfuse docs: Prompt Management overview (Langfuse (ClickHouse)) <https://raw.githubusercontent.com/langfuse/langfuse-docs/main/content/docs/prompt-management/overview.mdx> accessed 2026-10-07; A7-S072: Langfuse docs: Prompt version control (Langfuse (ClickHouse)) <https://raw.githubusercontent.com/langfuse/langfuse-docs/main/content/docs/prompt-management/features/prompt-version-control.mdx> accessed 2026-10-07; A7-S074: Langfuse docs: Enterprise License Key (self-hosted) (Langfuse (ClickHouse)) <https://raw.githubusercontent.com/langfuse/langfuse-docs/main/content/self-hosting/license-key.mdx> accessed 2026-10-07

¹³⁰⁹ Verified fact — A7-S071: Langfuse docs: Prompt Management overview (Langfuse (ClickHouse)) <https://raw.githubusercontent.com/langfuse/langfuse-docs/main/content/docs/prompt-management/overview.mdx> accessed 2026-10-07

¹³¹⁰ Verified fact — A7-S071: Langfuse docs: Prompt Management overview (Langfuse (ClickHouse)) <https://raw.githubusercontent.com/langfuse/langfuse-docs/main/content/docs/prompt-management/overview.mdx> accessed 2026-10-07; A7-S072: Langfuse docs: Prompt version control (Langfuse (ClickHouse)) <https://raw.githubusercontent.com/langfuse/langfuse-docs/main/content/docs/prompt-management/features/prompt-version-control.mdx> accessed 2026-10-07

¹³¹¹ Verified fact — A7-S072: Langfuse docs: Prompt version control (Langfuse (ClickHouse)) <https://raw.githubusercontent.com/langfuse/langfuse-docs/main/content/docs/prompt-management/features/prompt-version-control.mdx> accessed 2026-10-07

¹³¹² Verified fact — A7-S073: Langfuse Security and Compliance overview (Langfuse (ClickHouse)) <https://raw.githubusercontent.com/langfuse/langfuse-docs/main/content/security/index.mdx> accessed 2026-10-07; A7-S074: Langfuse docs: Enterprise License Key (self-hosted) (Langfuse (ClickHouse)) <https://raw.githubusercontent.com/langfuse/langfuse-docs/main/content/self-hosting/license-key.mdx> accessed 2026-10-07

¹³¹³ Verified fact — A7-S074: Langfuse docs: Enterprise License Key (self-hosted) (Langfuse (ClickHouse)) <https://raw.githubusercontent.com/langfuse/langfuse-docs/main/content/self-hosting/license-key.mdx> accessed 2026-10-07

¹³¹⁴ Verified fact — A7-S075: Langfuse README (Langfuse (ClickHouse)) <https://raw.githubusercontent.com/langfuse/langfuse/main/README.md> accessed 2026-10-07; V2-S041: ClickHouse welcomes Langfuse (with langfuse.com 'Langfuse joins ClickHouse', Orrick) <https://clickhouse.com/blog/clickhouse-acquires-langfuse-open-source-llm-observability> accessed 2026-10-08 [search-tool extract]

- Langfuse-specific prompt API and SDK ¹³¹⁵

Choose when

- Langfuse is already the L9 platform of record and you want prompt versions and traces in one place ^{AJ}
- You need a self-hosted registry inside the estate ^{AJ}

Avoid when

- You will not buy the Enterprise licence, because without protected labels any project member with write access can move the production label ^{AJ}
- Your change process requires approvals to happen in the source-control system ^{AJ}

Nearest competitors: C5-langsmith-prompts, C5-promptlayer, C5-prompts-as-code

Regulated-FS note. Use as the runtime delivery and trace-linkage layer, fed by CI from Git after the eval gate; make the production label protected and restrict label moves to the release pipeline's identity ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Langfuse (part of ClickHouse, Inc. since January 2026, per the Langfuse README; LICENSE copyright 'ClickHouse, Inc.')	Verified fact	A7-S075, V2-S041
Category	Prompt management / prompt registry (feature of an open-source LLM engineering platform)	Verified fact	A7-S071
Version / lineup	Docs labelled 'Version: v4'; Python SDK langfuse 4.17.0 released 5 October 2026	Verified fact	A7-S009, A7-S074
Licence	Open core: core features (including prompt management) MIT; 'Protected Prompt Labels', project-level RBAC, audit logs, SCIM and data-retention policies require an Enterprise licence key (ee folders)	Verified fact	A7-S074, A7-S075
Status events	16 January 2026: ClickHouse announced its acquisition of Langfuse alongside a US\$400m Series D; Langfuse states no planned licensing changes (closing date not stated)	Verified fact	A7-S075, V2-S041
Strategic direction	Positions prompt management as decoupling prompt updates from code deployment so non-engineers can change prompts in the UI; prompts linked to traces to analyse performance by prompt version	Verified fact	A7-S071
What it does	Stores, versions and serves prompts centrally instead of hard-coding them. Each prompt version gets a version ID; labels (e.g. production, staging, tenant or experiment labels) select which version the SDK fetches. SDKs cache prompts client-side, so retrieval adds no network latency on the hot path.	Verified fact	A7-S071, A7-S072
Stack position	Control-plane capability (C5) embedded in the L9 observability platform; prompt versions are linked to traces. The original graphic shows	Verified fact	A7-S071

¹³¹⁵ Verified fact — A7-S072: Langfuse docs: Prompt version control (Langfuse (ClickHouse)) <https://raw.githubusercontent.com/langfuse/langfuse-docs/main/content/docs/prompt-management/features/prompt-version-control.mdx> accessed 2026-10-07

Field	Value	Label	Sources
	Langfuse only in L9 and has no prompt-management layer.	d fact	
Integration	Langfuse SDKs (e.g. Python create_prompt/update_prompt, fetch by label) and UI; prompts cached by SDK	Verified fact	A7-S071, A7-S072
Dependencies	Same infrastructure as Langfuse Cloud when self-hosted: Postgres, ClickHouse, Redis and S3-compatible object storage; Langfuse Cloud runs on AWS and ClickHouse Cloud	Verified fact	A7-S073, A7-S074
Certifications	SOC 2 Type II and ISO 27001 (annual audits), external penetration tests, HIPAA-ready region; reports on request	Verified fact	A7-S073
GDPR / residency	GDPR compliant; DPA available; Langfuse Cloud regions US (us-west-2), EU (eu-west-1 Ireland), JP (ap-northeast-1) and HIPAA (us-west-2); self-hosted in any region	Verified fact	A7-S073
Security features	Encryption, project-scoped tenant isolation with RBAC checks before queries, masking, data retention and deletion controls	Verified fact	A7-S073
Access controls	Project-level RBAC roles, protected prompt labels (restricting who can change production labels), audit logs, SCIM and Org Management API are Enterprise (licence-key) features when self-hosted	Verified fact	A7-S074
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Not publicly verified	Not publicly verified	
Infrastructure cost	Self-hosting runs the same stack as the cloud service (Postgres, ClickHouse, Redis, object storage); 'no scalability limitations between the different versions'	Verified fact	A7-S074
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	langfuse Python package first published on PyPI 12 July 2023; latest 4.17.0 (5 October 2026)	Verified fact	A7-S009, V2-S028
Maturity	Platform at major version 4 (docs); SDK releases frequent (4.17.0 on 5 October 2026)	Verified fact	A7-S009, A7-S074
Deployment	saas: Yes; managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes; on_prem: Yes (self-hosted can run fully offline/air-gapped)		

Prompts-as-code pattern: Git-versioned prompt files (Prompty, Dotprompt) tested by config-driven evals (e.g. Promptfoo) (c5-prompts-as-code)

Tier: Strategic • **Flags:** Acquired • **Original graphic label:** not in graphic

Rationale: The only option whose change record lives in the firm’s own controls and survives any vendor exit; the Acquired flag refers to Promptfoo as the example eval CLI, which is replaceable ^{AJ}.

Criterion	Score	Rationale
Technical	3	Versioning, review, diff and CI evaluation are strong; runtime delivery, segmentation, instant rollback and trace linkage have to be built.
Enterprise readiness	4	Rubric rule 2: scored on what the pattern enables in-estate. It inherits the firm’s source-control SSO, RBAC, mandatory approvals and audit; scored at the rule 2 maximum of 4 (no commercial support for the formats) because approval and audit are the whole function of this control, unlike OPA or OpenLineage (3).
Security and compliance	3	Rubric rule 2: clear permissive licences; security policies, signed releases and CVE handling for the format libraries not verified; inherits host controls.
Deployment flexibility	5	Files in the customer’s repository; runs anywhere, including air-gapped.
Ecosystem	3	Two vendor-originated formats plus CLI tooling, langchain-prompty and LangSmith GitHub sync; no single standard.
Reliability and maturity	3	Git practice is mature; the formats are young (Prompty v2, Dotprompt 0.2.0) and Promptfoo is pre-1.0 and changing ownership.
Cost / TCO	4	Free and open source; engineering effort to build runtime delivery and trace linkage.
Lock-in / portability	4	Base 5 (plain files under customer control, permissive licences) reduced by 1 because the example eval tool’s ownership is changing and the formats are not under neutral governance.
Total (generic / FS)	3.60 / 3.65	

Evidence rules applied: Rubric rule 2 (self-hosted libraries and open formats): enterprise readiness and security scored on what the pattern enables and on project hygiene, each capped at 4; inherits host controls

Capabilities. Prompts as files in the application repository: Prompty (.prompty, markdown with YAML front matter for model, connection and template settings) or Dotprompt (executable Handlebars-based templates, language- and provider-agnostic); changes go through code review and CI, where a CLI such as Promptfoo evaluates the prompt files ^{ver¹³¹⁶}.

Strengths

- Uses the firm’s existing change-management controls (review, approval, branch protection, release tagging) without a new vendor ^{AJ}

¹³¹⁶ Verified fact — A7-S067: Prompty README (Microsoft) <https://raw.githubusercontent.com/microsoft/prompty/main/README.md> accessed 2026-10-07; A7-S066: Dotprompt README (Google) <https://raw.githubusercontent.com/google/dotprompt/main/README.md> accessed 2026-10-07; A7-S068: Promptfoo README (Promptfoo (OpenAI)) <https://raw.githubusercontent.com/promptfoo/promptfoo/main/README.md> accessed 2026-10-07

- Open formats with permissive licences: Prompty MIT, Dotprompt Apache-2.0, Promptfoo MIT ^{vr1317}
- Multi-language runtimes: Prompty for Python, TypeScript, Rust and C#; Dotprompt for JS/TS, Python, Go, Rust and Java ^{vr1318}
- One artefact can carry prompt, model and parameter settings, so the release pins them together ^{vr1319} ^{AI}

Limitations and risks

- No runtime label switching, segmentation or instant rollback without a deployment; trace linkage must be built ^{AI}
- Two competing formats, both pre-1.0 or recently re-versioned (Dotprompt Python 0.2.0; Prompty v2) ^{vr1320}
- Promptfoo: OpenAI announced its acquisition on 9 March 2026, closing date not published; Promptfoo states it is part of OpenAI ^{vr1321}
- Non-engineers cannot change prompts without a pull request, which is a feature for control and a cost for agility ^{AI}

Choose when

- Always, as the system of record for approved prompts and configuration in a regulated firm ^{Rec}

Avoid when

- As the only mechanism when business users must test prompt variants daily; pair it with a registry that CI publishes to ^{AI}

Nearest competitors: C5-langsmith-prompts, C5-langfuse-prompts, C5-promptlayer

Regulated-FS note. Make the repository the change record that model-risk and audit teams inspect: pull-request approval by a second person, CODEOWNERS for the risk owner, eval results attached to the PR, and a signed release tag pinning prompt, model and retrieval versions ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Pattern; formats from Microsoft (Prompty), Google (Dotprompt); Promptfoo is now part of OpenAI	Verified fact	A7-S067, A7-S066, A7-S068, V2-S042
Category	Named pattern and open file formats	Verified fact	A7-S067, A7-S066
Version /	Prompty v2 (Python prompty 2.0.2, 16 September 2026; runtimes for	Verified fact	A7-S008,

¹³¹⁷ Verified fact — A7-S008: prompty on PyPI (JSON API) (PyPI / Microsoft Prompty) <https://pypi.org/pypi/prompty/json> accessed 2026-10-07; A7-S065: dotpromptz on PyPI (PyPI / Google) <https://pypi.org/pypi/dotpromptz/json> accessed 2026-10-07; A7-S068: Promptfoo README (Promptfoo (OpenAI)) <https://raw.githubusercontent.com/promptfoo/promptfoo/main/README.md> accessed 2026-10-07

¹³¹⁸ Verified fact — A7-S067: Prompty README (Microsoft) <https://raw.githubusercontent.com/microsoft/prompty/main/README.md> accessed 2026-10-07; A7-S066: Dotprompt README (Google) <https://raw.githubusercontent.com/google/dotprompt/main/README.md> accessed 2026-10-07

¹³¹⁹ Verified fact — A7-S067: Prompty README (Microsoft) <https://raw.githubusercontent.com/microsoft/prompty/main/README.md> accessed 2026-10-07

¹³²⁰ Verified fact — A7-S065: dotpromptz on PyPI (PyPI / Google) <https://pypi.org/pypi/dotpromptz/json> accessed 2026-10-07; A7-S067: Prompty README (Microsoft) <https://raw.githubusercontent.com/microsoft/prompty/main/README.md> accessed 2026-10-07

¹³²¹ Verified fact — A7-S068: Promptfoo README (Promptfoo (OpenAI)) <https://raw.githubusercontent.com/promptfoo/promptfoo/main/README.md> accessed 2026-10-07; V2-S042: OpenAI to acquire Promptfoo (with TechCrunch via Yahoo Finance, SDxCentral, Security Boulevard) <https://openai.com/blog/openai-to-acquire-promptfoo> accessed 2026-10-08 [search-tool extract]

Field	Value	Label	Sources
lineup	Python, TypeScript, Rust, C#); Dotprompt (Python dotpromptz 0.2.0, 5 October 2026; JS/TS, Python, Go, Rust, Java); promptfoo 0.124.0 on npm (6 October 2026)	d fact	A7-S067, A7-S065, A7-S066, A7-S069
Licence	Open source: Prompty MIT; Dotprompt Apache-2.0; Promptfoo MIT	Verified fact	A7-S008, A7-S065, A7-S068
Status events	Promptfoo 'is now part of OpenAI' and 'remains open source and MIT licensed' (README); OpenAI announced its agreement to acquire Promptfoo on 9 March 2026 (terms undisclosed; closing date not published)	Verified fact	A7-S068, A7-S022, V2-S042
Strategic direction	Prompty v2 aligns the .prompty format across runtimes with generated model types and conformance vectors; Dotprompt ships IDE extensions and editors; Promptfoo remains open source after joining OpenAI	Verified fact	A7-S067, A7-S066, A7-S068
What it does	Prompts live as files in the application repository: .prompty is a markdown format with YAML front matter for model, connection and template settings; Dotprompt is an executable prompt template format extending Handlebars, agnostic to language and model provider. Changes go through code review and CI, where a CLI such as promptfoo runs evaluations against the prompt files.	Verified fact	A7-S067, A7-S066, A7-S068
Stack position	Alternative or complement to registry-based C5 tools: Git is the system of record and release follows the software change process	Architectural judgement	
Integration	Libraries in multiple languages, VS Code extension (Prompty), IDE/editor plug-ins (Dotprompt), promptfoo CLI via npm, brew or pip (Python wrapper promptfoo 0.2.0, 18 September 2026)	Verified fact	A7-S067, A7-S066, A7-S068, A7-S095
Dependencies	Source control and CI; language runtime packages for the chosen format	Verified fact	A7-S067, A7-S066
Certifications	Not applicable (pattern and open formats)	Architectural judgement	
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Free open-source formats and tools	Verified fact	A7-S008, A7-S065, A7-S068
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	LangSmith documents synchronising prompts with a GitHub repository, bridging registry and Git approaches; langchain-prompty integration package exists	Verified fact	A7-S076
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Not publicly verified	Not publicly verified	
Deployment	saas: Not applicable (pattern); managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes (files in the customer's repository); on_prem: Not publicly verified		

PromptLayer (Prompt Registry) (c5-promptlayer)

Tier: Tactical · **Flags:** none · **Original graphic label:** not in graphic

Rationale: Strong identity controls and rollout features, but security evidence is vendor-stated, pricing is unverified and it duplicates the registry built into the L9 platforms ^{AJ}.

Criterion	Score	Rationale
Technical	4	Versioned registry, release labels, staged rollout, segmentation and A/B testing, evals and tracing; approval workflow not documented.
Enterprise readiness	4	SSO, SCIM, group-role mapping, SSO enforcement, audit log and granular RBAC verified on Enterprise; SLA not verified.
Security and	3	Vendor-stated SOC 2 Type 2 with a DPA committing to annual SOC 2 Type II

Criterion	Score	Rationale
compliance		audits and penetration tests; no ISO 27001; report not seen.
Deployment flexibility	4	SaaS, customer AWS account and self-hosted (Enterprise); air-gap and on-premises not verified.
Ecosystem	3	REST API, Python/JS SDKs, OTel tracing for major provider SDKs, webhooks; smaller ecosystem than the platform vendors.
Reliability and maturity	3	Independent, continuous releases on the 1.5.x SDK line since 2022; funding not verified.
Cost / TCO	2	Pricing not publicly verified; self-hosting and identity features need an Enterprise contract.
Lock-in / portability	3	Proprietary template API, but templates are text retrievable by API; no Git sync documented.
Total (generic / FS)	3.40 / 3.40	

Capabilities. Prompt Registry with versioned templates; release labels (e.g. 'prod') select the served version and support staged rollouts, user segmentation and A/B testing via dynamic release labels; SDK fetches templates and can proxy provider calls for logging; evals and OpenTelemetry tracing alongside ^{VF1322}.

Strengths

- Release labels support staged rollout and segmentation of users to prompt versions, closest to feature-flag practice among the registries ^{VF1323}
- Enterprise Identity: SSO (SAML/OIDC via WorkOS), SCIM, group-to-role mapping, SSO enforcement, org-scoped audit log and granular RBAC ^{VF1324}
- Deployable into the customer's AWS account or self-hosted on Enterprise ^{VF1325}
- Independent; no acquisition found ^{VF1326}

Limitations and risks

- SOC 2 Type 2 is a vendor statement; no report or ISO 27001 seen ^{VF1327}
- Pricing and EU region not publicly verified ^{NPV}
- Proprietary template API; no Git synchronisation documented ^{VF1328 NPV}
- Small vendor; funding not publicly verified ^{NPV}

¹³²² Verified fact — A7-S004: promptlayer on PyPI (JSON API) (PyPI / PromptLayer) <https://pypi.org/pypi/promptlayer/json> accessed 2026-10-07; A7-S088: PromptLayer docs: Release Labels <https://raw.githubusercontent.com/MagnivOrg/prompt-layer-docs/main/features/prompt-registry/release-labels.mdx> accessed 2026-10-07

¹³²³ Verified fact — A7-S088: PromptLayer docs: Release Labels <https://raw.githubusercontent.com/MagnivOrg/prompt-layer-docs/main/features/prompt-registry/release-labels.mdx> accessed 2026-10-07

¹³²⁴ Verified fact — A7-S086: PromptLayer docs: Enterprise Identity overview <https://raw.githubusercontent.com/MagnivOrg/prompt-layer-docs/main/why-promptlayer/enterprise-identity/overview.mdx> accessed 2026-10-07

¹³²⁵ Verified fact — A7-S085: PromptLayer docs: Self-Hosted PromptLayer <https://raw.githubusercontent.com/MagnivOrg/prompt-layer-docs/main/self-hosted.mdx> accessed 2026-10-07; A7-S087: PromptLayer docs: Deploy PromptLayer on AWS <https://raw.githubusercontent.com/MagnivOrg/prompt-layer-docs/main/enterprise-deployments/aws.mdx> accessed 2026-10-07

¹³²⁶ Verified fact — A7-S004: promptlayer on PyPI (JSON API) (PyPI / PromptLayer) <https://pypi.org/pypi/promptlayer/json> accessed 2026-10-07

¹³²⁷ Verified fact — B-C5-S003: PromptLayer FAQ; Pricing; Data Processing Addendum 2026 <https://docs.promptlayer.com/features/faq> accessed 2026-10-08 [search-tool extract]

¹³²⁸ Verified fact — A7-S004: promptlayer on PyPI (JSON API) (PyPI / PromptLayer) <https://pypi.org/pypi/promptlayer/json> accessed 2026-10-07

Choose when

- You want a standalone, model- and framework-neutral prompt registry with segmentation, and can run it in your own AWS account ^{AJ}

Avoid when

- ISO 27001 is a procurement gate ^{AJ}
- You already run Langfuse or LangSmith as the L9 platform, which would duplicate the registry ^{AJ}

Nearest competitors: C5-langfuse-prompts, C5-langsmith-prompts, C5-launchdarkly-ai-configs

Regulated-FS note. Request the SOC 2 Type II report and subprocessor list before use with client data, prefer the customer-AWS deployment, and do not use the SDK proxy mode for regulated traffic unless the gateway (C1) is bypassed by design ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	PromptLayer (GitHub organisation 'MagnivOrg'); independent - no acquisition found	Verified fact	A7-S004
Category	Prompt registry and LLM engineering workbench (versioning, evals, tracing)	Verified fact	A7-S004
Version / lineup	promptlayer Python SDK 1.5.16 released 19 August 2026	Verified fact	A7-S004, V2-S028
Licence	Proprietary platform; Python SDK Apache-2.0; self-hosting requires an Enterprise licence	Verified fact	A7-S004, A7-S085
Strategic direction	Extends from prompt registry to agents and evals ('Version, test, and monitor every prompt and agent'); ships coding-agent skills and a Docs MCP server; OpenTelemetry tracing auto-instrumentation	Verified fact	A7-S004
What it does	Prompt Registry stores versioned prompt templates; release labels (e.g. 'prod') select the version served and support staged rollouts and user segmentation. The SDK fetches templates and can proxy provider SDK calls for logging; evals and tracing sit alongside.	Verified fact	A7-S004, A7-S088
Stack position	Control-plane prompt management (C5) with overlap into L9 tracing/evals	Verified fact	A7-S004
Integration	REST API and Python/JS SDKs; proxy wrapper around OpenAI and other SDKs; OpenTelemetry tracing for OpenAI, Anthropic, Google GenAI and AWS Bedrock SDKs; webhooks	Verified fact	A7-S004
Dependencies	Self-hosted: Python Flask backend, PostgreSQL 15, object storage (Amazon S3 or Google Cloud Storage), Valkey 8.1.0; AWS reference deployment uses EKS, RDS, ElastiCache and OpenSearch via OpenTofu and Helm	Verified fact	A7-S085, A7-S087
Certifications	Claims SOC 2, HIPAA and GDPR compliance (self-hosted 'inherits the same SOC 2, HIPAA, and GDPR compliance'); SOC 2 type not stated on the page fetched	Verified fact	A7-S085
GDPR / residency	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
Security features	Not publicly verified	Not publicly verified	
Access controls	Enterprise Identity: SSO (SAML/OIDC via WorkOS), SCIM directory sync, group-to-role mappings, SSO enforcement, org-scoped audit log; RBAC with granular permissions	Verified fact	A7-S086
Enterprise support	Enterprise plan required for self-hosting and Enterprise Identity; contact sales	Verified fact	A7-S085, A7-S086
Pricing	Not publicly verified	Not publicly verified	
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	promptlayer package first published on PyPI 29 December 2022	Verified fact	A7-S004
Maturity	SDK on 1.5.x line; continuous releases (1.5.16 on 19 August 2026)	Verified fact	A7-S004
Deployment	saas: Yes; managed_cloud: Not publicly verified; vpc_byoc: Yes (deploy into customer AWS account); private_cloud: Not publicly verified; self_hosted: Yes (Enterprise only); on_prem: Not publicly verified		

LaunchDarkly AgentControl (formerly AI Configs; API unchanged) (c5-launchdarkly-ai-configs)

Tier: Tactical • **Flags:** Renamed • **Original graphic label:** not in graphic

Rationale: Strongest certifications and rollout mechanics in the control, but SaaS-only with unverified residency scope and proprietary configuration delivery ^{AJ}.

Criterion	Score	Rationale
Technical	4	Runtime configuration of model, parameters and messages, per-context targeting, defaults, metrics, online evals and approvals; prompt diffing and Git-based review not documented.
Enterprise	4	SAML SSO, Enterprise custom roles, audit log API and Enterprise uptime SLA; SSO

Criterion	Score	Rationale
readiness		and roles evidence is a search extract of older pages (medium confidence); SCIM not verified.
Security and compliance	4	SOC 2 Type II, ISO 27001, ISO 27701 and FedRAMP Moderate are company-level; AgentControl's coverage is not stated, so one below 5 (CP2 Q4).
Deployment flexibility	2	Multi-tenant SaaS, an EU-hosted offering without region detail, and a US federal instance; no self-host or VPC.
Ecosystem	3	Established feature-management platform with an Apache-2.0 AI SDK; AI-specific integrations not evidenced beyond the Python SDK.
Reliability and maturity	3	Parent platform long-established, but the AI product is about 16 months GA and was renamed and repositioned in May 2026.
Cost / TCO	3	Transparent pricing (US\$10 per service connection per month with 5,000 AI runs; US\$5 per extra 1,000), but metered on every model call.
Lock-in / portability	2	Proprietary service and SDK; code-side defaults reduce runtime dependency but no export path is documented.
Total (generic / FS)	3.30 / 3.20	

Capabilities. AgentControl (formerly AI Configs; API unchanged) delivers model name, parameters and prompt messages per user context from the LaunchDarkly feature-management service, with variable interpolation, a code-side default if the service is unavailable, and a tracker recording tokens, latency and success per configuration; online evals with custom judges GA 11 March 2026; agents, trends and approvals added ^{vr-1329}.

Strengths

- Feature-flag style targeting and rollout of model and prompt configuration per user context, with a code-side default as the failure mode ^{vr-1330}
- SOC 2 Type II, ISO 27001, ISO 27701 and FedRAMP Moderate (Federal instance); Enterprise uptime SLA ^{vr-1331}
- SAML SSO with IdP-managed roles, Enterprise custom roles and an audit log API with before-and-after versions of each change ^{vr-1332}
- Published per-run pricing ^{vr-1333}

Limitations and risks

¹³²⁹ Verified fact — A7-S089: LaunchDarkly Server-Side AI SDK for Python README <https://raw.githubusercontent.com/launchdarkly/python-server-sdk-ai/main/README.md> accessed 2026-10-07; A7-S117: LaunchDarkly: Introducing AgentControl; AI Configs is now GA; AgentControl API docs <https://launchdarkly.com/blog/introducing-agentcontrol/> accessed 2026-10-07 [search-tool extract]; V2-S045: AgentControl API docs (LaunchDarkly) <https://launchdarkly.com/docs/api/agentcontrol> accessed 2026-10-08 [search-tool extract]

¹³³⁰ Verified fact — A7-S089: LaunchDarkly Server-Side AI SDK for Python README <https://raw.githubusercontent.com/launchdarkly/python-server-sdk-ai/main/README.md> accessed 2026-10-07

¹³³¹ Verified fact — A7-S119: LaunchDarkly Trust, security, and privacy; Platform overview; Security Program Addendum <https://launchdarkly.com/security/> accessed 2026-10-07 [search-tool extract]; V2-S065: LaunchDarkly Trust, security, and privacy (with Security Program Addendum, docs) <https://launchdarkly.com/security/> accessed 2026-10-08 [search-tool extract]

¹³³² Verified fact — B-C5-S001: Using SSO with LaunchDarkly; Configure SAML SSO; Creating custom roles <https://launchdarkly.com/docs/guides/account/sso> accessed 2026-10-08 [search-tool extract]; B-C5-S002: LaunchDarkly API: Audit log (list/search/get entries) <https://launchdarkly.com/docs/api/audit-log> accessed 2026-10-08 [search-tool extract]

¹³³³ Verified fact — A7-S118: LaunchDarkly Pricing; subscriptions and plans; calculating billing <https://launchdarkly.com/pricing/> accessed 2026-10-07 [search-tool extract]

- Multi-tenant SaaS only; EU residency offered but regions and data scope not detailed ^{VF-1334}
- Proprietary configuration service and SDK; no export path verified ^{VF-1335 NIPV}
- Renamed and repositioned twice in about a year (GA May 2025, AgentControl May 2026) ^{VF-1336}
- Every model call and judge run counts as a billable AI run ^{VF-1337}

Choose when

- LaunchDarkly is already the firm's feature-management standard and you need per-segment rollout and kill switches for model and prompt choices ^{AJ}

Avoid when

- Configuration must be held in the UK or EU with documented regions, or in-estate ^{AJ}
- Volume is high and per-run metering would dominate the cost ^{AJ}

Nearest competitors: C5-promptlayer, C5-langfuse-prompts, C5-prompts-as-code

Regulated-FS note. Keep the approved prompt text in Git and use AgentControl only to choose between approved, pinned variants; record the variation key on every trace, confirm EU data scope and audit-log retention on the contracted plan ^{REC}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	LaunchDarkly	Verified fact	A7-S059
Category	Runtime configuration for model, parameters and prompt messages, delivered through a feature-management platform	Verified fact	A7-S089
Version / lineup	AI Configs GA 28 May 2025; online evals GA 11 March 2026; renamed AgentControl (launch post 12 May 2026; rebrand noted by 1 April 2026); Python AI SDK launchdarkly-server-sdk-ai 1.2.0 (17 July 2026)	Verified fact	A7-S117, A7-S059
Licence	Proprietary SaaS; AI SDK Apache-2.0	Verified fact	A7-S059
Status events	28 May 2025: AI Configs GA; 11 March 2026: online evals GA; 12 May 2026: 'Introducing AgentControl' (AI Configs renamed AgentControl)	Verified fact	A7-S117
Strategic direction	Repositioned from prompt/model configuration to an operational layer for agents in production: agents, tools, trends, approvals, online evals with custom judges	Verified fact	A7-S117
What it does	Applications request an AI Config per user context; LaunchDarkly returns the model name, parameters and messages, with variable interpolation and a code-side default used if the service is unavailable. A tracker records token usage, latency and success rates per	Verified fact	A7-S089

¹³³⁴ Verified fact — A7-S119: LaunchDarkly Trust, security, and privacy; Platform overview; Security Program Addendum <https://launchdarkly.com/security/> accessed 2026-10-07 [search-tool extract]

¹³³⁵ Verified fact — A7-S089: LaunchDarkly Server-Side AI SDK for Python README <https://raw.githubusercontent.com/launchdarkly/python-server-sdk-ai/main/README.md> accessed 2026-10-07

¹³³⁶ Verified fact — A7-S117: LaunchDarkly: Introducing AgentControl; AI Configs is now GA; AgentControl API docs <https://launchdarkly.com/blog/introducing-agentcontrol/> accessed 2026-10-07 [search-tool extract]

¹³³⁷ Verified fact — A7-S118: LaunchDarkly Pricing; subscriptions and plans; calculating billing <https://launchdarkly.com/pricing/> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
	configuration.		
Stack position	Control-plane configuration (C5) on top of the existing LaunchDarkly feature-flag service	Verified fact	A7-S089
Integration	Server-side AI SDK (Python shown) wrapping LDClient; completion_config() with defaults; ManagedModel helper with automatic metrics	Verified fact	A7-S089
Dependencies	Requires the LaunchDarkly server-side SDK (LDClient) and an SDK key	Verified fact	A7-S089
Certifications	SOC 2 Type II, ISO 27001, ISO 27701, FedRAMP Moderate (Federal instance), EU-U.S. Data Privacy Framework; annual SOC 2 Type 2 audit and twice-yearly penetration tests per Security Program Addendum	Verified fact	A7-S119, V2-S065
GDPR / residency	GDPR/CCPA privacy programme; EU-hosted offering supporting EU data residency advertised (regions not detailed); multi-tenant with exceptions for large and US federal customers	Verified fact	A7-S119
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Enterprise plan with custom pricing; 'Enterprise uptime SLA' advertised	Verified fact	A7-S118, A7-S119
Pricing	Developer: US\$0 (limited AgentControl run entitlements); Foundation: US\$10 per service connection/month incl. 5,000 AI runs, US\$5 per additional 1,000 runs; Enterprise: custom. An AI run = each model call or judge run (as read 7 October 2026; cached snippets of different ages)	Verified fact	A7-S118
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	Vendor states LaunchDarkly 'serves over 100 billion feature flags daily' (undated README claim)	Verified fact	A7-S089
Maturity	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
Deployment	saas: Yes; managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Yes (dedicated US federal instance, FedRAMP Moderate); self_hosted: Not publicly verified; on_prem: Not publicly verified		

C6: AI FinOps

FinOps Open Cost and Usage Specification (FOCUS) (c6-finops-focus)

Tier: Strategic · **Flags:** none · **Original graphic label:** not in graphic

Rationale: Open, foundation-governed schema that removes tool lock-in from cost data; strategic as the data model, not as a product ^{AJ}.

Criterion	Score	Rationale
Technical	3	Covers cost and usage normalisation; AI model identity not yet ratified and token type deferred.
Enterprise readiness	3	Rubric rule 2: enables a common cost dataset and chargeback inside the estate; no commercial support of its own.
Security and compliance	3	Rubric rule 2: a specification with clear CC BY 4.0 licence and a validator; inherits the controls of the data platform it lands in.
Deployment flexibility	5	Specification; implemented anywhere.
Ecosystem	4	FinOps Foundation, validator and vendor mappings (e.g. Vantage); AI-provider adoption not verified.
Reliability and maturity	4	Ratified releases on a semi-annual cadence since 2024 under the Joint Development Foundation; AI scope still in progress.
Cost / TCO	5	Free open specification.
Lock-in / portability	5	Open standard, permissive licence, neutral governance.
Total (generic / FS)	3.80	
	/	

Criterion	Score	Rationale
	3.85	

Evidence rules applied: Rubric rule 2 (open specification): enterprise readiness and security scored on what the specification enables and on hygiene, each capped at 4; inherits host controls

Capabilities. Open specification for cost and usage data. FOCUS 1.4 was ratified on 4 June 2026; 1.2 added virtual-currency columns for credits and tokens; tokens are carried today through SKU IDs, ConsumedUnit and ConsumedQuantity. FOCUS 1.5 (no ratification date) is scoped to add AI pricing dimensions on SkuPriceDetails and four model-identity properties (ModelDeveloper, ModelFamily, ModelId, ModelVersion) with no new columns; a first-class token-type (input/output) column is deferred ^{VF1338}.

Strengths

- Vendor-neutral target schema for normalising gateway logs, provider invoices and cloud bills ^{VF1339 AJ}
- Free, CC BY 4.0, with a conformance validator (2.2.1, August 2026) ^{VF1340}
- Steady cadence: six public releases since June 2023 ^{VF1341}
- FinOps Framework 2026 adds a FinOps for AI category; vendors such as Vantage map to FOCUS ^{VF1342}

Limitations and risks

- No AI-specific columns today; model identity arrives only with 1.5, which has no date, and token type is deferred ^{VF1343}
- It is a data model only: no metering, budgets or enforcement ^{AJ}
- Adoption by AI providers not publicly verified ^{NPV}

Choose when

- Always, as the target schema for the firm's AI cost dataset ^{REC}

Avoid when

- Do not wait for 1.5 to start: map tokens to SKU and ConsumedQuantity now ^{REC}

¹³³⁸ Verified fact — A7-S049: FOCUS Specification CHANGELOG (FinOps Foundation (FOCUS Working Group)) https://raw.githubusercontent.com/FinOps-Open-Cost-and-Usage-Spec/FOCUS_Spec/working_draft/CHANGELOG.md accessed 2026-10-07; A7-S116: FOCUS 1.5 Release Scope; FOCUS for AI; Introducing FOCUS 1.2; Introducing FOCUS 1.4 (FinOps Foundation (FOCUS)) <https://focus.finops.org/focus-1-5-release-scope/> accessed 2026-10-07 [search-tool extract]; V2-S046: FOCUS 1.5 Release Scope: Confirmed and Stretch Features (with Introducing FOCUS 1.4; FOCUS for AI) (FinOps Foundation (FOCUS)) <https://focus.finops.org/focus-1-5-release-scope/> accessed 2026-10-08 [search-tool extract]

¹³³⁹ Verified fact — A7-S048: FOCUS Specification README (FinOps Foundation (FOCUS Working Group)) https://raw.githubusercontent.com/FinOps-Open-Cost-and-Usage-Spec/FOCUS_Spec/working_draft/README.md accessed 2026-10-07

¹³⁴⁰ Verified fact — A7-S096: FOCUS Specification licence (FinOps Foundation (FOCUS Working Group)) https://raw.githubusercontent.com/FinOps-Open-Cost-and-Usage-Spec/FOCUS_Spec/working_draft/license.md accessed 2026-10-07; A7-S050: focus-validator on PyPI (PyPI / FinOps Foundation) <https://pypi.org/pypi/focus-validator/json> accessed 2026-10-07

¹³⁴¹ Verified fact — A7-S049: FOCUS Specification CHANGELOG (FinOps Foundation (FOCUS Working Group)) https://raw.githubusercontent.com/FinOps-Open-Cost-and-Usage-Spec/FOCUS_Spec/working_draft/CHANGELOG.md accessed 2026-10-07

¹³⁴² Verified fact — A7-S115: FinOps Framework 2026; State of FinOps 2026; FinOps for AI technology category; Tokenomics working group; Who sets the AI budget? (FinOps Foundation) <https://www.finops.org/insights/2026-finops-framework/> accessed 2026-10-07 [search-tool extract]; A7-S113: Vantage blog: Managed AI Tags; Support for Anthropic Costs; Support for Claude Enterprise Costs; OpenAI cost tools <https://www.vantage.sh/blog/managed-ai-tags> accessed 2026-10-07 [search-tool extract]

¹³⁴³ Verified fact — A7-S116: FOCUS 1.5 Release Scope; FOCUS for AI; Introducing FOCUS 1.2; Introducing FOCUS 1.4 (FinOps Foundation (FOCUS)) <https://focus.finops.org/focus-1-5-release-scope/> accessed 2026-10-07 [search-tool extract]; V2-S046: FOCUS 1.5 Release Scope: Confirmed and Stretch Features (with Introducing FOCUS 1.4; FOCUS for AI) (FinOps Foundation (FOCUS)) <https://focus.finops.org/focus-1-5-release-scope/> accessed 2026-10-08 [search-tool extract]

Nearest competitors: C6-vantage, C6-cloudzero, C6-gateway-cost-attribution

Regulated-FS note. A FOCUS-shaped AI cost dataset makes provider spend comparable for concentration analysis and lets cost allocation feed the outsourcing register without vendor-specific mapping ^{Rec.}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	FinOps Foundation FOCUS Working Group (copyright Joint Development Foundation Projects, LLC, FOCUS Series)	Verified fact	A7-S048, A7-S096
Category	Open specification (billing/cost and usage data schema)	Verified fact	A7-S048
Version / lineup	v1.4 ratified by the FOCUS Steering Committee on 4 June 2026 (adds BillingPeriod and InvoiceDetail datasets, 47 columns); v1.3 December 2025; v1.2 June 2025; v1.1 November 2024; v1.0 20 June 2024	Verified fact	A7-S049, V2-S046
Licence	Open specification; documents under Creative Commons Attribution 4.0 International	Verified fact	A7-S096
Status events	4 June 2026: FOCUS 1.4 ratified (Invoice Detail and Billing Period datasets; zero incompatible changes); validator support for 1.4 expected later in Q3 2026	Verified fact	V2-S046
Strategic direction	FinOps Framework 2026 adds a 'FinOps for AI' technology category; State of FinOps 2026 ranks FinOps for AI the top forward-looking priority (98% of respondents manage AI spend); FOCUS 1.5 (in progress; no ratification date) adds a Price Sheet and, for AI, confirmed scope of AI pricing dimensions (cached vs fresh tokens, global vs regional serving) as properties on SkuPriceDetails plus four model properties (ModelDeveloper, ModelFamily, ModelId, ModelVersion) with no new columns; a first-class input/output token-type column is deferred, so token splits stay in separate SKUs (SkuMeter) and Consu	Verified fact	A7-S115, A7-S116, V2-S046
What it does	Defines standard datasets, columns (dimensions and metrics) and requirements so billing and usage data from different providers are interoperable and comparable.	Verified fact	A7-S048
Stack position	Data-model standard underneath C6: target schema for normalising gateway token logs and provider invoices before allocation	Architectural judgement	

Field	Value	Label	Sources
Integration	Spec plus validation tooling; focus-validator 2.2.1 released 5 August 2026	Verified fact	A7-S048, A7-S050
Dependencies	Not publicly verified	Not publicly verified	
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Free open specification	Verified fact	A7-S096
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Tokens are representable today via FOCUS 1.2 virtual-currency columns and SKU/ConsumedUnit/ConsumedQuantity, with no AI-specific columns; AI-specific fields (model identity, input/output tokens; stretch: cached vs fresh tokens, global vs regional serving) are scoped for FOCUS 1.5 (release date not confirmed). Vendors such as Vantage map their schemas to FOCUS.	Verified fact	A7-S116, A7-S113
Adoption signals	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
Maturity	Six public releases since v0.5 (June 2023); semi-annual cadence	verified	A7-S049
Deployment	saas: Not publicly verified; managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Not applicable (specification); on_prem: Not publicly verified	Verified fact	

Gateway-native and provider-native token cost tracking, budgets and chargeback (pattern) (c6-gateway-cost-attribution)

Tier: Strategic • **Flags:** Acquired • **Original graphic label:** not in graphic

Rationale: The only place where AI spend can be attributed per request and stopped in real time; implementations are replaceable behind the OpenAI-compatible interface. Acquired refers to Portkey ^{AI}.

Criterion	Score	Rationale
Technical	4	Per-key/team/user/customer budgets, rate limits, spend tracking, provider and cloud reconciliation data; cost per task needs a trace join and some controls fail open by default.
Enterprise readiness	4	Rubric rule 2 (pattern): implementations offer SSO, RBAC and audit (LiteLLM Enterprise SSO, RBAC, SCIM and audit logs; cloud IAM under CP2 Q1); capped at 4.
Security and compliance	3	Depends on implementation: LiteLLM SOC 2 Type 2 (September 2026), Cloudflare AI Gateway named in SOC 2 scope; LiteLLM's March 2026 supply-chain incident; Admin API keys are powerful credentials.
Deployment flexibility	5	Self-hosted gateways, managed cloud gateways and SaaS gateways; runs in-estate or air-gapped.
Ecosystem	4	OpenAI-compatible gateway endpoints and provider admin APIs; FOCUS mapping not native.
Reliability and maturity	3	Cloud cost tags are mature, but Foundry project attribution is preview, LiteLLM had a supply-chain incident, and Portkey changed owner.
Cost / TCO	4	Open-source core; a database is needed for budgets and spend tracking.
Lock-in / portability	3	Base 4 (open interfaces, OSS gateways) reduced by 1 because a named

Criterion	Score	Rationale
		implementation (Portkey) changed owner; cost data models differ per gateway.
Total (generic / FS)	3.85 / 3.70	

Evidence rules applied: Rubric rule 2 (pattern): enterprise readiness and security scored on what implementations enable and on their hygiene, each capped at 4; platform controls presumed for cloud gateways (CP2 Q1); confirm per service

Capabilities. Pattern: the gateway (C1) issues virtual keys per team, project, user or customer and records tokens and cost per request, enforcing budgets and rate limits (LiteLLM: budgets at key, user, team and customer level, TPM/RPM limits); provider admin APIs (Anthropic Usage & Cost Admin API) and cloud billing tags (Bedrock application inference profiles, Microsoft Foundry project tags) are the reconciliation sources^{vf1344}.

Strengths

- The gateway sees every model call, so it is the only point that can both attribute and stop spend in real time^{AI}
- Mature open-source implementation with budgets, limits and spend tracking (LiteLLM MIT core) and managed equivalents in the cloud gateways (token limits and quotas in Azure APIM, Apigee and Kong; spend limits in Cloudflare)^{vf1345}
- Provider-side data breaks tokens into uncached input, cached input, cache creation and output, by API key, workspace, model and service tier, for reconciliation^{vf1346}
- Cloud billing tags carry AI spend into the existing cost tools (Cost Explorer and CUR; Azure cost analysis)^{vf1347}

Limitations and risks

- Some LiteLLM controls fail open or are off by default: max_budget without a database, and a proxy admin key that bypasses budget checks^{vf1348}

¹³⁴⁴ Verified fact — A7-S010: litellm on PyPI (JSON API) (PyPI / BerriAI) <https://pypi.org/pypi/litellm/json> accessed 2026-10-07; A6-S015: LiteLLM docs: MCP Overview, Agent Gateway (A2A), Caching, Budgets/Rate Limits, v1.89.0 release notes (BerriAI (LiteLLM)) <https://docs.litellm.ai/docs/mcp> accessed 2026-10-07 [search-tool extract]; A7-S070: Usage and Cost Admin API (Claude Platform docs) (Anthropic) <https://platform.claude.com/docs/en/manage-claude/usage-cost-api> accessed 2026-10-07; B-C6-S004: Amazon Bedrock user guide: Application inference profiles; Best practices for cost attribution (Amazon Web Services) <https://docs.aws.amazon.com/bedrock/latest/userguide/cost-mgmt-application-inference-profiles.html> accessed 2026-10-08 [search-tool extract]; B-C6-S005: Plan and manage costs - Microsoft Foundry <https://learn.microsoft.com/en-us/azure/foundry/concepts/manage-costs> accessed 2026-10-08 [search-tool extract]

¹³⁴⁵ Verified fact — A6-S015: LiteLLM docs: MCP Overview, Agent Gateway (A2A), Caching, Budgets/Rate Limits, v1.89.0 release notes (BerriAI (LiteLLM)) <https://docs.litellm.ai/docs/mcp> accessed 2026-10-07 [search-tool extract]; A6-S053: AI gateway capabilities in Azure API Management (Microsoft Learn source, ms.date 29 May 2026) <https://raw.githubusercontent.com/MicrosoftDocs/azure-docs/main/articles/api-management/genai-gateway-capabilities.md> accessed 2026-10-07; A6-S024: Apigee AI solutions: gateway to enterprise-ready AI (Google Cloud) <https://cloud.google.com/solutions/apigee-ai> accessed 2026-10-07; A6-S016: Kong AI Gateway 2.0 Is Now GA — And It's Already Moving Faster (Kong Inc.) <https://konghq.com/blog/product-releases/kong-ai-gateway-2-0-ga> accessed 2026-10-07 [search-tool extract]; A6-S019: Cloudflare AI Gateway docs: Pricing, Features, Changelog; Cloudflare blog spend limits (Cloudflare, Inc.) <https://developers.cloudflare.com/ai-gateway/reference/pricing/> accessed 2026-10-07 [search-tool extract]

¹³⁴⁶ Verified fact — A7-S070: Usage and Cost Admin API (Claude Platform docs) (Anthropic) <https://platform.claude.com/docs/en/manage-claude/usage-cost-api> accessed 2026-10-07

¹³⁴⁷ Verified fact — B-C6-S004: Amazon Bedrock user guide: Application inference profiles; Best practices for cost attribution (Amazon Web Services) <https://docs.aws.amazon.com/bedrock/latest/userguide/cost-mgmt-application-inference-profiles.html> accessed 2026-10-08 [search-tool extract]; B-C6-S005: Plan and manage costs - Microsoft Foundry <https://learn.microsoft.com/en-us/azure/foundry/concepts/manage-costs> accessed 2026-10-08 [search-tool extract]

- Cloud billing tags are aggregated per usage type per day, not per request; per-request attribution needs invocation logs or gateway data ^{VF-1349}
- Gateway consolidation: Portkey acquired by Palo Alto Networks (completed 29 May 2026) and folded into Prisma AIRS ^{VF-1350}
- LiteLLM suffered a PyPI supply-chain compromise on 24 March 2026 ^{VF-1351}
- Cost per task needs a join between gateway records and traces (L9); no product does this end to end in the fact base ^{AJ}

Choose when

- Always, as the metering and enforcement point for model spend ^{Rec}

Avoid when

- Never as the only record: reconcile against provider invoices and cloud bills monthly ^{Rec}

Nearest competitors: C6-helicone, C6-vantage, C6-cloudzero

Regulated-FS note. Budget enforcement is an operational-resilience control against runaway agents; test that budgets fail closed, restrict the admin key, and reconcile gateway spend to invoices for the outsourcing register ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Pattern; implementations include LiteLLM (BerriAI), Portkey (acquired by Palo Alto Networks), Helicone, and model-provider admin APIs such as Anthropic's Usage & Cost Admin API	Verified fact	A7-S010, A7-S016, A7-S046, A7-S070
Category	Named pattern (AI FinOps: metering, attribution, budgets, showback/chargeback)	Verified fact	A7-S010, A7-S070
Version / lineup	Examples as of 7 October 2026: LiteLLM 1.104.1 (7 October 2026); portkey-ai SDK 2.3.4 (23 July 2026); Anthropic Usage & Cost Admin API (current docs)	Verified fact	A7-S010, A7-S081, A7-S070
Licence	Mixed: LiteLLM MIT (open core gateway); Portkey SDK MIT; provider APIs proprietary	Verified fact	A7-S010, A7-S081, A7-S070
Status events	29 May 2026: Portkey, Inc. acquired by Palo Alto Networks (Form 10-Q); 16 July 2026: Palo Alto Networks AI Gateway GA (product page)	Verified fact	A7-S016, A7-S028, V2-S025,

¹³⁴⁸ Verified fact — A6-S015: LiteLLM docs: MCP Overview, Agent Gateway (A2A), Caching, Budgets/Rate Limits, v1.89.0 release notes (BerriAI (LiteLLM)) <https://docs.litellm.ai/docs/mcp> accessed 2026-10-07 [search-tool extract]

¹³⁴⁹ Verified fact — B-C6-S004: Amazon Bedrock user guide: Application inference profiles; Best practices for cost attribution (Amazon Web Services) <https://docs.aws.amazon.com/bedrock/latest/userguide/cost-mgmt-application-inference-profiles.html> accessed 2026-10-08 [search-tool extract]

¹³⁵⁰ Verified fact — A6-S011: Palo Alto Networks Completes Acquisition of Portkey to Secure AI Agents (and 30 Apr 2026 announcement) <https://www.paloaltonetworks.com/company/press/2026/palo-alto-networks-completes-acquisition-of-portkey-to-secure-ai-agents> accessed 2026-10-07 [search-tool extract]; V2-S025: Palo Alto Networks Completes Acquisition of Portkey to Secure AI Agents (with PANW FY2026 10-K and 10-Q, SDxCentral) <https://www.paloaltonetworks.com/company/press/2026/palo-alto-networks-completes-acquisition-of-portkey-to-secure-ai-agents> accessed 2026-10-08 [search-tool extract]

¹³⁵¹ Verified fact — A6-S008: Security Update: Suspected Supply Chain Incident (and Security Townhall Updates) (BerriAI (LiteLLM)) <https://docs.litellm.ai/blog/security-update-march-2026> accessed 2026-10-07 [search-tool extract]; V2-S027: LiteLLM PyPI supply chain attack (with NetSPI, Comet, Snyk/Datadog reporting) (FutureSearch) <https://futuresearch.ai/blog/litellm-pypi-supply-chain-attack/> accessed 2026-10-08 [search-tool extract]

Field	Value	Label	Sources
			V2-S048
Strategic direction	Gateways are being absorbed by security vendors: Palo Alto Networks acquired Portkey (29 May 2026) 'to enhance our Prisma AIRS capabilities' and made its AI Gateway generally available on 16 July 2026	Verified fact	A7-S016, A7-S028
What it does	A gateway issues virtual keys per team, project or user and records token usage and cost per request, enabling budgets and multi-tenant spend reports (LiteLLM: 'multi-tenant cost tracking and spend management per project/user'). Model providers expose organisation-level usage and cost data by API key, workspace, model and service tier (Anthropic Usage & Cost Admin API), which can be reconciled with gateway records.	Verified fact	A7-S010, A7-S070
Stack position	Spans C1 (gateway) and C6; the gateway is the natural metering point because every model call passes through it, while provider billing APIs and cloud bills are the reconciliation source	Architectural judgement	
Integration	OpenAI-compatible gateway endpoints with virtual keys; provider admin REST APIs (Anthropic Admin API key required; workspace keys do not work)	Verified fact	A7-S010, A7-S070
Dependencies	Not publicly verified	Not publicly verified	
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
Pricing	Depends on implementation; open-source gateways have no licence fee for the core	Architectural judgement	
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	FOCUS provides a vendor-neutral cost/usage schema (AI token fields scoped for FOCUS 1.5); cost platforms such as Vantage and CloudZero ingest provider token data via admin APIs for allocation	Verified fact	A7-S048, A7-S116, A7-S113, A7-S114
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Not publicly verified	Not publicly verified	
Deployment	saas: Yes (e.g. hosted gateways, provider APIs); managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes (e.g. LiteLLM proxy, Helicone self-hosted); on_prem: Not publicly verified		

Vantage (cloud cost management platform) (c6-vantage)

Tier: Tactical · **Flags:** none · **Original graphic label:** not in graphic

Rationale: Useful reporting layer for organisations already using it; SaaS-only with unverified residency and vendor-described AI features ^{A1}.

Criterion	Score	Rationale
Technical	3	Provider integrations, AI tag normalisation and hierarchical allocation; billing-level granularity only, and features described in vendor blogs.
Enterprise readiness	3	SAML SSO verified (lifts NPV cap to max 3 under CP2 Q2); RBAC and audit logs not verified; SAML plan availability conflicts between pages.
Security and compliance	3	SOC 2 Type 2 and SOC 1 Type 2, reports on request; no ISO 27001 found.
Deployment flexibility	2	SaaS only; regions not verified.
Ecosystem	4	AI-provider and cloud billing integrations, API, MCP server and FOCUS-mapped schema.

Criterion	Score	Rationale
Reliability and maturity	3	Established cloud cost vendor; AI features added from 2025; ownership and funding not verified.
Cost / TCO	3	Managed AI Tags at no extra cost; AI spend counts toward quota tiers whose prices were not verified.
Lock-in / portability	3	Proprietary SaaS, but FOCUS-mapped data and an API make export feasible.
Total (generic / FS)	2.95 / 2.90	

Capabilities. Cloud cost management platform that ingests cloud and AI-provider billing (Anthropic tokens by model, workspace, API key and service tier; Claude Enterprise; OpenAI tokens by model and operation), allocates it via virtual tagging and hierarchical allocation, normalises AI spend with Managed AI Tags (model, provider, token type) and exposes data through an API and an MCP server; its internal schema maps to FOCUS ^{VF1352}.

Strengths

- Direct AI-provider integrations, with Anthropic GA September 2025 and Claude Enterprise added July 2026 ^{VF1353}
- Managed AI Tags normalise model, provider and token type across providers at no extra cost ^{VF1354}
- SOC 1 Type 2 and SOC 2 Type 2 (reports on request) and self-service SAML SSO ^{VF1355}
- Schema mapped to FOCUS ^{VF1356}

Limitations and risks

- AI cost features are described in vendor blog posts ^{VF1357}
- Anthropic integration uses an Admin API key, which has revocable read-write access ^{VF1358}
- SaaS only; EU region, RBAC and audit logs not publicly verified ^{NPV}
- Billing data is per provider account, not per task; cost per commentary still needs gateway and trace data ^{AJ}

Choose when

¹³⁵² Verified fact — A7-S113: Vantage blog: Managed AI Tags; Support for Anthropic Costs; Support for Claude Enterprise Costs; OpenAI cost tools <https://www.vantage.sh/blog/managed-ai-tags> accessed 2026-10-07 [search-tool extract]; A7-S090: Vantage MCP Server README <https://raw.githubusercontent.com/vantage-sh/vantage-mcp-server/main/README.md> accessed 2026-10-07

¹³⁵³ Verified fact — A7-S113: Vantage blog: Managed AI Tags; Support for Anthropic Costs; Support for Claude Enterprise Costs; OpenAI cost tools <https://www.vantage.sh/blog/managed-ai-tags> accessed 2026-10-07 [search-tool extract]

¹³⁵⁴ Verified fact — A7-S113: Vantage blog: Managed AI Tags; Support for Anthropic Costs; Support for Claude Enterprise Costs; OpenAI cost tools <https://www.vantage.sh/blog/managed-ai-tags> accessed 2026-10-07 [search-tool extract]

¹³⁵⁵ Verified fact — B-C6-S001: Vantage Security and Compliance; Vantage docs: Single Sign-On; SOC 1 Type 2 blog <https://www.vantage.sh/legal/security> accessed 2026-10-08 [search-tool extract]

¹³⁵⁶ Verified fact — A7-S113: Vantage blog: Managed AI Tags; Support for Anthropic Costs; Support for Claude Enterprise Costs; OpenAI cost tools <https://www.vantage.sh/blog/managed-ai-tags> accessed 2026-10-07 [search-tool extract]

¹³⁵⁷ Verified fact — A7-S113: Vantage blog: Managed AI Tags; Support for Anthropic Costs; Support for Claude Enterprise Costs; OpenAI cost tools <https://www.vantage.sh/blog/managed-ai-tags> accessed 2026-10-07 [search-tool extract]

¹³⁵⁸ Verified fact — A7-S113: Vantage blog: Managed AI Tags; Support for Anthropic Costs; Support for Claude Enterprise Costs; OpenAI cost tools <https://www.vantage.sh/blog/managed-ai-tags> accessed 2026-10-07 [search-tool extract]

- Vantage is already the cloud FinOps tool and you want AI provider spend in the same allocation and showback model ^{AJ}

Avoid when

- You cannot accept a third party holding a read-write provider admin key ^{AJ}
- You need per-request or per-task attribution, which belongs to the gateway ^{AJ}

Nearest competitors: C6-cloudzero, C6-gateway-cost-attribution, C6-finops-focus

Regulated-FS note. Treat it as the showback and chargeback reporting layer, fed by provider billing and gateway exports; assess the admin-key access under C4 and C7 and the platform under the outsourcing register ^{Rec}.

Facts (as of 2026-10-07)

Field	Value	Label	Sources
Company	Vantage (vantage.sh)	Verified fact	A7-S090
Category	Cloud cost management / FinOps platform	Verified fact	A7-S090
Version / lineup	Not publicly verified	Not publicly verified	
Licence	Not publicly verified	Not publicly verified	
Strategic direction	AI cost management: native Anthropic API (GA September 2025), Claude Enterprise (July 2026) and OpenAI integrations; Managed AI Tags (vntg:ai: model, provider, token type) normalise AI spend across providers; hosted MCP server (OAuth 2.1) for cost queries	Verified fact	A7-S113, A7-S090
What it does	Ingests cloud and AI-provider billing (token counts by model, workspace, API key, service tier for Anthropic; by model and operation for OpenAI) and allocates costs via virtual tagging and hierarchical allocation; exposes data through API and MCP.	Verified fact	A7-S113, A7-S090

Field	Value	Label	Sources
Stack position	Not publicly verified	Not publicly verified	
Integration	Provider admin API keys (Anthropic Admin API, Claude Enterprise Analytics API), cloud billing integrations; Vantage API and MCP server; internal VQL schema maps to FOCUS	Verified fact	A7-S113, A7-S090
Dependencies	Not publicly verified	Not publicly verified	
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Managed AI Tags at no additional cost; AI provider costs count toward Vantage quota tiers (tier prices not verified)	Verified fact	A7-S113
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
Adoption signals	Not publicly verified	verified	Not publicly verified
Maturity	Not publicly verified	Not publicly verified	
Deployment	saas: Yes; managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Not publicly verified; on_prem: Not publicly verified		

CloudZero (C6-cloudzero)

Tier: Tactical · Flags: none · Original graphic label: not in graphic

Rationale: Credible allocation features and SOC 2, but SaaS-only, proprietary allocation logic, unverified pricing and vendor-described AI claims ^{AJ}.

Criterion	Score	Rationale
Technical	3	Token-level ingestion across four AI sources and allocation by customer, feature and team; claims are vendor blog posts.
Enterprise readiness	3	SAML SSO and role-based data access verified; audit logs not verified; max 3 under CP2 Q2.
Security and compliance	3	SOC 2 Type 2 and SOC 1 Type 2 listed; no ISO 27001 found.
Deployment flexibility	2	SaaS receiving telemetry; regions not verified.
Ecosystem	3	Integrations for major AI providers and clouds plus a Kubernetes agent; FOCUS not evidenced.
Reliability and maturity	3	Established cost vendor with SOC 1 since 2022; ownership and funding not verified.
Cost / TCO	2	Pricing not publicly verified.

Criterion	Score	Rationale
Lock-in / portability	2	Proprietary allocation engine (CostFormation); export and FOCUS support not verified.
Total (generic / FS)	2.70 / 2.65	

Capabilities. Cloud cost intelligence platform that ingests token-level usage and cost from OpenAI, the Anthropic API, Amazon Bedrock (including Claude Platform on AWS) and Azure OpenAI, and allocates it by customer, feature, team, product and environment through its CostFormation engine; a Kubernetes agent supplies container telemetry ^{VF-1359}.

Strengths

- Anthropic ingestion breaks out uncached input, cache hits, output and tool usage by model, workspace and API key (vendor statement) ^{VF-1360}
- Allocation by customer and feature supports unit economics ^{VF-1361}
- SOC 1 Type 2 and SOC 2 Type 2 listed in the Trust Center; SAML SSO with custom roles that scope data access and map to IdP groups ^{VF-1362}

Limitations and risks

- AI cost claims are from CloudZero's own blog posts ^{VF-1363}
- No ISO 27001 found; SOC reports under NDA ^{VF-1364}
- Pricing, EU region and audit logs not publicly verified ^{NPV}
- Proprietary allocation engine; FOCUS support not evidenced ^{NPV}

Choose when

- CloudZero is already the cost platform and you want AI spend allocated by customer or product next to Kubernetes costs ^{AJ}

Avoid when

- Residency of billing metadata must be documented in the UK or EU ^{AJ}
- You want allocation rules in an open schema ^{AJ}

Nearest competitors: C6-vantage, C6-gateway-cost-attribution, C6-finops-focus

¹³⁵⁹ Verified fact — A7-S114: CloudZero blog posts on AI cost (token-based pricing, Bedrock, Anthropic, Azure OpenAI) <https://www.cloudzero.com/blog/token-based-pricing/> accessed 2026-10-07 [search-tool extract]; A7-S091: CloudZero Agent README <https://raw.githubusercontent.com/Cloudzero/cloudzero-agent/develop/README.md> accessed 2026-10-07

¹³⁶⁰ Verified fact — A7-S114: CloudZero blog posts on AI cost (token-based pricing, Bedrock, Anthropic, Azure OpenAI) <https://www.cloudzero.com/blog/token-based-pricing/> accessed 2026-10-07 [search-tool extract]

¹³⁶¹ Verified fact — A7-S114: CloudZero blog posts on AI cost (token-based pricing, Bedrock, Anthropic, Azure OpenAI) <https://www.cloudzero.com/blog/token-based-pricing/> accessed 2026-10-07 [search-tool extract]

¹³⁶² Verified fact — B-C6-S002: CloudZero Trust Center; CloudZero security page <https://trust.cloudzero.com/> accessed 2026-10-08 [search-tool extract]; B-C6-S003: CloudZero docs: Users & Permissions; SSO with SAML; View and Manage Roles <https://docs.cloudzero.com/docs/users-and-permissions> accessed 2026-10-08 [search-tool extract]

¹³⁶³ Verified fact — A7-S114: CloudZero blog posts on AI cost (token-based pricing, Bedrock, Anthropic, Azure OpenAI) <https://www.cloudzero.com/blog/token-based-pricing/> accessed 2026-10-07 [search-tool extract]

¹³⁶⁴ Verified fact — B-C6-S002: CloudZero Trust Center; CloudZero security page <https://trust.cloudzero.com/> accessed 2026-10-08 [search-tool extract]

Regulated-FS note. Use for showback where already contracted; keep the allocation rules documented outside the tool so chargeback survives an exit ^{Rec.}.

Facts (as of 2026-10-07)

Field	Value	Label	Sources
Company	CloudZero	Verified fact	A7-S091
Category	Cloud cost intelligence / allocation platform	Verified fact	A7-S091
Version / lineup	Not publicly verified	Not publicly verified	
Licence	Not publicly verified	Not publicly verified	
Strategic direction	Publishes extensive AI-pricing content and positions on AI cost allocation: integrations for OpenAI, Anthropic direct API, Amazon Bedrock, Claude Platform on AWS and Azure OpenAI	Verified fact	A7-S114
What it does	Ingests token-level usage and cost from AI providers and cloud bills and allocates it by customer, feature, team, product and environment via the CostFormation engine; a Kubernetes agent supplies container telemetry.	Verified fact	A7-S114, A7-S091
Stack position	Not publicly verified	Not publicly verified	
Integration	Kubernetes agent (Prometheus-compatible collector) shipping to CloudZero via upload API and S3	Verified fact	A7-S091
Dependencies	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Not publicly verified	Not publicly verified	
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	Vendor claims to be the first cloud cost platform to integrate directly with Anthropic; reports customers' Bedrock spend typically 1.5x-2x initial estimates	Verified fact	A7-S114
Maturity	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
Deployment	saas: Yes (platform receives telemetry); managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Not publicly verified; on_prem: Not publicly verified		

Helicone (AI Gateway and LLM Observability Platform) (C6-helicone)

Tier: Experimental • **Flags:** Acquired, Not recommended • **Original graphic label:** not in graphic

Rationale: Maintenance mode after the Mintlify acquisition makes it unsuitable as a dependency; not recommended for new use ^{AJ}.

Criterion	Score	Rationale
Technical	3	Per-request, session and user cost and latency tracking with a gateway and prompt versioning; no budgets or chargeback features evidenced.
Enterprise readiness	2	SSO, RBAC and audit logs not publicly verified; NPV cap.
Security and compliance	2	SOC 2 claimed without type in a README; NPV cap.
Deployment flexibility	3	SaaS and self-hosted (Docker; Helm on request for enterprise).
Ecosystem	3	OpenAI-compatible base URL, integrations for major providers and frameworks, PostHog export.
Reliability and maturity	1	Maintenance mode after the Mintlify acquisition (3 March 2026); feature development wound down.
Cost / TCO	4	Apache-2.0; free tier of 10,000 requests per month.
Lock-in / portability	3	Base 4 (Apache-2.0, base-URL integration) reduced by 1 for the ownership change; governance is company-controlled.
Total (generic / FS)	2.60 / 2.50	

Evidence rules applied: enterprise_readiness capped at 2: SSO, RBAC and audit logs NPV; security_compliance capped at 2: SOC 2 type not stated, no trust-centre report

Capabilities. Proxy or logging layer for LLM requests that tracks cost, latency and quality per request, session and user, with prompt versioning and an OpenAI-compatible AI gateway with fallbacks ^{vr1365}.

Strengths

- Apache-2.0 and self-hostable with Docker ^{vr1366}
- Simple integration by changing the base URL ^{vr1367}

Limitations and risks

¹³⁶⁵ Verified fact — A7-S046: Helicone README <https://raw.githubusercontent.com/Helicone/helicone/main/README.md> accessed 2026-10-07

¹³⁶⁶ Verified fact — A7-S046: Helicone README <https://raw.githubusercontent.com/Helicone/helicone/main/README.md> accessed 2026-10-07

¹³⁶⁷ Verified fact — A7-S046: Helicone README <https://raw.githubusercontent.com/Helicone/helicone/main/README.md> accessed 2026-10-07

- Acquired by Mintlify on 3 March 2026; in maintenance mode with security fixes, bug fixes and new model support only ^{VF-1368}
- No npm helper release since 7 November 2025 ^{VF-1369}
- 'SOC 2 and GDPR compliant' stated without SOC 2 type; access controls not publicly verified ^{VF-1370} ^{NPV}
- A competitor states new sign-ups are disabled (single rival source) ^{NPV}

Choose when

- Only as a short-term bridge for an existing deployment while migrating ^{AJ}

Avoid when

- Any new deployment ^{Rec}

Nearest competitors: C6-gateway-cost-attribution, L9-langfuse, C1-litellm

Regulated-FS note. Plan migration of existing usage to the gateway (C1) and the L9 platform; record the change of ownership and maintenance status in third-party risk files ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Helicone, acquired by Mintlify (announced and completed 3 March 2026)	Verified fact	A7-S112, V2-S043
Category	AI gateway plus LLM observability with cost and latency tracking	Verified fact	A7-S046
Version / lineup	No versioned platform release found; @helicone/helpers npm package last published 1.8.3 on 7 November 2025; legacy Python package 'helicone' last released 1.0.14 on 3 November 2023	Verified fact	A7-S047, A7-S005
Licence	Open source, Apache-2.0	Verified fact	A7-S046, A7-S005
Status events	3 March 2026: acquired by Mintlify; product in maintenance mode (security and bug fixes, new model support; standalone feature development wound down)	Verified fact	A7-S112, V2-S043
Strategic direction	After acquisition, founders moved to Mintlify to work on AI knowledge infrastructure; Helicone in maintenance mode	Verified fact	A7-S112
What it does	Proxies or logs LLM requests, then tracks cost, latency and quality per request, session and user; supports prompt versioning and an AI gateway with fallbacks.	Verified fact	A7-S046
Stack position	Overlaps C1 (gateway), L9 (observability) and C6 (cost tracking)	Verified fact	A7-S046

¹³⁶⁸ Verified fact — A7-S112: Helicone Blog: Helicone acquired by Mintlify (and Mintlify announcement) (Helicone / Mintlify) <https://www.helicone.ai/blog> accessed 2026-10-07 [search-tool extract]; V2-S043: Helicone Joins Mintlify (with mintlify.com 'Mintlify acquires Helicone'; PostHog comparison) (Helicone / Mintlify) <https://www.helicone.ai/blog/joining-mintlify> accessed 2026-10-08 [search-tool extract]

¹³⁶⁹ Verified fact — A7-S047: @helicone/helpers on npm (npm / Helicone) <https://registry.npmjs.org/@helicone/helpers> accessed 2026-10-07

¹³⁷⁰ Verified fact — A7-S046: Helicone README <https://raw.githubusercontent.com/Helicone/helicone/main/README.md> accessed 2026-10-07

Field	Value	Label	Sources
Integration	Change base URL to the Helicone gateway (OpenAI-compatible); integrations for OpenAI, Anthropic, Gemini, LangChain, Vercel AI SDK; export to PostHog	Verified fact	A7-S046
Dependencies	Self-hosting via docker-compose; production Helm chart available to enterprise customers on request	Verified fact	A7-S046
Certifications	Claims 'SOC 2 and GDPR compliant' (README; SOC 2 type not stated)	Verified fact	A7-S046
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Free tier of 10,000 requests per month, no credit card; paid plans not verified	Verified fact	A7-S046
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	Helicone states 14.2 trillion tokens processed and 16,000 organisations served over three years	Verified fact	A7-S112
Maturity	npm helper packages show no release since 7 November 2025	Verified fact	A7-S047
Deployment	saas: Yes; managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes (Docker; Helm for enterprise); on_prem: Not publicly verified		

C7: AI security

HashiCorp Vault (IBM Vault Self-Managed; HCP Vault Dedicated) (c7-hashicorp-vault)

Tier: Strategic • **Flags:** Acquired • **Original graphic label:** not in graphic

Rationale: Strategic for secrets and agent credential brokerage, conditional on accepting BUSL and IBM ownership (lock-in 2) ^{AJ}.

Criterion	Score	Rationale
Technical	4	Leading depth in secrets and agent credential brokerage; not a prompt-injection or supply-chain control.
Enterprise readiness	4	Policy ACLs, namespaces (multi-tenancy), SCIM 2.0, audit devices and IBM support lifecycle; human SSO and SLA not verified in the dataset.
Security and compliance	4	SOC 2 Type 2 and ISO 27001 with IBM Vault in scope (anchor 4); no CMK/BYOK, ISO 42001 or FedRAMP evidence for 5.
Deployment flexibility	4	HCP Vault Dedicated plus self-managed on-prem; air-gap not verified.
Ecosystem	4	API, CLI, Terraform provider and validated IdPs (IBM Verify, Auth0, PingFederate, Entra, Okta).
Reliability and maturity	4	Years of production use and IBM lifecycle; IBM ownership change (February 2025) and 2.0 major-version transition noted.
Cost / TCO	3	BUSL community use free; Enterprise needed for agentic IAM; HCP hourly pricing published.
Lock-in / portability	2	Source-available BUSL (base 3), reduced by 1 for the 2025 acquisition (rule 3); licence risk under rule 4.

Criterion	Score	Rationale
Total (generic / FS)	3.80 / 3.65	

Capabilities. Secrets storage and brokerage, dynamic credentials and certificates, identity-based authorisation; agentic IAM (GA in Vault Enterprise 2.1, 1 September 2026) registers agents, validates IdP OAuth JWTs, enforces user, agent-ceiling and request-scoped authorisation, and records user and agent in audit logs ^{VF1371}.

Strengths

- Keeps credentials out of the model with short-lived, request-scoped issuance and dual attribution ^{VF1372 AJ}
- SOC 2 Type 2 and ISO 27001/27017/27018 with IBM Vault in scope; FIPS 140-2 for Enterprise ^{VF1373}
- Namespaces, SCIM 2.0, policy ACLs and audit devices ^{VF1374}
- IBM lifecycle: two years support plus extensions ^{VF1375}

Limitations and risks

- BUSL 1.1 (IBM licensor) restricts competing hosted or embedded offerings ^{VF1376}
- Agentic IAM is Enterprise-only ^{VF1377}
- Owned by IBM since 27 February 2025 ^{VF1378}
- Version caveat: changelog stops at 2.1.1 but a v2.1.2 tag exists ^{VF1379}

Choose when

- Multi-cloud or hybrid estate needing one broker for human, workload and agent credentials ^{AJ}

Avoid when

¹³⁷¹ Verified fact — A7-S034: HashiCorp Vault agentic IAM is now generally available (HashiCorp (IBM)) <https://www.hashicorp.com/en/blog/hashicorp-vault-agentic-iam-is-now-generally-available> accessed 2026-10-07 [search-tool extract]; A7-S061: HashiCorp Vault CHANGELOG (HashiCorp / IBM) <https://raw.githubusercontent.com/hashicorp/vault/main/CHANGELOG.md> accessed 2026-10-07

¹³⁷² Verified fact — A7-S034: HashiCorp Vault agentic IAM is now generally available (HashiCorp (IBM)) <https://www.hashicorp.com/en/blog/hashicorp-vault-agentic-iam-is-now-generally-available> accessed 2026-10-07 [search-tool extract]

¹³⁷³ Verified fact — A7-S035: HashiCorp Compliance Overview / Security at HashiCorp (HashiCorp (IBM)) <https://www.hashicorp.com/en/trust/compliance> accessed 2026-10-07 [search-tool extract]

¹³⁷⁴ Verified fact — A7-S033: Vault release notes 2.x; IBM Vault Self-Managed 2.x.x lifecycle page (HashiCorp (IBM)) <https://developer.hashicorp.com/vault/docs/updates/release-notes> accessed 2026-10-07 [search-tool extract]; A7-S034: HashiCorp Vault agentic IAM is now generally available (HashiCorp (IBM)) <https://www.hashicorp.com/en/blog/hashicorp-vault-agentic-iam-is-now-generally-available> accessed 2026-10-07 [search-tool extract]

¹³⁷⁵ Verified fact — A7-S033: Vault release notes 2.x; IBM Vault Self-Managed 2.x.x lifecycle page (HashiCorp (IBM)) <https://developer.hashicorp.com/vault/docs/updates/release-notes> accessed 2026-10-07 [search-tool extract]

¹³⁷⁶ Verified fact — A7-S060: HashiCorp Vault LICENSE (BUSL 1.1) (HashiCorp / IBM) <https://raw.githubusercontent.com/hashicorp/vault/main/LICENSE> accessed 2026-10-07

¹³⁷⁷ Verified fact — A7-S034: HashiCorp Vault agentic IAM is now generally available (HashiCorp (IBM)) <https://www.hashicorp.com/en/blog/hashicorp-vault-agentic-iam-is-now-generally-available> accessed 2026-10-07 [search-tool extract]

¹³⁷⁸ Verified fact — A7-S032: IBM Completes Acquisition of HashiCorp, Creates Comprehensive, End-to-End Hybrid Cloud Platform <https://newsroom.ibm.com/2025-02-27-ibm-completes-acquisition-of-hashicorp-creates-comprehensive-end-to-end-hybrid-cloud-platform> accessed 2026-10-07 [search-tool extract]

¹³⁷⁹ Verified fact — V2-S061: Go module proxy @latest: OPA, SPIRE, agentgateway, envoyproxy/ai-gateway (Agent Router), hashicorp/vault (incl. v2 resolution error) and Vault CHANGELOG heading (Go module proxy (proxy.golang.org) / GitHub raw) <https://proxy.golang.org/github.com/open-policy-agent/opa/@latest> accessed 2026-10-08 [search-tool extract]

- A single cloud's native secrets service suffices and agent delegation is not needed ^{AI}

Nearest competitors: Cloud-native secrets managers, C4-entra-agent-id, C4-okta-auth0-ai-agents

Regulated-FS note. Use audit devices as the authoritative record of which agent obtained which credential on whose behalf; include IBM in concentration analysis if watsonx.governance is also chosen ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	HashiCorp, an IBM company (IBM acquisition closed 27 February 2025)	Verified fact	A7-S032
Category	Secrets management, identity-based access and encryption; agentic IAM for AI agents	Verified fact	A7-S034
Version / lineup	Vault 2.1.1 (16 September 2026); 2.1.0 (1 September 2026) made agentic IAM GA in Vault Enterprise; Vault 2.0 GA 14 April 2026 (first major version since 1.0, aligned to IBM lifecycle)	Verified fact	A7-S061, A7-S033, A7-S034, V2-S061
Licence	Source-available Business Source License 1.1 (licensor IBM) for Vault 1.15.0 and later; production use permitted except competing hosted/embedded offerings; IBM Vault Self-Managed sold under IBM International Program License Agreement; Enterprise features need a licence	Verified fact	A7-S060, A7-S033, A7-S036
Status events	27 February 2025: IBM completed acquisition of HashiCorp; June/August 2025: HCP Vault Secrets end of sale (30 June 2025) and end of life for pay-as-you-go customers (27 August 2025); 14 April 2026: Vault 2.0 GA under IBM support lifecycle; 1 September 2026: agentic IAM GA (Vault Enterprise 2.1)	Verified fact	A7-S032, A7-S036, A7-S033, A7-S061
Strategic direction	Positions Vault as the governance layer for AI agent identities: agent registry, OAuth request-scoped authorisation (authorization_details), on-behalf-of delegation, audit attribution to both user and agent; SPIFFE JWT-SVID and SCIM 2.0 in 2.0	Verified fact	A7-S034, A7-S033
What it does	Stores and brokers secrets, issues dynamic credentials and certificates,	Verified	A7-S034

Field	Value	Label	Sources
	and authorises access by identity. Agentic IAM registers AI agents, validates OAuth JWTs from IdPs, enforces the intersection of user permissions, agent ceiling policies and request-scoped authorisation details, and records both user and agent in audit logs.	d fact	
Stack position	C7 secrets and C4 agent identity/authorisation; not shown in the original graphic	Verified fact	A7-S034
Integration	API, CLI, Terraform Vault provider (vault_agent_registration, vault_oauth_resource_server_config_profile); validated IdPs: IBM Verify, Auth0, PingFederate, Microsoft Entra, Okta	Verified fact	A7-S034
Dependencies	Not publicly verified	Not publicly verified	
Certifications	SOC 2 Type 2; ISO 27001, 27017, 27018 (scope includes IBM Vault); Vault Enterprise FIPS 140-2; SOC 2 report under NDA via customertrust@ibm.com	Verified fact	A7-S035
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Policy-based ACLs, namespaces, SCIM 2.0 provisioning, agent registry UI, audit devices with user+agent attribution	Verified fact	A7-S033, A7-S034, A7-S061
Enterprise support	IBM lifecycle for 2.x: 2 years support + 1 year critical-fix extension + 3 years usage/existing fixes	Verified fact	A7-S033
Pricing	HCP Vault Dedicated: trial, pay-as-you-go or contract; hourly base cost by tier/size/region plus per-client hourly charge (Essentials/Standard); self-managed: contact sales	Verified fact	A7-S036
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
Maturity	Not publicly verified	verified	Not publicly verified
Deployment	saas: Yes (HCP Vault Dedicated); managed_cloud: Yes (HCP Vault Dedicated); vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes (Vault Community / IBM Vault Self-Managed); on_prem: Yes (self-managed)		

Model and package supply-chain scanning (pattern): ModelScan, picklescan, fickling, Hugging Face Hub scanning, safetensors (c7-model-supply-chain-scanning)

Tier: Strategic • **Flags:** Acquired • **Original graphic label:** not in graphic

Rationale: Mandatory baseline pattern (safetensors by default plus scanning gate); ModelScan's owner was acquired ¹⁰.

Criterion	Score	Rationale
Technical	3	Covers executable model formats competently; pattern-matching limits and no package or agent-artefact coverage.
Enterprise readiness	3	Rule 2: CI libraries inherit host controls; no commercial support for the open tools (commercial successors are separate products).
Security and compliance	3	Rule 2 hygiene: security policy for ModelScan, clear licences; owner acquired and components pre-1.0.
Deployment flexibility	5	Runs anywhere, including air-gapped CI.
Ecosystem	4	Hugging Face Hub uses picklescan; commercial platforms build on the same approach.
Reliability and	3	Mostly pre-1.0; ModelScan cadence slowed after acquisition.

Criterion	Score	Rationale
maturity		
Cost / TCO	5	Free open source with light operations.
Lock-in / portability	4	Permissive licences and an open format (base 5), reduced by 1 because ModelScan is vendor-owned after a 2025 acquisition (rule 3).
Total (generic / FS)	3.65	
	/	
	3.60	

Capabilities. Pattern: safe serialisation (safetensors) plus open-source scanners (ModelScan, picklescan, fickling) in CI and at registry promotion, complemented by Hugging Face Hub-side scanning ^{vf¹³⁸⁰}.

Strengths

- safetensors removes pickle's code-execution risk ('Pickle: Unsafe, runs arbitrary code') ^{vf¹³⁸¹}
- Free, local, CI-native and air-gap friendly ^{vf¹³⁸² AJ}
- ModelScan publishes a security policy with GitHub advisories ^{vf¹³⁸³}

Limitations and risks

- picklescan only pattern-matches module names; scanning reduces but does not remove risk ^{vf¹³⁸⁴}
- ModelScan's owner Protect AI is now Palo Alto Networks; README directs users to Guardian; last release February 2026 ^{vf¹³⁸⁵}
- Most components pre-1.0 ^{vf¹³⁸⁶}
- Does not address package compromise (LiteLLM case) ^{vf¹³⁸⁷ AJ}

Choose when

¹³⁸⁰ Verified fact — A7-S082: ModelScan README (Protect AI (Palo Alto Networks)) <https://raw.githubusercontent.com/protectai/modelscan/main/README.md> accessed 2026-10-07; A7-S083: picklescan README (picklescan maintainers) <https://raw.githubusercontent.com/mmaitre314/picklescan/main/README.md> accessed 2026-10-07; A7-S064: fickling on PyPI (PyPI / Trail of Bits) <https://pypi.org/pypi/fickling/json> accessed 2026-10-07; A7-S007: safetensors on PyPI (JSON API) (PyPI / Hugging Face) <https://pypi.org/pypi/safetensors/json> accessed 2026-10-07; A7-S037: Hugging Face Hub Security docs (malware scanning, pickle scanning, third-party scanners Protect AI and JFrog) <https://huggingface.co/docs/hub/security> accessed 2026-10-07 [search-tool extract]

¹³⁸¹ Verified fact — B-C7-S006: safetensors README (Hugging Face) <https://raw.githubusercontent.com/huggingface/safetensors/main/README.md> accessed 2026-10-08

¹³⁸² Verified fact — A7-S002: modelscan on PyPI (JSON API) (PyPI / Protect AI) <https://pypi.org/pypi/modelscan/json> accessed 2026-10-07; A7-S006: picklescan on PyPI (JSON API) <https://pypi.org/pypi/picklescan/json> accessed 2026-10-07

¹³⁸³ Verified fact — B-C7-S007: modelscan Security Policy (Protect AI (Palo Alto Networks)) <https://raw.githubusercontent.com/protectai/modelscan/main/SECURITY.md> accessed 2026-10-08

¹³⁸⁴ Verified fact — A7-S037: Hugging Face Hub Security docs (malware scanning, pickle scanning, third-party scanners Protect AI and JFrog) <https://huggingface.co/docs/hub/security> accessed 2026-10-07 [search-tool extract]

¹³⁸⁵ Verified fact — A7-S014: Palo Alto Networks Completes Acquisition of Protect AI <https://www.paloaltonetworks.com/company/press/2025/palo-alto-networks-completes-acquisition-of-protect-ai> accessed 2026-10-07 [search-tool extract]; A7-S082: ModelScan README (Protect AI (Palo Alto Networks)) <https://raw.githubusercontent.com/protectai/modelscan/main/README.md> accessed 2026-10-07; A7-S002: modelscan on PyPI (JSON API) (PyPI / Protect AI) <https://pypi.org/pypi/modelscan/json> accessed 2026-10-07

¹³⁸⁶ Verified fact — A7-S002: modelscan on PyPI (JSON API) (PyPI / Protect AI) <https://pypi.org/pypi/modelscan/json> accessed 2026-10-07; A7-S064: fickling on PyPI (PyPI / Trail of Bits) <https://pypi.org/pypi/fickling/json> accessed 2026-10-07; A7-S007: safetensors on PyPI (JSON API) (PyPI / Hugging Face) <https://pypi.org/pypi/safetensors/json> accessed 2026-10-07

¹³⁸⁷ Verified fact — A6-S008: Security Update: Suspected Supply Chain Incident (and Security Townhall Updates) (BerriAI (LiteLLM)) <https://docs.litellm.ai/blog/security-update-march-2026> accessed 2026-10-07 [search-tool extract]

- Always, as the minimum gate for self-hosted weights ^{Rec}

Avoid when

- Never as the only supply-chain control ^{AI}

Nearest competitors: C7-prisma-airs, C7-hiddenlayer, C7-openssf-model-signing

Regulated-FS note. Record scan results and artefact hashes in the model inventory (C8) so validation evidence refers to the exact weights scanned ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Pattern; ModelScan by Protect AI (now Palo Alto Networks); picklescan (community, used by Hugging Face); fickling by Trail of Bits; safetensors by Hugging Face	Verified fact	A7-S082, A7-S014, A7-S037, A7-S064, A7-S007
Category	Named pattern plus open-source scanners (e.g. picklescan: 'Security scanner detecting Python Pickle files performing suspicious actions') and a safe serialisation format	Verified fact	A7-S082, A7-S083, A7-S007
Version / lineup	modelscan 0.8.8 (18 February 2026); picklescan 1.0.5 (1 July 2026); fickling 0.1.12 (26 June 2026); safetensors 0.8.0 (9 June 2026)	Verified fact	A7-S002, A7-S006, A7-S064, A7-S007, V2-S028
Licence	Open source: ModelScan Apache-2.0; picklescan MIT; fickling LGPLv3+; safetensors Apache-2.0	Verified fact	A7-S002, A7-S006, A7-S064, A7-S007
Status events	22 July 2025: Protect AI (ModelScan, Guardian) acquired by Palo Alto Networks	Verified fact	A7-S014
Strategic direction	Commercial successors bundle scanning into AI security platforms (Prisma AIRS AI Model Security, HiddenLayer Model Scanner); Hugging Face layers ClamAV, picklescan, trufflehog and third-party scanners (Protect AI, JFrog) on the Hub	Verified fact	A7-S039, A7-S029, A7-S037

Field	Value	Label	Sources
What it does	Scans serialised model files (pickle, H5, SavedModel and others) for code that executes on load before models enter registries or runtime; safetensors avoids executable pickle formats altogether. Hugging Face notes picklescan only pattern-matches module names, and scanning reduces but does not remove risk.	Verified fact	A7-S082, A7-S037
Stack position	Pre-deployment gate in CI and model registry promotion (C7), upstream of L1/L2 self-hosted models	Architectural judgement	
Integration	CLIs and Python libraries for CI pipelines; Hub-side automatic scanning of public repositories	Verified fact	A7-S082, A7-S037
Dependencies	Not publicly verified	Not publicly verified	
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Open-source tools free; commercial scanners priced by vendor	Verified fact	A7-S002, A7-S006, A7-S031
Infrastructure cost	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	Protect AI reported 4.47 million model versions in 1.41 million Hub repositories scanned by 1 April 2025, flagging 352,000 issues across 51,700 models	Verified fact	A7-S037
Maturity	Not publicly verified	Not publicly verified	
Deployment	saas: Yes (Hugging Face Hub scanning; commercial platforms); managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes (open-source CLIs); on_prem: Not publicly verified		

OpenSSF Model Signing (OMS) specification and model-signing library (c7-openssf-model-signing)

Tier: Tactical · Flags: none · Original graphic label: not in graphic

Rationale: Right standard for internally produced weights, but narrow and with unverified adoption ^{AJ}.

Criterion	Score	Rationale
Technical	3	Does integrity and signer verification well; narrow scope by design.
Enterprise readiness	3	Rule 2: library inherits host controls; PKCS#11 and private Sigstore enable enterprise key custody.
Security and compliance	3	Rule 2 hygiene: foundation-governed, Apache-2.0; no security policy file found.
Deployment flexibility	4	Self-hosted library; offline key or certificate signing possible; private Sigstore supported.
Ecosystem	2	Adoption by hubs and registries not publicly verified.
Reliability and	3	v1.0 spec (April 2025), library 1.1.1; no release in the last twelve months found.

Criterion	Score	Rationale
maturity		
Cost / TCO	5	Free.
Lock-in / portability	5	Open specification, Apache-2.0, neutral governance.
Total (generic / FS)	3.35	
	/	
	3.50	

Capabilities. Open specification and Apache-2.0 library to sign and verify ML models of any format with a detached Sigstore bundle (DSSE/in-toto), using Sigstore, key pairs, certificates or PKCS#11 devices, with optional private Sigstore instances and a transparency log ^{vf¹³⁸⁸}.

Strengths

- Vendor-neutral integrity and signer verification under OpenSSF (Linux Foundation) ^{vf¹³⁸⁹}
- HSM (PKCS#11) and private Sigstore support suit regulated key custody ^{vf¹³⁹⁰}
- Free ^{vf¹³⁹¹}

Limitations and risks

- Proves integrity and signer, not provenance or safety ^{aj}
- Hub and registry adoption not publicly verified ^{npv}
- No library release on PyPI since 1.1.1 (10 October 2025) ^{vf¹³⁹²}; no security policy file found

Choose when

- You fine-tune or distil models internally and must prove served weights equal validated weights ^{aj}

Avoid when

- You only consume hosted model APIs ^{aj}

Nearest competitors: C7-model-supply-chain-scanning, C7-hiddenlayer

Regulated-FS note. Sign with a firm-controlled key or HSM and verify the signature at model-server start-up ^{rec}.

Facts (as of 2026-10-08)

¹³⁸⁸ Verified fact — A7-S040: Launch of Model Signing v1.0: OpenSSF AI/ML Working Group Secures the Machine Learning Supply Chain (OpenSSF (Linux Foundation)) <https://openssf.org/blog/2025/04/04/launch-of-model-signing-v1-0-openssf-ai-ml-working-group-secures-the-machine-learning-supply-chain/> accessed 2026-10-07 [search-tool extract]; A7-S038: model-signing on PyPI (JSON API) (PyPI / Sigstore model-transparency (OpenSSF)) <https://pypi.org/pypi/model-signing/json> accessed 2026-10-07; B-C7-S005: Model Transparency README (model-signing) (Sigstore / OpenSSF) <https://raw.githubusercontent.com/sigstore/model-transparency/main/README.md> accessed 2026-10-08

¹³⁸⁹ Verified fact — A7-S040: Launch of Model Signing v1.0: OpenSSF AI/ML Working Group Secures the Machine Learning Supply Chain (OpenSSF (Linux Foundation)) <https://openssf.org/blog/2025/04/04/launch-of-model-signing-v1-0-openssf-ai-ml-working-group-secures-the-machine-learning-supply-chain/> accessed 2026-10-07 [search-tool extract]

¹³⁹⁰ Verified fact — B-C7-S005: Model Transparency README (model-signing) (Sigstore / OpenSSF) <https://raw.githubusercontent.com/sigstore/model-transparency/main/README.md> accessed 2026-10-08

¹³⁹¹ Verified fact — A7-S038: model-signing on PyPI (JSON API) (PyPI / Sigstore model-transparency (OpenSSF)) <https://pypi.org/pypi/model-signing/json> accessed 2026-10-07

¹³⁹² Verified fact — A7-S038: model-signing on PyPI (JSON API) (PyPI / Sigstore model-transparency (OpenSSF)) <https://pypi.org/pypi/model-signing/json> accessed 2026-10-07

Field	Value	Label	Sources
Company	OpenSSF AI/ML Working Group (Linux Foundation); reference implementation in Sigstore model-transparency; contributors include Google, NVIDIA and HiddenLayer	Verified fact	A7-S040, A7-S038
Category	Open specification and library for signing and verifying ML models	Verified fact	A7-S040
Version / lineup	v1.0 launched 4 April 2025; model-signing 1.1.1 on PyPI (10 October 2025); OpenSSF podcast (July 2026) refers to v1.1 and v1.2 iterations	Verified fact	A7-S040, A7-S038, V2-S028
Licence	Open source, Apache-2.0 (library)	Verified fact	A7-S038
Strategic direction	Not publicly verified	Not publicly verified	
What it does	Signs a model of any format and size with a detached signature (Sigstore bundle) stored with the model, using Sigstore, self-signed certificates or key pairs, so consumers can verify integrity and signer before loading.	Verified fact	A7-S040
Stack position	C7 supply-chain control complementing scanning: verifies integrity and signer, not provenance or safety	Architectural judgement	
Integration	Python library and CLI (sign/verify)	Verified fact	A7-S038, A7-S040
Dependencies	Not publicly verified	Not publicly verified	
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Free open source	Verified fact	A7-S038
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Not publicly verified	Not publicly verified	
Deployment	saas: Not publicly verified; managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes; on_prem: Not publicly verified		

Prisma AIRS (AI Runtime Security) 3.0 (c7-prisma-airs)

Tier: Tactical · **Flags:** none · **Original graphic label:** not in graphic

Rationale: Strongest runtime platform technically, but product-scoped certification is unconfirmed and credit licensing plus a proprietary SDK create lock-in; Strategic candidate only in Palo Alto estates after due diligence ^{AJ}.

Criterion	Score	Rationale
Technical	5	Leading breadth across C7 research questions: supply chain, agent artefacts, red-teaming, runtime and gateway.

Criterion	Score	Rationale
Enterprise readiness	3	Tenant RBAC roles, AIRS API role and red-team override audit trail verified (B-C7-S003); AIRS-specific SSO and admin audit logs not documented, so 3.
Security and compliance	3	Company-level SOC 2 Type II and ISO 27001 with product scope not stated: one below the anchor of 4 (rule 8).
Deployment flexibility	4	Managed SaaS, AWS Marketplace, private-cloud firewall and local scans; on-prem and air-gap not verified.
Ecosystem	4	Registry, cloud-AI-platform and agent-tool connectors.
Reliability and maturity	3	GA with monthly releases, but three acquisitions in ten months and conflicting region documentation.
Cost / TCO	3	Published but complex pricing (token-metered credits, PAYG SKU).
Lock-in / portability	2	Proprietary SDK licence and credit licensing tied to Strata Cloud Manager; Palo Alto is the acquirer, so no rule 3 reduction.
Total (generic / FS)	3.60 / 3.35	

Capabilities. AI security platform (3.0, 23 March 2026): AI Model Security (35+ formats, 25+ threat categories, local scans), AI Red Teaming (including agents), AI Runtime by API or network intercept, Agent Artifact Scanning (agent code, MCP servers, skills), AI Skill Security and AI Gateway (GA 16 July 2026) ^{vr1393}.

Strengths

- Broadest coverage across supply chain, red-teaming, runtime and gateway ^{vr1394}
- Registry integrations (JFrog Artifactory, GitLab Model Registry) and connectors for Microsoft Foundry, Copilot Studio and others ^{vr1395}
- Tenant RBAC roles and audit trail of red-team verdict overrides; scan logs to SIEM ^{vr1396}
- Private-cloud firewall deployment (ESXi, KVM, OpenShift, Rancher) ^{vr1397}

Limitations and risks

¹³⁹³ Verified fact — A7-S027: Palo Alto Networks Secures Agentic AI with Prisma AIRS 3.0 <https://www.paloaltonetworks.com/company/press/2026/palo-alto-networks-secures-agentic-ai-with-prisma-airs-3-0> accessed 2026-10-07 [search-tool extract]; A7-S028: New Features - Prisma AIRS (January-August 2026 release notes) (Palo Alto Networks) <https://docs.paloaltonetworks.com/ai-runtime-security/new-features/by-date/prisma-airs/august-2026> accessed 2026-10-07 [search-tool extract]; A7-S039: Prisma AIRS AI Model Security (Palo Alto Networks) <https://www.paloaltonetworks.com/ai-security/ai-model-security> accessed 2026-10-07 [search-tool extract]; V2-S048: Announcing the General Availability of Prisma AIRS AI Gateway (with Calcalist/SDxCentral on Koi completion) (Palo Alto Networks) <https://www.paloaltonetworks.com/blog/2026/07/announcing-general-availability-of-prisma-airs-ai-gateway/> accessed 2026-10-08 [search-tool extract]

¹³⁹⁴ Verified fact — A7-S027: Palo Alto Networks Secures Agentic AI with Prisma AIRS 3.0 <https://www.paloaltonetworks.com/company/press/2026/palo-alto-networks-secures-agentic-ai-with-prisma-airs-3-0> accessed 2026-10-07 [search-tool extract]; A7-S028: New Features - Prisma AIRS (January-August 2026 release notes) (Palo Alto Networks) <https://docs.paloaltonetworks.com/ai-runtime-security/new-features/by-date/prisma-airs/august-2026> accessed 2026-10-07 [search-tool extract]

¹³⁹⁵ Verified fact — A7-S028: New Features - Prisma AIRS (January-August 2026 release notes) (Palo Alto Networks) <https://docs.paloaltonetworks.com/ai-runtime-security/new-features/by-date/prisma-airs/august-2026> accessed 2026-10-07 [search-tool extract]; A7-S031: Prisma AIRS Licenses; Token-Based Licensing; Managed AIRS for AWS Pricing; Prisma AIRS setup prerequisites (Palo Alto Networks) <https://docs.paloaltonetworks.com/ai-runtime-security/activation-and-onboarding/activate-your-ai-runtime-security-license> accessed 2026-10-07 [search-tool extract]

¹³⁹⁶ Verified fact — B-C7-S003: About Roles and Permissions Through Common Services; Prisma AIRS API onboarding; AI Red Teaming IAM; API scan logs (Palo Alto Networks) <https://docs.paloaltonetworks.com/common-services/identity-and-access-access-management/manage-identity-and-access/about-roles-and-permissions> accessed 2026-10-08 [search-tool extract]

¹³⁹⁷ Verified fact — A7-S031: Prisma AIRS Licenses; Token-Based Licensing; Managed AIRS for AWS Pricing; Prisma AIRS setup prerequisites (Palo Alto Networks) <https://docs.paloaltonetworks.com/ai-runtime-security/activation-and-onboarding/activate-your-ai-runtime-security-license> accessed 2026-10-07 [search-tool extract]

- Company-level SOC 2 Type II and ISO 27001:2022; Prisma AIRS scope not stated ^{VF-1398}
- Not FedRAMP; API intercept unavailable in FedRAMP environments ^{VF-1399}
- EU-Germany region routes some functions through the Netherlands; several modules Americas-only ^{VF-1400}
- Credit-based licensing, one-billion-token monthly minimum and proprietary SDK licence ^{VF-1401}
- Integration churn after Protect AI, Koi and Portkey acquisitions ^{VF-1402 AJ}

Choose when

- Palo Alto Networks is already your network security platform ^{AJ}
- You want one supplier for model scanning, red-teaming and runtime inspection ^{AJ}

Avoid when

- You need all processing in one EU jurisdiction ^{AJ}
- You do not want your AI gateway and AI security vendor to be the same company ^{AJ}

Nearest competitors: C7-hiddenlayer, C7-lakera, C7-model-supply-chain-scanning

Regulated-FS note. Map processing region per module before onboarding; if the AI Gateway is adopted, record gateway plus security as one concentration point in the register of information ^{REC}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Palo Alto Networks (acquired Protect AI, July 2025; Koi, April 2026; Portkey, May 2026)	Verified fact	A7-S014, A7-S016
Category	AI security platform: model scanning, posture, AI red teaming, runtime protection, agent security, AI gateway	Verified fact	A7-S014, A7-S027, A7-S028
Version / lineup	Prisma AIRS 3.0 launched 23 March 2026 (2.0 on 28 October 2025); monthly feature releases through August 2026; modules: AI Model Security, AI Red Teaming, AI Runtime (API and network intercept), Agent Artifact Scanning, AI Skill Security, AI Gateway (GA 16 July 2026)	Verified fact	A7-S027, A7-S015, A7-S028, A7-S031

¹³⁹⁸ Verified fact — B-C7-S002: Palo Alto Networks Trust Center <https://trust.paloaltonetworks.com/> accessed 2026-10-08 [search-tool extract]

¹³⁹⁹ Verified fact — A7-S120: Prisma AIRS API Intercept Supported Regions; AI Runtime Firewall prerequisites; Palo Alto FedRAMP Authorized Services page (Palo Alto Networks) <https://docs.paloaltonetworks.com/ai-runtime-security/activation-and-onboarding/prisma-airis-supported-regions> accessed 2026-10-07 [search-tool extract]; A7-S031: Prisma AIRS Licenses; Token-Based Licensing; Managed AIRS for AWS Pricing; Prisma AIRS setup prerequisites (Palo Alto Networks) <https://docs.paloaltonetworks.com/ai-runtime-security/activation-and-onboarding/activate-your-ai-runtime-security-license> accessed 2026-10-07 [search-tool extract]

¹⁴⁰⁰ Verified fact — A7-S120: Prisma AIRS API Intercept Supported Regions; AI Runtime Firewall prerequisites; Palo Alto FedRAMP Authorized Services page (Palo Alto Networks) <https://docs.paloaltonetworks.com/ai-runtime-security/activation-and-onboarding/prisma-airis-supported-regions> accessed 2026-10-07 [search-tool extract]

¹⁴⁰¹ Verified fact — A7-S031: Prisma AIRS Licenses; Token-Based Licensing; Managed AIRS for AWS Pricing; Prisma AIRS setup prerequisites (Palo Alto Networks) <https://docs.paloaltonetworks.com/ai-runtime-security/activation-and-onboarding/activate-your-ai-runtime-security-license> accessed 2026-10-07 [search-tool extract]; A7-S063: pan-aisecurity on PyPI (PyPI / Palo Alto Networks) <https://pypi.org/pypi/pan-aisecurity/json> accessed 2026-10-07

¹⁴⁰² Verified fact — A7-S014: Palo Alto Networks Completes Acquisition of Protect AI <https://www.paloaltonetworks.com/company/press/2025/palo-alto-networks-completes-acquisition-of-protect-ai> accessed 2026-10-07 [search-tool extract]; A7-S016: Palo Alto Networks Inc Form 10-Q for quarter ended 30 April 2026 (Palo Alto Networks / US SEC) <https://www.sec.gov/Archives/edgar/data/0001327567/000132756726000015/panw-20260430.htm> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
Licence	Proprietary; BYOL funded with Software NGFW credits; Python API-intercept SDK (pan-aisecurity 0.11.0) under proprietary licence	Verified fact	A7-S031, A7-S063
Status events	28 April 2025: intent to acquire Protect AI announced; 22 July 2025: Protect AI acquisition completed; 28 October 2025: Prisma AIRS 2.0 completes Protect AI integration; 23 March 2026: Prisma AIRS 3.0; 14 April 2026: Koi acquired (agentic endpoint security, to enhance Prisma AIRS); 29 May 2026: Portkey acquired (AI gateway); 16 July 2026: AI Gateway GA	Verified fact	A7-S014, A7-S015, A7-S027, A7-S016, A7-S028, V2-S039, V2-S048, V2-S025
Strategic direction	Moving from observing AI to 'authorising autonomous execution': agent artifact scanning (agent code, MCP servers, skills), agent red teaming (tool chaining, privilege misuse, memory poisoning), AI gateway from Portkey, integration with Anthropic inference hooks and OpenAI Codex Enterprise	Verified fact	A7-S027, A7-S028, A7-S016
What it does	Scans models and agent artefacts for malicious payloads and unsafe permissions, red-teams AI apps and agents, and inspects prompts/responses at runtime via API or network intercept for injection, data leakage and policy violations. AI Model Security analyses 35+ model file types for 25+ threat categories and can run scans locally.	Verified fact	A7-S027, A7-S039, A7-S031
Stack position	Cross-cutting C7 platform spanning supply chain (pre-deployment), runtime (C2/C3 overlap) and gateway (C1 overlap)	Verified fact	A7-S027, A7-S028
Integration	API intercept (SDK/REST), network intercept (AI runtime firewall), model registry scan sources (JFrog Artifactory, GitLab Model Registry), connectors for Microsoft Foundry, Copilot Studio, n8n, Anthropic inference hooks, OpenAI Codex Enterprise	Verified fact	A7-S028, A7-S031
Dependencies	Not publicly verified	Not publicly verified	
Certifications	Prisma AIRS not listed among Palo Alto Networks' FedRAMP-authorised offerings (Prisma Cloud, Prisma Access, Prisma SASE are); API intercept unavailable in FedRAMP environments. SOC 2/ISO for AIRS not verified.	Verified fact	A7-S120, A7-S031
GDPR / residency	API Intercept regions: Americas, EU-Germany, India, Singapore, Japan (region determines processing and storage), but in EU-Germany URL category detection and contextual grounding run via the Netherlands; older firewall setup pages state US-only inspection; AI Model Security in US, EU-Netherlands, Japan, Singapore; AI Skill Security and AI Discovery Americas only	Verified fact	A7-S120, A7-S031
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	BYOL credits; AI Runtime API metered in billions of tokens per month (1 token = 4 characters, minimum 1 billion) from February 2026; Managed AIRS for AWS PAYG premium SKU US\$3.00/hour plus traffic from US\$0.065/GB	Verified fact	A7-S031
Infrastructure cost	Network intercept needs at least 4 vCPUs; capacity up to 10K AI transactions per day per vCPU	Verified fact	A7-S031
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Not publicly verified	Not publicly verified	
Deployment	saas: Yes (Managed SaaS firewall on AWS, Azure, GCP; Strata Cloud Manager); managed_cloud: Yes (Managed AIRS for AWS via AWS Marketplace); vpc_byoc: Not publicly verified; private_cloud: Yes (runtime firewall on ESXi, KVM, OpenShift, Rancher); self_hosted: Yes (model scans run locally; VM-series firewall images); on_prem: Not publicly verified		

HiddenLayer AI Security Platform (c7-hiddenlayer)

Tier: Tactical · **Flags:** none · **Original graphic label:** not in graphic

Rationale: Capable independent specialist, but dated certification evidence, unverified deployment breadth and opaque pricing ^{AI}.

Criterion	Score	Rationale
Technical	4	Model scanning plus agent runtime and coding-agent protection; red-teaming not evidenced.
Enterprise readiness	3	SAML SSO and RBAC verified (B-C7-S004); audit logs not documented, so 3 (rule 7).
Security and	3	ISO 27001 and SOC 2 Type 2 at company level, scope not stated and evidence 20

Criterion	Score	Rationale
compliance		months old: one below anchor (rule 8).
Deployment flexibility	3	Customer AWS VPC verified; SaaS and on-prem only reported.
Ecosystem	3	REST API, Python SDK, AWS Marketplace, Databricks Unity Catalog.
Reliability and maturity	3	Independent and funded; agentic modules launched in 2026.
Cost / TCO	2	Contract-only pricing, not published.
Lock-in / portability	3	Proprietary platform with Apache-2.0 SDK; independent ownership.
Total (generic / FS)	3.10 / 3.10	

Capabilities. Model supply-chain scanning for registries and storage (S3, EFS, container registries), AI Runtime Security for agents against prompt injection, secret exposure and unsafe commands, and Agent Harness Security for coding agents (3 August 2026) ^{vr1403}.

Strengths

- Independent; US\$100m Series B on 2 September 2026 ^{vr1404}
- ISO 27001 and SOC 2 Type 2 announced February 2025 ^{vr1405}
- SAML SSO with RBAC and tenant user management ^{vr1406}
- Deploys inside the customer's AWS environment with private networking ^{vr1407}; co-contributor to OpenSSF Model Signing ^{vr1408}

Limitations and risks

- Certification evidence dated February 2025; no trust centre found ^{vr1409}
- SaaS and on-prem or air-gapped deployment rest on vendor text in a third-party listing ^{vr1410}

¹⁴⁰³ Verified fact — A7-S017: HiddenLayer Raises \$100M Series B to Advance AI Security <https://www.hiddenlayer.com/news/hiddenlayer-100m-series-b-ai-security> accessed 2026-10-07 [search-tool extract]; A7-S029: HiddenLayer on AWS / AWS Marketplace listing / Gartner Peer Insights listing <https://hiddenlayer.com/aws> accessed 2026-10-07 [search-tool extract]

¹⁴⁰⁴ Verified fact — A7-S017: HiddenLayer Raises \$100M Series B to Advance AI Security <https://www.hiddenlayer.com/news/hiddenlayer-100m-series-b-ai-security> accessed 2026-10-07 [search-tool extract]; V2-S047: HiddenLayer Raises \$100 Million Series B Led By Delta-v Capital (with unite.ai, AI Weekly) (Pulse 2.0) <https://pulse2.com/hiddenlayer-raises-100-million-series-b-led-by-delta-v-capital/> accessed 2026-10-08 [search-tool extract]

¹⁴⁰⁵ Verified fact — A7-S030: HiddenLayer Achieves ISO 27001 and Renews SOC 2 Type 2 Compliance <https://www.hiddenlayer.com/insight/hiddenlayer-achieves-iso-27001-and-renews-soc-2-type-2-compliance> accessed 2026-10-07 [search-tool extract]

¹⁴⁰⁶ Verified fact — B-C7-S004: HiddenLayer announces new features (SAML SSO and RBAC) <https://www.hiddenlayer.com/news/hiddenlayer-announces-new-features-to-safeguard-enterprise-ai-models-with-improved-risk-detection> accessed 2026-10-08 [search-tool extract]

¹⁴⁰⁷ Verified fact — A7-S029: HiddenLayer on AWS / AWS Marketplace listing / Gartner Peer Insights listing <https://hiddenlayer.com/aws> accessed 2026-10-07 [search-tool extract]

¹⁴⁰⁸ Verified fact — A7-S040: Launch of Model Signing v1.0: OpenSSF AI/ML Working Group Secures the Machine Learning Supply Chain (OpenSSF (Linux Foundation)) <https://openssf.org/blog/2025/04/04/launch-of-model-signing-v1-0-openssf-ai-ml-working-group-secures-the-machine-learning-supply-chain/> accessed 2026-10-07 [search-tool extract]

¹⁴⁰⁹ Verified fact — A7-S030: HiddenLayer Achieves ISO 27001 and Renews SOC 2 Type 2 Compliance <https://www.hiddenlayer.com/insight/hiddenlayer-achieves-iso-27001-and-renews-soc-2-type-2-compliance> accessed 2026-10-07 [search-tool extract]

¹⁴¹⁰ Reported — A7-S029: HiddenLayer on AWS / AWS Marketplace listing / Gartner Peer Insights listing <https://hiddenlayer.com/aws> accessed 2026-10-07 [search-tool extract]

- No customer audit-log documentation found ^{VF1411}
- Contract-only pricing ^{VF1412}

Choose when

- You want an independent model-scanning and agent-runtime specialist outside a network-security suite ^{AJ}

Avoid when

- You need verified on-prem or EU-region deployment today ^{AJ}

Nearest competitors: C7-prisma-air, C7-lakera, C7-model-supply-chain-scanning

Regulated-FS note. Obtain current SOC 2 and ISO certificates and a change-of-control exit clause; independents in this market have tended to be acquired ^{ROC}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	HiddenLayer (independent; US\$100 million Series B in September 2026, total raised over US\$155 million)	Verified fact	A7-S017
Category	AI security platform: model supply-chain scanning, AI runtime security for agents, AI threat detection and response	Verified fact	A7-S017, A7-S029
Version / lineup	Modules: AI Supply Chain Security (Model Scanner), AI Runtime Security (agentic capabilities, March 2026), Agent Harness Security for coding agents (3 August 2026); hiddenlayer-sdk 3.10.0 (10 September 2026)	Verified fact	A7-S017, A7-S029, A7-S062
Licence	Proprietary platform; Python client SDK Apache-2.0	Verified fact	A7-S062
Status events	23 March 2026: next-generation AI Runtime Security for agents; 3 August 2026: Agent Harness Security launched; 2 September 2026: US\$100m Series B (Delta-v Capital lead)	Verified fact	A7-S017, V2-S047
Strategic direction	Focus on agentic runtime security and AI coding-agent protection; Databricks Unity Catalog integration being extended	Verified fact	A7-S017
What it does	Scans model files for malicious code and vulnerabilities in registries and storage (S3, EFS, container registries), and protects agents at runtime against prompt injection, secret exposure and unsafe commands.	Verified fact	A7-S029, A7-S017

¹⁴¹¹ Verified fact — B-C7-S004: HiddenLayer announces new features (SAML SSO and RBAC) <https://www.hiddenlayer.com/news/hiddenlayer-announces-new-features-to-safeguard-enterprise-ai-models-with-improved-risk-detection> accessed 2026-10-08 [search-tool extract]

¹⁴¹² Verified fact — A7-S029: HiddenLayer on AWS / AWS Marketplace listing / Gartner Peer Insights listing <https://hiddenlayer.com/aws> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
Stack position	C7 across model supply chain and runtime	Verified fact	A7-S029, A7-S017
Integration	REST API and Python SDK; AWS Marketplace; Databricks Unity Catalog integration	Verified fact	A7-S062, A7-S029, A7-S017
Dependencies	Not publicly verified	Not publicly verified	
Certifications	ISO 27001 and SOC 2 Type 2 (all five trust services criteria) announced 13 February 2025	Verified fact	A7-S030
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Encryption at rest and in transit (security page)	Verified fact	A7-S030
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Custom/contract-based; available through AWS Marketplace private offers	Verified fact	A7-S029
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	Investors include Morgan Stanley and Microsoft's venture fund (Series B); co-contributor to OpenSSF Model Signing v1.0	Verified fact	A7-S017, A7-S040
Maturity	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
		d	
Deployment	saas: Yes; managed_cloud: Not publicly verified; vpc_byoc: Yes (inside customer AWS environment with private networking); private_cloud: Not publicly verified; self_hosted: Not publicly verified; on_prem: Yes (on-prem, air-gapped or hybrid per vendor listing text)		

Check Point AI Guardrails (formerly Lakera Guard), part of the Check Point AI Defense Plane; AI Agent Security in early access (c7-lakera)

Tier: Tactical • **Flags:** Acquired, Renamed • **Original graphic label:** not in graphic

Rationale: Focused, self-hostable runtime detector; acquisition, renaming, proprietary API and default prompt logging keep it a replaceable component ^{AJ}.

Criterion	Score	Rationale
Technical	3	Covers runtime injection, leakage and content screening and early-access agent discovery; no supply-chain scanning or red-teaming.
Enterprise readiness	3	SSO, RBAC (three roles) and SIEM export verified on Enterprise (rule 7); no SCIM or SLA found, so 3.
Security and compliance	3	SOC 2 Type II stated in vendor docs plus encryption (B-C7-S001); no ISO 27001; anchor 3.
Deployment flexibility	4	SaaS, self-hosted, private cloud and on-prem; air-gap not verified.
Ecosystem	3	REST API and SIEM export; broader integrations not verified.
Reliability and maturity	3	GA product; ownership change (Check Point, 22 October 2025) and naming in flux.
Cost / TCO	3	Published free tier; Enterprise quote-based.

Criterion	Score	Rationale
Lock-in / portability	2	Proprietary Guard API (base 3), reduced by 1 for the 2025 acquisition (rule 3).
Total (generic / FS)	3.10 / 3.00	

Capabilities. Runtime screening of prompts and responses through the Guard API for prompt attacks, data leakage (including PII), content violations and off-policy agent behaviour; AI Agent Security (early access from 10 April 2026) adds agent discovery and configuration risk assessment inside the Check Point AI Defense Plane ^{VF1413}.

Strengths

- Self-hostable on-premises or in a private cloud, with EU residency on SaaS ^{VF1414}
- SOC 2 Type II stated in vendor docs; encryption at rest and in transit ^{VF1415}
- SSO, RBAC and SIEM export on Enterprise ^{VF1416}
- Free Community tier for evaluation (10,000 requests/month) ^{VF1417}

Limitations and risks

- Acquired by Check Point (completed 22 October 2025); naming in flux between Lakera and Check Point ^{VF1418}
- Dashboard records all prompts and model outputs by default ^{VF1419}
- No ISO 27001 certificate found ^{VF1420}
- Proprietary Guard API; no supply-chain scanning or red-teaming in the product ^{VF1421 AI}

¹⁴¹³ Verified fact — A7-S025: Lakera docs changelog and 'The Check Point AI Defense Plane' introduction (Lakera (Check Point)) <https://docs.lakera.ai/changelog> accessed 2026-10-07 [search-tool extract]; A7-S026: Check Point Launches AI Defense Plane to Secure the Agentic Enterprise at Scale (Check Point Software) <https://www.checkpoint.com/press-releases/check-point-launches-ai-defense-plane-to-secure-the-agentic-enterprise-at-scale/> accessed 2026-10-07 [search-tool extract]

¹⁴¹⁴ Verified fact — A7-S023: Lakera Guard Platform - Pricing (Lakera (Check Point)) <https://platform.lakera.ai/pricing> accessed 2026-10-07 [search-tool extract]; A7-S024: Lakera Dashboard / Self-hosting AI Guardrails (docs.lakera.ai) (Lakera (Check Point)) <https://docs.lakera.ai/docs/platform> accessed 2026-10-07 [search-tool extract]

¹⁴¹⁵ Verified fact — B-C7-S001: AI Guardrails Dashboard (Lakera / Check Point docs) (Lakera (Check Point)) <https://docs.lakera.ai/docs/platform> accessed 2026-10-08 [search-tool extract]

¹⁴¹⁶ Verified fact — A7-S023: Lakera Guard Platform - Pricing (Lakera (Check Point)) <https://platform.lakera.ai/pricing> accessed 2026-10-07 [search-tool extract]; A7-S024: Lakera Dashboard / Self-hosting AI Guardrails (docs.lakera.ai) (Lakera (Check Point)) <https://docs.lakera.ai/docs/platform> accessed 2026-10-07 [search-tool extract]

¹⁴¹⁷ Verified fact — A7-S023: Lakera Guard Platform - Pricing (Lakera (Check Point)) <https://platform.lakera.ai/pricing> accessed 2026-10-07 [search-tool extract]

¹⁴¹⁸ Verified fact — A7-S013: Check Point Software Technologies Completes Acquisition of Lakera to Secure the World's AI Transformation (CXOtoday (republished Check Point press release)) <https://cxotoday.com/press-release/check-point-software-technologies-completes-acquisition-of-lakera-to-secure-the-worlds-ai-transformation/> accessed 2026-10-07 [search-tool extract]; V2-S038: Check Point Software Reports 2025 Third Quarter Financial Results (with 'Check Point Completes Acquisition of Lakera' release via cxotoday) (Check Point Software Technologies) <https://www.checkpoint.com/press-releases/check-point-software-reports-2025-third-quarter-financial-results/> accessed 2026-10-08 [search-tool extract]; V2-S067: Check Point AI Guardrails (Lakera Guard) privacy datasheet (with docs.lakera.ai 'The Check Point AI Defense Plane'; Check Point AI Defense Plane press release 2 (Check Point Software Technologies) <https://www.checkpoint.com/downloads/company/check-point-ai-guardrails-lakera-guard-privacy-datasheet.pdf> accessed 2026-10-08 [search-tool extract]

¹⁴¹⁹ Verified fact — B-C7-S001: AI Guardrails Dashboard (Lakera / Check Point docs) (Lakera (Check Point)) <https://docs.lakera.ai/docs/platform> accessed 2026-10-08 [search-tool extract]

¹⁴²⁰ Verified fact — B-C7-S001: AI Guardrails Dashboard (Lakera / Check Point docs) (Lakera (Check Point)) <https://docs.lakera.ai/docs/platform> accessed 2026-10-08 [search-tool extract]

Choose when

- You need a self-hostable runtime injection and leakage detector ^{AJ}
- Check Point is already your security standard ^{AJ}

Avoid when

- You need model scanning or red-teaming from the same product ^{AJ}
- You cannot control the detector's own prompt logging and retention ^{AJ}

Nearest competitors: C7-prisma-air, C7-hiddenlayer

Regulated-FS note. Self-host for client data or switch off default prompt logging and align retention to the records policy; refresh due diligence after the change of control ^{Rec}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Check Point Software Technologies (acquired Lakera AI AG, Zurich)	Verified fact	A7-S012, A7-S013
Category	AI runtime security: prompt-attack, data-leakage and policy guardrails for LLM apps and agents	Verified fact	A7-S025
Version / lineup	AI Guardrails (runtime, Guard API; standalone tier available); AI Agent Security (early access from 10 April 2026); umbrella 'Check Point AI Defense Plane' launched 23 March 2026	Verified fact	A7-S025, A7-S026
Licence	Proprietary (SaaS and self-hosted licence)	Verified fact	A7-S023, A7-S024
Status events	16 September 2025: Check Point announced agreement to acquire Lakera (value not disclosed; reported c. US\$300 million); 22 October 2025: completion (confirmed in Check Point Q3 2025 results); completion press release circulated 11 November 2025; FY2025 20-F reportedly gives consideration of about US\$201.8m (second-hand); 23 March 2026: Check Point AI Defense Plane launched; 10 April 2026: AI Agent Security in early access	Verified fact	A7-S012, A7-S013, A7-S026, A7-S025, V2-S038, V2-S067
Strategic direction	Check Point uses Lakera as the foundation of its Global Center of Excellence for AI Security, with Zurich as global AI security R&D centre; extending from guardrails to agent discovery, configuration risk assessment and runtime protection	Verified fact	A7-S012, A7-S025
What it does	Screens prompts and responses through the Guard API to detect prompt attacks, data leakage (including PII), content violations and off-policy agent behaviour. AI Agent Security adds discovery of agents across platforms and risk assessment of their configuration.	Verified fact	A7-S025
Stack position	Runtime control at the application/gateway boundary (C7, overlapping C2 guardrails and C3 DLP)	Verified fact	A7-S025
Integration	Guard REST API; SIEM log export (Enterprise)	Verified fact	A7-S025, A7-S024
Dependencies	Not publicly verified	Not publicly	

¹⁴²¹ Verified fact — A7-S025: Lakera docs changelog and 'The Check Point AI Defense Plane' introduction (Lakera (Check Point)) <https://docs.lakera.ai/changelog> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
		verified	
Certifications	Claims SOC 2 and GDPR compliance (pricing page; SOC 2 type not stated)	Verified fact	A7-S023
GDPR / residency	Community tier: EU data residency; Enterprise: EU or US; self-hosting for full residency control	Verified fact	A7-S023, A7-S024
Security features	Encryption in transit and at rest (pricing page)	Verified fact	A7-S023
Access controls	SSO, RBAC (three roles) and SIEM log export are Enterprise-only	Verified fact	A7-S023, A7-S024
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Community: US\$0/month, 10,000 requests/month, 8k-token prompt limit, SaaS, community support; Enterprise: quote-based (as read 7 October 2026; page undated)	Verified fact	A7-S023
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Not publicly verified	Not publicly verified	
Deployment	saas: Yes; managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Yes; self_hosted: Yes (AI Guardrails self-hosting); on_prem: Yes		

C8: Model risk, governance and auditability

OpenLineage (with Marquez reference implementation) (c8-openLineage)

Tier: Strategic · Flags: none · Original graphic label: not in graphic

Rationale: The neutral lineage interchange standard; the firm must add GenAI facets ^{AJ}.

Criterion	Score	Rationale
Technical	3	Competent lineage model and protocol; no GenAI facets.
Enterprise readiness	3	Rule 2: specification and clients inherit host controls; no commercial support verified.
Security and compliance	3	Rule 2 hygiene: Apache-2.0, foundation governance, monthly releases; no security policy file found.
Deployment flexibility	5	Self-hosted consumers anywhere; cloud lineage transports.
Ecosystem	5	De-facto open lineage standard across Airflow, Spark, dbt, Flink, Collibra and GCP.
Reliability and maturity	4	Graduate project with roughly monthly releases since 2021; Marquez slower.
Cost / TCO	4	Free; a backend (Marquez or catalogue) must be operated.
Lock-in / portability	5	Open specification, Apache-2.0, neutral governance.
Total (generic / FS)	3.80 / 3.85	

Capabilities. Open specification for runtime data-lineage metadata (run, job and dataset entities extended by facets, and an event protocol), with Python, Java and SQL clients and the Marquez reference implementation; 1.53.0 adds explicit dataset-, field- and job-level lineage facets ^{vr1422}.

Strengths

- Neutral, foundation-governed (LF AI & Data Graduate) and Apache-2.0 ^{vr1423}
- Airflow provider, Spark, dbt and Flink integrations; GCP Lineage transport; Collibra integration ^{vr1424}
- Tag facets can carry classification labels ^{vr1425}

¹⁴²² Verified fact — A7-S041: OpenLineage README (OpenLineage (LF AI & Data)) <https://raw.githubusercontent.com/OpenLineage/OpenLineage/main/README.md> accessed 2026-10-07; A7-S043: Marquez README (Marquez Project (LF AI & Data)) <https://raw.githubusercontent.com/MarquezProject/marquez/main/README.md> accessed 2026-10-07; A7-S042: OpenLineage CHANGELOG (OpenLineage (LF AI & Data)) <https://raw.githubusercontent.com/OpenLineage/OpenLineage/main/CHANGELOG.md> accessed 2026-10-07

¹⁴²³ Verified fact — A7-S041: OpenLineage README (OpenLineage (LF AI & Data)) <https://raw.githubusercontent.com/OpenLineage/OpenLineage/main/README.md> accessed 2026-10-07

¹⁴²⁴ Verified fact — A7-S045: apache-airflow-providers-openlineage on PyPI (PyPI / Apache Software Foundation) <https://pypi.org/pypi/apache-airflow-providers-openlineage/json> accessed 2026-10-07; A7-S042: OpenLineage CHANGELOG (OpenLineage (LF AI & Data)) <https://raw.githubusercontent.com/OpenLineage/OpenLineage/main/CHANGELOG.md> accessed 2026-10-07; A7-S122: Collibra Platform Cloud infrastructure and base components; Collibra Cloud Sites and OpenLineage integration blog; Collibra homepage https://productresources.collibra.com/docs/collibra/latest/Content/Cloud/Intro/co_cloud-introduction.htm accessed 2026-10-07 [search-tool extract]

¹⁴²⁵ Verified fact — A7-S042: OpenLineage CHANGELOG (OpenLineage (LF AI & Data)) <https://raw.githubusercontent.com/OpenLineage/OpenLineage/main/CHANGELOG.md> accessed 2026-10-07

Limitations and risks

- No GenAI, LLM, embedding or vector facets ^{VF1426}
- Marquez releases slowed (0.51.1, March 2025) ^{VF1427}
- Does not capture run-time LLM calls (OTel’s job) ^{AJ}

Choose when

- Always, for L8 ingestion and index-build pipelines ^{REC}

Avoid when

- You expect it to trace run-time LLM calls ^{AJ}

Nearest competitors: C8-collibra-ai-governance, OTel GenAI semantic conventions

Regulated-FS note. Define and publish a firm GenAI facet set (source document IDs, classification tags, chunking and embedding model versions, index version) ^{REC}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	LF AI & Data Foundation Graduate project (OpenLineage); Marquez is a separate LF AI & Data Graduated project	Verified fact	A7-S041, A7-S043
Category	Open specification for runtime data-lineage metadata, with client libraries and integrations	Verified fact	A7-S041
Version / lineup	openlineage-python 1.53.0 (1 September 2026); spec version 2-0-2 (Marquez compatibility table); Marquez latest image 0.51.1 (27 March 2025)	Verified fact	A7-S001, A7-S043, A7-S044, V2-S028
Licence	Open specification and code, Apache-2.0 (OpenLineage and Marquez)	Verified fact	A7-S041, A7-S043, A7-S001
Strategic direction	1.53.0 adds explicit lineage facets (exact dataset-, field- and job-level relationships), more Spark/Flink/dbt coverage and GCP Lineage transport retries; user-supplied tags facets in Python and Java clients	Verified fact	A7-S042
What it does	Defines a generic model of run, job and dataset entities with consistent naming, extended by facets, and an event protocol for emitting lineage as jobs run; Marquez collects, stores and visualises these events.	Verified fact	A7-S041, A7-S043

¹⁴²⁶ Verified fact — A7-S042: OpenLineage CHANGELOG (OpenLineage (LF AI & Data))
<https://raw.githubusercontent.com/OpenLineage/OpenLineage/main/CHANGELOG.md> accessed 2026-10-07

¹⁴²⁷ Verified fact — A7-S044: Docker Hub marquezproject/marquez tags (Docker Hub / Marquez Project)
https://hub.docker.com/v2/repositories/marquezproject/marquez/tags?page_size=5&ordering=last_updated accessed 2026-10-07

Field	Value	Label	Sources
Stack position	Lineage backbone for C8 and L8 ingestion (H7); GenAI/RAG steps (chunking, embedding, indexing) would need custom facets	Architectural judgment	
Integration	OpenAPI-defined spec; Python, Java and SQL clients; integrations for Spark, Airflow, dbt, Flink; GCP Lineage and GCS transports	Verified fact	A7-S041, A7-S042
Dependencies	Not publicly verified	Not publicly verified	
Certifications	Not applicable (open specification)	Architectural judgment	
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Free open source	Verified fact	A7-S041
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Apache Airflow ships an OpenLineage provider (apache-airflow-providers-openlineage 2.20.2, 29 September 2026); Collibra documents an OpenLineage integration; Egeria listed as related project	Verified fact	A7-S045, A7-S122, A7-S041
Adoption signals	Marquez Docker image pulled about 1.12 million times; openlineage-python first released May 2021	Verified fact	A7-S044, A7-S001
Maturity	OpenLineage releases roughly monthly (1.50-1.53 in 2026); Marquez	Verified	A7-S042,

Field	Value	Label	Sources
	releases slowed (0.51.1 in March 2025)	d fact	A7-S044
Deployment	saas: Not publicly verified; managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes (Marquez or other OpenLineage consumers); on_prem: Not publicly verified		

IBM watsonx.governance (c8-watsonx-governance)

Tier: Tactical · **Flags:** none · **Original graphic label:** not in graphic

Rationale: Highest technical coverage; Tactical only because certification scope is not publicly verified; conditional Strategic for IBM estates after due diligence ^{AJ}.

Criterion	Score	Rationale
Technical	5	Leading coverage: inventory, evaluation of models, prompts and agents, discovery, thresholds, approvals, compliance mapping.
Enterprise readiness	3	IAM-controlled access, console roles and approval audit trail verified (B-C8-S003); SSO federation and SCIM not verified, so 3. Rule 6 not applied (IBM service on AWS, not an AWS service).
Security and compliance	2	Capped at 2: product-scoped certifications not publicly verified and FedRAMP status conflicting.
Deployment flexibility	4	SaaS, self-managed in customer cloud, on-prem and hybrid; air-gap not verified.
Ecosystem	4	Python SDK, Orchestrate, OpenPages and SageMaker integration.
Reliability and maturity	4	IBM product with continuous releases (SDK 1.5.2, September 2026).
Cost / TCO	3	Published indicative pricing including free tier.
Lock-in / portability	3	Proprietary, IBM-licensed; factsheets exportable.
Total (generic / FS)	3.60	

Criterion	Score	Rationale
	/	
	3.40	

Evidence rules applied: security_compliance capped at 2: product-scoped SOC 2/ISO 27001 not publicly verified; FedRAMP status conflicting (B-C8-S002)

Capabilities. Inventories AI use cases and models; evaluates models, prompt templates and agents; monitors in production and ties metrics to risks, controls and approvals; Compliance Accelerators; AI Asset Discovery (9 July 2026) for unmanaged agents, tools, MCP servers and models; Enforcement Tracking (11 August 2026) for agent metrics against thresholds; exportable factsheets ^{vr1428}.

Strengths

- Broadest agent-aware governance; consumes evaluation and Guardium security evidence automatically ^{vr1429} AI
- SaaS (IBM Cloud, AWS), self-managed in AWS/Azure, on-prem and hybrid ^{vr1430}
- Published, indicative pricing (Lite free; US\$0.64 per evaluation) ^{vr1431}
- IAM and Governance-console roles; approval-workflow audit trail ^{vr1432}

Limitations and risks

- No product-scoped SOC 2 or ISO 27001 found; FedRAMP wording conflicts ^{vr1433}
- SR 11-7 accelerator maps to a superseded instrument; no SS1/23 or SR 26-2 claim ^{vr1434} AI
- Coupling to IBM's agent runtime (watsonx Orchestrate) ^{vr1435}
- Pricing table in source garbled; confirm ^{vr1436}

Choose when

¹⁴²⁸ Verified fact — A7-S103: IBM watsonx.governance product, pricing and model-governance pages; Compliance Accelerators announcement; IBM Cloud catalog <https://www.ibm.com/products/watsonx-governance/pricing> accessed 2026-10-07 [search-tool extract]; A7-S111: IBM announcements 2026: Enforcement Tracking; AI Asset Discovery; security metrics and agent monitoring; Think 2026 AI assurance <https://www.ibm.com/new/announcements/from-governance-policies-to-governance-proof-with-enforcement-tracking-for-watsonx-orchestrate> accessed 2026-10-07 [search-tool extract]; A7-S058: ibm-watsonx-gov on PyPI (PyPI / IBM) <https://pypi.org/pypi/ibm-watsonx-gov/json> accessed 2026-10-07; B-C8-S003: Managing IAM access for watsonx.governance as a Service on AWS; factsheets; Enforcement Tracking (IBM) <https://dataplatfom.cloud.ibm.com/docs/content/wsj/openpages/wxgov-aws-iam.html?context=wx> accessed 2026-10-08 [search-tool extract]

¹⁴²⁹ Verified fact — A7-S111: IBM announcements 2026: Enforcement Tracking; AI Asset Discovery; security metrics and agent monitoring; Think 2026 AI assurance <https://www.ibm.com/new/announcements/from-governance-policies-to-governance-proof-with-enforcement-tracking-for-watsonx-orchestrate> accessed 2026-10-07 [search-tool extract]

¹⁴³⁰ Verified fact — A7-S103: IBM watsonx.governance product, pricing and model-governance pages; Compliance Accelerators announcement; IBM Cloud catalog <https://www.ibm.com/products/watsonx-governance/pricing> accessed 2026-10-07 [search-tool extract]

¹⁴³¹ Verified fact — A7-S103: IBM watsonx.governance product, pricing and model-governance pages; Compliance Accelerators announcement; IBM Cloud catalog <https://www.ibm.com/products/watsonx-governance/pricing> accessed 2026-10-07 [search-tool extract]

¹⁴³² Verified fact — B-C8-S003: Managing IAM access for watsonx.governance as a Service on AWS; factsheets; Enforcement Tracking (IBM) <https://dataplatfom.cloud.ibm.com/docs/content/wsj/openpages/wxgov-aws-iam.html?context=wx> accessed 2026-10-08 [search-tool extract]

¹⁴³³ Verified fact — A7-S103: IBM watsonx.governance product, pricing and model-governance pages; Compliance Accelerators announcement; IBM Cloud catalog <https://www.ibm.com/products/watsonx-governance/pricing> accessed 2026-10-07 [search-tool extract]; B-C8-S002: Introducing FedRAMP-authorized watsonx services on AWS GovCloud; IBM Watson privacy and security POV (2019) <https://www.ibm.com/new/announcements/introducing-fedramp-authorized-watsonx-services-on-aws-govcloud> accessed 2026-10-08 [search-tool extract]

¹⁴³⁴ Verified fact — A7-S103: IBM watsonx.governance product, pricing and model-governance pages; Compliance Accelerators announcement; IBM Cloud catalog <https://www.ibm.com/products/watsonx-governance/pricing> accessed 2026-10-07 [search-tool extract]

¹⁴³⁵ Verified fact — A7-S111: IBM announcements 2026: Enforcement Tracking; AI Asset Discovery; security metrics and agent monitoring; Think 2026 AI assurance <https://www.ibm.com/new/announcements/from-governance-policies-to-governance-proof-with-enforcement-tracking-for-watsonx-orchestrate> accessed 2026-10-07 [search-tool extract]

- You run watsonx Orchestrate or OpenPages, or want one platform for discovery, evaluation and governance of agents ^{AJ}

Avoid when

- You want governance independent of your agent runtime vendor ^{AJ}

Nearest competitors: C8-validmind, C8-modelop, C8-credo-ai

Regulated-FS note. Obtain SOC 2 and ISO scope letters naming watsonx.governance; record IBM as one concentration point if Vault (C7) is also used ^{Rec}.

Facts (as of 2026-10-07)

Field	Value	Label	Sources
Company	IBM	Verified fact	A7-S058
Category	AI governance, evaluation and model risk platform	Verified fact	A7-S058
Version / lineup	Continuous SaaS updates (IBM Cloud catalog updated 20 July 2026); v2.2.0 introduced policy packs; 2026 features: AI Asset Discovery (9 July 2026), Enforcement Tracking (11 August 2026), Guardium security metrics; SDK ibm-watsonx-gov 1.5.2 (18 September 2026)	Verified fact	A7-S103, A7-S111, A7-S058
Licence	Proprietary; SDK under IBM International License Agreement for Non-Warranted Programs	Verified fact	A7-S058
Strategic direction	Agentic governance: automatic retrieval of agent evaluation metrics against governance thresholds, discovery of unmanaged agents, tools, MCP servers and models, regulatory horizon scanning (CUBE) previewed at Think 2026, integration with watsonx Orchestrate	Verified fact	A7-S111
What it does	Inventories AI use cases and models, evaluates models, prompt templates and agents, monitors them in production and ties metrics to risks, controls and approvals; Compliance Accelerators map obligations to controls.	Verified fact	A7-S103, A7-S111, A7-S058
Stack position	C8 governance consuming L9 evaluation/monitoring evidence and C7 security findings (Guardium)	Verified fact	A7-S111
Integration	Python SDK with sample notebooks	Verified fact	A7-S058, A7-S094
Dependencies	Not publicly verified	Not publicly verified	
Certifications	FedRAMP-authorised watsonx.governance on AWS GovCloud (IBM page); SOC 2/ISO not verified	Verified fact	A7-S103
GDPR / residency	Not publicly verified	Not publicly verified	

¹⁴³⁶ Verified fact — A7-S103: IBM watsonx.governance product, pricing and model-governance pages; Compliance Accelerators announcement; IBM Cloud catalog <https://www.ibm.com/products/watsonx-governance/pricing> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
Security features	Not publicly verified	Not publicly verified	A7-S103
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Lite free tier; Essentials pay-as-you-go (model evaluation US\$0.64 per evaluation, about 100 free); GRC per-instance and per-concurrent-user charges; AWS SaaS bundle US\$38,160 (1 instance, 12,000 evaluations/year, 5 use cases, 25 concurrent users); self-managed priced per virtual processor core; indicative, varies by country	Verified fact	
Infrastructure cost	Not publicly verified	Not publicly verified	A7-S103
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	IBM states Leader in 2026 IDC MarketScape for AI-enabled financial GRC (with OpenPages)	Verified fact	
Maturity	Not publicly verified	Not publicly verified	
Deployment	saas: Yes (IBM Cloud, AWS); managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes (software in customer-managed AWS/Azure; licences on both marketplaces); on_prem: Yes (on-premises and hybrid)		

Credo AI AI governance platform (AI Registry, policy packs, Govern AI Assistant) (c8-credo-ai)

Tier: Tactical · **Flags:** none · **Original graphic label:** not in graphic

Rationale: Strong deployment and identity controls for policy-led governance; certification evidence unconfirmed and validation depth limited ^{AI}.

Criterion	Score	Rationale
Technical	3	Inventory, policy and evidence strong; validation and monitoring workflows not evidenced.
Enterprise readiness	4	SSO, SCIM, audit-log APIs and role scopes verified.
Security and compliance	2	Capped at 2: SOC 2 Type II report not re-confirmed (V2 section 4 item 9).
Deployment flexibility	5	EU SaaS, self-hosted and air-gapped.
Ecosystem	4	30+ partner integrations including Azure, watsonx, Databricks, Collibra.
Reliability and maturity	3	Independent; last disclosed round July 2024; Lens deprecated.
Cost / TCO	2	Pricing not published.
Lock-in / portability	3	Proprietary but independent; audit-log APIs give data access.
Total (generic / FS)	3.30 / 3.25	

Evidence rules applied: security_compliance capped at 2: SOC 2 Type II report not re-confirmed (V2 verification log section 4)

Capabilities. AI governance platform: AI Registry with auto-discovery, policy packs (EU AI Act, NIST AI RMF, ISO 42001, SOC 2) from a knowledge graph of 160+ policies and 110+ controls, audit-ready evidence and audit trails; moving to agent registry and runtime governance; AIUC-1 integrated ^{vf1437}.

Strengths

- Widest deployment choice: SaaS on AWS (US) or Azure (EU), self-hosted Kubernetes, air-gapped ^{vf1438}
- SSO via OIDC (SAML via dex), SCIM, audit logs and audit-log APIs, role scopes ^{vf1439}
- Integrations with Azure, watsonx, Databricks, Collibra and 30+ partners ^{vf1440}

Limitations and risks

- SOC 2 Type II report listed but not re-confirmed by the verifier ^{vf1441}; no ISO 27001/42001 found

¹⁴³⁷ Verified fact — A7-S102: Credo AI - Product; Credo AI homepage; AI Governance Platform & Services <https://www.credo.ai/product> accessed 2026-10-07 [search-tool extract]

¹⁴³⁸ Verified fact — A7-S107: Trust Center - Credo AI; Credo AI SOC 2 Type II page <https://trust.credo.ai/> accessed 2026-10-07 [search-tool extract]; A7-S123: Credo AI self-hosted documentation (installation, requirements, SSO with Okta OIDC/SAML, FAQ) <https://docs.selfhost.credo.ai/docs/self-hosted/installation/common/> accessed 2026-10-07 [search-tool extract]

¹⁴³⁹ Verified fact — A7-S123: Credo AI self-hosted documentation (installation, requirements, SSO with Okta OIDC/SAML, FAQ) <https://docs.selfhost.credo.ai/docs/self-hosted/installation/common/> accessed 2026-10-07 [search-tool extract]

¹⁴⁴⁰ Verified fact — A7-S102: Credo AI - Product; Credo AI homepage; AI Governance Platform & Services <https://www.credo.ai/product> accessed 2026-10-07 [search-tool extract]; A7-S110: ModelOp: SR 11-7 Model Risk Management; Collibra blogs (OSFI E-23, SR 26-2); Credo AI blog (financial services) (ModelOp / Collibra / Credo AI) <https://www.modelop.com/ai-governance/ai-regulations-standards/sr-11-7> accessed 2026-10-07 [search-tool extract]

¹⁴⁴¹ Verified fact — A7-S107: Trust Center - Credo AI; Credo AI SOC 2 Type II page <https://trust.credo.ai/> accessed 2026-10-07 [search-tool extract]; V2-S064: Collibra strengthens AI governance leadership with ISO 42001 certification (22 Jan 2025); Credo AI SOC 2 Type II page (Collibra / Credo AI) <https://www.collibra.com/company/newsroom/press-releases/collibra-strengthens-ai-governance-leadership-with-iso-42001-certification-ai-pact->

- No SR 26-2 or SS1/23 mapping found; light on model validation ^{VF¹⁴⁴² AJ}
- Open-source Lens deprecated (last release May 2023) ^{VF¹⁴⁴³}
- Pricing not published ^{NPV}

Choose when

- The programme is policy- and compliance-led across many business units ^{AJ}

Avoid when

- The main need is quantitative model validation ^{AJ}

Nearest competitors: C8-collibra-ai-governance, C8-watsonx-governance, C8-modelop

Regulated-FS note. Self-host or use the Azure EU instance; obtain the current SOC 2 report directly ^{REC}.

Facts (as of 2026-10-07)

Field	Value	Label	Sources
Company	Credo AI	Verified fact	A7-S092, A7-S093
Category	AI governance platform (registry, policy packs, evidence, third-party AI risk)	Verified fact	A7-S102, A7-S110
Version / lineup	Not publicly verified	Not publicly verified	
Licence	Proprietary (SaaS and self-hosted); open-source Lens deprecated	Verified fact	A7-S107, A7-S123, A7-S092
Status events	Open-source ‘Lens’ responsible-AI assessment framework marked deprecated / ‘no longer maintained’; last PyPI release 1.1.8 on 3 May 2023	Verified fact	A7-S092, A7-S093
Strategic direction	Moving to agent governance: Agent Registry, runtime governance and Govern AI Assistant (GA in 2026); AIUC-1 agent standard integrated	Verified fact	A7-S102

^{commitment} accessed 2026-10-08 [search-tool extract]

¹⁴⁴² Verified fact — A7-S110: ModelOp: SR 11-7 Model Risk Management; Collibra blogs (OSFI E-23, SR 26-2); Credo AI blog (financial services) (ModelOp / Collibra / Credo AI) <https://www.modelop.com/ai-governance/ai-regulations-standards/sr-11-7> accessed 2026-10-07 [search-tool extract]

¹⁴⁴³ Verified fact — A7-S092: Lens by Credo AI README https://raw.githubusercontent.com/credo-ai/credoai_lens/develop/README.md accessed 2026-10-07; A7-S093: credoai-lens on PyPI (PyPI / Credo AI) <https://pypi.org/pypi/credoai-lens/json> accessed 2026-10-07

Field	Value	Label	Sources
	(Credo does not issue certification); policy packs for OMB M-25, Colorado ADMT, NAIC AI		
What it does	Auto-discovers and catalogues AI systems in an AI Registry, applies pre-built policy packs (EU AI Act, NIST AI RMF, ISO 42001, SOC 2) from a knowledge graph of 160+ policies and 110+ controls, and generates audit-ready evidence and audit trails.	Verified fact	A7-S102
Stack position	C8 enterprise AI governance (policy/compliance evidence) rather than model-level validation; vendor contrasts this with SR 11-7 model risk management	Verified fact	A7-S110
Integration	Integrations with Microsoft Azure, IBM watsonx, Databricks, Collibra and 30+ partners; audit-log APIs; SCIM provisioning	Verified fact	A7-S102, A7-S110, A7-S123
Dependencies	Not publicly verified	Not publicly verified	
Certifications	SOC 2 Type II (report dated 31 December 2025; Q3 2026 bridge letter) covering security, availability, confidentiality; no ISO 27001/42001 certificate found	Verified fact	A7-S107
GDPR / residency	EU hosting on Microsoft Azure in Europe; US hosting on AWS	Verified fact	A7-S107
Security features	Not publicly verified	Not publicly verified	
Access controls	SSO via OIDC (SAML via dex) for self-hosted; SCIM; audit logs and audit-log APIs; role scopes in tokens	Verified fact	A7-S123
Enterprise support	Advisory services; vendor claims governance workflows stood up in 4 weeks	Verified fact	A7-S102
Pricing	Not publicly verified	Not publicly verified	
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption	Vendor-stated: Visionary in Gartner's inaugural Magic Quadrant for AI	Verified	A7-S102

Field	Value	Label	Sources
signals	Governance Platforms; Leader per Forrester and IDC; Fast Company Most Innovative 2026	d fact	
Maturity	Not publicly verified	Not publicly verified	
Deployment	saas: Yes (AWS US; Azure EU); managed_cloud: Not publicly verified; vpc_byoc: Not publicly verified; private_cloud: Not publicly verified; self_hosted: Yes (Kubernetes, EKS validated, Replicated); on_prem: Yes (air-gapped install supported)		

Collibra AI Governance / AI Command Center (Collibra ‘Enterprise AI Control Plane’) (c8-collibra-ai-governance)

Tier: Tactical · Flags: none · Original graphic label: not in graphic

Rationale: Strong certifications and the lineage bridge, but SaaS-only deployment and no validation workflow

Criterion	Score	Rationale
Technical	4	Inventory, assessments, controls mapping and lineage link; no validation or monitoring workflow.
Enterprise readiness	3	SAML SSO and roles verified (B-C8-S004); audit logs not documented, so 3 (rule 7).
Security and compliance	4	Company-level certifications meeting the 5 anchor, product scope not stated: 4 (rule 8).
Deployment flexibility	2	SaaS on AWS and GCP only; no self-host or VPC evidence.
Ecosystem	4	SageMaker, Azure AI Foundry and Databricks connectors; CLI; OpenLineage.

Criterion	Score	Rationale
Reliability and maturity	3	Established vendor; AI module repositioned and acquisition just announced.
Cost / TCO	2	Pricing not published.
Lock-in / portability	3	Proprietary SaaS; OpenLineage integration eases lineage portability.
Total (generic / FS)	3.20 / 3.20	

Capabilities. AI governance on Collibra's data intelligence platform: AI use-case register, model and agent asset domains linked to model versions and agents from integrated AI platforms, EU AI Act and NIST AI RMF assessment templates, 46 controls mapped to EU AI Act, NIST and BCBS 239; OpenLineage integration; trail ML acquisition (announced 5 October 2026) for continuous assessment and runtime enforcement ^{VF1444}.

Strengths

- Only product joining the AI inventory to the data catalogue and lineage (OpenLineage) ^{VF1445 AI}
- Company-level SOC 1/2, ISO 27001/27017/27018, ISO 42001, FedRAMP, HIPAA ^{VF1446}
- SAML 2.0 SSO with IdP group mapping and global roles ^{VF1447}
- BCBS 239 control mapping useful for bank-owned managers ^{VF1448}

Limitations and risks

- SaaS only on current evidence (AWS and GCP); self-hosting and EU region list not verified ^{VF1449 NPV}
- No model validation or monitoring workflow ^{AI}
- trail ML deal three days old; integration roadmap open ^{VF1450}
- Pricing not published ^{NPV}

¹⁴⁴⁴ Verified fact — A7-S099: About Collibra AI Governance (product docs) and AI Command Center product page https://productresources.collibra.com/docs/collibra/latest/Content/AIGovernance/co_about-ai-governance.htm accessed 2026-10-07 [search-tool extract]; A7-S122: Collibra Platform Cloud infrastructure and base components; Collibra Cloud Sites and OpenLineage integration blog; Collibra homepage https://productresources.collibra.com/docs/collibra/latest/Content/Cloud/Intro/co_cloud-introduction.htm accessed 2026-10-07 [search-tool extract]; A7-S101: Collibra acquires trail ML to automate AI governance from policy to production <https://www.collibra.com/company/newsroom/press-releases/collibra-acquires-trail-ml-to-automate-ai-governance-from-policy-to-production> accessed 2026-10-07 [search-tool extract]

¹⁴⁴⁵ Verified fact — A7-S122: Collibra Platform Cloud infrastructure and base components; Collibra Cloud Sites and OpenLineage integration blog; Collibra homepage https://productresources.collibra.com/docs/collibra/latest/Content/Cloud/Intro/co_cloud-introduction.htm accessed 2026-10-07 [search-tool extract]; A7-S099: About Collibra AI Governance (product docs) and AI Command Center product page https://productresources.collibra.com/docs/collibra/latest/Content/AIGovernance/co_about-ai-governance.htm accessed 2026-10-07 [search-tool extract]

¹⁴⁴⁶ Verified fact — A7-S106: Collibra trust center: Security and compliance commitment <https://www.collibra.com/company/trust-center> accessed 2026-10-07 [search-tool extract]; A7-S100: Collibra Strengthens AI Governance Leadership with ISO 42001 Certification, AI Pact Commitment and the Delivery of the EU AI Act Assessment Tool <https://www.collibra.com/company/newsroom/press-releases/collibra-strengthens-ai-governance-leadership-with-iso-42001-certification-ai-pact-commitment> accessed 2026-10-07 [search-tool extract]; V2-S064: Collibra strengthens AI governance leadership with ISO 42001 certification (22 Jan 2025); Credo AI SOC 2 Type II page (Collibra / Credo AI) <https://www.collibra.com/company/newsroom/press-releases/collibra-strengthens-ai-governance-leadership-with-iso-42001-certification-ai-pact-commitment> accessed 2026-10-08 [search-tool extract]

¹⁴⁴⁷ Verified fact — B-C8-S004: Collibra Single Sign-On (SSO) documentation https://productresources.collibra.com/docs/collibra/latest/Content/Console/Infrastructure/DGCSecurityConfiguration/co_single-sign-on.htm accessed 2026-10-08 [search-tool extract]

¹⁴⁴⁸ Verified fact — A7-S099: About Collibra AI Governance (product docs) and AI Command Center product page https://productresources.collibra.com/docs/collibra/latest/Content/AIGovernance/co_about-ai-governance.htm accessed 2026-10-07 [search-tool extract]

¹⁴⁴⁹ Verified fact — A7-S122: Collibra Platform Cloud infrastructure and base components; Collibra Cloud Sites and OpenLineage integration blog; Collibra homepage https://productresources.collibra.com/docs/collibra/latest/Content/Cloud/Intro/co_cloud-introduction.htm accessed 2026-10-07 [search-tool extract]

Choose when

- Collibra is already the data catalogue of record ^{AJ}

Avoid when

- You need self-hosting or a model-validation workflow ^{AJ}

Nearest competitors: C8-credo-ai, C8-watsonx-governance, C8-modelop

Regulated-FS note. Use the BCBS 239 mapping for bank-owned managers; confirm EU hosting and the trail ML integration roadmap ^{REC}.

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	Collibra (acquired trail ML, Munich, announced 5 October 2026)	Verified fact	A7-S105, A7-S101
Category	AI governance on a data intelligence/governance platform (AI use-case registry, assessments, controls)	Verified fact	A7-S099, A7-S122
Version / lineup	Not publicly verified	Not publicly verified	
Licence	Proprietary SaaS	Verified fact	A7-S122
Status events	January 2025: ISO 42001 certification (Schellman); AI Pact signatory; 5 October 2026: acquisition of trail ML announced (price undisclosed)	Verified fact	A7-S100, A7-S105, V2-S044, V2-S064
Strategic direction	Repositioned as 'The Enterprise AI Control Plane'; adding agent-powered continuous assessment and runtime enforcement via trail ML; code-first registry CLI; AIUC-1 templates (May 2026)	Verified fact	A7-S122, A7-S105, A7-S099
What it does	Provides an AI use-case register and model/agent asset domains, links use cases to model versions and agents from integrated AI platforms, runs EU AI Act and NIST AI RMF assessment templates and 46 controls mapped to EU AI Act, NIST and BCBS 239 articles.	Verified fact	A7-S099
Stack position	C8 governance anchored in the data catalogue and lineage layer (bridges H7 data lineage and AI inventory)	Verified fact	A7-S099, A7-S122
Integration	Platform-agnostic connectors to AWS SageMaker, Azure AI Foundry, Databricks; developer CLI manifests; OpenLineage integration	Verified fact	A7-S122, A7-S099
Dependencies	Not publicly verified	Not publicly verified	
Certifications	SOC 1, SOC 2, ISO 27001, 27017, 27018, ISO 42001, FedRAMP, ITAR, HIPAA, TISAX; CSA STAR self-assessment	Verified fact	A7-S106, A7-S100,

¹⁴⁵⁰ Verified fact — V2-S044: Collibra acquires Munich's trail ML to automate AI governance (with unite.ai, citybiz) (The Next Web) <https://thenextweb.com/news/collibra-trail-ml-eu-ai-act-market> accessed 2026-10-08 [search-tool extract]

Field	Value	Label	Sources
			V2-S064
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Not publicly verified	Not publicly verified	
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	
Adoption signals	Not publicly verified	Not publicly verified	
Maturity	Not publicly verified	Not publicly verified	
Deployment	saas: Yes (managed on AWS and GCP); managed_cloud: Yes (Cloud Sites); vpc_byoc: Not publicly verified; private_cloud: Not publicly		

Field	Value	Label	Sources
	verified; self_hosted: Not publicly verified; on_prem: Not publicly verified		

ModelOp Center (positioned as ‘Enterprise AI Command Center’) (c8-modelop)

Tier: Tactical · **Flags:** none · **Original graphic label:** not in graphic

Rationale: Credible automation claims but the thinnest public evidence in the set ^{AJ}.

Criterion	Score	Rationale
Technical	4	Inventory, intake, tiering, approvals, templates, tests and blocking workflows (vendor-described).
Enterprise readiness	3	SAML/OAuth2 IdP integration and role-based access verified (rule 7, maximum 3); audit logs not documented.
Security and compliance	2	Capped at 2: no certifications publicly verified.
Deployment flexibility	4	Self-hosted, on-prem, hybrid and EKS; SaaS not verified.
Ecosystem	3	Connectors and REST APIs; SageMaker, S3, Redshift.
Reliability and maturity	2	Maturity and release cadence not publicly verified; small disclosed funding.
Cost / TCO	2	List pricing not published.
Lock-in / portability	3	Proprietary but self-hostable; independent.
Total (generic / FS)	3.00 / 2.95	

Evidence rules applied: security_compliance capped at 2: certifications not publicly verified (B-C8-S001)

Capabilities. AI governance and lifecycle automation (‘Enterprise AI Command Center’): inventory with self-service intake, risk tiering and approvals; 25+ governance process templates and 100+ tests and controls; risk-based workflows that can block non-compliant actions; claimed inline protections for agents ^{VF¹⁴⁵¹}.

Strengths

- Lifecycle automation that can enforce, not only record ^{VF¹⁴⁵² AJ}
- Self-hosted, on-prem, hybrid and EKS via AWS Marketplace ^{VF¹⁴⁵³}
- Identity-provider integration via OAuth2 and SAML with role-based access ^{VF¹⁴⁵⁴}

¹⁴⁵¹ Verified fact — A7-S098: ModelOp: Automate AI Governance Workflows & Enforce Compliance; Enterprise AI Command Center; AI Lifecycle Automation <https://www.modelop.com/ai-governance-software/controls> accessed 2026-10-07 [search-tool extract]

¹⁴⁵² Verified fact — A7-S098: ModelOp: Automate AI Governance Workflows & Enforce Compliance; Enterprise AI Command Center; AI Lifecycle Automation <https://www.modelop.com/ai-governance-software/controls> accessed 2026-10-07 [search-tool extract]

¹⁴⁵³ Verified fact — A7-S109: ModelOp Now Available on AWS Marketplace; ModelOp homepage <https://www.modelop.com/blog/modelop-now-available-on-aws-marketplace> accessed 2026-10-07 [search-tool extract]

¹⁴⁵⁴ Verified fact — B-C8-S001: ModelOp AI Governance Software Integrations <https://www.modelop.com/ai-governance-software/integrations> accessed 2026-10-08 [search-tool extract]

Limitations and risks

- No SOC 2 or ISO 27001 evidence found ^{VF1455} ^{NPV}
- Maturity, release cadence and SaaS availability not publicly verified ^{NPV}
- Small disclosed funding (US\$10m, date not confirmed) ^{VF1456}
- SR 11-7 templates map to a superseded instrument ^{VF1457} ^{AJ}

Choose when

- You want self-hosted governance automation with blocking workflows and will do full due diligence ^{AJ}

Avoid when

- Certification evidence is a shortlisting gate ^{AJ}

Nearest competitors: C8-watsonx-governance, C8-validmind, C8-credo-ai

Regulated-FS note. Require SOC 2 Type II and audit-log documentation before any pilot with production records ^{REC}.

Facts (as of 2026-10-07)

Field	Value	Label	Sources
Company	ModelOp (privately held, Chicago; raised US\$10m led by Baird Capital - date not confirmed)	Verified fact	A7-S104
Category	AI governance and AI lifecycle automation software	Verified fact	A7-S098
Version / lineup	Not publicly verified	Not publicly verified	
Licence	Proprietary	Verified fact	A7-S109
Strategic direction	Governs ML, GenAI, agentic, internal and third-party AI from intake to retirement; inline protections for agentic systems (prompt injection, PII leakage, unsafe tool use); vendor claims Visionary in 2026 Gartner Magic Quadrant for AI Governance Platforms	Verified fact	A7-S098, A7-S104

¹⁴⁵⁵ Verified fact — B-C8-S001: ModelOp AI Governance Software Integrations <https://www.modelop.com/ai-governance-software/integrations> accessed 2026-10-08 [search-tool extract]

¹⁴⁵⁶ Verified fact — A7-S104: ModelOp Raises \$10 Million to accelerate the growth of its leading AI Governance software, led by investor Baird Capital; ModelOp Center product page <https://www.modelop.com/blog/a-milestone-moment-modelop-raises-10-million-to-accelerate-the-growth-of-its-leading-ai-governance-software> accessed 2026-10-07 [search-tool extract]

¹⁴⁵⁷ Verified fact — A7-S110: ModelOp: SR 11-7 Model Risk Management; Collibra blogs (OSFI E-23, SR 26-2); Credo AI blog (financial services) (ModelOp / Collibra / Credo AI) <https://www.modelop.com/ai-governance/ai-regulations-standards/sr-11-7> accessed 2026-10-07 [search-tool extract]

Field	Value	Label	Sources
What it does	Maintains an inventory of AI systems with self-service intake, risk tiering and approvals; applies controls from internal policies and regulations via 25+ governance process templates and 100+ tests/controls; risk-based workflows can block non-compliant actions.	Verified fact	A7-S098
Stack position	C8 governance control plane with lifecycle automation; vendor positions Collibra as data governance and Credo as compliance-focused	Verified fact	A7-S098, A7-S110
Integration	Out-of-the-box connectors and REST APIs; AWS SageMaker, S3, Redshift, DocumentDB	Verified fact	A7-S104, A7-S109
Dependencies	Not publicly verified	Not publicly verified	
Certifications	Not publicly verified	Not publicly verified	
GDPR / residency	Not publicly verified	Not publicly verified	
Security features	Not publicly verified	Not publicly verified	
Access controls	Not publicly verified	Not publicly verified	
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Available through AWS Marketplace; list pricing not published	Verified fact	A7-S109
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Not publicly verified	Not publicly verified	

Field	Value	Label	Sources
Adoption signals	Not publicly verified	verified	Not publicly verified
Maturity	Not publicly verified	Not publicly verified	
Deployment	saas: Not publicly verified; managed_cloud: Not publicly verified; vpc_byoc: Yes (Kubernetes/Amazon EKS via AWS Marketplace); private_cloud: Not publicly verified; self_hosted: Yes; on_prem: Yes (homepage lists Cloud, On-Premise, Hybrid)		

ValidMind AI risk platform (ValidMind Platform and ValidMind Library) (c8-validmind)

Tier: Tactical · **Flags:** none · **Original graphic label:** not in graphic

Rationale: Best MRM workflow fit and the only SS1/23 claimant, held back by unverified certification evidence and unpublished pricing ^{AJ}.

Criterion	Score	Rationale
Technical	4	Inventory, documentation, validation, approvals, monitoring thresholds and attestation; lineage and agent discovery not evidenced.
Enterprise readiness	3	RBAC and audit logs verified; SSO not verified, so maximum 3 (rule 7).
Security and compliance	2	Capped at 2: SOC 2 type not stated and no report found; vendor statement only.
Deployment flexibility	3	Multi-tenant SaaS plus single-tenant VPV in the major clouds; self-hosting not verified.
Ecosystem	3	Python Library with LLM, PyTorch and Hugging Face extras; data-source connectors.
Reliability and	3	Frequent Library releases since May 2023; funding not verified.

Criterion	Score	Rationale
maturity		
Cost / TCO	2	Pricing not published.
Lock-in / portability	3	Proprietary platform holding inventory and documentation; export not verified; AGPL library noted.
Total (generic / FS)	2.95	
	/	
	2.90	

Evidence rules applied: security_compliance capped at 2: SOC 2 Type II not publicly verified (vendor claim, type not stated)

Capabilities. Model risk management platform: Python Library logs tests and documentation artefacts; Platform holds a customisable model inventory and runs documentation, validation workflows, approvals and ongoing monitoring with thresholds, alerts and workflow triggers; LLM test extras and LLM features; point-in-time model attestation ^{vf1458}.

Strengths

- Closest fit to bank-style MRM (inventory, documentation, validation, monitoring) with GenAI tests ^{aj}
- Only vendor in the set claiming PRA SS1/23 mapping; also SR 26-2, EU AI Act Article 9 and OSFI E-23 (vendor claims) ^{vf1459}
- Single-tenant Virtual Private ValidMind on AWS, GCP or Azure with private connectivity ^{vf1460}
- RBAC, audit logs, data isolation and encryption documented ^{vf1461}

Limitations and risks

- SOC 2 claimed without type; no SOC 2 Type II report or ISO certificate found ^{vf1462}
- SSO and self-hosting not publicly verified ^{nrv}
- Library is AGPL-3.0 or commercial (licence consideration under rule 4) ^{vf1463}

¹⁴⁵⁸ Verified fact — A7-S003: validmind on PyPI (JSON API) (PyPI / ValidMind) <https://pypi.org/pypi/validmind/json> accessed 2026-10-07; A7-S054: ValidMind docs: deployment available options https://raw.githubusercontent.com/validmind/documentation/main/site/about/deployment/_deployment-available-options.qmd accessed 2026-10-07; A7-S056: ValidMind docs: ongoing monitoring <https://raw.githubusercontent.com/validmind/documentation/main/site/guide/monitoring/ongoing-monitoring.qmd> accessed 2026-10-07; A7-S097: ValidMind docs: Large language model features <https://raw.githubusercontent.com/validmind/documentation/main/site/about/overview-llm-features.qmd> accessed 2026-10-07; B-C8-S005: ValidMind docs: Working with attestations; platform page; privacy FAQ <https://docs.validmind.ai/guide/attestation/working-with-attestations.html> accessed 2026-10-08 [search-tool extract]

¹⁴⁵⁹ Verified fact — A7-S051: ValidMind docs: SR 26-2 use case <https://raw.githubusercontent.com/validmind/documentation/main/site/about/use-cases/sr-26-2.qmd> accessed 2026-10-07; A7-S052: ValidMind docs: SS1/23 use case <https://raw.githubusercontent.com/validmind/documentation/main/site/about/use-cases/ss1-23.qmd> accessed 2026-10-07; A7-S053: ValidMind docs: EU AI Act use case <https://raw.githubusercontent.com/validmind/documentation/main/site/about/use-cases/eu-ai-act.qmd> accessed 2026-10-07; A7-S057: ValidMind docs: OSFI E-23 use case <https://raw.githubusercontent.com/validmind/documentation/main/site/about/use-cases/e-23.qmd> accessed 2026-10-07

¹⁴⁶⁰ Verified fact — A7-S054: ValidMind docs: deployment available options https://raw.githubusercontent.com/validmind/documentation/main/site/about/deployment/_deployment-available-options.qmd accessed 2026-10-07; A7-S084: ValidMind docs: Virtual Private ValidMind <https://raw.githubusercontent.com/validmind/documentation/main/site/about/deployment/virtual-private-validmind.qmd> accessed 2026-10-07

¹⁴⁶¹ Verified fact — A7-S055: ValidMind docs: multi-tenant cloud <https://raw.githubusercontent.com/validmind/documentation/main/site/about/deployment/multi-tenant-cloud.qmd> accessed 2026-10-07

¹⁴⁶² Verified fact — A7-S121: ValidMind AI Governance Platform page; ValidMind blog (NIST, RiskTech100) <https://validmind.com/platform/> accessed 2026-10-07 [search-tool extract]; B-C8-S005: ValidMind docs: Working with attestations; platform page; privacy FAQ <https://docs.validmind.ai/guide/attestation/working-with-attestations.html> accessed 2026-10-08 [search-tool extract]

¹⁴⁶³ Verified fact — A7-S003: validmind on PyPI (JSON API) (PyPI / ValidMind) <https://pypi.org/pypi/validmind/json> accessed 2026-10-07

- Pricing not published ^{W-1464}

Choose when

- A second-line MRM function wants GenAI in the same validation workflow as other models ^{AJ}

Avoid when

- You need a broad AI policy and third-party AI register more than validation, or cannot obtain a SOC 2 Type II report ^{AJ}

Nearest competitors: C8-watsonx-governance, C8-modelop, C8-credo-ai

Regulated-FS note. Request the SOC 2 Type II report before loading risk records; check AGPL obligations for any modified Library code you distribute ^{Rec.}

Facts (as of 2026-10-08)

Field	Value	Label	Sources
Company	ValidMind Inc.	Verified fact	A7-S051
Category	Model risk management and AI governance platform (inventory, documentation, validation, monitoring)	Verified fact	A7-S003, A7-S054
Version / lineup	ValidMind Library 2.13.14 released 3 September 2026; Platform is cloud-hosted	Verified fact	A7-S003, V2-S028
Licence	Open core: Library dual-licensed AGPL-3.0 or ValidMind Commercial Licence; Platform proprietary	Verified fact	A7-S003, A7-S051
Strategic direction	Use-case guides map platform workflows to SR 26-2 (successor to SR 11-7), PRA SS1/23, EU AI Act and OSFI E-23; LLM features for test interpretation and document checking; LLM test extras in the Library	Verified fact	A7-S051, A7-S052, A7-S053, A7-S057, A7-S097, A7-S003
What it does	The Library runs tests and logs documentation artefacts from Python model-development environments; the Platform holds a customisable model inventory and runs documentation, validation workflows, approvals and ongoing monitoring with thresholds, alerts and workflow triggers on breach.	Verified fact	A7-S003, A7-S054, A7-S056
Stack position	C8 governance system of record fed by development tests and monitoring results	Verified fact	A7-S003, A7-S056
Integration	Python Library and ValidMind API to Platform; private connectivity via AWS PrivateLink, Google Cloud Private Service Connect, Azure Private Link	Verified fact	A7-S054
Dependencies	Python environment for the Library; data sources such as GCS, S3, Snowflake in customer environment	Verified fact	A7-S003, A7-S054
Certifications	Claims to meet SOC 2, GDPR and CCPA requirements (platform page; SOC 2 type not stated); no ISO 27001/42001 certificate found	Verified fact	A7-S121
GDPR / residency	Not publicly verified	Not publicly	

¹⁴⁶⁴ Verified fact — A7-S003: validmind on PyPI (JSON API) (PyPI / ValidMind) <https://pypi.org/pypi/validmind/json> accessed 2026-10-07

Field	Value	Label	Sources
		verified	
Security features	Data isolation, encryption in transit and at rest, private link; states PII and customer data are not stored in documentation	Verified fact	A7-S055, A7-S054
Access controls	Role-based access control; audit logs; continuous monitoring and logging	Verified fact	A7-S055
Enterprise support	Not publicly verified	Not publicly verified	
Pricing	Free account registration to start; commercial pricing not published	Verified fact	A7-S003
Infrastructure cost	Not publicly verified	Not publicly verified	
Ecosystem	Library extras for LLMs, PyTorch and Hugging Face Transformers	Verified fact	A7-S003
Adoption signals	validmind package first published on PyPI 11 May 2023; vendor states Chartis ranked it No. 1 AI Governance Platform in RiskTech100 2026	Verified fact	A7-S003, A7-S121
Maturity	Frequent Library releases (2.13.x in September 2026)	Verified fact	A7-S003
Deployment	saas: Yes (multi-tenant cloud); managed_cloud: Not publicly verified; vpc_byoc: Yes (Virtual Private ValidMind, single-tenant, on AWS, GCP or Azure); private_cloud: Not publicly verified; self_hosted: Not publicly verified; on_prem: Not publicly verified		